A DIRECT OBSTACLE PROOF OF A DIMENSION-FREE WEAK-TYPE $(1,1)$ BOUND FOR THE GAUSSIAN RIESZ TRANSFORMS: BLUEPRINT OF THE LEAN FORMALIZATION

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Abstract. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of the main results of C. E. Gutiérrez and J. J. Manfredi, *A direct obstacle proof of a dimension-free weak-type* $(1,1)$ *bound for the Gaussian Riesz transforms* (arXiv:2610.26478). Each subsection corresponds to the Lean declaration of the same name in the directory `GaussianRiesz/`; cross-references point to the results that a proof uses. Section 3 lists the formalized results of the paper: Theorem 3.1, Theorem 5.1, Proposition 5.3 and Corollaries 5.4 and 5.5. Appendix A records the one place where the formalization departs from the paper's argument: the proof of Proposition 5.3 does not assume the maximal estimate (5.15) of Fabes, Gutiérrez and Scotto, and proves a localized form of it instead. Within each section, items appear in dependency order.

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1. Definitions

1.1. `Preamble`.

Definition 1.1 (The probabilists' Hermite polynomials)**.** For $k \in \mathbb{N}_0$ let $\mathrm{He}_k \in \mathbb{Z}[X]$ be the probabilists' Hermite polynomials, defined by the recurrence

$$\mathrm{He}_0 = 1, \qquad \mathrm{He}_{k+1} = X\,\mathrm{He}_k - \mathrm{He}_k',$$

where $p \mapsto p'$ is formal differentiation of polynomials. Thus $\mathrm{He}_1 = X$, $\mathrm{He}_2 = X^2 - 1$, and in general He_k is monic of degree k with integer coefficients. This recurrence is the definition of the family: it is the imported background object on which the tablet's Hermite constructions are built, and nothing beyond the recurrence and the two displayed initial values is assumed here. These are the polynomials orthogonal with respect to the weight $e^{-x^2/2}$ on $\mathbb{R}$; the rescaling to the weight e^{-x^2} used by the reference paper, and the normalization of the resulting family, are carried out in the tablet's own definition nodes and are not background.

Proposition 1.2 (Tonelli, Fubini, and Fubini for null sets)**.** *Let (α, μ) and (β, ν) be σ-finite measure spaces and let $f : \alpha \times \beta \to \mathbb{R}$ be measurable for the product σ-algebra.*

1. *(Tonelli) If $f \geq 0$ then $\int f\, d(\mu \otimes \nu) = \iint f(x,y)\, d\nu(y)\, d\mu(x) = \iint f(x,y)\, d\mu(x)\, d\nu(y)$ as elements of $[0, +\infty]$.*
2. *(Fubini) If f is $(\mu \otimes \nu)$-integrable then for μ-a.e. x the slice $y \mapsto f(x,y)$ is ν-integrable, the function $x \mapsto \int f(x,y)\, d\nu(y)$ is μ-integrable, and the three integrals in (1) agree as real numbers.*
3. *(Fubini for null sets) If $N \subseteq \alpha \times \beta$ is measurable with $(\mu \otimes \nu)(N) = 0$, then for μ-a.e. x the slice $N_x = \{y : (x,y) \in N\}$ satisfies $\nu(N_x) = 0$.*

Proposition 1.3 (Dominated convergence)**.** *Let (α, μ) be a measure space, let $f_k : \alpha \to \mathbb{R}$ be measurable with $f_k \to f$ μ-a.e., and suppose there is a μ-integrable $g \geq 0$ with $|f_k| \leq g$ μ-a.e. for every k. Then f and every f_k are μ-integrable, $\int |f_k - f|\, d\mu \to 0$, and $\int f_k\, d\mu \to \int f\, d\mu$.*

Proposition 1.4 (The scalar Gaussian integral)**.** *$\int_{\mathbb{R}} e^{-x^2}\, dx = \sqrt{\pi}$, and more generally $\int_{\mathbb{R}} e^{-bx^2}\, dx = \sqrt{\pi/b}$ for every real $b > 0$.*

Proposition 1.5 (Gaussian exponential moments)**.** *For every $n \geq 1$ and every $t \in \mathbb{R}$,*

$$\int_{\mathbb{R}^n} e^{t|x|}\, e^{-|x|^2}\, dx < \infty,$$

the integral being taken with respect to Lebesgue measure. More generally $\int_{\mathbb{R}^n} |P(x)|\, e^{t|x|}\, e^{-|x|^2}\, dx < \infty$ for every polynomial P, since $|P(x)| e^{t|x|} \leq C_P e^{(t+1)|x|}$ for a constant C_P.

Proposition 1.6 (Cauchy–Schwarz)**.** *Let H be a real inner-product space. Then $|\langle x, y\rangle| \leq \|x\|\, \|y\|$ for all $x, y \in H$. In integral form: if (α, μ) is a measure space and $f, g \in L^2(\mu)$ are real valued, then the product fg is μ-integrable and $\left|\int fg\, d\mu\right| \leq \|f\|_{L^2(\mu)}\, \|g\|_{L^2(\mu)}$.*

Proposition 1.7 (Jensen's inequality for a probability measure)**.** *Let μ be a probability measure on α, let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex, and let $f \in L^1(\mu)$ be real valued with $\varphi \circ f \in L^1(\mu)$. Then $\varphi\big(\int f\, d\mu\big) \leq \int \varphi \circ f\, d\mu$.*

Proposition 1.8 (Riesz representation in a Hilbert space)**.** *Let H be a real Hilbert space and let $\Lambda : H \to \mathbb{R}$ be a continuous linear functional. Then there is a unique $y \in H$ with $\Lambda(x) = \langle y, x\rangle$ for every $x \in H$, and $\|\Lambda\|_{H^*} = \|y\|_H$.*

Proposition 1.9 (Uniqueness of characteristic functions)**.** *Let μ and ν be finite Borel measures on $\mathbb{R}^n$ with*

$$\int_{\mathbb{R}^n} e^{i\langle \xi, x\rangle}\, d\mu(x) = \int_{\mathbb{R}^n} e^{i\langle \xi, x\rangle}\, d\nu(x) \qquad \text{for every } \xi \in \mathbb{R}^n.$$

Then $\mu = \nu$.

Proposition 1.10 (Smoothness of a mollification, and the du Bois-Reymond lemma)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be locally integrable for Lebesgue measure.*

(1) *If $\rho \in C_c^\infty(\mathbb{R}^n)$, then the convolution*

$$(\rho * f)(x) = \int_{\mathbb{R}^n} \rho(t)\, f(x - t)\, dt$$

is defined for every $x \in \mathbb{R}^n$ and belongs to $C^\infty(\mathbb{R}^n)$.

(2) *If $\int_{\mathbb{R}^n} \varphi(x)\, f(x)\, dx = 0$ for every $\varphi \in C_c^\infty(\mathbb{R}^n)$, then $f = 0$ Lebesgue-almost everywhere.*

Item (2) is the fundamental lemma of the calculus of variations (du Bois-Reymond). No L^p convergence of mollifications is asserted here; that is a separate statement with a proof of its own, and it has its own tablet node.

Proposition 1.11 (Borel–Cantelli)**.** *Let μ be a measure on α and let $(A_k)_{k\in\mathbb{N}}$ be measurable sets with $\sum_k \mu(A_k) < \infty$. Then $\mu\big(\limsup_k A_k\big) = 0$; equivalently, μ-almost every point lies in only finitely many A_k.*

Proposition 1.12 (Fatou's lemma)**.** *Let (α, μ) be a measure space and let $f_k : \alpha \to [0, +\infty]$ be measurable. Then $\int \liminf_k f_k\, d\mu \le \liminf_k \int f_k\, d\mu$.*

Proposition 1.13 (Entire functions and uniqueness of power-series coefficients)**.** *Let $f : \mathbb{C} \to \mathbb{C}$ be entire. Then f is represented on all of $\mathbb{C}$ by its Taylor series at 0, $f(z) = \sum_{k\ge 0} c_k z^k$ with $c_k = f^{(k)}(0)/k!$, the series converging absolutely and locally uniformly. Moreover the coefficients are unique: if $\sum_k c_k z^k = \sum_k d_k z^k$ for all z in some neighbourhood of 0, with both series convergent there, then $c_k = d_k$ for every k.*

Proposition 1.14 (σ-finiteness of Lebesgue measure)**.** *Lebesgue measure on $\mathbb{R}^n$ is σ-finite: $\mathbb{R}^n$ is the union of the countably many balls $B(0, k)$, $k \in \mathbb{N}$, each of finite Lebesgue measure. The same therefore holds for any measure obtained from Lebesgue measure by multiplying by a locally bounded density.*

Proposition 1.15 (Density of compactly supported continuous functions in L^p)**.** *Let μ be a regular Borel measure on $\mathbb{R}^n$ (Lebesgue measure is one), let $1 \le p < \infty$ and let $f \in L^p(\mu)$ be real valued. For every $\eta > 0$ there is $g : \mathbb{R}^n \to \mathbb{R}$ which is continuous, has compact support, lies in $L^p(\mu)$, and satisfies $\|f - g\|_{L^p(\mu)} \le \eta$. Any such g is bounded and uniformly continuous on $\mathbb{R}^n$: it is continuous, and if S is a compact set containing* $\operatorname{supp} g$ *in its interior then g is uniformly continuous on the compact set S by the Heine–Cantor theorem and vanishes identically outside* $\operatorname{supp} g$*, which gives a single modulus of continuity valid on all of $\mathbb{R}^n$.*

Proposition 1.16 (Translation invariance of Lebesgue measure)**.** *Let λ denote Lebesgue measure on $\mathbb{R}^n$ and let $t \in \mathbb{R}^n$. Then $\lambda(A + t) = \lambda(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}^n$, and consequently, for every measurable $h : \mathbb{R}^n \to [0, +\infty]$ and every measurable A,*

$$\int_A h(x - t)\, dx = \int_{A-t} h(y)\, dy,$$

both sides being equal in $[0, +\infty]$; the same identity holds for λ-integrable real-valued h. Lebesgue measure is also invariant under the reflection $x \mapsto -x$, so $\int h(-x)\, dx = \int h(x)\, dx$ for every such h; combining the two invariances, the map $t \mapsto x - t$ preserves λ for every fixed x, and $\int h(x - t)\, d\lambda(t) = \int h\, d\lambda$.

Proposition 1.17 (Summable families of reals, and the extended-real bridge)**.** *Let I be a countable index set and let $(c_i)_{i\in I}$, $(d_i)_{i\in I}$ be families of real numbers.*

1. *(Comparison) If $0 \le c_i \le d_i$ for every i and (d_i) is summable, then (c_i) is summable and $\sum_i c_i \le \sum_i d_i$.*
2. *(Linearity) If (c_i) and (d_i) are summable and $a, b \in \mathbb{R}$, then $(ac_i + bd_i)_i$ is summable with $\sum_i (ac_i + bd_i) = a\sum_i c_i + b\sum_i d_i$.*
3. *(Extended-real bridge) Suppose $c_i \ge 0$ for every i, and read $\sum_i c_i$ as an element of $[0, +\infty]$ by taking the supremum of the finite partial sums. Then $\sum_i c_i < \infty$ if and only if (c_i) is summable as a family of real numbers, and in that case the two readings of $\sum_i c_i$ agree.*
4. *(Vanishing) If $e_i \in [0, +\infty]$ for every i and $\sum_i e_i = 0$, then $e_i = 0$ for every i; this is immediate from $e_{i_0} \le \sum_i e_i$ for each fixed i_0, the terms being nonnegative.*

Clause (1) applied together with the pointwise inequality $(s+t)^2 \le 2s^2 + 2t^2$ and clause (2) shows that, for any fixed weights $w_i \ge 0$, the families (c_i) with $\sum_i w_i c_i^2 < \infty$ form a linear subspace of $\mathbb{R}^I$.

Proposition 1.18 (Weak-$*$ sequential compactness of dual balls, separable case)**.** *Let E be a normed space over $\mathbb{R}$ which is* separable*, i.e. has a countable dense subset, and let E^* denote its dual: the continuous linear functionals $E \to \mathbb{R}$, normed by $\|\Lambda\|_{E^*} = \sup_{\|w\|\le 1} |\Lambda(w)|$. Let $R \ge 0$ and let $(\Lambda_k)_{k\ge 1}$ be a sequence in E^* with $\|\Lambda_k\|_{E^*} \le R$ for every k. Then there are indices $k_1 < k_2 < \cdots$ and $\Lambda \in E^*$ with $\|\Lambda\|_{E^*} \le R$ such that*

$$\Lambda_{k_j}(w) \longrightarrow \Lambda(w) \qquad \text{as } j \to \infty, \text{ for every } w \in E.$$

The two hypotheses are exactly separability of E and completeness of the scalar field; nothing is asserted about sequential compactness in E itself, and no reflexivity is used. Transporting this statement to a Hilbert space, and verifying separability of the Hilbert space in question, are not part of this item.

Proposition 1.19 (Elementary facts about $\liminf$)**.** *Let (x_k), (y_k), (z_k) be sequences.*

1. *(Monotonicity) If the terms take values in $[0, +\infty]$, or if they are real and both sequences are bounded, and $x_k \le y_k$ for every k, then $\liminf_k x_k \le \liminf_k y_k$.*
2. *(Convergent sequences) If $x_k \to x$, with values in $\mathbb{R}$ or in $[0, +\infty]$, then $\liminf_k x_k = x$.*
3. *(Adding a convergent sequence) If (y_k) is a bounded real sequence and $z_k \to z$ in $\mathbb{R}$, then $(y_k + z_k)$ is bounded and $\liminf_k (y_k + z_k) = \big(\liminf_k y_k\big) + z$.*

Proposition 1.20 (Monotonicity and subadditivity of a measure)**.** *Let μ be a measure on a measurable space $(\alpha, \mathcal{A})$.*

1. *(Monotonicity) If $A \subseteq B$ then $\mu(A) \le \mu(B)$.*
2. *(Subadditivity) $\mu(A \cup B) \le \mu(A) + \mu(B)$ for all A, B, and more generally $\mu\big(\bigcup_k A_k\big) \le \sum_k \mu(A_k)$ for any countable family (A_k).*
3. *(Monotonicity up to null sets) If $A \subseteq B \cup N$ with $\mu(N) = 0$, then $\mu(A) \le \mu(B)$; this follows from (1) and (2), since $\mu(A) \le \mu(B \cup N) \le \mu(B) + \mu(N) = \mu(B)$.*

All assertions are as elements of $[0, +\infty]$.

Proposition 1.21 (Compact sets: extreme values and finite Lebesgue measure)**.** (1) *(Extreme value theorem) Let K be a nonempty compact subset of $\mathbb{R}^n$ and let $\varrho : \mathbb{R}^n \to \mathbb{R}$ be continuous. Then ϱ attains both a minimum and a maximum on K: there are $x_-, x_+ \in K$ with $\varrho(x_-) \le \varrho(x) \le \varrho(x_+)$ for every $x \in K$.*

2. *(Finite Lebesgue measure) Every compact $K \subseteq \mathbb{R}^n$ is bounded, hence $K \subseteq B(0, r)$ for some $r \in \mathbb{N}$, and therefore $\lambda(K) \le \lambda(B(0, r)) < \infty$ by monotonicity of λ and the finiteness of the measure of a ball.*

1.2. `Hermite1`.

Definition 1.22 (The normalized one-dimensional Hermite functions)**.** Let $\mathrm{He}_k \in \mathbb{Z}[X]$ be the probabilists' Hermite polynomials, determined by

$$\mathrm{He}_0 = 1, \qquad \mathrm{He}_{k+1} = X\,\mathrm{He}_k - \mathrm{He}_k' \quad (k \ge 0).$$

For $k \in \mathbb{N}_0$ define $h_k : \mathbb{R} \to \mathbb{R}$ by

$$h_k(x) = \frac{\mathrm{He}_k(\sqrt{2}\,x)}{\sqrt{k!}}.$$

This rescaling is dictated by the variance $1/2$ of the one-dimensional measure $\pi^{-1/2}e^{-x^2}\,dx$: the substitution $y = \sqrt{2}\,x$ carries that measure to the standard Gaussian, against which He_k is orthogonal with $\int \mathrm{He}_k^2\,d\mathcal{N}(0,1) = k!$. Equivalently, h_k is the classical positive-leading-coefficient Hermite polynomial H_k for the weight e^{-x^2} divided by $(2^k k!)^{1/2}$, which is the normalization used in the reference paper; the two powers of 2 in $H_k(x) = 2^{k/2}\,\mathrm{He}_k(\sqrt{2}\,x)$ and in $(2^k k!)^{1/2}$ cancel exactly.

1.3. `MultiIndexSum`.

Definition 1.23 (Multi-indices and their order)**.** Write $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, so that a multi-index is an element $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$, i.e. a function $\alpha : \{1, \ldots, n\} \to \mathbb{N}_0$. Its order is the nonnegative integer

$$|\alpha| = \alpha_1 + \cdots + \alpha_n = \sum_{j=1}^{n} \alpha_j.$$

Thus $|\alpha| = 0$ holds exactly for the zero multi-index $\alpha = 0$, and $|\alpha| \geq 1$ for every $\alpha \neq 0$. We write e_j for the multi-index whose jth entry is 1 and whose other entries vanish, and $\alpha - e_j$ for the multi-index obtained from α by decreasing its jth entry by one, with the convention that this entry is left at 0 when $\alpha_j = 0$.

1.4. `GaussMeasure`.

Definition 1.24 (The Gaussian measure γ_n)**.** Fix $n \geq 1$ and work on $\mathbb{R}^n$, equipped with the Euclidean norm $|x| = (\sum_{j=1}^n x_j^2)^{1/2}$. The Gaussian measure γ_n is the Borel measure on $\mathbb{R}^n$ that is absolutely continuous with respect to Lebesgue measure with density

$$\frac{d\gamma_n}{dx}(x) = \pi^{-n/2}e^{-|x|^2}.$$

Equivalently, for every Borel set $A \subseteq \mathbb{R}^n$,

$$\gamma_n(A) = \int_A \pi^{-n/2}e^{-|x|^2}\,dx.$$

This is the measure denoted γ in the reference paper, with the normalization $d\gamma(x) = \pi^{-n/2}e^{-|x|^2}\,dx$; its covariance is $\frac{1}{2}I$, i.e. each coordinate has variance $1/2$. All function spaces, integrals and inner products in the tablet are real and taken with respect to γ_n unless stated otherwise.

1.5. `HermiteFun`.

Definition 1.25 (The n-dimensional Hermite functions)**.** For a multi-index $\alpha \in \mathbb{N}_0^n$ as in `MultiIndexSum`, define $h_\alpha : \mathbb{R}^n \to \mathbb{R}$ as the product of the one-dimensional normalized Hermite functions of `Hermite1` in the separate coordinates:

$$h_\alpha(x) = \prod_{j=1}^{n} h_{\alpha_j}(x_j), \qquad x = (x_1, \ldots, x_n) \in \mathbb{R}^n.$$

Each h_α is a polynomial in x of total degree $|\alpha|$. This is the Hermite family of the reference paper, built from the measure γ_n of `GaussMeasure`, which factorizes as the n-fold product of the one-dimensional measure $\pi^{-1/2}e^{-t^2}\,dt$.

1.6. `HermiteL2`.

Definition 1.26 (The Hermite family as elements of $L^2(\gamma_n)$)**.** By `HermiteFunMemLp` each Hermite function h_α of `HermiteFun` is square integrable with respect to γ_n. We write again $h_\alpha \in L^2(\gamma_n)$ for the almost-everywhere equivalence class it determines, so that

$$\{h_\alpha : \alpha \in \mathbb{N}_0^n\} \subseteq L^2(\gamma_n)$$

is an $\mathbb{N}_0^n$-indexed family of vectors in the real Hilbert space $L^2(\gamma_n)$, whose inner product is $\langle v, w\rangle = \int vw\,d\gamma_n$.

1.7. `ToL1`.

Definition 1.27 (The inclusion $L^2(\gamma_n) \hookrightarrow L^1(\gamma_n)$)**.** Let γ_n be as in `GaussMeasure`; it is a probability measure by `GaussIsProb`, hence finite. Consequently every $v \in L^2(\gamma_n)$ is also γ_n-integrable, and we write $\iota v \in L^1(\gamma_n)$ for the class of v in $L^1(\gamma_n)$. The map $\iota : L^2(\gamma_n) \to L^1(\gamma_n)$ is the identity on representatives: ιv and v are the same function, viewed in the two different Lebesgue spaces. It is linear and injective, and it is the map used whenever an $L^2(\gamma_n)$ input is regarded as an $L^1(\gamma_n)$ input, in particular in the statement that the Riesz transform extends from $L^2(\gamma_n)$ to $L^1(\gamma_n)$.

1.8. `HermiteBasis`.

Definition 1.28 (The orthonormal Hermite basis of $L^2(\gamma_n)$)**.** By `HermiteL2Orthonormal` the family $\{h_\alpha\}_{\alpha\in\mathbb{N}_0^n}$ of `HermiteL2` is orthonormal in $L^2(\gamma_n)$, and by `HermiteL2Complete` its closed linear span is all of $L^2(\gamma_n)$. It is therefore an orthonormal basis of the real Hilbert space $L^2(\gamma_n)$, indexed by $\mathbb{N}_0^n$. We fix this basis once and for all and write

$$v_\alpha = \langle v, h_\alpha \rangle = \int v\, h_\alpha \, d\gamma_n, \qquad \alpha \in \mathbb{N}_0^n,$$

for the Hermite coefficients of $v \in L^2(\gamma_n)$. Parseval's identity then holds in the form $\|v\|_2^2 = \sum_\alpha |v_\alpha|^2$, the coefficient map $v \mapsto (v_\alpha)_\alpha$ is an isometric isomorphism of $L^2(\gamma_n)$ onto $\ell^2(\mathbb{N}_0^n)$, and every square-summable family of coefficients is realized by a unique element of $L^2(\gamma_n)$.

1.9. `FracPowCoeff`.

Definition 1.29 (The spectral multiplier of order s)**.** For $s \in \mathbb{R}$ and a family of real numbers $c = (c_\alpha)_{\alpha\in\mathbb{N}_0^n}$, define a new family $m_s(c) = (m_s(c)_\alpha)_{\alpha\in\mathbb{N}_0^n}$ by

$$m_s(c)_\alpha = \begin{cases} 0, & \alpha = 0, \\ |\alpha|^s\, c_\alpha, & \alpha \neq 0, \end{cases}$$

with $|\alpha|$ as in `MultiIndexSum`. The convention $m_s(c)_0 = 0$ is built into the multiplier for every s, including $s \geq 0$; it records the fact that the constant Hermite mode $\alpha = 0$ is annihilated, which is exactly the pseudoinverse convention of the reference paper for $s < 0$ and, for $s = 0$, the removal of the constant mode. Note $|\alpha|^s$ is well-defined for $\alpha \neq 0$ because then $|\alpha| \geq 1 > 0$.

1.10. `FracPowDomain`.

Definition 1.30 (The spectral domains $\mathrm{Dom}(L^s)$)**.** Let $v \in L^2(\gamma_n)$ have Hermite coefficients $v_\alpha = \langle v, h_\alpha \rangle$ relative to the basis of `HermiteBasis`. For $s \in \mathbb{R}$ set

$$\mathrm{Dom}(L^s) = \Big\{ v \in L^2(\gamma_n) : \sum_{\alpha\neq 0} |\alpha|^{2s}\, |v_\alpha|^2 < \infty \Big\},$$

i.e. $v \in \mathrm{Dom}(L^s)$ exactly when the multiplied coefficient family $m_s\big((v_\alpha)_\alpha\big)$ of `FracPowCoeff` is square summable. This is the spectral domain criterion of the reference paper. For $s \geq 0$ the multiplier $|\alpha|^{2s}$ is at least 1 on every nonconstant mode, while for $s \leq 0$ it is at most 1. The two cases used below are $s = 1/2$, where $\mathrm{Dom}(L^{1/2}) = \{v : \sum_{\alpha\neq 0} |\alpha||v_\alpha|^2 < \infty\}$ is the operator domain of $B = L^{1/2}$, and $s = 1/4$, whose domain is the fractional form domain.

1.11. `Energy`.

Definition 1.31 (The fractional energy and the form domain $\mathcal{V}$)**.** For $v \in L^2(\gamma_n)$ with Hermite coefficients v_α relative to `HermiteBasis`, define the fractional energy as the value in $[0, +\infty]$

$$\mathcal{E}(v) = \sum_{\alpha\neq 0} |\alpha|^{1/2}\, |v_\alpha|^2,$$

with $|\alpha|$ as in `MultiIndexSum`. All summands are nonnegative, so the sum is unconditionally defined in $[0, +\infty]$; it is finite exactly when v lies in the fractional form domain

$$\mathcal{V} = \{ v \in L^2(\gamma_n) : \mathcal{E}(v) < \infty \} = \mathrm{Dom}(L^{1/4}),$$

the second equality being the case $s = 1/4$ of FracPowDomain, since the multiplier there is $|\alpha|^{2\cdot(1/4)} = |\alpha|^{1/2}$. Thus $\mathcal{E}(v) = \|L^{1/4}v\|_2^2$ for $v \in \mathcal{V}$, and the convention $\mathcal{E}(v) = +\infty$ for $v \in L^2(\gamma_n) \setminus \mathcal{V}$ is built into the definition, as in the reference paper. The space $\mathcal{V}$ carries the Hilbert norm $\|v\|_{\mathcal{V}}^2 = \|v\|_2^2 + \mathcal{E}(v)$. Constants belong to $\mathcal{V}$ and have zero energy, since the index $\alpha = 0$ is omitted from the sum; no mean-zero condition is imposed.

1.12. OUSemigroup.

Definition 1.32 (The Ornstein–Uhlenbeck semigroup, spectrally)**.** For $s \geq 0$ and $v \in L^2(\gamma_n)$ with Hermite coefficients v_α relative to HermiteBasis, define

$$P_s v = \sum_{\alpha \in \mathbb{N}_0^n} e^{-s|\alpha|}\, v_\alpha\, h_\alpha,$$

with $|\alpha|$ as in MultiIndexSum. The series converges in $L^2(\gamma_n)$ for every $v \in L^2(\gamma_n)$ and every $s \geq 0$, because $0 < e^{-s|\alpha|} \leq 1$ makes the multiplied coefficient family square summable; $P_s v$ is then the unique element of $L^2(\gamma_n)$ with Hermite coefficients $e^{-s|\alpha|} v_\alpha$. (For the degenerate case in which the multiplied family fails to be square summable, which cannot occur for $s \geq 0$, we set $P_s v = 0$; this convention is never invoked below.) Thus $P_0 = I$, the spectral action is $P_s h_\alpha = e^{-s|\alpha|} h_\alpha$, the constant mode is fixed ($P_s h_0 = h_0$, so $P_s \mathbf{1} = \mathbf{1}$ by Hermite1Zero), and $P_{s+t} = P_s P_t$.

1.13. EnergyBil.

Definition 1.33 (The fractional energy form)**.** For $v, w \in L^2(\gamma_n)$ with Hermite coefficients v_α, w_α relative to HermiteBasis, define the real number

$$\mathcal{E}(v, w) = \sum_{\alpha \neq 0} |\alpha|^{1/2}\, v_\alpha\, w_\alpha$$

whenever the family on the right is summable, and $\mathcal{E}(v, w) = 0$ otherwise (a convention never invoked in any statement below). The family is absolutely summable as soon as both v and w have finite energy in the sense of Energy: by Cauchy–Schwarz,

$$\sum_{\alpha \neq 0} |\alpha|^{1/2}\, |v_\alpha w_\alpha| \leq \Big(\sum_{\alpha \neq 0} |\alpha|^{1/2} |v_\alpha|^2\Big)^{1/2} \Big(\sum_{\alpha \neq 0} |\alpha|^{1/2} |w_\alpha|^2\Big)^{1/2} = \mathcal{E}(v)^{1/2}\, \mathcal{E}(w)^{1/2}.$$

On $\mathcal{V}$ the map $(v, w) \mapsto \mathcal{E}(v, w)$ is therefore a well-defined symmetric bilinear form with $\mathcal{E}(v, v) = \mathcal{E}(v)$, and it satisfies the Cauchy–Schwarz bound just displayed. It is the energy form of the reference paper, equal to $\langle L^{1/4}v, L^{1/4}w\rangle$ on $\mathcal{V} = \mathrm{Dom}(L^{1/4})$. It vanishes whenever either argument is constant, since the index $\alpha = 0$ is omitted.

1.14. ObstacleFunctional.

Definition 1.34 (The obstacle functional J)**.** Let $F \in L^2(\gamma_n)$ and $a \in \mathbb{R}$. For v in the form domain $\mathcal{V}$ of Energy define

$$J(v) = \tfrac{1}{2}\mathcal{E}(v) + \int (a - F)\, v\, d\gamma_n.$$

Both terms are finite real numbers for $v \in \mathcal{V}$: the energy because $\mathcal{E}(v) < \infty$ there, and the linear term because Cauchy–Schwarz gives $\big|\int (a - F) v\, d\gamma_n\big| \leq \|a\mathbf{1} - F\|_2\, \|v\|_2 < \infty$, using that γ_n is a probability measure so that the constant a lies in $L^2(\gamma_n)$. Thus J is real-valued on $\mathcal{V}$, and the same formula is used for $v \in L^2(\gamma_n) \setminus \mathcal{V}$ with the convention $\mathcal{E}(v)$ replaced by 0 there; that convention is never invoked, since every statement about J below restricts to $v \in \mathcal{V}$.

1.15. ObstacleCone.

Definition 1.35 (The admissible cone $\mathcal{K}$)**.** With $\mathcal{V} = \{v \in L^2(\gamma_n) : \mathcal{E}(v) < \infty\}$ the form domain of Energy, set

$$\mathcal{K} = \{v \in \mathcal{V} : v \geq 0 \ \ \gamma_n\text{-a.e.}\}.$$

This is the admissible set of the obstacle problem: the nonnegative elements of the fractional form domain. It is nonempty, since $0 \in \mathcal{K}$, and it is a convex cone: if $v, w \in \mathcal{K}$ and $\lambda, \mu \geq 0$ then

$\lambda v + \mu w \geq 0$ a.e., and $\mathcal{E}(\lambda v + \mu w) < \infty$ because $\mathcal{V}$ is a linear subspace of $L^2(\gamma_n)$. Constants $c \geq 0$ belong to $\mathcal{K}$, with $\mathcal{E}(c) = 0$.

1.16. `ObstacleMultiplier`.

Definition 1.36 (The obstacle multiplier η as a functional on the form domain)**.** Let $F \in L^2(\gamma_n)$, let $a \in \mathbb{R}$ and let $u \in \mathcal{V}$. Define, for test functions $\varphi \in \mathcal{V}$,

$$\langle \eta, \varphi \rangle = \mathcal{E}(u, \varphi) - \int (F - a)\, \varphi \, d\gamma_n,$$

with $\mathcal{E}(\cdot, \cdot)$ as in `EnergyBil`. Both terms are finite real numbers for $u, \varphi \in \mathcal{V}$: the first by the Cauchy–Schwarz bound $|\mathcal{E}(u, \varphi)| \leq \mathcal{E}(u)^{1/2} \mathcal{E}(\varphi)^{1/2}$ recorded in `EnergyBil`, the second because $|\int (F - a)\varphi \, d\gamma_n| \leq \|F - a\mathbf{1}\|_2 \|\varphi\|_2$. Thus η is a linear functional on $\mathcal{V}$, bounded for the graph norm $\|\varphi\|_{\mathcal{V}}^2 = \|\varphi\|_2^2 + \mathcal{E}(\varphi)$, and it is completely determined by u, F and a. It is the first variation of the functional J of `ObstacleFunctional` at u: $J(u + t\varphi) - J(u) = t\langle \eta, \varphi \rangle + \frac{t^2}{2} \mathcal{E}(\varphi)$ for every $t \in \mathbb{R}$. At this stage η is a functional only; the shorthand $\eta = Bu + a - F$ is to be read in this form sense and is not yet a pointwise formula for a function.

1.17. `OUCoupling`.

Definition 1.37 (The Ornstein–Uhlenbeck coupling measure ν_s)**.** For $s > 0$ let X and Y be independent $\mathbb{R}^n$-valued random vectors, each with law γ_n as in `GaussMeasure`; concretely, let $\gamma_n \otimes \gamma_n$ be the product measure on $\mathbb{R}^n \times \mathbb{R}^n$. Define ν_s as the push-forward of $\gamma_n \otimes \gamma_n$ under the affine map

$$(x, y) \longmapsto \big(x,\ e^{-s} x + \sqrt{1 - e^{-2s}}\ y\big),$$

i.e. ν_s is the joint law of the pair $\big(X,\ e^{-s} X + \sqrt{1 - e^{-2s}}\, Y\big)$. Thus ν_s is a centred Gaussian measure on $\mathbb{R}^{2n}$. It replaces the explicit Mehler kernel $M_s(x, y)$ of the reference paper: the paper uses only that M_s is symmetric, positive, and of unit mass in each variable, and these three properties correspond exactly to ν_s being a symmetric probability measure whose two marginals are both γ_n.

1.18. `FracPow`.

Definition 1.38 (The spectral powers L^s, the projection Π_0, and $L^{-1/2}\Pi_0$)**.** Let $v \in L^2(\gamma_n)$ with Hermite coefficients v_α relative to the basis of `HermiteBasis`. For $s \in \mathbb{R}$ define

$$L^s v = \sum_{\alpha \neq 0} |\alpha|^s \, v_\alpha \, h_\alpha \qquad \text{whenever } v \in \mathrm{Dom}(L^s),$$

the series converging in $L^2(\gamma_n)$ precisely because the coefficient family $m_s\big((v_\alpha)_\alpha\big)$ of `FracPowCoeff` is then square summable, by `FracPowDomain`; and set $L^s v = 0$ when $v \notin \mathrm{Dom}(L^s)$, a convention that makes L^s a total map and is never invoked in any statement below. Thus $L^s v$ is the unique element of $L^2(\gamma_n)$ whose Hermite coefficients are 0 at $\alpha = 0$ and $|\alpha|^s v_\alpha$ at $\alpha \neq 0$.

Three instances carry names. For $s = 0$ the operator is the projection onto the mean-zero subspace,

$$\Pi_0 v = L^0 v = \sum_{\alpha \neq 0} v_\alpha h_\alpha,$$

defined on all of $L^2(\gamma_n)$. For $s = 1/2$ it is the operator $B = L^{1/2}$ with domain $\mathrm{Dom}(L^{1/2})$. For $s = -1/2$ it is the pseudoinverse

$$L^{-1/2}\Pi_0 v = \sum_{\alpha \neq 0} |\alpha|^{-1/2} v_\alpha h_\alpha,$$

defined on all of $L^2(\gamma_n)$; the projection Π_0 is already incorporated, because the multiplier of `FracPowCoeff` sends the constant mode to zero. This reflects that L annihilates constants, so an inverse square root cannot be defined on the constant mode by taking the reciprocal of its eigenvalue; instead the constant mode is removed and the reciprocal square root $|\alpha|^{-1/2}$ of the eigenvalue $|\alpha|$ is applied on each remaining mode. That the index $|\alpha|$ is indeed the eigenvalue of the Ornstein–Uhlenbeck differential operator $-\frac{1}{2}\Delta + x \cdot \nabla$ on h_α is the content of `HermiteFunOU`.

1.19. `HasGaussWeakGrad`.

Definition 1.39 (Weak Euclidean gradient)**.** Let $v : \mathbb{R}^n \to \mathbb{R}$ and $G = (G_1, \dots, G_n) : \mathbb{R}^n \to \mathbb{R}^n$ be measurable. We say that G is a *weak Euclidean gradient* of v, written $G = \nabla v$ weakly, if for every index $j \in \{1, \dots, n\}$ and every test function $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\int_{\mathbb{R}^n} v(x)\, \partial_j \varphi(x)\, dx = -\int_{\mathbb{R}^n} G_j(x)\, \varphi(x)\, dx.$$

Both integrals are taken with respect to *Lebesgue* measure, not against the Gaussian weight; this is the ordinary distributional derivative. The integrals are finite in every situation in which the notion is used below, because the Gaussian density of `GaussMeasure` is smooth and strictly positive, so every function in $L^2(\gamma_n)$ is Lebesgue-integrable on compact sets and $\varphi, \partial_j \varphi$ are bounded with compact support.

1.20. `SobolevW12`.

Definition 1.40 (The Gaussian Sobolev space $W^{1,2}(\gamma_n)$)**.** Set

$$W^{1,2}(\gamma_n) = \big\{ v \in L^2(\gamma_n) : v \text{ has a weak Euclidean gradient } \nabla v \text{ with } 2^{-1/2} \nabla v \in L^2(\gamma_n; \mathbb{R}^n) \big\},$$

the weak gradient being understood in the sense of `HasGaussWeakGrad` and $L^2(\gamma_n; \mathbb{R}^n)$ denoting the space of $\mathbb{R}^n$-valued functions square integrable against the measure γ_n of `GaussMeasure`, with the Euclidean norm on the target. Equivalently, $v \in W^{1,2}(\gamma_n)$ iff $v \in L^2(\gamma_n)$ and there is $\Gamma \in L^2(\gamma_n; \mathbb{R}^n)$ with $\sqrt{2}\,\Gamma$ a weak Euclidean gradient of v; the factor $\sqrt{2}$ records the energy normalization $D = 2^{-1/2}\nabla$ of the reference paper, so that Γ is Dv. This is the characterization of $W^{1,2}(\gamma_n)$ used throughout: the functions in $L^2(\gamma_n)$ whose weak Euclidean derivatives again lie in $L^2(\gamma_n)$. The reference paper notes that this coincides with the completion of $C_c^\infty(\mathbb{R}^n)$ in the norm $\|v\|^2 = \|v\|_2^2 + \frac{1}{2}\|\nabla v\|_2^2$; that equivalence is not used anywhere below, and only the weak-derivative description is taken as the definition.

1.21. `GaussD`.

Definition 1.41 (The energy-normalized gradient operator D)**.** For $v \in W^{1,2}(\gamma_n)$ in the sense of `SobolevW12`, let

$$Dv = 2^{-1/2} \nabla v \in L^2(\gamma_n; \mathbb{R}^n),$$

that is, the unique element Γ of $L^2(\gamma_n; \mathbb{R}^n)$ such that $\sqrt{2}\,\Gamma$ is a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`; uniqueness as an element of $L^2(\gamma_n; \mathbb{R}^n)$ is `GaussWeakGradUnique`. Its jth coordinate is $D_j v = 2^{-1/2} \partial_j v$, with ∂_j understood in the distributional sense. For $v \in L^2(\gamma_n) \setminus W^{1,2}(\gamma_n)$ we set $Dv = 0$, a convention that makes D a total map on $L^2(\gamma_n)$ and that is never invoked in any statement below: every statement mentioning Dv carries the hypothesis that v lies in $W^{1,2}(\gamma_n)$ or in a subspace of it.

1.22. `RieszTransform`.

Definition 1.42 (The energy-normalized vector Riesz transform)**.** For $f \in L^2(\gamma_n)$ set

$$R_\gamma f = D\, L^{-1/2} \Pi_0 f \in L^2(\gamma_n; \mathbb{R}^n),$$

with $L^{-1/2}\Pi_0$ the pseudoinverse of `FracPow` and $D = 2^{-1/2}\nabla$ the energy-normalized weak gradient of `GaussD`. Its jth coordinate is $R_{\gamma,j} f = D_j L^{-1/2} \Pi_0 f$. The arbitrary L^2 input f is never differentiated: one first forms the potential $v = L^{-1/2}\Pi_0 f$, which lies in $\mathrm{Dom}(L^{1/2})$ and hence in $W^{1,2}(\gamma_n)$ by `PseudoInverseSqrt` and `SqrtDomainSobolev`, so its weak gradient exists in $L^2(\gamma_n; \mathbb{R}^n)$. Consequently $R_\gamma f$ is well defined for every $f \in L^2(\gamma_n)$, with no differentiability assumption on f, and $R_\gamma : L^2(\gamma_n) \to L^2(\gamma_n; \mathbb{R}^n)$ is linear.

1.23. `ComplexRieszTransform`.

Definition 1.43 (The complex Riesz transform $R_\gamma^{\mathbb{C}}$)**.** Write $\iota : \mathbb{R}^n \to \mathbb{C}^n$ for the coordinatewise inclusion $\iota(y)_j = y_j$ of a real vector as a complex one; it is a bounded $\mathbb{R}$-linear map, and it is a real-linear isometry for the Euclidean norms, since $|\iota(y)|_{\mathbb{C}^n}^2 = \sum_j |y_j|^2 = |y|_{\mathbb{R}^n}^2$.

Let $v \in L^2(\gamma_n; \mathbb{C})$. Composing v pointwise with the two bounded $\mathbb{R}$-linear maps $\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$ produces the real and imaginary parts

$$\mathrm{Re}\, v,\ \mathrm{Im}\, v \ \in\ L^2(\gamma_n; \mathbb{R}), \qquad (\mathrm{Re}\, v)(x) = \mathrm{Re}\big(v(x)\big), \quad (\mathrm{Im}\, v)(x) = \mathrm{Im}\big(v(x)\big),$$

each of them square γ_n-integrable because composition with a bounded linear map preserves membership in a Lebesgue space. Define the *complex Riesz transform* of v as the element of $L^2(\gamma_n; \mathbb{C}^n)$

$$R_\gamma^{\mathbb{C}} v \ =\ \iota\big(R_\gamma(\mathrm{Re}\, v)\big) \ +\ i\, \iota\big(R_\gamma(\mathrm{Im}\, v)\big),$$

where R_γ is the energy-normalized vector Riesz transform of `RieszTransform` and $i\,(\cdot)$ is multiplication by i in the complex vector space $L^2(\gamma_n; \mathbb{C}^n)$. Equivalently, coordinate by coordinate and γ_n-almost everywhere,

$$\big(R_\gamma^{\mathbb{C}} v\big)_j = R_{\gamma,j}(\mathrm{Re}\, v) + i\, R_{\gamma,j}(\mathrm{Im}\, v), \qquad j \in \{1, \dots, n\}.$$

The domain of the functions stays real: the underlying measure space is $(\mathbb{R}^n, \gamma_n)$ throughout, and only the scalars of the values become complex. The definition is unconditional — every $v \in L^2(\gamma_n; \mathbb{C})$ has real and imaginary parts in $L^2(\gamma_n; \mathbb{R})$, to which R_γ applies with no differentiability or domain hypothesis by `RieszTransform`, and ι carries $L^2(\gamma_n; \mathbb{R}^n)$ into $L^2(\gamma_n; \mathbb{C}^n)$ — so $R_\gamma^{\mathbb{C}}$ is a map $L^2(\gamma_n; \mathbb{C}) \to L^2(\gamma_n; \mathbb{C}^n)$ defined on all of $L^2(\gamma_n; \mathbb{C})$.

On real inputs it agrees with R_γ up to the inclusion ι: if $\mathrm{Im}\, v = 0$ then $R_\gamma^{\mathbb{C}} v = \iota(R_\gamma(\mathrm{Re}\, v))$, since $R_\gamma 0 = 0$. This is the operator the reference paper calls the complex-linear extension of R_γ and writes again as R_γ when the input is complex, its values being measured in the norm $|\cdot|_{\mathbb{C}^n}$. That $R_\gamma^{\mathbb{C}}$ really is $\mathbb{C}$-linear is not asserted here. It is not formal either: the real and imaginary parts of $c\, v$ are not c times those of v — for $c = i$ they are exchanged, up to a sign — so complex homogeneity has to be derived from the $\mathbb{R}$-linearity of R_γ, and that derivation belongs to a separate proof-bearing node rather than to this definition.

1.24. `ComplexToL1`.

Definition 1.44 (The inclusion $L^2(\gamma_n; \mathbb{C}) \hookrightarrow L^1(\gamma_n; \mathbb{C})$)**.** Let γ_n be as in `GaussMeasure`; it is a probability measure by `GaussIsProb`, hence finite. Consequently every $v \in L^2(\gamma_n; \mathbb{C})$, that is, every γ_n-a.e. class of measurable $\mathbb{C}$-valued functions on $\mathbb{R}^n$ with $\int |v|^2 d\gamma_n < \infty$, is also γ_n-integrable, and we write $\iota_{\mathbb{C}} v \in L^1(\gamma_n; \mathbb{C})$ for the class of v in $L^1(\gamma_n; \mathbb{C})$. The map $\iota_{\mathbb{C}} : L^2(\gamma_n; \mathbb{C}) \to L^1(\gamma_n; \mathbb{C})$ is the identity on representatives: $\iota_{\mathbb{C}} v$ and v are the same function, viewed in the two different Lebesgue spaces. It is $\mathbb{C}$-linear and injective, and

$$\|\iota_{\mathbb{C}} v\|_{L^1(\gamma_n;\mathbb{C})} = \int_{\mathbb{R}^n} |v(x)|\, d\gamma_n(x),$$

since the L^1 norm of a class is the integral of the modulus of any of its representatives and v is one of them.

This is the complex-scalar inclusion. It is a different map from the tablet's real inclusion $\iota : L^2(\gamma_n; \mathbb{R}) \to L^1(\gamma_n; \mathbb{R})$, which acts between spaces of real-valued classes, whereas $\iota_{\mathbb{C}}$ acts between the corresponding spaces of complex-valued classes; neither is a special case of the other, since a complex L^2 class is not a real one. The complex-valued estimate of the reference paper is stated for complex L^1 inputs, so the complex inclusion is what its statement needs, and it is introduced as its own object because three later statements use it: the norm-control and L^1-convergence clauses of the complex in-measure limit lemma, and the clause "agrees with the L^2 transform" in each of the complex L^1 extension lemma and the complex endpoint theorem.

1.25. `LocalMaximalFunction`.

Definition 1.45 (The local Hardy–Littlewood maximal function M_2)**.** Fix $n \in \mathbb{N}_0$ and let λ denote Lebesgue measure on $\mathbb{R}^n$. For $w : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$ define

$$M_2 w(x) = \sup_{0<\rho\le 2} \frac{1}{\lambda\big(\overline{B}(x,\rho)\big)} \int_{\overline{B}(x,\rho)} |w(y)|\, dy \ \in\ [0,+\infty],$$

the supremum being over real radii ρ in the half-open interval $(0,2]$ and $\overline{B}(x,\rho) = \{y : |y-x| \le \rho\}$ the closed ball. The integral is the Lebesgue (lower) integral of the nonnegative function $|w|$ over $\overline{B}(x,\rho)$ with respect to λ, taken in $[0,+\infty]$.

No Gaussian measure occurs here. Both the integral and the normalizing volume are taken against Lebesgue measure, with no Gaussian weight in the integrand and no Gaussian measure in the normalization, even though this object is used inside the tablet's Gaussian development. Its role there is to control the *local* Gaussian maximal operator on a fixed ball, where the Gaussian density is bounded above and below by positive constants, so that no covering theory for a non-doubling measure is needed.

The radius cap at 2 *is part of the definition, not a convenience.* The covering step used to prove the weak-type bound for M_2 extracts a disjoint subfamily from a family of closed balls whose radii admit a common upper bound; without a cap on ρ no such bound exists. Every downstream use is on balls of radius at most 2, so nothing is lost.

Values in $[0,+\infty]$*, and no hypotheses on* w*.* The supremum is taken in the extended nonnegative reals, so $M_2 w(x)$ is defined for every w and every x with no measurability, local integrability or boundedness hypothesis: an unbounded family of averages gives the value $+\infty$ rather than an undefined or conventional value. This is deliberate. A real-valued supremum would silently take a junk value where the averages are unbounded, and a statement bounding such a junk value above would then be satisfiable without any analytic content; the extended-real-valued definition makes every bound on $M_2 w$ an honest bound. For the same reason the integrals are lower Lebesgue integrals of a nonnegative function, which are defined for every w, rather than Bochner integrals, which would be 0 by convention on a non-integrable integrand.

Finally, for every x and every $\rho \in (0,2]$ the normalizing factor $\lambda(\overline{B}(x,\rho))$ is a strictly positive real number: this is clear for $n \ge 1$, and for $n = 0$ the space $\mathbb{R}^0$ is a single point, every closed ball is the whole space, and λ is the unit mass. So the quotient inside the supremum never involves division by 0 or by $+\infty$.

1.26. `GaussRieszKernel`.

Definition 1.46 (The normalized Gaussian Riesz kernel $K_{\gamma,j}$)**.** Fix $n \ge 1$ and $j \in \{1,\dots,n\}$. Define, for $x, y \in \mathbb{R}^n$,

$$K_{\gamma,j}(x,y) = \frac{\sqrt{2}}{\pi^{(n+1)/2}}\, k_j^{\mathrm{raw}}(x,y), \qquad k_j^{\mathrm{raw}}(x,y) = \int_0^1 \frac{y_j - r x_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big) dr,$$

the r-integral being taken over the open interval $(0,1)$ with respect to Lebesgue measure. This is the off-diagonal Lebesgue kernel of the jth coordinate $R_{\gamma,j}$ of the tablet's energy-normalized Gaussian Riesz transform, as displayed in the reference paper. The displayed formula is the whole of the definition: it is an explicit function of (x,y) and depends on no other tablet node, which is why this node imports only the shared preamble root.

Where the formula comes from. It is the time integral of the x-derivative of the Lebesgue kernel, reparametrized by $r = e^{-s}$. Writing the Riesz transform as $(1/\sqrt{\pi}) \int_0^\infty D_j^x P_s\, ds/\sqrt{s}$ and inserting the derivative identity $D_j^x p_s(x,y) = \sqrt{2}\, r(y_j - r x_j)(1-r^2)^{-1} p_s(x,y)$ for the Lebesgue kernel p_s of the Ornstein–Uhlenbeck semigroup, the substitution $r = e^{-s}$, under which $s = -\log r$, $ds = -dr/r$ and $s \in (0,\infty)$ corresponds to $r \in (0,1)$, produces exactly the displayed integrand: the factor r from the derivative cancels the $1/r$ of ds, the $\pi^{-n/2}$ of p_s combines with the $1/\sqrt{\pi}$ into $\pi^{-(n+1)/2}$, the two powers of $1-r^2$ combine into $(1-r^2)^{-n/2-1}$, and $s^{-1/2}$ becomes $(-\log r)^{-1/2}$. This paragraph records provenance only; the object defined here is the displayed formula, and the identification of $K_{\gamma,j}$ with the operator is a separate assertion proved elsewhere.

Scope of the definition. The reference paper displays the formula for $x \neq y$, which is the only range in which it is used. For $x = y$ the r-integral need not converge, and the definition then takes the value the Bochner integral convention assigns to a non-integrable integrand, namely 0. No statement built on this definition refers to the value on the diagonal. The integrand is also left unconstrained at the two endpoints $r = 0$ and $r = 1$, which the open interval excludes; they form a Lebesgue-null set and do not affect the integral.

The normalization is part of the definition, not a convention that can be dropped. The function k_j^{raw} alone is the kernel of the historical reference, and the prefactor $\sqrt{2}\,\pi^{-(n+1)/2}$ is what makes $K_{\gamma,j}$ the kernel of the energy-normalized $R_{\gamma,j}$ rather than of a constant multiple of it; in the same letters, the kernel of the historical spectral operator $\mathcal{R}_j$ is $-2\pi^{-(n+1)/2}k_j^{\mathrm{raw}}$. As the paper states, a claim about dimension-free constants must specify these factors, since an unnormalized kernel is a different family of operators.

2. Lemmas

2.1. `GaussIsProb`.

Lemma 2.1 (γ_n is a probability measure)**.** *For every $n \geq 1$ the measure γ_n of `GaussMeasure` is a probability measure: $\gamma_n(\mathbb{R}^n) = 1$.*

Proof. By `GaussMeasure`, γ_n is Lebesgue measure on $\mathbb{R}^n$ weighted by the density $\rho(x) = \pi^{-n/2}e^{-|x|^2}$, so

$$\gamma_n(\mathbb{R}^n) = \int_{\mathbb{R}^n} \pi^{-n/2}e^{-|x|^2}\,dx.$$

Since $|x|^2 = \sum_{j=1}^n x_j^2$, the exponential factorizes:

$$\pi^{-n/2}e^{-|x|^2} = \prod_{j=1}^n \pi^{-1/2}e^{-x_j^2}.$$

The integrand is nonnegative and measurable, and Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of Lebesgue measure on $\mathbb{R}$, so Tonelli's theorem applies and gives

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \pi^{-1/2}e^{-x_j^2}\,dx = \prod_{j=1}^n \int_{\mathbb{R}} \pi^{-1/2}e^{-s^2}\,ds.$$

The classical Gaussian integral is $\int_{\mathbb{R}} e^{-s^2}\,ds = \sqrt{\pi}$, so each of the n factors equals $\pi^{-1/2}\cdot\sqrt{\pi} = 1$. Hence the product is 1 and $\gamma_n(\mathbb{R}^n) = 1$. In particular the total mass is finite, so γ_n is a finite measure, and being of total mass exactly 1 it is a probability measure. □

2.2. `HermiteFunMemLp`.

Lemma 2.2 (Hermite functions are square integrable for γ_n)**.** *For every $n \geq 1$ and every multi-index $\alpha \in \mathbb{N}_0^n$, the function h_α of `HermiteFun` is Borel measurable and satisfies*

$$\int_{\mathbb{R}^n} |h_\alpha|^2\,d\gamma_n < \infty,$$

so h_α determines an element of $L^2(\gamma_n)$.

Proof. By `Hermite1`, each h_{α_j} is a polynomial in one variable (a rescaled Hermite polynomial divided by a positive constant), and by `HermiteFun`, $h_\alpha(x) = \prod_{j=1}^n h_{\alpha_j}(x_j)$. A finite product of polynomials in the separate coordinates is a polynomial in $x_1, \ldots, x_n$, hence continuous and therefore Borel measurable. The same is true of h_α^2.

It remains to show that polynomials are square integrable against γ_n. Expand h_α^2 as a finite linear combination of monomials $x^\beta = \prod_j x_j^{\beta_j}$, say $h_\alpha^2 = \sum_{\beta\in S} c_\beta\, x^\beta$ with S finite. Then

$$\int |h_\alpha|^2\,d\gamma_n \leq \sum_{\beta\in S} |c_\beta| \int \prod_{j=1}^n |x_j|^{\beta_j}\,d\gamma_n,$$

so it suffices that each of the finitely many integrals on the right is finite. By `GaussMeasure` the density of γ_n factorizes as $\pi^{-n/2}e^{-|x|^2} = \prod_{j=1}^n \pi^{-1/2}e^{-x_j^2}$, the integrand is nonnegative, and Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of Lebesgue measure on $\mathbb{R}$; so Tonelli's theorem gives

$$\int_{\mathbb{R}^n} \prod_{j=1}^n |x_j|^{\beta_j}\, d\gamma_n = \prod_{j=1}^n \int_{\mathbb{R}} |s|^{\beta_j}\, \pi^{-1/2}e^{-s^2}\, ds.$$

Each one-dimensional factor is finite: the integrand is continuous on $\mathbb{R}$, and for $|s| \geq 1$ it is bounded by $\pi^{-1/2}|s|^{\beta_j}e^{-s^2} \leq \pi^{-1/2}e^{\beta_j|s|}e^{-s^2}$, which is integrable because $\beta_j|s| - s^2 \to -\infty$ at least quadratically. Hence every factor, and so the whole finite sum, is finite.

Finally γ_n is a probability measure by `GaussIsProb`, so the underlying measure space is finite and the class of h_α in $L^2(\gamma_n)$ is well defined. □

2.3. `HermiteDerivative`.

Lemma 2.3 (Derivative recurrence for the probabilists' Hermite polynomials). *Let* $\mathrm{He}_k \in \mathbb{Z}[X]$ *denote the probabilists' Hermite polynomials, defined by* $\mathrm{He}_0 = 1$ *and the recurrence*

$$\mathrm{He}_{k+1} = X\,\mathrm{He}_k - \mathrm{He}_k', \qquad k \geq 0.$$

Then for every $k \geq 0$,

$$\mathrm{He}_{k+1}' = (k+1)\,\mathrm{He}_k.$$

Proof. We argue by induction on k. Throughout, $'$ is the formal derivative on $\mathbb{Z}[X]$, which is additive and satisfies the Leibniz rule $(Xp)' = p + Xp'$.

Base case $k = 0$. Since $\mathrm{He}_0 = 1$ is constant, $\mathrm{He}_0' = 0$, so the defining recurrence gives $\mathrm{He}_1 = X\,\mathrm{He}_0 - \mathrm{He}_0' = X$. Hence $\mathrm{He}_1' = 1 = 1 \cdot \mathrm{He}_0 = (0+1)\,\mathrm{He}_0$, as required.

Inductive step. Let $k \geq 0$ and assume

$$\mathrm{He}_{k+1}' = (k+1)\,\mathrm{He}_k. \tag{IH}$$

We must show $\mathrm{He}_{k+2}' = (k+2)\,\mathrm{He}_{k+1}$. Apply the defining recurrence at index $k+1$ and then substitute (IH):

$$\mathrm{He}_{k+2} = X\,\mathrm{He}_{k+1} - \mathrm{He}_{k+1}' = X\,\mathrm{He}_{k+1} - (k+1)\,\mathrm{He}_k.$$

Differentiating this identity and using the Leibniz rule on the first term,

$$\mathrm{He}_{k+2}' = \mathrm{He}_{k+1} + X\,\mathrm{He}_{k+1}' - (k+1)\,\mathrm{He}_k'.$$

Substitute (IH) for He_{k+1}' in the middle term:

$$\mathrm{He}_{k+2}' = \mathrm{He}_{k+1} + (k+1)\,X\,\mathrm{He}_k - (k+1)\,\mathrm{He}_k' = \mathrm{He}_{k+1} + (k+1)\big(X\,\mathrm{He}_k - \mathrm{He}_k'\big).$$

By the defining recurrence at index k, the bracket is exactly He_{k+1}. Therefore

$$\mathrm{He}_{k+2}' = \mathrm{He}_{k+1} + (k+1)\,\mathrm{He}_{k+1} = (k+2)\,\mathrm{He}_{k+1},$$

which completes the induction. Note the step used only (IH) and the defining recurrence at indices k and $k+1$; no separate information about He_k' was needed. □

2.4. `HermiteGeneratingComplex`.

Lemma 2.4 (Complex generating function of the probabilists' Hermite polynomials). *Let* He_k *be the probabilists' Hermite polynomials of the shared background, and for* $\zeta \in \mathbb{C}$ *write* $\mathrm{He}_k(\zeta)$ *for the evaluation of* He_k *at* ζ *under the canonical ring map* $\mathbb{Z}[X] \to \mathbb{C}$. *Then for all* $\zeta, \xi \in \mathbb{C}$ *the family*

$$k \longmapsto \mathrm{He}_k(\zeta)\,\frac{\xi^k}{k!} \qquad (k \in \mathbb{N}_0)$$

is unconditionally summable in $\mathbb{C}$ *with sum*

$$\sum_{k=0}^{\infty} \mathrm{He}_k(\zeta)\,\frac{\xi^k}{k!} = \exp\big(\zeta\xi - \tfrac{1}{2}\xi^2\big).$$

Proof. Fix $\zeta, \xi \in \mathbb{C}$ and put

$$G(v) = \exp\bigl(-\tfrac{1}{2}v^2\bigr), \qquad v \in \mathbb{C}.$$

Step 1 (G is entire, with $G'(w) = -w\,G(w)$). The map $v \mapsto -\frac{1}{2}v^2$ is a polynomial, hence everywhere complex differentiable with derivative $-w$ at w (indeed $-\frac{1}{2}v^2 = (-\frac{1}{2})v^2$ and $\frac{d}{dv}v^2 = 2w$ at w, so the derivative is $(-\frac{1}{2})(2w) = -w$). Since exp is entire with $\exp' = \exp$, the chain rule gives that G is complex differentiable at every $w \in \mathbb{C}$ with

$$G'(w) = -w\,\exp\bigl(-\tfrac{1}{2}w^2\bigr) = -w\,G(w).$$

In particular G is entire, and so are all of its iterated derivatives.

Step 2 (Rodrigues' identity over $\mathbb{C}$). We claim that for every $k \in \mathbb{N}_0$ and every $w \in \mathbb{C}$,

$$G^{(k)}(w) = (-1)^k\,\mathrm{He}_k(w)\,\exp\bigl(-\tfrac{1}{2}w^2\bigr). \tag{R_k}$$

We argue by induction on k. For $k = 0$ the left side is $G(w)$ and the right side is $\mathrm{He}_0(w)\exp(-w^2/2) = \exp(-w^2/2)$, since $\mathrm{He}_0 = 1$; so (R_0) holds.

Assume (R_m) holds for all w, i.e. $G^{(m)} = \bigl(v \mapsto (-1)^m\,\mathrm{He}_m(v)\,\exp(-v^2/2)\bigr)$ as functions on $\mathbb{C}$. Then $G^{(m+1)}$ is the derivative of that function. The polynomial evaluation $v \mapsto \mathrm{He}_m(v)$ is entire with derivative $\mathrm{He}_m'(w)$ at w, where He_m' is the formal derivative of He_m (the derivative of a polynomial function agrees with the evaluation of the formal derivative). Hence by the product rule together with Step 1,

$$G^{(m+1)}(w) = (-1)^m\,\mathrm{He}_m'(w)\,\exp\bigl(-\tfrac{1}{2}w^2\bigr) + (-1)^m\,\mathrm{He}_m(w)\cdot\bigl(-w\,\exp(-\tfrac{1}{2}w^2)\bigr) = (-1)^m\bigl(\mathrm{He}_m'(w) - w\,\mathrm{He}_m(w)\bigr)\exp$$

By the defining recurrence of the Hermite family, which is shared background, $\mathrm{He}_{m+1} = X\,\mathrm{He}_m - \mathrm{He}_m'$, so evaluating at w gives $\mathrm{He}_m'(w) - w\,\mathrm{He}_m(w) = -\mathrm{He}_{m+1}(w)$. Therefore

$$G^{(m+1)}(w) = (-1)^m \cdot \bigl(-\mathrm{He}_{m+1}(w)\bigr)\exp\bigl(-\tfrac{1}{2}w^2\bigr) = (-1)^{m+1}\,\mathrm{He}_{m+1}(w)\,\exp\bigl(-\tfrac{1}{2}w^2\bigr),$$

which is (R_{m+1}). This completes the induction.

Step 3 (Taylor expansion of G about ζ). Since G is entire, so is $v \mapsto G(\zeta + v)$, whose k-th derivative at 0 equals $G^{(k)}(\zeta)$. Applying the shared background fact that an entire function is represented on all of $\mathbb{C}$ by its Taylor series at 0, with the series converging absolutely, to this translate and evaluating at $v = -\xi$ yields: the family

$$k \longmapsto \frac{1}{k!}\,(-\xi)^k\,G^{(k)}(\zeta)$$

is unconditionally (equivalently, absolutely) summable in $\mathbb{C}$, with

$$\sum_{k=0}^{\infty} \frac{1}{k!}\,(-\xi)^k\,G^{(k)}(\zeta) = G(\zeta - \xi). \tag{T}$$

Step 4 (identification of the terms). Fix k. Substituting (R_k) at $w = \zeta$ into the k-th term of (T) and using $(-1)^k(-1)^k = 1$,

$$\frac{1}{k!}(-\xi)^k G^{(k)}(\zeta) = \frac{1}{k!}\,(-1)^k\xi^k \cdot (-1)^k\,\mathrm{He}_k(\zeta)\,\exp\bigl(-\tfrac{1}{2}\zeta^2\bigr) = \Bigl(\mathrm{He}_k(\zeta)\frac{\xi^k}{k!}\Bigr)\exp\bigl(-\tfrac{1}{2}\zeta^2\bigr).$$

Step 5 (identification of the sum). Expanding the square, $-\frac{1}{2}(\zeta - \xi)^2 = \bigl(\zeta\xi - \frac{1}{2}\xi^2\bigr) + \bigl(-\frac{1}{2}\zeta^2\bigr)$, so by the functional equation $\exp(a + b) = \exp a \exp b$,

$$G(\zeta - \xi) = \exp\bigl(-\tfrac{1}{2}(\zeta - \xi)^2\bigr) = \exp\bigl(\zeta\xi - \tfrac{1}{2}\xi^2\bigr)\,\exp\bigl(-\tfrac{1}{2}\zeta^2\bigr).$$

Step 6 (conclusion). Combining Steps 3–5, the family $k \mapsto \bigl(\mathrm{He}_k(\zeta)\xi^k/k!\bigr)\exp(-\zeta^2/2)$ is unconditionally summable with sum $\exp(\zeta\xi - \xi^2/2)\exp(-\zeta^2/2)$. The constant $c = \exp(-\frac{1}{2}\zeta^2)$ is nonzero, because the complex exponential never vanishes. Dividing an unconditionally summable family and its sum by the fixed nonzero constant c preserves unconditional summability and divides the sum by c; since $\bigl(\mathrm{He}_k(\zeta)\xi^k/k!\bigr)c/c = \mathrm{He}_k(\zeta)\xi^k/k!$ for every k and $\exp(\zeta\xi - \xi^2/2)\,c/c = \exp(\zeta\xi - \xi^2/2)$, we conclude that $k \mapsto \mathrm{He}_k(\zeta)\xi^k/k!$ is unconditionally summable with sum $\exp(\zeta\xi - \frac{1}{2}\xi^2)$, as asserted. □

2.5. `HermiteGenerating`.

Lemma 2.5 (Generating function of the probabilists' Hermite polynomials)**.** *Let* He_k *be the probabilists' Hermite polynomials of* `HermiteDerivative`*. For all real* y *and* z *the series below converges absolutely and*

$$\sum_{k=0}^{\infty} \mathrm{He}_k(y)\,\frac{z^k}{k!} = e^{\,yz-z^2/2}.$$

Proof. Fix $y \in \mathbb{R}$ and set $G(z) = e^{yz-z^2/2}$, an entire function of $z \in \mathbb{C}$ restricted here to real z. Let $c_k = G^{(k)}(0)$ be its Taylor coefficients at $z = 0$, so that

$$G(z) = \sum_{k=0}^{\infty} c_k \frac{z^k}{k!}$$

with absolute convergence for every z, because G is entire (the exponential of a polynomial) and hence equal to its everywhere-absolutely-convergent Taylor series. It therefore suffices to prove $c_k = \mathrm{He}_k(y)$ for every k.

Step 1 (a recurrence for the c_k). Differentiating G gives $G'(z) = (y - z)\,G(z)$. Writing both sides as power series,

$$G'(z) = \sum_{k=0}^{\infty} c_{k+1}\frac{z^k}{k!}, \qquad (y-z)G(z) = \sum_{k=0}^{\infty} y\,c_k\frac{z^k}{k!} - \sum_{k=0}^{\infty} c_k\frac{z^{k+1}}{k!} = \sum_{k=0}^{\infty}\big(y\,c_k - k\,c_{k-1}\big)\frac{z^k}{k!},$$

where in the last step the second sum was reindexed by $k \mapsto k+1$ and $\frac{1}{k!}\cdot k = \frac{1}{(k-1)!}$ was used, with the convention that the term with $k = 0$ is absent (equivalently c_{-1} is multiplied by 0). Two everywhere-convergent power series are equal only if their coefficients agree, so

$$c_{k+1} = y\,c_k - k\,c_{k-1}, \qquad k \geq 0,$$

with the $k = 0$ instance reading $c_1 = y\,c_0$.

Step 2 (the same recurrence for $\mathrm{He}_k(y)$). By the defining recurrence of the probabilists' Hermite polynomials, which is shared context, $\mathrm{He}_{k+1} = X\,\mathrm{He}_k - \mathrm{He}_k'$; and by `HermiteDerivative`, applied at index $k-1$, $\mathrm{He}_k' = k\,\mathrm{He}_{k-1}$ for $k \geq 1$, while $\mathrm{He}_0' = 0$ because He_0 is the constant polynomial 1. Hence, evaluating at y,

$$\mathrm{He}_{k+1}(y) = y\,\mathrm{He}_k(y) - k\,\mathrm{He}_{k-1}(y), \qquad k \geq 0,$$

with the same convention at $k = 0$, where it reads $\mathrm{He}_1(y) = y\,\mathrm{He}_0(y)$.

Step 3 (matching initial data and concluding). We have $c_0 = G(0) = 1$ and $\mathrm{He}_0(y) = 1$. Both families therefore satisfy the identical two-term recurrence of Steps 1–2 with the identical value at $k = 0$; note the recurrence determines c_{k+1} from c_k and c_{k-1} alone, and at $k = 0$ determines c_1 from c_0 alone. By induction on k, $c_k = \mathrm{He}_k(y)$ for every $k \geq 0$. Substituting into the Taylor expansion of G gives the asserted identity, with the absolute convergence inherited from that expansion. □

2.6. `HermiteAbsMajorant`.

Lemma 2.6 (Absolute-value majorant for the Hermite generating series)**.** *Let* He_k *be the probabilists' Hermite polynomials of* `HermiteDerivative`*. For all real* w *and* z *the series* $\sum_{k\geq 0}|\mathrm{He}_k(w)|\,|z|^k/k!$ *converges, and*

$$\sum_{k=0}^{\infty}\big|\mathrm{He}_k(w)\big|\,\frac{|z|^k}{k!} \;\leq\; \exp\big(|w|\,|z| + z^2/2\big).$$

Note the sign of the z^2 *term: it is* $+z^2/2$*, not* $-z^2/2$*.*

Proof. Step 0: the route taken, and why. The asserted bound cannot be obtained by inserting absolute values into a signed generating identity for He_k: from any identity of the form $\sum_k \mathrm{He}_k(w)z^k/k! = \Phi(w,z)$ the triangle inequality yields $|\Phi(w,z)| \leq \sum_k |\mathrm{He}_k(w)|\,|z|^k/k!$, which is a lower bound for the quantity to be estimated rather than the upper bound asserted. No such

identity is used anywhere below. Instead the proof builds an all-plus comparison family $\widehat{\mathrm{He}}_k$ from the defining recurrence, bounds $|\mathrm{He}_k|$ by it, and computes its generating function directly.

Step 1: the three-term recurrence for He_k. By the defining recurrence of the probabilists' Hermite polynomials, which is shared context, $\mathrm{He}_0 = 1$ and $\mathrm{He}_{k+1} = X\,\mathrm{He}_k - \mathrm{He}_k'$ for every $k \geq 0$. Since He_0 is constant, $\mathrm{He}_0' = 0$ and hence $\mathrm{He}_1 = X$. For $k \geq 1$ the conclusion of `HermiteDerivative`, applied at index $k-1$, gives $\mathrm{He}_k' = k\,\mathrm{He}_{k-1}$. Substituting,

$$\mathrm{He}_{k+1} = X\,\mathrm{He}_k - k\,\mathrm{He}_{k-1}, \qquad k \geq 1, \tag{1}$$

together with $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = X$.

Step 2: the all-plus companion family. Define real polynomials $\widehat{\mathrm{He}}_k$ by

$$\widehat{\mathrm{He}}_0 = 1, \qquad \widehat{\mathrm{He}}_1 = X, \qquad \widehat{\mathrm{He}}_{k+1} = X\,\widehat{\mathrm{He}}_k + k\,\widehat{\mathrm{He}}_{k-1} \quad (k \geq 1),$$

i.e. the recurrence (1) with the minus sign replaced by a plus. By induction on k, every $\widehat{\mathrm{He}}_k$ has nonnegative coefficients: this holds for $k = 0, 1$, and if it holds at $k-1$ and k then $X\widehat{\mathrm{He}}_k$ and $k\widehat{\mathrm{He}}_{k-1}$ both have nonnegative coefficients, hence so does their sum. Consequently

$$\widehat{\mathrm{He}}_k(W) \geq 0 \qquad \text{for every } W \geq 0. \tag{2}$$

Step 3: $|\mathrm{He}_k(w)| \leq \widehat{\mathrm{He}}_k(|w|)$. We prove this for all real w by induction on k, using the two preceding values. For $k = 0$ both sides equal 1, and for $k = 1$ both sides equal $|w|$. Let $k \geq 1$ and assume the bound at $k-1$ and at k. Evaluating (1) at w and applying the triangle inequality,

$$\big|\mathrm{He}_{k+1}(w)\big| \leq |w|\,\big|\mathrm{He}_k(w)\big| + k\,\big|\mathrm{He}_{k-1}(w)\big| \leq |w|\,\widehat{\mathrm{He}}_k(|w|) + k\,\widehat{\mathrm{He}}_{k-1}(|w|) = \widehat{\mathrm{He}}_{k+1}(|w|),$$

where the middle step uses the two inductive hypotheses together with $|w| \geq 0$ and $k \geq 0$ (both multipliers are nonnegative, so the inequalities are preserved), and the last step is the defining recurrence of Step 2 evaluated at $|w| \geq 0$. This completes the induction.

Step 4: the generating function of the all-plus family. Fix $W \in \mathbb{R}$ and set $\widehat{G}(z) = e^{Wz + z^2/2}$, an entire function of z (the exponential of a polynomial), so $\widehat{G}$ equals its Taylor series about 0 with absolute convergence for every z: $\widehat{G}(z) = \sum_{k \geq 0} \hat{c}_k z^k/k!$ with $\hat{c}_k = \widehat{G}^{(k)}(0)$. Differentiating, $\widehat{G}'(z) = (W + z)\,\widehat{G}(z)$. Writing both sides as power series,

$$\widehat{G}'(z) = \sum_{k \geq 0} \hat{c}_{k+1}\frac{z^k}{k!}, \qquad (W+z)\widehat{G}(z) = \sum_{k \geq 0} W\hat{c}_k\frac{z^k}{k!} + \sum_{k \geq 0} \hat{c}_k\frac{z^{k+1}}{k!} = \sum_{k \geq 0}\big(W\hat{c}_k + k\,\hat{c}_{k-1}\big)\frac{z^k}{k!},$$

where the second sum was reindexed by $k \mapsto k+1$ using $\frac{1}{k!} \cdot k = \frac{1}{(k-1)!}$, the $k = 0$ term being absent. Two everywhere-convergent power series are equal only if their coefficients agree, so $\hat{c}_{k+1} = W\hat{c}_k + k\,\hat{c}_{k-1}$ for $k \geq 0$, with the $k = 0$ instance reading $\hat{c}_1 = W\hat{c}_0$. Also $\hat{c}_0 = \widehat{G}(0) = 1$. The numbers $\widehat{\mathrm{He}}_k(W)$ satisfy the identical recurrence and the identical initial value by Step 2, so by induction on k, $\hat{c}_k = \widehat{\mathrm{He}}_k(W)$ for every k. Therefore

$$\sum_{k=0}^{\infty} \widehat{\mathrm{He}}_k(W)\,\frac{z^k}{k!} = e^{Wz + z^2/2}, \qquad W, z \in \mathbb{R}, \tag{3}$$

with absolute convergence.

Step 5: conclusion by comparison. Apply (3) with $W = |w| \geq 0$ and with $|z|$ in place of z, noting $|z|^2 = z^2$:

$$\sum_{k=0}^{\infty} \widehat{\mathrm{He}}_k(|w|)\,\frac{|z|^k}{k!} = e^{|w|\,|z| + z^2/2} < \infty.$$

By (2) every term of this series is nonnegative, and by Step 3 we have the termwise bound

$$0 \;\leq\; \big|\mathrm{He}_k(w)\big|\,\frac{|z|^k}{k!} \;\leq\; \widehat{\mathrm{He}}_k(|w|)\,\frac{|z|^k}{k!} \qquad \text{for every } k.$$

A series of nonnegative terms dominated termwise by a convergent series of nonnegative terms converges, with sum at most that of the dominating series. Hence $\sum_k |\mathrm{He}_k(w)|\,|z|^k/k!$ converges and

$$\sum_{k=0}^{\infty} \big|\mathrm{He}_k(w)\big|\,\frac{|z|^k}{k!} \;\le\; e^{\,|w|\,|z|+z^2/2},$$

which is the assertion. □

2.7. `Hermite1Orthonormal`.

Lemma 2.7 (One-dimensional Hermite orthonormality)**.** *With h_k as in `Hermite1`, the family $\{h_k\}_{k\ge 0}$ is orthonormal in $L^2(\pi^{-1/2}e^{-x^2}\,dx)$: for all $k,l \in \mathbb{N}_0$ the integral below converges absolutely and*

$$\int_{\mathbb{R}} \pi^{-1/2}e^{-x^2}\,h_k(x)\,h_l(x)\,dx = \begin{cases} 1, & k=l,\\ 0, & k\neq l. \end{cases}$$

Proof. Throughout write $d\gamma_1(x) = \pi^{-1/2}e^{-x^2}\,dx$. No change of variables is made anywhere in this proof: every integral below is taken against γ_1 in the variable x, so the Gaussian background items are used at the weight e^{-x^2} at which they are stated.

Step 0: γ_1 is a probability measure, and its moments. By the scalar Gaussian integral of the shared context, $\int_{\mathbb{R}} e^{-x^2}\,dx = \sqrt{\pi}$, so $\gamma_1(\mathbb{R}) = 1$ and γ_1 is a probability measure. By the Gaussian exponential moment item of the shared context, taken with $n = 1$,

$$\int_{\mathbb{R}} e^{c|x|}\,d\gamma_1(x) < \infty \quad\text{and}\quad \int_{\mathbb{R}} |P(x)|\,e^{c|x|}\,d\gamma_1(x) < \infty \tag{0}$$

for every real c and every real polynomial P. In particular every polynomial is γ_1-integrable.

By `Hermite1`, $h_k(x) = \mathrm{He}_k(\sqrt{2}\,x)/\sqrt{k!}$ for every k and every real x, where He_k is the probabilists' Hermite polynomial of degree k. Since $x \mapsto \mathrm{He}_k(\sqrt{2}x)\,\mathrm{He}_l(\sqrt{2}x)$ is a polynomial, the second part of (0) (with $c = 0$) shows that the integral in the statement converges absolutely, and

$$\int_{\mathbb{R}} h_k\,h_l\,d\gamma_1 = \frac{a_{kl}}{\sqrt{k!\,l!}}, \qquad \text{where} \qquad a_{kl} := \int_{\mathbb{R}} \mathrm{He}_k(\sqrt{2}x)\,\mathrm{He}_l(\sqrt{2}x)\,d\gamma_1(x). \tag{1}$$

By (1) it suffices to prove

$$a_{kl} = \begin{cases} k!, & k=l,\\ 0, & k\neq l, \end{cases} \tag{$*$}$$

since then $a_{kl}/\sqrt{k!\,l!}$ equals $k!/\sqrt{k!\,k!} = 1$ when $k = l$ and 0 otherwise.

Step 1: the moment generating function of γ_1. For every real c,

$$\int_{\mathbb{R}} e^{cx}\,d\gamma_1(x) = e^{c^2/4}. \tag{2}$$

Indeed $cx - x^2 = -\big(x - c/2\big)^2 + c^2/4$, so $e^{cx}\pi^{-1/2}e^{-x^2} = e^{c^2/4}\,\pi^{-1/2}e^{-(x-c/2)^2}$ pointwise; by the translation invariance of Lebesgue measure (shared context, with $t = c/2$ and $h(y) = \pi^{-1/2}e^{-y^2} \ge 0$), $\int \pi^{-1/2}e^{-(x-c/2)^2}\,dx = \int \pi^{-1/2}e^{-y^2}\,dy = 1$ by Step 0. Integrating this pointwise identity between nonnegative measurable functions and pulling the constant $e^{c^2/4}$ out of the integral gives (2); the left side is finite because $e^{cx} \le e^{|c|\,|x|}$ and $\int e^{|c||x|}d\gamma_1 < \infty$ by (0).

Step 2: the two generating series and their absolute majorants. Fix real z and w for the whole of Steps 2–5. For $K \in \mathbb{N}$ put

$$S_K^z(x) = \sum_{k=0}^{K} \mathrm{He}_k(\sqrt{2}x)\,\frac{z^k}{k!}, \qquad S_K^w(x) = \sum_{l=0}^{K} \mathrm{He}_l(\sqrt{2}x)\,\frac{w^l}{l!}, \qquad T_K(x) = S_K^z(x)\,S_K^w(x).$$

Each S_K^z, S_K^w and T_K is a polynomial in x, hence continuous and measurable. By `HermiteGenerating` applied at the point $\sqrt{2}x$ with parameter z, and again at the point $\sqrt{2}x$ with parameter w,

$$S_K^z(x) \xrightarrow[K\to\infty]{} e^{\sqrt{2}xz - z^2/2}, \qquad S_K^w(x) \xrightarrow[K\to\infty]{} e^{\sqrt{2}xw - w^2/2} \tag{3}$$

for every real x, and therefore $T_K(x) \to e^{\sqrt{2}x(z+w)-(z^2+w^2)/2}$ pointwise in x.

For the majorant, apply `HermiteAbsMajorant` twice: once with its first argument $\sqrt{2}x$ and its second argument z, and once with its first argument $\sqrt{2}x$ and its second argument w. Since $|\sqrt{2}x| = \sqrt{2}\,|x|$, this gives, for every real x,

$$\sum_{k=0}^{\infty} \bigl|\mathrm{He}_k(\sqrt{2}x)\bigr| \,\frac{|z|^k}{k!} \le e^{\sqrt{2}|x|\,|z|+z^2/2}, \qquad \sum_{l=0}^{\infty} \bigl|\mathrm{He}_l(\sqrt{2}x)\bigr| \,\frac{|w|^l}{l!} \le e^{\sqrt{2}|x|\,|w|+w^2/2}, \tag{4}$$

both series converging. (These are genuine upper bounds on the absolute-value series; they are *not* obtained by putting absolute values on the signed identity of `HermiteGenerating`, which would bound the absolute-value series from below instead.) Truncating the sums in (4) at K and using $|\sum| \le \sum|\cdot|$ for the *finite* sums S_K^z, S_K^w,

$$|T_K(x)| \;\le\; \Bigl(\sum_{k=0}^{\infty} \bigl|\mathrm{He}_k(\sqrt{2}x)\bigr| \frac{|z|^k}{k!}\Bigr)\Bigl(\sum_{l=0}^{\infty} \bigl|\mathrm{He}_l(\sqrt{2}x)\bigr| \frac{|w|^l}{l!}\Bigr) \;\le\; \Phi(x) := e^{(z^2+w^2)/2}\, e^{\sqrt{2}(|z|+|w|)\,|x|} \tag{5}$$

for every K and every x. By (0) with $c = \sqrt{2}(|z|+|w|)$, the function Φ is γ_1-integrable.

Step 3: passing to the limit, and the value e^{zw}. By dominated convergence (shared context), applied to the sequence (T_K) with dominating function Φ and pointwise limit $x \mapsto e^{\sqrt{2}x(z+w)-(z^2+w^2)/2}$,

$$\int T_K \, d\gamma_1 \;\xrightarrow[K\to\infty]{}\; e^{-(z^2+w^2)/2} \int e^{\sqrt{2}(z+w)x}\, d\gamma_1(x) \;=\; e^{-(z^2+w^2)/2}\, e^{(z+w)^2/2} \;=\; e^{zw},$$

where the middle equality is (2) with $c = \sqrt{2}(z+w)$, so that $c^2/4 = 2(z+w)^2/4 = (z+w)^2/2$, and the last uses $(z+w)^2 - z^2 - w^2 = 2zw$.

On the other hand T_K is the finite product of two finite sums, so expanding and using linearity of the integral over the finitely many terms — each $x \mapsto \mathrm{He}_k(\sqrt{2}x)\mathrm{He}_l(\sqrt{2}x)$ being γ_1-integrable by (0) — gives

$$\int T_K \, d\gamma_1 = \sum_{k=0}^{K}\sum_{l=0}^{K} b_{kl}, \qquad \text{where} \qquad b_{kl} := a_{kl}\,\frac{z^k w^l}{k!\,l!}. \tag{6}$$

Step 4: absolute summability of the double family, and its sum. We first record the master bound: for all $\rho, \sigma \ge 0$,

$$\sum_{k,l\ge 0} |a_{kl}|\,\frac{\rho^k \sigma^l}{k!\,l!} \;<\; \infty. \tag{7}$$

Indeed $|a_{kl}| \le \int |\mathrm{He}_k(\sqrt{2}x)|\,|\mathrm{He}_l(\sqrt{2}x)|\,d\gamma_1(x)$, so by Tonelli (shared context, with counting measure on $\mathbb{N}\times\mathbb{N}$ and γ_1, both σ-finite, applied to the nonnegative integrand $((k,l),x) \mapsto |\mathrm{He}_k(\sqrt{2}x)||\mathrm{He}_l(\sqrt{2}x)|\rho^k\sigma^l/(k!l!)$),

$$\sum_{k,l} |a_{kl}|\frac{\rho^k\sigma^l}{k!\,l!} \le \int \Bigl(\sum_k \bigl|\mathrm{He}_k(\sqrt{2}x)\bigr|\frac{\rho^k}{k!}\Bigr)\Bigl(\sum_l \bigl|\mathrm{He}_l(\sqrt{2}x)\bigr|\frac{\sigma^l}{l!}\Bigr) d\gamma_1(x) \le \int e^{(\rho^2+\sigma^2)/2} e^{\sqrt{2}(\rho+\sigma)|x|}\, d\gamma_1(x),$$

the factorization of the double sum into a product of two single sums being Tonelli for the two counting measures, and the last inequality being (4) with z, w replaced by ρ, σ. The final integral is finite by (0), which proves (7).

Taking $\rho = |z|$, $\sigma = |w|$ in (7) shows that the family $(b_{kl})_{(k,l)\in\mathbb{N}^2}$ of (6) is absolutely summable; let S denote its sum. The square partial sums in (6) converge to S: given $\varepsilon > 0$, fix an enumeration of $\mathbb{N}\times\mathbb{N}$ and let F be the set of its first N terms, with N so large that the corresponding tail of $\sum |b_{kl}|$ is at most ε; choose K_0 with $F \subseteq \{0,\dots,K_0\}^2$ (possible since F is finite); then for every $K \ge K_0$ the finite set $\{0,\dots,K\}^2$ contains F, so $\bigl|\sum_{k,l\le K} b_{kl} - S\bigr| \le \sum_{(k,l)\notin F} |b_{kl}| \le \varepsilon$. Comparing with Step 3,

$$S = \sum_{k,l\ge 0} a_{kl}\,\frac{z^k w^l}{k!\,l!} = e^{zw}. \tag{8}$$

Next, regroup (8) by the second index. For $l \in \mathbb{N}$ set

$$c_l(z) := \sum_{k\geq 0} a_{kl}\,\frac{z^k}{k!\,l!}, \tag{9}$$

the series converging absolutely for each fixed l by (7) (take $\rho = |z|$, $\sigma = 1$, and drop all indices but l in the second slot). By Fubini for the absolutely summable family (b_{kl}) — that is, the Fubini item of the shared context applied with counting measure on $\mathbb{N}$ in each factor, legitimate because $\sum_{k,l}|b_{kl}| < \infty$ by (7) — we may sum first in k and then in l:

$$\sum_{l\geq 0} c_l(z)\,w^l = S = e^{zw}, \tag{10}$$

and moreover, for every $\sigma \geq 0$, $\sum_l |c_l(z)|\sigma^l \leq \sum_{k,l}|a_{kl}|\,|z|^k\sigma^l/(k!\,l!) < \infty$ by (9) and (7).

Step 5: a one-variable coefficient-comparison principle, applied twice. We use the following elementary principle (U).

(U). *Let $(p_m)_{m\geq 0}$ and $(q_m)_{m\geq 0}$ be real sequences with $\sum_m |p_m|\rho^m < \infty$ and $\sum_m |q_m|\rho^m < \infty$ for every $\rho \geq 0$, and suppose $\sum_m p_m t^m = \sum_m q_m t^m$ for every real t. Then $p_m = q_m$ for every m.*

This is the real-variable instance of the uniqueness of power-series coefficients recorded in the shared context; since that item is stated for a neighbourhood of the origin in $\mathbb{C}$ and the hypotheses here give agreement only for real t, we derive (U) directly. Put $e_m = p_m - q_m$, so $\sum_m |e_m|\rho^m < \infty$ for every $\rho \geq 0$ and, the two series being absolutely convergent, $E(t) := \sum_m e_m t^m = 0$ for every real t. Argue by induction on m: suppose $L \geq 0$ and $e_0 = \cdots = e_{L-1} = 0$ (vacuous when $L = 0$). For $0 < |t| \leq 1$,

$$0 = E(t) = \sum_{m\geq L} e_m t^m = t^L\Big(e_L + R(t)\Big), \qquad R(t) := \sum_{m\geq L+1} e_m t^{m-L},$$

and $|R(t)| \leq |t|\sum_{m\geq L+1}|e_m|\,|t|^{m-L-1} \leq |t|\sum_{m\geq L+1}|e_m| = C|t|$, where $C < \infty$ by the hypothesis with $\rho = 1$ and $|t|^{m-L-1} \leq 1$. Dividing by $t^L \neq 0$ gives $e_L = -R(t)$, hence $|e_L| \leq C|t|$ for all $0 < |t| \leq 1$; letting $t \to 0$ gives $e_L = 0$. This completes the induction and proves (U).

First application, in the variable w. Fix real z. The exponential function is entire, so by the power-series item of the shared context it equals its everywhere-absolutely-convergent Taylor series at 0; evaluating that series at zw gives $e^{zw} = \sum_{m\geq 0}(zw)^m/m! = \sum_{l\geq 0}(z^l/l!)\,w^l$, with $\sum_l |z^l/l!|\sigma^l < \infty$ for every $\sigma \geq 0$. By (10) the two series $\sum_l c_l(z)w^l$ and $\sum_l (z^l/l!)w^l$ agree for every real w, and both satisfy the summability hypothesis of (U) (the first by the bound at the end of Step 4). Hence (U) gives

$$c_l(z) = \frac{z^l}{l!} \qquad \text{for every } l \in \mathbb{N} \text{ and every real } z. \tag{11}$$

Second application, in the variable z. Fix $l \in \mathbb{N}$ and multiply (11) by $l!$; by (9) this reads

$$\sum_{k\geq 0} \frac{a_{kl}}{k!}\,z^k \;=\; z^l \qquad \text{for every real } z.$$

The left-hand coefficients satisfy $\sum_k |a_{kl}/k!|\rho^k < \infty$ for every $\rho \geq 0$, by (7) with $\sigma = 1$ and the single index l retained; the right-hand side is the power series with coefficients δ_{kl}, trivially absolutely convergent. By (U), now in the variable z, $a_{kl}/k! = \delta_{kl}$ for every k, i.e. $a_{kl} = k!$ when $k = l$ and $a_{kl} = 0$ when $k \neq l$. This is $(*)$, and by (1) the statement follows. □

2.8. `HermiteL2Orthonormal`.

Lemma 2.8 (The n-dimensional Hermite family is orthonormal)**.** *The family $\{h_\alpha : \alpha \in \mathbb{N}_0^n\}$ of `HermiteL2` is orthonormal in $L^2(\gamma_n)$:*

$$\int_{\mathbb{R}^n} h_\alpha h_\beta\,d\gamma_n = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \qquad \alpha, \beta \in \mathbb{N}_0^n.$$

Proof. By `GaussMeasure` the density of γ_n factorizes as $\pi^{-n/2}e^{-|x|^2} = \prod_{j=1}^n \pi^{-1/2}e^{-x_j^2}$, and Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of Lebesgue measure on $\mathbb{R}$; hence γ_n is the n-fold product of the one-dimensional measure $d\gamma_1(s) = \pi^{-1/2}e^{-s^2}\,ds$.

By `HermiteFun`,

$$h_\alpha(x)\,h_\beta(x) = \prod_{j=1}^n \big(h_{\alpha_j}(x_j)\,h_{\beta_j}(x_j)\big),$$

a product of functions of the separate coordinates. The product is γ_n-integrable: by `HermiteFunMemLp` both h_α and h_β lie in $L^2(\gamma_n)$, so their product is in $L^1(\gamma_n)$ by Cauchy–Schwarz. Fubini's theorem for the product measure therefore applies and gives

$$\int_{\mathbb{R}^n} h_\alpha h_\beta\,d\gamma_n = \prod_{j=1}^n \int_{\mathbb{R}} \pi^{-1/2}e^{-s^2}\,h_{\alpha_j}(s)\,h_{\beta_j}(s)\,ds.$$

By `Hermite1Orthonormal` the jth factor equals 1 if $\alpha_j = \beta_j$ and 0 otherwise. Consequently the product equals 1 exactly when $\alpha_j = \beta_j$ for every $j \in \{1,\dots,n\}$, i.e. when $\alpha = \beta$, and it contains a zero factor, hence vanishes, as soon as $\alpha_j \neq \beta_j$ for some j, i.e. when $\alpha \neq \beta$. □

2.9. `L2IntoL1`.

Lemma 2.9 (L^2 controls L^1 for a probability measure)**.** *For every $v \in L^2(\gamma_n)$ one has, with ι the inclusion of `ToL1`,*

$$\|\iota v\|_{L^1(\gamma_n)} \le \|v\|_{L^2(\gamma_n)}, \qquad \textit{that is,} \qquad \int |v|\,d\gamma_n \le \Big(\int |v|^2\,d\gamma_n\Big)^{1/2}.$$

Proof. Since γ_n is a probability measure by `GaussIsProb`, we have $\gamma_n(\mathbb{R}^n) = 1$. By `ToL1`, ιv has the same representative function as v, so $\|\iota v\|_{L^1(\gamma_n)} = \int |v|\,d\gamma_n$. The Cauchy–Schwarz inequality applied to the pair of functions $|v|$ and the constant function 1 gives

$$\int |v|\cdot 1\,d\gamma_n \le \Big(\int |v|^2\,d\gamma_n\Big)^{1/2}\Big(\int 1^2\,d\gamma_n\Big)^{1/2} = \|v\|_{L^2(\gamma_n)}\cdot \gamma_n(\mathbb{R}^n)^{1/2} = \|v\|_{L^2(\gamma_n)}.$$

This is the asserted inequality. □

2.10. `Hermite1Zero`.

Lemma 2.10 (The zeroth Hermite function is the constant one)**.** *With h_k as in `Hermite1`, one has $h_0(x) = 1$ for every $x \in \mathbb{R}$.*

Proof. By the defining recurrence recorded in `Hermite1`, He_0 is the constant polynomial 1, so $\mathrm{He}_0(\sqrt{2}\,x) = 1$ for every $x \in \mathbb{R}$. Moreover $0! = 1$, hence $\sqrt{0!} = 1$. Substituting both into the formula of `Hermite1` gives

$$h_0(x) = \frac{\mathrm{He}_0(\sqrt{2}\,x)}{\sqrt{0!}} = \frac{1}{1} = 1.$$

□

2.11. `HermiteL2Complete`.

Lemma 2.11 (Completeness of the Hermite family in $L^2(\gamma_n)$)**.** *The closed linear span of the family $\{h_\alpha : \alpha \in \mathbb{N}_0^n\}$ of `HermiteL2` is all of $L^2(\gamma_n)$:*

$$\overline{\mathrm{span}_{\mathbb{R}}\{h_\alpha : \alpha \in \mathbb{N}_0^n\}} = L^2(\gamma_n).$$

Proof. The closed span S is a closed linear subspace of the Hilbert space $L^2(\gamma_n)$, so $S = L^2(\gamma_n)$ if and only if $S^\perp = \{0\}$. Let therefore $f \in L^2(\gamma_n)$ satisfy $\langle f, h_\alpha\rangle = \int f h_\alpha\,d\gamma_n = 0$ for every $\alpha \in \mathbb{N}_0^n$; we must show $f = 0$ γ_n-a.e.

Step 1: the Hermite functions of order $\le N$ span the polynomials of degree $\le N$. By `Hermite1` and the recurrence recorded there, He_k is monic of degree k; hence $h_k(x) = \mathrm{He}_k(\sqrt{2}x)/\sqrt{k!}$ is a

one-variable polynomial of degree exactly k with leading coefficient $2^{k/2}/\sqrt{k!} \neq 0$. Consequently, by HermiteFun,

$$h_\alpha(x) = \prod_{j=1}^{n} h_{\alpha_j}(x_j) = c_\alpha\, x^\alpha + \sum_\beta d_{\alpha\beta}\, x^\beta, \qquad c_\alpha = \prod_j \frac{2^{\alpha_j/2}}{\sqrt{\alpha_j!}} \neq 0,$$

where every β occurring in the sum satisfies $\beta_j \le \alpha_j$ for all j and $\beta \neq \alpha$, hence $|\beta| < |\alpha|$. Let P_N denote the finite-dimensional space of real polynomials on $\mathbb{R}^n$ of total degree at most N, with basis $\{x^\alpha : |\alpha| \le N\}$. The display shows $h_\alpha \in P_{|\alpha|}$, so $\operatorname{span}\{h_\alpha : |\alpha| \le N\} \subseteq P_N$. Conversely, we show $x^\alpha \in \operatorname{span}\{h_\beta : |\beta| \le |\alpha|\}$ by induction on $|\alpha|$: for $|\alpha| = 0$ this is $x^0 = 1 = h_0$ up to the nonzero constant c_0, and for $|\alpha| = N > 0$ the display gives $x^\alpha = c_\alpha^{-1}\big(h_\alpha - \sum_\beta d_{\alpha\beta} x^\beta\big)$ with all $|\beta| < N$, each of which lies in the span by the inductive hypothesis. Hence $\operatorname{span}\{h_\alpha : |\alpha| \le N\} = P_N$ for every N.

Since f is orthogonal to every h_α, it is orthogonal to their span, hence to every P_N, hence to every monomial:

$$\int f\, x^\beta\, d\gamma_n = 0 \qquad \text{for every } \beta \in \mathbb{N}_0^n. \tag{1}$$

These integrals are finite. Indeed, by the span identity just established, x^β is a finite linear combination of the functions h_α with $|\alpha| \le |\beta|$, and each of those lies in $L^2(\gamma_n)$ by HermiteFunMemLp; since $L^2(\gamma_n)$ is a vector space, it follows that $x^\beta \in L^2(\gamma_n)$, and therefore $fx^\beta \in L^1(\gamma_n)$ by Cauchy–Schwarz.

Step 2: two finite measures with the same moments. Write $f = f^+ - f^-$ with $f^\pm = \max\{\pm f, 0\}$, so $0 \le f^\pm \le |f|$ and $f^\pm \in L^2(\gamma_n)$. Let $\mu^\pm$ be the measures with densities $f^\pm$ with respect to γ_n, i.e. $\mu^\pm(A) = \int_A f^\pm d\gamma_n$. They are finite, since $\mu^\pm(\mathbb{R}^n) = \int f^\pm d\gamma_n \le \|f^\pm\|_{L^2(\gamma_n)} < \infty$ by L2IntoL1. Since $f = f^+ - f^-$, identity (1) says

$$\int x^\beta\, d\mu^+ = \int x^\beta\, d\mu^- \qquad \text{for every } \beta \in \mathbb{N}_0^n. \tag{2}$$

Step 3: exponential moments. Fix $c > 0$. The function $x \mapsto e^{c|x|}$ lies in $L^2(\gamma_n)$: by GaussMeasure, $\int e^{2c|x|}\, d\gamma_n = \int \pi^{-n/2} e^{2c|x| - |x|^2}\, dx$, and the integrand is continuous and, for $|x| \ge 4c$, bounded by $\pi^{-n/2} e^{-|x|^2/2}$ (because then $2c|x| \le |x|^2/2$). That majorant is Lebesgue integrable: substituting $x = \sqrt{2}\, y$, whose Jacobian determinant is $2^{n/2}$,

$$\int_{\mathbb{R}^n} \pi^{-n/2} e^{-|x|^2/2}\, dx = 2^{n/2} \int_{\mathbb{R}^n} \pi^{-n/2} e^{-|y|^2}\, dy = 2^{n/2}\, \gamma_n(\mathbb{R}^n) = 2^{n/2},$$

where the middle equality is the density formula of GaussMeasure and the last is GaussIsProb. On the complementary region $\{|x| \le 4c\}$ the integrand is continuous, hence bounded, and that region has finite Lebesgue measure, so it contributes a finite amount as well. Hence $\int e^{2c|x|}\, d\gamma_n < \infty$, i.e. $e^{c|\cdot|} \in L^2(\gamma_n)$. Therefore, by Cauchy–Schwarz,

$$\int e^{c|x|}\, d\mu^\pm = \int e^{c|x|}\, f^\pm\, d\gamma_n \le \|e^{c|\cdot|}\|_{L^2(\gamma_n)}\, \|f^\pm\|_{L^2(\gamma_n)} < \infty. \tag{3}$$

Step 4: equal characteristic functions. Fix $\xi \in \mathbb{R}^n$ and expand the exponential as a power series: $e^{i\langle \xi, x\rangle} = \sum_{m \ge 0} \frac{i^m}{m!} \langle \xi, x\rangle^m$. Since $|\langle \xi, x\rangle| \le |\xi|\, |x|$,

$$\sum_{m \ge 0} \frac{1}{m!} \int \big|\langle \xi, x\rangle\big|^m\, d\mu^\pm \le \sum_{m \ge 0} \frac{|\xi|^m}{m!} \int |x|^m\, d\mu^\pm = \int e^{|\xi|\,|x|}\, d\mu^\pm < \infty$$

by (3) with $c = |\xi|$, the interchange of sum and integral for the nonnegative terms being Tonelli's theorem for counting measure times $\mu^\pm$. This absolute convergence permits the same interchange for the signed/complex series, so

$$\int e^{i\langle \xi, x\rangle}\, d\mu^\pm(x) = \sum_{m \ge 0} \frac{i^m}{m!} \int \langle \xi, x\rangle^m\, d\mu^\pm(x).$$

By the multinomial theorem, $\langle \xi, x\rangle^m$ is a finite linear combination of monomials x^β with $|\beta| = m$, with coefficients depending only on ξ and m. Hence by (2) each term $\int \langle \xi, x\rangle^m d\mu^+$ equals $\int \langle \xi, x\rangle^m d\mu^-$, and therefore

$$\int e^{i\langle \xi, x\rangle}\, d\mu^+(x) = \int e^{i\langle \xi, x\rangle}\, d\mu^-(x) \qquad \text{for every } \xi \in \mathbb{R}^n.$$

That is, μ^+ and μ^- are finite Borel measures on $\mathbb{R}^n$ with the same characteristic function. By the uniqueness theorem for characteristic functions of finite Borel measures on $\mathbb{R}^n$, $\mu^+ = \mu^-$.

Step 5: conclusion. Let $A = \{f > 0\}$, a Borel set. On A we have $f^- = 0$, so $\mu^-(A) = 0$, and therefore $\mu^+(A) = 0$ as well, i.e. $\int_A f^+ \, d\gamma_n = 0$. But $f^+ = f > 0$ everywhere on A, and the integral of a strictly positive function over A vanishes only if $\gamma_n(A) = 0$. Hence $\gamma_n(\{f > 0\}) = 0$. Symmetrically, taking $A' = \{f < 0\}$ gives $\mu^+(A') = 0$, hence $\mu^-(A') = 0$, hence $\gamma_n(\{f < 0\}) = 0$. Therefore $f = 0$ γ_n-almost everywhere, i.e. $f = 0$ in $L^2(\gamma_n)$.

Thus $S^\perp = \{0\}$ and $S = L^2(\gamma_n)$. □

2.12. `OUSemigroupSpectral`.

Lemma 2.12 (The semigroup acts by the exponential multiplier)**.** *For every $s \geq 0$, every $v \in L^2(\gamma_n)$ and every $\alpha \in \mathbb{N}_0^n$, the element $P_s v$ of `OUSemigroup` has Hermite coefficients*

$$(P_s v)_\alpha = e^{-s|\alpha|}\, v_\alpha,$$

relative to the basis of `HermiteBasis`. In particular the degenerate convention in `OUSemigroup` is never in force for $s \geq 0$, and $P_s h_\alpha = e^{-s|\alpha|} h_\alpha$, $P_s \mathbf{1} = \mathbf{1}$.

Proof. Let $s \geq 0$ and let $(v_\alpha)_\alpha$ be the coefficient family of v, which is square summable with $\sum_\alpha |v_\alpha|^2 = \|v\|_2^2$ by Parseval (`HermiteBasis`). Since $|\alpha| \geq 0$ and $s \geq 0$, we have $-s|\alpha| \leq 0$, hence $0 < e^{-s|\alpha|} \leq 1$ for every α. Therefore

$$\sum_\alpha \left| e^{-s|\alpha|} v_\alpha \right|^2 \leq \sum_\alpha |v_\alpha|^2 = \|v\|_2^2 < \infty,$$

so the multiplied family is square summable. By `HermiteBasis` the coefficient map is an isometric isomorphism onto $\ell^2(\mathbb{N}_0^n)$, so this family is the coefficient family of a unique element of $L^2(\gamma_n)$, and by the definition in `OUSemigroup` that element is $P_s v$. Hence $(P_s v)_\alpha = e^{-s|\alpha|} v_\alpha$ for every α.

Taking $v = h_\beta$, whose coefficients are 1 at $\alpha = \beta$ and 0 elsewhere, gives $P_s h_\beta = e^{-s|\beta|} h_\beta$. For $\beta = 0$ we get $P_s h_0 = h_0$ since $|0| = 0$, and $h_0 = \mathbf{1}$ by `Hermite1Zero`. □

2.13. `OUSelfAdjointContraction`.

Lemma 2.13 (Self-adjointness and L^2 contractivity of P_s)**.** *For every $s \geq 0$ and all $v, w \in L^2(\gamma_n)$,*

$$\langle P_s v, w\rangle = \langle v, P_s w\rangle \qquad \text{and} \qquad \|P_s v\|_{L^2(\gamma_n)} \leq \|v\|_{L^2(\gamma_n)},$$

with P_s as in `OUSemigroup`.

Proof. By `OUSemigroupSpectral` the coefficients of $P_s v$ and $P_s w$ are $e^{-s|\alpha|} v_\alpha$ and $e^{-s|\alpha|} w_\alpha$. Parseval's identity from `HermiteBasis` applies to both inner products, giving

$$\langle P_s v, w\rangle = \sum_\alpha e^{-s|\alpha|} v_\alpha w_\alpha = \langle v, P_s w\rangle,$$

all three series being absolutely convergent by Cauchy–Schwarz in $\ell^2(\mathbb{N}_0^n)$ together with $0 < e^{-s|\alpha|} \leq 1$ (valid because $s \geq 0$ and $|\alpha| \geq 0$). The two outer expressions coincide because the middle series is symmetric in the roles of v and w.

For the contraction, Parseval and the same bound $e^{-s|\alpha|} \leq 1$ give

$$\|P_s v\|_2^2 = \sum_\alpha e^{-2s|\alpha|} |v_\alpha|^2 \leq \sum_\alpha |v_\alpha|^2 = \|v\|_2^2,$$

and taking square roots finishes the proof. □

2.14. `EnergyFormBilinear`.

Lemma 2.14 (The energy form is a real symmetric bilinear form with quadratic form $\mathcal{E}$)**.** *Let $\mathcal{E}(\cdot)$ be the $[0,+\infty]$-valued energy of `Energy`, let $\mathcal{E}(\cdot,\cdot)$ be the real-valued form of `EnergyBil`, and write v_α for the Hermite coefficients of `HermiteBasis` and $|\alpha|$ for the multi-index weight of `MultiIndexSum`. Then:*

(1) *for all $v, w \in L^2(\gamma_n)$ with $\mathcal{E}(v) < \infty$ and $\mathcal{E}(w) < \infty$, the family $\big(|\alpha|^{1/2}\, v_\alpha\, w_\alpha\big)_{\alpha\neq 0}$ is summable;*

(2) *for every $v \in L^2(\gamma_n)$ with $\mathcal{E}(v) < \infty$, $\mathcal{E}(v,v) = \mathcal{E}(v)$, the right-hand side being read as a real number;*

(3) *for all $v_1, v_2, w \in L^2(\gamma_n)$ with $\mathcal{E}(v_1) < \infty$, $\mathcal{E}(v_2) < \infty$ and $\mathcal{E}(w) < \infty$, and for every $c \in \mathbb{R}$,*

$$\mathcal{E}(v_1 + v_2,\, w) = \mathcal{E}(v_1, w) + \mathcal{E}(v_2, w), \qquad \mathcal{E}(c\, v_1,\, w) = c\, \mathcal{E}(v_1, w), \qquad \mathcal{E}(v_1, w) = \mathcal{E}(w, v_1);$$

(4) *for all $v_1, v_2 \in L^2(\gamma_n)$ with $\mathcal{E}(v_1) < \infty$ and $\mathcal{E}(v_2) < \infty$, and every $c \in \mathbb{R}$, also $\mathcal{E}(v_1+v_2) < \infty$ and $\mathcal{E}(c\, v_1) < \infty$.*

Clause (2) is the bridge between the $[0,+\infty]$-valued energy, whose $+\infty$ convention off $\mathcal{V}$ is built into `Energy`, and the real-valued form; clauses (3) and (4) say that $\mathcal{E}(\cdot,\cdot)$ is a symmetric bilinear form on the linear subspace $\mathcal{V} = \{v : \mathcal{E}(v) < \infty\}$.

Proof. Throughout write, for $\alpha \neq 0$,

$$e_\alpha(v) = |\alpha|^{1/2}\, v_\alpha^2 \geq 0,$$

the α-th summand of $\mathcal{E}$; the inequality holds because $|\alpha|$ is a nonnegative integer by `MultiIndexSum`, so $|\alpha|^{1/2} \geq 0$, and $v_\alpha^2 \geq 0$. By `Energy`,

$$\mathcal{E}(v) = \sum_{\alpha\neq 0} e_\alpha(v) \qquad \text{as an element of } [0,+\infty],$$

and by `EnergyBil`, $\mathcal{E}(v,w) = \sum_{\alpha\neq 0} |\alpha|^{1/2} v_\alpha w_\alpha$.

Step 1: finiteness of the energy is summability of its summands. Since every $e_\alpha(v)$ is nonnegative, the extended-real bridge of the shared context applies to the family $(e_\alpha(v))_{\alpha\neq 0}$: the sum $\sum_{\alpha\neq 0} e_\alpha(v)$ is finite in $[0,+\infty]$ if and only if the family is summable as a family of real numbers, and in that case the two readings of the sum agree. Hence

$$\mathcal{E}(v) < \infty \iff \big(e_\alpha(v)\big)_{\alpha\neq 0} \text{ is summable}, \quad \text{and then} \quad \mathcal{E}(v) = \sum_{\alpha\neq 0} e_\alpha(v) \in \mathbb{R}. \tag{1}$$

Step 2: clause (1). Let $\mathcal{E}(v), \mathcal{E}(w) < \infty$. For every $\alpha \neq 0$, the elementary inequality $|s|\,|t| \leq \tfrac{1}{2}(s^2 + t^2)$ — which is $0 \leq (|s| - |t|)^2$ rearranged — applied with $s = v_\alpha$, $t = w_\alpha$ and multiplied by $|\alpha|^{1/2} \geq 0$ gives

$$\big||\alpha|^{1/2} v_\alpha w_\alpha\big| = |\alpha|^{1/2}\, |v_\alpha|\, |w_\alpha| \leq \tfrac{1}{2}\big(e_\alpha(v) + e_\alpha(w)\big). \tag{2}$$

By Step 1 both $(e_\alpha(v))$ and $(e_\alpha(w))$ are summable, so by linearity of summation (shared context) the right-hand side of (2) is a summable family of nonnegative reals; by the comparison test (shared context) the family of absolute values on the left is then summable, and a family whose absolute values are summable is itself summable. This is clause (1).

Step 3: clause (2). Let $\mathcal{E}(v) < \infty$. Termwise $|\alpha|^{1/2} v_\alpha v_\alpha = e_\alpha(v)$, so the two series defining $\mathcal{E}(v,v)$ and $\sum_{\alpha\neq 0} e_\alpha(v)$ have identical terms and hence identical sums. By the second half of (1) that common value is $\mathcal{E}(v)$ read as a real number, which is clause (2).

Step 4: clause (3). By `HermiteBasis` the coefficient map $v \mapsto (v_\alpha)_\alpha$ is linear, so $(v_1 + v_2)_\alpha = (v_1)_\alpha + (v_2)_\alpha$ and $(c\, v_1)_\alpha = c\, (v_1)_\alpha$ for every α. Therefore, termwise,

$$|\alpha|^{1/2}(v_1 + v_2)_\alpha w_\alpha = |\alpha|^{1/2}(v_1)_\alpha w_\alpha + |\alpha|^{1/2}(v_2)_\alpha w_\alpha, \qquad |\alpha|^{1/2}(c\, v_1)_\alpha w_\alpha = c\, |\alpha|^{1/2}(v_1)_\alpha w_\alpha.$$

By Step 2 the two families $(|\alpha|^{1/2}(v_i)_\alpha w_\alpha)_{\alpha\neq 0}$, $i = 1, 2$, are summable, so by linearity of summation (shared context) the sum of the first display is the sum of the two sums, giving additivity,

and the sum of the second display is c times the sum, giving homogeneity. Symmetry is immediate termwise from $v_\alpha w_\alpha = w_\alpha v_\alpha$, the two series again having identical terms.

Step 5: clause (4). Let $\mathcal{E}(v_1), \mathcal{E}(v_2) < \infty$ and $c \in \mathbb{R}$. From $0 \le (s-t)^2$ we get $(s+t)^2 \le 2s^2 + 2t^2$, so multiplying by $|\alpha|^{1/2} \ge 0$,

$$0 \le e_\alpha(v_1 + v_2) = |\alpha|^{1/2}\big((v_1)_\alpha + (v_2)_\alpha\big)^2 \le 2\, e_\alpha(v_1) + 2\, e_\alpha(v_2),$$

using linearity of the coefficient map as in Step 4. By Step 1 the right-hand side is summable, so by the comparison test (shared context) $(e_\alpha(v_1 + v_2))$ is summable, and by the other direction of (1), $\mathcal{E}(v_1 + v_2) < \infty$. Likewise $e_\alpha(c\, v_1) = c^2\, e_\alpha(v_1)$, a scalar multiple of a summable family, hence summable, so $\mathcal{E}(c\, v_1) < \infty$. □

2.15. `SubordinationIntegrandIntegrable`.

Lemma 2.15 (Integrability of the subordination integrand on $(0,\infty)$)**.** *Let $\lambda \in \mathbb{R}$ with $\lambda \ge 0$. Then the function*

$$s \longmapsto \frac{1 - e^{-s\lambda}}{s^{3/2}}$$

is integrable on the half-line $(0,\infty)$ with respect to Lebesgue measure.

Proof. Write $f(s) = (1 - e^{-s\lambda})\, s^{-3/2}$ for $s > 0$. Since $(0,\infty) = (0,1] \cup (1,\infty)$ is a disjoint union of measurable sets, f is integrable on $(0,\infty)$ if and only if it is integrable on each of the two pieces, so it suffices to treat the two regimes separately. They genuinely need different arguments: near the origin the numerator is small and must be used, while at infinity the numerator is merely bounded and it is the decay of $s^{-3/2}$ that is used.

Two elementary facts about the exponential are used in both regimes. First, from the convexity inequality $1 + x \le e^x$ applied at $x = -u$,

$$1 - e^{-u} \le u \qquad \text{for every } u \in \mathbb{R}. \tag{E1}$$

Second, if $u \ge 0$ then $-u \le 0$, so $e^{-u} \le 1$ and hence

$$0 \le 1 - e^{-u} \qquad \text{for every } u \ge 0. \tag{E2}$$

Note also that f is measurable on $(0,\infty)$, being built from the continuous functions $s \mapsto e^{-s\lambda}$ and $s \mapsto s^{-3/2}$ by arithmetic operations, the latter continuous and positive on $(0,\infty)$. In each regime we therefore only need a dominating integrable majorant.

Regime 1: the interval $(0,1]$. Let $0 < s \le 1$. Then $s\lambda \ge 0$ because $s > 0$ and $\lambda \ge 0$, so (E2) at $u = s\lambda$ gives $0 \le 1 - e^{-s\lambda}$, and therefore $f(s) \ge 0$ and $|f(s)| = f(s)$. By (E1) at $u = s\lambda$ we have $1 - e^{-s\lambda} \le s\lambda$, and dividing by the positive number $s^{3/2}$ preserves the inequality:

$$|f(s)| = \frac{1 - e^{-s\lambda}}{s^{3/2}} \le \frac{s\lambda}{s^{3/2}} = \lambda\, s^{1-3/2} = \lambda\, s^{-1/2},$$

where $s/s^{3/2} = s^{-1/2}$ because $s^{-1/2}\, s^{3/2} = s^1$ for $s > 0$. The function $s \mapsto s^p$ is integrable on $(0,1]$ whenever $p > -1$, and $p = -\frac{1}{2} > -1$, so $s \mapsto \lambda\, s^{-1/2}$ is integrable on $(0,1]$. A measurable function dominated in absolute value by an integrable function is integrable, so f is integrable on $(0,1]$.

Regime 2: the interval $(1,\infty)$. Let $s > 1$. Again $s\lambda \ge 0$, so (E2) gives $0 \le 1 - e^{-s\lambda}$ and $|f(s)| = f(s)$. This time we use only that the numerator is bounded above by 1: since $e^{-s\lambda} > 0$, we have $1 - e^{-s\lambda} \le 1$, and hence

$$|f(s)| = \frac{1 - e^{-s\lambda}}{s^{3/2}} \le \frac{1}{s^{3/2}} = s^{-3/2}.$$

The function $s \mapsto s^a$ is integrable on $(1,\infty)$ whenever $a < -1$, and $a = -\frac{3}{2} < -1$, so $s \mapsto s^{-3/2}$ is integrable on $(1,\infty)$. By the same domination principle, f is integrable on $(1,\infty)$.

Conclusion. As f is integrable on $(0,1]$ and on $(1,\infty)$, it is integrable on their union $(0,\infty)$, which is the assertion.

Two remarks. The case $\lambda = 0$ is not exceptional: both (E1) and (E2) hold at $\lambda = 0$, where the numerator $1 - e^0 = 0$ and the integrand vanishes identically, so the two bounds above degenerate

harmlessly to $0 \le 0$ and $0 \le s^{-3/2}$. Also, the exponent $3/2$ is what makes both ends work at once: for a general exponent $a > 0$ in place of $3/2$, the bound near the origin is λs^{1-a}, which is integrable on $(0,1]$ exactly when $1-a > -1$, that is $a < 2$, while the bound at infinity is s^{-a}, which is integrable on $(1,\infty)$ exactly when $a > 1$. Both regimes therefore require $1 < a < 2$, and $a = 3/2$ is the midpoint of that range. □

2.16. `ScalarFractional`.

Lemma 2.16 (Subordination identity for the square root)**.** *For every real $\lambda \ge 0$ the integral below converges absolutely and*

$$\frac{1}{2\sqrt{\pi}} \int_0^\infty \bigl(1 - e^{-s\lambda}\bigr)\, \frac{ds}{s^{3/2}} = \sqrt{\lambda}.$$

Proof. The integrand is nonnegative, since $1 - e^{-s\lambda} \ge 0$ for $s, \lambda \ge 0$, so absolute convergence is the same as finiteness of the integral, which the computation below establishes.

If $\lambda = 0$ the integrand vanishes identically and both sides are 0.

Assume $\lambda > 0$. Near $s = 0$ we have $1 - e^{-s\lambda} \le s\lambda$, so the integrand is at most $\lambda s^{-1/2}$, which is integrable on $(0,1]$; near infinity the integrand is at most $s^{-3/2}$, which is integrable on $[1,\infty)$. Hence the integral is finite.

Integrate by parts on $[\varepsilon, R] \subset (0,\infty)$ using $\frac{d}{ds}\bigl(-2s^{-1/2}\bigr) = s^{-3/2}$ and $\frac{d}{ds}\bigl(1 - e^{-s\lambda}\bigr) = \lambda e^{-s\lambda}$:

$$\int_\varepsilon^R \bigl(1 - e^{-s\lambda}\bigr) \frac{ds}{s^{3/2}} = \Bigl[-2s^{-1/2}\bigl(1 - e^{-s\lambda}\bigr)\Bigr]_\varepsilon^R + 2\lambda \int_\varepsilon^R e^{-s\lambda}\, s^{-1/2}\, ds.$$

As $\varepsilon \downarrow 0$ the boundary term at ε tends to 0, because $1 - e^{-\varepsilon\lambda} \le \varepsilon\lambda$ gives $2\varepsilon^{-1/2}(1 - e^{-\varepsilon\lambda}) \le 2\lambda\varepsilon^{1/2} \to 0$; as $R \to \infty$ the boundary term at R tends to 0 because $0 \le 1 - e^{-R\lambda} \le 1$ and $R^{-1/2} \to 0$. Therefore

$$\int_0^\infty \bigl(1 - e^{-s\lambda}\bigr) \frac{ds}{s^{3/2}} = 2\lambda \int_0^\infty e^{-s\lambda} s^{-1/2}\, ds.$$

In the last integral substitute $s = u/\lambda$, so $ds = du/\lambda$ and $s^{-1/2} = \lambda^{1/2} u^{-1/2}$:

$$\int_0^\infty e^{-s\lambda} s^{-1/2} ds = \lambda^{-1/2} \int_0^\infty e^{-u} u^{-1/2}\, du = \lambda^{-1/2}\, \Gamma(1/2) = \lambda^{-1/2}\sqrt{\pi}.$$

Combining, $\int_0^\infty (1 - e^{-s\lambda}) s^{-3/2} ds = 2\lambda \cdot \lambda^{-1/2}\sqrt{\pi} = 2\sqrt{\pi}\,\sqrt{\lambda}$, and dividing by $2\sqrt{\pi}$ gives the assertion. □

2.17. `BilinearEnergyFormula`.

Lemma 2.17 (The energy form as a semigroup integral)**.** *Let $v, w \in \mathcal{V}$, i.e. $\mathcal{E}(v) < \infty$ and $\mathcal{E}(w) < \infty$ in the sense of* ***Energy****. Then the integral below converges absolutely and*

$$\mathcal{E}(v,w) = \frac{1}{2\sqrt{\pi}} \int_0^\infty \bigl\langle (I - P_s)v,\, w \bigr\rangle\, \frac{ds}{s^{3/2}},$$

with $\mathcal{E}(\cdot,\cdot)$ as in ***EnergyBil*** *and P_s as in* ***OUSemigroup****.*

Proof. Write v_α, w_α for the Hermite coefficients of v, w relative to `HermiteBasis`.

Step 1 (pointwise in s). Fix $s > 0$. By `OUSemigroupSpectral` the coefficients of $P_s v$ are $e^{-s|\alpha|} v_\alpha$, so Parseval (`HermiteBasis`) gives

$$\bigl\langle (I - P_s)v, w \bigr\rangle = \sum_{\alpha \ne 0} \bigl(1 - e^{-s|\alpha|}\bigr) v_\alpha w_\alpha,$$

the index $\alpha = 0$ dropping out because $1 - e^0 = 0$. The series converges absolutely since $0 \le 1 - e^{-s|\alpha|} \le 1$ and $(v_\alpha), (w_\alpha)$ are square summable. (The expression is symmetric in v and w, as it must be by `OUSelfAdjointContraction`.)

Step 2 (absolute convergence of the s-integral). All terms below are nonnegative, so Tonelli's theorem applies to counting measure on $\mathbb{N}_0^n \setminus \{0\}$ and $s^{-3/2}ds$ on $(0,\infty)$:

$$\frac{1}{2\sqrt{\pi}}\int_0^\infty \sum_{\alpha\neq 0}\bigl(1-e^{-s|\alpha|}\bigr)|v_\alpha w_\alpha|\frac{ds}{s^{3/2}} = \sum_{\alpha\neq 0}|v_\alpha w_\alpha|\cdot\frac{1}{2\sqrt{\pi}}\int_0^\infty \bigl(1-e^{-s|\alpha|}\bigr)\frac{ds}{s^{3/2}}.$$

For $\alpha \neq 0$ we have $|\alpha| \geq 1 > 0$ by `MultiIndexSum`, so `ScalarFractional` with $\lambda = |\alpha|$ evaluates each inner integral as $|\alpha|^{1/2}$. Hence the double integral equals $\sum_{\alpha\neq 0}|\alpha|^{1/2}|v_\alpha w_\alpha|$, which by the Cauchy–Schwarz bound recorded in `EnergyBil` is at most $\mathcal{E}(v)^{1/2}\mathcal{E}(w)^{1/2}$, finite by hypothesis. This is exactly where the assumptions $\mathcal{E}(v), \mathcal{E}(w) < \infty$ are used. In particular $s \mapsto \langle (I-P_s)v, w\rangle\, s^{-3/2}$ is absolutely integrable on $(0,\infty)$.

Step 3 (Fubini). By Step 2 the double family is absolutely integrable, so Fubini's theorem allows exchanging sum and integral in the signed identity of Step 1:

$$\frac{1}{2\sqrt{\pi}}\int_0^\infty \langle (I-P_s)v, w\rangle\frac{ds}{s^{3/2}} = \sum_{\alpha\neq 0} v_\alpha w_\alpha \cdot \frac{1}{2\sqrt{\pi}}\int_0^\infty \bigl(1-e^{-s|\alpha|}\bigr)\frac{ds}{s^{3/2}} = \sum_{\alpha\neq 0}|\alpha|^{1/2} v_\alpha w_\alpha,$$

again by `ScalarFractional`. By `EnergyBil` the last sum is $\mathcal{E}(v,w)$, its summability hypothesis being met by Step 2. □

2.18. `CompIntegrationByParts`.

Lemma 2.18 (Integration by parts for a C^1 composition against a test function)**.** *Fix $n \in \mathbb{N}_0$ and write λ for Lebesgue measure on $\mathbb{R}^n$. Let $q, q' : \mathbb{R} \to \mathbb{R}$ be such that q is differentiable at every $t \in \mathbb{R}$ with derivative $q'(t)$, and such that q' is continuous, so $q \in C^1(\mathbb{R})$. Fix a coordinate index $j \in \{1,\dots,n\}$, and let $w : \mathbb{R}^n \to \mathbb{R}$ be differentiable at every point of $\mathbb{R}^n$ and such that its jth partial derivative $x \mapsto \partial_j w(x)$ is continuous on $\mathbb{R}^n$. Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support. Then*

$$\int_{\mathbb{R}^n} q(w(x))\,\partial_j\varphi(x)\,d\lambda(x) = -\int_{\mathbb{R}^n}\bigl(q'(w(x))\,\partial_j w(x)\bigr)\,\varphi(x)\,d\lambda(x),$$

where ∂_j denotes the Fréchet derivative evaluated at the j-th standard basis vector e_j. Both integrals converge, and the identity is one of real numbers.

The hypothesis on w is C^1 in the jth direction, and no more. *Everywhere differentiability plus continuity of the single partial $\partial_j w$ is exactly what the argument below consumes; neither higher smoothness of w nor continuity of the full Fréchet derivative Dw is used, and neither is assumed. That is why the index j is quantified ahead of the hypothesis rather than after it: for each separate direction the identity has its own, separate, minimal hypothesis. A smooth w satisfies both conditions for every j, so the smooth case is contained in this one.*

Every integral in this statement is taken against Lebesgue measure. No Gaussian weight appears: although this lemma is used inside the tablet's Gaussian development, the statement and its proof are purely classical and involve γ_n nowhere. Correspondingly it cites no other tablet node at all: it rests only on the shared background calculus, which is what makes it a self-contained classical identity worth isolating.

Proof. Abbreviate $f = q \circ w$, so $f(x) = q(w(x))$.

Step 1 (the chain rule for f). By hypothesis w is differentiable at every x, with Fréchet derivative $Dw(x)$, and q is differentiable at every real number, so the composite is differentiable at every x with

$$Df(x) = q'(w(x))\,Dw(x)$$

as continuous linear maps $\mathbb{R}^n \to \mathbb{R}$, the scalar $q'(w(x))$ multiplying the linear map $Dw(x)$. Evaluating at e_j gives the pointwise identity

$$\partial_j f(x) = q'(w(x))\,\partial_j w(x) \qquad \text{for every } x \in \mathbb{R}^n. \tag{C}$$

Step 2 (continuity of the four integrands' factors). q is continuous, being differentiable everywhere, and w is continuous, being differentiable everywhere, so $f = q \circ w$ is continuous as a composite of continuous maps. The map $x \mapsto \partial_j w(x)$ is continuous by hypothesis, and

$x \mapsto \partial_j \varphi(x)$ is continuous because φ is smooth, so its derivative is continuous and evaluation at e_j is continuous. By (C), $x \mapsto \partial_j f(x)$ is then continuous as well, being the product of the continuous $q' \circ w$ with the continuous $\partial_j w$. Finally $\partial_j \varphi$ has compact support, the support of a derivative being contained in the support of the function.

Step 3 (integrability of the three products). Integration by parts requires the three products

$$\partial_j f \cdot \varphi, \qquad f \cdot \partial_j \varphi, \qquad f \cdot \varphi$$

to be λ-integrable. Each is a continuous function with compact support: in the first and third the factor φ has compact support, and in the second the factor $\partial_j \varphi$ does, while the other factor is continuous by Step 2. A continuous function with compact support is λ-integrable, so all three integrals converge.

Step 4 (integration by parts). Both f and φ are differentiable at every point, in particular at every point of the closed support of the other, so the classical integration-by-parts identity in the direction e_j applies to the pair (f, φ) with the integrability of Step 3 as its hypothesis, and yields

$$\int_{\mathbb{R}^n} f(x)\, \partial_j \varphi(x)\, d\lambda(x) = -\int_{\mathbb{R}^n} \partial_j f(x)\, \varphi(x)\, d\lambda(x).$$

There is no boundary term because φ is compactly supported.

Step 5 (substituting the chain rule). By (C) the integrands on the right agree at every point: $\partial_j f(x)\, \varphi(x) = \big(q'(w(x))\, \partial_j w(x)\big)\varphi(x)$ for every x, hence in particular λ-almost everywhere, so the two integrals are equal. Substituting into Step 4 and recalling $f = q \circ w$ gives

$$\int_{\mathbb{R}^n} q(w(x))\, \partial_j \varphi(x)\, d\lambda(x) = -\int_{\mathbb{R}^n} \big(q'(w(x))\, \partial_j w(x)\big)\varphi(x)\, d\lambda(x),$$

which is the assertion. □

2.19. `VariationalInequality`.

Lemma 2.19 (The variational inequality satisfied by the minimizer)**.** *Let $F \in L^2(\gamma_n)$, $a \in \mathbb{R}$, and let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`. Then for every $v \in \mathcal{K}$,*

$$\mathcal{E}(u,\, v-u) \;\geq\; \int (F-a)\,(v-u)\, d\gamma_n,$$

with $\mathcal{E}(\cdot,\cdot)$ the energy form of `EnergyBil`.

Proof. Fix $v \in \mathcal{K}$ and put $h = v - u$. For $0 \leq \theta \leq 1$ the function

$$u + \theta h = (1-\theta)u + \theta v$$

is a convex combination of u and v, hence lies in $\mathcal{K}$ because $\mathcal{K}$ is convex by `ObstacleCone`. It is therefore an admissible competitor, and minimality of u gives $q(\theta) := J(u+\theta h) \geq q(0) = J(u)$ for all such θ.

Both u and v lie in $\mathcal{V}$, hence so does h, and $\mathcal{E}(h) < \infty$. Using `ObstacleFunctional`, the identity $\mathcal{E}(w) = \mathcal{E}(w,w)$ and bilinearity and symmetry of $\mathcal{E}(\cdot,\cdot)$ from `EnergyBil`, we compute for $0 < \theta \leq 1$:

$$q(\theta) - q(0) = \tfrac{1}{2}\big(\mathcal{E}(u+\theta h) - \mathcal{E}(u)\big) + \theta \int (a-F)h\, d\gamma_n = \theta\Big(\mathcal{E}(u,h) + \int (a-F)h\, d\gamma_n\Big) + \frac{\theta^2}{2}\mathcal{E}(h),$$

all terms being finite real numbers. Dividing the inequality $q(\theta) - q(0) \geq 0$ by $\theta > 0$ gives

$$\mathcal{E}(u,h) + \int (a-F)h\, d\gamma_n + \frac{\theta}{2}\mathcal{E}(h) \;\geq\; 0.$$

Letting $\theta \downarrow 0$, the last term tends to 0 because $\mathcal{E}(h)$ is a fixed finite number. Hence

$$\mathcal{E}(u,h) + \int (a-F)h\, d\gamma_n \geq 0,$$

that is, $\mathcal{E}(u, v-u) \geq \int (F-a)(v-u)\, d\gamma_n$. □

2.20. `MultiplierVanishesOnPotential`.

Lemma 2.20 (Dual complementarity: the multiplier annihilates the potential)**.** *Let $F \in L^2(\gamma_n)$, $a \in \mathbb{R}$, and let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`. Let η be the multiplier of `ObstacleMultiplier` associated with u, F and a. Then*

$$\langle \eta, u \rangle = 0,$$

equivalently $\mathcal{E}(u) = \int (F - a)\, u \, d\gamma_n$.

Proof. Both $v = 0$ and $v = 2u$ belong to $\mathcal{K}$: the first by `ObstacleCone`, the second because $\mathcal{K}$ is a convex cone there, so it is stable under multiplication by the nonnegative scalar 2.

Apply `VariationalInequality` with $v = 0$. Then $v - u = -u$, and the inequality reads

$$\mathcal{E}(u, -u) \;\geq\; \int (F - a)(-u)\, d\gamma_n,$$

i.e., multiplying by -1 and using bilinearity of $\mathcal{E}(\cdot, \cdot)$ from `EnergyBil`,

$$\mathcal{E}(u, u) - \int (F - a) u \, d\gamma_n \;\leq\; 0,$$

which by `ObstacleMultiplier` says $\langle \eta, u \rangle \leq 0$.

Apply `VariationalInequality` with $v = 2u$. Then $v - u = u$, and the inequality reads

$$\mathcal{E}(u, u) \;\geq\; \int (F - a) u \, d\gamma_n,$$

which by `ObstacleMultiplier` says $\langle \eta, u \rangle \geq 0$.

Combining the two, $\langle \eta, u \rangle = 0$. Since $\langle \eta, u \rangle = \mathcal{E}(u, u) - \int (F - a) u \, d\gamma_n$ and $\mathcal{E}(u, u) = \mathcal{E}(u)$ for $u \in \mathcal{V}$ by `EnergyBil`, this is the stated identity. □

2.21. `MultiplierPositive`.

Lemma 2.21 (Dual-order positivity of the multiplier)**.** *Let $F \in L^2(\gamma_n)$, $a \in \mathbb{R}$, and let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`. Let η be the multiplier of `ObstacleMultiplier` associated with u, F and a. Then*

$$\langle \eta, \varphi \rangle \geq 0 \qquad \text{for every } \varphi \in \mathcal{K},$$

i.e. η takes nonnegative values on every nonnegative test function of finite energy. This is the statement $\eta \geq 0$ in the dual order; it is not yet a pointwise inequality for a function.

Proof. Let $\varphi \in \mathcal{K}$, so $\varphi \geq 0$ γ_n-a.e. and $\mathcal{E}(\varphi) < \infty$. Put $v = u + \varphi$. Then $v \geq 0$ a.e., because $u \geq 0$ and $\varphi \geq 0$ a.e., and $\mathcal{E}(v) < \infty$ because $\mathcal{V}$ is a linear subspace of $L^2(\gamma_n)$; hence $v \in \mathcal{K}$ by `ObstacleCone`.

Apply `VariationalInequality` with this competitor v. Since $v - u = \varphi$, it gives

$$\mathcal{E}(u, \varphi) \;\geq\; \int (F - a)\, \varphi \, d\gamma_n.$$

Rearranging,

$$\mathcal{E}(u, \varphi) - \int (F - a) \varphi \, d\gamma_n \;\geq\; 0,$$

and by `ObstacleMultiplier` the left-hand side is exactly $\langle \eta, \varphi \rangle$. □

2.22. `GaussCharFun`.

Lemma 2.22 (Characteristic function of the Gaussian measure γ_n)**.** *Fix $n \in \mathbb{N}_0$ and let γ_n be the Gaussian measure of `GaussMeasure` on $\mathbb{R}^n$, with density $\pi^{-n/2} e^{-|x|^2}$ with respect to Lebesgue measure. Then for every $t \in \mathbb{R}^n$ the characteristic function of γ_n, namely $\widehat{\gamma_n}(t) = \int_{\mathbb{R}^n} e^{\,i\langle x, t \rangle}\, d\gamma_n(x)$, is given by*

$$\int_{\mathbb{R}^n} e^{\,i\langle x, t \rangle}\, d\gamma_n(x) = e^{-|t|^2/4}.$$

Note the denominator 4 *rather than the* 2 *of the standard normal law: with the normalization* $d\gamma_n = \pi^{-n/2} e^{-|x|^2}\,dx$ *of* `GaussMeasure` *each coordinate has variance* $\frac{1}{2}$ *rather than* 1*, and the characteristic function of a centred Gaussian of covariance* $\sigma^2 I$ *is* $e^{-\sigma^2|t|^2/2}$*, which at* $\sigma^2 = \frac{1}{2}$ *is* $e^{-|t|^2/4}$*.*

Proof. Fix $t \in \mathbb{R}^n$.

Step 1 (unfolding the characteristic function and the density). By definition the characteristic function of a measure μ on an inner-product space at t is $\int e^{\,i\langle x,t\rangle}\,d\mu(x)$, so the quantity to compute is $\int_{\mathbb{R}^n} e^{\,i\langle x,t\rangle}\,d\gamma_n(x)$. By `GaussMeasure`, γ_n is Lebesgue measure weighted by the density

$$\rho(x) = \pi^{-n/2} e^{-|x|^2},$$

which is measurable and everywhere finite (indeed $0 < \rho(x) \le \pi^{-n/2}$). Integration against a weighted measure is integration of the weighted integrand, so

$$\int_{\mathbb{R}^n} e^{\,i\langle x,t\rangle}\,d\gamma_n(x) = \int_{\mathbb{R}^n} \rho(x)\, e^{\,i\langle x,t\rangle}\,dx,$$

the right-hand integral being with respect to Lebesgue measure on $\mathbb{R}^n$.

Step 2 (rewriting the integrand). For every $x \in \mathbb{R}^n$, using $e^a e^b = e^{a+b}$, the symmetry $\langle x,t\rangle = \langle t,x\rangle$ of the real inner product, and the factorization $\rho(x) = \pi^{-n/2}\cdot e^{-|x|^2}$,

$$\rho(x)\, e^{\,i\langle x,t\rangle} = \pi^{-n/2}\, e^{-|x|^2}\, e^{\,i\langle t,x\rangle} = \pi^{-n/2}\, e^{\,-|x|^2+i\langle t,x\rangle}.$$

Since the two integrands agree at every point, their Lebesgue integrals agree, and the constant $\pi^{-n/2}$ may be taken outside the integral by linearity:

$$\int_{\mathbb{R}^n} e^{\,i\langle x,t\rangle}\,d\gamma_n(x) = \pi^{-n/2} \int_{\mathbb{R}^n} e^{\,-|x|^2+i\langle t,x\rangle}\,dx.$$

Step 3 (the Gaussian Fourier integral). The Gaussian Fourier transform in the form valid on a real inner-product space V of finite dimension d states that for complex b with $\operatorname{Re} b > 0$, complex c and $w \in V$,

$$\int_V e^{\,-b|x|^2+c\,\langle w,x\rangle}\,dx = \Big(\frac{\pi}{b}\Big)^{d/2} \exp\Big(\frac{c^2|w|^2}{4b}\Big).$$

Apply this with $V = \mathbb{R}^n$, so $d = \dim_{\mathbb{R}} V = n$, and with $b = 1$ (whose real part is $1 > 0$), $c = i$ and $w = t$. Since $i^2 = -1$,

$$\int_{\mathbb{R}^n} e^{\,-|x|^2+i\langle t,x\rangle}\,dx = \pi^{n/2} \exp\Big(\frac{i^2|t|^2}{4}\Big) = \pi^{n/2}\, e^{-|t|^2/4}.$$

Step 4 (conclusion). Substituting Step 3 into Step 2 and cancelling the powers of π, which is legitimate because $\pi \neq 0$ and $\pi^{-n/2}\pi^{n/2} = \pi^{-n/2+n/2} = \pi^0 = 1$,

$$\int_{\mathbb{R}^n} e^{\,i\langle x,t\rangle}\,d\gamma_n(x) = \pi^{-n/2}\cdot \pi^{n/2}\, e^{-|t|^2/4} = e^{-|t|^2/4},$$

which is the assertion. As a check on the normalization, $t = 0$ gives $\gamma_n(\mathbb{R}^n) = 1$, and the exponent $-|t|^2/4$ is the one recorded in the statement. □

2.23. `GaussMixPushforward`.

Lemma 2.23 (Stability of γ_n under an orthogonal mixing of two independent copies)**.** *Fix* $n \in \mathbb{N}$ *and let* $a, b \in \mathbb{R}$ *satisfy*

$$a \ge 0, \qquad b \ge 0, \qquad a^2 + b^2 = 1.$$

Let γ_n *be the Gaussian measure of* `GaussMeasure` *on* $\mathbb{R}^n$*, and consider the map*

$$\Phi_{a,b} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n, \qquad \Phi_{a,b}(x,y) = a\,x + b\,y.$$

Then the pushforward of the product measure $\gamma_n \otimes \gamma_n$ *along* $\Phi_{a,b}$ *is* γ_n *again:*

$$(\Phi_{a,b})_*\big(\gamma_n \otimes \gamma_n\big) = \gamma_n.$$

Every measure here is Gaussian: the source is the product of two copies of γ_n on $\mathbb{R}^n \times \mathbb{R}^n$ and the target is γ_n itself, with no Lebesgue measure appearing except inside the density that `GaussMeasure` *already carries. In probabilistic language: if X and Y are independent with law γ_n and $a^2 + b^2 = 1$, then $aX + bY$ again has law γ_n. This is the property that lets a γ_n-distributed variable be split as a contraction of itself plus an independent Gaussian of the complementary variance, which is what makes the Ornstein–Uhlenbeck evolution expressible as an average against an independent copy; the explicit size of the resulting kernel is never needed, only that such a representation exists.*

The hypotheses $a \geq 0$ and $b \geq 0$ are what the proof below actually uses, at the one point where $\|a\,t\|$ is replaced by $a\,\|t\|$; see the remark at the end of the proof.

Proof. Step 1: it suffices to compare characteristic functions. The map $\Phi_{a,b}$ is measurable, being built from scalar multiplication and addition. By `GaussIsProb` the measure γ_n is a probability measure, hence so is the product $\gamma_n \otimes \gamma_n$, and hence so is its pushforward along a measurable map. Two probability measures on $\mathbb{R}^n$ with the same characteristic function are equal, so it is enough to prove

$$\widehat{(\Phi_{a,b})_*(\gamma_n \otimes \gamma_n)}(t) = \widehat{\gamma_n}(t) \qquad \text{for every } t \in \mathbb{R}^n. \tag{1}$$

Step 2: the left-hand side as an integral over the product. By the change-of-variables formula for a pushforward along a measurable map, the characteristic function of the pushforward is the integral of the same exponential composed with $\Phi_{a,b}$:

$$\widehat{(\Phi_{a,b})_*(\gamma_n \otimes \gamma_n)}(t) = \int_{\mathbb{R}^n \times \mathbb{R}^n} e^{\,i\langle a\,x + b\,y,\ t\rangle}\, d(\gamma_n \otimes \gamma_n)(x,y).$$

Step 3: the integrand factorizes. For all x, y, t, bilinearity of the inner product gives

$$\langle a\,x + b\,y,\ t\rangle = a\,\langle x, t\rangle + b\,\langle y, t\rangle = \langle x,\ a\,t\rangle + \langle y,\ b\,t\rangle,$$

so, since $e^{z+w} = e^z e^w$,

$$e^{\,i\langle ax+by,\ t\rangle} = e^{\,i\langle x,\ at\rangle} \cdot e^{\,i\langle y,\ bt\rangle}. \tag{2}$$

The two factors depend on x alone and on y alone respectively.

Step 4: Fubini for the product measure. Each factor in (2) has modulus 1, hence is integrable against the probability measure γ_n in its own variable. The integral of a product of two such one-variable functions against $\gamma_n \otimes \gamma_n$ is therefore the product of the two integrals, and each of those is a characteristic function of γ_n:

$$\int_{\mathbb{R}^n \times \mathbb{R}^n} e^{\,i\langle x, at\rangle} e^{\,i\langle y, bt\rangle}\, d(\gamma_n \otimes \gamma_n)(x,y) = \widehat{\gamma_n}(a\,t)\ \widehat{\gamma_n}(b\,t).$$

Step 5: evaluate, and use $a^2 + b^2 = 1$. By `GaussCharFun`, $\widehat{\gamma_n}(u) = e^{-|u|^2/4}$ for every $u \in \mathbb{R}^n$. Now $|a\,t| = |a|\,|t| = a\,|t|$, the last equality because $a \geq 0$, and likewise $|b\,t| = b\,|t|$ because $b \geq 0$. Hence

$$\widehat{\gamma_n}(at)\ \widehat{\gamma_n}(bt) = e^{-a^2|t|^2/4}\, e^{-b^2|t|^2/4} = e^{-(a^2+b^2)|t|^2/4} = e^{-|t|^2/4} = \widehat{\gamma_n}(t),$$

using $e^z e^w = e^{z+w}$ for the middle step and $a^2 + b^2 = 1$ for the next. This is (1), and Step 1 then gives $(\Phi_{a,b})_*(\gamma_n \otimes \gamma_n) = \gamma_n$.

Remark. The conclusion in fact holds for arbitrary real a, b with $a^2 + b^2 = 1$, since Step 5 would then read $|a\,t| = |a|\,|t|$ and $|a|^2 = a^2$ leaves the computation unchanged. The sign hypotheses are stated because they are what this proof route consumes — they are used only to rewrite $|a|$ as a and $|b|$ as b — and not because the statement needs them. □

2.24. `ExpNegSqIntegrationByParts`.

Lemma 2.24 (Integration by parts against the weight $e^{-\sigma^2}$ on the line)**.** *Let $g, g' : \mathbb{R} \to \mathbb{R}$ be such that g is differentiable at every $\sigma \in \mathbb{R}$ with derivative $g'(\sigma)$. Assume the three integrability conditions*

(1) *$\sigma \mapsto e^{-\sigma^2}\, g'(\sigma)$ is integrable on $\mathbb{R}$;*
(2) *$\sigma \mapsto e^{-\sigma^2}\,\big(\sigma\, g(\sigma)\big)$ is integrable on $\mathbb{R}$;*

(3) $\sigma \mapsto e^{-\sigma^2}\, g(\sigma)$ *is integrable on* $\mathbb{R}$.

Then

$$\int_{\mathbb{R}} e^{-\sigma^2}\, \sigma\, g(\sigma)\, d\sigma = \frac{1}{2} \int_{\mathbb{R}} e^{-\sigma^2}\, g'(\sigma)\, d\sigma.$$

Both integrals are ordinary Lebesgue integrals over $\mathbb{R}$ *with respect to* $d\sigma$*: the weight* $e^{-\sigma^2}$ *sits inside the integrand, and the measure is Lebesgue measure, unnormalised. In particular this is not an identity about the Gaussian measure* γ_1 *of the surrounding development, which carries an extra normalising factor* $\pi^{-1/2}$ *appearing nowhere above; no Gaussian measure occurs in this statement at all. Note also that no decay hypothesis on* g *is assumed beyond the three integrability conditions — there is no requirement that* g *vanish at infinity, and no boundary term appears in the conclusion.*

Proof. Set

$$G(\sigma) = -\tfrac{1}{2}\, e^{-\sigma^2},$$

which is the point of the argument: G is an explicit antiderivative of the function $\sigma \mapsto \sigma\, e^{-\sigma^2}$ appearing on the left.

Step 1 ($G'(\sigma) = \sigma e^{-\sigma^2}$). The map $s \mapsto -s^2$ is differentiable with derivative -2σ at σ, so by the chain rule $s \mapsto e^{-s^2}$ is differentiable with derivative $-2\sigma e^{-\sigma^2}$, and multiplying by the constant $-\frac{1}{2}$,

$$G'(\sigma) = -\tfrac{1}{2} \cdot \left(-2\sigma\, e^{-\sigma^2}\right) = \sigma\, e^{-\sigma^2} \qquad (\sigma \in \mathbb{R}).$$

In particular G is differentiable at every point, as is g by hypothesis.

Step 2 (recasting the integrability hypotheses). The integration-by-parts theorem to be applied asks for the integrability of the three products $g' \cdot G$, $g \cdot G'$ and $g \cdot G$. Each is a constant multiple of one of the given integrands, by Step 1 and the definition of G: $g' \cdot G$ is $-\frac{1}{2}$ times the integrand of (1), $g \cdot G'$ is the integrand of (2) rearranged, and $g \cdot G$ is $-\frac{1}{2}$ times the integrand of (3). Since a constant multiple of an integrable function is integrable, all three products are integrable.

Step 3 (integration by parts). Both g and G are differentiable at every real number, so in particular each is differentiable at every point of the closed support of the other, and the integration-by-parts identity on $\mathbb{R}$ applies with the integrability of Step 2 as its hypothesis:

$$\int_{\mathbb{R}} g(\sigma)\, G'(\sigma)\, d\sigma = -\int_{\mathbb{R}} g'(\sigma)\, G(\sigma)\, d\sigma.$$

It is the integrability assumptions, not any decay of g, that licenses this; no boundary contribution survives.

Step 4 (reading off the two sides). By Step 1 the left-hand integrand is $g(\sigma)\, \sigma\, e^{-\sigma^2}$, which is the integrand of the asserted left-hand side, so

$$\int_{\mathbb{R}} e^{-\sigma^2}\, \sigma\, g(\sigma)\, d\sigma = \int_{\mathbb{R}} g(\sigma)\, G'(\sigma)\, d\sigma.$$

On the right, substituting $G(\sigma) = -\frac{1}{2} e^{-\sigma^2}$ and taking the constant out of the integral,

$$-\int_{\mathbb{R}} g'(\sigma)\, G(\sigma)\, d\sigma = -\int_{\mathbb{R}} g'(\sigma) \cdot \left(-\tfrac{1}{2} e^{-\sigma^2}\right) d\sigma = \frac{1}{2} \int_{\mathbb{R}} e^{-\sigma^2}\, g'(\sigma)\, d\sigma.$$

Chaining these two identities with Step 3 gives the assertion.

Remark. The identity is the Gaussian integration-by-parts formula in disguise: dividing both sides by $\sqrt{\pi}$ turns the weight $e^{-\sigma^2}\, d\sigma$ into the probability measure γ_1, and the statement becomes $\int \sigma\, g\, d\gamma_1 = \frac{1}{2} \int g'\, d\gamma_1$. That normalised form is not what is proved here, and the factor $\frac{1}{2}$ above reflects the variance $\frac{1}{2}$ of the weight rather than any choice of normalisation. □

2.25. `Hermite1Reproduction`.

Lemma 2.25 (One-dimensional Hermite reproduction identity)**.** *Let h_k be the normalized one-dimensional Hermite functions of* `Hermite1`*, let $0 \le r \le 1$ and write $\tau = \sqrt{1-r^2}$. Then for every $k \in \mathbb{N}_0$ and every $t \in \mathbb{R}$ the integral below converges absolutely and*

$$\int_{\mathbb{R}} \pi^{-1/2} e^{-\sigma^2}\, h_k\big(r\,t + \tau\,\sigma\big)\, d\sigma = r^k\, h_k(t).$$

Proof. Throughout write $d\gamma_1(\sigma) = \pi^{-1/2} e^{-\sigma^2}\, d\sigma$ and $\tau = \sqrt{1-r^2}$. No change of variables is made anywhere in this proof: every integral below is taken against γ_1 in the variable σ, so the Gaussian background items are used at the weight $e^{-\sigma^2}$ at which they are stated. Note at the outset that $0 \le r \le 1$ gives $0 \le \tau \le 1$, since $0 \le 1 - r^2 \le 1$.

Step 0: γ_1 is a probability measure, and its moments. By the scalar Gaussian integral of the shared context, $\int_{\mathbb{R}} e^{-\sigma^2}\, d\sigma = \sqrt{\pi}$, so $\gamma_1(\mathbb{R}) = 1$. By the Gaussian exponential moment item of the shared context, taken with $n = 1$,

$$\int_{\mathbb{R}} e^{c|\sigma|}\, d\gamma_1(\sigma) < \infty \quad \text{and} \quad \int_{\mathbb{R}} |P(\sigma)|\, e^{c|\sigma|}\, d\gamma_1(\sigma) < \infty \tag{0}$$

for every real c and every real polynomial P. In particular every polynomial is γ_1-integrable.

By `Hermite1`, $h_k(x) = \mathrm{He}_k(\sqrt{2}\,x)/\sqrt{k!}$ for every k and every real x, where He_k is the probabilists' Hermite polynomial of degree k. Applying this at $x = rt + \tau\sigma$, the integrand of the statement is $\pi^{-1/2} e^{-\sigma^2}\, \mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))/\sqrt{k!}$; since $\sigma \mapsto \mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))$ is a polynomial in σ, the second part of (0) with $c = 0$ shows that the integral in the statement converges absolutely, and that

$$\int_{\mathbb{R}} \pi^{-1/2} e^{-\sigma^2}\, h_k(rt + \tau\sigma)\, d\sigma = \frac{a_k}{\sqrt{k!}}, \qquad \text{where} \qquad a_k := \int_{\mathbb{R}} \mathrm{He}_k\big(\sqrt{2}(rt+\tau\sigma)\big)\, d\gamma_1(\sigma). \tag{1}$$

By (1) and the same application of `Hermite1` to the right-hand side of the statement, namely $r^k h_k(t) = r^k\, \mathrm{He}_k(\sqrt{2}\,t)/\sqrt{k!}$, it suffices to prove

$$a_k = r^k\, \mathrm{He}_k(\sqrt{2}\,t) \qquad \text{for every } k \in \mathbb{N}_0. \tag{$*$}$$

Step 1: the moment generating function of γ_1. For every real c,

$$\int_{\mathbb{R}} e^{c\sigma}\, d\gamma_1(\sigma) = e^{c^2/4}. \tag{2}$$

Indeed $c\sigma - \sigma^2 = -\big(\sigma - c/2\big)^2 + c^2/4$, so $e^{c\sigma}\pi^{-1/2}e^{-\sigma^2} = e^{c^2/4}\,\pi^{-1/2}e^{-(\sigma-c/2)^2}$ pointwise; by the translation invariance of Lebesgue measure (shared context, with $t = c/2$ and $h(y) = \pi^{-1/2}e^{-y^2} \ge 0$), $\int \pi^{-1/2}e^{-(\sigma-c/2)^2}\, d\sigma = \int \pi^{-1/2}e^{-y^2}\, dy = 1$ by Step 0. Integrating the pointwise identity between nonnegative measurable functions and pulling the constant $e^{c^2/4}$ out of the integral gives (2); the left side is finite because $e^{c\sigma} \le e^{|c|\,|\sigma|}$ and $\int e^{|c||\sigma|} d\gamma_1 < \infty$ by (0).

Step 2: the generating series at the shifted point, and its majorant. Fix a real z for the whole of Steps 2–4. For $K \in \mathbb{N}$ put

$$S_K(\sigma) = \sum_{k=0}^{K} \mathrm{He}_k\big(\sqrt{2}(rt+\tau\sigma)\big)\, \frac{z^k}{k!},$$

a polynomial in σ, hence continuous and measurable. By `HermiteGenerating` applied at the point $\sqrt{2}(rt+\tau\sigma)$ with parameter z,

$$S_K(\sigma) \xrightarrow[K\to\infty]{} \exp\big(\sqrt{2}(rt+\tau\sigma)z - z^2/2\big) \qquad \text{for every real } \sigma. \tag{3}$$

For the majorant, apply `HermiteAbsMajorant` with its first argument $\sqrt{2}(rt+\tau\sigma)$ and its second argument z: for every real σ,

$$\sum_{k=0}^{\infty} \big|\mathrm{He}_k\big(\sqrt{2}(rt+\tau\sigma)\big)\big|\, \frac{|z|^k}{k!} \;\le\; \exp\big(\sqrt{2}\,|rt+\tau\sigma|\,|z| + z^2/2\big), \tag{4}$$

the series converging. (This is a genuine upper bound on the absolute-value series; it is *not* obtained by putting absolute values on the signed identity of `HermiteGenerating`, which would bound that series from below instead.) Since $0 \le r \le 1$ and $0 \le \tau \le 1$, $|rt + \tau\sigma| \le |t| + |\sigma|$, so the right-hand side of (4) is at most

$$\Phi(\sigma) := e^{\sqrt{2}\,|t|\,|z| + z^2/2}\; e^{\sqrt{2}\,|z|\,|\sigma|}.$$

Truncating the sum in (4) at K and using $|\sum| \le \sum|\cdot|$ for the *finite* sum S_K,

$$\big|S_K(\sigma)\big| \le \Phi(\sigma) \qquad \text{for every } K \text{ and every } \sigma, \tag{5}$$

and Φ is γ_1-integrable by (0) with $c = \sqrt{2}\,|z|$.

Step 3: passing to the limit, and the value of the limit. By dominated convergence (shared context), applied to the sequence (S_K) with dominating function Φ and pointwise limit (3),

$$\int S_K\, d\gamma_1 \;\xrightarrow[K\to\infty]{}\; \int \exp\big(\sqrt{2}(rt+\tau\sigma)z - z^2/2\big)\, d\gamma_1(\sigma).$$

The right-hand side is computed from (2): pulling out the factor $e^{\sqrt{2}\,rtz - z^2/2}$, which does not depend on σ, and applying (2) with $c = \sqrt{2}\,\tau z$, so that $c^2/4 = 2\tau^2 z^2/4 = (1-r^2)z^2/2$,

$$\int \exp\big(\sqrt{2}(rt+\tau\sigma)z - z^2/2\big)\, d\gamma_1(\sigma) = e^{\sqrt{2}\,rtz - z^2/2 + (1-r^2)z^2/2} = \exp\big(\sqrt{2}\,t\,(rz) - (rz)^2/2\big), \tag{6}$$

using $\tau^2 = 1 - r^2$ and $-z^2/2 + (1-r^2)z^2/2 = -r^2z^2/2 = -(rz)^2/2$. Applying `HermiteGenerating` once more, now at the point $\sqrt{2}\,t$ with the parameter rz in place of z, the right-hand side of (6) equals

$$\sum_{k=0}^{\infty} \mathrm{He}_k(\sqrt{2}\,t)\,\frac{(rz)^k}{k!} = \sum_{k=0}^{\infty} r^k\,\mathrm{He}_k(\sqrt{2}\,t)\,\frac{z^k}{k!}. \tag{7}$$

On the other hand S_K is a *finite* sum, and each of its terms $\sigma \mapsto \mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))\,z^k/k!$ is γ_1-integrable by (0), so additivity of the integral over finitely many integrable summands together with the fact that the integral commutes with multiplication by the constant $z^k/k!$ gives

$$\int S_K\, d\gamma_1 = \sum_{k=0}^{K} a_k\,\frac{z^k}{k!} \qquad \text{for every } K. \tag{8}$$

Combining the display above with (6), (7) and (8): for every real z the partial sums $\sum_{k\le K} a_k z^k/k!$ converge as $K \to \infty$, with

$$\sum_{k=0}^{\infty} a_k\,\frac{z^k}{k!} = \sum_{k=0}^{\infty} r^k\,\mathrm{He}_k(\sqrt{2}\,t)\,\frac{z^k}{k!}. \tag{9}$$

Step 4: absolute summability of both coefficient families. Let $\rho \ge 0$. First, $|a_k| \le \int |\mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))|\, d\gamma_1(\sigma)$, so by Tonelli (shared context, with counting measure on $\mathbb{N}_0$ and γ_1, both σ-finite, applied to the nonnegative integrand $(k,\sigma) \mapsto |\mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))|\rho^k/k!$) and then by (4) with $|z|$ replaced by ρ,

$$\sum_{k=0}^{\infty} |a_k|\,\frac{\rho^k}{k!} \le \int \Big(\sum_{k=0}^{\infty} \big|\mathrm{He}_k(\sqrt{2}(rt+\tau\sigma))\big|\,\frac{\rho^k}{k!}\Big)\, d\gamma_1(\sigma) \le \int e^{\sqrt{2}|t|\rho + \rho^2/2}\, e^{\sqrt{2}\rho|\sigma|}\, d\gamma_1(\sigma) < \infty, \tag{10}$$

the last integral being finite by (0). Second, since $0 \le r \le 1$ we have $|r^k| \le 1$, so by `HermiteAbsMajorant` applied with first argument $\sqrt{2}\,t$ and second argument ρ,

$$\sum_{k=0}^{\infty} \big|r^k\,\mathrm{He}_k(\sqrt{2}\,t)\big|\,\frac{\rho^k}{k!} \le \sum_{k=0}^{\infty} \big|\mathrm{He}_k(\sqrt{2}\,t)\big|\,\frac{\rho^k}{k!} \le e^{\sqrt{2}|t|\rho + \rho^2/2} < \infty. \tag{11}$$

Step 5: a one-variable coefficient-comparison principle, derived here. We use the following elementary principle (U).

(U). *Let $(p_m)_{m\ge 0}$ and $(q_m)_{m\ge 0}$ be real sequences with $\sum_m |p_m|\rho^m < \infty$ and $\sum_m |q_m|\rho^m < \infty$ for every $\rho \ge 0$, and suppose $\sum_m p_m z^m = \sum_m q_m z^m$ for every real z. Then $p_m = q_m$ for every m.*

This is the real-variable instance of the uniqueness of power-series coefficients recorded in the shared context; since that item is stated for a neighbourhood of the origin in $\mathbb{C}$ and the hypotheses here give agreement only for real z, it is derived here rather than assumed. Put $e_m = p_m - q_m$, so $\sum_m |e_m|\rho^m < \infty$ for every $\rho \ge 0$ and, the two series being absolutely convergent, $E(z) := \sum_m e_m z^m = 0$ for every real z. Argue by induction on m: suppose $L \ge 0$ and $e_0 = \cdots = e_{L-1} = 0$ (vacuous when $L = 0$). For $0 < |z| \le 1$,

$$0 = E(z) = \sum_{m\ge L} e_m z^m = z^L\big(e_L + R(z)\big), \qquad R(z) := \sum_{m\ge L+1} e_m z^{m-L},$$

and $|R(z)| \le |z| \sum_{m\ge L+1} |e_m|\,|z|^{m-L-1} \le |z| \sum_{m\ge L+1} |e_m| = C|z|$, where $C < \infty$ by the hypothesis with $\rho = 1$ and $|z|^{m-L-1} \le 1$. Dividing by $z^L \ne 0$ gives $e_L = -R(z)$, hence $|e_L| \le C|z|$ for all $0 < |z| \le 1$; letting $z \to 0$ gives $e_L = 0$. This completes the induction and proves (U).

Step 6: conclusion. Apply (U) with $p_k = a_k/k!$ and $q_k = r^k\,\mathrm{He}_k(\sqrt{2}\,t)/k!$. Its summability hypotheses are exactly (10) and (11), and its agreement hypothesis is (9). Therefore $a_k/k! = r^k\,\mathrm{He}_k(\sqrt{2}\,t)/k!$ for every k, and multiplying by $k! \ne 0$ gives $(*)$. By (1) this is the assertion of the lemma. □

2.26. `HermiteFunReproduction`.

Lemma 2.26 (n-dimensional Hermite reproduction identity)**.** *Fix $n \in \mathbb{N}$ and let $a \in \mathbb{R}$ satisfy $0 \le a \le 1$. Write $\tau = \sqrt{1-a^2}$. Then for every multi-index $\alpha \in \mathbb{N}_0^n$ and every $x \in \mathbb{R}^n$,*

$$\int_{\mathbb{R}^n} h_\alpha\big(a\,x + \tau\,y\big)\,d\gamma_n(y) = a^{|\alpha|}\,h_\alpha(x),$$

where h_α is the n-dimensional Hermite function of `HermiteFun`, $|\alpha| = \alpha_1 + \cdots + \alpha_n$ is the order of `MultiIndexSum`, and γ_n is the Gaussian measure of `GaussMeasure`.

The integral on the left is taken against the Gaussian *measure γ_n, which already carries the normalising factor $\pi^{-n/2}$ and the weight $e^{-|y|^2}$ in its density; no further weight appears in the integrand. This is the direct multi-index companion of `Hermite1Reproduction`, whose one-dimensional statement is instead written as a Lebesgue integral $d\sigma$ with the weight $\pi^{-1/2}e^{-\sigma^2}$ displayed in the integrand. The two conventions describe the same object in dimension one, and the proof below passes between them explicitly.*

Proof. Step 1: unfold γ_n as a weighted Lebesgue integral. By `GaussMeasure`, γ_n is Lebesgue measure on $\mathbb{R}^n$ with density $\rho(y) = \pi^{-n/2}e^{-|y|^2}$. That density is measurable and finite at every point, so integration against γ_n is integration of the density-weighted integrand against Lebesgue measure:

$$\int_{\mathbb{R}^n} h_\alpha(ax+\tau y)\,d\gamma_n(y) = \int_{\mathbb{R}^n} \pi^{-n/2}e^{-|y|^2}\,h_\alpha(ax+\tau y)\,dy.$$

Step 2: the integrand factorizes over the coordinates. Fix $y \in \mathbb{R}^n$. Three factorizations combine. First, $|y|^2 = \sum_{j=1}^n y_j^2$, so $e^{-|y|^2} = \prod_{j=1}^n e^{-y_j^2}$. Second, the constant splits as $\pi^{-n/2} = \prod_{j=1}^n \pi^{-1/2}$, since the product of n equal factors $\pi^{-1/2}$ is $\pi^{-n/2}$. Third, by `HermiteFun` the n-dimensional Hermite function is the product of one-dimensional ones over the coordinates, and the jth coordinate of $ax + \tau y$ is $ax_j + \tau y_j$, so

$$h_\alpha(ax+\tau y) = \prod_{j=1}^n h_{\alpha_j}\big(ax_j + \tau y_j\big)$$

with h_k the one-dimensional Hermite function of `Hermite1`. Multiplying the three and regrouping the factors coordinatewise,

$$\pi^{-n/2}e^{-|y|^2}\,h_\alpha(ax+\tau y) = \prod_{j=1}^n \Big(\pi^{-1/2}e^{-y_j^2}\,h_{\alpha_j}\big(ax_j+\tau y_j\big)\Big). \tag{1}$$

Step 3: the integral of a product splits. Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of Lebesgue measure on $\mathbb{R}$, the identification of $\mathbb{R}^n$ with the product being measure preserving. Each factor in (1) depends on one coordinate only, so Fubini's theorem for a finite product of measures applies and gives

$$\int_{\mathbb{R}^n} \prod_{j=1}^{n} \Big(\pi^{-1/2} e^{-y_j^2}\, h_{\alpha_j}(a x_j + \tau y_j) \Big)\, dy = \prod_{j=1}^{n} \int_{\mathbb{R}} \pi^{-1/2} e^{-\sigma^2}\, h_{\alpha_j}(a x_j + \tau \sigma)\, d\sigma.$$

Step 4: evaluate each one-dimensional factor. The hypotheses $0 \le a \le 1$ are exactly what `Hermite1Reproduction` requires of its contraction parameter, and $\tau = \sqrt{1-a^2}$ is the spreading factor that lemma pairs with it. Applying it with $r = a$, $k = \alpha_j$ and $t = x_j$ evaluates the jth factor as

$$\int_{\mathbb{R}} \pi^{-1/2} e^{-\sigma^2}\, h_{\alpha_j}(a x_j + \tau \sigma)\, d\sigma = a^{\alpha_j}\, h_{\alpha_j}(x_j).$$

Step 5: multiply. Taking the product over j and separating the two kinds of factor,

$$\prod_{j=1}^{n} a^{\alpha_j}\, h_{\alpha_j}(x_j) = \Big(\prod_{j=1}^{n} a^{\alpha_j} \Big) \prod_{j=1}^{n} h_{\alpha_j}(x_j) = a^{\sum_j \alpha_j}\, h_\alpha(x) = a^{|\alpha|}\, h_\alpha(x),$$

where the exponents were collected using $\prod_j a^{\alpha_j} = a^{\sum_j \alpha_j}$, the remaining product was recognised as $h_\alpha(x)$ by `HermiteFun`, and $\sum_j \alpha_j = |\alpha|$ by `MultiIndexSum`. Chaining Steps 1 through 5 gives the assertion. □

2.27. `MehlerFormula`.

Lemma 2.27 (Mehler averaging formula and positivity preservation)**.** *Let $s > 0$ and $v \in L^2(\gamma_n)$, and write $r = e^{-s} \in (0,1)$ and $\tau = \sqrt{1-e^{-2s}} = \sqrt{1-r^2} \in (0,1]$. Then the spectrally defined semigroup P_s of `OUSemigroup` is given by the probabilistic averaging formula*

$$P_s v(x) = \int_{\mathbb{R}^n} v\big(r\,x + \tau\,y\big)\, d\gamma_n(y) \qquad \text{for } \gamma_n\text{-a.e. } x \in \mathbb{R}^n,$$

the inner integral being absolutely convergent for γ_n-a.e. x. Consequently P_s is positivity preserving: if $v \ge 0$ γ_n-a.e. then $P_s v \ge 0$ γ_n-a.e.

Proof. Define, for $v \in L^2(\gamma_n)$,

$$(Q_s v)(x) = \int_{\mathbb{R}^n} v(rx + \tau y)\, d\gamma_n(y).$$

Step 1: the affine map pushes $\gamma_n \otimes \gamma_n$ to a measure with second marginal γ_n. By `GaussMeasure` the density of γ_n factorizes over the coordinates as $\prod_j \pi^{-1/2} e^{-x_j^2}$, and γ_n is a probability measure by `GaussIsProb`; thus γ_n is the n-fold product of the one-dimensional measure $d\gamma_1(t) = \pi^{-1/2} e^{-t^2} dt$, which is the centred Gaussian law of variance $1/2$ on $\mathbb{R}$. Consequently, if X and Y are independent with law γ_n, then each coordinate of $rX + \tau Y$ is a sum of two independent centred Gaussians of variances $r^2/2$ and $\tau^2/2$, hence centred Gaussian of variance $(r^2+\tau^2)/2 = 1/2$, and distinct coordinates remain independent. Therefore

$$rX + \tau Y \ \text{ has law } \ \gamma_n. \tag{1}$$

Step 2: Q_s is a well-defined contraction on $L^2(\gamma_n)$. By Tonelli and (1),

$$\iint |v(rx+\tau y)|^2\, d\gamma_n(y)\, d\gamma_n(x) = \int |v|^2\, d\gamma_n = \|v\|_2^2 < \infty.$$

Hence for γ_n-a.e. x the function $y \mapsto v(rx+\tau y)$ is in $L^2(\gamma_n)$, so in $L^1(\gamma_n)$ by `L2IntoL1`; thus $Q_s v(x)$ is defined by an absolutely convergent integral for a.e. x. Jensen's inequality for the probability measure γ_n gives $|Q_s v(x)|^2 \le \int |v(rx+\tau y)|^2 d\gamma_n(y)$, and integrating in x and using the display above,

$$\|Q_s v\|_{L^2(\gamma_n)} \le \|v\|_{L^2(\gamma_n)}. \tag{2}$$

In particular Q_s is a bounded linear operator on $L^2(\gamma_n)$. The argument of this step used nothing about v beyond $v \in L^2(\gamma_n)$, so its conclusions hold with any element of $L^2(\gamma_n)$ in place of v.

Step 3: Q_s and P_s agree on each Hermite function. Fix $\alpha \in \mathbb{N}_0^n$. By `HermiteFun`, $h_\alpha(z) = \prod_j h_{\alpha_j}(z_j)$, and the jth coordinate of $rx + \tau y$ is $rx_j + \tau y_j$. Hence

$$h_\alpha(rx + \tau y) = \prod_{j=1}^{n} h_{\alpha_j}(rx_j + \tau y_j),$$

a product of functions of the separate coordinates y_j.

We justify the factorization of the integral. By `HermiteFunMemLp`, $h_\alpha \in L^2(\gamma_n)$, so Step 2 applied with $v = h_\alpha$ shows that for γ_n-a.e. x the function $y \mapsto h_\alpha(rx + \tau y)$ is γ_n-integrable; fix such an x. Tonelli's theorem for the product measure $\gamma_n = \gamma_1^{\otimes n}$ of Step 1, applied to the nonnegative function $y \mapsto \prod_j |h_{\alpha_j}(rx_j + \tau y_j)|$, gives

$$\int \prod_{j=1}^{n} \bigl|h_{\alpha_j}(rx_j + \tau y_j)\bigr| \, d\gamma_n(y) = \prod_{j=1}^{n} \int_{\mathbb{R}} \bigl|h_{\alpha_j}(rx_j + \tau\sigma)\bigr| \, d\gamma_1(\sigma)$$

as an identity in $[0, +\infty]$, and the left-hand side is finite by the previous sentence. No factor on the right vanishes: by `Hermite1` and the recurrence recorded there, He_{α_j} is monic of degree α_j, so h_{α_j} is a nonzero polynomial; since $\tau > 0$, the map $\sigma \mapsto h_{\alpha_j}(rx_j + \tau\sigma)$ is again a nonzero polynomial, hence nonzero off a finite set, and γ_1 has a strictly positive density by `GaussMeasure`, so its integral against γ_1 is strictly positive. A product of elements of $[0, +\infty]$ with no vanishing factor is finite only if every factor is finite, so each one-dimensional integral above is finite. Fubini's theorem for the product measure therefore gives

$$(Q_s h_\alpha)(x) = \prod_{j=1}^{n} \int_{\mathbb{R}} \pi^{-1/2} e^{-\sigma^2} \, h_{\alpha_j}(rx_j + \tau\sigma) \, d\sigma.$$

Since $0 < r < 1$, `Hermite1Reproduction` applies with this r (note $\tau = \sqrt{1 - r^2}$ as required there) and evaluates the jth factor as $r^{\alpha_j} h_{\alpha_j}(x_j)$. Multiplying,

$$(Q_s h_\alpha)(x) = r^{\sum_j \alpha_j} \prod_j h_{\alpha_j}(x_j) = r^{|\alpha|} \, h_\alpha(x) = e^{-s|\alpha|} \, h_\alpha(x),$$

using `MultiIndexSum` and $r = e^{-s}$. On the other hand `OUSemigroupSpectral` (applicable since $s > 0 \geq 0$) gives $P_s h_\alpha = e^{-s|\alpha|} h_\alpha$, because the coefficients of h_β are $\delta_{\alpha\beta}$ by `HermiteL2Orthonormal`. Hence $Q_s h_\alpha = P_s h_\alpha$ for every α.

Step 4: $Q_s = P_s$ on all of $L^2(\gamma_n)$. Both operators are linear and bounded on $L^2(\gamma_n)$: Q_s by (2), and P_s by `OUSelfAdjointContraction`, which gives $\|P_s v\|_2 \leq \|v\|_2$. They agree on every h_α by Step 3, hence on the linear span of $\{h_\alpha\}$ by linearity, hence on the closure of that span by continuity. By `HermiteL2Complete` that closure is all of $L^2(\gamma_n)$. Therefore $Q_s v = P_s v$ in $L^2(\gamma_n)$, i.e. $P_s v(x) = \int v(rx + \tau y) \, d\gamma_n(y)$ for γ_n-a.e. x, which is the asserted formula.

Step 5: positivity preservation. Suppose $v \geq 0$ γ_n-a.e., and let N be the γ_n-null set where this fails. By (1) the map $(x, y) \mapsto rx + \tau y$ pushes $\gamma_n \otimes \gamma_n$ forward to γ_n, so the set of pairs (x, y) with $rx + \tau y \in N$ has $(\gamma_n \otimes \gamma_n)$-measure $\gamma_n(N) = 0$. By Fubini for this null set, for γ_n-a.e. x we have $rx + \tau y \notin N$ for γ_n-a.e. y, and then $v(rx + \tau y) \geq 0$ for γ_n-a.e. y. For such x the integral defining $Q_s v(x)$ is an integral of an a.e. nonnegative integrable function, hence $Q_s v(x) \geq 0$. Combining with Step 4, $P_s v \geq 0$ γ_n-a.e. □

2.28. `SemigroupEnergy`.

Lemma 2.28 (The energy as a semigroup integral)**.** *For every $v \in L^2(\gamma_n)$,*

$$\mathcal{E}(v) = \frac{1}{2\sqrt{\pi}} \int_0^\infty \bigl\langle (I - P_s)v, \, v \bigr\rangle \, \frac{ds}{s^{3/2}},$$

an identity in $[0, +\infty]$: the integrand is nonnegative, and the right-hand side is $+\infty$ exactly when v lies outside the form domain $\mathcal{V}$, in which case $\mathcal{E}(v) = +\infty$ by the convention of `Energy`. Here P_s is the semigroup of `OUSemigroup`.

Proof. Step 1 (spectral expansion of the integrand). Fix $s > 0$. By `OUSemigroupSpectral` the Hermite coefficients of $P_s v$ are $e^{-s|\alpha|} v_\alpha$, so by Parseval (`HermiteBasis`)

$$\big\langle (I - P_s)v, v \big\rangle = \sum_{\alpha} \big(1 - e^{-s|\alpha|}\big) |v_\alpha|^2 = \sum_{\alpha \neq 0} \big(1 - e^{-s|\alpha|}\big) |v_\alpha|^2,$$

the index $\alpha = 0$ contributing 0 because $|0| = 0$ by `MultiIndexSum` and hence $1 - e^0 = 0$. Every summand is nonnegative, since $0 \le 1 - e^{-s|\alpha|} \le 1$, and the sum lies between 0 and $\|v\|_2^2$.

Step 2 (Tonelli). For each fixed α the map $s \mapsto (1 - e^{-s|\alpha|}) s^{-3/2} |v_\alpha|^2$ is continuous and nonnegative on $(0, \infty)$, hence measurable. Tonelli's theorem, applied to counting measure on $\mathbb{N}_0^n \setminus \{0\}$ and the measure $s^{-3/2}\, ds$ on $(0, \infty)$, allows exchanging the sum and the integral without any finiteness assumption, both sides being values in $[0, +\infty]$:

$$\frac{1}{2\sqrt{\pi}} \int_0^\infty \langle (I - P_s)v, v \rangle \, \frac{ds}{s^{3/2}} = \sum_{\alpha \neq 0} |v_\alpha|^2 \cdot \frac{1}{2\sqrt{\pi}} \int_0^\infty \big(1 - e^{-s|\alpha|}\big) \frac{ds}{s^{3/2}}.$$

Step 3 (evaluating each factor). For $\alpha \neq 0$ we have $|\alpha| \ge 1 > 0$ by `MultiIndexSum`, so `ScalarFractional` applies with $\lambda = |\alpha| \ge 0$ and gives

$$\frac{1}{2\sqrt{\pi}} \int_0^\infty \big(1 - e^{-s|\alpha|}\big) \frac{ds}{s^{3/2}} = |\alpha|^{1/2}.$$

Substituting into Step 2,

$$\frac{1}{2\sqrt{\pi}} \int_0^\infty \langle (I - P_s)v, v \rangle \frac{ds}{s^{3/2}} = \sum_{\alpha \neq 0} |\alpha|^{1/2} |v_\alpha|^2,$$

and by `Energy` the right-hand side is exactly $\mathcal{E}(v)$, as an identity in $[0, +\infty]$. In particular the integral is finite precisely when $\mathcal{E}(v) < \infty$, i.e. when $v \in \mathcal{V}$. □

2.29. `OUCouplingKernel`.

Lemma 2.29 (ν_s is a symmetric coupling of γ_n with itself representing P_s)**.** *Let $s > 0$ and let ν_s be the measure of `OUCoupling`. Then*

(1) *ν_s is a probability measure on $\mathbb{R}^n \times \mathbb{R}^n$;*
(2) *ν_s is symmetric, i.e. invariant under the coordinate swap $(x, y) \mapsto (y, x)$;*
(3) *both marginals of ν_s equal γ_n; and*
(4) *for all $v, w \in L^2(\gamma_n)$,*

$$\langle P_s v, w \rangle_{L^2(\gamma_n)} = \iint v(y)\, w(x)\, d\nu_s(x, y),$$

the integral being absolutely convergent, with P_s as in `OUSemigroup`.

Proof. Write $r = e^{-s} \in (0, 1)$ and $\tau = \sqrt{1 - r^2} \in (0, 1]$, and let $T(x, y) = (x,\, rx + \tau y)$, so that $\nu_s = T_*(\gamma_n \otimes \gamma_n)$ by `OUCoupling`. By `GaussMeasure` and `GaussIsProb`, γ_n is the n-fold product of the centred Gaussian law of variance $1/2$ on $\mathbb{R}$; in particular γ_n is a centred Gaussian measure on $\mathbb{R}^n$ with covariance $\frac{1}{2} I$.

(1). γ_n is a probability measure by `GaussIsProb`, hence so is the product $\gamma_n \otimes \gamma_n$ on $\mathbb{R}^n \times \mathbb{R}^n$. The map T is continuous, hence Borel measurable, and the push-forward of a probability measure under a measurable map is a probability measure. So ν_s is a probability measure.

(3). The first marginal of ν_s is the push-forward of $\gamma_n \otimes \gamma_n$ under $(x, y) \mapsto x$, which is γ_n by definition of the product measure. The second marginal is the law of $rX + \tau Y$ for independent X, Y of law γ_n, which is γ_n: coordinatewise, $rX_j + \tau Y_j$ is a sum of independent centred Gaussians of variances $r^2/2$ and $\tau^2/2$, hence centred Gaussian of variance $(r^2 + \tau^2)/2 = 1/2$, and distinct coordinates remain independent.

(2). ν_s is the push-forward of the centred Gaussian measure $\gamma_n \otimes \gamma_n$ on $\mathbb{R}^{2n}$ under the linear map T, hence is itself a centred Gaussian measure on $\mathbb{R}^{2n}$, and is therefore determined by its covariance. With (ξ, η) denoting the two blocks of coordinates, that covariance has blocks

$$\mathrm{Cov}(X, X) = \tfrac{1}{2} I, \qquad \mathrm{Cov}(rX + \tau Y,\, rX + \tau Y) = \tfrac{1}{2} I, \qquad \mathrm{Cov}(X,\, rX + \tau Y) = r \cdot \tfrac{1}{2} I,$$

the last because X and Y are independent and centred, so the Y-term contributes 0. Thus the covariance matrix is $\frac{1}{2}\begin{pmatrix} I & rI \\ rI & I \end{pmatrix}$, which is invariant under interchanging the two blocks. Two centred Gaussian measures on $\mathbb{R}^{2n}$ with the same covariance coincide; applying this to ν_s and its push-forward under the swap (whose covariance is the block-swapped matrix, i.e. the same matrix) gives the asserted symmetry.

(4). Let $v, w \in L^2(\gamma_n)$. By `MehlerFormula`, for γ_n-a.e. x,

$$P_s v(x) = \int_{\mathbb{R}^n} v(rx + \tau y)\, d\gamma_n(y).$$

First, absolute convergence of the double integral. By Tonelli and part (3) (second marginal), and then Cauchy–Schwarz on $L^2(\gamma_n)$,

$$\iint |v(rx+\tau y)\, w(x)|\, d\gamma_n(y) d\gamma_n(x) \le \Big(\iint |v(rx+\tau y)|^2\Big)^{1/2} \Big(\iint |w(x)|^2\Big)^{1/2} = \|v\|_2\, \|w\|_2 < \infty,$$

where $\iint |v(rx+\tau y)|^2 d\gamma_n(y) d\gamma_n(x) = \int |v|^2 d\gamma_n$ because $(x,y) \mapsto rx + \tau y$ pushes $\gamma_n \otimes \gamma_n$ to γ_n (the computation in (3)), and $\iint |w(x)|^2 d\gamma_n(y) d\gamma_n(x) = \|w\|_2^2$ since γ_n is a probability measure.

Fubini's theorem therefore applies and gives

$$\langle P_s v, w\rangle = \int w(x) \Big(\int v(rx + \tau y)\, d\gamma_n(y) \Big) d\gamma_n(x) = \iint v(rx + \tau y)\, w(x)\, d(\gamma_n \otimes \gamma_n)(x, y).$$

By the definition of the push-forward and of T, for any $(\gamma_n \otimes \gamma_n)$-integrable Ψ one has $\int \Psi \circ T\, d(\gamma_n \otimes \gamma_n) = \int \Psi\, d\nu_s$. Apply this with $\Psi(x', y') = v(y')\, w(x')$: then $\Psi(T(x,y)) = v(rx + \tau y)\, w(x)$, so the last double integral equals $\iint v(y') w(x')\, d\nu_s(x', y')$, as asserted. □

2.30. `DifferenceEnergy`.

Lemma 2.30 (The energy as a difference integral against the coupling)**.** *Let $v \in L^2(\gamma_n)$, let $\mathcal{E}$ be the energy of `Energy` and let ν_s be the coupling of `OUCoupling`. Then:*

(1) *for every $s > 0$ the function $(x, y) \mapsto |v(x) - v(y)|^2$ is ν_s-integrable;*

(2) *for every $s > 0$,*

$$\iint |v(x) - v(y)|^2\, d\nu_s(x, y) \;\le\; 4\, \|v\|_2^2;$$

(3) *and*

$$\mathcal{E}(v) = \frac{1}{4\sqrt{\pi}} \int_0^\infty \Big(\iint |v(x) - v(y)|^2\, d\nu_s(x, y) \Big) \frac{ds}{s^{3/2}},$$

an identity in $[0, +\infty]$.

In particular the inner double integral in (3) is a finite real number for every $s > 0$, so the outer integral is an integral of a genuine nonnegative function of s.

Proof. Steps (1) and (2): integrability and the bound. Fix $s > 0$. By `OUCouplingKernel`, ν_s is a probability measure whose two marginals both equal γ_n. Applying the marginal property to the nonnegative measurable function $x \mapsto v(x)^2$ in each variable separately,

$$\iint v(x)^2\, d\nu_s(x, y) = \int v^2\, d\gamma_n = \|v\|_2^2, \qquad \iint v(y)^2\, d\nu_s(x, y) = \int v^2\, d\gamma_n = \|v\|_2^2,$$

both finite since $v \in L^2(\gamma_n)$. Expanding $|v(x) - v(y)|^2 = v(x)^2 + v(y)^2 - 2v(x)v(y)$ and using $2|v(x)v(y)| \le v(x)^2 + v(y)^2$ pointwise, we get the pointwise bound

$$\big|\, |v(x) - v(y)|^2 \,\big| = |v(x) - v(y)|^2 \le 2\big(v(x)^2 + v(y)^2\big),$$

whose ν_s-integral is at most $2(\|v\|_2^2 + \|v\|_2^2) = 4\|v\|_2^2 < \infty$. A measurable function dominated in absolute value by a ν_s-integrable function is ν_s-integrable, which proves (1); and monotonicity of the integral then gives

$$\iint |v(x) - v(y)|^2\, d\nu_s(x, y) \le 4\, \|v\|_2^2,$$

which is (2). In particular the mixed term $\iint v(x)v(y)\, d\nu_s$ is absolutely integrable as well.

Step (3): the identity. By part (4) of `OUCouplingKernel` applied with $w = v$,

$$\iint v(x)\, v(y)\, d\nu_s(x,y) = \langle P_s v, v\rangle.$$

Expanding $|v(x)-v(y)|^2 = v(x)^2 + v(y)^2 - 2v(x)v(y)$ and using the three displayed identities,

$$\tfrac{1}{2}\iint |v(x)-v(y)|^2\, d\nu_s(x,y) = \tfrac{1}{2}\big(2\|v\|_2^2 - 2\langle P_s v, v\rangle\big) = \big\langle (I-P_s)v,\, v\big\rangle.$$

Substituting this into the identity of `SemigroupEnergy`, i.e. replacing $\langle (I-P_s)v, v\rangle$ by $\frac{1}{2}\iint |v(x)-v(y)|^2 d\nu_s$ and absorbing the factor $\frac{1}{2}$ into the prefactor $\frac{1}{2\sqrt{\pi}}$, gives

$$\mathcal{E}(v) = \frac{1}{4\sqrt{\pi}} \int_0^\infty \Big(\iint |v(x)-v(y)|^2 d\nu_s(x,y)\Big) \frac{ds}{s^{3/2}},$$

as an identity in $[0,+\infty]$ since all integrands are nonnegative. □

2.31. `LatticeContraction`.

Lemma 2.31 (Lattice property: 1-Lipschitz maps do not increase the energy)**.** *Let $\Phi : \mathbb{R} \to \mathbb{R}$ be 1-Lipschitz with $\Phi(0) = 0$, let $v \in \mathcal{V}$, i.e. $\mathcal{E}(v) < \infty$ in the sense of `Energy`, and let $w \in L^2(\gamma_n)$ satisfy $w = \Phi \circ v$ γ_n-almost everywhere. Then*

$$\mathcal{E}(w) < \infty \qquad \text{and} \qquad \mathcal{E}(w) \le \mathcal{E}(v).$$

In particular, taking $\Phi(t) = t^+ = \max\{t, 0\}$, $\Phi(t) = t^- = \max\{-t, 0\}$ and $\Phi(t) = |t|$, all of which are 1-Lipschitz and vanish at 0, the functions v^+, v^- and $|v|$ belong to $\mathcal{V}$ and have energy at most $\mathcal{E}(v)$.

Proof. First, w is indeed in $L^2(\gamma_n)$ by hypothesis; note that this is consistent, since $|\Phi(t)| = |\Phi(t) - \Phi(0)| \le |t|$ by the Lipschitz bound and $\Phi(0) = 0$, so $|w| \le |v|$ a.e.

Fix $s > 0$. Since $w = \Phi \circ v$ γ_n-a.e. and both marginals of the coupling ν_s of `OUCoupling` equal γ_n by `OUCouplingKernel`, the set where $w \neq \Phi \circ v$ is ν_s-negligible in each variable separately, hence $w(x) = \Phi(v(x))$ and $w(y) = \Phi(v(y))$ for ν_s-almost every (x, y). The Lipschitz bound then gives

$$|w(x) - w(y)|^2 = |\Phi(v(x)) - \Phi(v(y))|^2 \le |v(x) - v(y)|^2 \qquad \nu_s\text{-a.e.}$$

Integrating this pointwise inequality against the nonnegative measure ν_s,

$$\iint |w(x) - w(y)|^2\, d\nu_s(x,y) \le \iint |v(x) - v(y)|^2\, d\nu_s(x,y) \qquad \text{for every } s > 0.$$

Both sides are finite for each s, being at most $4\|w\|_2^2$ and $4\|v\|_2^2$ respectively, as recorded in `DifferenceEnergy`.

Now integrate in s against the nonnegative measure $s^{-3/2}ds$ on $(0,\infty)$, which preserves the inequality, and multiply by the positive constant $(4\sqrt{\pi})^{-1}$. By `DifferenceEnergy`, applied once to w and once to v, the two resulting quantities are $\mathcal{E}(w)$ and $\mathcal{E}(v)$, as values in $[0,+\infty]$. Hence $\mathcal{E}(w) \le \mathcal{E}(v)$. Since $\mathcal{E}(v) < \infty$ by hypothesis, also $\mathcal{E}(w) < \infty$, i.e. $w \in \mathcal{V}$.

For the three special cases, $t \mapsto t^+$, $t \mapsto t^-$ and $t \mapsto |t|$ are each 1-Lipschitz (they are compositions and maxima of 1-Lipschitz maps, and $|a^+ - b^+| \le |a-b|$, $|a^- - b^-| \le |a-b|$, $||a| - |b|| \le |a-b|$) and each sends 0 to 0. □

2.32. `L2InnerGaussIntegral`.

Lemma 2.32 (The $L^2(\gamma_n)$ inner product is the tested Gaussian integral)**.** *Fix $n \in \mathbb{N}_0$ and let γ_n be the Gaussian measure of `GaussMeasure` on $\mathbb{R}^n$. Let $x, y \in L^2(\gamma_n)$ be two elements of the real Hilbert space of (equivalence classes modulo γ_n-null sets of) square-γ_n-integrable real-valued functions on $\mathbb{R}^n$, and write again $x, y : \mathbb{R}^n \to \mathbb{R}$ for the representative functions they determine. Then the product xy is γ_n-integrable and the inner product of x and y is the Gaussian integral of their pointwise product:*

$$\langle x, y\rangle = \int_{\mathbb{R}^n} x(z)\, y(z)\, d\gamma_n(z).$$

Proof. Both sides are unchanged if x or y is replaced by another representative of the same class: the inner product is a function of the classes alone, and two representatives of x agree γ_n-almost everywhere, so the two integrands agree γ_n-almost everywhere and the integrals coincide. The asserted identity is therefore well posed, and it suffices to prove it for the chosen representatives.

The space $L^2(\gamma_n)$ of real-valued functions is the Lebesgue space of order 2 over the measure γ_n, whose Hilbert-space structure is the standard one: by construction of the inner product on a Lebesgue space of order 2 with values in a real inner-product space, which is shared background,

$$\langle x, y\rangle = \int_{\mathbb{R}^n} \langle x(z), y(z)\rangle_{\mathbb{R}}\, d\gamma_n(z),$$

the integral existing because the pointwise inner product is dominated in absolute value by $\frac{1}{2}\big(x(z)^2 + y(z)^2\big)$, which is γ_n-integrable as $x, y \in L^2(\gamma_n)$. (This domination also gives the asserted integrability of xy; alternatively it is the integral form of Cauchy–Schwarz, which is shared background.)

It remains to identify the integrand pointwise. The scalar field here is $\mathbb{R}$, regarded as a one-dimensional real inner-product space over itself, whose inner product is $\langle a, b\rangle_{\mathbb{R}} = \bar{a}\, b = a\, b$, complex conjugation being the identity on $\mathbb{R}$. Hence for every $z \in \mathbb{R}^n$,

$$\langle x(z), y(z)\rangle_{\mathbb{R}} = x(z)\, y(z),$$

using in addition the commutativity of multiplication in $\mathbb{R}$ to present the product in the stated order $x(z)\, y(z)$ rather than $y(z)\, x(z)$. Substituting this pointwise identity — valid at every z, hence in particular γ_n-almost everywhere — into the displayed formula for $\langle x, y\rangle$ gives

$$\langle x, y\rangle = \int_{\mathbb{R}^n} x(z)\, y(z)\, d\gamma_n(z),$$

which is the assertion. □

2.33. `DisjointCrossTermSign`.

Lemma 2.33 (Sign of the cross term for disjoint nonnegative functions)**.** *Let $\mathcal{E}(\cdot,\cdot)$ be the form of `EnergyBil` and $\mathcal{E}(\cdot)$ the energy of `Energy`.*

1. *If $v, w \in \mathcal{V}$ are nonnegative γ_n-a.e. and $vw = 0$ γ_n-a.e., then $\mathcal{E}(v, w) \le 0$.*
2. *Consequently, for every $z \in \mathcal{V}$ and every $z^- \in L^2(\gamma_n)$ with $z^- = \max\{-z, 0\}$ γ_n-a.e.,*

$$\mathcal{E}(z, z^-) \le -\mathcal{E}(z^-).$$

Proof. (1) Fix $s > 0$. Since $v \ge 0$ a.e., `MehlerFormula` gives $P_s v \ge 0$ γ_n-a.e. As also $w \ge 0$ a.e., the product $(P_s v)\, w$ is nonnegative a.e., and it is integrable because $P_s v, w \in L^2(\gamma_n)$. Hence

$$\langle P_s v, w\rangle = \int (P_s v)\, w\, d\gamma_n \ge 0.$$

On the other hand $\langle v, w\rangle = \int vw\, d\gamma_n = 0$, because $vw = 0$ a.e. by hypothesis. Therefore

$$\big\langle (I - P_s)v,\, w\big\rangle = \langle v, w\rangle - \langle P_s v, w\rangle = -\langle P_s v, w\rangle \le 0 \qquad \text{for every } s > 0.$$

Both v and w lie in $\mathcal{V}$, so `BilinearEnergyFormula` applies and expresses $\mathcal{E}(v, w)$ as the positive constant $(2\sqrt{\pi})^{-1}$ times the absolutely convergent integral of the nonpositive function $s \mapsto \langle (I - P_s)v, w\rangle\, s^{-3/2}$ over $(0, \infty)$. An absolutely convergent integral of a nonpositive function is nonpositive, so $\mathcal{E}(v, w) \le 0$.

(2) By hypothesis $z \in \mathcal{V}$, i.e. $\mathcal{E}(z) < \infty$, which is the finite-energy hypothesis `LatticeContraction` requires of its input. Let $z^+ \in L^2(\gamma_n)$ be (a representative of) $\max\{z, 0\}$. The maps $t \mapsto t^+$ and $t \mapsto t^-$ are 1-Lipschitz and vanish at 0, so `LatticeContraction`, applied to this z with each of them in turn, gives $z^+, z^- \in \mathcal{V}$. Pointwise a.e. we have $z = z^+ - z^-$, $z^+ \ge 0$, $z^- \ge 0$ and $z^+ z^- = 0$, so by bilinearity of $\mathcal{E}(\cdot,\cdot)$ on $\mathcal{V}$ (`EnergyBil`)

$$\mathcal{E}(z, z^-) = \mathcal{E}(z^+, z^-) - \mathcal{E}(z^-, z^-) = \mathcal{E}(z^+, z^-) - \mathcal{E}(z^-).$$

Part (1), applied to the pair (z^+, z^-), gives $\mathcal{E}(z^+, z^-) \le 0$. Hence $\mathcal{E}(z, z^-) \le -\mathcal{E}(z^-)$, and all quantities are finite real numbers because $z, z^+, z^- \in \mathcal{V}$. □

2.34. `PosPartConstSubMemLp`.

Lemma 2.34 (The positive part of a constant minus an $L^2(\gamma_n)$ function is $L^2(\gamma_n)$)**.** *Fix $n \in \mathbb{N}_0$ and let γ_n be the Gaussian measure of `GaussMeasure`. Let $F \in L^2(\gamma_n)$ and let $a \in \mathbb{R}$ be any real number. Then the positive part of $a - F$,*

$$(a - F)^+(x) = \max\big(a - F(x),\, 0\big),$$

again lies in $L^2(\gamma_n)$.

All three memberships here are with respect to the Gaussian measure γ_n: F is square γ_n-integrable and so is the conclusion's function, the integrals being taken against γ_n and not against Lebesgue measure. Note also that no sign condition on a is assumed or needed — a ranges over all of $\mathbb{R}$, and the argument below uses nothing about whether a is positive.

Proof. Step 1 (the constant is square integrable). A constant function is square integrable for any finite measure, and γ_n is a probability measure by `GaussIsProb`, hence finite. So $x \mapsto a$ lies in $L^2(\gamma_n)$.

Step 2 ($a - F$ is square integrable). $L^2(\gamma_n)$ is a vector space, so the difference of the constant a of Step 1 and the given $F \in L^2(\gamma_n)$ lies in $L^2(\gamma_n)$:

$$x \longmapsto a - F(x) \quad \in L^2(\gamma_n). \tag{D}$$

Step 3 (pointwise domination of the positive part). For every $x \in \mathbb{R}^n$,

$$\big|\max(a - F(x), 0)\big| \le \big|a - F(x)\big|. \tag{P}$$

To see this, write $t = a - F(x)$ and note first that $\max(t, 0) \ge 0$, so the left-hand side is $\max(t, 0)$ itself. Now distinguish two cases, which together are exhaustive. If $t \le 0$ then $\max(t, 0) = 0$, and the right-hand side $|t|$ is nonnegative, so the inequality reads $0 \le |t|$ and holds. If $t \ge 0$ then $\max(t, 0) = t$, and $t \le |t|$ holds because a real number is at most its own absolute value. In both cases (P) follows. This is the only step that touches the shape of the positive part, and it is where the absence of any hypothesis on a shows: the comparison is between $\max(t, 0)$ and $|t|$ for an arbitrary real t, with no reference to how t arose.

Step 4 (measurability of the positive part). The function F is γ_n-almost-everywhere strongly measurable, being an element of $L^2(\gamma_n)$, hence so is $x \mapsto a - F(x)$ by Step 2. Taking the pointwise maximum with the constant 0 preserves almost-everywhere measurability, since the maximum of two almost-everywhere measurable real functions is almost-everywhere measurable, and for real-valued functions almost-everywhere measurability and almost-everywhere strong measurability agree. So $x \mapsto \max(a - F(x), 0)$ is γ_n-almost-everywhere strongly measurable.

Step 5 (conclusion). A function that is γ_n-almost-everywhere strongly measurable and dominated in absolute value, γ_n-almost everywhere, by a function of $L^2(\gamma_n)$ is itself in $L^2(\gamma_n)$. Step 4 supplies the measurability, (P) supplies the domination — indeed at every point, so in particular γ_n-almost everywhere — and (D) says the dominating function $x \mapsto a - F(x)$ is in $L^2(\gamma_n)$. Hence $x \mapsto \max(a - F(x), 0)$ lies in $L^2(\gamma_n)$, as asserted. □

2.35. `LewyStampacchia`.

Lemma 2.35 (Lewy–Stampacchia order estimate on the full form domain)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., let $a \in \mathbb{R}$, and let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`. Let η be the multiplier of `ObstacleMultiplier` associated with u, F, a, and put $k = (a - F)^+ = \max\{a - F, 0\}$. Then*

$$\langle \eta, \varphi \rangle \;\le\; \int k\, \varphi \, d\gamma_n \qquad \text{for every } \varphi \in \mathcal{K}.$$

Thus the multiplier is dominated, in the order of the dual of the form domain, by the functional induced by the bounded function $(a - F)^+$. The estimate holds for every nonnegative form-domain test function, not merely for smooth compactly supported ones.

Proof. Fix $\varphi \in \mathcal{K}$ and $\varepsilon > 0$. Put

$$z = u - \varepsilon\varphi, \qquad z^- = \max\{-z, 0\}, \qquad v = z^+ = \max\{z, 0\}.$$

Since $u, \varphi \in \mathcal{V}$ and $\mathcal{V}$ is a linear subspace of $L^2(\gamma_n)$ (its defining condition in `Energy` is the finiteness of a weighted ℓ^2 sum, and the coefficient map of `HermiteBasis` is linear, so finiteness is preserved by linear combinations via the triangle inequality in the weighted ℓ^2 space), we have $z \in \mathcal{V}$. The maps $t \mapsto t^+$ and $t \mapsto t^-$ are 1-Lipschitz and vanish at 0, so `LatticeContraction` applied to each (with $v = z$ and w the class of the composite) gives $z^+, z^- \in \mathcal{V}$. As $z^+ \geq 0$, we get $v = z^+ \in \mathcal{K}$ by `ObstacleCone`, so v is an admissible competitor.

Step 1: substituting the competitor into the variational inequality. Pointwise $z = z^+ - z^-$, and $u = z + \varepsilon\varphi$, so

$$v - u = z^+ - (z + \varepsilon\varphi) = z^- - \varepsilon\varphi.$$

`VariationalInequality`, applied to the minimizer u and the competitor v, states $\int (F-a)(v-u)\, d\gamma_n \leq \mathcal{E}(u, v-u)$. Substituting the display and using bilinearity of $\mathcal{E}(\cdot,\cdot)$ (`EnergyBil`) and linearity of the integral,

$$\int (F-a) z^-\, d\gamma_n - \varepsilon \int (F-a)\varphi\, d\gamma_n \;\leq\; \mathcal{E}(u, z^-) - \varepsilon\, \mathcal{E}(u, \varphi).$$

Rearranging so that the ε-terms stand alone on the left, and using the definition $\langle \eta, \varphi \rangle = \mathcal{E}(u,\varphi) - \int (F-a)\varphi\, d\gamma_n$ of `ObstacleMultiplier`,

$$\varepsilon\, \langle \eta, \varphi \rangle \;\leq\; \mathcal{E}(u, z^-) + \int (a - F)\, z^-\, d\gamma_n. \tag{1}$$

Step 2: the energy term. Since $u = z + \varepsilon\varphi$, bilinearity gives $\mathcal{E}(u, z^-) = \mathcal{E}(z, z^-) + \varepsilon\, \mathcal{E}(\varphi, z^-)$. Part (2) of `DisjointCrossTermSign`, applied to $z \in \mathcal{V}$ and its negative part z^-, bounds the first summand by $-\mathcal{E}(z^-)$; and the Cauchy–Schwarz bound recorded in `EnergyBil` bounds the second by $\varepsilon\, \mathcal{E}(\varphi)^{1/2} \mathcal{E}(z^-)^{1/2}$. Putting $p = \mathcal{E}(\varphi)^{1/2}$ and $q = \mathcal{E}(z^-)^{1/2}$, both finite nonnegative reals,

$$\mathcal{E}(u, z^-) \;\leq\; -q^2 + \varepsilon p q = -\Big(q - \frac{\varepsilon p}{2}\Big)^2 + \frac{\varepsilon^2 p^2}{4} \;\leq\; \frac{\varepsilon^2}{4}\, \mathcal{E}(\varphi). \tag{2}$$

Step 3: the load term. Because $u \geq 0$ and $\varphi \geq 0$ γ_n-a.e. (both lie in $\mathcal{K}$, see `ObstacleCone`), we have pointwise a.e.

$$z^- = \max\{\varepsilon\varphi - u,\, 0\} \leq \varepsilon\varphi, \qquad z^- \geq 0.$$

Now $k = (a - F)^+$ satisfies $a - F \leq k$ and $k \geq 0$ pointwise, so multiplying the first inequality by the nonnegative number z^- and then using $z^- \leq \varepsilon\varphi$ and $k \geq 0$,

$$(a - F)\, z^- \;\leq\; k\, z^- \;\leq\; \varepsilon\, k\, \varphi \qquad \gamma_n\text{-a.e.}$$

Integrating,

$$\int (a - F)\, z^-\, d\gamma_n \;\leq\; \varepsilon \int k\, \varphi\, d\gamma_n. \tag{3}$$

The right-hand integral is finite: since $F \geq 0$ a.e. we have $0 \leq k = (a-F)^+ \leq \max\{a, 0\}$, so k is bounded, and $\varphi \in L^2(\gamma_n) \subseteq L^1(\gamma_n)$ by `L2IntoL1`. The left-hand integral is finite for the same reason, z^- being in $L^2(\gamma_n)$.

Step 4: conclusion. Combining (1), (2) and (3),

$$\varepsilon\, \langle \eta, \varphi \rangle \;\leq\; \frac{\varepsilon^2}{4} \mathcal{E}(\varphi) + \varepsilon \int k\varphi\, d\gamma_n.$$

Dividing by $\varepsilon > 0$,

$$\langle \eta, \varphi \rangle \;\leq\; \frac{\varepsilon}{4} \mathcal{E}(\varphi) + \int k\varphi\, d\gamma_n.$$

The test function φ is fixed with $\mathcal{E}(\varphi) < \infty$, and $\varepsilon > 0$ was arbitrary, so letting $\varepsilon \downarrow 0$ removes the first term and gives $\langle \eta, \varphi \rangle \leq \int k\varphi\, d\gamma_n$. □

2.36. `HermiteL2ZeroAEOne`.

Lemma 2.36 (The zeroth Hermite element of $L^2(\gamma_n)$ is almost everywhere the constant one)**.** *Fix $n \in \mathbb{N}_0$ and let $\alpha = 0$ be the zero multi-index in $\mathbb{N}_0^n$. Let $h_0 \in L^2(\gamma_n)$ be the element of `HermiteL2` determined by the Hermite function h_0 of `HermiteFun`. Then the representative function of that L^2 class agrees γ_n-almost everywhere with the constant function 1:*

$$h_0 = 1 \qquad \gamma_n\text{-almost everywhere on } \mathbb{R}^n.$$

The two sides of this identity are of different kinds, and the statement is exactly that they agree off a γ_n-null set: on the left is the function $\mathbb{R}^n \to \mathbb{R}$ obtained by coercing the $L^2(\gamma_n)$ equivalence class h_0 to a representative, and on the right is the genuine function $x \mapsto 1$. The conclusion is an almost-everywhere identity rather than a pointwise one precisely because the left-hand side is only determined up to a γ_n-null set.

Proof. Step 1 (the zeroth Hermite function is identically one). By `HermiteFun`, the Hermite function attached to a multi-index α is the product of one-dimensional normalized Hermite functions over the coordinates, $h_\alpha(x) = \prod_{j=1}^n h_{\alpha_j}(x_j)$. For the zero multi-index every entry is $\alpha_j = 0$, so

$$h_0(x) = \prod_{j=1}^{n} h_0(x_j) \qquad (x \in \mathbb{R}^n),$$

where the factors on the right are the one-dimensional zeroth Hermite functions. By `Hermite1Zero` each of those factors equals 1 at every real argument, so the product is a product of n copies of 1, namely 1. Hence

$$h_0(x) = 1 \qquad \text{for every } x \in \mathbb{R}^n, \tag{P}$$

a genuinely pointwise identity, holding everywhere and not merely almost everywhere.

Step 2 (passing from the function to its L^2 class). By `HermiteFunMemLp` the function h_0 is square integrable for γ_n, so it determines an element of $L^2(\gamma_n)$, and this is the element named in `HermiteL2`. Coercing that class back to a function returns a representative, which by construction of the Lebesgue space agrees γ_n-almost everywhere with the function h_0 one started from. Writing $\widetilde{h_0}$ for the representative, this says

$$\widetilde{h_0} = h_0 \qquad \gamma_n\text{-almost everywhere.} \tag{R}$$

Step 3 (conclusion). An identity that holds at every point holds in particular γ_n-almost everywhere, so (P) gives $h_0 = 1$ γ_n-almost everywhere. Chaining this with (R) and using that γ_n-almost-everywhere equality is transitive — the union of two γ_n-null sets being γ_n-null — yields

$$\widetilde{h_0} = 1 \qquad \gamma_n\text{-almost everywhere,}$$

which is the assertion. □

2.37. `ObstacleMultiplierLinearOnFormDomain`.

Lemma 2.37 (The obstacle multiplier is an energy form minus a fixed inner product, hence linear on the form domain)**.** *Fix $n \in \mathbb{N}$, let $F \in L^2(\gamma_n)$ with γ_n the Gaussian measure of `GaussMeasure`, let $a \in \mathbb{R}$, and let $u \in L^2(\gamma_n)$ have finite energy, $\mathcal{E}(u) \neq +\infty$ in the sense of `Energy`. Write*

$$g = F - a\,h_0 \ \in L^2(\gamma_n),$$

where h_0 is the zeroth Hermite mode of `HermiteL2`; note that g is an element of $L^2(\gamma_n)$ and not merely a function. Write $\langle \eta, \cdot \rangle$ for the obstacle multiplier of `ObstacleMultiplier` at this F, a and u, and $\mathcal{E}(\cdot,\cdot)$ for the bilinear energy form of `EnergyBil`. Then all three of the following hold.

(1) *For* every $\varphi \in L^2(\gamma_n)$,

$$\langle \eta, \varphi \rangle = \mathcal{E}(u, \varphi) - \langle g, \varphi \rangle_{L^2(\gamma_n)}.$$

(2) *For all* $v, w \in L^2(\gamma_n)$ *with* $\mathcal{E}(v) \neq +\infty$ *and* $\mathcal{E}(w) \neq +\infty$,
$$\langle \eta, v + w \rangle = \langle \eta, v \rangle + \langle \eta, w \rangle.$$
(3) *For every* $r \in \mathbb{R}$ *and every* $v \in L^2(\gamma_n)$ *with* $\mathcal{E}(v) \neq +\infty$,
$$\langle \eta, r\, v \rangle = r\, \langle \eta, v \rangle.$$

Every measure here is Gaussian: the inner product in (1) is the one of $L^2(\gamma_n)$*, the load integral it replaces is a* γ_n*-integral, and the energy form is the Gaussian one of* `EnergyBil`*.*

Two asymmetries in the hypotheses are deliberate and worth stating, because a reader would otherwise expect them to match. Clause (1) carries no *finite-energy hypothesis on* φ*: both sides are defined for every element of* $L^2(\gamma_n)$*, so the identity holds on all of* $L^2(\gamma_n)$ *and not merely on the form domain. Clauses (2) and (3) do need finite energy on their arguments, and they are also the only clauses that use the finite-energy hypothesis on* u *– it enters through the symmetry of* $\mathcal{E}$*, which* `EnergyFormBilinear` *supplies only when all the arguments involved have finite energy.*

Proof. Step 1: g *agrees with* $F - a$ *almost everywhere.* By `HermiteL2ZeroAEOne` the zeroth Hermite mode satisfies $h_0 = 1$ γ_n-almost everywhere. Hence, for γ_n-almost every x,
$$g(x) = F(x) - a\, h_0(x) = F(x) - a,$$
the first equality because the coercion of a difference of L^2 elements is almost everywhere the difference of the coercions, and of a scalar multiple almost everywhere the scalar multiple.

Step 2: clause (1). Let $\varphi \in L^2(\gamma_n)$ be arbitrary. By `ObstacleMultiplier`,
$$\langle \eta, \varphi \rangle = \mathcal{E}(u, \varphi) - \int_{\mathbb{R}^n} (F - a)\, \varphi \, d\gamma_n.$$
By `L2InnerGaussIntegral` the inner product of two elements of $L^2(\gamma_n)$ is the γ_n-integral of their product, so $\langle g, \varphi \rangle = \int g\, \varphi \, d\gamma_n$. By Step 1 the two integrands $g\, \varphi$ and $(F - a)\, \varphi$ agree γ_n-almost everywhere, so their integrals agree, and subtracting from $\mathcal{E}(u, \varphi)$ gives clause (1). No hypothesis on φ was used.

Step 3: the inputs from `EnergyFormBilinear`*.* That lemma supplies two things used below. First, the form domain is closed under the vector space operations: if $\mathcal{E}(v_1), \mathcal{E}(v_2) \neq +\infty$ then $\mathcal{E}(v_1 + v_2) \neq +\infty$ and $\mathcal{E}(c\, v_1) \neq +\infty$ for every $c \in \mathbb{R}$. Second, on the form domain $\mathcal{E}$ is additive and homogeneous in its first argument and symmetric: $\mathcal{E}(v_1 + v_2, w) = \mathcal{E}(v_1, w) + \mathcal{E}(v_2, w)$, $\mathcal{E}(c\, v_1, w) = c\, \mathcal{E}(v_1, w)$ and $\mathcal{E}(v_1, w) = \mathcal{E}(w, v_1)$, whenever the arguments involved have finite energy.

Step 4: clause (2). Let $\mathcal{E}(v), \mathcal{E}(w) \neq +\infty$. Then $\mathcal{E}(v + w) \neq +\infty$ by closure, so clause (1) applies at v, at w and at $v + w$. Using symmetry to move u into the second slot, then additivity in the first, then symmetry back:
$$\mathcal{E}(u, v + w) = \mathcal{E}(v + w, u) = \mathcal{E}(v, u) + \mathcal{E}(w, u) = \mathcal{E}(u, v) + \mathcal{E}(u, w),$$
each application being legitimate because u, v, w and $v + w$ all have finite energy. The inner product is additive in its second argument, $\langle g, v + w \rangle = \langle g, v \rangle + \langle g, w \rangle$. Substituting both into clause (1) at $v + w$ and regrouping gives $\langle \eta, v + w \rangle = \langle \eta, v \rangle + \langle \eta, w \rangle$.

Step 5: clause (3). Let $r \in \mathbb{R}$ and $\mathcal{E}(v) \neq +\infty$. Then $\mathcal{E}(r\, v) \neq +\infty$ by closure, so clause (1) applies at v and at $r\, v$. As in Step 4,
$$\mathcal{E}(u, r\, v) = \mathcal{E}(r\, v, u) = r\, \mathcal{E}(v, u) = r\, \mathcal{E}(u, v),$$
and the inner product is homogeneous in its second argument over the reals, $\langle g, r\, v \rangle = r\, \langle g, v \rangle$. Substituting both into clause (1) at $r\, v$ and factoring out r gives $\langle \eta, r\, v \rangle = r\, \langle \eta, v \rangle$. □

2.38. `MultiplierAPrioriL2Bound`.

Lemma 2.38 (A-priori $L^2(\gamma_n)$ bound on the obstacle multiplier)**.** *Fix* $n \in \mathbb{N}$*. Let* $F \in L^2(\gamma_n)$ *with* $F \geq 0$ γ_n*-almost everywhere, let* $a \in \mathbb{R}$ *with* $a > 0$*, and let* $u \in L^2(\gamma_n)$ *satisfy*

- $u \in \mathcal{K}$*, the obstacle cone of* `ObstacleCone`*, and*
- $J(u) \leq J(w)$ *for every* $w \in \mathcal{K}$*, with* $J =$ *the functional of* `ObstacleFunctional` *for this* F *and* a*.*

Then for every $\psi \in L^2(\gamma_n)$ of finite energy, i.e. with $\mathcal{E}(\psi) \neq +\infty$ in the sense of ***Energy****,*

$$\big|T(\psi)\big| \le a\,\|\psi\|_{L^2(\gamma_n)}, \qquad T(\psi) = \text{the multiplier } \textit{ObstacleMultiplier} \text{ at } \psi.$$

The bound is uniform in ψ: the same constant serves every ψ in the form domain $\{\mathcal{E} \neq +\infty\}$, with no dependence on ψ beyond its $L^2(\gamma_n)$ norm. And the constant is a itself, not a multiple of it – there is no implicit C, and no factor depending on n, on F or on u. Every measure here is the Gaussian measure γ_n of ***GaussMeasure****: the norm is the $L^2(\gamma_n)$ norm, the almost-everywhere nonnegativity of F is γ_n-almost everywhere, and the integrals appearing in the proof are γ_n-integrals. All four hypotheses are used: $F \ge 0$ and $a > 0$ in Step 5, and the cone membership and minimality of u in Steps 3 and 4, through* ***MultiplierPositive*** *and* ***LewyStampacchia*** *respectively.*

This is the a priori estimate that lets the multiplier extend from the form domain to a bounded linear functional on $L^2(\gamma_n)$, which is what makes a representing function available at all.

Proof. Step 0: the multiplier as a difference of a form and an inner product. By `HermiteL2ZeroAEOne` the zeroth Hermite mode h_0 of `HermiteL2` equals the constant 1 γ_n-almost everywhere, so the element $g = F - a\,h_0$ of $L^2(\gamma_n)$ satisfies $g = F - a$ γ_n-almost everywhere. Hence, by `L2InnerGaussIntegral` rewriting the load integral as an inner product,

$$T(\varphi) = \mathcal{B}(u,\varphi) - \langle g, \varphi\rangle \qquad \text{for every } \varphi \in L^2(\gamma_n), \tag{0}$$

with $\mathcal{B}$ the bilinear form of `EnergyBil`.

Step 1: T is additive and homogeneous on the form domain. By `EnergyFormBilinear` the set $\{\mathcal{E} \neq +\infty\}$ is closed under sums and scalar multiples, and on it $\mathcal{B}$ is symmetric and linear in each argument. Combining this with (0) and the linearity of the inner product in its second argument: if $\mathcal{E}(v), \mathcal{E}(w) \neq +\infty$ and $r \in \mathbb{R}$ then

$$T(v+w) = T(v) + T(w), \qquad T(r\,v) = r\,T(v). \tag{1}$$

Step 2: the positive and negative parts of ψ lie in the cone. Let $\psi^+ = \max(\psi, 0)$ and $\psi^- = \max(-\psi, 0)$. Both are almost everywhere strongly measurable and satisfy $|\psi^{\pm}| \le |\psi|$ pointwise, so both lie in $L^2(\gamma_n)$ by domination. Both are nonnegative by construction. For the energy, the maps $t \mapsto \max(t,0)$ and $t \mapsto \max(-t,0)$ are 1-Lipschitz on $\mathbb{R}$ and send 0 to 0, so `LatticeContraction` applies and gives $\mathcal{E}(\psi^+) \neq +\infty$ and $\mathcal{E}(\psi^-) \neq +\infty$ from $\mathcal{E}(\psi) \neq +\infty$. By `ObstacleCone` – finite energy together with almost everywhere nonnegativity – we conclude $\psi^+, \psi^- \in \mathcal{K}$.

Step 3: $|T(\psi)| \le T(|\psi|)$. Pointwise $\psi = \psi^+ - \psi^-$, hence $\psi = \psi^+ + (-1)\psi^-$ in $L^2(\gamma_n)$, and (1) gives

$$T(\psi) = T(\psi^+) - T(\psi^-).$$

Since u lies in the cone and minimises J there, `MultiplierPositive` applies to each of $\psi^+, \psi^- \in \mathcal{K}$ and gives $T(\psi^+) \ge 0$ and $T(\psi^-) \ge 0$. A difference of two nonnegative reals has absolute value at most their sum, so

$$|T(\psi)| = |T(\psi^+) - T(\psi^-)| \le T(\psi^+) + T(\psi^-) = T(\psi^+ + \psi^-),$$

the last equality by (1) again. Pointwise $\psi^+ + \psi^- = |\psi|$, so this is the claim.

Step 4: the Lewy–Stampacchia bound at $|\psi|$. The element $\psi^+ + \psi^-$ is nonnegative almost everywhere, being a sum of two such, and has finite energy because the form domain is closed under sums (`EnergyFormBilinear`); so $\psi^+ + \psi^- \in \mathcal{K}$ by `ObstacleCone`. The hypotheses on F, a and u are exactly those of `LewyStampacchia`, which applied at this test element gives

$$T(\psi^+ + \psi^-) \le \int_{\mathbb{R}^n} (a - F)^+ \left(\psi^+ + \psi^-\right) d\gamma_n = \int_{\mathbb{R}^n} (a - F)^+ \,|\psi|\, d\gamma_n, \tag{2}$$

writing $(a-F)^+ = \max(a - F, 0)$ and using $\psi^+ + \psi^- = |\psi|$ almost everywhere.

Step 5: $0 \le (a-F)^+ \le a$ almost everywhere. Fix x with $F(x) \ge 0$, which is almost every x. If $a - F(x) \le 0$ then $(a - F(x))^+ = 0 \le a$, using $a > 0$ for the second inequality. If $a - F(x) \ge 0$

then $(a - F(x))^+ = a - F(x) \le a$, using $F(x) \ge 0$. In both cases $0 \le (a - F(x))^+ \le a$. Multiplying by $|\psi(x)| \ge 0$,

$$(a - F(x))^+ \, |\psi(x)| \le a \, |\psi(x)| \qquad \text{for } \gamma_n\text{-almost every } x,$$

and both sides are γ_n-integrable: the left because $(a-F)^+$ lies in $L^2(\gamma_n)$ by `PosPartConstSubMemLp` and the product of two $L^2(\gamma_n)$ functions is integrable, and the right because $\psi \in L^2(\gamma_n)$ is integrable against the probability measure γ_n (`GaussIsProb`). Monotonicity of the integral therefore gives

$$\int_{\mathbb{R}^n} (a - F)^+ \, |\psi| \, d\gamma_n \le \int_{\mathbb{R}^n} a \, |\psi| \, d\gamma_n = a \int_{\mathbb{R}^n} |\psi| \, d\gamma_n. \tag{3}$$

Step 6: from the L^1 integral to the L^2 norm. The $L^1(\gamma_n)$ norm of the image of ψ under the inclusion of `ToL1` is $\int |\psi| \, d\gamma_n$, since that inclusion does not change the underlying function. By `L2IntoL1`, $\|\psi\|_{L^1(\gamma_n)} \le \|\psi\|_{L^2(\gamma_n)}$, so

$$\int_{\mathbb{R}^n} |\psi| \, d\gamma_n \le \|\psi\|_{L^2(\gamma_n)}, \qquad \text{hence} \qquad a \int_{\mathbb{R}^n} |\psi| \, d\gamma_n \le a \, \|\psi\|_{L^2(\gamma_n)}, \tag{4}$$

the last step multiplying by $a \ge 0$.

Conclusion. Chaining Step 3, (2), (3) and (4),

$$|T(\psi)| \le T(\psi^+ + \psi^-) \le \int_{\mathbb{R}^n} (a - F)^+ |\psi| \, d\gamma_n \le a \int_{\mathbb{R}^n} |\psi| \, d\gamma_n \le a \, \|\psi\|_{L^2(\gamma_n)}.$$

Since ψ was an arbitrary element of the form domain, the bound holds uniformly there with the stated constant a. □

2.39. `HermiteBasisVectorFacts`.

Lemma 2.39 (The basis vectors of the Hermite basis: identification, coefficients, finite energy)**.** *Fix $n \in \mathbb{N}_0$ and let $\{h_\alpha\}_{\alpha \in \mathbb{N}_0^n}$ be the orthonormal Hermite basis of the real Hilbert space $L^2(\gamma_n)$ fixed in `HermiteBasis`, whose vectors are the elements of `HermiteL2`. Write v_α for the α-th Hermite coefficient of $v \in L^2(\gamma_n)$ and $\mathcal{E}$ for the fractional energy of `Energy`. Then the following three statements hold, each for every multi-index $\beta \in \mathbb{N}_0^n$.*

1. *(Identification) The β-th vector of the basis is the β-th Hermite element: the basis, indexed by $\mathbb{N}_0^n$, sends β to h_β.*
2. *(Coefficients) The Hermite coefficients of a basis vector form a Kronecker delta: for every $\alpha \in \mathbb{N}_0^n$,*
$$(h_\beta)_\alpha = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{otherwise.} \end{cases}$$
3. *(Finite energy) $\mathcal{E}(h_\beta) \ne +\infty$.*

The statement contains no integral, and the only measure entering it is the one implicit in the ambient space $L^2(\gamma_n)$; nothing further about measures is asserted here. The energy of clause (3) is a sum of nonnegative terms valued in $[0, +\infty]$, so the assertion is exactly that this sum is finite.

Proof. Fix $\beta \in \mathbb{N}_0^n$ throughout; the three clauses are proved in turn, and the proof of the third uses the second.

Clause (1). The basis of `HermiteBasis` is by construction the orthonormal basis assembled from the family of `HermiteL2` once that family is known to be orthonormal, by `HermiteL2Orthonormal`, and to have dense span, by `HermiteL2Complete`. Unfolding that construction, the underlying indexed family of the basis is the Hermite family itself, so evaluating the basis at β returns h_β.

Clause (2). Rewriting the basis vector by clause (1), the claim becomes a statement about the coefficients of a basis vector of an orthonormal basis. For such a basis the coefficient map sends the β-th basis vector to the indicator sequence supported at β with value 1 there: this is the defining property of an orthonormal basis' coordinate isomorphism applied to a basis vector. Reading off the α-th entry of that sequence gives 1 when $\alpha = \beta$ and 0 otherwise, which is the asserted Kronecker delta.

Clause (3). By Energy,

$$\mathcal{E}(h_\beta) = \sum_{\alpha\neq 0} |\alpha|^{1/2} \left|(h_\beta)_\alpha\right|^2,$$

a sum over the multi-indices $\alpha \neq 0$ of nonnegative terms in $[0,+\infty]$. By clause (2) the coefficient $(h_\beta)_\alpha$ vanishes unless $\alpha = \beta$, so

$$|\alpha|^{1/2} \left|(h_\beta)_\alpha\right|^2 = 0 \qquad \text{for every } \alpha \neq 0 \text{ with } \alpha \neq \beta. \tag{Z}$$

Now distinguish two cases, which are true for genuinely different reasons.

Case $\beta = 0$. The summation index α ranges only over multi-indices with $\alpha \neq 0$, so every α occurring in the sum satisfies $\alpha \neq 0 = \beta$, and (Z) applies to all of them. Hence every term vanishes and the sum is a sum of zeros: $\mathcal{E}(h_0) = 0$, which is certainly not $+\infty$. Note that in this case the surviving term of the other case is simply absent from the index set, which is why the energy is literally 0 rather than merely finite.

Case $\beta \neq 0$. Then β itself occurs in the index set, and (Z) says every other term vanishes, so the sum reduces to its single term at $\alpha = \beta$:

$$\mathcal{E}(h_\beta) = |\beta|^{1/2} \left|(h_\beta)_\beta\right|^2.$$

That term is the image in $[0,+\infty]$ of a real number — a real power of the natural number $|\beta|$ times the square of a real coefficient — and the canonical map from the reals into $[0,+\infty]$ never takes the value $+\infty$. Hence $\mathcal{E}(h_\beta) \neq +\infty$.

The two cases are exhaustive, so clause (3) holds for every β, and with it the conjunction of the three clauses. □

2.40. `MultiplierL2Representation`.

Lemma 2.40 (The multiplier is represented by an L^2 function)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., let $a > 0$, and let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`. Let η be the multiplier functional of `ObstacleMultiplier`. Then there exists $\eta_2 \in L^2(\gamma_n)$ with*

$$\langle \eta, \varphi\rangle = \int \eta_2\, \varphi \, d\gamma_n \qquad \textit{for every } \varphi \in \mathcal{V}.$$

Proof. Step 1: an L^2 bound for the functional on $\mathcal{V}$. Let $\psi \in \mathcal{V}$. The maps $t \mapsto t^+$, $t \mapsto t^-$ and $t \mapsto |t|$ are 1-Lipschitz and vanish at 0, so LatticeContraction, applied to ψ with each of these in turn, gives $\psi^+, \psi^-, |\psi| \in \mathcal{V}$. All three are nonnegative, so by ObstacleCone they lie in $\mathcal{K}$.

The map $\varphi \mapsto \langle \eta, \varphi\rangle$ is linear on $\mathcal{V}$, by inspection of the formula in ObstacleMultiplier (the form $\mathcal{E}(u,\cdot)$ is linear in its second argument and the integral is linear). Writing $\psi = \psi^+ - \psi^-$ and using MultiplierPositive, which gives $\langle \eta, \psi^+\rangle \geq 0$ and $\langle \eta, \psi^-\rangle \geq 0$,

$$\left|\langle \eta, \psi\rangle\right| = \left|\langle \eta, \psi^+\rangle - \langle \eta, \psi^-\rangle\right| \leq \langle \eta, \psi^+\rangle + \langle \eta, \psi^-\rangle = \langle \eta, \psi^+ + \psi^-\rangle = \langle \eta, |\psi|\rangle,$$

the last step because $\psi^+ + \psi^- = |\psi|$ pointwise. By LewyStampacchia applied to the test function $|\psi| \in \mathcal{K}$,

$$\langle \eta, |\psi|\rangle \leq \int k\,|\psi|\, d\gamma_n, \qquad k = (a - F)^+.$$

Since $F \geq 0$ γ_n-a.e. and $a > 0$, we have $0 \leq k = (a-F)^+ \leq a$ a.e. Therefore, using L2IntoL1,

$$\left|\langle \eta, \psi\rangle\right| \;\leq\; a \int |\psi|\, d\gamma_n \;=\; a\, \|\psi\|_{L^1(\gamma_n)} \;\leq\; a\, \|\psi\|_{L^2(\gamma_n)} \qquad \text{for every } \psi \in \mathcal{V}. \tag{1}$$

Step 2: $\mathcal{V}$ is dense in $L^2(\gamma_n)$. Every finite linear combination $\sum_{\alpha\in S} c_\alpha h_\alpha$ with $S \subseteq \mathbb{N}_0^n$ finite lies in $\mathcal{V}$: by HermiteL2Orthonormal its Hermite coefficients vanish outside S, so the defining series of Energy has finitely many nonzero terms and is finite. By HermiteL2Complete the closed linear span of $\{h_\alpha\}$ is all of $L^2(\gamma_n)$, so the set of such finite combinations is dense; hence so is the larger set $\mathcal{V}$.

Step 3: extension and Riesz representation. By (1) the linear functional $\psi \mapsto \langle \eta, \psi\rangle$ on the dense subspace $\mathcal{V} \subseteq L^2(\gamma_n)$ is bounded for the $L^2(\gamma_n)$-norm, with norm at most a. A bounded linear functional on a dense subspace of a normed space extends uniquely to a bounded linear

functional on the whole space, with the same norm: for $\psi \in L^2(\gamma_n)$ choose $\psi_j \in \mathcal{V}$ with $\psi_j \to \psi$ in $L^2(\gamma_n)$ (Step 2), note that $(\langle \eta, \psi_j \rangle)_j$ is Cauchy in $\mathbb{R}$ by (1) applied to $\psi_j - \psi_i \in \mathcal{V}$, and define the extension as the limit; it is independent of the chosen sequence by the same estimate, and it is linear and bounded by $a\|\cdot\|_2$.

$L^2(\gamma_n)$ is a Hilbert space, so the Riesz representation theorem supplies a unique $\eta_2 \in L^2(\gamma_n)$ with $\|\eta_2\|_2 \le a$ representing this extension, i.e.

$$\langle \eta, \psi \rangle = \langle \eta_2, \psi \rangle = \int \eta_2 \, \psi \, d\gamma_n \qquad \text{for every } \psi \in \mathcal{V},$$

where we used that the extension agrees with $\langle \eta, \cdot \rangle$ on $\mathcal{V}$. This is the assertion. □

2.41. `OneLeMultiIndexSum`.

Lemma 2.41 (A nonzero multi-index has order at least one). *Fix $n \in \mathbb{N}_0$ and let $\alpha \in \mathbb{N}_0^n$ be a multi-index with order $|\alpha| = \sum_{j=1}^n \alpha_j$ as in `MultiIndexSum`. If $\alpha \neq 0$, that is, if α is not the multi-index all of whose entries vanish, then*

$$1 \le |\alpha|,$$

an inequality between real numbers: the order $|\alpha|$ is a natural number, and it is its image under the canonical cast $\mathbb{N}_0 \to \mathbb{R}$ that is compared with the real number 1.

Proof. Step 1 (the order is a nonzero natural number). We claim $|\alpha| \neq 0$ as a natural number. Suppose instead that $|\alpha| = 0$, that is $\sum_{j=1}^n \alpha_j = 0$. Every summand α_j is a natural number and so is nonnegative, and a finite sum of nonnegative integers vanishes only if every one of its terms vanishes; hence $\alpha_j = 0$ for every index $j \in \{1, \dots, n\}$. Two multi-indices that agree at every index are equal, so $\alpha = 0$, contradicting the hypothesis $\alpha \neq 0$. Therefore $|\alpha| \neq 0$.

Step 2 (a nonzero natural number is at least one). In $\mathbb{N}_0$ the conditions $m \neq 0$ and $1 \le m$ are equivalent, since the natural numbers below 1 are exactly $\{0\}$. Applying this to $m = |\alpha|$ and using Step 1 gives

$$1 \le |\alpha| \qquad \text{in } \mathbb{N}_0.$$

Step 3 (transfer to the reals). The canonical cast $\mathbb{N}_0 \to \mathbb{R}$ is an order embedding: for natural numbers m and m' one has $m \le m'$ if and only if the corresponding reals satisfy the same inequality, and the cast sends 1 to the real number 1. Applying the forward direction to the inequality of Step 2 therefore yields $1 \le |\alpha|$ as an inequality in $\mathbb{R}$, which is the assertion. This cast is the only place the statement mixes the two number systems: the hypothesis and Steps 1–2 are statements about natural numbers, and only the conclusion is read in $\mathbb{R}$. □

2.42. `SemigroupRegularity`.

Lemma 2.42 (Instantaneous form-domain regularity and strong continuity of P_s). *Let $v \in L^2(\gamma_n)$. Then*

1. *for every $s > 0$ one has $\mathcal{E}(P_s v) < \infty$, i.e. $P_s v$ lies in the form domain $\mathcal{V}$ of `Energy`; and*
2. *$P_s v \to v$ in $L^2(\gamma_n)$ as $s \downarrow 0$.*

Here P_s is the semigroup of `OUSemigroup`.

Proof. Write v_α for the Hermite coefficients of v relative to `HermiteBasis`, so $\sum_\alpha |v_\alpha|^2 = \|v\|_2^2 < \infty$ by Parseval. By `OUSemigroupSpectral`, for every $s \ge 0$,

$$(P_s v)_\alpha = e^{-s|\alpha|} \, v_\alpha, \tag{1}$$

with $|\alpha|$ as in `MultiIndexSum`.

Part (1). Fix $s > 0$. By `Energy` and (1),

$$\mathcal{E}(P_s v) = \sum_{\alpha \neq 0} |\alpha|^{1/2} \, e^{-2s|\alpha|} \, |v_\alpha|^2.$$

For $\alpha \neq 0$ the number $\ell = |\alpha|$ is an integer with $\ell \ge 1$, by `MultiIndexSum`. Consider $F_s(\ell) = \ell^{1/2} e^{-2s\ell}$ for real $\ell \ge 1$. It is continuous on $[1, \infty)$ and tends to 0 as $\ell \to \infty$ (exponential decay beats the square root), so it is bounded there; set $C_s = \sup_{\ell \ge 1} \ell^{1/2} e^{-2s\ell} < \infty$. (Concretely

$C_s \le \sup_{\ell\ge 1} \ell^{1/2} e^{-2s\ell}$ is attained either at $\ell = 1$ or at the critical point $\ell = 1/(4s)$, and is finite in either case.) Then, termwise,

$$|\alpha|^{1/2} e^{-2s|\alpha|} |v_\alpha|^2 \le C_s\, |v_\alpha|^2 \qquad (\alpha \ne 0),$$

and summing the nonnegative terms,

$$\mathcal{E}(P_s v) \le C_s \sum_{\alpha\ne 0} |v_\alpha|^2 \le C_s\, \|v\|_2^2 < \infty.$$

Hence $P_s v \in \mathcal{V}$.

Part (2). By Parseval (`HermiteBasis`) and (1),

$$\|P_s v - v\|_2^2 = \sum_\alpha \left|e^{-s|\alpha|} - 1\right|^2 |v_\alpha|^2 = \sum_\alpha \left(1 - e^{-s|\alpha|}\right)^2 |v_\alpha|^2.$$

For every $s > 0$ and every α we have $0 \le 1 - e^{-s|\alpha|} \le 1$, so each summand is bounded by $|v_\alpha|^2$, uniformly in s; and the dominating family $(|v_\alpha|^2)_\alpha$ is summable with sum $\|v\|_2^2$. Moreover, for each fixed α, $e^{-s|\alpha|} \to 1$ as $s \downarrow 0$, so each summand tends to 0. The dominated convergence theorem for sums (counting measure on $\mathbb{N}_0^n$) therefore gives $\|P_s v - v\|_2^2 \to 0$ as $s \downarrow 0$, i.e. $P_s v \to v$ in $L^2(\gamma_n)$. □

2.43. `MultiplierPointwiseBound`.

Lemma 2.43 (Pointwise Lewy–Stampacchia bounds for the representing function)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., let $a > 0$, let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`, and let $\eta_2 \in L^2(\gamma_n)$ satisfy $\langle \eta, \varphi\rangle = \int \eta_2 \varphi\, d\gamma_n$ for every $\varphi \in \mathcal{V}$, where η is the multiplier of `ObstacleMultiplier`. Then*

$$0 \le \eta_2 \le (a - F)^+ \qquad \gamma_n\text{-a.e.}$$

In particular $0 \le \eta_2 \le a$ a.e., since $F \ge 0$ a.e. and $a > 0$.

Proof. Write $k = (a - F)^+$. Then $k \in L^2(\gamma_n)$, since $0 \le k \le a$ γ_n-a.e. (using $F \ge 0$ a.e. and $a > 0$) and γ_n is a probability measure (`GaussIsProb`).

Step 1: smoothed indicators are admissible test functions. Let $A \subseteq \mathbb{R}^n$ be an arbitrary Borel set. Its indicator $\mathbf{1}_A$ lies in $L^2(\gamma_n)$, because $\int \mathbf{1}_A^2\, d\gamma_n = \gamma_n(A) \le 1$ by `GaussIsProb`. For $s > 0$ put $\varphi_s = P_s \mathbf{1}_A$ with P_s the semigroup of `OUSemigroup`. By part (1) of `SemigroupRegularity`, $\mathcal{E}(\varphi_s) < \infty$, i.e. $\varphi_s \in \mathcal{V}$. By `MehlerFormula`, applied with this $s > 0$ and the nonnegative function $\mathbf{1}_A$, we get $\varphi_s \ge 0$ γ_n-a.e. Hence $\varphi_s \in \mathcal{K}$ by `ObstacleCone`.

Step 2: two inequalities for each $s > 0$. Apply `MultiplierPositive` with the test function $\varphi_s \in \mathcal{K}$: it gives $\langle \eta, \varphi_s\rangle \ge 0$. Apply `LewyStampacchia` with the same test function: it gives $\langle \eta, \varphi_s\rangle \le \int k\, \varphi_s\, d\gamma_n$. Rewriting the middle quantity by the representation hypothesis (legitimate since $\varphi_s \in \mathcal{V}$),

$$0 \;\le\; \int \eta_2\, \varphi_s\, d\gamma_n \;\le\; \int k\, \varphi_s\, d\gamma_n \qquad \text{for every } s > 0. \tag{1}$$

Step 3: letting $s \downarrow 0$. By part (2) of `SemigroupRegularity`, $\varphi_s = P_s \mathbf{1}_A \to \mathbf{1}_A$ in $L^2(\gamma_n)$ as $s \downarrow 0$. Since $\eta_2, k \in L^2(\gamma_n)$, the two maps $\varphi \mapsto \int \eta_2 \varphi\, d\gamma_n$ and $\varphi \mapsto \int k\varphi\, d\gamma_n$ are continuous linear functionals on $L^2(\gamma_n)$ by Cauchy–Schwarz. Passing to the limit in (1) therefore gives

$$0 \;\le\; \int_A \eta_2\, d\gamma_n \;\le\; \int_A k\, d\gamma_n. \tag{2}$$

Both integrals are finite, since $\eta_2, k \in L^2(\gamma_n) \subseteq L^1(\gamma_n)$ by `L2IntoL1`.

Step 4: from arbitrary sets to pointwise bounds. The Borel set A in (2) was arbitrary.

Take $A = \{\eta_2 < 0\}$. The left inequality of (2) gives $\int_A \eta_2\, d\gamma_n \ge 0$. But $\eta_2 < 0$ everywhere on A, so if $\gamma_n(A) > 0$ then $\int_A \eta_2\, d\gamma_n < 0$: indeed $A = \bigcup_{j\ge 1} \{\eta_2 < -1/j\}$, so some $A_j = \{\eta_2 < -1/j\}$ has $\gamma_n(A_j) > 0$, and then $\int_A \eta_2 d\gamma_n \le \int_{A_j} \eta_2 d\gamma_n \le -\gamma_n(A_j)/j < 0$ (the first step because $\eta_2 \le 0$ on $A \setminus A_j$). This contradiction forces $\gamma_n(A) = 0$, i.e. $\eta_2 \ge 0$ γ_n-a.e.

Take $A' = \{\eta_2 - k > 0\}$. The right inequality of (2) gives $\int_{A'} (\eta_2 - k)\, d\gamma_n \le 0$, while $\eta_2 - k > 0$ everywhere on A'. By the same decomposition argument, $\gamma_n(A') = 0$, i.e. $\eta_2 \le k$ γ_n-a.e.

Thus $0 \le \eta_2 \le k = (a - F)^+$ γ_n-a.e., and $k \le a$ a.e. as noted at the outset. □

2.44. `Complementarity`.

Lemma 2.44 (Pointwise complementarity on the contact set)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., let $a > 0$, let $u \in \mathcal{K}$ minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`, and let $\eta_2 \in L^2(\gamma_n)$ satisfy $\langle \eta, \varphi \rangle = \int \eta_2 \varphi \, d\gamma_n$ for every $\varphi \in \mathcal{V}$, where η is the multiplier of `ObstacleMultiplier`. Then*

$$\int \eta_2 \, u \, d\gamma_n = 0 \qquad \text{and} \qquad \eta_2 = 0 \quad \gamma_n\text{-a.e. on } \Omega = \{u > 0\}.$$

Equivalently, with $g = a\mathbf{1} - \eta_2$, one has $g = a$ a.e. on Ω.

Proof. Since $u \in \mathcal{K} \subseteq \mathcal{V}$, we may take $\varphi = u$ in the representation hypothesis, which gives

$$\langle \eta, u \rangle = \int \eta_2 \, u \, d\gamma_n,$$

the integral being finite by Cauchy–Schwarz since $\eta_2, u \in L^2(\gamma_n)$. By `MultiplierVanishesOnPotential`, $\langle \eta, u \rangle = 0$. Hence $\int \eta_2 u \, d\gamma_n = 0$.

By `MultiplierPointwiseBound` we have $\eta_2 \ge 0$ γ_n-a.e., and $u \ge 0$ γ_n-a.e. by `ObstacleCone`. Therefore the product $\eta_2 u$ is nonnegative a.e. and integrable with integral zero, so $\eta_2 u = 0$ γ_n-almost everywhere.

Consequently, at γ_n-almost every point where $u > 0$ we must have $\eta_2 = 0$; that is, $\eta_2 = 0$ a.e. on $\Omega = \{u > 0\}$. With $g = a\mathbf{1} - \eta_2$ this reads $g = a$ a.e. on Ω. No regularity or openness of Ω is used. □

2.45. `HermiteFunContDiff`.

Lemma 2.45 (The Hermite functions are smooth)**.** *For every $n \ge 1$ and every multi-index $\alpha \in \mathbb{N}_0^n$, the Hermite function h_α of `HermiteFun` is infinitely differentiable on $\mathbb{R}^n$: $h_\alpha \in C^\infty(\mathbb{R}^n)$. In particular h_α is differentiable at every point, and so is each of its partial derivatives of every order.*

Proof. Step 1: each one-dimensional factor is smooth. Fix $k \in \mathbb{N}_0$. By `Hermite1`,

$$h_k(s) = \frac{\mathrm{He}_k(\sqrt{2}\, s)}{\sqrt{k!}},$$

so h_k is the composition of three maps, each smooth on $\mathbb{R}$: the linear map $s \mapsto \sqrt{2}\, s$; the polynomial function $y \mapsto \mathrm{He}_k(y)$, which is smooth because a polynomial function on $\mathbb{R}$ is a finite sum of monomials $y \mapsto c\, y^m$ and each monomial is smooth; and multiplication by the constant $1/\sqrt{k!}$, which is a nonzero real number since $k! \ge 1$. A composition of smooth maps is smooth and a constant multiple of a smooth map is smooth, so $h_k \in C^\infty(\mathbb{R})$.

Step 2: the product over the coordinates is smooth. By `HermiteFun`,

$$h_\alpha(x) = \prod_{j=1}^{n} h_{\alpha_j}(x_j).$$

For each j the coordinate projection $\pi_j : \mathbb{R}^n \to \mathbb{R}$, $\pi_j(x) = x_j$, is linear, hence smooth; composing with Step 1 shows that $x \mapsto h_{\alpha_j}(x_j) = (h_{\alpha_j} \circ \pi_j)(x)$ is smooth on $\mathbb{R}^n$. A finite product of real-valued smooth functions is smooth, by induction on the number of factors using the fact that a product of two smooth real-valued functions is smooth. Since the product above has exactly n factors, $h_\alpha \in C^\infty(\mathbb{R}^n)$.

The two stated consequences are immediate: a C^∞ function is differentiable at every point, and each of its partial derivatives of any order is again C^∞, hence differentiable everywhere. □

2.46. `HermiteFunDeriv`.

Lemma 2.46 (Action of the energy-normalized gradient on Hermite functions)**.** *Let $D = 2^{-1/2}\nabla$, so that $D_j = 2^{-1/2}\partial_j$. For every multi-index $\alpha \in \mathbb{N}_0^n$, every $j \in \{1,\dots,n\}$ and every $x \in \mathbb{R}^n$, the Hermite function h_α of `HermiteFun` is differentiable at x and*

$$\partial_j h_\alpha(x) = \sqrt{2}\,\sqrt{\alpha_j}\,h_{\alpha-e_j}(x), \qquad \text{equivalently} \qquad D_j h_\alpha = \sqrt{\alpha_j}\,h_{\alpha-e_j},$$

where $\alpha - e_j$ is the multi-index of `MultiIndexSum` obtained by decreasing the jth entry of α by one, with the convention that this entry stays at 0 when $\alpha_j = 0$ (in which case the right-hand side vanishes anyway, because of the factor $\sqrt{\alpha_j} = 0$).

Proof. Step 1: the one-dimensional identity. We first prove

$$h_k'(s) = \sqrt{2}\,\sqrt{k}\,h_{k-1}(s), \qquad k \in \mathbb{N}_0,\ s \in \mathbb{R}, \tag{1}$$

with the right-hand side read as 0 when $k = 0$.

For $k = 0$: by `Hermite1Zero`, $h_0 \equiv 1$, so $h_0' = 0$; and the right-hand side carries the factor $\sqrt{0} = 0$. Both sides vanish.

For $k \geq 1$ write $k = m+1$. By `Hermite1`, $h_{m+1}(s) = \mathrm{He}_{m+1}(\sqrt{2}\,s)/\sqrt{(m+1)!}$, a polynomial in s, hence differentiable, and the chain rule gives

$$h_{m+1}'(s) = \frac{\sqrt{2}\,\mathrm{He}_{m+1}'(\sqrt{2}\,s)}{\sqrt{(m+1)!}}.$$

By `HermiteDerivative`, $\mathrm{He}_{m+1}' = (m+1)\,\mathrm{He}_m$, so

$$h_{m+1}'(s) = \frac{\sqrt{2}\,(m+1)\,\mathrm{He}_m(\sqrt{2}\,s)}{\sqrt{(m+1)!}} = \sqrt{2}\,\frac{m+1}{\sqrt{m+1}}\cdot\frac{\mathrm{He}_m(\sqrt{2}\,s)}{\sqrt{m!}} = \sqrt{2}\,\sqrt{m+1}\,h_m(s),$$

where the middle step used $(m+1)! = (m+1)\,m!$, so that $\sqrt{(m+1)!} = \sqrt{m+1}\,\sqrt{m!}$, and the last step used `Hermite1` again. This is (1) with $k = m+1$.

Step 2: differentiability. By `HermiteFunContDiff`, $h_\alpha \in C^\infty(\mathbb{R}^n)$. A C^∞ function is in particular differentiable at every point, so h_α is differentiable at x; this is the first assertion, and it also licenses the use of $\partial_j h_\alpha(x)$ and of the product rule below.

Step 3: the n-dimensional identity. By `HermiteFun`, $h_\alpha(x) = \prod_{i=1}^n h_{\alpha_i}(x_i)$. Only the $i = j$ factor depends on x_j, so the product rule gives

$$\partial_j h_\alpha(x) = h_{\alpha_j}'(x_j)\prod_{i\neq j} h_{\alpha_i}(x_i).$$

Substituting (1) with $k = \alpha_j$ and $s = x_j$,

$$\partial_j h_\alpha(x) = \sqrt{2}\,\sqrt{\alpha_j}\,h_{\alpha_j-1}(x_j)\prod_{i\neq j} h_{\alpha_i}(x_i).$$

The multi-index $\alpha - e_j$ agrees with α in every entry $i \neq j$ and has jth entry $\alpha_j - 1$, so by `HermiteFun` the product on the right is exactly $h_{\alpha-e_j}(x)$. Hence $\partial_j h_\alpha(x) = \sqrt{2}\sqrt{\alpha_j}\,h_{\alpha-e_j}(x)$. When $\alpha_j = 0$ the prefactor is 0, so the identity holds regardless of how $\alpha - e_j$ is interpreted, which justifies the stated convention. Dividing by $\sqrt{2}$ gives the form for $D_j = 2^{-1/2}\partial_j$. □

2.47. `HermiteFunOU`.

Lemma 2.47 (Hermite functions are eigenfunctions of the Ornstein–Uhlenbeck operator)**.** *Let $L = -\frac{1}{2}\Delta + x\cdot\nabla$ be the Ornstein–Uhlenbeck differential operator, i.e.*

$$(Lv)(x) = -\tfrac{1}{2}\sum_{j=1}^n \partial_j^2 v(x) + \sum_{j=1}^n x_j\,\partial_j v(x).$$

For every multi-index $\alpha \in \mathbb{N}_0^n$, the Hermite function h_α of `HermiteFun` is twice continuously differentiable on $\mathbb{R}^n$, i.e. $h_\alpha \in C^2(\mathbb{R}^n)$; and for every $x \in \mathbb{R}^n$ it satisfies

$$-\tfrac{1}{2}\Delta h_\alpha(x) + x\cdot\nabla h_\alpha(x) = |\alpha|\,h_\alpha(x),$$

with $|\alpha|$ as in `MultiIndexSum`. Thus h_α is an eigenfunction of L with eigenvalue $|\alpha|$; this is the identity that ties the Hermite index $|\alpha|$ to the differential operator L of the reference paper.

Proof. Step 0: the regularity assertion. By `HermiteFunContDiff`, $h_\alpha \in C^\infty(\mathbb{R}^n)$, and a C^∞ function is in particular C^2; this is the first assertion. It also makes every expression below meaningful: h_α is differentiable at every point, each first partial derivative $\partial_j h_\alpha$ is again C^∞ hence differentiable at every point, so the second partial derivatives $\partial_j^2 h_\alpha$ exist and Δh_α is defined pointwise.

Step 1: the three-term recurrence for the normalized h_k. By the defining recurrence recorded in `Hermite1`, $\mathrm{He}_{m+1} = X\,\mathrm{He}_m - \mathrm{He}_m'$, and by `HermiteDerivative`, $\mathrm{He}_m' = m\,\mathrm{He}_{m-1}$ for $m \geq 1$ while $\mathrm{He}_0' = 0$. Hence for every $m \geq 0$, evaluating at y,

$$\mathrm{He}_{m+1}(y) = y\,\mathrm{He}_m(y) - m\,\mathrm{He}_{m-1}(y),$$

the last term being absent when $m = 0$. Put $y = \sqrt{2}\,s$ and divide by $\sqrt{m!}$. Using `Hermite1` in the form $\mathrm{He}_m(\sqrt{2}s) = \sqrt{m!}\,h_m(s)$, together with $\sqrt{(m+1)!} = \sqrt{m+1}\sqrt{m!}$ and $m/\sqrt{m}\,\sqrt{(m-1)!}/\sqrt{m!}\cdot\sqrt{m} = \sqrt{m}$ (concretely, $m\sqrt{(m-1)!}/\sqrt{m!} = m/\sqrt{m} = \sqrt{m}$), this becomes

$$\sqrt{2}\,s\,h_m(s) = \sqrt{m+1}\,h_{m+1}(s) + \sqrt{m}\,h_{m-1}(s), \qquad m \geq 0, \tag{1}$$

with the convention that a term carrying the factor $\sqrt{0}$ is zero.

Step 2: the one-dimensional eigenvalue relation. We prove

$$-\tfrac{1}{2}h_k''(s) + s\,h_k'(s) = k\,h_k(s), \qquad k \in \mathbb{N}_0,\ s \in \mathbb{R}. \tag{2}$$

By `HermiteFunDeriv` in dimension one (the case $n = 1$, $j = 1$, where the multi-index is the single integer k), $h_k' = \sqrt{2}\sqrt{k}\,h_{k-1}$. Applying the same identity once more to h_{k-1},

$$h_k'' = \sqrt{2}\sqrt{k}\,h_{k-1}' = \sqrt{2}\sqrt{k}\cdot\sqrt{2}\sqrt{k-1}\,h_{k-2} = 2\sqrt{k(k-1)}\,h_{k-2},$$

with the usual convention that a term with a vanishing square-root factor is zero. Therefore

$$-\tfrac{1}{2}h_k''(s) + s\,h_k'(s) = -\sqrt{k(k-1)}\,h_{k-2}(s) + \sqrt{k}\cdot\sqrt{2}\,s\,h_{k-1}(s).$$

Apply (1) with $m = k-1$ (legitimate for $k \geq 1$; for $k = 0$ both sides of (2) are 0 because $h_0' = h_0'' = 0$ by `Hermite1Zero`): $\sqrt{2}\,s\,h_{k-1}(s) = \sqrt{k}\,h_k(s) + \sqrt{k-1}\,h_{k-2}(s)$. Substituting,

$$-\tfrac{1}{2}h_k'' + s\,h_k' = -\sqrt{k(k-1)}\,h_{k-2} + \sqrt{k}\big(\sqrt{k}\,h_k + \sqrt{k-1}\,h_{k-2}\big) = k\,h_k,$$

since $\sqrt{k}\sqrt{k-1} = \sqrt{k(k-1)}$ and the two h_{k-2} terms cancel. This is (2).

Step 3: assembling the n-dimensional identity. By `HermiteFun`, $h_\alpha(x) = \prod_i h_{\alpha_i}(x_i)$. Only the $i = j$ factor depends on x_j, so

$$\partial_j h_\alpha(x) = h_{\alpha_j}'(x_j)\prod_{i\neq j} h_{\alpha_i}(x_i), \qquad \partial_j^2 h_\alpha(x) = h_{\alpha_j}''(x_j)\prod_{i\neq j} h_{\alpha_i}(x_i).$$

Hence, summing over j,

$$-\tfrac{1}{2}\Delta h_\alpha(x) + x\cdot\nabla h_\alpha(x) = \sum_{j=1}^{n}\Big(-\tfrac{1}{2}h_{\alpha_j}''(x_j) + x_j\,h_{\alpha_j}'(x_j)\Big)\prod_{i\neq j} h_{\alpha_i}(x_i).$$

By (2) with $k = \alpha_j$ and $s = x_j$, the jth bracket equals $\alpha_j\,h_{\alpha_j}(x_j)$, so the jth summand equals $\alpha_j\prod_i h_{\alpha_i}(x_i) = \alpha_j\,h_\alpha(x)$. Therefore

$$-\tfrac{1}{2}\Delta h_\alpha(x) + x\cdot\nabla h_\alpha(x) = \Big(\sum_{j=1}^{n}\alpha_j\Big)h_\alpha(x) = |\alpha|\,h_\alpha(x),$$

the last equality being the definition of $|\alpha|$ in `MultiIndexSum`. □

2.48. `PseudoInverseSqrt`.

Lemma 2.48 (The pseudoinverse lands in the operator domain and inverts $L^{1/2}$)**.** *For every $f \in L^2(\gamma_n)$ put $v = L^{-1/2}\Pi_0 f$, the element supplied by `FracPow` with $s = -1/2$. Then*

$$v \in \mathrm{Dom}(L^{1/2}) \qquad \text{and} \qquad L^{1/2}v = \Pi_0 f \quad \text{in } L^2(\gamma_n),$$

with $\mathrm{Dom}(L^{1/2})$ as in `FracPowDomain` and $\Pi_0 = L^0$ as in `FracPow`.

Proof. Write $f_\alpha = \langle f, h_\alpha\rangle$ for the Hermite coefficients of f relative to `HermiteBasis`; by Parseval, $\sum_\alpha |f_\alpha|^2 = \|f\|_2^2 < \infty$.

Step 1: $f \in \mathrm{Dom}(L^{-1/2})$, so v is given by the series. For $\alpha \neq 0$ we have $|\alpha| \geq 1$ by `MultiIndexSum`, hence $|\alpha|^{-1} \leq 1$. By `FracPowCoeff` with $s = -1/2$ the multiplied family is 0 at $\alpha = 0$ and $|\alpha|^{-1/2} f_\alpha$ at $\alpha \neq 0$, so

$$\sum_{\alpha\neq 0} \big||\alpha|^{-1/2} f_\alpha\big|^2 = \sum_{\alpha\neq 0} |\alpha|^{-1}|f_\alpha|^2 \leq \sum_{\alpha\neq 0} |f_\alpha|^2 \leq \|f\|_2^2 < \infty.$$

Thus the family is square summable, which is exactly the criterion of `FracPowDomain` for $s = -1/2$; so $f \in \mathrm{Dom}(L^{-1/2})$ and the defining formula of `FracPow` gives $v = L^{-1/2}\Pi_0 f$ the coefficients

$$v_0 = 0, \qquad v_\alpha = |\alpha|^{-1/2} f_\alpha \quad (\alpha \neq 0).$$

Step 2: $v \in \mathrm{Dom}(L^{1/2})$. Using Step 1 and $|\alpha| \cdot |\alpha|^{-1} = 1$ for $\alpha \neq 0$,

$$\sum_{\alpha\neq 0} |\alpha|\,|v_\alpha|^2 = \sum_{\alpha\neq 0} |\alpha| \cdot |\alpha|^{-1}|f_\alpha|^2 = \sum_{\alpha\neq 0} |f_\alpha|^2,$$

which is finite, being at most $\|f\|_2^2$. By `FracPowDomain` with $s = 1/2$, whose criterion is precisely $\sum_{\alpha\neq 0} |\alpha||v_\alpha|^2 < \infty$, this gives $v \in \mathrm{Dom}(L^{1/2})$. The weight $|\alpha|$ has exactly cancelled the squared multiplier $|\alpha|^{-1}$.

Step 3: $L^{1/2}v = \Pi_0 f$. By Step 2 the formula of `FracPow` with $s = 1/2$ applies to v and gives $L^{1/2}v$ the coefficients 0 at $\alpha = 0$ and, for $\alpha \neq 0$,

$$|\alpha|^{1/2} v_\alpha = |\alpha|^{1/2}\,|\alpha|^{-1/2} f_\alpha = f_\alpha.$$

On the other hand, by `FracPowCoeff` and `FracPow` with $s = 0$, the element $\Pi_0 f$ has exactly those coefficients: 0 at $\alpha = 0$ and f_α at $\alpha \neq 0$ (its domain condition holds since the multiplied family is dominated by $(|f_\alpha|^2)_\alpha$). The coefficient map of `HermiteBasis` is injective, so $L^{1/2}v = \Pi_0 f$. □

2.49. `GaussL2LocallyIntegrable`.

Lemma 2.49 ($L^2(\gamma_n)$ lies in L^1_{loc} and in L^2_{loc} for Lebesgue measure)**.** *Let γ_n be as in `GaussMeasure`, write λ for Lebesgue measure on $\mathbb{R}^n$, and let $v : \mathbb{R}^n \to \mathbb{R}$ be γ_n-measurable with $\int_{\mathbb{R}^n} |v|^2\, d\gamma_n < \infty$, i.e. $v \in L^2(\gamma_n)$ as a function. Then:*

1. *v is locally λ-integrable, i.e. every point of $\mathbb{R}^n$ has a neighbourhood U with $\int_U |v|\, d\lambda < \infty$;*
2. *for every compact $K \subseteq \mathbb{R}^n$ the restriction of v to K lies in $L^2(K, \lambda)$, i.e. $\int_K |v|^2\, d\lambda < \infty$;*
3. *(quantitative form, uniform in the function) for every compact $K \subseteq \mathbb{R}^n$ there is a constant $C_K \geq 0$, depending only on K and on n, such that* every *γ_n-measurable $g : \mathbb{R}^n \to \mathbb{R}$ with $\int_{\mathbb{R}^n} |g|^2\, d\gamma_n < \infty$ is λ-integrable on K and satisfies*

$$\int_K |g|\, d\lambda \;\leq\; C_K\, \|g\|_{L^2(\gamma_n)}.$$

All three conclusions are statements about Lebesgue measure, not about γ_n; they are the comparison that lets a function of the Gaussian space be fed to test identities written against Lebesgue measure. Conclusion (3) differs from (1) in that its constant does not depend on the function, which is what permits passing to a limit along an $L^2(\gamma_n)$-convergent sequence inside an integral taken against λ.

Proof. Step 0: measurability transfers between the two measures. By `GaussMeasure`, γ_n is absolutely continuous with respect to λ with Radon–Nikodym density $\varrho(x) = \pi^{-n/2}e^{-|x|^2}$, which is continuous and strictly positive at every point; in particular $\int h\,d\gamma_n = \int h\,\varrho\,d\lambda$ for every measurable $h : \mathbb{R}^n \to [0,+\infty]$, the case $h = \mathbf{1}_A$ being the set-level formula displayed there. Consequently a set is λ-null if and only if it is γ_n-null: if $\lambda(A) = 0$ then $\gamma_n(A) = \int_A \varrho\,d\lambda = 0$; and if $\gamma_n(A) = 0$ then, since $\varrho > 0$ everywhere, $\int_A \varrho\,d\lambda = 0$ forces $\lambda(A) = 0$. Hence v, which is γ_n-almost everywhere equal to a measurable function, is also λ-almost everywhere equal to that same function, so v is λ-measurable. All integrals below are therefore meaningful.

Step 1: the density comparison on a compact set. Let $K \subseteq \mathbb{R}^n$ be compact and nonempty (for $K = \emptyset$ both conclusions are trivial). Since ϱ is continuous and K is compact and nonempty, ϱ attains a minimum on K by the extreme value theorem, which is shared context; call it $c_K = \min_{x\in K}\varrho(x) = \varrho(x_-)$ for a minimizing $x_- \in K$. As $\varrho > 0$ everywhere, $c_K > 0$. Also $\lambda(K) < \infty$, again shared context for compact sets. Now let $h : \mathbb{R}^n \to [0,+\infty]$ be measurable. On K we have $h \le c_K^{-1}\,h\,\varrho$ pointwise, because $\varrho \ge c_K$ there and $h \ge 0$. Integrating this pointwise inequality over K and then using the density formula of `GaussMeasure`,

$$\int_K h\,d\lambda \;\le\; \frac{1}{c_K}\int_K h\,\varrho\,d\lambda \;=\; \frac{1}{c_K}\int_K h\,d\gamma_n. \tag{1}$$

Step 2: conclusion (2). Apply (1) with $h = |v|^2$:

$$\int_K |v|^2\,d\lambda \;\le\; \frac{1}{c_K}\int_K |v|^2\,d\gamma_n \;\le\; \frac{1}{c_K}\int_{\mathbb{R}^n} |v|^2\,d\gamma_n \;=\; \frac{1}{c_K}\,\|v\|^2_{L^2(\gamma_n)} \;<\; \infty,$$

the middle inequality because $|v|^2 \ge 0$, and finiteness by the hypothesis $\int |v|^2 d\gamma_n < \infty$. So $v \in L^2(K,\lambda)$, which is conclusion (2).

Step 3: conclusion (1). Fix $x_0 \in \mathbb{R}^n$ and put $K = \overline{B}(x_0,1)$, a compact neighbourhood of x_0; $\lambda(K) < \infty$ by the shared-context item on compact sets. By Cauchy–Schwarz, which is shared context, applied on the measure space (K,λ) to the pair of functions $|v|$ and the constant function 1 — both in $L^2(K,\lambda)$, the first by Step 2 and the second because $\lambda(K) < \infty$ —

$$\int_K |v|\,d\lambda \;\le\; \Big(\int_K |v|^2\,d\lambda\Big)^{1/2}\Big(\int_K 1\,d\lambda\Big)^{1/2} \;=\; \Big(\int_K |v|^2\,d\lambda\Big)^{1/2}\lambda(K)^{1/2},$$

which is finite by Step 2 and $\lambda(K) < \infty$. So v is λ-integrable on the neighbourhood K of x_0. As x_0 was arbitrary, v is locally λ-integrable, which is conclusion (1).

Step 4: conclusion (3), with an explicit constant. Let K be compact. If K is empty put $C_K = 0$; the asserted inequality then reads $0 \le 0$. Otherwise set

$$C_K := \Big(\frac{\lambda(K)}{c_K}\Big)^{1/2},$$

with $c_K = \min_{x\in K}\varrho(x) > 0$ as in Step 1; this is a finite nonnegative number depending only on K and on n, because K compact gives $\lambda(K) < \infty$ and Step 1 gives $c_K > 0$. Let g be any γ_n-measurable function with $\int |g|^2\,d\gamma_n < \infty$. Steps 0 and 1 apply verbatim with g in place of v — Step 0 used only γ_n-measurability of v, and Step 1 is a statement about an arbitrary nonnegative measurable h and does not mention v at all — so g is λ-measurable and, by (1) with $h = |g|^2$ together with $|g|^2 \ge 0$,

$$\int_K |g|^2\,d\lambda \;\le\; \frac{1}{c_K}\int_K |g|^2\,d\gamma_n \;\le\; \frac{1}{c_K}\int_{\mathbb{R}^n} |g|^2\,d\gamma_n \;=\; \frac{1}{c_K}\,\|g\|^2_{L^2(\gamma_n)}.$$

Taking square roots, which preserves the inequality because $t \mapsto t^{1/2}$ is nondecreasing on $[0,+\infty)$ and both sides are nonnegative and finite, gives $\big(\int_K |g|^2 d\lambda\big)^{1/2} \le c_K^{-1/2}\|g\|_{L^2(\gamma_n)}$. By Cauchy–Schwarz, which is shared context, on the measure space (K,λ) applied to $|g|$ and the constant function 1 — both in $L^2(K,\lambda)$, the first by the display just proved and the second because $\lambda(K) < \infty$ —

$$\int_K |g|\,d\lambda \;\le\; \Big(\int_K |g|^2\,d\lambda\Big)^{1/2}\lambda(K)^{1/2} \;\le\; \frac{\lambda(K)^{1/2}}{c_K^{1/2}}\,\|g\|_{L^2(\gamma_n)} \;=\; C_K\,\|g\|_{L^2(\gamma_n)},$$

the second step multiplying the previous bound by $\lambda(K)^{1/2} \geq 0$. Finally, g is λ-integrable on K: it is λ-measurable by Step 0, and the Cauchy–Schwarz bound just displayed is precisely the finiteness of $\int_K |g|\, d\lambda$, so g restricted to K lies in $L^1(K, \lambda)$. Since g was arbitrary, this is conclusion (3). □

2.50. `GaussCoordMemLp`.

Lemma 2.50 (Coordinates of a vector-valued $L^2(\gamma_n)$ element are scalar $L^2(\gamma_n)$)**.** *Fix $n \in \mathbb{N}_0$ and let γ_n be the Gaussian measure of `GaussMeasure` on $\mathbb{R}^n$. Let $W \in L^2(\gamma_n; \mathbb{R}^n)$ be a square-γ_n-integrable $\mathbb{R}^n$-valued function on $\mathbb{R}^n$, and let i be a coordinate index, $1 \leq i \leq n$. Then the scalar function*

$$x \longmapsto \big(W(x)\big)_i$$

is square γ_n-integrable, that is, it belongs to $L^2(\gamma_n)$.

Proof. Write $W_i(x) = (W(x))_i$ for the i-th coordinate function. Membership in $L^2(\gamma_n)$ is the conjunction of two assertions, almost-everywhere strong measurability and finiteness of the L^2 seminorm, and we verify them in turn.

Measurability. The i-th coordinate projection $\pi_i : \mathbb{R}^n \to \mathbb{R}$, $\pi_i(y) = y_i$, is linear and continuous, being a bounded linear functional on a finite-dimensional normed space. As W is an element of $L^2(\gamma_n; \mathbb{R}^n)$ it is γ_n-almost-everywhere strongly measurable, and the composition of a continuous map with an almost-everywhere strongly measurable map is again almost-everywhere strongly measurable. Since $W_i = \pi_i \circ W$, the function W_i is γ_n-almost-everywhere strongly measurable.

Pointwise domination. For every $y \in \mathbb{R}^n$ and every coordinate index i,

$$|y_i| \leq |y|,$$

where $|y| = \big(\sum_{m=1}^n |y_m|^2\big)^{1/2}$ is the Euclidean norm. This is the coordinate-versus-norm inequality for the ℓ^2 norm, and it is where the exponent enters: it holds because a single summand of a sum of nonnegative terms is at most the whole sum, so $|y_i|^2 \leq \sum_{m=1}^n |y_m|^2$, and the square root is monotone on $[0, \infty)$, whence $|y_i| = (|y_i|^2)^{1/2} \leq \big(\sum_m |y_m|^2\big)^{1/2} = |y|$. (The same argument works verbatim for the ℓ^p norm with $1 \leq p < \infty$; no feature special to $p = 2$ is used.) Applying this at $y = W(x)$ gives

$$|W_i(x)| \leq |W(x)| \qquad \text{for every } x \in \mathbb{R}^n,$$

and in particular for γ_n-almost every x.

Conclusion. The function $x \mapsto |W(x)|$ lies in $L^2(\gamma_n)$, since by hypothesis $W \in L^2(\gamma_n; \mathbb{R}^n)$, which is precisely the statement that its pointwise norm is square γ_n-integrable. A function that is almost-everywhere strongly measurable and is dominated almost everywhere in absolute value by a function of $L^2(\gamma_n)$ is itself in $L^2(\gamma_n)$: monotonicity of the integral gives

$$\int_{\mathbb{R}^n} |W_i|^2\, d\gamma_n \leq \int_{\mathbb{R}^n} |W|^2\, d\gamma_n < \infty.$$

Combining this with the measurability established above, $W_i \in L^2(\gamma_n)$, as asserted. □

2.51. `GaussWeakGradUnique`.

Lemma 2.51 (Uniqueness of the weak gradient in $L^2(\gamma_n)$)**.** *Let $v : \mathbb{R}^n \to \mathbb{R}$ be measurable and let $G, H \in L^2(\gamma_n; \mathbb{R}^n)$ both be weak Euclidean gradients of v in the sense of `HasGaussWeakGrad`. Then $G = H$ as elements of $L^2(\gamma_n; \mathbb{R}^n)$, i.e. $G = H$ γ_n-almost everywhere.*

Proof. Fix $j \in \{1, \dots, n\}$ and put $w = G_j - H_j$.

Step 1: w annihilates every test function. By `HasGaussWeakGrad` applied to the pair (v, G) and then to the pair (v, H), for every $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\int v\, \partial_j \varphi\, dx = -\int G_j\, \varphi\, dx, \qquad \int v\, \partial_j \varphi\, dx = -\int H_j\, \varphi\, dx,$$

the integrals being with respect to Lebesgue measure. Subtracting,

$$\int_{\mathbb{R}^n} w\,\varphi\,dx = 0 \qquad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^n). \tag{1}$$

Step 2: w is locally Lebesgue integrable. Both G_j and H_j lie in $L^2(\gamma_n)$, since G and H lie in $L^2(\gamma_n;\mathbb{R}^n)$. By `GaussL2LocallyIntegrable`, applied to each of them, each is locally integrable for Lebesgue measure. A difference of two locally Lebesgue-integrable functions is locally Lebesgue integrable, so $w = G_j - H_j \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ for Lebesgue measure.

Step 3: $w = 0$ Lebesgue-a.e. The du Bois-Reymond item of shared context says exactly that a function which is locally integrable for Lebesgue measure and integrates to zero against every $\varphi \in C_c^\infty(\mathbb{R}^n)$ vanishes Lebesgue-almost everywhere. Its two hypotheses are in hand: local Lebesgue integrability of w is Step 2, and the vanishing of $\int w\varphi\,dx$ for every such φ is (1) from Step 1. Hence $w = 0$ Lebesgue-almost everywhere.

Step 4: transfer to γ_n. Again by `GaussMeasure`, γ_n has a strictly positive density with respect to Lebesgue measure, so a Lebesgue-null set is γ_n-null and conversely. Therefore $G_j = H_j$ γ_n-almost everywhere.

As j was arbitrary, $G = H$ γ_n-almost everywhere, i.e. $G = H$ in $L^2(\gamma_n;\mathbb{R}^n)$. □

2.52. `GaussWeakGradAECongr`.

Lemma 2.52 (The weak-gradient relation depends only on the γ_n-a.e. classes)**.** *Let $v, v' : \mathbb{R}^n \to \mathbb{R}$ and $F, F' : \mathbb{R}^n \to \mathbb{R}^n$ be measurable, and suppose $v = v'$ γ_n-almost everywhere and $F = F'$ γ_n-almost everywhere, with γ_n the measure of `GaussMeasure`. If F is a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`, then F' is a weak Euclidean gradient of v'.*

Proof. Step 1: γ_n-null sets are Lebesgue-null sets. By `GaussMeasure`, $\gamma_n(A) = \int_A \rho\,dx$ for every Borel $A \subseteq \mathbb{R}^n$, where $\rho(x) = \pi^{-n/2}e^{-|x|^2}$; the function ρ is Borel measurable and satisfies $\rho(x) > 0$ for every x, since $\pi^{-n/2} > 0$ and the real exponential is strictly positive. Let $A \subseteq \mathbb{R}^n$ be Borel with $\gamma_n(A) = 0$, i.e. $\int_A \rho\,dx = 0$. The integrand ρ is nonnegative and the integral of a nonnegative measurable function over A vanishes only if that function is 0 Lebesgue-almost everywhere on A; since $\rho > 0$ everywhere, the set where $\rho = 0$ is empty, and therefore A has Lebesgue measure 0. Consequently every property that holds γ_n-almost everywhere holds Lebesgue-almost everywhere.

Applying this to the hypotheses, we get

$$v = v' \quad \text{Lebesgue-a.e.}, \qquad F = F' \quad \text{Lebesgue-a.e.} \tag{1}$$

In particular, for each fixed $j \in \{1,\dots,n\}$ the jth coordinates satisfy $F_j = F'_j$ Lebesgue-almost everywhere, because equality of the vectors at a point implies equality of their coordinates at that point.

Step 2: the test identities agree. Fix $j \in \{1,\dots,n\}$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$. By `HasGaussWeakGrad` applied to the pair (v, F),

$$\int_{\mathbb{R}^n} v\,\partial_j\varphi\,dx = -\int_{\mathbb{R}^n} F_j\,\varphi\,dx, \tag{2}$$

both integrals being with respect to Lebesgue measure. By (1) the integrands $v\,\partial_j\varphi$ and $v'\,\partial_j\varphi$ agree Lebesgue-almost everywhere, and likewise $F_j\,\varphi$ and $F'_j\,\varphi$ agree Lebesgue-almost everywhere; functions agreeing almost everywhere have equal integrals, so

$$\int_{\mathbb{R}^n} v'\,\partial_j\varphi\,dx = \int_{\mathbb{R}^n} v\,\partial_j\varphi\,dx, \qquad \int_{\mathbb{R}^n} F'_j\,\varphi\,dx = \int_{\mathbb{R}^n} F_j\,\varphi\,dx.$$

Substituting both into (2) gives

$$\int_{\mathbb{R}^n} v'\,\partial_j\varphi\,dx = -\int_{\mathbb{R}^n} F'_j\,\varphi\,dx.$$

Since j and φ were arbitrary, this is exactly the defining condition of `HasGaussWeakGrad` for the pair (v', F'), i.e. F' is a weak Euclidean gradient of v'. □

2.53. `GaussDWitness`.

Lemma 2.53 (Characterization of the chosen energy-normalized gradient)**.** *Let $v \in L^2(\gamma_n)$ and let $\Gamma \in L^2(\gamma_n;\mathbb{R}^n)$, and suppose $\sqrt{2}\,\Gamma$ is a weak Euclidean gradient of v in the sense of* `HasGaussWeakGrad`*. Then $v \in W^{1,2}(\gamma_n)$ in the sense of* `SobolevW12`*, and $Dv = \Gamma$ with D the operator of* `GaussD`*. The conjunction is the whole content: by* `GaussD` *the value Dv is set to 0 by convention when $v \notin W^{1,2}(\gamma_n)$, so the equality $Dv = \Gamma$ on its own asserts nothing about Γ unless the membership is also asserted.*

Proof. The two conclusions are proved from different inputs: the membership comes from the existential form of `SobolevW12`, which the hypothesis witnesses directly, while the identification $Dv = \Gamma$ requires `GaussWeakGradUnique`, which is what upgrades "Γ is *a* field with the defining property" to "Γ is *the* field with it".

Step 1: membership. By `SobolevW12`, $v \in W^{1,2}(\gamma_n)$ means precisely that $v \in L^2(\gamma_n)$ and that there exists some $\Gamma' \in L^2(\gamma_n;\mathbb{R}^n)$ for which $\sqrt{2}\,\Gamma'$ is a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`. The hypotheses of the present lemma supply $v \in L^2(\gamma_n)$ and exhibit $\Gamma' = \Gamma$ as such a witness. Hence $v \in W^{1,2}(\gamma_n)$.

Step 2: the selected gradient is itself a weak gradient. By `GaussD`, for $v \in W^{1,2}(\gamma_n)$ the element Dv of $L^2(\gamma_n;\mathbb{R}^n)$ is characterized by the property that $\sqrt{2}\,(Dv)$ is a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`. Since Step 1 has established $v \in W^{1,2}(\gamma_n)$, this property is available:

$$\sqrt{2}\,(Dv) \text{ is a weak Euclidean gradient of } v. \tag{1}$$

In reading (1) and the hypothesis we are free to replace the pointwise scalar multiple $x \mapsto \sqrt{2}\,(Dv)(x)$ by the element $\sqrt{2}\,(Dv)$ of $L^2(\gamma_n;\mathbb{R}^n)$, and likewise for Γ: the two agree γ_n-almost everywhere, and by `GaussWeakGradAECongr` the weak-gradient relation is unchanged when either argument is replaced by a γ_n-almost-everywhere equal function.

Step 3: uniqueness. Both $\sqrt{2}\,(Dv)$ and $\sqrt{2}\,\Gamma$ lie in $L^2(\gamma_n;\mathbb{R}^n)$, being scalar multiples of elements of that space, and by (1) and the hypothesis both are weak Euclidean gradients of the same function v. Therefore `GaussWeakGradUnique`, applied to the function v and to the two fields $\sqrt{2}\,(Dv)$ and $\sqrt{2}\,\Gamma$, gives

$$\sqrt{2}\,(Dv) = \sqrt{2}\,\Gamma \qquad \text{in } L^2(\gamma_n;\mathbb{R}^n).$$

Since $\sqrt{2} \neq 0$ and $L^2(\gamma_n;\mathbb{R}^n)$ is a real vector space, scalar multiplication by $\sqrt{2}$ is injective on it, so $Dv = \Gamma$. □

2.54. `HermiteGradGram`.

Lemma 2.54 (Coordinates and Gram matrix of the Hermite gradients)**.** *Fix $n \in \mathbb{N}$ and let $D = 2^{-1/2}\nabla$ be the operator of* `GaussD`*. Then both of the following hold.*

(i) (Membership and coordinate formula.) *For every multi-index $\alpha \in \mathbb{N}_0^n$, the Hermite function h_α of* `HermiteL2`*, viewed as an element of $L^2(\gamma_n)$, lies in $W^{1,2}(\gamma_n)$ in the sense of* `SobolevW12`*, and its energy-normalized gradient satisfies*

$$Dh_\alpha(x) = \big(\sqrt{\alpha_1}\,h_{\alpha-e_1}(x), \dots, \sqrt{\alpha_n}\,h_{\alpha-e_n}(x)\big) \qquad \text{for } \gamma_n\text{-almost every } x \in \mathbb{R}^n,$$

with h_β the Hermite function of `HermiteFun` *and $\alpha - e_j$ the multi-index obtained by decreasing the jth entry of α by one, under the convention of* `MultiIndexSum` *that this entry stays at 0 when $\alpha_j = 0$.*

(ii) (Gram matrix.) *For all multi-indices $\alpha, \beta \in \mathbb{N}_0^n$,*

$$\langle Dh_\alpha,\, Dh_\beta\rangle_{L^2(\gamma_n;\mathbb{R}^n)} = \begin{cases} |\alpha|, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

where $|\alpha| = \alpha_1 + \cdots + \alpha_n$ is the order of `MultiIndexSum`*.*

The equality in (i) is an identity γ_n-almost everywhere, as it must be: Dh_α is an element of $L^2(\gamma_n;\mathbb{R}^n)$, that is, an equivalence class, while the right-hand side is a genuine function. Both assertions are stated directly about Dh_α, and not about some auxiliary L^2 representative of the

right-hand side; it is this that lets (ii) be used without carrying the membership proof of (i) around with it.

Proof. Throughout, write

$$F_\alpha(x) = \big(\sqrt{\alpha_1}\, h_{\alpha-e_1}(x), \dots, \sqrt{\alpha_n}\, h_{\alpha-e_n}(x)\big) \in \mathbb{R}^n, \qquad x \in \mathbb{R}^n,$$

for the field appearing on the right of (i). The proof establishes (i) by exhibiting $\sqrt{2}\, F_\alpha$ as a weak Euclidean gradient of h_α and appealing to `GaussDWitness`, and then reads (ii) off the coordinate formula.

Step 1: $F_\alpha \in L^2(\gamma_n; \mathbb{R}^n)$. Each coordinate of F_α is a constant multiple of a Hermite function, which is continuous by `HermiteFunContDiff`, so F_α is continuous and in particular measurable. For the square-integrability, bound the Euclidean norm by the sum of the absolute values of the coordinates: for every x,

$$|F_\alpha(x)| = \Big(\sum_{j=1}^n \big|\sqrt{\alpha_j}\, h_{\alpha-e_j}(x)\big|^2\Big)^{1/2} \le \sum_{j=1}^n \big|\sqrt{\alpha_j}\, h_{\alpha-e_j}(x)\big|,$$

the inequality because $\sum_j a_j^2 \le (\sum_j a_j)^2$ for nonnegative reals a_j and the square root is monotone. By `HermiteFunMemLp` each $h_{\alpha-e_j}$ lies in $L^2(\gamma_n)$, hence so does each $|\sqrt{\alpha_j}\, h_{\alpha-e_j}|$, and hence so does the finite sum on the right. A measurable function dominated pointwise in absolute value by an $L^2(\gamma_n)$ function is itself in $L^2(\gamma_n)$, so $F_\alpha \in L^2(\gamma_n; \mathbb{R}^n)$.

Step 2: $\sqrt{2}\, F_\alpha$ is a weak Euclidean gradient of h_α. Fix j and a test function $\varphi \in C_c^\infty(\mathbb{R}^n)$. By `HermiteFunDeriv`,

$$\partial_j h_\alpha(x) = \sqrt{2}\,\sqrt{\alpha_j}\, h_{\alpha-e_j}(x) = \sqrt{2}\,\big(F_\alpha(x)\big)_j \qquad (x \in \mathbb{R}^n), \tag{1}$$

and h_α is differentiable at every point. By `HermiteFunContDiff`, h_α and $\partial_j h_\alpha$ are continuous, and φ and $\partial_j \varphi$ are continuous with compact support; hence each of the three products $(\partial_j h_\alpha)\varphi$, $h_\alpha\, \partial_j \varphi$ and $h_\alpha \varphi$ is continuous with compact support, and therefore Lebesgue integrable on $\mathbb{R}^n$. With h_α and φ differentiable everywhere and these three integrability conditions in hand, the integration-by-parts formula on $\mathbb{R}^n$ in the direction e_j applies and gives

$$\int_{\mathbb{R}^n} h_\alpha(x)\, \partial_j \varphi(x)\, dx = -\int_{\mathbb{R}^n} \partial_j h_\alpha(x)\, \varphi(x)\, dx = -\int_{\mathbb{R}^n} \sqrt{2}\,\big(F_\alpha(x)\big)_j\, \varphi(x)\, dx,$$

the last equality by (1). As j and φ were arbitrary, this is exactly the defining identity of `HasGaussWeakGrad` for the pair $(h_\alpha, \sqrt{2}\, F_\alpha)$. Note that these are Lebesgue integrals, as `HasGaussWeakGrad` requires; no Gaussian weight enters at this step, and none is needed, since all three integrands have compact support.

Step 3: proof of (i). By Step 1 the field F_α determines an element of $L^2(\gamma_n; \mathbb{R}^n)$, which we again call F_α; it agrees γ_n-almost everywhere with the function F_α, so by `GaussWeakGradAECongr` the conclusion of Step 2 holds for that element as well. The same applies on the other side: h_α as an element of $L^2(\gamma_n)$ via `HermiteL2` agrees γ_n-almost everywhere with the function h_α of `HermiteFun`. Thus $\sqrt{2}\, F_\alpha$ is a weak Euclidean gradient of the $L^2(\gamma_n)$ element h_α, and `GaussDWitness` applies and yields both

$$h_\alpha \in W^{1,2}(\gamma_n) \qquad \text{and} \qquad Dh_\alpha = F_\alpha \ \text{ in } L^2(\gamma_n; \mathbb{R}^n). \tag{2}$$

The second equality of (2) is an equality of L^2 elements, hence says that Dh_α agrees γ_n-almost everywhere with the function F_α, which is the coordinate formula of (i).

Step 4: proof of (ii). Let $\alpha, \beta \in \mathbb{N}_0^n$. The inner product of $L^2(\gamma_n; \mathbb{R}^n)$ pairs matching coordinates and integrates against γ_n, so by (2) applied to α and to β,

$$\langle Dh_\alpha, Dh_\beta \rangle = \sum_{j=1}^n \int_{\mathbb{R}^n} \big(F_\alpha(x)\big)_j \big(F_\beta(x)\big)_j\, d\gamma_n(x) = \sum_{j=1}^n \sqrt{\alpha_j}\sqrt{\beta_j}\, \langle h_{\alpha-e_j}, h_{\beta-e_j} \rangle, \tag{3}$$

the interchange of the finite sum with the integral being legitimate because each summand is integrable: it is a product of two $L^2(\gamma_n)$ functions by `HermiteFunMemLp`. By `HermiteL2Orthonormal` the inner product in the jth summand of (3) equals 1 when $\alpha - e_j = \beta - e_j$ and 0 otherwise.

The degenerate indices must be disposed of first, because under the convention of `MultiIndexSum` — the jth entry is left at 0 when $\alpha_j = 0$ — the decremented multi-indices can coincide even for $\alpha \neq \beta$: for instance $n = 2$, $j = 1$, $\alpha = (0, 5)$, $\beta = (1, 5)$ give $\alpha - e_1 = \beta - e_1 = (0, 5)$. So it is the prefactor, not the Gram entry, that must kill those summands.

(4a) If $\alpha_j = 0$ or $\beta_j = 0$, the jth summand of (3) is 0. Its prefactor is $\sqrt{\alpha_j}\sqrt{\beta_j}$ and one of the two factors is $\sqrt{0} = 0$, so the prefactor is 0; the Gram entry is a real number, hence the product vanishes whatever that entry is.

(4b) If $\alpha_j \geq 1$ and $\beta_j \geq 1$, then $\alpha - e_j = \beta - e_j$ forces $\alpha = \beta$. By `MultiIndexSum` the multi-index $\alpha - e_j$ agrees with α at every entry $i \neq j$ and has jth entry $\alpha_j - 1$, and likewise for β; the convention plays no role here, since both jth entries are at least 1 and the decrement is therefore genuine. So $\alpha - e_j = \beta - e_j$ gives $\alpha_i = \beta_i$ for every $i \neq j$ together with $\alpha_j - 1 = \beta_j - 1$, and the latter yields $\alpha_j = \beta_j$ because both α_j and β_j are at least 1. Hence $\alpha = \beta$.

Suppose now $\alpha \neq \beta$, and let j be arbitrary. If $\alpha_j = 0$ or $\beta_j = 0$, the jth summand is 0 by (4a). Otherwise $\alpha_j \geq 1$ and $\beta_j \geq 1$, and then $\alpha - e_j \neq \beta - e_j$ — equality would force $\alpha = \beta$ by (4b), contradicting $\alpha \neq \beta$ — so the Gram entry is 0 and the jth summand is 0 again. Every summand of (3) vanishes, so $\langle Dh_\alpha, Dh_\beta \rangle = 0$.

Suppose instead $\alpha = \beta$. The jth summand of (3) is $\sqrt{\alpha_j}\sqrt{\alpha_j}\,\langle h_{\alpha - e_j}, h_{\alpha - e_j} \rangle$, and this equals α_j: if $\alpha_j \geq 1$ the two decremented indices are literally the same multi-index, so the Gram entry is 1 and $\sqrt{\alpha_j}\sqrt{\alpha_j} = \alpha_j$; if $\alpha_j = 0$ the summand is 0 by (4a) and $\alpha_j = 0$ as well. Summing over j and using `MultiIndexSum` gives $\sum_{j=1}^{n} \alpha_j = |\alpha|$. This is (ii). □

2.55. `GaussDLinear`.

Lemma 2.55 (D is linear on $W^{1,2}(\gamma_n)$ and annihilates constants)**.** *Let $W^{1,2}(\gamma_n)$ be as in `SobolevW12` and D as in `GaussD`. Then:*

(1) *for all $u, v \in W^{1,2}(\gamma_n)$, also $u + v \in W^{1,2}(\gamma_n)$ and $D(u + v) = Du + Dv$;*
(2) *for every $c \in \mathbb{R}$ and $u \in W^{1,2}(\gamma_n)$, also $c\,u \in W^{1,2}(\gamma_n)$ and $D(c\,u) = c\,Du$;*
(3) *for every $c \in \mathbb{R}$ the constant function $c\mathbf{1} = c\,h_0$ of `HermiteL2` and `Hermite1Zero` lies in $W^{1,2}(\gamma_n)$ and $D(c\mathbf{1}) = 0$.*

Proof. Throughout, recall from `SobolevW12` that $w \in W^{1,2}(\gamma_n)$ means precisely that there is $\Gamma \in L^2(\gamma_n; \mathbb{R}^n)$ such that $\sqrt{2}\,\Gamma$ is a weak Euclidean gradient of w in the sense of `HasGaussWeakGrad`, and from `GaussD` that Dw is then that Γ; it is unique as an element of $L^2(\gamma_n; \mathbb{R}^n)$ by `GaussWeakGradUnique`. Note also that the defining identity of `HasGaussWeakGrad`, namely

$$\int_{\mathbb{R}^n} w\,\partial_j\varphi\,dx = -\int_{\mathbb{R}^n} G_j\,\varphi\,dx \qquad (j = 1, \dots, n,\ \varphi \in C_c^\infty(\mathbb{R}^n)),$$

is, for each fixed j and φ, linear in the pair (w, G): both sides are linear in (w, G) separately.

(1) Let $u, v \in W^{1,2}(\gamma_n)$ and put $G = \sqrt{2}\,Du$, $H = \sqrt{2}\,Dv$, the weak Euclidean gradients of u and v. Then $G + H \in L^2(\gamma_n; \mathbb{R}^n)$, and adding the two defining identities for the same j and φ gives

$$\int (u + v)\,\partial_j\varphi\,dx = -\int (G_j + H_j)\,\varphi\,dx,$$

so $G + H$ is a weak Euclidean gradient of $u + v$. Writing $G + H = \sqrt{2}\,(Du + Dv)$ with $Du + Dv \in L^2(\gamma_n; \mathbb{R}^n)$, the criterion of `SobolevW12` is met with the witness $Du + Dv$, so $u + v \in W^{1,2}(\gamma_n)$; and since by `GaussD` the element $D(u + v)$ is a witness too, `GaussWeakGradUnique` applied to the function $u + v$ and the two fields $\sqrt{2}\,D(u + v)$ and $\sqrt{2}(Du + Dv)$ gives $D(u + v) = Du + Dv$.

(2) Identical, multiplying the single defining identity for u by the scalar c: $c\,G = \sqrt{2}\,(c\,Du)$ is a weak Euclidean gradient of $c\,u$, so $c\,u \in W^{1,2}(\gamma_n)$ with witness $c\,Du$, and `GaussWeakGradUnique` gives $D(c\,u) = c\,Du$.

(3) By `Hermite1Zero` and `HermiteL2`, h_0 is the constant function 1, so $c\,h_0$ is the constant function c; it lies in $L^2(\gamma_n)$ because γ_n is a probability measure (`GaussIsProb`). We claim the zero field is a weak Euclidean gradient of it. Indeed, let $j \in \{1, \dots, n\}$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$. Then $\partial_j\varphi$ is smooth with compact support, and integrating in the x_j variable first (Fubini applies, the

integrand being continuous with compact support, hence absolutely integrable) the fundamental theorem of calculus gives

$$\int_{\mathbb{R}} \partial_j \varphi(x)\, dx_j = 0 \qquad \text{for every fixed value of the other coordinates,}$$

because φ vanishes outside a compact set, so the inner integral is $\varphi\big|_{x_j=+\infty} - \varphi\big|_{x_j=-\infty} = 0$. Hence $\int_{\mathbb{R}^n} \partial_j \varphi\, dx = 0$ and therefore

$$\int_{\mathbb{R}^n} c\, \partial_j \varphi\, dx = 0 = -\int_{\mathbb{R}^n} 0 \cdot \varphi\, dx,$$

which is the defining identity of `HasGaussWeakGrad` for the function $c\mathbf{1}$ and the field $G = 0$. Since $0 = \sqrt{2} \cdot 0$ with $0 \in L^2(\gamma_n; \mathbb{R}^n)$, the criterion of `SobolevW12` holds with witness 0, so $c\mathbf{1} \in W^{1,2}(\gamma_n)$; and `GaussWeakGradUnique`, applied to $c\mathbf{1}$ and the two fields $\sqrt{2}\, D(c\mathbf{1})$ and 0, gives $D(c\mathbf{1}) = 0$. □

2.56. `GaussWeakGradClosed`.

Lemma 2.56 (Closedness of the weak gradient under $L^2(\gamma_n)$ limits)**.** *Let $v_k \to v$ in $L^2(\gamma_n)$ and $G_k \to G$ in $L^2(\gamma_n; \mathbb{R}^n)$, and suppose that for every k the field G_k is a weak Euclidean gradient of v_k in the sense of `HasGaussWeakGrad`. Then G is a weak Euclidean gradient of v.*

Proof. Fix $j \in \{1, \dots, n\}$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$, and let $K = \operatorname{supp} \varphi$, a compact set. By hypothesis and `HasGaussWeakGrad`, for every k,

$$\int_{\mathbb{R}^n} v_k\, \partial_j \varphi\, dx = -\int_{\mathbb{R}^n} (G_k)_j\, \varphi\, dx. \tag{1}$$

It suffices to pass to the limit on both sides, since the limiting identity is exactly the required one for the pair (v, G).

Step 1: a uniform local estimate. By `GaussMeasure` the density $\rho(x) = \pi^{-n/2} e^{-|x|^2}$ is continuous and strictly positive, so $c_K := \min_K \rho > 0$. Let ψ be any bounded measurable function supported in K, and let $w \in L^2(\gamma_n)$. Then

$$\Big|\int w\, \psi\, dx\Big| \le \|\psi\|_\infty \int_K |w|\, dx \le \frac{\|\psi\|_\infty}{c_K} \int_K |w|\, \rho\, dx = \frac{\|\psi\|_\infty}{c_K} \int_K |w|\, d\gamma_n \le \frac{\|\psi\|_\infty}{c_K}\, \|w\|_{L^2(\gamma_n)},$$

where the last step is `L2IntoL1` together with $\int_K |w|\, d\gamma_n \le \int |w|\, d\gamma_n = \|w\|_{L^1(\gamma_n)}$, valid because γ_n is a probability measure (`GaussIsProb`).

Step 2: the left-hand sides converge. Apply Step 1 with $\psi = \partial_j \varphi$, which is bounded and supported in K, and $w = v_k - v \in L^2(\gamma_n)$:

$$\Big|\int v_k\, \partial_j \varphi\, dx - \int v\, \partial_j \varphi\, dx\Big| \le \frac{\|\partial_j \varphi\|_\infty}{c_K}\, \|v_k - v\|_{L^2(\gamma_n)} \xrightarrow[k\to\infty]{} 0,$$

since $v_k \to v$ in $L^2(\gamma_n)$.

Step 3: the right-hand sides converge. Apply Step 1 with $\psi = \varphi$ and $w = (G_k)_j - G_j$. The jth coordinate of a field is dominated by its Euclidean norm, so $\|(G_k)_j - G_j\|_{L^2(\gamma_n)} \le \|G_k - G\|_{L^2(\gamma_n; \mathbb{R}^n)}$, whence

$$\Big|\int (G_k)_j\, \varphi\, dx - \int G_j\, \varphi\, dx\Big| \le \frac{\|\varphi\|_\infty}{c_K}\, \|G_k - G\|_{L^2(\gamma_n; \mathbb{R}^n)} \xrightarrow[k\to\infty]{} 0.$$

Step 4: conclusion. Letting $k \to \infty$ in (1) and using Steps 2–3,

$$\int_{\mathbb{R}^n} v\, \partial_j \varphi\, dx = -\int_{\mathbb{R}^n} G_j\, \varphi\, dx.$$

As $j \in \{1, \dots, n\}$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$ were arbitrary, this is precisely the assertion that G is a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`. □

2.57. `HermiteZeroModeConstant`.

Lemma 2.57 (The zeroth Hermite mode is the constant function 1)**.** *Fix $n \in \mathbb{N}$ and write h_0 for the Hermite function of index $\alpha = 0$, regarded through `HermiteL2` as an element of $L^2(\gamma_n)$, with γ_n the Gaussian measure of `GaussMeasure`. Then all of the following hold.*

(1) *For every function $f : \mathbb{R}^n \to \mathbb{R}$,*
$$\int_{\mathbb{R}^n} f(x)\, h_0(x)\, d\gamma_n(x) = \int_{\mathbb{R}^n} f(x)\, d\gamma_n(x).$$
(2) *For every $v \in L^2(\gamma_n)$, $\langle v, h_0\rangle = \int_{\mathbb{R}^n} v\, d\gamma_n$.*
(3) *For every $v \in L^2(\gamma_n)$, $\langle h_0, v\rangle = \int_{\mathbb{R}^n} v\, d\gamma_n$.*
(4) $\langle h_0, h_0\rangle = 1$.
(5) $\|h_0\|_{L^2(\gamma_n)} = 1$.

Here $\langle\cdot,\cdot\rangle$ is the real inner product of $L^2(\gamma_n)$. In short: h_0 is the constant function 1, so integrating against it is integrating against γ_n, and it is a unit vector.

Two points about the form of the statement. First, h_0 is an element of $L^2(\gamma_n)$, that is, an equivalence class, so the identity $h_0 = 1$ holds only γ_n-almost everywhere; every clause above is therefore obtained from a congruence of integrands almost everywhere, and never from a pointwise substitution. Clause (1) carries no integrability hypothesis on f for exactly this reason: it is the equality of the integrals of two functions that agree γ_n-almost everywhere, which holds whatever the integrability status of f. Second, (2) and (3) differ only by the symmetry of the real inner product and are both recorded because consumers meet the pairing in both argument orders.

Proof. By `HermiteL2ZeroAEOne`,
$$(*) \qquad h_0(x) = 1 \qquad \text{for } \gamma_n\text{-almost every } x \in \mathbb{R}^n.$$
This is the only input about h_0 used below; everything else is formal.

Clause (1). Let $f : \mathbb{R}^n \to \mathbb{R}$. By $(*)$ the two integrands agree almost everywhere: for γ_n-almost every x we have $f(x)\, h_0(x) = f(x)\cdot 1 = f(x)$. Integrals of functions that agree γ_n-almost everywhere are equal, so $\int f\, h_0\, d\gamma_n = \int f\, d\gamma_n$. Note that this is a congruence of integrands and not a rewriting of h_0 by 1 at a point, which $(*)$ does not license.

Clause (2). Let $v \in L^2(\gamma_n)$. By `L2InnerGaussIntegral` the inner product of two elements of $L^2(\gamma_n)$ is the γ_n-integral of their product, so
$$\langle v, h_0\rangle = \int_{\mathbb{R}^n} v(x)\, h_0(x)\, d\gamma_n(x) = \int_{\mathbb{R}^n} v\, d\gamma_n,$$
the second equality being clause (1) applied to $f = v$.

Clause (3). The inner product of the real Hilbert space $L^2(\gamma_n)$ is symmetric, so $\langle h_0, v\rangle = \langle v, h_0\rangle$, and clause (2) finishes.

Clause (4). Apply clause (2) with $v = h_0$, which is legitimate since $h_0 \in L^2(\gamma_n)$:
$$\langle h_0, h_0\rangle = \int_{\mathbb{R}^n} h_0\, d\gamma_n.$$
By $(*)$ again the integrand agrees γ_n-almost everywhere with the constant function 1, so this integral equals $\int_{\mathbb{R}^n} 1\, d\gamma_n = \gamma_n(\mathbb{R}^n)$, which is 1 because γ_n is a probability measure by `GaussIsProb`. Hence $\langle h_0, h_0\rangle = 1$.

Clause (5). In a real inner product space $\langle w, w\rangle = \|w\|^2$, so clause (4) gives $\|h_0\|^2 = 1$. Since $\|h_0\| \geq 0$ and the only nonnegative real whose square is 1 is 1 itself, $\|h_0\| = 1$. □

2.58. `FracPowLinear`.

Lemma 2.58 (L^s is linear on its domain, and the domain is everything for $s \leq 0$)**.** *Let $s \in \mathbb{R}$, let $\mathrm{Dom}(L^s)$ be as in `FracPowDomain` and let L^s be as in `FracPow`. Then:*

(1) *if $s \leq 0$, then every $v \in L^2(\gamma_n)$ lies in $\mathrm{Dom}(L^s)$;*

(2) *for all* $v, w \in \mathrm{Dom}(L^s)$*, also* $v + w \in \mathrm{Dom}(L^s)$ *and* $L^s(v + w) = L^s v + L^s w$*;*
(3) *for every* $c \in \mathbb{R}$ *and* $v \in \mathrm{Dom}(L^s)$*, also* $c\,v \in \mathrm{Dom}(L^s)$ *and* $L^s(c\,v) = c\,L^s v$*.*

Proof. Throughout we use the following three facts, in the form in which they are stated.

(a) The coefficient map. By `HermiteBasis` the map $v \mapsto (v_\alpha)_\alpha$, $v_\alpha = \langle v, h_\alpha \rangle$, is an isometric isomorphism of $L^2(\gamma_n)$ onto $\ell^2(\mathbb{N}_0^n)$. Being an isomorphism of real Hilbert spaces it is in particular linear and injective, so for all $v, w \in L^2(\gamma_n)$ and $c \in \mathbb{R}$

$$(v + w)_\alpha = v_\alpha + w_\alpha, \qquad (c\,v)_\alpha = c\,v_\alpha \qquad (\alpha \in \mathbb{N}_0^n),$$

every $v \in L^2(\gamma_n)$ has $\sum_\alpha |v_\alpha|^2 = \|v\|_2^2 < \infty$, and two elements of $L^2(\gamma_n)$ with the same coefficient family are equal.

(b) The domain criterion. By `FracPowDomain`, $v \in \mathrm{Dom}(L^s)$ holds precisely when the multiplied family $m_s\big((v_\alpha)_\alpha\big)$ of `FracPowCoeff` is square summable, where by `FracPowCoeff`

$$m_s(c)_\alpha = \begin{cases} 0, & \alpha = 0, \\ |\alpha|^s c_\alpha, & \alpha \neq 0. \end{cases}$$

(c) The value. By `FracPow`, for $v \in \mathrm{Dom}(L^s)$ the element $L^s v$ is the unique element of $L^2(\gamma_n)$ whose coefficient family is $m_s\big((v_\alpha)_\alpha\big)$.

Proof of (1). Let $s \leq 0$ and $v \in L^2(\gamma_n)$. Fix $\alpha \neq 0$. By `MultiIndexSum` we have $|\alpha| \geq 1$, so $\log|\alpha| \geq 0$ and hence, since $s \leq 0$,

$$|\alpha|^s = e^{s \log|\alpha|} \leq e^0 = 1, \qquad \text{and also } |\alpha|^s > 0.$$

Therefore, by (b),

$$\big|m_s\big((v_\alpha)_\alpha\big)_\alpha\big| = |\alpha|^s\,|v_\alpha| \leq |v_\alpha| \qquad (\alpha \neq 0),$$

while for $\alpha = 0$ the left-hand side is $0 \leq |v_0|$. Thus the family $m_s((v_\alpha)_\alpha)$ is dominated in absolute value, term by term, by the family $(v_\alpha)_\alpha$, whence

$$\sum_\alpha \big|m_s\big((v_\alpha)_\alpha\big)_\alpha\big|^2 \leq \sum_\alpha |v_\alpha|^2 < \infty$$

by (a) and the comparison test for series of nonnegative terms. So $m_s((v_\alpha)_\alpha)$ is square summable, and $v \in \mathrm{Dom}(L^s)$ by (b).

Proof of (2). Let $v, w \in \mathrm{Dom}(L^s)$. We first check that m_s is additive on coefficient families. Indeed, using $(v + w)_\alpha = v_\alpha + w_\alpha$ from (a), at $\alpha = 0$ both $m_s((v + w)_\alpha)_0 = 0$ and $m_s((v_\alpha))_0 + m_s((w_\alpha))_0 = 0 + 0 = 0$, while for $\alpha \neq 0$

$$m_s\big(((v + w)_\alpha)_\alpha\big)_\alpha = |\alpha|^s(v_\alpha + w_\alpha) = |\alpha|^s v_\alpha + |\alpha|^s w_\alpha = m_s\big((v_\alpha)_\alpha\big)_\alpha + m_s\big((w_\alpha)_\alpha\big)_\alpha.$$

Hence

$$(*) \qquad m_s\big(((v + w)_\alpha)_\alpha\big) = m_s\big((v_\alpha)_\alpha\big) + m_s\big((w_\alpha)_\alpha\big)$$

as families indexed by $\mathbb{N}_0^n$. Both summands on the right are square summable by (b), because $v, w \in \mathrm{Dom}(L^s)$; and a sum of two square-summable families is square summable, since $|a+b|^2 \leq 2|a|^2 + 2|b|^2$ termwise. So the left-hand side of $(*)$ is square summable, i.e. $v + w \in \mathrm{Dom}(L^s)$ by (b).

It remains to identify $L^s(v + w)$. By (c) the coefficient family of $L^s(v + w)$ is the left-hand side of $(*)$, and the coefficient families of $L^s v$ and $L^s w$ are the two summands on the right. By linearity of the coefficient map (a), the coefficient family of $L^s v + L^s w$ is the sum of those two, i.e. the right-hand side of $(*)$. So $L^s(v + w)$ and $L^s v + L^s w$ are two elements of $L^2(\gamma_n)$ with the same coefficient family; by the injectivity in (a) they are equal.

Proof of (3). Let $c \in \mathbb{R}$ and $v \in \mathrm{Dom}(L^s)$. Using $(c\,v)_\alpha = c\,v_\alpha$ from (a), at $\alpha = 0$ both $m_s(((cv)_\alpha))_0 = 0$ and $c\,m_s((v_\alpha))_0 = c \cdot 0 = 0$, while for $\alpha \neq 0$

$$m_s\big(((c\,v)_\alpha)_\alpha\big)_\alpha = |\alpha|^s(c\,v_\alpha) = c\,\big(|\alpha|^s v_\alpha\big) = c\,m_s\big((v_\alpha)_\alpha\big)_\alpha.$$

Hence

$$(**) \qquad m_s\big(((c\,v)_\alpha)_\alpha\big) = c\,m_s\big((v_\alpha)_\alpha\big)$$

as families. The right-hand side is square summable, because $\sum_\alpha |c\,x_\alpha|^2 = c^2 \sum_\alpha |x_\alpha|^2$ and $m_s((v_\alpha)_\alpha)$ is square summable by (b). So $c\,v \in \mathrm{Dom}(L^s)$ by (b). Finally, by (c) the coefficient family of $L^s(c\,v)$ is the left-hand side of $(**)$, and by (c) together with the linearity in (a) the coefficient family of $c\,L^s v$ is the right-hand side of $(**)$; the two agree, so the injectivity in (a) gives $L^s(c\,v) = c\,L^s v$. □

2.59. `ProjZeroCentering`.

Lemma 2.59 (The spectral projection removes exactly the Gaussian mean)**.** *Write* $\mathfrak{m}(v) = \int v\,d\gamma_n$ *for* $v \in L^2(\gamma_n)$*. Then for every* $v \in L^2(\gamma_n)$*,*

$$\Pi_0 v = v - \mathfrak{m}(v)\,h_0 \qquad \text{in } L^2(\gamma_n),$$

where $\Pi_0 = L^0$ *is the operator of* `FracPow` *and* $h_0 \in L^2(\gamma_n)$ *is the Hermite element of index* 0 *from* `HermiteL2`*. Since* h_0 *is the constant function* 1 *by* `Hermite1Zero`*, this is the identity* $\Pi_0 v = v - \mathfrak{m}(v)\mathbf{1}$ *of the reference paper.*

Proof. Both sides are elements of $L^2(\gamma_n)$. By `HermiteBasis` the Hermite coefficient map $w \mapsto (w_\beta)_\beta$, $w_\beta = \langle w, h_\beta\rangle$, is an injective linear isometry onto $\ell^2(\mathbb{N}_0^n)$; so it suffices to check that the two sides have the same coefficient at every $\beta \in \mathbb{N}_0^n$.

Step 1: the zeroth coefficient of v *is its mean.* By `Hermite1Zero` the function h_0 is the constant 1, and γ_n is a probability measure by `GaussIsProb`. Hence

$$v_0 = \langle v, h_0\rangle = \int v \cdot 1\,d\gamma_n = \mathfrak{m}(v),$$

the integral being finite because $v \in L^2(\gamma_n) \subseteq L^1(\gamma_n)$ by `L2IntoL1`.

Step 2: the coefficients of the right-hand side. By `HermiteL2Orthonormal`, $\langle h_0, h_\beta\rangle = 1$ for $\beta = 0$ and $= 0$ for $\beta \neq 0$. By linearity of the coefficient map, the element $v - \mathfrak{m}(v)h_0$ therefore has coefficients

$$\big(v - \mathfrak{m}(v)h_0\big)_0 = v_0 - \mathfrak{m}(v) = 0, \qquad \big(v - \mathfrak{m}(v)h_0\big)_\beta = v_\beta \quad (\beta \neq 0),$$

using Step 1 for the first equality.

Step 3: the coefficients of the left-hand side. By `FracPowCoeff` with $s = 0$, the multiplied coefficient family is

$$m_0\big((v_\beta)_\beta\big)_\beta = \begin{cases} 0, & \beta = 0, \\ |\beta|^0 v_\beta = v_\beta, & \beta \neq 0. \end{cases}$$

This family is square summable, being dominated termwise by $(|v_\beta|^2)_\beta$, which is summable with sum $\|v\|_2^2$ by Parseval (`HermiteBasis`). By `FracPowDomain` with $s = 0$ this says $v \in \mathrm{Dom}(L^0)$, so the defining formula of `FracPow` applies and gives $\Pi_0 v$ precisely those coefficients: 0 at $\beta = 0$ and v_β at $\beta \neq 0$.

Steps 2 and 3 produce the same coefficient family, so the two elements of $L^2(\gamma_n)$ coincide. □

2.60. `RieszSqrtIdentity`.

Lemma 2.60 (The structural identity $R_\gamma L^{1/2} u = Du$)**.** *For every* $u \in \mathrm{Dom}(L^{1/2})$ *in the sense of* `FracPowDomain`*,*

$$R_\gamma\big(L^{1/2}u\big) = Du \qquad \text{in } L^2(\gamma_n; \mathbb{R}^n),$$

with R_γ *as in* `RieszTransform`*,* $L^{1/2}$ *as in* `FracPow` *and* D *as in* `GaussD`*. This is the identity that the endpoint argument exploits: applying* R_γ *to the bad part* $L^{1/2}u$ *of a decomposition produces the ordinary gradient of the potential* u*.*

Proof. Write u_α for the Hermite coefficients of u relative to `HermiteBasis`, and put $w = L^{1/2}u$.

Step 1: the coefficients of w*.* Since $u \in \mathrm{Dom}(L^{1/2})$, the formula of `FracPow` with $s = 1/2$ applies and gives

$$w_0 = 0, \qquad w_\alpha = |\alpha|^{1/2}\,u_\alpha \quad (\alpha \neq 0).$$

Step 2: applying the pseudoinverse to w. By `FracPowCoeff` with $s = -1/2$ the multiplied family for w is 0 at $\alpha = 0$ and, for $\alpha \neq 0$,

$$|\alpha|^{-1/2}\, w_\alpha = |\alpha|^{-1/2}|\alpha|^{1/2}u_\alpha = u_\alpha,$$

using $|\alpha| \geq 1 > 0$ from `MultiIndexSum`. This family is square summable, being dominated by $(|u_\alpha|^2)_\alpha$ which sums to $\|u\|_2^2$ by Parseval (`HermiteBasis`); so $w \in \mathrm{Dom}(L^{-1/2})$ by `FracPowDomain` and, by `FracPow`, $L^{-1/2}\Pi_0 w$ has coefficients 0 at $\alpha = 0$ and u_α at $\alpha \neq 0$.

By `FracPowCoeff` and `FracPow` with $s = 0$, those are exactly the coefficients of $\Pi_0 u$, and the coefficient map of `HermiteBasis` is injective, so

$$L^{-1/2}\Pi_0\big(L^{1/2}u\big) = \Pi_0 u.$$

By `ProjZeroCentering`, $\Pi_0 u = u - \mathfrak{m}(u)\mathbf{1}$ where $\mathfrak{m}(u) = \int u\, d\gamma_n$.

Step 3: applying D. By `RieszTransform`, $R_\gamma w = D\big(L^{-1/2}\Pi_0 w\big)$, which by Step 2 equals $D\big(u - \mathfrak{m}(u)\mathbf{1}\big)$.

Now $u \in W^{1,2}(\gamma_n)$ by `SqrtDomainSobolev`, and $-\mathfrak{m}(u)\mathbf{1} \in W^{1,2}(\gamma_n)$ with $D(-\mathfrak{m}(u)\mathbf{1}) = 0$ by part (3) of `GaussDLinear`. Part (1) of `GaussDLinear`, applied to the pair u and $-\mathfrak{m}(u)\mathbf{1}$, therefore gives $u - \mathfrak{m}(u)\mathbf{1} \in W^{1,2}(\gamma_n)$ and

$$D\big(u - \mathfrak{m}(u)\mathbf{1}\big) = Du + D\big(-\mathfrak{m}(u)\mathbf{1}\big) = Du + 0 = Du.$$

Combining with Step 3, $R_\gamma(L^{1/2}u) = Du$. □

2.61. `TransformedLocalization`.

Lemma 2.61 (Exact localization of the transformed bad part)**.** *Let $f, g \in L^2(\gamma_n)$ and $u \in \mathrm{Dom}(L^{1/2})$ satisfy $f = g + Bu$ in $L^2(\gamma_n)$ with $B = L^{1/2}$, let $t \in \mathbb{R}$, and let $\Omega \subseteq \mathbb{R}^n$ be an arbitrary subset with $Du = 0$ γ_n-a.e. on Ω^c; no measurability of Ω is required, and $\gamma_n(\Omega)$ below means the outer measure of Ω, which agrees with the measure of Ω whenever Ω is measurable. Then, with R_γ as in `RieszTransform` and D as in `GaussD`,*

$$R_\gamma f = R_\gamma g + Du \quad \gamma_n\text{-a.e.},$$

and consequently

$$\gamma_n\big(\{|R_\gamma f| > t\}\big) \;\leq\; \gamma_n(\Omega) + \gamma_n\big(\{|R_\gamma g| > t\}\big).$$

There is no splitting of the level into $t/2 + t/2$: off Ω the two outputs $R_\gamma f$ and $R_\gamma g$ are exactly equal.

Proof. Step 1: applying $L^{-1/2}\Pi_0$ to the decomposition. By `RieszTransform`, $R_\gamma w = D\big(L^{-1/2}\Pi_0 w\big)$ for every $w \in L^2(\gamma_n)$, where $L^{-1/2}\Pi_0$ is the spectral power L^s of `FracPow` at $s = -1/2$. Since $-1/2 \leq 0$, clause (1) of `FracPowLinear` applies with $s = -1/2$ and shows that every element of $L^2(\gamma_n)$ lies in $\mathrm{Dom}(L^{-1/2})$; in particular g and $Bu = L^{1/2}u$ do, both being elements of $L^2(\gamma_n)$. Clause (2) of `FracPowLinear`, applied with $s = -1/2$ to the pair (g, Bu) whose two domain hypotheses have just been verified, gives

$$L^{-1/2}\Pi_0(g + Bu) = L^{-1/2}\Pi_0 g + L^{-1/2}\Pi_0(Bu),$$

and the hypothesis $f = g + Bu$ turns the left-hand side into $L^{-1/2}\Pi_0 f$. So

$$L^{-1/2}\Pi_0 f = L^{-1/2}\Pi_0 g + L^{-1/2}\Pi_0(Bu). \tag{1}$$

Step 2: applying D to that sum. `PseudoInverseSqrt` says that $L^{-1/2}\Pi_0 w \in \mathrm{Dom}(L^{1/2})$ for every $w \in L^2(\gamma_n)$; applying it once with $w = g$ and once with $w = Bu$ gives

$$L^{-1/2}\Pi_0 g \in \mathrm{Dom}(L^{1/2}), \qquad L^{-1/2}\Pi_0(Bu) \in \mathrm{Dom}(L^{1/2}).$$

Membership in $\mathrm{Dom}(L^{1/2})$ is exactly the hypothesis `SqrtDomainSobolev` requires, so both elements lie in $W^{1,2}(\gamma_n)$. That pair of memberships is exactly what clause (1) of `GaussDLinear` requires of its two inputs, so applying it to $L^{-1/2}\Pi_0 g$ and $L^{-1/2}\Pi_0(Bu)$ and combining with (1) gives

$$R_\gamma f = D\big(L^{-1/2}\Pi_0 f\big) = D\big(L^{-1/2}\Pi_0 g\big) + D\big(L^{-1/2}\Pi_0(Bu)\big) = R_\gamma g + R_\gamma(Bu), \tag{2}$$

the first and last equalities again by `RieszTransform`. (This is the only place linearity of R_γ is used, and it has now been reduced to the linearity of its two factors; no linearity of R_γ itself is assumed.)

Step 3: identifying $R_\gamma(Bu)$. By hypothesis $u \in \mathrm{Dom}(L^{1/2})$, which is precisely the hypothesis of `RieszSqrtIdentity`; that node therefore applies to this u and gives $R_\gamma\big(L^{1/2}u\big) = Du$, i.e. $R_\gamma(Bu) = Du$. Substituting into (2),

$$R_\gamma f = R_\gamma g + Du$$

as elements of $L^2(\gamma_n;\mathbb{R}^n)$. Two elements of $L^2(\gamma_n;\mathbb{R}^n)$ that are equal are in particular equal γ_n-almost everywhere as functions, which is the first assertion.

Step 4: the measure estimate. Let

$$N = \{x \in \Omega^c : Du(x) \neq 0\} \cup \{x : R_\gamma f(x) \neq R_\gamma g(x) + Du(x)\}.$$

The first set is γ_n-null by the hypothesis that $Du = 0$ a.e. on Ω^c, and the second is γ_n-null by Step 3; so $\gamma_n(N) = 0$. Let $x \notin \Omega \cup N$. Then $Du(x) = 0$ and $R_\gamma f(x) = R_\gamma g(x) + Du(x) = R_\gamma g(x)$. Consequently, if moreover $|R_\gamma f(x)| > t$, then $|R_\gamma g(x)| > t$. This shows

$$\{|R_\gamma f| > t\} \ \subseteq\ \Omega \cup N \cup \{|R_\gamma g| > t\},$$

hence $\{|R_\gamma f| > t\} \subseteq \Omega \cup \{|R_\gamma g| > t\}$ up to the γ_n-null set N. Monotonicity of γ_n up to null sets followed by its subadditivity therefore give

$$\gamma_n(\{|R_\gamma f| > t\}) \le \gamma_n\big(\Omega \cup \{|R_\gamma g| > t\}\big) \le \gamma_n(\Omega) + \gamma_n(\{|R_\gamma g| > t\}).$$

□

2.62. `ComplexTransformedLocalization`.

Lemma 2.62 (Exact localization of the transformed bad part, complex inputs)**.** *Let $v, w \in L^2(\gamma_n;\mathbb{C})$, let $g_1, g_2, u_1, u_2 \in L^2(\gamma_n;\mathbb{R})$ with $u_1, u_2 \in \mathrm{Dom}(L^{1/2})$ in the sense of `FracPowDomain`, and suppose that, with $B = L^{1/2}$ as in `FracPow`,*

$$\mathrm{Re}\, v = g_1 + Bu_1, \qquad \mathrm{Im}\, v = g_2 + Bu_2 \qquad \text{in } L^2(\gamma_n;\mathbb{R}),$$

and that w has these good parts as its own real and imaginary parts:

$$\mathrm{Re}\, w = g_1, \qquad \mathrm{Im}\, w = g_2.$$

Let $t \in \mathbb{R}$ and let $\Omega \subseteq \mathbb{R}^n$ be an arbitrary subset such that $Du_1 = 0$ and $Du_2 = 0$ γ_n-a.e. on Ω^c, with D as in `GaussD`; no measurability of Ω is required, and $\gamma_n(\Omega)$ below is the outer measure, which agrees with the measure when Ω is measurable. Then, with $R_\gamma^{\mathbb{C}}$ as in `ComplexRieszTransform`:

(1) *for γ_n-almost every x with $x \notin \Omega$, $R_\gamma^{\mathbb{C}} v(x) = R_\gamma^{\mathbb{C}} w(x)$; and*

(2)
$$\gamma_n\big(\{|R_\gamma^{\mathbb{C}} v|_{\mathbb{C}^n} > t\}\big) \ \le\ \gamma_n(\Omega) + \gamma_n\big(\{|R_\gamma^{\mathbb{C}} w|_{\mathbb{C}^n} > t\}\big).$$

This is the reference paper's sentence "The transformed bad part vanishes outside Ω" in the complex setting, and it is the complex counterpart of `TransformedLocalization`. It is not a restatement of that node: here there are two *bad parts u_1, u_2, sharing a* single *exceptional set Ω, and the conclusion is about the complex operator $R_\gamma^{\mathbb{C}}$ and the norm of $\mathbb{C}^n$, which is not a level set of either real transform. As in the real node, there is no splitting of the level into $t/2 + t/2$: off Ω the two complex outputs are exactly equal.*

Proof. Write $\iota : \mathbb{R}^n \to \mathbb{C}^n$ for the coordinatewise inclusion of `ComplexRieszTransform`, so that by that node

$$\text{(D)} \qquad R_\gamma^{\mathbb{C}} v = \iota\big(R_\gamma(\mathrm{Re}\, v)\big) + i\,\iota\big(R_\gamma(\mathrm{Im}\, v)\big), \qquad R_\gamma^{\mathbb{C}} w = \iota\big(R_\gamma(\mathrm{Re}\, w)\big) + i\,\iota\big(R_\gamma(\mathrm{Im}\, w)\big).$$

By the hypotheses $\mathrm{Re}\, w = g_1$ and $\mathrm{Im}\, w = g_2$, the second formula reads $R_\gamma^{\mathbb{C}} w = \iota(R_\gamma g_1) + i\,\iota(R_\gamma g_2)$.

Step 1: the two real localizations. Apply `TransformedLocalization` with the data $(f, g, u) = (\mathrm{Re}\, v,\ g_1,\ u_1)$, this t and this Ω. Its hypotheses are met: $u_1 \in \mathrm{Dom}(L^{1/2})$ is hypothesis hu_1

here, $\operatorname{Re} v = g_1 + Bu_1$ is hypothesis h_1, and $Du_1 = 0$ a.e. on Ω^c is the first locality hypothesis. Its first clause gives

$$R_\gamma(\operatorname{Re} v) = R_\gamma g_1 + Du_1 \qquad \gamma_n\text{-a.e.} \tag{1}$$

Applying the same node with $(f, g, u) = (\operatorname{Im} v,\, g_2,\, u_2)$, the same t and the same Ω, whose hypotheses are hu_2, h_2 and the second locality hypothesis, gives

$$R_\gamma(\operatorname{Im} v) = R_\gamma g_2 + Du_2 \qquad \gamma_n\text{-a.e.} \tag{2}$$

Step 2: a single null set. Let N be the union of the following five γ_n-null sets: the set where $x \notin \Omega$ but $Du_1(x) \neq 0$; the set where $x \notin \Omega$ but $Du_2(x) \neq 0$; the sets where (1) and where (2) fail; and the set where the two formulas (D) fail for the chosen representatives (each of the operations in (D) — applying ι pointwise, adding, and multiplying by i — is performed on representatives and so induces an identity valid outside a null set). A finite union of null sets is null (subadditivity of a measure, shared context), so $\gamma_n(N) = 0$.

Step 3: clause (1). Let $x \notin \Omega \cup N$. Then $Du_1(x) = 0$ and $Du_2(x) = 0$, so (1) and (2) at this x give

$$R_\gamma(\operatorname{Re} v)(x) = R_\gamma g_1(x), \qquad R_\gamma(\operatorname{Im} v)(x) = R_\gamma g_2(x).$$

Substituting into the first formula of (D) at this x, and then reading the second formula backwards,

$$R^{\mathbb{C}}_\gamma v(x) = \iota\big(R_\gamma g_1(x)\big) + i\,\iota\big(R_\gamma g_2(x)\big) = R^{\mathbb{C}}_\gamma w(x).$$

Since N is null, this holds for γ_n-almost every x with $x \notin \Omega$, which is clause (1).

Step 4: clause (2). Let x satisfy $|R^{\mathbb{C}}_\gamma v(x)|_{\mathbb{C}^n} > t$ and $x \notin \Omega \cup N$. By Step 3, $R^{\mathbb{C}}_\gamma v(x) = R^{\mathbb{C}}_\gamma w(x)$, so $|R^{\mathbb{C}}_\gamma w(x)|_{\mathbb{C}^n} > t$ as well. Hence

$$\{|R^{\mathbb{C}}_\gamma v|_{\mathbb{C}^n} > t\} \;\subseteq\; \Omega \cup N \cup \{|R^{\mathbb{C}}_\gamma w|_{\mathbb{C}^n} > t\},$$

that is, $\{|R^{\mathbb{C}}_\gamma v| > t\}$ is contained in $\Omega \cup \{|R^{\mathbb{C}}_\gamma w| > t\}$ up to the null set N. By monotonicity of γ_n up to null sets and then subadditivity (both shared context),

$$\gamma_n\big(\{|R^{\mathbb{C}}_\gamma v| > t\}\big) \le \gamma_n\big(\Omega \cup \{|R^{\mathbb{C}}_\gamma w| > t\}\big) \le \gamma_n(\Omega) + \gamma_n\big(\{|R^{\mathbb{C}}_\gamma w| > t\}\big),$$

which is clause (2). □

2.63. `RieszL2Identity`.

Lemma 2.63 (The exact L^2 identity for R_γ)**.** *For every $f \in L^2(\gamma_n)$,*

$$\|R_\gamma f\|^2_{L^2(\gamma_n;\mathbb{R}^n)} = \|\Pi_0 f\|^2_{L^2(\gamma_n)} \le \|f\|^2_{L^2(\gamma_n)},$$

with R_γ as in `RieszTransform` and Π_0 as in `FracPow`.

Proof. Put $v = L^{-1/2}\Pi_0 f$. By `PseudoInverseSqrt`, $v \in \operatorname{Dom}(L^{1/2})$ and $L^{1/2}v = \Pi_0 f$. By `SqrtDomainSobolev` applied to v, we have $v \in W^{1,2}(\gamma_n)$ and

$$\|Dv\|_{L^2(\gamma_n;\mathbb{R}^n)} = \|L^{1/2}v\|_{L^2(\gamma_n)}.$$

By `RieszTransform`, $R_\gamma f = Dv$. Combining the two displayed facts,

$$\|R_\gamma f\| = \|Dv\| = \|L^{1/2}v\| = \|\Pi_0 f\|,$$

which is the first assertion. For the second, by `FracPow` with $s = 0$ the element $\Pi_0 f$ has Hermite coefficients 0 at $\alpha = 0$ and f_α at $\alpha \neq 0$, so Parseval's identity from `HermiteBasis` gives

$$\|\Pi_0 f\|_2^2 = \sum_{\alpha \neq 0} |f_\alpha|^2 \le \sum_{\alpha} |f_\alpha|^2 = \|f\|_2^2.$$

Taking square roots and combining with the first assertion gives $\|R_\gamma f\| \le \|f\|$. □

2.64. `ComplexRieszL2Identity`.

Lemma 2.64 (The complex L^2 identity for $R_\gamma^{\mathbb{C}}$)**.** *Let $v \in L^2(\gamma_n;\mathbb{C})$, let $\operatorname{Re} v, \operatorname{Im} v \in L^2(\gamma_n;\mathbb{R})$ be its real and imaginary parts and let $R_\gamma^{\mathbb{C}}$ be the complex Riesz transform of `ComplexRieszTransform`. Then:*

(1) *(the L^2 identity)*
$$\big\|R_\gamma^{\mathbb{C}} v\big\|^2_{L^2(\gamma_n;\mathbb{C}^n)} = \big\|R_\gamma(\operatorname{Re} v)\big\|^2_{L^2(\gamma_n;\mathbb{R}^n)} + \big\|R_\gamma(\operatorname{Im} v)\big\|^2_{L^2(\gamma_n;\mathbb{R}^n)};$$
(2) *(the contraction bound)* $\big\|R_\gamma^{\mathbb{C}} v\big\|_{L^2(\gamma_n;\mathbb{C}^n)} \le \|v\|_{L^2(\gamma_n;\mathbb{C})}$*;*
(3) *(the pointwise form) for γ_n-almost every $x \in \mathbb{R}^n$,*
$$\big|R_\gamma^{\mathbb{C}} v(x)\big|^2_{\mathbb{C}^n} = \big|R_\gamma(\operatorname{Re} v)(x)\big|^2_{\mathbb{R}^n} + \big|R_\gamma(\operatorname{Im} v)(x)\big|^2_{\mathbb{R}^n}.$$

This is the reference paper's complex L^2 identity, the one its proof of the complex-valued estimate obtains "by applying (2.10) of the paper to real and imaginary parts"; `RieszL2Identity` is the node supplying that real L^2 bound. Clause (3) is the pointwise statement, kept because the Chebyshev step of the complex weak-type argument is applied to the level sets of $|R_\gamma^{\mathbb{C}} v|_{\mathbb{C}^n}$ and therefore needs the identity at almost every point, not only after integration. Clause (2) is not a formal consequence of clause (1) together with the real bound alone: it also uses the identity $\|v\|^2 = \|\operatorname{Re} v\|^2 + \|\operatorname{Im} v\|^2$ in $L^2(\gamma_n;\mathbb{C})$, which is what converts the sum of the two real bounds back into the single complex norm.

Proof. Write $u_1 = R_\gamma(\operatorname{Re} v)$ and $u_2 = R_\gamma(\operatorname{Im} v)$, both in $L^2(\gamma_n;\mathbb{R}^n)$; they are defined for these inputs with no side condition by `RieszTransform`. Throughout we use that the norm of a Lebesgue space of order 2 over γ_n with values in a normed space E is given by $\|w\|^2 = \int |w(x)|_E^2 \, d\gamma_n(x)$, which is the definition of that norm, and that the operations on such a space — composition with a bounded linear map, addition, and multiplication by a scalar — are defined on representatives, so each induces an identity valid at γ_n-almost every point.

Step 1: the scalar identity. For real numbers a, b we have $|a+ib|^2 = a^2 + b^2$, since $|a+ib|^2 = (a+ib)\overline{(a+ib)} = (a+ib)(a-ib) = a^2 + b^2$.

Step 2: clause (3). By `ComplexRieszTransform`, $R_\gamma^{\mathbb{C}} v = \iota(u_1) + i\,\iota(u_2)$ with ι the coordinatewise inclusion $\mathbb{R}^n \to \mathbb{C}^n$ of that node. Each of the three operations involved — applying ι pointwise, adding, and multiplying by i — holds at almost every point for the chosen representatives, and the intersection of the three resulting full-measure sets is again of full measure. So for γ_n-almost every x and every $j \in \{1,\dots,n\}$,
$$\big(R_\gamma^{\mathbb{C}} v\big)(x)_j = u_1(x)_j + i\,u_2(x)_j,$$
$u_1(x)_j$ and $u_2(x)_j$ being real. For such an x, by Step 1 applied in each coordinate,
$$\big|R_\gamma^{\mathbb{C}} v(x)\big|^2_{\mathbb{C}^n} = \sum_{j=1}^n \big|u_1(x)_j + i\,u_2(x)_j\big|^2 = \sum_{j=1}^n \Big(u_1(x)_j^2 + u_2(x)_j^2\Big) = |u_1(x)|^2_{\mathbb{R}^n} + |u_2(x)|^2_{\mathbb{R}^n},$$
the last equality by splitting the finite sum. This is clause (3).

Step 3: clause (1). The three functions $x \mapsto |R_\gamma^{\mathbb{C}} v(x)|^2$, $x \mapsto |u_1(x)|^2$ and $x \mapsto |u_2(x)|^2$ are nonnegative and γ_n-integrable, their integrals being the squared L^2 norms of $R_\gamma^{\mathbb{C}} v$, u_1 and u_2, which are finite. Integrating the identity of Step 2 — legitimate because it holds almost everywhere and because the integral of a sum of two integrable functions is the sum of the integrals —
$$\big\|R_\gamma^{\mathbb{C}} v\big\|^2 = \int |u_1(x)|^2 \, d\gamma_n(x) + \int |u_2(x)|^2 \, d\gamma_n(x) = \|u_1\|^2 + \|u_2\|^2,$$
which is clause (1).

Step 4: the real and imaginary parts of v. By `ComplexRieszTransform` the parts $\operatorname{Re} v$ and $\operatorname{Im} v$ are obtained by composing v pointwise with Re and Im, so for γ_n-almost every x, $(\operatorname{Re} v)(x) = \operatorname{Re}(v(x))$ and $(\operatorname{Im} v)(x) = \operatorname{Im}(v(x))$. Writing $v(x) = \operatorname{Re}(v(x)) + i\operatorname{Im}(v(x))$ and applying Step 1,
$$|v(x)|^2 = \big(\operatorname{Re} v\big)(x)^2 + \big(\operatorname{Im} v\big)(x)^2 \qquad \text{for } \gamma_n\text{-a.e. } x.$$

Integrating, exactly as in Step 3 and with the same justification (all three integrands are nonnegative with finite integral, those integrals being the squared L^2 norms of v, $\operatorname{Re} v$ and $\operatorname{Im} v$),

$$(*) \qquad \|v\|^2_{L^2(\gamma_n;\mathbb{C})} = \|\operatorname{Re} v\|^2_{L^2(\gamma_n;\mathbb{R})} + \|\operatorname{Im} v\|^2_{L^2(\gamma_n;\mathbb{R})}.$$

Step 5: clause (2). Apply `RieszL2Identity` twice, once to the input $\operatorname{Re} v$ and once to $\operatorname{Im} v$; its second assertion gives $\|u_1\| \le \|\operatorname{Re} v\|$ and $\|u_2\| \le \|\operatorname{Im} v\|$. All four quantities are nonnegative, so squaring preserves the inequalities: $\|u_1\|^2 \le \|\operatorname{Re} v\|^2$ and $\|u_2\|^2 \le \|\operatorname{Im} v\|^2$. Adding these and then using clause (1) on the left and $(*)$ on the right,

$$\left\|R^{\mathbb{C}}_\gamma v\right\|^2 = \|u_1\|^2 + \|u_2\|^2 \le \|\operatorname{Re} v\|^2 + \|\operatorname{Im} v\|^2 = \|v\|^2.$$

Both $\|R^{\mathbb{C}}_\gamma v\|$ and $\|v\|$ are nonnegative and the square root is nondecreasing on $[0,\infty)$, so $\|R^{\mathbb{C}}_\gamma v\| \le \|v\|$, which is clause (2). □

2.65. `RieszTransformLinear`.

Lemma 2.65 (The Riesz transform is linear on $L^2(\gamma_n)$)**.** *Fix $n \in \mathbb{N}_0$ and let $R_\gamma : L^2(\gamma_n) \to L^2(\gamma_n;\mathbb{R}^n)$ be the energy-normalized vector Riesz transform of `RieszTransform`. Then R_γ is linear over the real scalars, in the two clauses:*

(1) *(additivity) for all $v, w \in L^2(\gamma_n)$, $R_\gamma(v+w) = R_\gamma v + R_\gamma w$;*
(2) *(homogeneity) for every $c \in \mathbb{R}$ and every $v \in L^2(\gamma_n)$, $R_\gamma(c\,v) = c\,R_\gamma v$.*

Both clauses are unconditional: they hold for every pair of elements and every real scalar, with no domain restriction and no differentiability assumption on the arguments. The scalars are real throughout, $L^2(\gamma_n)$ being a real Hilbert space, so clause (2) is homogeneity over $\mathbb{R}$ and not merely over the nonnegative reals.

Proof. By `RieszTransform`, R_γ is the composite $R_\gamma f = D\,L^{-1/2}\Pi_0 f$: one first forms the potential $v = L^{-1/2}\Pi_0 f$ and then takes its energy-normalized weak gradient D. So it suffices to see that each of the two factors is linear and that the intermediate domain conditions are satisfied automatically, after which the composite of two linear maps is linear.

Step 1 (the first factor is everywhere defined). The exponent here is $s = -1/2 \le 0$, so by the first clause of `FracPowLinear` every $v \in L^2(\gamma_n)$ lies in $\operatorname{Dom}(L^{-1/2})$. Hence no domain hypothesis has to be carried: $L^{-1/2}\Pi_0$ is defined on all of $L^2(\gamma_n)$, which is what makes the two clauses above unconditional.

Step 2 (the first factor is linear). By the remaining two clauses of `FracPowLinear`, applied at $s = -1/2$ and using Step 1 to discharge their domain hypotheses,

$$L^{-1/2}\Pi_0(v+w) = L^{-1/2}\Pi_0 v + L^{-1/2}\Pi_0 w, \qquad L^{-1/2}\Pi_0(c\,v) = c\,L^{-1/2}\Pi_0 v$$

for all $v, w \in L^2(\gamma_n)$ and all $c \in \mathbb{R}$.

Step 3 (the potentials are differentiable). For every $v \in L^2(\gamma_n)$ the potential $L^{-1/2}\Pi_0 v$ lies in $\operatorname{Dom}(L^{1/2})$ by `PseudoInverseSqrt`, hence in $W^{1,2}(\gamma_n)$ by `SqrtDomainSobolev`. So the second factor D may be applied to every potential that arises, and again no side condition survives into the statement.

Step 4 (the second factor is linear). By `GaussDLinear`, D is additive and real-homogeneous on $W^{1,2}(\gamma_n)$: for u, u' in that space and $c \in \mathbb{R}$ one has $D(u+u') = Du + Du'$ and $D(c\,u) = c\,Du$, the domain hypotheses being supplied by Step 3.

Step 5 (composing). Fix $v, w \in L^2(\gamma_n)$. Unfolding the definition of R_γ three times and applying Step 2 and then Step 4,

$$R_\gamma(v+w) = D\,L^{-1/2}\Pi_0(v+w) = D\big(L^{-1/2}\Pi_0 v + L^{-1/2}\Pi_0 w\big) = D\,L^{-1/2}\Pi_0 v + D\,L^{-1/2}\Pi_0 w = R_\gamma v + R_\gamma w,$$

which is clause (1). Likewise, for $c \in \mathbb{R}$ and $v \in L^2(\gamma_n)$,

$$R_\gamma(c\,v) = D\,L^{-1/2}\Pi_0(c\,v) = D\big(c\,L^{-1/2}\Pi_0 v\big) = c\,D\,L^{-1/2}\Pi_0 v = c\,R_\gamma v,$$

which is clause (2). As v, w and c were arbitrary, both clauses hold. □

2.66. `ComplexRieszTransformLinear`.

Lemma 2.66 (The complex Riesz transform is $\mathbb{C}$-linear, and unimodular rotations do not change the pointwise size of its values)**.** *Let $R_\gamma^{\mathbb{C}} : L^2(\gamma_n;\mathbb{C}) \to L^2(\gamma_n;\mathbb{C}^n)$ be the complex Riesz transform of `ComplexRieszTransform`. Then:*

(1) *(additivity) $R_\gamma^{\mathbb{C}}(v+w) = R_\gamma^{\mathbb{C}}v + R_\gamma^{\mathbb{C}}w$ for all $v, w \in L^2(\gamma_n;\mathbb{C})$;*
(2) *($\mathbb{C}$-homogeneity) $R_\gamma^{\mathbb{C}}(c\,v) = c\,R_\gamma^{\mathbb{C}}v$ for every $c \in \mathbb{C}$ and every $v \in L^2(\gamma_n;\mathbb{C})$;*
(3) *(unimodular invariance of the pointwise size) if $c \in \mathbb{C}$ with $|c| = 1$, then for every $v \in L^2(\gamma_n;\mathbb{C})$ and γ_n-almost every $x \in \mathbb{R}^n$,*

$$\big|R_\gamma^{\mathbb{C}}(c\,v)(x)\big|_{\mathbb{C}^n} = \big|R_\gamma^{\mathbb{C}}v(x)\big|_{\mathbb{C}^n}.$$

Clauses (1) and (2) say that $R_\gamma^{\mathbb{C}}$ is the complex-linear extension of R_γ that the reference paper's complex-valued estimate is about, and clause (3) is the sentence "Complex linearity implies $|R_\gamma(e^{i\theta}f)| = |R_\gamma f|$" that licenses the paper's rotation average; it is stated for an arbitrary unimodular c, of which $c = e^{i\theta}$ is the case the paper uses. Clause (3) is recorded here, rather than left to the reader of clause (2), because it is an assertion about the values of representatives at almost every point, whereas clause (2) is an identity between equivalence classes.

The scalars in clause (2) are the complex *ones. This is strictly more than the real linearity of R_γ in `RieszTransformLinear`: the real and imaginary parts of $c\,v$ are not c times those of v — already for $c = i$ one has $\operatorname{Re}(iv) = -\operatorname{Im} v$ and $\operatorname{Im}(iv) = \operatorname{Re} v$, so the two parts are exchanged — and clause (2) is therefore not a restatement of real homogeneity but a genuine consequence of it combined with the structure of `ComplexRieszTransform`.*

Proof. Throughout write $\iota : \mathbb{R}^n \to \mathbb{C}^n$ for the coordinatewise inclusion of `ComplexRieszTransform` and recall from that node the defining formula

$$R_\gamma^{\mathbb{C}}z = \iota\big(R_\gamma(\operatorname{Re} z)\big) + i\,\iota\big(R_\gamma(\operatorname{Im} z)\big), \qquad z \in L^2(\gamma_n;\mathbb{C}). \tag{D}$$

Three elementary facts are used repeatedly. First, the maps $z \mapsto \operatorname{Re} z$ and $z \mapsto \operatorname{Im} z$ from $L^2(\gamma_n;\mathbb{C})$ to $L^2(\gamma_n;\mathbb{R})$ are additive and $\mathbb{R}$-homogeneous, because they are composition with the $\mathbb{R}$-linear maps $\operatorname{Re}, \operatorname{Im} : \mathbb{C} \to \mathbb{R}$ and the vector operations on a Lebesgue space are performed on representatives. Second, the map $y \mapsto \iota(y)$ from $L^2(\gamma_n;\mathbb{R}^n)$ to $L^2(\gamma_n;\mathbb{C}^n)$ is additive and $\mathbb{R}$-homogeneous, for the same reason, ι being $\mathbb{R}$-linear. Third, $L^2(\gamma_n;\mathbb{C}^n)$ is a complex vector space whose real scalar action is the restriction of its complex one, so real and complex scalars may be combined freely.

Step 1: clause (1). Let $v, w \in L^2(\gamma_n;\mathbb{C})$. By the first fact, $\operatorname{Re}(v+w) = \operatorname{Re} v + \operatorname{Re} w$ and $\operatorname{Im}(v+w) = \operatorname{Im} v + \operatorname{Im} w$. By the additivity clause of `RieszTransformLinear`, applied to the two real inputs in each line,

$$R_\gamma\big(\operatorname{Re}(v+w)\big) = R_\gamma(\operatorname{Re} v) + R_\gamma(\operatorname{Re} w), \qquad R_\gamma\big(\operatorname{Im}(v+w)\big) = R_\gamma(\operatorname{Im} v) + R_\gamma(\operatorname{Im} w).$$

Substituting into (D) at $z = v+w$ and using the second and third facts to distribute ι and the factor i over the sums,

$$R_\gamma^{\mathbb{C}}(v+w) = \Big(\iota R_\gamma(\operatorname{Re} v) + i\,\iota R_\gamma(\operatorname{Im} v)\Big) + \Big(\iota R_\gamma(\operatorname{Re} w) + i\,\iota R_\gamma(\operatorname{Im} w)\Big) = R_\gamma^{\mathbb{C}}v + R_\gamma^{\mathbb{C}}w,$$

the regrouping of the four summands being commutativity and associativity of addition in $L^2(\gamma_n;\mathbb{C}^n)$, and the two bracketed expressions being $R_\gamma^{\mathbb{C}}v$ and $R_\gamma^{\mathbb{C}}w$ by (D).

Step 2: the parts of $c\,v$. Let $c \in \mathbb{C}$ and put $a = \operatorname{Re} c$, $b = \operatorname{Im} c$, so $c = a + ib$ with a, b real. For any $z \in \mathbb{C}$,

$$c\,z = (a+ib)(\operatorname{Re} z + i \operatorname{Im} z) = \big(a \operatorname{Re} z - b \operatorname{Im} z\big) + i\big(a \operatorname{Im} z + b \operatorname{Re} z\big),$$

and the two bracketed quantities are real, so $\operatorname{Re}(cz) = a \operatorname{Re} z - b \operatorname{Im} z$ and $\operatorname{Im}(cz) = a \operatorname{Im} z + b \operatorname{Re} z$. Applying this at $z = v(x)$ for almost every x, and using that the vector operations on L^2 are representativewise, we get the identities in $L^2(\gamma_n;\mathbb{R})$

$$\operatorname{Re}(c\,v) = a\ \operatorname{Re} v - b\ \operatorname{Im} v, \qquad \operatorname{Im}(c\,v) = a\ \operatorname{Im} v + b\ \operatorname{Re} v.$$

Step 3: transforming the parts. Write $u_1 = R_\gamma(\operatorname{Re} v)$ and $u_2 = R_\gamma(\operatorname{Im} v)$. Both clauses of `RieszTransformLinear` are unconditional, so they may be applied to the combinations of Step

2 with the real scalars a, $-b$ and b (additivity for the sum, homogeneity for each scalar multiple, a difference being the sum with the scalar multiple by -1):

$$R_\gamma\big(\mathrm{Re}(c\,v)\big) = a\,u_1 - b\,u_2, \qquad R_\gamma\big(\mathrm{Im}(c\,v)\big) = a\,u_2 + b\,u_1.$$

Step 4: clause (2). Substituting Step 3 into (D) at $z = c\,v$ and distributing ι and the factor i by the second and third facts,

$$R_\gamma^{\mathbb{C}}(c\,v) = \iota\big(a\,u_1 - b\,u_2\big) + i\,\iota\big(a\,u_2 + b\,u_1\big) = a\,\iota u_1 - b\,\iota u_2 + ia\,\iota u_2 + ib\,\iota u_1.$$

Collecting the two terms carrying ιu_1 and the two carrying ιu_2,

$$R_\gamma^{\mathbb{C}}(c\,v) = (a + ib)\,\iota u_1 + (ia - b)\,\iota u_2.$$

Now $ia - b = i(a + ib)$, because $i(a + ib) = ia + i^2 b = ia - b$. Hence

$$R_\gamma^{\mathbb{C}}(c\,v) = c\,\iota u_1 + i\,c\,\iota u_2 = c\,\big(\iota u_1 + i\,\iota u_2\big) = c\,R_\gamma^{\mathbb{C}} v,$$

the last equality by (D) at $z = v$. This is clause (2). Taking $c = i$, that is $a = 0$ and $b = 1$, the computation reads $R_\gamma^{\mathbb{C}}(iv) = -\iota u_2 + i\,\iota u_1 = i(\iota u_1 + i\iota u_2) = iR_\gamma^{\mathbb{C}} v$, and it is visibly the real homogeneity of R_γ — through $R_\gamma(-\operatorname{Im} v) = -u_2$ — that makes it work.

Step 5: clause (3). Let $|c| = 1$ and $v \in L^2(\gamma_n;\mathbb{C})$. By clause (2), $R_\gamma^{\mathbb{C}}(c\,v) = c\,R_\gamma^{\mathbb{C}} v$ as elements of $L^2(\gamma_n;\mathbb{C}^n)$, that is, as γ_n-a.e. equivalence classes. Scalar multiplication there is performed on representatives, so for γ_n-almost every x,

$$R_\gamma^{\mathbb{C}}(c\,v)(x) = c\,\big(R_\gamma^{\mathbb{C}} v(x)\big) \qquad \text{in } \mathbb{C}^n.$$

Taking norms at such an x and using absolute homogeneity of the norm of $\mathbb{C}^n$ over the complex scalars,

$$\big|R_\gamma^{\mathbb{C}}(c\,v)(x)\big|_{\mathbb{C}^n} = |c|\,\big|R_\gamma^{\mathbb{C}} v(x)\big|_{\mathbb{C}^n} = \big|R_\gamma^{\mathbb{C}} v(x)\big|_{\mathbb{C}^n},$$

since $|c| = 1$. This is clause (3). □

2.67. `RotationAverageL1`.

Lemma 2.67 (The rotation average of the two real L^1 norms)**.** *Let $v \in L^2(\gamma_n;\mathbb{C})$. For $\theta \in \mathbb{R}$ put $c_\theta = e^{i\theta}$ and let*

$$f_1(\theta) = \mathrm{Re}(c_\theta v), \qquad f_2(\theta) = \mathrm{Im}(c_\theta v)$$

be the real and imaginary parts of the rotated function, that is, the γ_n-a.e. classes obtained by composing $c_\theta v$ pointwise with the bounded $\mathbb{R}$-linear maps $\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$; they are elements of $L^2(\gamma_n;\mathbb{R})$, since composition with a bounded linear map preserves membership in a Lebesgue space. Set

$$S_\theta = \|f_1(\theta)\|_{L^1(\gamma_n)} + \|f_2(\theta)\|_{L^1(\gamma_n)},$$

the two L^1 norms being taken through the inclusion of `ToL1`. Then:

(1) *the function $\theta \mapsto S_\theta$ is integrable on $[0, 2\pi]$; and*
(2)

$$\frac{1}{2\pi}\int_0^{2\pi} S_\theta\,d\theta = \frac{4}{\pi}\int_{\mathbb{R}^n} |v(x)|\,d\gamma_n(x).$$

This is the reference paper's displayed rotation average $\frac{1}{2\pi}\int_0^{2\pi} S_\theta\,d\theta = \frac{4}{\pi}\|f\|_1$, with $S_\theta = \|\mathrm{Re}(e^{i\theta}f)\|_1 + \|\mathrm{Im}(e^{i\theta}f)\|_1$, and it is the single place the constant $8/\pi$ of the complex estimate comes from: the $4/\pi$ here is $2 \cdot \frac{2}{\pi}$, one $\frac{2}{\pi}$ for the cosine and one for the sine. The right-hand side is written as the integral $\int |v|\,d\gamma_n$ rather than as a norm in a complex L^1 space; this is the same real number as the paper's $\|f\|_{L^1(\gamma;\mathbb{C})}$, and writing it this way avoids introducing a complex L^1 inclusion that the tablet does not yet have. Clause (1) is not decoration: it is what Tonelli's theorem supplies alongside the identity, and it is what licenses integrating an inequality in θ against the constant function, which is how the average is used.

Proof. Write $M = \int_{\mathbb{R}^n} |v(x)|\, d\gamma_n(x)$ and fix a representative of v, so that $v : \mathbb{R}^n \to \mathbb{C}$ is measurable and square γ_n-integrable.

Step 0: M is finite. By `GaussIsProb` the measure γ_n is a probability measure, so the constant function 1 lies in $L^2(\gamma_n)$; by the integral form of Cauchy–Schwarz (shared context) applied to $|v|$ and 1, $M \le \big(\int |v|^2 d\gamma_n\big)^{1/2} \cdot 1 = \|v\|_{L^2} < \infty$.

Step 1: the one-variable integral of $|\cos|$ over a period. We claim $\int_0^{2\pi} |\cos\theta|\, d\theta = 4$. Split the interval at $\pi/2$ and $3\pi/2$, and use throughout that $\sin' = \cos$, so that the fundamental theorem of calculus gives $\int_a^b \cos = \sin b - \sin a$ for all $a \le b$. On $[0, \pi/2]$ we have $\cos \ge 0$, so $\int_0^{\pi/2} |\cos| = \int_0^{\pi/2} \cos = \sin(\pi/2) - \sin 0 = 1$. On $[\pi/2, 3\pi/2]$ we have $\cos \le 0$, so $\int_{\pi/2}^{3\pi/2} |\cos| = -\int_{\pi/2}^{3\pi/2} \cos = -(\sin(3\pi/2) - \sin(\pi/2)) = -(-1-1) = 2$. On $[3\pi/2, 2\pi]$ we have $\cos \ge 0$ again, so $\int_{3\pi/2}^{2\pi} |\cos| = \sin(2\pi) - \sin(3\pi/2) = 0 - (-1) = 1$. The integrand is continuous, hence integrable on each piece, and adding the three pieces gives $1 + 2 + 1 = 4$.

Step 2: the same integral after a phase shift. For every $p \in \mathbb{R}$,

$$\int_0^{2\pi} \big|\cos(\theta + p)\big|\, d\theta = 4, \qquad \int_0^{2\pi} \big|\sin(\theta + p)\big|\, d\theta = 4.$$

For the first: substituting $\theta \mapsto \theta - p$ turns the integral into $\int_p^{p+2\pi} |\cos u|\, du$, and $|\cos|$ is 2π-periodic (because $\cos(u + 2\pi) = \cos u$), so the integral of $|\cos|$ over any interval of length 2π is the same as over $[0, 2\pi]$, which is 4 by Step 1. For the second: $\sin s = \cos(s - \pi/2)$, so $|\sin(\theta + p)| = |\cos(\theta + (p - \pi/2))|$ for every θ, and the first identity applies with the phase $p - \pi/2$.

Step 3: the pointwise rotation identity. Fix $x \in \mathbb{R}^n$ and put $\rho = |v(x)| \ge 0$. If $\rho = 0$ then $v(x) = 0$ and both sides of the identity below are 0. Otherwise write $v(x) = \rho e^{i\varphi}$ with $\varphi = \arg v(x)$, so that $c_\theta v(x) = \rho e^{i(\theta+\varphi)}$ and therefore

$$\mathrm{Re}\big(c_\theta v(x)\big) = \rho \cos(\theta + \varphi), \qquad \mathrm{Im}\big(c_\theta v(x)\big) = \rho \sin(\theta + \varphi).$$

Taking absolute values, using $\rho \ge 0$, and integrating in θ with Step 2 at the phase $p = \varphi$,

$$\int_0^{2\pi} \Big(\big|\mathrm{Re}(c_\theta v(x))\big| + \big|\mathrm{Im}(c_\theta v(x))\big|\Big) d\theta = \rho \cdot 4 + \rho \cdot 4 = 8\,|v(x)|. \tag{1}$$

In the degenerate case $\rho = 0$ the two sides of (1) are 0 as well, so (1) holds for every x.

Step 4: joint measurability. Let $H(\theta, x) = \big|\mathrm{Re}(c_\theta v(x))\big| + \big|\mathrm{Im}(c_\theta v(x))\big| \ge 0$ on $[0, 2\pi] \times \mathbb{R}^n$. The map $(\theta, x) \mapsto c_\theta = e^{i\theta}$ is continuous, hence measurable; the map $(\theta, x) \mapsto v(x)$ is measurable, being the composition of the measurable v with the projection; their product is measurable; and Re, Im and $|\cdot|$ are continuous. So H is measurable for the product σ-algebra, and it is nonnegative.

Step 5: Tonelli. Lebesgue measure restricted to $[0, 2\pi]$ is finite and γ_n is a probability measure by `GaussIsProb`, so both are σ-finite and Tonelli's theorem (shared context) applies to H: the two iterated integrals agree in $[0, +\infty]$. Computing in the order $d\theta$ first and using (1),

$$\int_{\mathbb{R}^n} \Big(\int_0^{2\pi} H(\theta, x)\, d\theta\Big) d\gamma_n(x) = \int_{\mathbb{R}^n} 8\,|v(x)|\, d\gamma_n(x) = 8M,$$

which is finite by Step 0. Hence also

$$\int_0^{2\pi} \Big(\int_{\mathbb{R}^n} H(\theta, x)\, d\gamma_n(x)\Big) d\theta = 8M < \infty, \tag{2}$$

and Tonelli further gives that the inner integral $\theta \mapsto \int H(\theta, x)\, d\gamma_n(x)$ is a measurable function of θ on $[0, 2\pi]$.

Step 6: the inner integral is S_θ. Fix θ. By the description in the statement, $f_1(\theta)$ and $f_2(\theta)$ are the γ_n-a.e. classes of the functions $x \mapsto \mathrm{Re}(c_\theta v(x))$ and $x \mapsto \mathrm{Im}(c_\theta v(x))$. By `ToL1` the $L^1(\gamma_n)$ class of a real $L^2(\gamma_n)$ function has the same representative, so its L^1 norm is the integral of its

absolute value. Therefore

$$\int_{\mathbb{R}^n} H(\theta,x)\,d\gamma_n(x) = \int \bigl|\mathrm{Re}(c_\theta v)\bigr|d\gamma_n + \int \bigl|\mathrm{Im}(c_\theta v)\bigr|d\gamma_n = \|f_1(\theta)\|_{L^1(\gamma_n)} + \|f_2(\theta)\|_{L^1(\gamma_n)} = S_\theta,$$

the first equality by linearity of the integral for the two nonnegative measurable summands of H.

Step 7: conclusion. By Step 6 and Step 5, $\theta \mapsto S_\theta$ is measurable on $[0,2\pi]$ and nonnegative with $\int_0^{2\pi} S_\theta\,d\theta = 8M < \infty$ by (2); a nonnegative measurable function with finite integral is integrable, which is clause (1). Dividing the identity $\int_0^{2\pi} S_\theta\,d\theta = 8M$ by $2\pi > 0$,

$$\frac{1}{2\pi}\int_0^{2\pi} S_\theta\,d\theta = \frac{8M}{2\pi} = \frac{4}{\pi}M = \frac{4}{\pi}\int |v|\,d\gamma_n,$$

which is clause (2). □

2.68. `FractionalGap`.

Lemma 2.68 (The fractional spectral gap). *For every $v \in L^2(\gamma_n)$,*

$$\|\Pi_0 v\|_2^2 \le \mathcal{E}(v)$$

as an inequality in $[0,+\infty]$, where Π_0 is as in `FracPow` and $\mathcal{E}$ as in `Energy`. Equivalently, by `ProjZeroCentering`, $\|v-\mathfrak{m}(v)\mathbf{1}\|_2^2 \le \mathcal{E}(v)$ with $\mathfrak{m}(v) = \int v\,d\gamma_n$; in particular the inequality is vacuously true when $\mathcal{E}(v) = +\infty$, and it is the spectral gap estimate of the reference paper when v lies in the form domain $\mathcal{V}$.

Proof. Step 1: the left-hand side as a coefficient sum. By `FracPowCoeff` and `FracPow` with $s = 0$, the element $\Pi_0 v$ has Hermite coefficients 0 at $\alpha = 0$ and v_α at $\alpha \ne 0$ (the domain condition of `FracPowDomain` being satisfied, as the multiplied family is dominated by the square-summable family $(|v_\alpha|^2)_\alpha$). Parseval's identity from `HermiteBasis` therefore gives

$$\|\Pi_0 v\|_2^2 = \sum_{\alpha\ne 0} |v_\alpha|^2.$$

Step 2: a termwise comparison. Let $\alpha \ne 0$. By `MultiIndexSum`, $|\alpha| = \sum_j \alpha_j$ is a nonnegative integer which vanishes only for $\alpha = 0$; hence $|\alpha| \ge 1$, and so $|\alpha|^{1/2} \ge 1$. Multiplying the nonnegative number $|v_\alpha|^2$ by this gives

$$|v_\alpha|^2 \le |\alpha|^{1/2}\,|v_\alpha|^2.$$

Step 3: summation. All terms on both sides are nonnegative, so summing the Step-2 inequalities over $\alpha \in \mathbb{N}_0^n \setminus \{0\}$ is legitimate in $[0,+\infty]$ and preserves the inequality:

$$\sum_{\alpha\ne 0} |v_\alpha|^2 \;\le\; \sum_{\alpha\ne 0} |\alpha|^{1/2}|v_\alpha|^2.$$

By `Energy` the right-hand side is exactly $\mathcal{E}(v)$, interpreted in $[0,+\infty]$. Combining with Step 1 gives $\|\Pi_0 v\|_2^2 \le \mathcal{E}(v)$.

Finally, by `ProjZeroCentering`, $\Pi_0 v = v - \mathfrak{m}(v)h_0$ with $h_0 = \mathbf{1}$, so the left-hand side also equals $\|v-\mathfrak{m}(v)\mathbf{1}\|_2^2$, giving the centred form. □

2.69. `EnergyConstantInvariance`.

Lemma 2.69 (Constants are null directions of the energy). *Let $h_0 \in L^2(\gamma_n)$ be the Hermite element of index 0 from `HermiteL2`, which is the constant function $\mathbf{1}$ by `Hermite1Zero`. Then for every $v \in L^2(\gamma_n)$ and every $c \in \mathbb{R}$,*

$$\mathcal{E}(v,\mathbf{1}) = 0 \qquad \text{and} \qquad \mathcal{E}(v + c\mathbf{1}) = \mathcal{E}(v),$$

where $\mathcal{E}(\cdot,\cdot)$ is the form of `EnergyBil` and $\mathcal{E}(\cdot)$ the energy of `Energy`, the second identity holding as an equality in $[0,+\infty]$. In particular adding a constant to v changes neither its membership in the form domain $\mathcal{V}$ nor its energy.

Proof. Step 0: the coefficients of h_0. By `HermiteL2Orthonormal` the family $\{h_\beta\}$ is orthonormal, so the Hermite coefficients of h_0 itself are $(h_0)_0 = \langle h_0, h_0\rangle = 1$ and $(h_0)_\beta = \langle h_0, h_\beta\rangle = 0$ for every $\beta \neq 0$.

Step 1: $\mathcal{E}(v, \mathbf{1}) = 0$. By `EnergyBil` the form is the sum $\mathcal{E}(v, w) = \sum_{\alpha\neq 0} |\alpha|^{1/2} v_\alpha w_\alpha$, taken over indices $\alpha \neq 0$ only, whenever the family is summable. Taking $w = h_0 = \mathbf{1}$, every term carries the factor $(h_0)_\alpha$ with $\alpha \neq 0$, which is 0 by Step 0. Hence every term of the family is 0; the family is therefore (absolutely) summable with sum 0, so the summability hypothesis in `EnergyBil` is met and $\mathcal{E}(v, \mathbf{1}) = 0$.

Step 2: $\mathcal{E}(v + c\mathbf{1}) = \mathcal{E}(v)$. The Hermite coefficient map is linear (`HermiteBasis`), so by Step 0 the element $v + ch_0$ has coefficients

$$(v + ch_0)_0 = v_0 + c, \qquad (v + ch_0)_\alpha = v_\alpha \quad \text{for every } \alpha \neq 0.$$

By `Energy` the defining series $\mathcal{E}(w) = \sum_{\alpha\neq 0} |\alpha|^{1/2} |w_\alpha|^2$ omits the index $\alpha = 0$ entirely, so it depends only on the coefficients at indices $\alpha \neq 0$. Those are unchanged in passing from v to $v + ch_0$, so the two series are termwise identical. As both are sums of nonnegative terms, they have the same value in $[0, +\infty]$: $\mathcal{E}(v + c\mathbf{1}) = \mathcal{E}(v)$.

In particular $\mathcal{E}(v + c\mathbf{1}) < \infty$ if and only if $\mathcal{E}(v) < \infty$, i.e. $v + c\mathbf{1} \in \mathcal{V}$ if and only if $v \in \mathcal{V}$. □

2.70. `ObstacleCoercivity`.

Lemma 2.70 (Coercivity of J and finiteness of its infimum)**.** *Let $F \in L^2(\gamma_n)$, put $m = \int F\, d\gamma_n$ and let $a > m$. No sign condition on F is assumed. Let J be the functional of `ObstacleFunctional` and $\mathcal{K}$ the cone of `ObstacleCone`. Then:*

(1) *for every $v \in \mathcal{K}$, writing $c = \int v\, d\gamma_n$,*

$$J(v) \;\geq\; \tfrac{1}{4}\mathcal{E}(v) + (a - m)\, c - \|F - m\mathbf{1}\|_2^2;$$

(2) *the infimum $I = \inf_{v\in\mathcal{K}} J(v)$ is finite and satisfies*

$$-\|F - m\mathbf{1}\|_2^2 \;\leq\; I \;\leq\; 0.$$

Here $\mathbf{1} = h_0$ is the constant function of `HermiteL2` and `Hermite1Zero`.

Proof. Let $v \in \mathcal{K}$, so by `ObstacleCone` we have $\mathcal{E}(v) < \infty$ and $v \geq 0$ γ_n-a.e. Put $c = \int v\, d\gamma_n$; the integral is finite by `L2IntoL1`, and $c \geq 0$ because $v \geq 0$ a.e. Put $w = v - c\mathbf{1}$. Since γ_n is a probability measure (`GaussIsProb`), $\int w\, d\gamma_n = c - c = 0$.

Step 1: the energy is unchanged by centring, and controls $\|w\|_2$. By `EnergyConstantInvariance` applied with the constant $-c$, $\mathcal{E}(w) = \mathcal{E}(v - c\mathbf{1}) = \mathcal{E}(v)$. By `ProjZeroCentering`, $\Pi_0 v = v - \mathfrak{m}(v)\mathbf{1} = w$ since $\mathfrak{m}(v) = c$. Hence `FractionalGap` gives

$$\|w\|_2^2 = \|\Pi_0 v\|_2^2 \leq \mathcal{E}(v), \qquad \text{so} \qquad \|w\|_2 \leq \mathcal{E}(v)^{1/2}. \tag{1}$$

Step 2: splitting the linear term. Using $v = c\mathbf{1} + w$, $\gamma_n(\mathbb{R}^n) = 1$ (`GaussIsProb`), $\int w\, d\gamma_n = 0$ and $m = \int F\, d\gamma_n$, and noting all integrals are finite by Cauchy–Schwarz and `L2IntoL1`,

$$\int (a - F)\, v\, d\gamma_n = c \int (a - F)\, d\gamma_n + \int (a - F)\, w\, d\gamma_n = (a - m)\, c - \int (F - m)\, w\, d\gamma_n,$$

where we used $\int (a - F) d\gamma_n = a - m$ for the first term, and for the second $\int a\, w\, d\gamma_n = a \int w\, d\gamma_n = 0$ together with $\int F w\, d\gamma_n = \int (F - m) w\, d\gamma_n$ (again because $\int w\, d\gamma_n = 0$).

Step 3: estimating the cross term. By Cauchy–Schwarz and (1),

$$\left| \int (F - m)\, w\, d\gamma_n \right| \leq \|F - m\mathbf{1}\|_2\, \|w\|_2 \leq \|F - m\mathbf{1}\|_2\, \mathcal{E}(v)^{1/2}.$$

Young's inequality in the form $pq \leq p^2 + \frac{1}{4}q^2$ (equivalent to $(p - \frac{1}{2}q)^2 \geq 0$), applied with $p = \|F - m\mathbf{1}\|_2 \geq 0$ and $q = \mathcal{E}(v)^{1/2} \geq 0$, gives

$$\|F - m\mathbf{1}\|_2\, \mathcal{E}(v)^{1/2} \leq \|F - m\mathbf{1}\|_2^2 + \tfrac{1}{4}\, \mathcal{E}(v). \tag{2}$$

Step 4: proof of (1) of the statement. By `ObstacleFunctional` and Step 2,

$$J(v) = \tfrac{1}{2}\mathcal{E}(v) + (a - m)c - \int (F - m) w\, d\gamma_n.$$

Bounding the last term below by $-\big(\|F - m\mathbf{1}\|_2^2 + \frac{1}{4}\mathcal{E}(v)\big)$ via Step 3 and (2),

$$J(v) \geq \tfrac{1}{2}\mathcal{E}(v) + (a-m)c - \|F - m\mathbf{1}\|_2^2 - \tfrac{1}{4}\mathcal{E}(v) = \tfrac{1}{4}\mathcal{E}(v) + (a-m)c - \|F - m\mathbf{1}\|_2^2,$$

since $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. This is assertion (1). (All quantities are finite real numbers: $\mathcal{E}(v) < \infty$ since $v \in \mathcal{K}$, and the linear term is finite by Step 2.)

Step 5: proof of (2) of the statement. In the inequality of Step 4 we have $\frac{1}{4}\mathcal{E}(v) \geq 0$ and $(a-m)c \geq 0$, the latter because $a - m > 0$ by hypothesis and $c \geq 0$. Hence

$$J(v) \geq -\|F - m\mathbf{1}\|_2^2 \qquad \text{for every } v \in \mathcal{K},$$

so the same lower bound holds for $I = \inf_{\mathcal{K}} J$. Conversely, $0 \in \mathcal{K}$ by `ObstacleCone` (the zero function is nonnegative and has $\mathcal{E}(0) = 0 < \infty$ by `Energy`), and `ObstacleFunctional` gives $J(0) = \frac{1}{2}\mathcal{E}(0) + \int (a - F) \cdot 0 \, d\gamma_n = 0$. Hence $I \leq J(0) = 0$. Together, $-\|F - m\mathbf{1}\|_2^2 \leq I \leq 0$; in particular I is a finite real number. □

2.71. `ObstacleSublevelBounded`.

Lemma 2.71 (Sublevel sets of J are bounded in the form norm)**.** *Let $F \in L^2(\gamma_n)$, $m = \int F \, d\gamma_n$, $a > m$, and let $C \in \mathbb{R}$. No sign condition on F is assumed. Put $M_C = C + \|F - m\mathbf{1}\|_2^2$. Then every $v \in \mathcal{K}$ with $J(v) \leq C$ satisfies*

$$\|v\|_2^2 + \mathcal{E}(v) \;\leq\; \frac{M_C^2}{(a-m)^2} + 8M_C,$$

with $\mathcal{K}$ as in `ObstacleCone`, J as in `ObstacleFunctional` and $\mathcal{E}$ as in `Energy`. The bound depends only on C, a, m and F, not on v.

Proof. Let $v \in \mathcal{K}$ with $J(v) \leq C$, and put $c = \int v \, d\gamma_n$, which is finite by `L2IntoL1` and satisfies $c \geq 0$ because $v \geq 0$ γ_n-a.e. by `ObstacleCone`.

Step 1: the two separate bounds. Part (1) of `ObstacleCoercivity` gives $\frac{1}{4}\mathcal{E}(v) + (a-m)c - \|F - m\mathbf{1}\|_2^2 \leq J(v) \leq C$, i.e.

$$\tfrac{1}{4}\,\mathcal{E}(v) + (a-m)\,c \;\leq\; C + \|F - m\mathbf{1}\|_2^2 = M_C. \tag{1}$$

Both summands on the left are nonnegative: $\mathcal{E}(v) \geq 0$ by `Energy`, and $(a-m)c \geq 0$ since $a - m > 0$ and $c \geq 0$. Hence $M_C \geq 0$, and dropping either summand gives

$$\mathcal{E}(v) \leq 4\,M_C, \qquad 0 \leq c \leq \frac{M_C}{a-m}. \tag{2}$$

Step 2: Pythagoras for the splitting $v = c\mathbf{1} + w$. Put $w = v - c\mathbf{1}$, where $\mathbf{1} = h_0$ as in `HermiteL2` and `Hermite1Zero`. Since γ_n is a probability measure (`GaussIsProb`), $\int w \, d\gamma_n = 0$, i.e. $\langle w, \mathbf{1}\rangle = 0$; and $\|\mathbf{1}\|_2^2 = \gamma_n(\mathbb{R}^n) = 1$. Hence $c\mathbf{1}$ and w are orthogonal in $L^2(\gamma_n)$ and

$$\|v\|_2^2 = \|c\mathbf{1}\|_2^2 + \|w\|_2^2 = c^2 + \|w\|_2^2.$$

By `ProjZeroCentering`, $w = \Pi_0 v$, so `FractionalGap` gives $\|w\|_2^2 \leq \mathcal{E}(v)$. Therefore

$$\|v\|_2^2 \leq c^2 + \mathcal{E}(v). \tag{3}$$

Step 3: conclusion. Adding $\mathcal{E}(v)$ to both sides of (3) and then applying the two bounds of (2),

$$\|v\|_2^2 + \mathcal{E}(v) \;\leq\; c^2 + 2\,\mathcal{E}(v) \;\leq\; \frac{M_C^2}{(a-m)^2} + 2 \cdot 4\,M_C \;=\; \frac{M_C^2}{(a-m)^2} + 8\,M_C,$$

which is the assertion. □

2.72. `NonnegTestCharacterization`.

Lemma 2.72 (Nonnegativity tested against nonnegative L^2 functions)**.** *For $v \in L^2(\gamma_n)$ the following are equivalent:*

(1) *$v \geq 0$ γ_n-almost everywhere;*
(2) *$\int v\varphi \, d\gamma_n \geq 0$ for every $\varphi \in L^2(\gamma_n)$ with $\varphi \geq 0$ γ_n-almost everywhere.*

Proof. (1) $\Rightarrow$ (2). If $v \geq 0$ and $\varphi \geq 0$ a.e., then $v\varphi \geq 0$ a.e.; the product is integrable by Cauchy–Schwarz since $v, \varphi \in L^2(\gamma_n)$, and the integral of an a.e. nonnegative integrable function is nonnegative.

(2) $\Rightarrow$ (1). Let $v^- = \max\{-v, 0\}$. Then v^- is measurable and $0 \leq v^- \leq |v|$, so $v^- \in L^2(\gamma_n)$ and $v^- \geq 0$ a.e. Taking $\varphi = v^-$ in (2) gives

$$0 \leq \int v\, v^- \, d\gamma_n.$$

Pointwise, $v\, v^- = -(v^-)^2$: at points where $v \geq 0$ we have $v^- = 0$ so both sides vanish, and at points where $v < 0$ we have $v^- = -v$ so $vv^- = -v^2 = -(v^-)^2$. Hence $0 \leq \int vv^- d\gamma_n = -\|v^-\|_2^2 \leq 0$, so $\|v^-\|_2 = 0$ and therefore $v^- = 0$ γ_n-a.e., i.e. $v \geq 0$ γ_n-a.e. □

2.73. `ConeWeaklyClosed`.

Lemma 2.73 (Weak closedness in $L^2(\gamma_n)$ of the nonnegativity constraint)**.** *Let $(v_k)_{k\geq 1} \subseteq L^2(\gamma_n)$ satisfy $v_k \geq 0$ γ_n-almost everywhere for every k, and suppose $v_k \rightharpoonup v$ weakly in $L^2(\gamma_n)$, i.e. $\langle v_k, \varphi\rangle \to \langle v, \varphi\rangle$ for every $\varphi \in L^2(\gamma_n)$. Then $v \geq 0$ γ_n-almost everywhere. Consequently the admissible cone $\mathcal{K}$ of `ObstacleCone` is closed under weak $L^2(\gamma_n)$ limits of sequences with uniformly bounded energy, and no pointwise convergence is needed.*

Proof. Let $\varphi \in L^2(\gamma_n)$ with $\varphi \geq 0$ γ_n-a.e. By the implication (1) $\Rightarrow$ (2) of `NonnegTestCharacterization`, applied to each v_k, we have

$$\langle v_k, \varphi\rangle = \int v_k\varphi \, d\gamma_n \geq 0 \qquad \text{for every } k.$$

By the weak convergence hypothesis, $\langle v_k, \varphi\rangle \to \langle v, \varphi\rangle$. A limit of a sequence of nonnegative reals is nonnegative, so $\int v\varphi \, d\gamma_n = \langle v, \varphi\rangle \geq 0$.

Since φ was an arbitrary nonnegative element of $L^2(\gamma_n)$, the implication (2) $\Rightarrow$ (1) of `NonnegTestCharacterization` applies to v and yields $v \geq 0$ γ_n-almost everywhere. □

2.74. `EnergyWeaklyLSC`.

Lemma 2.74 (Weak lower semicontinuity of the fractional energy on $L^2(\gamma_n)$)**.** *Let $(v_k)_{k\geq 1} \subseteq L^2(\gamma_n)$ and $v \in L^2(\gamma_n)$ with $v_k \rightharpoonup v$ weakly in $L^2(\gamma_n)$, i.e. $\langle v_k, \varphi\rangle \to \langle v, \varphi\rangle$ for every $\varphi \in L^2(\gamma_n)$. Then, as an inequality in $[0, +\infty]$,*

$$\mathcal{E}(v) \leq \liminf_{k\to\infty} \mathcal{E}(v_k),$$

with $\mathcal{E}$ as in `Energy`.

Proof. For a finite set $S \subseteq \mathbb{N}_0^n \setminus \{0\}$ put

$$\mathcal{E}_S(w) = \sum_{\alpha\in S} |\alpha|^{1/2} \, |w_\alpha|^2, \qquad w \in L^2(\gamma_n),$$

where $w_\alpha = \langle w, h_\alpha\rangle$ are the Hermite coefficients relative to `HermiteBasis` and $|\alpha|$ is as in `MultiIndexSum`. Each $\mathcal{E}_S$ is a finite sum of squares of the weakly continuous linear functionals $w \mapsto \langle w, h_\alpha\rangle$, so weak convergence gives, for each fixed finite S,

$$\mathcal{E}_S(v_k) \longrightarrow \mathcal{E}_S(v) \qquad (k \to \infty).$$

On the other hand $\mathcal{E}_S(w) \leq \mathcal{E}(w)$ for every w and every finite S, because the defining series of $\mathcal{E}$ in `Energy` has nonnegative terms and contains those of $\mathcal{E}_S$. Hence

$$\mathcal{E}_S(v) = \lim_k \mathcal{E}_S(v_k) \leq \liminf_k \mathcal{E}(v_k).$$

The right-hand side does not depend on S. Taking the supremum over all finite subsets $S \subseteq \mathbb{N}_0^n \setminus \{0\}$ and using that a sum of nonnegative terms is the supremum of its finite partial sums, i.e. $\mathcal{E}(v) = \sup_S \mathcal{E}_S(v)$ in $[0, +\infty]$, we obtain $\mathcal{E}(v) \le \liminf_k \mathcal{E}(v_k)$. □

2.75. `GaussL2WeakSeqCompact.`

Lemma 2.75 (Bounded sequences in $L^2(\gamma_n)$ have weakly convergent subsequences)**.** *Let $(v_k)_{k\ge 0}$ be a sequence in $L^2(\gamma_n)$ and let $R \in \mathbb{R}$ satisfy $\|v_k\|_{L^2(\gamma_n)} \le R$ for every k. Then there are indices $k_0 < k_1 < k_2 < \cdots$ and an element $u \in L^2(\gamma_n)$ with $\|u\|_{L^2(\gamma_n)} \le R$ such that*

$$\langle v_{k_j},\, \varphi\rangle_{L^2(\gamma_n)} \longrightarrow \langle u,\, \varphi\rangle_{L^2(\gamma_n)} \qquad \text{as } j \to \infty, \text{ for every } \varphi \in L^2(\gamma_n).$$

This is weak sequential compactness of bounded sets in the single Hilbert space $L^2(\gamma_n)$. No weak topology on $\mathcal{V} = \mathrm{Dom}(L^{1/4})$ is used, no reflexivity of $\mathcal{V}$ is invoked, and no compactness of any embedding and no dependence on the dimension n is asserted.

Proof. Write $H = L^2(\gamma_n)$, a real Hilbert space, and H^* for its dual. The proof has two steps: H is separable, and the weak-$*$ statement for H^* recorded in the shared context transports to H through the Riesz representation. If $R < 0$ the hypothesis is vacuous, since $\|v_0\| \ge 0 > R$ is impossible; so $R \ge 0$ throughout.

Step 1: the index set $\mathbb{N}_0^n$ is countable. $\mathbb{N}_0^n$ is the set of functions $\{1, \dots, n\} \to \mathbb{N}_0$, i.e. a product of n copies of the countable set $\mathbb{N}_0$. A product of finitely many countable sets is countable, by induction on the number of factors: the empty product is a one-point set, and if A is countable and B is countable then $A \times B$ is countable because $\mathbb{N}_0 \times \mathbb{N}_0$ is countable (e.g. by the injection $(i, j) \mapsto 2^i 3^j$) and a set injecting into a countable set is countable. Hence $\mathbb{N}_0^n$ is countable, and we may fix an enumeration of it.

Step 2: H is separable. Let

$$D = \Big\{ \textstyle\sum_{\alpha\in F} q_\alpha\, h_\alpha \;:\; F \subseteq \mathbb{N}_0^n \text{ finite},\ q_\alpha \in \mathbb{Q} \text{ for } \alpha \in F \Big\} \subseteq H,$$

with h_α the Hermite functions of `HermiteL2`.

D is countable. By Step 1 the set of finite subsets of $\mathbb{N}_0^n$ is countable (the finite subsets of a countable set form a countable union, over $m \in \mathbb{N}_0$, of the images of the countable sets $(\mathbb{N}_0^n)^m$ under the map sending an m-tuple to the set of its entries), and for each fixed finite F the set of families $(q_\alpha)_{\alpha\in F} \in \mathbb{Q}^F$ is countable, $\mathbb{Q}$ being countable and F finite. So D is a countable union of countable sets, hence countable.

D is dense in H. Let $v \in H$ and $\varepsilon > 0$. By `HermiteL2Complete` the closed linear span of $\{h_\alpha : \alpha \in \mathbb{N}_0^n\}$ is all of H, so v lies in the closure of the set of finite real linear combinations of the h_α; hence there are a finite $F \subseteq \mathbb{N}_0^n$ and reals $(c_\alpha)_{\alpha\in F}$ with

$$\Big\| v - \sum_{\alpha\in F} c_\alpha h_\alpha \Big\| < \frac{\varepsilon}{2}.$$

If $F = \emptyset$ the element $0 \in D$ already satisfies $\|v - 0\| < \varepsilon/2$, so assume $F \ne \emptyset$ and let $N = |F| \ge 1$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, choose for each $\alpha \in F$ a rational q_α with $|c_\alpha - q_\alpha| < \varepsilon/(2N)$, and put $w = \sum_{\alpha\in F} q_\alpha h_\alpha \in D$. By `HermiteL2Orthonormal` each h_α is a unit vector, $\|h_\alpha\| = 1$, so by the triangle inequality in H

$$\Big\| \sum_{\alpha\in F} c_\alpha h_\alpha - w \Big\| = \Big\| \sum_{\alpha\in F} (c_\alpha - q_\alpha) h_\alpha \Big\| \le \sum_{\alpha\in F} |c_\alpha - q_\alpha|\, \|h_\alpha\| < N \cdot \frac{\varepsilon}{2N} = \frac{\varepsilon}{2}.$$

Adding the two displays, $\|v - w\| < \varepsilon$ with $w \in D$. As v and ε were arbitrary, D is a countable dense subset of H, i.e. H is separable.

Step 3: the sequence of associated functionals. For each k define $\Lambda_k : H \to \mathbb{R}$ by $\Lambda_k(w) = \langle v_k, w\rangle$. Each Λ_k is linear in w by linearity of the inner product in its second argument, and by Cauchy–Schwarz, which is shared context, $|\Lambda_k(w)| \le \|v_k\|\, \|w\|$ for every w, so Λ_k is continuous with $\|\Lambda_k\|_{H^*} \le \|v_k\|$. Thus $\Lambda_k \in H^*$, and applying the Riesz representation item of the shared

context to Λ_k produces a *unique* $y \in H$ with $\Lambda_k = \langle y, \cdot \rangle$ and $\|\Lambda_k\|_{H^*} = \|y\|$; since v_k is such an element by the definition of Λ_k, uniqueness gives $y = v_k$ and therefore

$$\|\Lambda_k\|_{H^*} = \|v_k\| \le R \qquad \text{for every } k. \tag{1}$$

Step 4: extraction, and transport back to H. By Step 2 the normed space H is separable, and by (1) the sequence (Λ_k) lies in the ball of radius R of H^*. The weak-$*$ sequential compactness item of the shared context, applied to $E = H$, this $R \ge 0$ and this sequence, therefore yields indices $k_0 < k_1 < k_2 < \cdots$ and $\Lambda \in H^*$ with $\|\Lambda\|_{H^*} \le R$ such that

$$\Lambda_{k_j}(w) \longrightarrow \Lambda(w) \qquad \text{for every } w \in H. \tag{2}$$

Applying the Riesz representation item once more, now to Λ, gives $u \in H$ with $\Lambda(w) = \langle u, w \rangle$ for every $w \in H$ and $\|u\| = \|\Lambda\|_{H^*} \le R$.

Finally, let $\varphi \in H$ be arbitrary. By the definition of Λ_{k_j}, by (2) with $w = \varphi$, and by the representation of Λ,

$$\langle v_{k_j},\, \varphi \rangle = \Lambda_{k_j}(\varphi) \longrightarrow \Lambda(\varphi) = \langle u,\, \varphi \rangle.$$

So the indices $k_0 < k_1 < \cdots$ and the element u, with $\|u\| \le R$, are as asserted. □

2.76. `ObstacleMinimizerExists`.

Lemma 2.76 (Existence of a minimizer of J over the cone)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., put $m = \int F \, d\gamma_n$ and let $a > m$. Then there exists $u \in \mathcal{K}$ with*

$$J(u) \le J(v) \qquad \textit{for every } v \in \mathcal{K},$$

where $\mathcal{K}$ is the cone of `ObstacleCone` and J the functional of `ObstacleFunctional`.

Proof. By part (2) of `ObstacleCoercivity`, the infimum $I = \inf_{v \in \mathcal{K}} J(v)$ is a finite real number with $I \le 0$.

Step 1: a bounded minimizing sequence. By the definition of the infimum, for each integer $k \ge 1$ there is $v_k \in \mathcal{K}$ with

$$I \le J(v_k) < I + \tfrac{1}{k}.$$

This uses only that I is finite and does not presuppose that it is attained. Since $I \le 0$ we get $J(v_k) < I + 1 \le 1$ for every k, so `ObstacleSublevelBounded` applies with $C = 1$ and yields a finite constant $R^2 = M_1^2/(a-m)^2 + 8M_1$, $M_1 = 1 + \|F - m\mathbf{1}\|_2^2$, independent of k, with

$$\|v_k\|_2^2 + \mathcal{E}(v_k) \le R^2 \qquad \text{for every } k. \tag{1}$$

In particular $\|v_k\|_2 \le R$ and $\mathcal{E}(v_k) \le R^2$. Also $J(v_k) \to I$, since $I \le J(v_k) < I + 1/k$.

Step 2: a weakly convergent subsequence in $L^2(\gamma_n)$. By (1) and the nonnegativity of $\mathcal{E}(v_k)$ (which holds by `Energy`, the energy being a sum of nonnegative terms) we have $\|v_k\|_2^2 \le R^2$, hence $\|v_k\|_2 \le R$ for every k, taking $R \ge 0$ to be the nonnegative square root of R^2. Apply `GaussL2WeakSeqCompact` to the sequence (v_k) with this bound R: it supplies indices $k_1 < k_2 < \cdots$ and $u \in L^2(\gamma_n)$ with $\|u\|_2 \le R$ and

$$\langle v_{k_j}, \varphi \rangle \longrightarrow \langle u, \varphi \rangle \qquad \text{for every } \varphi \in L^2(\gamma_n). \tag{2}$$

No compact embedding, and no compactness uniform in n, is used; in particular no weak topology on $\mathcal{V}$ and no reflexivity of $\mathcal{V}$ is invoked, which is where this proof departs from the reference paper's route.

Step 3: the limit is admissible. Each v_{k_j} satisfies $v_{k_j} \ge 0$ γ_n-a.e. by `ObstacleCone`, so `ConeWeaklyClosed`, applied to the sequence $(v_{k_j})_j$ and the limit u with hypothesis (2), gives $u \ge 0$ γ_n-a.e. Moreover `EnergyWeaklyLSC`, applied to the same sequence and limit, gives

$$\mathcal{E}(u) \le \liminf_{j \to \infty} \mathcal{E}(v_{k_j}) \le R^2 < \infty, \tag{3}$$

using (1). By `ObstacleCone` these two facts say $u \in \mathcal{K}$.

Step 4: the linear term is weakly continuous. Since γ_n is a probability measure (`GaussIsProb`), the constant function $\mathbf{1}$ lies in $L^2(\gamma_n)$, hence so does $a\mathbf{1} - F$. Therefore, by (2) applied with $\varphi = a\mathbf{1} - F$,

$$\int (a - F)\, v_{k_j}\, d\gamma_n = \langle v_{k_j},\, a\mathbf{1} - F\rangle \longrightarrow \langle u,\, a\mathbf{1} - F\rangle = \int (a - F)\, u\, d\gamma_n. \tag{4}$$

Step 5: $J(u) \le I$. By `ObstacleFunctional` and then (3) and (4),

$$J(u) = \tfrac{1}{2}\mathcal{E}(u) + \int (a - F)u\, d\gamma_n \le \tfrac{1}{2} \liminf_j \mathcal{E}(v_{k_j}) + \lim_j \int (a - F)v_{k_j}\, d\gamma_n = \liminf_j J(v_{k_j}),$$

where the last equality uses that the liminf of a sum of a sequence with a liminf and a sequence with a limit is the sum of the liminf and the limit, all quantities being finite reals by (1) and (3). Since $J(v_k) \to I$ by Step 1, the subsequence also satisfies $J(v_{k_j}) \to I$, so $\liminf_j J(v_{k_j}) = I$ and hence $J(u) \le I$.

Step 6: conclusion. By Step 3, $u \in \mathcal{K}$, so the definition of I as the infimum over $\mathcal{K}$ gives $I \le J(u)$. Combined with Step 5, $J(u) = I$, and therefore $J(u) = I \le J(v)$ for every $v \in \mathcal{K}$. □

2.77. `ObstacleMinimizerUnique`.

Lemma 2.77 (Uniqueness of the minimizer, including its constant mode)**.** *Let $F \in L^2(\gamma_n)$, put $m = \int F\, d\gamma_n$ and let $a > m$. If $u_1, u_2 \in \mathcal{K}$ both minimize the functional J of `ObstacleFunctional` over the cone $\mathcal{K}$ of `ObstacleCone`, then $u_1 = u_2$ in $L^2(\gamma_n)$.*

Proof. Let $I = J(u_1) = J(u_2)$, a real number (both values are finite because $u_1, u_2 \in \mathcal{K}$ have finite energy, and the linear term is finite by Cauchy–Schwarz with $a\mathbf{1} - F \in L^2(\gamma_n)$, using `GaussIsProb`). Each is the minimum because each u_i minimizes and the other is admissible.

Step 0: the midpoint is admissible. By `ObstacleCone`, $\mathcal{K}$ is convex, so $\frac{1}{2}(u_1 + u_2) \in \mathcal{K}$.

Step 1: the parallelogram identity for J. For $v, w \in \mathcal{V}$ the energy is the quadratic form of the symmetric bilinear form $\mathcal{E}(\cdot,\cdot)$ of `EnergyBil`, i.e. $\mathcal{E}(v) = \mathcal{E}(v, v)$ (immediate from comparing the defining series in `Energy` and `EnergyBil` at $w = v$, both being absolutely convergent when $\mathcal{E}(v) < \infty$). Expanding bilinearly,

$$\mathcal{E}\Big(\frac{u_1 + u_2}{2}\Big) = \tfrac{1}{4}\big(\mathcal{E}(u_1) + 2\mathcal{E}(u_1, u_2) + \mathcal{E}(u_2)\big), \qquad \mathcal{E}(u_1 - u_2) = \mathcal{E}(u_1) - 2\mathcal{E}(u_1, u_2) + \mathcal{E}(u_2),$$

and adding $\frac{1}{4}$ of the second to the first gives

$$\mathcal{E}\Big(\frac{u_1 + u_2}{2}\Big) = \frac{\mathcal{E}(u_1) + \mathcal{E}(u_2)}{2} - \frac{1}{4}\,\mathcal{E}(u_1 - u_2).$$

The remaining term of J is linear in its argument, so by `ObstacleFunctional`

$$J\Big(\frac{u_1 + u_2}{2}\Big) = \frac{J(u_1) + J(u_2)}{2} - \frac{1}{8}\,\mathcal{E}(u_1 - u_2). \tag{1}$$

Step 2: the difference has zero energy, hence is constant. By Step 0 and minimality, $I \le J(\frac{1}{2}(u_1 + u_2))$, so (1) with $J(u_1) = J(u_2) = I$ gives $I \le I - \frac{1}{8}\mathcal{E}(u_1 - u_2)$, i.e. $\mathcal{E}(u_1 - u_2) \le 0$. By `Energy` the energy is a sum of nonnegative terms, so $\mathcal{E}(u_1 - u_2) = 0$ and in fact each term vanishes: $|\alpha|^{1/2}\,|(u_1 - u_2)_\alpha|^2 = 0$ for every $\alpha \ne 0$. Since $|\alpha| \ge 1$ for $\alpha \ne 0$ by `MultiIndexSum`, the factor $|\alpha|^{1/2}$ is nonzero, so $(u_1 - u_2)_\alpha = 0$ for every $\alpha \ne 0$. By Parseval and injectivity of the coefficient map (`HermiteBasis`), and since $h_0 = \mathbf{1}$ by `Hermite1Zero`, this forces

$$u_1 - u_2 = d\,\mathbf{1}, \qquad d = (u_1 - u_2)_0 \in \mathbb{R}.$$

Step 3: the strict height inequality pins d. By `EnergyConstantInvariance` applied with the constant d, $\mathcal{E}(u_1) = \mathcal{E}(u_2 + d\mathbf{1}) = \mathcal{E}(u_2)$. Hence, subtracting the two expressions for J in `ObstacleFunctional`, the energy terms cancel and

$$0 = J(u_1) - J(u_2) = \int (a - F)\,(u_1 - u_2)\, d\gamma_n = d \int (a - F)\, d\gamma_n = d\,(a - m),$$

using $u_1 - u_2 = d\mathbf{1}$, $\gamma_n(\mathbb{R}^n) = 1$ (`GaussIsProb`) and $m = \int F\, d\gamma_n$; the integrals are finite by `L2IntoL1` and Cauchy–Schwarz. Since $a - m > 0$ by hypothesis, $d = 0$, so $u_1 = u_2$.

Thus the energy determines the mean-zero part of the minimizer and the strict height inequality determines its constant part. □

2.78. `EnergyBilIsOperator`.

Lemma 2.78 (The energy form represents the operator $B = L^{1/2}$ on its domain)**.** *Let $v \in \mathrm{Dom}(L^{1/2})$ in the sense of `FracPowDomain` and let $w \in L^2(\gamma_n)$ be arbitrary. Then*

$$\mathcal{E}(v,w) = \langle L^{1/2}v, w\rangle = \int (L^{1/2}v)\, w\, d\gamma_n,$$

with $\mathcal{E}(\cdot,\cdot)$ as in `EnergyBil` and $L^{1/2}$ as in `FracPow`. This is the identity $\mathcal{E}(v,w) = \langle Bv, w\rangle$ of the reference paper. It requires only that v lie in the operator domain $\mathrm{Dom}(B)$; no condition whatever is imposed on w, so in particular the identity holds for every w in the form domain $\mathcal{V}$, which is the case the reference paper uses. It is not asserted for general $v \in \mathcal{V}$.

Proof. Write v_α, w_α for the Hermite coefficients of v, w relative to `HermiteBasis`.

Step 1: the coefficients of $L^{1/2}v$. Since $v \in \mathrm{Dom}(L^{1/2})$, the defining formula of `FracPow` with $s = 1/2$ applies (its domain hypothesis being exactly the criterion of `FracPowDomain`) and gives $L^{1/2}v$ the coefficients

$$(L^{1/2}v)_0 = 0, \qquad (L^{1/2}v)_\alpha = |\alpha|^{1/2}\, v_\alpha \quad (\alpha \neq 0),$$

with $|\alpha|$ as in `MultiIndexSum`; moreover $\sum_{\alpha\neq 0} |\alpha|\, |v_\alpha|^2 < \infty$ by `FracPowDomain`, i.e. the family $\big(|\alpha|^{1/2} v_\alpha\big)_{\alpha\neq 0}$ is square summable.

Step 2: Parseval. Both $L^{1/2}v$ and w lie in $L^2(\gamma_n)$, so Parseval's identity for the orthonormal basis of `HermiteBasis` expresses their inner product as the absolutely convergent sum of coefficient products:

$$\langle L^{1/2}v, w\rangle = \sum_{\alpha} (L^{1/2}v)_\alpha\, w_\alpha = \sum_{\alpha\neq 0} |\alpha|^{1/2}\, v_\alpha\, w_\alpha,$$

the index $\alpha = 0$ dropping out by Step 1. Absolute convergence holds by Cauchy–Schwarz in ℓ^2, since $\big(|\alpha|^{1/2} v_\alpha\big)_{\alpha\neq 0}$ is square summable by Step 1 and $(w_\alpha)_\alpha$ is square summable with sum $\|w\|_2^2$.

Step 3: identification with the form. The last series in Step 2 is literally the defining series of $\mathcal{E}(v,w)$ in `EnergyBil`, and the summability hypothesis in that definition is met by the absolute convergence just established. Hence $\mathcal{E}(v,w) = \langle L^{1/2}v, w\rangle$, which is the assertion. Note that no finiteness assumption on $\mathcal{E}(w)$ was used: the absolute convergence in Step 2 came from square summability of $\big(|\alpha|^{1/2} v_\alpha\big)_{\alpha\neq 0}$, which is the hypothesis on v, paired with square summability of $(w_\alpha)_\alpha$, which holds for every $w \in L^2(\gamma_n)$ by Parseval. □

2.79. `DomainUpgrade`.

Lemma 2.79 (Upgrading the potential from the form domain to the operator domain)**.** *Let $F \in L^2(\gamma_n)$, $a \in \mathbb{R}$, $u \in \mathcal{K}$ as in `ObstacleCone`, and suppose $\eta_2 \in L^2(\gamma_n)$ represents the multiplier functional of `ObstacleMultiplier`, i.e.*

$$\mathcal{E}(u,\varphi) - \int (F-a)\varphi\, d\gamma_n = \int \eta_2\, \varphi\, d\gamma_n \qquad \text{for every } \varphi \in \mathcal{V}.$$

Put $g = a\mathbf{1} - \eta_2 \in L^2(\gamma_n)$. Then

$$u \in \mathrm{Dom}(L^{1/2}) \qquad \text{and} \qquad L^{1/2}u = F - g \quad \text{in } L^2(\gamma_n).$$

In particular the form identity becomes an L^2 operator identity, and $F = g + Bu$ with $B = L^{1/2}$.

Proof. Set $h = F - g = F - a\mathbf{1} + \eta_2$, an element of $L^2(\gamma_n)$ (the constant $\mathbf{1}$ lies in $L^2(\gamma_n)$ because γ_n is a probability measure, `GaussIsProb`). Rearranging the representation hypothesis, it says exactly

$$(*) \qquad \mathcal{E}(u,\varphi) = \int h\, \varphi\, d\gamma_n = \langle h, \varphi\rangle \qquad \text{for every } \varphi \in \mathcal{V}.$$

Step 1: the Hermite elements are admissible test functions. For each $\beta \in \mathbb{N}_0^n$ the element h_β of `HermiteL2` lies in $\mathcal{V}$: by `HermiteL2Orthonormal` its coefficients are $\delta_{\beta\cdot}$, so the defining series of `Energy` reduces to the single term $|\beta|^{1/2}$ when $\beta \neq 0$ and to the empty sum when $\beta = 0$; either way $\mathcal{E}(h_\beta) < \infty$.

Step 2: testing $(*)$ *on* h_β. Let $\beta \neq 0$. By `EnergyBil` and `HermiteL2Orthonormal`, the defining series of $\mathcal{E}(u, h_\beta)$ has exactly one nonzero term, namely the one at $\alpha = \beta$, so $\mathcal{E}(u, h_\beta) = |\beta|^{1/2}\, u_\beta$ where $u_\beta = \langle u, h_\beta\rangle$. Hence $(*)$ with $\varphi = h_\beta$ gives

$$\langle h, h_\beta\rangle = |\beta|^{1/2}\, u_\beta \qquad (\beta \neq 0). \tag{1}$$

For $\beta = 0$, the defining series of `EnergyBil` runs over indices $\alpha \neq 0$ only, and every term carries the factor $(h_0)_\alpha = 0$ (again `HermiteL2Orthonormal`), so $\mathcal{E}(u, h_0) = 0$ and $(*)$ gives

$$\langle h, h_0\rangle = 0. \tag{2}$$

Recall $h_0 = \mathbf{1}$ by `Hermite1Zero`.

Step 3: Parseval yields the domain criterion. By Parseval's identity for the orthonormal basis of `HermiteBasis`, applied to $h \in L^2(\gamma_n)$, and using (1) and (2),

$$\sum_{\beta\neq 0} |\beta|\, |u_\beta|^2 = \sum_{\beta\neq 0} \big|\langle h, h_\beta\rangle\big|^2 = \sum_{\beta} \big|\langle h, h_\beta\rangle\big|^2 = \|h\|_2^2 < \infty,$$

the middle equality because the $\beta = 0$ term vanishes by (2). The criterion of `FracPowDomain` for $s = 1/2$ is precisely $\sum_{\beta\neq 0} |\beta|\, |u_\beta|^2 < \infty$, so $u \in \mathrm{Dom}(L^{1/2})$.

Step 4: identifying $L^{1/2}u$. Now that $u \in \mathrm{Dom}(L^{1/2})$, the defining formula of `FracPow` with $s = 1/2$ applies and gives $L^{1/2}u$ the coefficients 0 at $\beta = 0$ and $|\beta|^{1/2} u_\beta$ at $\beta \neq 0$. By (1) and (2) these are exactly $\langle h, h_\beta\rangle$ for every β, including $\beta = 0$. Since the coefficient map of `HermiteBasis` is injective, $L^{1/2}u = h = F - g$. Consistently, Step 3 gives $u \in \mathrm{Dom}(L^{1/2})$, which is exactly the hypothesis `EnergyBilIsOperator` requires of its first argument; it imposes no condition on its second argument, so it applies to the pair (u, φ) for every $\varphi \in L^2(\gamma_n)$, and in particular for every $\varphi \in \mathcal{V}$, and identifies the left-hand side of $(*)$ as $\langle L^{1/2}u, \varphi\rangle$. □

2.80. `MassConservation`.

Lemma 2.80 (Conservation of Gaussian mass)**.** *Let $F \in L^2(\gamma_n)$ with $m = \int F\, d\gamma_n$, let $a \in \mathbb{R}$, let $u \in L^2(\gamma_n)$, and suppose $\eta_2 \in L^2(\gamma_n)$ represents the multiplier functional of `ObstacleMultiplier` on $\mathcal{V}$. Put $g = a\mathbf{1} - \eta_2$. Then*

$$\int g\, d\gamma_n = m, \qquad \textit{equivalently} \qquad \int \eta_2\, d\gamma_n = a - m.$$

Proof. The constant function $\mathbf{1} = h_0$ of `HermiteL2` and `Hermite1Zero` is an admissible form-domain test function: by `Energy` its energy is 0, since the defining series omits the index $\alpha = 0$ and all other coefficients of h_0 vanish; hence $\mathbf{1} \in \mathcal{V}$.

Apply the representation hypothesis with $\varphi = \mathbf{1}$. Its left-hand side is, by `ObstacleMultiplier`,

$$\mathcal{E}(u, \mathbf{1}) - \int (F - a)\, d\gamma_n.$$

By `EnergyConstantInvariance` the first term is $\mathcal{E}(u, \mathbf{1}) = 0$. Hence the hypothesis reads

$$-\int (F - a)\, d\gamma_n = \int \eta_2\, d\gamma_n,$$

all integrals being finite by `L2IntoL1` together with $\gamma_n(\mathbb{R}^n) = 1$ (`GaussIsProb`). Using $\int F\, d\gamma_n = m$ and $\int a\, d\gamma_n = a$, the left-hand side is $a - m$, so

$$\int \eta_2\, d\gamma_n = a - m.$$

Finally $\int g\, d\gamma_n = \int (a - \eta_2)\, d\gamma_n = a - (a - m) = m$. □

2.81. `ExceptionalSetBound`.

Lemma 2.81 (Measure bound for the exceptional set). *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., let $a > 0$ and $m \in \mathbb{R}$, and let $u, \eta_2 \in L^2(\gamma_n)$ satisfy*

(1) $\int (a - \eta_2)\, d\gamma_n = m$;
(2) $0 \le \eta_2 \le (a - F)^+$ γ_n-a.e.; *and*
(3) $\eta_2 = 0$ γ_n-a.e. *on* $\{u > 0\}$.

Put $g = a\mathbf{1} - \eta_2$ and $\Omega = \{u > 0\}$. Then

$$a\,\gamma_n(\Omega) \le m.$$

Proof. From (2), $0 \le \eta_2 \le (a - F)^+ \le a$ γ_n-a.e., the last inequality because $F \ge 0$ a.e. and $a > 0$. Hence

$$0 \le g = a - \eta_2 \le a \qquad \gamma_n\text{-a.e.}$$

In particular g is bounded, hence in $L^1(\gamma_n)$ since γ_n is a probability measure (`GaussIsProb`), and $g \ge 0$ a.e.

By (3), $g = a$ γ_n-a.e. on Ω. Therefore

$$a\,\gamma_n(\Omega) = \int_\Omega a\, d\gamma_n = \int_\Omega g\, d\gamma_n.$$

Since $g \ge 0$ a.e. on the complement Ω^c as well, discarding that contribution only increases the integral:

$$\int_\Omega g\, d\gamma_n \le \int_\Omega g\, d\gamma_n + \int_{\Omega^c} g\, d\gamma_n = \int g\, d\gamma_n.$$

Finally $\int g\, d\gamma_n = \int (a - \eta_2)\, d\gamma_n = m$ by (1). Combining the three displays gives $a\gamma_n(\Omega) \le m$. This is exactly the step where nonnegativity of g off the contact set is consumed; it in turn came from $F \ge 0$. □

2.82. `YoungL2Contraction`.

Lemma 2.82 (Young L^2 contraction for a nonnegative kernel of unit mass). *Fix $n \in \mathbb{N}$ and let $\kappa : \mathbb{R}^n \to \mathbb{R}$ be measurable with*

$$\kappa(t) \ge 0 \ \text{ for every } t, \qquad \kappa \in L^1(\mathbb{R}^n), \qquad \int_{\mathbb{R}^n} \kappa(t)\, dt = 1.$$

Let $h : \mathbb{R}^n \to \mathbb{R}$ be almost everywhere strongly measurable and let

$$(\kappa * h)(x) = \int_{\mathbb{R}^n} \kappa(t)\, h(x - t)\, dt.$$

Then

$$\|\kappa * h\|_{L^2(\mathbb{R}^n)} \le \|h\|_{L^2(\mathbb{R}^n)}.$$

Every measure in this statement is Lebesgue *measure on $\mathbb{R}^n$: no Gaussian measure and no Gaussian weight occurs anywhere, in the integrand or in the measure. The two norms are the extended-valued L^2 seminorms, taking values in $[0, +\infty]$, so the inequality is meaningful without assuming $h \in L^2$; when $\|h\|_{L^2} = +\infty$ it holds trivially, and the content of the lemma is the case $\|h\|_{L^2} < \infty$, where it says that convolution with κ is a contraction of $L^2(\mathbb{R}^n)$. Note also that the conclusion is global on $\mathbb{R}^n$: no compact set, and no thickening of one, enters.*

Proof. All integrals below are against Lebesgue measure and are taken in $[0, +\infty]$, as integrals of nonnegative measurable functions; this is what makes every step available without a prior integrability hypothesis on h.

Step 0: the mass hypothesis in the nonnegative calculus. Since κ is integrable and nonnegative, its upper integral agrees with its Lebesgue integral, so

$$\int_{\mathbb{R}^n} \kappa(t)\, dt = 1 \qquad \text{holds in } [0, +\infty] \text{ as well.} \tag{0}$$

Step 1: the triangle inequality under the convolution integral. Fix $x \in \mathbb{R}^n$. Taking absolute values inside the defining integral of the convolution and using $\kappa \geq 0$,

$$|(\kappa * h)(x)| \leq \int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|\,dt. \tag{1}$$

The integrand on the right is measurable in t: κ is measurable by hypothesis, and $t \mapsto h(x-t)$ is almost everywhere strongly measurable because $t \mapsto x - t$ preserves Lebesgue measure.

Step 2: Cauchy–Schwarz against the kernel, giving a pointwise Jensen bound. Still with x fixed, split the kernel as $\kappa(t) = \sqrt{\kappa(t)} \cdot \sqrt{\kappa(t)}$, which is legitimate because $\kappa(t) \geq 0$, and write

$$\kappa(t)\,|h(x-t)| = \sqrt{\kappa(t)} \cdot \Big(\sqrt{\kappa(t)}\,|h(x-t)|\Big).$$

Hölder's inequality with the conjugate pair $(2,2)$ applied to the two factors gives

$$\int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|\,dt \leq \Big(\int_{\mathbb{R}^n} \kappa(t)\,dt\Big)^{1/2} \Big(\int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|^2\,dt\Big)^{1/2},$$

where the first bracket arose from $\big(\sqrt{\kappa}\big)^2 = \kappa$ and the second from $\big(\sqrt{\kappa}\,|h(x-\cdot)|\big)^2 = \kappa\,|h(x-\cdot)|^2$, again by $\kappa \geq 0$. By (0) the first bracket is 1. Combining with (1) and squaring,

$$|(\kappa * h)(x)|^2 \leq \int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|^2\,dt \qquad \text{for every } x \in \mathbb{R}^n. \tag{2}$$

This is the Jensen inequality for the probability density κ, obtained here through Cauchy–Schwarz.

Step 3: the degenerate case. If $\int_{\mathbb{R}^n} |h|^2 = +\infty$ then $\|h\|_{L^2} = +\infty$ and the asserted inequality holds because the right-hand side is the top element of $[0, +\infty]$. So assume from now on

$$\int_{\mathbb{R}^n} |h(y)|^2\,dy < \infty. \tag{3}$$

This finiteness is used once, in Step 4, to move a constant out of an integral.

Step 4: integrate (2) *and interchange.* Integrating (2) in x over $\mathbb{R}^n$ and then exchanging the order of the two integrations,

$$\int_{\mathbb{R}^n} |(\kappa * h)(x)|^2\,dx \leq \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|^2\,dt\,dx = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|^2\,dx\,dt.$$

The interchange is Tonelli's theorem, which applies because the integrand is nonnegative and jointly measurable on $\mathbb{R}^n \times \mathbb{R}^n$: it is the composition of the jointly measurable map $(z_1, z_2) \mapsto \kappa(z_2)\,|h(z_1)|^2$ with the measure-preserving substitution $(x,t) \mapsto (x-t, t)$ of $\mathbb{R}^n \times \mathbb{R}^n$.

For each fixed t, the inner integral is computed by translation invariance of Lebesgue measure:

$$\int_{\mathbb{R}^n} \kappa(t)\,|h(x-t)|^2\,dx = \kappa(t) \int_{\mathbb{R}^n} |h(x-t)|^2\,dx = \kappa(t) \int_{\mathbb{R}^n} |h(y)|^2\,dy,$$

the constant $\kappa(t)$ coming out because it is finite. Integrating in t and using (3) to pull the now-constant factor $\int |h|^2$ out, and then (0),

$$\int_{\mathbb{R}^n} |(\kappa * h)(x)|^2\,dx \leq \Big(\int_{\mathbb{R}^n} \kappa(t)\,dt\Big) \int_{\mathbb{R}^n} |h(y)|^2\,dy = \int_{\mathbb{R}^n} |h(y)|^2\,dy. \tag{4}$$

Step 5: take square roots. Both L^2 seminorms are the one-half power of the corresponding integral of squares, and $u \mapsto u^{1/2}$ is monotone on $[0, +\infty]$, so (4) gives $\|\kappa * h\|_{L^2} \leq \|h\|_{L^2}$, which is the assertion. □

2.83. `MollifierL2Local`.

Lemma 2.83 (Mollifications of an L^2_{loc} function converge in L^2 on compact sets)**.** *Write λ for Lebesgue measure on $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be such that $f \in L^2(K, \lambda)$ for every compact $K \subseteq \mathbb{R}^n$; that is, $f \in L^2_{\mathrm{loc}}(\mathbb{R}^n)$. Let $(\rho_\varepsilon)_{\varepsilon>0}$ be a mollifier family: for each $\varepsilon > 0$ the function*

$\rho_\varepsilon : \mathbb{R}^n \to \mathbb{R}$ is smooth, satisfies $\rho_\varepsilon \geq 0$, is supported in the closed ball $\{|t| \leq \varepsilon\}$, and has $\int_{\mathbb{R}^n} \rho_\varepsilon \, d\lambda = 1$. Write

$$(\rho_\varepsilon * f)(x) = \int_{\mathbb{R}^n} \rho_\varepsilon(t)\, f(x-t)\, d\lambda(t).$$

Then for every compact $K \subseteq \mathbb{R}^n$,

$$\big\|\rho_\varepsilon * f - f\big\|_{L^2(K,\lambda)} \longrightarrow 0 \qquad \text{as } \varepsilon \downarrow 0$$

through positive values of ε, the norms being read in $[0,+\infty]$. No continuity, boundedness or global integrability of f is assumed.

Proof. Fix the compact set K. For $r > 0$ put $K_r = \{x \in \mathbb{R}^n : \operatorname{dist}(x,K) \leq r\}$; each K_r is closed and bounded, hence compact, and $\lambda(K) \leq \lambda(K_r) < \infty$. Throughout, ε is restricted to $0 < \varepsilon \leq 1$. If $K = \emptyset$ every norm below is 0 and there is nothing to prove, so assume $K \neq \emptyset$.

Step 0: the convolution is defined on K. Let $0 < \varepsilon \leq 1$ and $x \in K$. Since $\operatorname{supp}\rho_\varepsilon \subseteq \{|t| \leq \varepsilon\}$, the integrand $t \mapsto \rho_\varepsilon(t) f(x-t)$ vanishes unless $|t| \leq \varepsilon \leq 1$, and for such t we have $x - t \in K_1$. Now $f \in L^2(K_1,\lambda)$ by hypothesis and $\lambda(K_1) < \infty$, so Cauchy–Schwarz (shared context) on the measure space (K_1,λ), applied to $|f|$ and the constant 1, gives $\int_{K_1} |f|\, d\lambda \leq \|f\|_{L^2(K_1,\lambda)} \lambda(K_1)^{1/2} < \infty$. As ρ_ε is continuous with compact support it is bounded, say by M_ε, so $\int |\rho_\varepsilon(t) f(x-t)|\, d\lambda(t) \leq M_\varepsilon \int_{K_1} |f|\, d\lambda < \infty$ after the change of variable $y = x - t$, which preserves λ by translation invariance and reflection invariance of Lebesgue measure (shared context). Hence $(\rho_\varepsilon * f)(x)$ is a well-defined real number for every $x \in K$ and every $0 < \varepsilon \leq 1$.

Step 1: a uniform local L^2 bound for the mollification. Claim. For every measurable $h : \mathbb{R}^n \to \mathbb{R}$ with $h \in L^2(K_1,\lambda)$ and every $0 < \varepsilon \leq 1$,

$$\|\rho_\varepsilon * h\|_{L^2(K,\lambda)} \;\leq\; \|h\|_{L^2(K_1,\lambda)}. \tag{1}$$

Proof of the claim. Fix $0 < \varepsilon \leq 1$. Because $\rho_\varepsilon \geq 0$ and $\int \rho_\varepsilon \, d\lambda = 1$, the measure $\mu_\varepsilon = \rho_\varepsilon \, d\lambda$ is a probability measure on $\mathbb{R}^n$. Fix $x \in K$. The function $t \mapsto h(x-t)$ is μ_ε-integrable by the computation of Step 0 applied to h, and $s \mapsto s^2$ is convex on $\mathbb{R}$, so Jensen's inequality for a probability measure (shared context) gives

$$\big((\rho_\varepsilon * h)(x)\big)^2 = \Big(\int h(x-t)\, d\mu_\varepsilon(t)\Big)^2 \;\leq\; \int h(x-t)^2 \, d\mu_\varepsilon(t) \;=\; \int \rho_\varepsilon(t)\, h(x-t)^2 \, d\lambda(t).$$

The function $(x,t) \mapsto \rho_\varepsilon(t) h(x-t)^2$ is nonnegative and jointly measurable, being the product of a measurable function of t with the composition of the measurable h^2 with the continuous map $(x,t) \mapsto x - t$. Integrating the displayed inequality over $x \in K$ and applying Tonelli's theorem (shared context) to interchange the order of integration,

$$\int_K \big((\rho_\varepsilon * h)(x)\big)^2 d\lambda(x) \;\leq\; \int \rho_\varepsilon(t) \Big(\int_K h(x-t)^2 \, d\lambda(x)\Big) d\lambda(t).$$

For $|t| \leq \varepsilon \leq 1$, translation invariance of Lebesgue measure (shared context) gives $\int_K h(x-t)^2 d\lambda(x) = \int_{K-t} h(y)^2 d\lambda(y)$, and $K - t \subseteq K_1$ because every $y \in K - t$ satisfies $\operatorname{dist}(y,K) \leq |t| \leq 1$; since $h^2 \geq 0$ this yields $\int_K h(x-t)^2 d\lambda(x) \leq \|h\|^2_{L^2(K_1,\lambda)}$. For $|t| > \varepsilon$ the factor $\rho_\varepsilon(t)$ vanishes. Hence the right-hand side is at most $\|h\|^2_{L^2(K_1,\lambda)} \int \rho_\varepsilon \, d\lambda = \|h\|^2_{L^2(K_1,\lambda)}$. Taking square roots gives (1) and proves the claim.

Step 2: the case of a continuous, compactly supported function. Let $g : \mathbb{R}^n \to \mathbb{R}$ be continuous with compact support. By the density item of shared context, such a g is bounded and uniformly continuous; let $\omega(\varepsilon) = \sup\{|g(y) - g(z)| : |y - z| \leq \varepsilon\}$ be its modulus of continuity, so $\omega(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$. For $0 < \varepsilon \leq 1$ and any x, using $\int \rho_\varepsilon \, d\lambda = 1$ to write $g(x) = \int \rho_\varepsilon(t) g(x) \, d\lambda(t)$ (legitimate because g is bounded and ρ_ε is integrable),

$$\big|(\rho_\varepsilon * g)(x) - g(x)\big| = \Big|\int \rho_\varepsilon(t) \big(g(x-t) - g(x)\big) d\lambda(t)\Big| \;\leq\; \int \rho_\varepsilon(t)\, \big|g(x-t) - g(x)\big|\, d\lambda(t) \;\leq\; \omega(\varepsilon),$$

the first inequality by Jensen's inequality (shared context) applied to the probability measure $\mu_\varepsilon = \rho_\varepsilon \, d\lambda$ of Step 1 and the convex function $s \mapsto |s|$, and the rest $\rho_\varepsilon(t) = 0$ unless $|t| \leq \varepsilon$, in

which case $|g(x-t)-g(x)| \le \omega(\varepsilon)$, and again $\int \rho_\varepsilon \, d\lambda = 1$. Therefore

$$\|\rho_\varepsilon * g - g\|_{L^2(K,\lambda)} \;\le\; \omega(\varepsilon)\,\lambda(K)^{1/2} \qquad (0<\varepsilon\le 1). \tag{2}$$

Step 3: approximation and conclusion. Let $\eta > 0$. Put $f_1 = \mathbf{1}_{K_2} f$. Then f_1 is measurable, vanishes off the compact set K_2, and $\int_{\mathbb{R}^n} |f_1|^2 d\lambda = \int_{K_2} |f|^2 d\lambda < \infty$ by hypothesis, so $f_1 \in L^2(\mathbb{R}^n,\lambda)$. Lebesgue measure is a regular Borel measure, so by the density item of shared context there is $g : \mathbb{R}^n \to \mathbb{R}$ continuous with compact support and

$$\|f_1 - g\|_{L^2(\mathbb{R}^n,\lambda)} \;\le\; \eta/3. \tag{3}$$

Next, for $0 < \varepsilon \le 1$ and $x \in K$ we have $(\rho_\varepsilon * f)(x) = (\rho_\varepsilon * f_1)(x)$: the integrand vanishes unless $|t| \le \varepsilon \le 1$, and then $x - t \in K_1 \subseteq K_2$, where $f_1 = f$. Also $f = f_1$ on $K \subseteq K_2$. Hence, pointwise on K,

$$\rho_\varepsilon * f - f = \rho_\varepsilon * (f_1 - g) \;+\; \big(\rho_\varepsilon * g - g\big) \;+\; \big(g - f_1\big),$$

where we used linearity of the convolution in its second argument, legitimate because $f_1 - g$ and g are each in $L^2(K_1,\lambda)$ (the first by (3), the second because g is bounded and $\lambda(K_1) < \infty$), so all three convolutions are defined by Step 0. Taking $L^2(K,\lambda)$ norms and using the triangle inequality for the norm of $L^2(K,\lambda)$,

$$\|\rho_\varepsilon * f - f\|_{L^2(K,\lambda)} \;\le\; \|\rho_\varepsilon * (f_1 - g)\|_{L^2(K,\lambda)} + \|\rho_\varepsilon * g - g\|_{L^2(K,\lambda)} + \|g - f_1\|_{L^2(K,\lambda)}.$$

By (1) applied to $h = f_1 - g$, the first term is at most $\|f_1 - g\|_{L^2(K_1,\lambda)} \le \|f_1 - g\|_{L^2(\mathbb{R}^n,\lambda)} \le \eta/3$ by (3). The third term is at most $\|g - f_1\|_{L^2(\mathbb{R}^n,\lambda)} \le \eta/3$, again by (3). By (2) the second term is at most $\omega(\varepsilon)\lambda(K)^{1/2}$, and since $\omega(\varepsilon) \to 0$ there is $\varepsilon_0 \in (0,1]$ with $\omega(\varepsilon)\lambda(K)^{1/2} \le \eta/3$ for all $0 < \varepsilon \le \varepsilon_0$. Altogether

$$\|\rho_\varepsilon * f - f\|_{L^2(K,\lambda)} \;\le\; \eta \qquad \text{for all } 0 < \varepsilon \le \varepsilon_0.$$

Since $\eta > 0$ was arbitrary, $\|\rho_\varepsilon * f - f\|_{L^2(K,\lambda)} \to 0$ as $\varepsilon \downarrow 0$ through positive values, which is the assertion. □

2.84. `ConvolutionFDerivOfWeakGrad`.

Lemma 2.84 (Mollification commutes with the weak gradient)**.** *Fix $n \in \mathbb{N}_0$ and write λ for Lebesgue measure on $\mathbb{R}^n$. Let $\rho : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $v : \mathbb{R}^n \to \mathbb{R}$ be locally λ-integrable, and let $G = (G_1, \dots, G_n) : \mathbb{R}^n \to \mathbb{R}^n$ be a weak Euclidean gradient of v in the sense of `HasGaussWeakGrad`. Then for every coordinate index $j \in \{1, \dots, n\}$ and every point $x \in \mathbb{R}^n$,*

$$\partial_j(\rho * v)(x) = (\rho * G_j)(x),$$

where $*$ denotes convolution against λ with the pointwise product as its bilinear pairing, $(\rho * w)(x) = \int_{\mathbb{R}^n} \rho(y)\, w(x-y)\, d\lambda(y)$, and ∂_j on the left is the Fréchet derivative of the mollification, evaluated at the j-th standard basis vector e_j. Both sides are real numbers, for each fixed j and x; the identity is pointwise in x, not an almost-everywhere one.

Every measure in this statement is Lebesgue measure: v is assumed locally λ-integrable, both convolutions are against λ, and the weak gradient of `HasGaussWeakGrad` is, by that node's own definition, the ordinary distributional gradient, its two integrals taken against λ and not against a Gaussian weight. No Gaussian density enters anywhere, and the γ_n of the surrounding development plays no role here.

Proof. Fix j and x. Write $L : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ for the pointwise-multiplication pairing, so that $\rho * w$ means convolution with respect to L and λ. Two preliminary remarks about ρ: being smooth it is in particular C^1, and its derivative $y \mapsto D\rho(y)$ is continuous and again compactly supported, the support of a derivative being contained in that of the function.

Step 0 (symmetry of the convolution). Because L is the symmetric pairing of multiplication on $\mathbb{R}$ — it equals its own flip, $L^{\mathrm{t}} = L$ — convolution against L is commutative where defined: $\rho * v = v * \rho$. We may therefore compute with $v * \rho$, which puts the smooth factor on the right, where the differentiation theorem wants it.

Step A (differentiating the mollification). Since ρ is C^1 with $D\rho$ continuous and compactly supported, and v is locally λ-integrable, the convolution $v * \rho$ is differentiable at x and its derivative may be computed by differentiating inside the integral, in the second factor:

$$D(v * \rho)(x) = \big(v * D\rho\big)(x),$$

an identity of continuous linear maps $\mathbb{R}^n \to \mathbb{R}$, where on the right $D\rho$ is paired with v through L in its scalar slot. This is the standard theorem that a convolution inherits the smoothness of whichever factor is smooth, and it is where the compact support of $D\rho$ is used, to make the defining integral converge for a merely locally integrable v.

Step B (evaluating in the direction e_j). Applying the linear map of Step A to e_j and exchanging that application with the integral, which is legitimate because evaluation at e_j is a continuous linear functional,

$$\partial_j(v * \rho)(x) = \big(a \mapsto (D\rho(a))(e_j)\big) * v\,(x) \quad \text{after a further use of Step 0,}$$

so the left-hand side of the lemma is the convolution of v against the scalar function $a \mapsto (D\rho(a))(e_j) = \partial_j\rho(a)$.

Step C (the weak-gradient identity with a reflected test function). Fix x and consider the function

$$\varphi(y) = \rho(x - y),$$

the reflection of ρ about x. It is smooth and compactly supported, hence an admissible test function, and its j-th partial derivative is $\partial_j\varphi(y) = -(\partial_j\rho)(x - y)$ by the chain rule for the reflection $y \mapsto x - y$, whose derivative is $-\mathrm{id}$. Applying `HasGaussWeakGrad` to this φ and this j gives

$$\int_{\mathbb{R}^n} v(y)\,\partial_j\varphi(y)\,d\lambda(y) = -\int_{\mathbb{R}^n} G_j(y)\,\varphi(y)\,d\lambda(y),$$

that is, after substituting the two displayed formulas for φ and $\partial_j\varphi$ and cancelling the two minus signs,

$$\int_{\mathbb{R}^n} v(y)\,(\partial_j\rho)(x - y)\,d\lambda(y) = \int_{\mathbb{R}^n} G_j(y)\,\rho(x - y)\,d\lambda(y).$$

The left-hand side is the convolution appearing at the end of Step B, and the right-hand side is $(G_j * \rho)(x)$.

Step D (conclusion). Chaining Step 0, Step B and Step C,

$$\partial_j(\rho * v)(x) = \partial_j(v * \rho)(x) = \big(\partial_j\rho\big) * v\,(x) = \big(G_j * \rho\big)(x) = \big(\rho * G_j\big)(x),$$

the last equality being commutativity of the convolution once more, available because L equals its own flip. Since j and x were arbitrary, this is the assertion. □

2.85. `NormIntegralLeOfProductDomination`.

Lemma 2.85 (An integral dominated by a product of two L^2 functions)**.** *Fix $n \in \mathbb{N}_0$ and write λ for Lebesgue measure on $\mathbb{R}^n$. Let $K \subseteq \mathbb{R}^n$ and let λ_K denote the restriction of λ to K, so that for a function f one writes $\|f\|_{L^2(K)} = \|f\|_{L^2(\lambda_K)}$ and $\int_K f\,d\lambda$ for the integral against λ_K. Let $h, a, b : \mathbb{R}^n \to \mathbb{R}$ and let $C \in \mathbb{R}$ with $C \geq 0$. Assume*

(1) *$a \in L^2(\lambda_K)$ and $b \in L^2(\lambda_K)$;*
(2) *$|h(x)| \leq C\,\big(|a(x)|\,|b(x)|\big)$ for every $x \in \mathbb{R}^n$.*

Then

$$\Big\|\int_K h\,d\lambda\Big\| \leq C\,\|a\|_{L^2(K)}\,\|b\|_{L^2(K)}.$$

Two points about the hypotheses, both of which are exactly as stated and not abbreviations of something stronger. First, every measure here is Lebesgue: λ_K is Lebesgue measure restricted to K and the $L^2(K)$ norms are taken against it, with no Gaussian weight anywhere, even though the lemma is used inside the tablet's Gaussian development. Second, nothing at all is assumed about h beyond the pointwise bound (2) — no integrability and no measurability — and no such hypothesis is needed: if h fails to be integrable then the Bochner integral $\int_K h\,d\lambda$ is by convention 0, while the right-hand side is nonnegative because $C \geq 0$ and norms are nonnegative, so the

inequality holds a fortiori. Correspondingly the lemma cites no other tablet node; it rests only on the shared background measure theory.

Proof. Step 1 (Cauchy–Schwarz on K). We first record that for $a, b \in L^2(\lambda_K)$,

$$\int_K |a|\,|b|\,d\lambda \le \|a\|_{L^2(K)}\,\|b\|_{L^2(K)}. \tag{CS}$$

Indeed, the product ab lies in $L^1(\lambda_K)$, by the Hölder pairing of the exponents 2 and 2 into 1, so it is λ_K-integrable. Since $\|a(x)b(x)\| = |a(x)|\,|b(x)|$ at every x, the integral on the left of (CS) is the $L^1(\lambda_K)$ norm of ab, read as a real number; and the Hölder inequality for the exponent pairing $1/2 + 1/2 = 1$ bounds that L^1 norm by the product of the two L^2 norms. Both sides are finite — the right side because a and b are in $L^2(\lambda_K)$, the left because of the bound itself — so passing from the inequality between extended-real norms to the inequality (CS) between their real values is legitimate, the monotonicity of $t \mapsto t$ on $[0,\infty]$ restricted to finite values being all that is used.

Step 2 (the integrand of the majorant is integrable). The function $x \mapsto |a(x)|\,|b(x)|$ is λ_K-integrable: it has the same absolute value as ab, which is integrable by Step 1, so integrability transfers.

Step 3 (dominating the integral). By hypothesis (2) the integrand h is dominated in absolute value by $C\,|a|\,|b|$, which is λ_K-integrable by Step 2 and by $C \ge 0$. The norm of an integral is at most the integral of the norm of the integrand, and the integral is monotone on integrable dominating pairs, so

$$\Big\|\int_K h\,d\lambda\Big\| \le \int_K |h|\,d\lambda \le \int_K C\,|a|\,|b|\,d\lambda.$$

(If h is not integrable the left side is 0 and the chain holds trivially, as noted in the statement.)

Step 4 (conclusion). Pulling the nonnegative constant C out of the last integral by linearity and applying (CS),

$$\int_K C\,|a|\,|b|\,d\lambda = C\int_K |a|\,|b|\,d\lambda \le C\,\|a\|_{L^2(K)}\,\|b\|_{L^2(K)},$$

the final inequality being (CS) multiplied by $C \ge 0$, which preserves the direction. Chaining with Step 3 gives the assertion. □

2.86. `GaussWeakGradChainRule`.

Lemma 2.86 (Chain rule for the weak Euclidean gradient)**.** *Fix $n \in \mathbb{N}_0$ and let γ_n be the Gaussian measure of* `GaussMeasure`*. Let $q, q' : \mathbb{R} \to \mathbb{R}$ and let $L_q \in \mathbb{R}$ satisfy $L_q \ge 0$, and assume*

1. *for every $t \in \mathbb{R}$ the function q is differentiable at t with derivative $q'(t)$;*
2. *q' is continuous (so $q \in C^1(\mathbb{R})$);*
3. *$|q'(t)| \le 1$ for every $t \in \mathbb{R}$;*
4. *q is 1-Lipschitz: $|q(s) - q(t)| \le |s - t|$ for all s, t;*
5. *q' is L_q-Lipschitz: $|q'(s) - q'(t)| \le L_q\,|s - t|$ for all s, t;*
6. *$q(0) = 0$.*

Let further $U : \mathbb{R}^n \to \mathbb{R}$ and $G = (G_1, \dots, G_n) : \mathbb{R}^n \to \mathbb{R}^n$ satisfy

7. *$U \in L^2(\gamma_n)$;*
8. *each coordinate function $x \mapsto G_j(x)$ lies in $L^2(\gamma_n)$, for every $j \in \{1, \dots, n\}$;*
9. *G is a weak Euclidean gradient of U in the sense of* `HasGaussWeakGrad`*.*

Then the composition $x \mapsto q(U(x))$ has weak Euclidean gradient $x \mapsto q'(U(x))\,G(x)$, again in the sense of `HasGaussWeakGrad`*; here $q'(U(x))\,G(x)$ is the scalar multiple of the vector $G(x) \in \mathbb{R}^n$ by the real number $q'(U(x))$, so its j-th coordinate is $q'(U(x))\,G_j(x)$. Explicitly: for every $j \in \{1, \dots, n\}$ and every test function $\varphi \in C_c^\infty(\mathbb{R}^n)$,*

$$\int_{\mathbb{R}^n} q(U(x))\,\partial_j\varphi(x)\,dx = -\int_{\mathbb{R}^n} q'(U(x))\,G_j(x)\,\varphi(x)\,dx,$$

both integrals being taken with respect to Lebesgue measure.

Proof. Fix $j \in \{1, \dots, n\}$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$; we must prove the displayed identity.

Step 0 (consequences of the hypotheses on q). Differentiability at every point gives continuity of q. Taking $t = 0$ in the 1-Lipschitz hypothesis and using $q(0) = 0$ gives

$$|q(t)| = |q(t) - q(0)| \le |t| \qquad (t \in \mathbb{R}), \tag{Q}$$

so q grows at most linearly; this is what makes $q \circ U$ as integrable as U itself.

Step 1 (passing to Lebesgue measure on a compact set). By `GaussL2LocallyIntegrable`, a function in $L^2(\gamma_n)$ is locally Lebesgue-integrable and lies in $L^2(K, \lambda)$ for every compact $K \subseteq \mathbb{R}^n$, λ denoting Lebesgue measure; the reason is that the Gaussian density of `GaussMeasure` is continuous and strictly positive, hence bounded below on K. Applying this to U and to G_j gives $U, G_j \in L^2(K, \lambda)$ for every compact K. Put

$$K = \operatorname{supp} \varphi \cup \operatorname{supp}(\partial_j \varphi),$$

a compact set, and note that φ and $\partial_j \varphi$ both vanish off K and are bounded, say $|\varphi| \le M_\varphi$. Consequently every integral appearing below may be restricted to K, and each is finite: by (Q) and Cauchy–Schwarz on (K, λ), $q(U)\, \partial_j \varphi$ is integrable because $|q(U)| \le |U|$ with $U, \partial_j \varphi \in L^2(K, \lambda)$, and $q'(U)\, G_j\, \varphi$ is integrable because $|q'(U)| \le 1$, φ is bounded and $G_j \in L^1(K, \lambda)$.

Step 2 (mollification). Fix a mollifier family $(\rho_\varepsilon)_{\varepsilon > 0}$ as in `MollifierL2Local`: each ρ_ε is smooth, nonnegative, supported in the closed ball of radius ε, with $\int \rho_\varepsilon \, d\lambda = 1$. Set

$$U_\varepsilon = \rho_\varepsilon * U, \qquad G_{j,\varepsilon} = \rho_\varepsilon * G_j.$$

Since U is locally integrable and ρ_ε is smooth with compact support, U_ε is smooth and $G_{j,\varepsilon}$ is continuous.

Step 3 (mollification commutes with the weak gradient). For every x,

$$\partial_j U_\varepsilon(x) = G_{j,\varepsilon}(x).$$

Indeed, differentiating under the convolution integral is legitimate here (ρ_ε is smooth and compactly supported, U locally integrable), and it turns $\partial_j(\rho_\varepsilon * U)(x)$ into the pairing of U against ∂_j of the translated and reflected bump $t \mapsto \rho_\varepsilon(x - t)$, which is itself a C_c^∞ test function. Hypothesis (9) applied to that test function replaces the pairing with U by the pairing with G_j, which is exactly $(\rho_\varepsilon * G_j)(x)$.

Step 4 (the identity for the mollified function). U_ε is smooth and $q \in C^1$, so the classical chain rule gives $\partial_j(q \circ U_\varepsilon) = q'(U_\varepsilon)\, \partial_j U_\varepsilon$ pointwise, and all the functions involved are continuous. Integration by parts against the compactly supported φ, with no boundary term, then yields

$$\int_{\mathbb{R}^n} q(U_\varepsilon(x))\, \partial_j \varphi(x)\, dx = -\int_{\mathbb{R}^n} q'(U_\varepsilon(x))\, \partial_j U_\varepsilon(x)\, \varphi(x)\, dx = -\int_{\mathbb{R}^n} q'(U_\varepsilon(x))\, G_{j,\varepsilon}(x)\, \varphi(x)\, dx,$$

the last step by Step 3. This is the asserted identity with (U, G_j) replaced by $(U_\varepsilon, G_{j,\varepsilon})$.

Step 5 (the two $L^2(K)$ convergences). By `MollifierL2Local` applied to U and to G_j, both of which lie in $L^2(K', \lambda)$ for every compact K' by Step 1,

$$\|U_\varepsilon - U\|_{L^2(K,\lambda)} \to 0, \qquad \|G_{j,\varepsilon} - G_j\|_{L^2(K,\lambda)} \to 0 \qquad (\varepsilon \downarrow 0). \tag{C}$$

In particular $\|G_{j,\varepsilon}\|_{L^2(K,\lambda)}$ is bounded for small ε, being convergent.

Step 6 (passing to the limit on the left). Using the 1-Lipschitz hypothesis pointwise and then Cauchy–Schwarz on (K, λ),

$$\Big| \int_K \big(q(U_\varepsilon) - q(U)\big) \partial_j \varphi \, d\lambda \Big| \le \int_K |U_\varepsilon - U|\, |\partial_j \varphi| \, d\lambda \le \|U_\varepsilon - U\|_{L^2(K,\lambda)}\, \|\partial_j \varphi\|_{L^2(K,\lambda)},$$

which tends to 0 by (C), the second factor being finite since $\partial_j \varphi$ is bounded with support in K and $\lambda(K) < \infty$. Hence the left-hand side of Step 4 converges to $\int q(U)\, \partial_j \varphi \, d\lambda$.

Step 7 (passing to the limit on the right). Split the difference of the integrands as

$$q'(U_\varepsilon) G_{j,\varepsilon} \varphi - q'(U) G_j \varphi = \big(q'(U_\varepsilon) - q'(U)\big) G_{j,\varepsilon} \varphi + q'(U) \big(G_{j,\varepsilon} - G_j\big) \varphi.$$

For the first term, the L_q-Lipschitz hypothesis on q' gives $|q'(U_\varepsilon)-q'(U)| \le L_q|U_\varepsilon-U|$ pointwise, so with $|\varphi| \le M_\varphi$ and Cauchy–Schwarz on (K,λ),

$$\Big|\int_K \big(q'(U_\varepsilon)-q'(U)\big)G_{j,\varepsilon}\varphi\,d\lambda\Big| \le L_q M_\varphi\,\|U_\varepsilon-U\|_{L^2(K,\lambda)}\,\|G_{j,\varepsilon}\|_{L^2(K,\lambda)},$$

which tends to 0 by (C) together with the boundedness of $\|G_{j,\varepsilon}\|_{L^2(K,\lambda)}$ noted in Step 5. For the second term, $|q'(U)| \le 1$ by hypothesis (3), so again by Cauchy–Schwarz

$$\Big|\int_K q'(U)\big(G_{j,\varepsilon}-G_j\big)\varphi\,d\lambda\Big| \le M_\varphi\,\|G_{j,\varepsilon}-G_j\|_{L^2(K,\lambda)}\,\lambda(K)^{1/2},$$

which tends to 0 by (C). Hence the right-hand side of Step 4 converges to $-\int q'(U)\,G_j\,\varphi\,d\lambda$.

Step 8 (conclusion). For each $\varepsilon>0$ the two sides of Step 4 are equal, and by Steps 6 and 7 they converge respectively to $\int q(U)\,\partial_j\varphi\,d\lambda$ and $-\int q'(U)\,G_j\,\varphi\,d\lambda$ as $\varepsilon\downarrow 0$. A convergent net of reals has a unique limit, so the two limits agree:

$$\int_{\mathbb{R}^n} q(U(x))\,\partial_j\varphi(x)\,dx = -\int_{\mathbb{R}^n} q'(U(x))\,G_j(x)\,\varphi(x)\,dx.$$

Since j and φ were arbitrary, and the j-th coordinate of $q'(U(x))\,G(x)$ is $q'(U(x))\,G_j(x)$, the function $x\mapsto q(U(x))$ has weak Euclidean gradient $x\mapsto q'(U(x))\,G(x)$, as asserted. □

2.87. `SobolevChainRule`.

Lemma 2.87 (Sobolev chain rule for the smooth regularized absolute value)**.** *For $\delta>0$ let $q_\delta(t)=\sqrt{t^2+\delta^2}-\delta$. Let $u\in W^{1,2}(\gamma_n)$ in the sense of `SobolevW12` and let $w\in L^2(\gamma_n)$ satisfy $w=q_\delta\circ u$ γ_n-almost everywhere. Then $w\in W^{1,2}(\gamma_n)$ and*

$$Dw = \frac{u}{\sqrt{u^2+\delta^2}}\,Du \qquad \gamma_n\text{-a.e.},$$

with D the energy-normalized weak gradient of `GaussD`. No regularity of the set $\{u>0\}$ or of its boundary is used; the statement concerns the local weak gradient only, not any locality of the nonlocal operator B.

Proof. Step 0: properties of q_δ and notation. q_δ is smooth on $\mathbb{R}$ with $q_\delta'(t)=t/\sqrt{t^2+\delta^2}$ and $q_\delta''(t)=\delta^2(t^2+\delta^2)^{-3/2}$. Hence $|q_\delta'|\le 1$ everywhere, q_δ is 1-Lipschitz, $q_\delta(0)=0$, and q_δ' is Lipschitz with constant $\sup|q_\delta''|=\delta^2\cdot\delta^{-3}=1/\delta$. Set

$$G=\sqrt{2}\,Du\in L^2(\gamma_n;\mathbb{R}^n), \qquad \Theta(x)=q_\delta'\big(u(x)\big)=\frac{u(x)}{\sqrt{u(x)^2+\delta^2}},$$

so G is a weak Euclidean gradient of u by `SobolevW12` and `GaussD`, and $|\Theta|\le 1$ pointwise. Then $\Theta\,G\in L^2(\gamma_n;\mathbb{R}^n)$, since $|\Theta G|\le|G|$ pointwise. Our goal is to show that $\Theta\,G$ is a weak Euclidean gradient of w; the conclusion then follows, because $\Theta G=\sqrt{2}\,(\Theta\,Du)$ with $\Theta\,Du\in L^2(\gamma_n;\mathbb{R}^n)$ gives $w\in W^{1,2}(\gamma_n)$ by `SobolevW12`, and `GaussD` together with `GaussWeakGradUnique` identifies $Dw=\Theta\,Du$, which is the asserted formula.

Since $w=q_\delta\circ u$ γ_n-a.e., and γ_n and Lebesgue measure have the same null sets (the density of `GaussMeasure` is continuous and strictly positive), we may and do replace w by $q_\delta\circ u$ throughout: changing a function on a null set changes neither side of the defining identity of `HasGaussWeakGrad`.

Step 1: reduction to a fixed test function and local mollification. Fix $j\in\{1,\dots,n\}$ and $\varphi\in C_c^\infty(\mathbb{R}^n)$; we must show

$$\int_{\mathbb{R}^n} q_\delta(u)\,\partial_j\varphi\,dx = -\int_{\mathbb{R}^n}\Theta\,G_j\,\varphi\,dx. \tag{1}$$

Let $K=\operatorname{supp}\varphi$, a compact subset of $\mathbb{R}^n$, let λ denote Lebesgue measure, and put $K_1=\{x:\operatorname{dist}(x,K)\le 1\}$, which is compact.

By `GaussL2LocallyIntegrable`, applied in turn to u and to each component G_j of G – all of these lie in $L^2(\gamma_n)$, u by hypothesis and G_j because $G\in L^2(\gamma_n;\mathbb{R}^n)$ – each of $u,G_1,\dots,G_n$ is locally λ-integrable and lies in $L^2(K',\lambda)$ for every compact $K'\subseteq\mathbb{R}^n$.

Fix a mollifier family $(\rho_\varepsilon)_{\varepsilon>0}$: each ρ_ε smooth, nonnegative, supported in $\{|t| \le \varepsilon\}$, with $\int \rho_\varepsilon \, d\lambda = 1$; for instance $\rho_\varepsilon(t) = \varepsilon^{-n}\rho(t/\varepsilon)$ for a fixed smooth nonnegative ρ supported in the unit ball with $\int \rho \, d\lambda = 1$. Put $u_\varepsilon = \rho_\varepsilon * u$, that is $u_\varepsilon(x) = \int \rho_\varepsilon(t)\, u(x-t)\, d\lambda(t)$. Since u is locally λ-integrable and $\rho_\varepsilon \in C_c^\infty(\mathbb{R}^n)$, the mollification item of shared context gives that u_ε is defined at every point and is C^∞ on $\mathbb{R}^n$.

The gradient identity. For every $x \in \mathbb{R}^n$, every j and every $\varepsilon > 0$,

$$\partial_j u_\varepsilon(x) = (\rho_\varepsilon * G_j)(x). \tag{1a}$$

Indeed, fix x and put $\psi(y) = \rho_\varepsilon(x-y)$, so that $\psi \in C_c^\infty(\mathbb{R}^n)$ and $\partial_{y_j}\psi(y) = -(\partial_j\rho_\varepsilon)(x-y)$. Differentiating the convolution in x moves the derivative onto the smooth factor, $\partial_j u_\varepsilon = (\partial_j \rho_\varepsilon) * u$, and rewriting that integral by the change of variable $y = x - t$, which preserves λ by translation and reflection invariance of Lebesgue measure (shared context), gives

$$\partial_j u_\varepsilon(x) = \int u(y)\,(\partial_j\rho_\varepsilon)(x-y)\, d\lambda(y) = -\int u\, \partial_j\psi \, d\lambda = \int G_j\, \psi \, d\lambda = (\rho_\varepsilon * G_j)(x),$$

the third equality being the defining identity of `HasGaussWeakGrad` for the pair (u, G) with test function ψ, and the fourth the same change of variable read backwards.

The convergences. Apply `MollifierL2Local` twice with the compact set K_1 and the fixed mollifier family: once to $f = u$ and once to $f = G_j$. Its hypothesis, that f lie in $L^2(K', \lambda)$ for every compact K', is exactly what `GaussL2LocallyIntegrable` supplied above for both functions. Hence, as $\varepsilon \downarrow 0$ through positive values,

$$u_\varepsilon \to u \quad \text{in } L^2(K_1, \lambda), \qquad \partial_j u_\varepsilon \to G_j \quad \text{in } L^2(K_1, \lambda),$$

the second statement using (1a) to identify $\partial_j u_\varepsilon$ with $\rho_\varepsilon * G_j$. Since $K \subseteq K_1$, the same convergences hold in $L^2(K, \lambda)$, because $\|w\|_{L^2(K,\lambda)} \le \|w\|_{L^2(K_1,\lambda)}$ for every w.

Step 2: the classical chain rule for the mollifications. Each u_ε is smooth on a neighbourhood of K and q_δ is smooth on $\mathbb{R}$, so the classical chain rule gives, pointwise on K,

$$\partial_j\big(q_\delta(u_\varepsilon)\big) = q_\delta'(u_\varepsilon)\, \partial_j u_\varepsilon.$$

Both sides are continuous on a neighbourhood of K, and φ is smooth with support in K, so ordinary integration by parts gives

$$\int q_\delta(u_\varepsilon)\, \partial_j\varphi \, dx = -\int q_\delta'(u_\varepsilon)\, \partial_j u_\varepsilon \, \varphi \, dx. \tag{2}$$

Step 3: passing to the limit in (2). Left-hand side: $|q_\delta(u_\varepsilon) - q_\delta(u)| \le |u_\varepsilon - u|$ pointwise, because q_δ is 1-Lipschitz; hence $q_\delta(u_\varepsilon) \to q_\delta(u)$ in $L^1(K)$ by Step 1, and since $\partial_j\varphi$ is bounded and supported in K, $\int q_\delta(u_\varepsilon)\partial_j\varphi\, dx \to \int q_\delta(u)\, \partial_j\varphi\, dx$.

Right-hand side: write

$$q_\delta'(u_\varepsilon)\, \partial_j u_\varepsilon - \Theta\, G_j = q_\delta'(u_\varepsilon)\big(\partial_j u_\varepsilon - G_j\big) + \big(q_\delta'(u_\varepsilon) - q_\delta'(u)\big)G_j.$$

The first term tends to 0 in $L^1(K)$ because $|q_\delta'| \le 1$ and $\partial_j u_\varepsilon \to G_j$ in $L^2(K, \lambda)$ and hence in $L^1(K, \lambda)$ (Step 1, with Cauchy–Schwarz on the finite-measure set K). For the second term, q_δ' is $1/\delta$-Lipschitz by Step 0, so $|q_\delta'(u_\varepsilon) - q_\delta'(u)| \le |u_\varepsilon - u|/\delta$, and by Cauchy–Schwarz on K,

$$\int_K \big|q_\delta'(u_\varepsilon) - q_\delta'(u)\big|\, |G_j|\, dx \le \frac{1}{\delta}\, \|u_\varepsilon - u\|_{L^2(K)}\, \|G_j\|_{L^2(K)} \longrightarrow 0.$$

Hence $\int q_\delta'(u_\varepsilon)\partial_j u_\varepsilon\, \varphi\, dx \to \int \Theta\, G_j\, \varphi\, dx$, φ being bounded with support in K.

Letting $\varepsilon \downarrow 0$ in (2) therefore yields (1).

Step 4: conclusion. Since j and φ were arbitrary, (1) says exactly that $\Theta\, G$ is a weak Euclidean gradient of $q_\delta \circ u = w$ in the sense of `HasGaussWeakGrad`. By Step 0 this gives $w \in W^{1,2}(\gamma_n)$ and $Dw = \Theta\, Du = \dfrac{u}{\sqrt{u^2 + \delta^2}}\, Du$ γ_n-a.e. □

2.88. `GradientLocality`.

Lemma 2.88 (The gradient vanishes off the positivity set)**.** *Let $u \in W^{1,2}(\gamma_n)$ in the sense of `SobolevW12` be nonnegative γ_n-almost everywhere. Then*

$$Du = \mathbf{1}_{\{u>0\}}\, Du \qquad \text{in } L^2(\gamma_n;\mathbb{R}^n),$$

that is, $Du = 0$ γ_n-almost everywhere on $\{u \le 0\}$, with D the energy-normalized weak gradient of `GaussD`.

Proof. For $\delta > 0$ let $q_\delta(t) = \sqrt{t^2+\delta^2} - \delta$ and let $w_\delta \in L^2(\gamma_n)$ be the class of $q_\delta \circ u$. This is a legitimate element of $L^2(\gamma_n)$: q_δ is 1-Lipschitz with $q_\delta(0) = 0$, so $|q_\delta(t)| \le |t|$ and hence $|w_\delta| \le |u|$ γ_n-a.e. By `SobolevChainRule`,

$$w_\delta \in W^{1,2}(\gamma_n), \qquad Dw_\delta = \Theta_\delta\, Du \ \ \gamma_n\text{-a.e.}, \qquad \Theta_\delta = \frac{u}{\sqrt{u^2+\delta^2}}. \tag{1}$$

Step 1: $w_\delta \to u$ in $L^2(\gamma_n)$ as $\delta \downarrow 0$. Fix a representative with $u \ge 0$ everywhere off a γ_n-null set. At such a point, $\sqrt{u^2+\delta^2} \ge \max\{u,\delta\}$, hence

$$0 \le u - q_\delta(u) = u + \delta - \sqrt{u^2+\delta^2} \le \delta,$$

because $\sqrt{u^2+\delta^2} \ge u$ gives the upper bound δ and $\sqrt{u^2+\delta^2} \le u+\delta$ (square both sides: $u^2+\delta^2 \le u^2 + 2u\delta + \delta^2$, true since $u \ge 0$) gives the lower bound 0. Therefore $|u - w_\delta| \le \delta$ γ_n-a.e., and since γ_n is a probability measure (`GaussIsProb`),

$$\|u - w_\delta\|^2_{L^2(\gamma_n)} \le \delta^2 \longrightarrow 0 \qquad (\delta \downarrow 0).$$

Step 2: $Dw_\delta \to \mathbf{1}_{\{u>0\}} Du$ in $L^2(\gamma_n;\mathbb{R}^n)$. Pointwise γ_n-a.e., using $u \ge 0$: if $u(x) > 0$ then $\Theta_\delta(x) = u(x)/\sqrt{u(x)^2+\delta^2} \to 1$ as $\delta \downarrow 0$; and if $u(x) = 0$ then $\Theta_\delta(x) = 0$ for every $\delta > 0$. Hence $\Theta_\delta \to \mathbf{1}_{\{u>0\}}$ γ_n-a.e. Moreover $|\Theta_\delta| \le 1$, so

$$\left|\Theta_\delta\, Du - \mathbf{1}_{\{u>0\}} Du\right|^2 \le 4\,|Du|^2 \qquad \gamma_n\text{-a.e.},$$

and $|Du|^2 \in L^1(\gamma_n)$ because $Du \in L^2(\gamma_n;\mathbb{R}^n)$. The dominated convergence theorem therefore gives $\Theta_\delta\, Du \to \mathbf{1}_{\{u>0\}} Du$ in $L^2(\gamma_n;\mathbb{R}^n)$ as $\delta \downarrow 0$; by (1) this is $Dw_\delta \to \mathbf{1}_{\{u>0\}} Du$.

Step 3: closedness of the weak gradient. Choose any sequence $\delta_k \downarrow 0$. By (1) and `SobolevW12`, $\sqrt{2}\, Dw_{\delta_k}$ is a weak Euclidean gradient of w_{δ_k} in the sense of `HasGaussWeakGrad`, for each k. By Steps 1 and 2, $w_{\delta_k} \to u$ in $L^2(\gamma_n)$ and $\sqrt{2}\, Dw_{\delta_k} \to \sqrt{2}\, \mathbf{1}_{\{u>0\}} Du$ in $L^2(\gamma_n;\mathbb{R}^n)$. Hence `GaussWeakGradClosed` applies and shows that $\sqrt{2}\, \mathbf{1}_{\{u>0\}} Du$ is a weak Euclidean gradient of u.

Step 4: conclusion. By hypothesis $u \in W^{1,2}(\gamma_n)$, so by `SobolevW12` and `GaussD` the field $\sqrt{2}\, Du$ is also a weak Euclidean gradient of u. Both $\sqrt{2}\, Du$ and $\sqrt{2}\, \mathbf{1}_{\{u>0\}} Du$ lie in $L^2(\gamma_n;\mathbb{R}^n)$ (the latter since $|\mathbf{1}_{\{u>0\}} Du| \le |Du|$), so `GaussWeakGradUnique` applied to the function u and these two fields gives

$$Du = \mathbf{1}_{\{u>0\}}\, Du \qquad \text{in } L^2(\gamma_n;\mathbb{R}^n).$$

Equivalently, at γ_n-almost every point with $u \le 0$ (hence $u = 0$, since $u \ge 0$ a.e.) the indicator vanishes and therefore $Du = 0$. □

2.89. `SignedDecomposition`.

Lemma 2.89 (The signed decomposition of a real L^2 input at a high level)**.** *Let $f \in L^2(\gamma_n)$ be real valued, put $M = \|f\|_{L^1(\gamma_n)}$ in the sense of `ToL1`, and let $t > M$. Then there exist $u \in \mathrm{Dom}(L^{1/2})$, a function $g \in L^2(\gamma_n)$ and a measurable set $\Omega \subseteq \mathbb{R}^n$ such that*

$$\begin{aligned}
&f = g + Bu \quad \text{in } L^2(\gamma_n), \qquad B = L^{1/2},\\
&t\,\gamma_n(\Omega) \le M,\\
&|g| \le t \quad \text{a.e.},\\
&\|g\|_{L^1(\gamma_n)} \le M,\\
&\|g\|^2_{L^2(\gamma_n)} \le t\,M,\\
&Du = 0 \quad \text{a.e. on } \Omega^c,
\end{aligned}$$

with D as in `GaussD`*. Note the sup bound is t, not $2t$.*

Proof. Step 0: the two nonnegative parts and the height choice. Put $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$, so $f = f^+ - f^-$ and $|f| = f^+ + f^-$ pointwise. Both are measurable with $0 \le f^\pm \le |f|$, hence lie in $L^2(\gamma_n)$ and are nonnegative γ_n-a.e. Set

$$m_+ = \int f^+ \, d\gamma_n, \qquad m_- = \int f^- \, d\gamma_n,$$

both finite by `L2IntoL1`, and $m_+ + m_- = \int |f| \, d\gamma_n = M$ by `ToL1`. As $m_\pm \ge 0$, this gives $m_\pm \le M < t$. Therefore the common obstacle height $a = t$ satisfies the strict hypothesis $a > \int F \, d\gamma_n$ of `ObstacleDecomposition` for each of the two inputs $F = f^+$ and $F = f^-$. This is a condition on the integral of the input, not on its pointwise size, which may well exceed t.

Step 1: two applications of the obstacle decomposition. Apply `ObstacleDecomposition` with $F = f^+$, $m = m_+$ and $a = t$, and then with $F = f^-$, $m = m_-$ and $a = t$. For each sign $\pm$ this produces $u_\pm \in \mathrm{Dom}(L^{1/2})$ and $g_\pm \in L^2(\gamma_n)$ with

$$\begin{aligned} &u_\pm \ge 0 \text{ a.e.}, \qquad f^\pm = g_\pm + L^{1/2} u_\pm, \\ &\min\{f^\pm, t\} \le g_\pm \le t \text{ a.e.}, \qquad \int g_\pm \, d\gamma_n = m_\pm, \\ &t\,\gamma_n(\{u_\pm > 0\}) \le m_\pm, \qquad Du_\pm = 0 \text{ a.e. on } \{u_\pm > 0\}^c. \end{aligned}$$

Since $f^\pm \ge 0$ a.e. and $t > 0$, the lower bound gives $g_\pm \ge \min\{f^\pm, t\} \ge 0$ a.e. The notation does not assert that u_+ and u_- are the positive and negative parts of one function; they are independently constructed, and likewise for g_+ and g_-.

Step 2: the signed objects. Define

$$g = g_+ - g_-, \qquad u = u_+ - u_-, \qquad \Omega = \{u_+ > 0\} \cup \{u_- > 0\}.$$

Ω is measurable, being the union of two measurable sets (the sets $\{u_\pm > 0\}$ are measurable because $u_\pm$ are measurable). Clearly $g \in L^2(\gamma_n)$. Also $u \in \mathrm{Dom}(L^{1/2})$: by `FracPowDomain` the condition is $\sum_{\alpha \ne 0} |\alpha| \, |u_\alpha|^2 < \infty$, the coefficient map of `HermiteBasis` is linear, and the set of families with that weighted square-sum finite is a linear subspace (triangle inequality in the weighted ℓ^2 space). Moreover $L^{1/2}$ is linear on its domain: by `FracPow` and `FracPowCoeff`, $L^{1/2} w$ has coefficients $|\alpha|^{1/2} w_\alpha$ off $\alpha = 0$ and 0 at $\alpha = 0$, which depend linearly on (w_α). Subtracting the two identities of Step 1 therefore gives

$$f = f^+ - f^- = (g_+ - g_-) + L^{1/2}(u_+ - u_-) = g + Bu \quad \text{in } L^2(\gamma_n).$$

Step 3: the measure of Ω. By subadditivity of γ_n and Step 1,

$$t\,\gamma_n(\Omega) \le t\,\gamma_n(\{u_+ > 0\}) + t\,\gamma_n(\{u_- > 0\}) \le m_+ + m_- = M.$$

Step 4: the sup bound. From $0 \le g_\pm \le t$ a.e. (Step 1) we get, a.e., $g = g_+ - g_- \le g_+ \le t$ and $g \ge -g_- \ge -t$, i.e. $|g| \le t$. The bound is t, not $2t$, precisely because each good part is individually squeezed between 0 and t.

Step 5: the L^1 bound. Since $g_\pm \ge 0$ a.e., $|g| = |g_+ - g_-| \le g_+ + g_-$ a.e. Hence, using the mass identities of Step 1 and $m_+ + m_- = M$,

$$\|g\|_{L^1(\gamma_n)} = \int |g| \, d\gamma_n \le \int (g_+ + g_-) \, d\gamma_n = m_+ + m_- = M.$$

The first step is an inequality, not an equality, because g_+ and g_- need not have disjoint supports.

Step 6: the L^2 bound. Combining Steps 4 and 5,

$$\|g\|^2_{L^2(\gamma_n)} = \int |g|^2 \, d\gamma_n = \int |g| \cdot |g| \, d\gamma_n \le t \int |g| \, d\gamma_n \le t\, M.$$

Step 7: locality of the signed gradient. By Step 1 and `SqrtDomainSobolev`, both $u_\pm$ lie in $W^{1,2}(\gamma_n)$. By parts (1) and (2) of `GaussDLinear`, $u = u_+ - u_- \in W^{1,2}(\gamma_n)$ and $Du = Du_+ - Du_-$. Now let $x \notin \Omega$. Then $u_+(x) \le 0$ and $u_-(x) \le 0$, so x lies in both $\{u_+ > 0\}^c$ and $\{u_- > 0\}^c$, where by Step 1 we have $Du_+ = 0$ and $Du_- = 0$ γ_n-a.e. Hence $Du = Du_+ - Du_- = 0$ γ_n-a.e. on Ω^c. □

2.90. `ComplexMainBound`.

Lemma 2.90 (The weak-type bound for complex L^2 inputs, with constant $8/\pi$)**.** *For every $v \in L^2(\gamma_n;\mathbb{C})$ and every real $t > 0$,*

$$t\,\gamma_n\big(\{x : |R_\gamma^{\mathbb{C}} v(x)|_{\mathbb{C}^n} > t\}\big) \;\le\; \frac{8}{\pi}\int_{\mathbb{R}^n} |v(x)|\,d\gamma_n(x),$$

with $R_\gamma^{\mathbb{C}}$ as in `ComplexRieszTransform` and the Euclidean norm of $\mathbb{C}^n$ on the vector output. The constant $8/\pi$ does not depend on n.

This is the estimate the reference paper proves for complex L^2 inputs before extending it to L^1, i.e. its displayed complex bound restricted to $f \in L^2$. The right-hand side is written as the integral $\int |v|\,d\gamma_n$ rather than as a norm in a complex L^1 space: that is the same real number as the paper's $\|f\|_{L^1(\gamma;\mathbb{C})}$, and writing it this way keeps the input in $L^2(\gamma_n;\mathbb{C})$ and avoids introducing a complex L^1 inclusion the tablet does not yet have. The constant is $8/\pi$ and not $2\sqrt{2}$: the paper obtains it by averaging the bound over rotations $e^{i\theta}$ rather than by the cruder comparison $S \le \sqrt{2}\,\|f\|_1$, and $8/\pi < 2\sqrt{2}$.

Proof. Put $M = \int_{\mathbb{R}^n} |v(x)|\,d\gamma_n(x)$. By `GaussIsProb` γ_n is a probability measure, so the constant function 1 is in $L^2(\gamma_n)$ and the integral form of Cauchy–Schwarz (shared context) applied to $|v|$ and 1 gives $M \le \|v\|_{L^2} < \infty$; also $M \ge 0$, the integrand being nonnegative. Every measure of a set below is at most $\gamma_n(\mathbb{R}^n) = 1$, so all the quantities appearing are finite real numbers. Note $8/\pi > 1$, since $\pi < 8$; hence $M \le \frac{8}{\pi}M$, because $M \ge 0$. ("Step 0" below refers to this paragraph.)

Case 1: $0 < t \le M$. By monotonicity of γ_n, $\gamma_n(\{|R_\gamma^{\mathbb{C}} v| > t\}) \le \gamma_n(\mathbb{R}^n) = 1$ using `GaussIsProb`, so, multiplying by $t > 0$,

$$t\,\gamma_n\big(\{|R_\gamma^{\mathbb{C}} v| > t\}\big) \le t \le M \le \frac{8}{\pi}M,$$

which is the assertion. (This is the paper's "smaller levels are bounded by $\|f\|_1$ since γ has mass one"; it includes the borderline level $t = M$, so no decomposition at a borderline height is needed.)

Case 2: $t > M$. Fix $\theta \in \mathbb{R}$ and put $c_\theta = e^{i\theta}$, so $|c_\theta| = 1$. Write

$$v_\theta = c_\theta\, v \in L^2(\gamma_n;\mathbb{C}), \qquad f_1 = \operatorname{Re} v_\theta, \quad f_2 = \operatorname{Im} v_\theta \;\in L^2(\gamma_n;\mathbb{R}), \qquad S_\theta = \|f_1\|_{L^1(\gamma_n)} + \|f_2\|_{L^1(\gamma_n)},$$

the L^1 norms through the inclusion of `ToL1`, which by that node has the same representative, so $\|h\|_{L^1(\gamma_n)} = \int |h|\,d\gamma_n$ for real $h \in L^2(\gamma_n)$.

Step 2.1: each part has L^1 norm at most M, hence less than t. For γ_n-almost every x we have $|f_1(x)| = |\operatorname{Re}(c_\theta v(x))| \le |c_\theta v(x)| = |c_\theta|\,|v(x)| = |v(x)|$, using that the real part of a complex number is bounded in absolute value by its modulus and that $|c_\theta| = 1$. Both $|f_1|$ and $|v|$ are γ_n-integrable: the first because f_1 has an $L^1(\gamma_n)$ class with the same representative by `ToL1`, and the second by Step 0 of this proof, where $M < \infty$ was checked. Integrating the pointwise inequality between these two nonnegative integrable functions,

$$\|f_1\|_{L^1(\gamma_n)} = \int |f_1|\,d\gamma_n \le \int |v|\,d\gamma_n = M < t,$$

and the same computation with Im in place of Re gives $\|f_2\|_{L^1(\gamma_n)} \le M < t$.

Step 2.2: two signed decompositions at height t. Step 2.1 is exactly the hypothesis of `SignedDecomposition` for each f_i at this level t. Applying it to f_1 gives $u_1 \in \mathrm{Dom}(L^{1/2})$, $g_1 \in L^2(\gamma_n)$ and a measurable $\Omega_1 \subseteq \mathbb{R}^n$ with

$$f_1 = g_1 + Bu_1, \qquad t\,\gamma_n(\Omega_1) \le \|f_1\|_{L^1(\gamma_n)}, \qquad \|g_1\|^2_{L^2(\gamma_n)} \le t\,\|f_1\|_{L^1(\gamma_n)}, \qquad Du_1 = 0 \text{ a.e. on } \Omega_1^c,$$

and applying it to f_2 gives u_2, g_2, Ω_2 with the same four properties, with the index 2 throughout. (The node also supplies $|g_i| \le t$ a.e. and $\|g_i\|_{L^1} \le \|f_i\|_{L^1}$; neither is used here.) These two applications are exactly the paper's "four nonnegative decompositions of $f_1^+, f_1^-, f_2^+, f_2^-$": `SignedDecomposition` is itself the signed statement obtained from the positive and negative parts of one real input, so two applications of it, to f_1 and to f_2, deliver all four.

Step 2.3: the shared exceptional set. Put $\Omega = \Omega_1 \cup \Omega_2$. Since $\Omega_1 \subseteq \Omega$ we have $\Omega^c \subseteq \Omega_1^c$, so $Du_1 = 0$ a.e. on Ω^c; likewise $Du_2 = 0$ a.e. on Ω^c. By subadditivity of γ_n (shared context) and the two measure bounds of Step 2.2, multiplied through by $t > 0$,

$$t\,\gamma_n(\Omega) \le t\,\gamma_n(\Omega_1) + t\,\gamma_n(\Omega_2) \le \|f_1\|_{L^1} + \|f_2\|_{L^1} = S_\theta, \qquad \text{so} \qquad \gamma_n(\Omega) \le \frac{S_\theta}{t}. \tag{1}$$

Adding the two L^2 bounds of Step 2.2,

$$\|g_1\|_{L^2}^2 + \|g_2\|_{L^2}^2 \le t\,\|f_1\|_{L^1} + t\,\|f_2\|_{L^1} = t\,S_\theta. \tag{2}$$

Step 2.4: the complex good function. Let w be the γ_n-a.e. class of the function $x \mapsto g_1(x) + i\,g_2(x)$. It is measurable, being a sum of a measurable function and i times a measurable function, and $|g_1(x) + ig_2(x)|^2 = g_1(x)^2 + g_2(x)^2$ is γ_n-integrable because $g_1, g_2 \in L^2(\gamma_n)$; hence $w \in L^2(\gamma_n;\mathbb{C})$. Taking real and imaginary parts pointwise, $\operatorname{Re} w = g_1$ and $\operatorname{Im} w = g_2$ in $L^2(\gamma_n;\mathbb{R})$. This is the paper's $g_1 + ig_2$.

Step 2.5: localization. Apply `ComplexTransformedLocalization` with the complex inputs v_θ and w, the real data g_1, g_2, u_1, u_2, this t and this Ω. Its hypotheses hold: $u_1, u_2 \in \operatorname{Dom}(L^{1/2})$ and $\operatorname{Re} v_\theta = f_1 = g_1 + Bu_1$, $\operatorname{Im} v_\theta = f_2 = g_2 + Bu_2$ by Step 2.2; $\operatorname{Re} w = g_1$ and $\operatorname{Im} w = g_2$ by Step 2.4; and $Du_1 = Du_2 = 0$ a.e. on Ω^c by Step 2.3. Its second clause gives

$$\gamma_n\big(\{|R_\gamma^{\mathbb{C}} v_\theta| > t\}\big) \le \gamma_n(\Omega) + \gamma_n\big(\{|R_\gamma^{\mathbb{C}} w| > t\}\big). \tag{3}$$

Step 2.6: Chebyshev on the good part. The function $|R_\gamma^{\mathbb{C}} w|^2$ is nonnegative and γ_n-integrable, its integral being $\|R_\gamma^{\mathbb{C}} w\|^2_{L^2(\gamma_n;\mathbb{C}^n)} < \infty$. Since $t > 0$, the set $\{|R_\gamma^{\mathbb{C}} w| > t\}$ is the set $\{|R_\gamma^{\mathbb{C}} w|^2 > t^2\}$, so Chebyshev's inequality (shared context — it is the integration of the pointwise bound $t^2\,\mathbf{1}_{\{|R_\gamma^{\mathbb{C}} w|^2 > t^2\}} \le |R_\gamma^{\mathbb{C}} w|^2$, both sides being nonnegative and the right-hand side integrable) gives

$$\gamma_n\big(\{|R_\gamma^{\mathbb{C}} w| > t\}\big) \le t^{-2}\,\big\|R_\gamma^{\mathbb{C}} w\big\|^2_{L^2(\gamma_n;\mathbb{C}^n)}.$$

By clause (1) of `ComplexRieszL2Identity` applied to w, and then $\operatorname{Re} w = g_1$, $\operatorname{Im} w = g_2$ from Step 2.4,

$$\big\|R_\gamma^{\mathbb{C}} w\big\|^2 = \big\|R_\gamma(\operatorname{Re} w)\big\|^2 + \big\|R_\gamma(\operatorname{Im} w)\big\|^2 = \|R_\gamma g_1\|^2 + \|R_\gamma g_2\|^2.$$

By `RieszL2Identity` applied to g_1 and to g_2, $\|R_\gamma g_i\|^2 \le \|g_i\|_{L^2}^2$ for $i = 1, 2$. Combining with (2),

$$\gamma_n\big(\{|R_\gamma^{\mathbb{C}} w| > t\}\big) \le t^{-2}\big(\|g_1\|_{L^2}^2 + \|g_2\|_{L^2}^2\big) \le t^{-2}\,t\,S_\theta = \frac{S_\theta}{t}. \tag{4}$$

Step 2.7: the bound at the rotation θ. Substituting (1) and (4) into (3),

$$\gamma_n\big(\{|R_\gamma^{\mathbb{C}} v_\theta| > t\}\big) \le \frac{S_\theta}{t} + \frac{S_\theta}{t} = \frac{2S_\theta}{t},$$

and multiplying by $t > 0$,

$$t\,\gamma_n\big(\{|R_\gamma^{\mathbb{C}} v_\theta| > t\}\big) \le 2\,S_\theta. \tag{5}$$

Step 2.8: the left-hand side does not depend on θ. Since $|c_\theta| = 1$, clause (3) of `ComplexRieszTransformLinear` applied with $c = c_\theta$ and this v, gives $|R_\gamma^{\mathbb{C}}(c_\theta v)(x)| = |R_\gamma^{\mathbb{C}} v(x)|$ for γ_n-almost every x. The two sets $\{|R_\gamma^{\mathbb{C}} v_\theta| > t\}$ and $\{|R_\gamma^{\mathbb{C}} v| > t\}$ therefore differ only inside a γ_n-null set, so by monotonicity up to null sets applied in both directions they have the same measure. Writing $A = t\,\gamma_n(\{|R_\gamma^{\mathbb{C}} v| > t\})$, which is a real number in $[0, t]$ because γ_n is a probability measure, (5) becomes

$$A \le 2\,S_\theta \qquad \text{for every } \theta \in \mathbb{R}. \tag{6}$$

This is the paper's "Complex linearity implies $|R_\gamma(e^{i\theta} f)| = |R_\gamma f|$".

Step 2.9: averaging over θ. By clause (1) of `RotationAverageL1` the function $\theta \mapsto S_\theta$ is integrable on $[0, 2\pi]$, hence so is $\theta \mapsto 2S_\theta$; the constant function A is integrable on $[0, 2\pi]$ as well. Both being integrable, the pointwise inequality (6) may be integrated over $[0, 2\pi]$ by monotonicity of the integral:

$$2\pi A = \int_0^{2\pi} A\,d\theta \le \int_0^{2\pi} 2S_\theta\,d\theta.$$

Dividing by $2\pi > 0$ and applying clause (2) of `RotationAverageL1`,

$$A \le 2 \cdot \frac{1}{2\pi} \int_0^{2\pi} S_\theta \, d\theta = 2 \cdot \frac{4}{\pi} \int |v| \, d\gamma_n = \frac{8}{\pi} M,$$

which is the assertion in Case 2. (It is exactly here that the integrability clause of `RotationAverageL1` is needed: without it the inequality (6), which holds at every θ, could not be integrated.)

In both cases $t\,\gamma_n(\{|R_\gamma^{\mathbb{C}} v| > t\}) \le \frac{8}{\pi} M$, which is the asserted bound. □

2.91. `ComplexRieszTransformAELimitUnique`.

Lemma 2.91 (The almost-everywhere limit of the complex Riesz images, and its uniqueness in measure)**.** *Let $f \in L^1(\gamma_n;\mathbb{C})$ and let $(v_k)_{k\in\mathbb{N}}$ be a sequence in $L^2(\gamma_n;\mathbb{C})$. Write $\iota_{\mathbb{C}}$ for the inclusion of `ComplexToL1` and $R_\gamma^{\mathbb{C}}$ for the transform of `ComplexRieszTransform`. Assume the three hypotheses*

(A) *for all $u, w \in L^2(\gamma_n;\mathbb{C})$ and every $\varepsilon > 0$,*

$$\gamma_n\big(\{x : \varepsilon \le |R_\gamma^{\mathbb{C}} u(x) - R_\gamma^{\mathbb{C}} w(x)|_{\mathbb{C}^n}\}\big) \le \frac{16}{\pi} \cdot \frac{\|\iota_{\mathbb{C}} u - \iota_{\mathbb{C}} w\|_{L^1}}{\varepsilon};$$

(L) *$\iota_{\mathbb{C}} v_k \to f$ in $L^1(\gamma_n;\mathbb{C})$;*

(C) *$\|\iota_{\mathbb{C}} v_j - \iota_{\mathbb{C}} v_{j+1}\|_{L^1} \le 8^{-j}$ for every j.*

Then there is an element g of $\mathcal{M}(\gamma_n;\mathbb{C}^n)$, the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{C}^n$-valued functions on $\mathbb{R}^n$, such that

(1) *$R_\gamma^{\mathbb{C}} v_k \to g$ pointwise γ_n-a.e.; and*

(2) *for* every *sequence (w_k) in $L^2(\gamma_n;\mathbb{C})$ with $\iota_{\mathbb{C}} w_k \to f$ in $L^1(\gamma_n;\mathbb{C})$, the sequence $R_\gamma^{\mathbb{C}} w_k$ converges to g in measure.*

Proof. Part 1: the almost-everywhere limit. Put $E_j = \{x : 2^{-j} \le |R_\gamma^{\mathbb{C}} v_j(x) - R_\gamma^{\mathbb{C}} v_{j+1}(x)|_{\mathbb{C}^n}\}$. By (A) with $\varepsilon = 2^{-j}$ and then (C),

$$\gamma_n(E_j) \le \frac{16}{\pi} \cdot \frac{\|\iota_{\mathbb{C}} v_j - \iota_{\mathbb{C}} v_{j+1}\|_{L^1}}{2^{-j}} \le \frac{16}{\pi} \cdot 2^j \cdot 8^{-j} = \frac{16}{\pi} \cdot 4^{-j}.$$

The geometric series $\sum_j \frac{16}{\pi} 4^{-j}$ converges, so $\sum_j \gamma_n(E_j) < \infty$ and the Borel–Cantelli lemma (shared context) gives: for γ_n-a.e. x the set $\{j : x \in E_j\}$ is finite. Fix such an x and let B bound that finite set. For $j \ge B + 1$ we have $|R_\gamma^{\mathbb{C}} v_j(x) - R_\gamma^{\mathbb{C}} v_{j+1}(x)| < 2^{-j}$, so all but finitely many terms of $\sum_j |R_\gamma^{\mathbb{C}} v_j(x) - R_\gamma^{\mathbb{C}} v_{j+1}(x)|$ are dominated by the convergent series $\sum_j 2^{-j}$; hence that series converges, and a sequence in the complete space $\mathbb{C}^n$ whose consecutive distances form a convergent series is Cauchy, hence convergent. Define $g(x)$ to be this limit at every x where $(R_\gamma^{\mathbb{C}} v_j(x))_j$ converges, and $g(x) = 0$ elsewhere. Then $R_\gamma^{\mathbb{C}} v_j \to g$ pointwise γ_n-a.e., which is clause (1); and g is measurable, being a γ_n-a.e. pointwise limit of the measurable functions $R_\gamma^{\mathbb{C}} v_j$, so it determines an element of $\mathcal{M}(\gamma_n;\mathbb{C}^n)$.

Part 2: every approximating sequence has the same limit in measure. Since γ_n is a finite measure (`GaussIsProb`), the almost everywhere convergence of Part 1 implies convergence in measure: for every $\delta > 0$,

$$\gamma_n\big(\{x : \delta \le |R_\gamma^{\mathbb{C}} v_k(x) - g(x)|_{\mathbb{C}^n}\}\big) \longrightarrow 0. \tag{E}$$

Let now (w_k) be any sequence in $L^2(\gamma_n;\mathbb{C})$ with $\iota_{\mathbb{C}} w_k \to f$ in L^1, and fix $\varepsilon > 0$. By the triangle inequality in $\mathbb{C}^n$, if $\varepsilon \le |R_\gamma^{\mathbb{C}} w_k(x) - g(x)|$ then at least one of $|R_\gamma^{\mathbb{C}} w_k(x) - R_\gamma^{\mathbb{C}} v_k(x)|$ and $|R_\gamma^{\mathbb{C}} v_k(x) - g(x)|$ is at least $\varepsilon/2$; so

$$\{\varepsilon \le |R_\gamma^{\mathbb{C}} w_k - g|\} \subseteq \{\tfrac{\varepsilon}{2} \le |R_\gamma^{\mathbb{C}} w_k - R_\gamma^{\mathbb{C}} v_k|\} \cup \{\tfrac{\varepsilon}{2} \le |R_\gamma^{\mathbb{C}} v_k - g|\},$$

and monotonicity and subadditivity of γ_n (shared context) give

$$\gamma_n\big(\{\varepsilon \le |R_\gamma^{\mathbb{C}} w_k - g|\}\big) \le \gamma_n\big(\{\tfrac{\varepsilon}{2} \le |R_\gamma^{\mathbb{C}} w_k - R_\gamma^{\mathbb{C}} v_k|\}\big) + \gamma_n\big(\{\tfrac{\varepsilon}{2} \le |R_\gamma^{\mathbb{C}} v_k - g|\}\big).$$

The second summand tends to 0 by (E) with $\delta = \varepsilon/2$. For the first, $\|\iota_{\mathbb{C}} w_k - \iota_{\mathbb{C}} v_k\|_{L^1} \to 0$, because both $\iota_{\mathbb{C}} w_k$ and $\iota_{\mathbb{C}} v_k$ converge to f in L^1 by hypothesis and by (L) respectively, so (A) at the level $\varepsilon/2$ bounds the first summand by $32\,\|\iota_{\mathbb{C}} w_k - \iota_{\mathbb{C}} v_k\|_{L^1}/(\pi\varepsilon) \to 0$. Both summands

are nonnegative and tend to 0, so their sum does, and the left-hand side, squeezed between 0 and that sum, tends to 0. That is convergence in measure of $R_\gamma^{\mathbb{C}} w_k$ to g, which is clause (2). □

2.92. `ComplexL1ApproxByL2Truncations`.

Lemma 2.92 (Complex L^1 approximation by L^2 radial truncations, with geometric decay)**.** *Let $f \in L^1(\gamma_n;\mathbb{C})$ and write $\iota_{\mathbb{C}}$ for the inclusion of `ComplexToL1`. Assume*

(N) *$\|\iota_{\mathbb{C}} u\|_{L^1} = \int |u|\, d\gamma_n$ for every $u \in L^2(\gamma_n;\mathbb{C})$, the L^1 norm of an inclusion written as the integral of the modulus of any representative.*

Then there is a sequence $(v_j)_{j\in\mathbb{N}}$ in $L^2(\gamma_n;\mathbb{C})$ such that

(1) *$\|\iota_{\mathbb{C}} v_k\|_{L^1} \le \|f\|_{L^1}$ for every k;*
(2) *$\iota_{\mathbb{C}} v_k \to f$ in $L^1(\gamma_n;\mathbb{C})$; and*
(3) *$\|\iota_{\mathbb{C}} v_j - \iota_{\mathbb{C}} v_{j+1}\|_{L^1} \le 8^{-j}$ for every j.*

Proof. Part 1: the radial truncations. Fix a representative $F : \mathbb{R}^n \to \mathbb{C}$ of f. For $k \in \mathbb{N}$ define $\varphi_k : \mathbb{C} \to \mathbb{C}$ by

$$\varphi_k(z) = \begin{cases} z, & |z| \le k, \\ k\, z/|z|, & |z| > k, \end{cases}$$

and put $\tau_k = \varphi_k \circ F$. (This is the complex analogue of the real two-sided truncation $\max\{-k, \min\{\cdot, k\}\}$: it cuts the modulus at k and leaves the argument alone.)

(1a) φ_k is continuous, so τ_k is measurable. On $\{|z| \le k\}$ and on $\{|z| \ge k\}$ the two formulas are continuous — the second because $z \mapsto |z|$ is continuous and nonzero there, as $k \ge 0$ and $|z| \ge k$ with $|z| > 0$ whenever $k > 0$; for $k = 0$ the second branch is the constant 0 — and on the overlap $\{|z| = k\}$ they agree, since $kz/|z| = z$ there. Hence φ_k is continuous on $\mathbb{C}$ by gluing on two closed sets, so Borel measurable, and $\tau_k = \varphi_k \circ F$ is measurable.

(1b) the two pointwise bounds. For every z, $|\varphi_k(z)| = \min\{|z|, k\}$: if $|z| \le k$ this is $|z|$, and otherwise it is $|kz/|z|| = k$. Hence for every x

$$|\tau_k(x)| \le k \qquad \text{and} \qquad |\tau_k(x)| \le |F(x)|. \tag{B}$$

(1c) τ_k lies in L^2, and clause (1). Since γ_n is a probability measure (`GaussIsProb`), the bound $|\tau_k| \le k$ makes τ_k square γ_n-integrable; let $\tilde{v}_k \in L^2(\gamma_n;\mathbb{C})$ be its class. By hypothesis (N), the second bound in (B) and monotonicity of the integral,

$$\|\iota_{\mathbb{C}} \tilde{v}_k\|_{L^1} = \int |\tau_k|\, d\gamma_n \le \int |F|\, d\gamma_n = \|f\|_{L^1},$$

which is clause (1) for this sequence.

(1d) L^1 convergence, i.e. clause (2). For every x, as soon as $k \ge |F(x)|$ we have $\varphi_k(F(x)) = F(x)$, so $\tau_k(x) \to F(x)$, the sequence being eventually constant at each point. By (B), $|\tau_k(x) - F(x)| \le |\tau_k(x)| + |F(x)| \le 2|F(x)|$, and $2|F|$ is γ_n-integrable because $f \in L^1(\gamma_n;\mathbb{C})$. Dominated convergence (shared context) therefore gives $\|\iota_{\mathbb{C}} \tilde{v}_k - f\|_{L^1} = \int |\tau_k - F|\, d\gamma_n \to 0$.

Part 2: a rapidly converging subsequence. By (1d) we may choose, for each $j \in \mathbb{N}$, an index $\sigma(j)$ with $\|\iota_{\mathbb{C}} \tilde{v}_{\sigma(j)} - f\|_{L^1} \le \frac{1}{2} 8^{-j}$. Put $v_j = \tilde{v}_{\sigma(j)}$. Clause (1) is inherited from (1c), and $\|\iota_{\mathbb{C}} v_j - f\|_{L^1} \le \frac{1}{2} 8^{-j} \to 0$ is clause (2) for (v_j). By the triangle inequality and $8^{-(j+1)} \le 8^{-j}$,

$$\|\iota_{\mathbb{C}} v_j - \iota_{\mathbb{C}} v_{j+1}\|_{L^1} \le \tfrac{1}{2} 8^{-j} + \tfrac{1}{2} 8^{-(j+1)} \le 8^{-j},$$

which is clause (3). □

2.93. `ComplexRieszL1Limit`.

Lemma 2.93 (A common in-measure limit for the complex Riesz images of L^1 approximating sequences)**.** *Let $f \in L^1(\gamma_n;\mathbb{C})$. Then there exist an element g of $\mathcal{M}(\gamma_n;\mathbb{C}^n)$, the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{C}^n$-valued functions on $\mathbb{R}^n$, and a sequence $(v_k)_{k\in\mathbb{N}}$ in $L^2(\gamma_n;\mathbb{C})$, such that*

(1) *$\|\iota_{\mathbb{C}} v_k\|_{L^1(\gamma_n;\mathbb{C})} \le \|f\|_{L^1(\gamma_n;\mathbb{C})}$ for every k, with $\iota_{\mathbb{C}}$ the inclusion of `ComplexToL1`;*
(2) *$\iota_{\mathbb{C}} v_k \to f$ in $L^1(\gamma_n;\mathbb{C})$;*
(3) *$R_\gamma^{\mathbb{C}} v_k \to g$ pointwise γ_n-a.e., with $R_\gamma^{\mathbb{C}}$ as in `ComplexRieszTransform`; and*

(4) *for* every *sequence* (w_k) *in* $L^2(\gamma_n;\mathbb{C})$ *with* $\iota_{\mathbb{C}} w_k \to f$ *in* $L^1(\gamma_n;\mathbb{C})$, *the sequence* $R^{\mathbb{C}}_\gamma w_k$ *converges to* g *in measure, i.e.* $\gamma_n\big(\{x : \varepsilon \le |R^{\mathbb{C}}_\gamma w_k(x) - g(x)|_{\mathbb{C}^n}\}\big) \to 0$ *as* $k \to \infty$, *for every* $\varepsilon > 0$.

This is the complex counterpart of the Cauchy-in-measure and a.e.-limit construction that the reference paper invokes in the last sentence of its proof of the complex-valued estimate, where it says that approximation by complex L^2 functions, with the Cauchy-in-measure and Fatou arguments of its L^1 extension subsection, gives the extension to L^1. Clause (4) is the part that does the real work downstream: it says the limit g depends only on f and not on the approximating sequence, which is what makes an extension operator well defined and what forces it to be linear.

Proof. Throughout, $\iota_{\mathbb{C}}$ is the inclusion of `ComplexToL1`, and we use from that node that $\iota_{\mathbb{C}} v$ and v have the same representative, so that $\|\iota_{\mathbb{C}} v\|_{L^1} = \int |v|\, d\gamma_n$, and that $\iota_{\mathbb{C}}$ is $\mathbb{C}$-linear.

Part A: the Cauchy-in-measure estimate. Let $v, w \in L^2(\gamma_n;\mathbb{C})$ and $\varepsilon > 0$. By clauses (1) and (2) of `ComplexRieszTransformLinear`, applied with the scalar $c = -1$ for the second,

$$R^{\mathbb{C}}_\gamma(v - w) = R^{\mathbb{C}}_\gamma v + R^{\mathbb{C}}_\gamma\big((-1)w\big) = R^{\mathbb{C}}_\gamma v - R^{\mathbb{C}}_\gamma w.$$

Apply `ComplexMainBound` to the complex L^2 function $v - w$ at the level $\varepsilon/2 > 0$:

$$\tfrac{\varepsilon}{2}\,\gamma_n\big(\{x : \tfrac{\varepsilon}{2} < |R^{\mathbb{C}}_\gamma(v-w)(x)|_{\mathbb{C}^n}\}\big) \le \frac{8}{\pi}\int |(v-w)(x)|\, d\gamma_n(x) = \frac{8}{\pi}\,\|\iota_{\mathbb{C}} v - \iota_{\mathbb{C}} w\|_{L^1},$$

the last equality because $\iota_{\mathbb{C}}(v-w) = \iota_{\mathbb{C}} v - \iota_{\mathbb{C}} w$ by $\mathbb{C}$-linearity and because the L^1 norm of that class is the integral of $|v-w|$. Since $\varepsilon/2 < \varepsilon$ and $R^{\mathbb{C}}_\gamma(v-w)$ agrees γ_n-a.e. with $R^{\mathbb{C}}_\gamma v - R^{\mathbb{C}}_\gamma w$, the set $\{x : \varepsilon \le |R^{\mathbb{C}}_\gamma v(x) - R^{\mathbb{C}}_\gamma w(x)|\}$ is contained, up to a γ_n-null set, in $\{x : \varepsilon/2 < |R^{\mathbb{C}}_\gamma(v-w)(x)|\}$; monotonicity of γ_n up to null sets (shared context) and division by $\varepsilon/2 > 0$ give

$$\gamma_n\big(\{x : \varepsilon \le |R^{\mathbb{C}}_\gamma v(x) - R^{\mathbb{C}}_\gamma w(x)|_{\mathbb{C}^n}\}\big) \le \frac{16}{\pi}\cdot\frac{\|\iota_{\mathbb{C}} v - \iota_{\mathbb{C}} w\|_{L^1}}{\varepsilon}. \tag{A}$$

Part B: the radial truncations. Fix a representative $F : \mathbb{R}^n \to \mathbb{C}$ of f. For $k \in \mathbb{N}$ define $\varphi_k : \mathbb{C} \to \mathbb{C}$ by

$$\varphi_k(z) = \begin{cases} z, & |z| \le k, \\ k\,z/|z|, & |z| > k, \end{cases}$$

and put $\tau_k = \varphi_k \circ F$. (This is the complex analogue of the real two-sided truncation $\max\{-k, \min\{\cdot, k\}\}$: it cuts the modulus at k and leaves the argument alone.)

(B1) φ_k is continuous, so τ_k is measurable. On $\{|z| \le k\}$ and on $\{|z| \ge k\}$ the two formulas are continuous — the second because $z \mapsto |z|$ is continuous and nonzero there, as $k \ge 0$ and $|z| \ge k$ with $|z| > 0$ whenever $k > 0$; for $k = 0$ the second branch is the constant 0 — and on the overlap $\{|z| = k\}$ they agree, since $kz/|z| = z$ there. Hence φ_k is continuous on $\mathbb{C}$ by gluing on two closed sets, so Borel measurable, and $\tau_k = \varphi_k \circ F$ is measurable.

(B2) the two pointwise bounds. For every z, $|\varphi_k(z)| = \min\{|z|, k\}$: if $|z| \le k$ this is $|z|$, and otherwise it is $|kz/|z|| = k$. Hence for every x

$$|\tau_k(x)| \le k \qquad \text{and} \qquad |\tau_k(x)| \le |F(x)|. \tag{B}$$

(B3) τ_k lies in L^2, and conclusion (1). Since γ_n is a probability measure (`GaussIsProb`), the bound $|\tau_k| \le k$ makes τ_k square γ_n-integrable; let $\tilde v_k \in L^2(\gamma_n;\mathbb{C})$ be its class. The second bound in (B) and monotonicity of the integral give

$$\|\iota_{\mathbb{C}} \tilde v_k\|_{L^1} = \int |\tau_k|\, d\gamma_n \le \int |F|\, d\gamma_n = \|f\|_{L^1},$$

which is conclusion (1) for this sequence.

(B4) L^1 convergence, i.e. conclusion (2). For every x, as soon as $k \ge |F(x)|$ we have $\varphi_k(F(x)) = F(x)$, so $\tau_k(x) \to F(x)$, the sequence being eventually constant at each point. By (B), $|\tau_k(x) - F(x)| \le |\tau_k(x)| + |F(x)| \le 2|F(x)|$, and $2|F|$ is γ_n-integrable because $f \in L^1(\gamma_n;\mathbb{C})$. Dominated convergence (shared context) therefore gives $\|\iota_{\mathbb{C}} \tilde v_k - f\|_{L^1} = \int |\tau_k - F|\, d\gamma_n \to 0$.

Part C: a rapidly converging subsequence. By (B4) we may choose, for each $j \in \mathbb{N}$, an index $\sigma(j)$ with $\|\iota_{\mathbb{C}}\tilde{v}_{\sigma(j)} - f\|_{L^1} \le \frac{1}{2}8^{-j}$. Put $v_j = \tilde{v}_{\sigma(j)}$. Conclusion (1) is inherited from (B3), and $\|\iota_{\mathbb{C}}v_j - f\|_{L^1} \le \frac{1}{2}8^{-j} \to 0$ is conclusion (2) for (v_j). By the triangle inequality and $8^{-(j+1)} \le 8^{-j}$,

$$\|\iota_{\mathbb{C}}v_j - \iota_{\mathbb{C}}v_{j+1}\|_{L^1} \le \tfrac{1}{2}8^{-j} + \tfrac{1}{2}8^{-(j+1)} \le 8^{-j}. \tag{C}$$

Parts B and C together are `ComplexL1ApproxByL2Truncations`, which states exactly clauses (1), (2) and the geometric decay (C) for any $f \in L^1(\gamma_n;\mathbb{C})$, taking as its one hypothesis the norm identity $\|\iota_{\mathbb{C}}u\|_{L^1} = \int |u|\,d\gamma_n$ that Part A supplies; it is what the formalization invokes here.

Parts D and E: conclusions (3) and (4). The sequence (v_k) now satisfies the three hypotheses of `ComplexRieszTransformAELimitUnique`: its hypothesis (A) is the Cauchy-in-measure estimate (A) of Part A, applied to arbitrary u, w and $\varepsilon > 0$; its hypothesis (L) is conclusion (2), established in Part B; and its hypothesis (C) is the geometric decay (C) of Part C. That node therefore supplies an element g of $\mathcal{M}(\gamma_n;\mathbb{C}^n)$ such that $R^{\mathbb{C}}_{\gamma}v_k \to g$ pointwise γ_n-a.e., which is conclusion (3), and such that for every sequence (w_k) in $L^2(\gamma_n;\mathbb{C})$ with $\iota_{\mathbb{C}}w_k \to f$ in $L^1(\gamma_n;\mathbb{C})$ the images $R^{\mathbb{C}}_{\gamma}w_k$ converge to g in measure, which is conclusion (4). Together with (1) and (2) from Parts B and C this proves the lemma. □

2.94. `ComplexL1Extension`.

Lemma 2.94 (Existence of the complex L^1 extension satisfying the $8/\pi$ weak bound)**.** *There is a $\mathbb{C}$-linear map*

$$T_{\mathbb{C}} : L^1(\gamma_n;\mathbb{C}) \longrightarrow \mathcal{M}(\gamma_n;\mathbb{C}^n),$$

where $\mathcal{M}(\gamma_n;\mathbb{C}^n)$ denotes the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{C}^n$-valued functions on $\mathbb{R}^n$, such that

(1) *$T_{\mathbb{C}}$ agrees with $R^{\mathbb{C}}_{\gamma}$ on $L^2(\gamma_n;\mathbb{C})$: for every $v \in L^2(\gamma_n;\mathbb{C})$ one has $T_{\mathbb{C}}(\iota_{\mathbb{C}}v) = R^{\mathbb{C}}_{\gamma}v$, with $\iota_{\mathbb{C}}$ the inclusion of `ComplexToL1` and $R^{\mathbb{C}}_{\gamma}$ as in `ComplexRieszTransform`; and*

(2) *$T_{\mathbb{C}}$ satisfies the weak bound*

$$t\,\gamma_n\big(\{x : |T_{\mathbb{C}}f(x)|_{\mathbb{C}^n} > t\}\big) \le \frac{8}{\pi}\,\|f\|_{L^1(\gamma_n;\mathbb{C})} \qquad \text{for every } f \in L^1(\gamma_n;\mathbb{C}) \text{ and } t > 0.$$

This is the existence half of the reference paper's last sentence, that approximation by complex L^2 functions gives a continuous complex-linear extension to L^1 — "continuous" being clause (2), which is boundedness from $L^1(\gamma_n;\mathbb{C})$ into weak L^1 with quasi-norm at most $8/\pi$. The scalars in the linearity are the complex ones, which is what makes the map an extension of the complex transform rather than merely of its real and imaginary parts.

Proof. Step 1: the definition of $T_{\mathbb{C}}$, and its characterizing property. Let $f \in L^1(\gamma_n;\mathbb{C})$. Apply `ComplexRieszL1Limit` to f and let $g_f \in \mathcal{M}(\gamma_n;\mathbb{C}^n)$ and $(v_k^f) \subseteq L^2(\gamma_n;\mathbb{C})$ be the element and the sequence it supplies; define $T_{\mathbb{C}}f = g_f$. Clause (4) of that node then reads:

$$(\star) \qquad \text{for every } (w_k) \subseteq L^2(\gamma_n;\mathbb{C}) \text{ with } \iota_{\mathbb{C}}w_k \to f \text{ in } L^1, \quad R^{\mathbb{C}}_{\gamma}w_k \to T_{\mathbb{C}}f \text{ in measure.}$$

Property $(\star)$ determines $T_{\mathbb{C}}f$ as an element of $\mathcal{M}(\gamma_n;\mathbb{C}^n)$: clause (2) of the same node supplies at least one such sequence, namely (v_k^f), and a limit in measure is unique up to a γ_n-null set, because if $A_k \to A$ and $A_k \to A'$ in measure then for every $\varepsilon > 0$, $\gamma_n(\{|A - A'| \ge 2\varepsilon\}) \le \gamma_n(\{|A - A_k| \ge \varepsilon\}) + \gamma_n(\{|A_k - A'| \ge \varepsilon\}) \to 0$, so that measure is 0 for every ε, and $\{A \ne A'\}$ is the countable union of the sets $\{|A - A'| \ge 2/j\}$. Everything below uses only $(\star)$ together with clauses (1)–(3) at the particular sequence (v_k^f).

Step 2: convergence in measure respects the vector operations. If $A_k \to A$ and $B_k \to B$ in measure and $c \in \mathbb{C}$, then $A_k + B_k \to A + B$ and $c\,A_k \to c\,A$ in measure. For the sum: if $|A_k + B_k - (A + B)| \ge \varepsilon$ then $|A_k - A| \ge \varepsilon/2$ or $|B_k - B| \ge \varepsilon/2$, so by monotonicity and subadditivity of γ_n (shared context) the measure of the first set is at most the sum of the measures of the other two, both of which tend to 0. For the scalar multiple: if $c = 0$ both sides are 0; otherwise $\{|cA_k - cA| \ge \varepsilon\} = \{|A_k - A| \ge \varepsilon/|c|\}$, whose measure tends to 0.

Step 3: $\mathbb{C}$-linearity. Let $f, h \in L^1(\gamma_n;\mathbb{C})$ and $c \in \mathbb{C}$, and let $(v_k^f), (v_k^h)$ be the sequences of Step 1 for f and for h. Since $\iota_{\mathbb{C}}$ is $\mathbb{C}$-linear (`ComplexToL1`), $\iota_{\mathbb{C}}(v_k^f + v_k^h) = \iota_{\mathbb{C}} v_k^f + \iota_{\mathbb{C}} v_k^h$, and by the triangle inequality in L^1,

$$\big\|\iota_{\mathbb{C}}(v_k^f + v_k^h) - (f+h)\big\|_{L^1} \le \|\iota_{\mathbb{C}} v_k^f - f\|_{L^1} + \|\iota_{\mathbb{C}} v_k^h - h\|_{L^1} \longrightarrow 0$$

by clause (2) of `ComplexRieszL1Limit` at f and at h; so $\iota_{\mathbb{C}}(v_k^f + v_k^h) \to f + h$ in L^1, and $(\star)$ at $f+h$ gives $R_\gamma^{\mathbb{C}}(v_k^f + v_k^h) \to T_{\mathbb{C}}(f+h)$ in measure. On the other hand clause (1) of `ComplexRieszTransformLinear` gives $R_\gamma^{\mathbb{C}}(v_k^f + v_k^h) = R_\gamma^{\mathbb{C}} v_k^f + R_\gamma^{\mathbb{C}} v_k^h$, and by $(\star)$ at f and at h the two summands converge in measure to $T_{\mathbb{C}} f$ and $T_{\mathbb{C}} h$, so by Step 2 their sum converges in measure to $T_{\mathbb{C}} f + T_{\mathbb{C}} h$. Uniqueness of limits in measure (Step 1) gives $T_{\mathbb{C}}(f+h) = T_{\mathbb{C}} f + T_{\mathbb{C}} h$.

Likewise $\iota_{\mathbb{C}}(c\, v_k^f) = c\, \iota_{\mathbb{C}} v_k^f$ by $\mathbb{C}$-linearity, and $\|c\, \iota_{\mathbb{C}} v_k^f - cf\|_{L^1} = |c|\, \|\iota_{\mathbb{C}} v_k^f - f\|_{L^1} \to 0$ by absolute homogeneity of the L^1 norm, so $\iota_{\mathbb{C}}(c\, v_k^f) \to cf$ in L^1 and $(\star)$ at cf gives $R_\gamma^{\mathbb{C}}(c\, v_k^f) \to T_{\mathbb{C}}(cf)$ in measure; clause (2) of `ComplexRieszTransformLinear` gives $R_\gamma^{\mathbb{C}}(c\, v_k^f) = c\, R_\gamma^{\mathbb{C}} v_k^f$, which converges in measure to $c\, T_{\mathbb{C}} f$ by $(\star)$ at f and Step 2. Hence $T_{\mathbb{C}}(cf) = c\, T_{\mathbb{C}} f$. So $T_{\mathbb{C}}$ is a $\mathbb{C}$-linear map $L^1(\gamma_n;\mathbb{C}) \to \mathcal{M}(\gamma_n;\mathbb{C}^n)$.

Step 4: clause (1), agreement on L^2. Let $v \in L^2(\gamma_n;\mathbb{C})$ and take the constant sequence $w_k = v$. Then $\iota_{\mathbb{C}} w_k = \iota_{\mathbb{C}} v$ for every k, so trivially $\iota_{\mathbb{C}} w_k \to \iota_{\mathbb{C}} v$ in L^1, and $(\star)$ at $f = \iota_{\mathbb{C}} v$ says that the constant sequence $R_\gamma^{\mathbb{C}} v$ converges in measure to $T_{\mathbb{C}}(\iota_{\mathbb{C}} v)$. A constant sequence converges in measure to its own value, so by uniqueness of limits in measure (Step 1) $T_{\mathbb{C}}(\iota_{\mathbb{C}} v) = R_\gamma^{\mathbb{C}} v$.

Step 5: clause (2), the weak bound. Fix $f \in L^1(\gamma_n;\mathbb{C})$ and $t > 0$, and let (v_k^f) be the sequence of Step 1, so that by clause (3) of `ComplexRieszL1Limit` we have $R_\gamma^{\mathbb{C}} v_k^f \to g_f = T_{\mathbb{C}} f$ pointwise γ_n-a.e. At a point x with $|T_{\mathbb{C}} f(x)| > t$ the margin $|T_{\mathbb{C}} f(x)| - t$ is strictly positive and $|R_\gamma^{\mathbb{C}} v_k^f(x)| \to |T_{\mathbb{C}} f(x)|$ by continuity of the norm, so $|R_\gamma^{\mathbb{C}} v_k^f(x)| > t$ for all large k. Hence γ_n-a.e.

$$\mathbf{1}_{\{|T_{\mathbb{C}} f| > t\}} \le \liminf_{k\to\infty} \mathbf{1}_{\{|R_\gamma^{\mathbb{C}} v_k^f| > t\}},$$

and Fatou's lemma (shared context), applied to these nonnegative measurable functions, gives

$$\gamma_n\big(\{|T_{\mathbb{C}} f| > t\}\big) \le \liminf_k\, \gamma_n\big(\{|R_\gamma^{\mathbb{C}} v_k^f| > t\}\big).$$

By `ComplexMainBound` applied to v_k^f at the level $t > 0$, and then by $\|\iota_{\mathbb{C}} v_k^f\|_{L^1} = \int |v_k^f|\, d\gamma_n$ (`ComplexToL1`) followed by clause (1) of `ComplexRieszL1Limit`,

$$t\, \gamma_n\big(\{|R_\gamma^{\mathbb{C}} v_k^f| > t\}\big) \le \frac{8}{\pi} \int |v_k^f|\, d\gamma_n = \frac{8}{\pi}\, \|\iota_{\mathbb{C}} v_k^f\|_{L^1} \le \frac{8}{\pi}\, \|f\|_{L^1},$$

so every term of the lim inf is at most $\frac{8}{\pi t}\|f\|_{L^1}$ and therefore $\gamma_n(\{|T_{\mathbb{C}} f| > t\}) \le \frac{8}{\pi t}\|f\|_{L^1}$. Multiplying by $t > 0$ gives clause (2). □

2.95. `SmallLevelBound`.

Lemma 2.95 (The small-level estimate, with constant one)**.** *Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be any measurable vector-valued function and let $0 < t \le M$. Then*

$$t\, \gamma_n\big(\{x : |G(x)| > t\}\big) \;\le\; M,$$

where $|\cdot|$ is the Euclidean norm on $\mathbb{R}^n$. The constant is 1, better than the target constant 2, and the estimate requires no property of G beyond measurability. The borderline level $t = M$ is included, so no decomposition at a borderline height is needed there.

Proof. The measure γ_n is a probability measure by `GaussIsProb`, so every measurable set has γ_n-measure at most $\gamma_n(\mathbb{R}^n) = 1$. In particular

$$\gamma_n\big(\{x : |G(x)| > t\}\big) \;\le\; 1.$$

Multiplying by $t > 0$ preserves the inequality, giving

$$t\, \gamma_n\big(\{|G| > t\}\big) \;\le\; t.$$

Finally $t \le M$ by hypothesis, so the left-hand side is at most M. □

2.96. `MainBound`.

Lemma 2.96 (The weak-type bound for real L^2 inputs)**.** *For every real $f \in L^2(\gamma_n)$ and every $t > 0$,*

$$t\,\gamma_n\big(\{x : |R_\gamma f(x)| > t\}\big) \;\le\; 2\,\|f\|_{L^1(\gamma_n)},$$

with R_γ as in `RieszTransform`, $\|\cdot\|_{L^1(\gamma_n)}$ as in `ToL1`, and the Euclidean norm on the vector output. The constant 2 does not depend on n.

Proof. Put $M = \|f\|_{L^1(\gamma_n)}$, which is finite by `L2IntoL1`.

Case 1: $0 < t \le M$. By `SmallLevelBound`, applied to the measurable vector-valued function $G = R_\gamma f$, we get $t\,\gamma_n(\{|R_\gamma f| > t\}) \le M \le 2M$, the last step because $M \ge 0$. (This includes the borderline level $t = M$, so no decomposition at a borderline height is needed.)

Case 2: $t > M$. By `SignedDecomposition` there are $u \in \mathrm{Dom}(L^{1/2})$, $g \in L^2(\gamma_n)$ and a measurable set Ω with

$$f = g + L^{1/2}u, \qquad t\,\gamma_n(\Omega) \le M, \qquad \|g\|^2_{L^2(\gamma_n)} \le tM, \qquad Du = 0 \text{ a.e. on } \Omega^c.$$

By `TransformedLocalization`, applied with these data and this t,

$$\gamma_n\big(\{|R_\gamma f| > t\}\big) \le \gamma_n(\Omega) + \gamma_n\big(\{|R_\gamma g| > t\}\big).$$

For the second term, Chebyshev's inequality for the nonnegative function $|R_\gamma g|^2 \in L^1(\gamma_n)$ gives

$$\gamma_n\big(\{|R_\gamma g| > t\}\big) \le t^{-2}\,\|R_\gamma g\|^2_{L^2(\gamma_n;\mathbb{R}^n)}.$$

By `RieszL2Identity`, $\|R_\gamma g\|_2 \le \|g\|_2$, so by the L^2 bound above

$$\gamma_n\big(\{|R_\gamma g| > t\}\big) \le t^{-2}\|g\|_2^2 \le t^{-2}\,tM = \frac{M}{t}.$$

Also $\gamma_n(\Omega) \le M/t$ from the displayed bound, since $t > 0$. Adding,

$$\gamma_n\big(\{|R_\gamma f| > t\}\big) \le \frac{M}{t} + \frac{M}{t} = \frac{2M}{t},$$

and multiplying by $t > 0$ gives $t\,\gamma_n(\{|R_\gamma f| > t\}) \le 2M$.

In both cases the asserted bound holds. □

2.97. `L1ApproxByL2Truncations`.

Lemma 2.97 ($L^1(\gamma_n)$ is approximated by truncations lying in $L^2(\gamma_n)$)**.** *Fix $n \in \mathbb{N}$ and let $f \in L^1(\gamma_n)$, with γ_n the Gaussian measure of `GaussMeasure`. Then there is a sequence $(v_k)_{k\in\mathbb{N}}$ in $L^2(\gamma_n)$ such that*

$$\|\iota\, v_k\|_{L^1(\gamma_n)} \le \|f\|_{L^1(\gamma_n)} \quad \text{for every } k, \qquad \text{and} \qquad \iota\, v_k \longrightarrow f \quad \text{in } L^1(\gamma_n) \quad \text{as } k \to \infty,$$

where $\iota : L^2(\gamma_n) \to L^1(\gamma_n)$ is the inclusion of `ToL1`.

Both spaces are taken with respect to the Gaussian *measure γ_n: $\|\cdot\|_{L^1(\gamma_n)}$ is the γ_n-integral of the absolute value and $\|\cdot\|_{L^2(\gamma_n)}$ the square root of the γ_n-integral of the square, and no Lebesgue measure appears. The uniform bound is by $\|f\|_{L^1(\gamma_n)}$ itself, not by a constant multiple of it; this is what makes the sequence usable where a norm budget has to be respected exactly.*

Proof. Let $F : \mathbb{R}^n \to \mathbb{R}$ be a representative of f, and define the level-k truncation

$$T_k(x) = \max\big(-k,\ \min(F(x),\ k)\big), \qquad k \in \mathbb{N},\ x \in \mathbb{R}^n.$$

Step 1: $|T_k| \le k$ everywhere. By construction $T_k(x) \ge -k$, since $T_k(x)$ is a maximum one of whose arguments is $-k$. For the upper bound, $T_k(x) \le k$ holds because both arguments of the maximum are at most k: the first is $-k \le k$ as $k \ge 0$, and the second is $\min(F(x), k) \le k$. Hence $|T_k(x)| \le k$.

Step 2: $|T_k| \le |F|$ everywhere. Again bound $T_k(x)$ from both sides. From below, $-|F(x)| \le \min(F(x), k)$, because $-|F(x)| \le F(x)$ and $-|F(x)| \le 0 \le k$; and $\min(F(x), k) \le T_k(x)$ since $T_k(x)$ is a maximum one of whose arguments is that minimum. From above, both arguments of the maximum are at most $|F(x)|$: the first because $-k \le 0 \le |F(x)|$ requires only $k \ge 0$, and the second because $\min(F(x), k) \le F(x) \le |F(x)|$. Hence $|T_k(x)| \le |F(x)|$.

Step 3: $T_k(x) \to F(x)$ for every x. Fix x and let $k \geq \lceil |F(x)| \rceil$, so that $|F(x)| \leq k$. Then $F(x) \leq k$, whence $\min(F(x), k) = F(x)$; and $-k \leq F(x)$, whence $\max(-k, F(x)) = F(x)$. So $T_k(x) = F(x)$ for all such k: the sequence is eventually constant equal to $F(x)$, and in particular converges to it.

Step 4: T_k defines an element v_k of $L^2(\gamma_n)$. F is almost everywhere strongly measurable, being a representative of an $L^1(\gamma_n)$ class, and T_k is obtained from it by taking a maximum and a minimum against constants, so T_k is almost everywhere strongly measurable. By Step 1 it is bounded by the constant k, and γ_n is a finite measure – indeed a probability measure, by `GaussIsProb` – so a bounded almost everywhere strongly measurable function is square integrable: $\int |T_k|^2 \, d\gamma_n \leq k^2 \, \gamma_n(\mathbb{R}^n) < \infty$. Let $v_k \in L^2(\gamma_n)$ be the class of T_k. By `ToL1` the inclusion ι does not change the underlying function, so T_k is also a representative of $\iota \, v_k \in L^1(\gamma_n)$.

Step 5: the uniform norm bound. The $L^1(\gamma_n)$ norm of a class is the γ_n-integral of the absolute value of any representative, so by Step 4 and then Step 2,

$$\|\iota \, v_k\|_{L^1(\gamma_n)} = \int_{\mathbb{R}^n} |T_k| \, d\gamma_n \leq \int_{\mathbb{R}^n} |F| \, d\gamma_n = \|f\|_{L^1(\gamma_n)},$$

the inequality by monotonicity of the integral, both integrands being integrable: $|F|$ because $f \in L^1(\gamma_n)$, and $|T_k|$ because it is dominated by $|F|$.

Step 6: convergence in $L^1(\gamma_n)$. Since T_k represents $\iota \, v_k$ and F represents f, the difference $\iota \, v_k - f$ is represented by $T_k - F$, so

$$\|\iota \, v_k - f\|_{L^1(\gamma_n)} = \int_{\mathbb{R}^n} |T_k - F| \, d\gamma_n.$$

Apply dominated convergence to the integrands $g_k = |T_k - F|$. They are almost everywhere strongly measurable, being absolute values of differences of such functions. They are dominated uniformly in k: by the triangle inequality and Step 2,

$$|T_k(x) - F(x)| \leq |T_k(x)| + |F(x)| \leq 2 \, |F(x)| \qquad \text{for every } x,$$

and $2|F|$ is γ_n-integrable because $f \in L^1(\gamma_n)$. And they tend to 0 pointwise: by Step 3, $T_k(x) - F(x) \to 0$ for every x, hence so does its absolute value. Dominated convergence therefore gives

$$\int_{\mathbb{R}^n} |T_k - F| \, d\gamma_n \longrightarrow 0 \qquad (k \to \infty),$$

that is, $\|\iota \, v_k - f\|_{L^1(\gamma_n)} \to 0$. Convergence of the norms of the differences to 0 is convergence in the normed space, so $\iota \, v_k \to f$ in $L^1(\gamma_n)$. Together with Step 5 this exhibits the required sequence. □

2.98. `RieszL1Limit`.

Lemma 2.98 (A common in-measure limit for the Riesz images of L^1 approximating sequences)**.** *Let $f \in L^1(\gamma_n; \mathbb{R})$. Then there exist an element g of $\mathcal{M}(\gamma_n; \mathbb{R}^n)$, the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions, and a sequence $(v_k)_{k \in \mathbb{N}}$ in $L^2(\gamma_n)$, such that*

1. *$\|\iota v_k\|_{L^1(\gamma_n)} \leq \|f\|_{L^1(\gamma_n)}$ for every k, where ι is the inclusion of `ToL1`;*
2. *$\iota v_k \to f$ in $L^1(\gamma_n)$;*
3. *$R_\gamma v_k \to g$ pointwise γ_n-a.e., with R_γ as in `RieszTransform`; and*
4. *for* every *sequence (w_k) in $L^2(\gamma_n)$ with $\iota w_k \to f$ in $L^1(\gamma_n)$, the sequence $R_\gamma w_k$ converges to g in measure, i.e. $\gamma_n\big(\{x : \varepsilon \leq |R_\gamma w_k(x) - g(x)|\}\big) \to 0$ as $k \to \infty$, for every $\varepsilon > 0$.*

Proof. Part A: R_γ is linear on $L^2(\gamma_n)$. By `RieszTransform`, $R_\gamma v = D L^{-1/2} v$. Part (1) of `FracPowLinear` with $s = -1/2 \leq 0$ says that $\mathrm{Dom}(L^{-1/2})$ is all of $L^2(\gamma_n)$, so parts (2) and (3) of `FracPowLinear` apply to every pair of elements and give $L^{-1/2}(v + w) = L^{-1/2} v + L^{-1/2} w$ and $L^{-1/2}(cv) = c \, L^{-1/2} v$ for all $v, w \in L^2(\gamma_n)$ and $c \in \mathbb{R}$. By `PseudoInverseSqrt`, $L^{-1/2} v \in \mathrm{Dom}(L^{1/2})$ for every v, and hence $L^{-1/2} v \in W^{1,2}(\gamma_n)$ by `SqrtDomainSobolev`. Parts (1) and

(2) of GaussDLinear then give $D(L^{-1/2}v + L^{-1/2}w) = DL^{-1/2}v + DL^{-1/2}w$ and $D(c\,L^{-1/2}v) = c\,DL^{-1/2}v$. Composing,

$$R_\gamma(v+w) = R_\gamma v + R_\gamma w, \qquad R_\gamma(cv) = c\,R_\gamma v, \tag{A}$$

and in particular, taking $c=-1$, $R_\gamma(v-w) = R_\gamma v - R_\gamma w$. Also $\iota(v-w) = \iota v - \iota w$, since by ToL1 both sides have the γ_n-a.e. representative $v-w$.

Part B: the Cauchy-in-measure estimate. Let $v, w \in L^2(\gamma_n)$ and $\varepsilon > 0$. Apply MainBound to the L^2 function $v-w$ at the level $\varepsilon/2 > 0$:

$$\tfrac{\varepsilon}{2}\,\gamma_n\big(\{x : \tfrac{\varepsilon}{2} < |R_\gamma(v-w)(x)|\}\big) \le 2\,\|\iota(v-w)\|_{L^1(\gamma_n)}.$$

By Part A, $R_\gamma(v-w) = R_\gamma v - R_\gamma w$ and $\iota(v-w) = \iota v - \iota w$, so γ_n-a.e. the function $R_\gamma(v-w)$ agrees with $R_\gamma v - R_\gamma w$. Since $\varepsilon/2 < \varepsilon$, the set $\{x : \varepsilon \le |R_\gamma v(x) - R_\gamma w(x)|\}$ is therefore contained, up to a γ_n-null set, in $\{x : \varepsilon/2 < |R_\gamma(v-w)(x)|\}$, and monotonicity of γ_n gives

$$\tfrac{\varepsilon}{2}\,\gamma_n\big(\{x : \varepsilon \le |R_\gamma v(x) - R_\gamma w(x)|\}\big) \le 2\,\|\iota v - \iota w\|_{L^1(\gamma_n)}.$$

Dividing by $\varepsilon/2 > 0$,

$$\gamma_n\big(\{x : \varepsilon \le |R_\gamma v(x) - R_\gamma w(x)|\}\big) \le \frac{4\,\|\iota v - \iota w\|_{L^1(\gamma_n)}}{\varepsilon}. \tag{B}$$

Part C: the truncations. Write F for a fixed representative of f and put $\tau_k(x) = \max\{-k, \min\{F(x), k\}\}$ for $k \in \mathbb{N}$. Each τ_k is measurable, being built from F by two lattice operations with constants. For every x and k we have, since $k \ge 0$,

$$|\tau_k(x)| \le k \qquad \text{and} \qquad |\tau_k(x)| \le |F(x)|. \tag{C1}$$

For the first: $-k \le \max\{-k, \min\{F(x), k\}\} \le \max\{-k, k\} = k$. For the second: $-|F(x)| \le \min\{F(x), k\}$ because $-|F(x)| \le F(x)$ and $-|F(x)| \le 0 \le k$, whence $-|F(x)| \le \tau_k(x)$; and $\tau_k(x) \le |F(x)|$ because $-k \le 0 \le |F(x)|$ and $\min\{F(x), k\} \le F(x) \le |F(x)|$. Since γ_n is a probability measure (GaussIsProb), the bound $|\tau_k| \le k$ makes τ_k an element of $L^2(\gamma_n)$; call it $\tilde{v}_k$. Moreover, for every x, as soon as $k \ge |F(x)|$ we have $-k \le F(x) \le k$, hence $\tau_k(x) = F(x)$; so $\tau_k(x) \to F(x)$ for every x, the sequence being eventually constant. The second bound in (C1) gives, by monotonicity of the integral, $\|\iota\tilde{v}_k\|_{L^1(\gamma_n)} = \int |\tau_k|\,d\gamma_n \le \int |F|\,d\gamma_n = \|f\|_{L^1(\gamma_n)}$, which is conclusion (1) for this sequence. Finally $|\tau_k(x) - F(x)| \le |\tau_k(x)| + |F(x)| \le 2|F(x)|$ by (C1), the dominating function $2|F|$ is integrable because $f \in L^1(\gamma_n)$, and $|\tau_k(x) - F(x)| \to 0$ for every x; so dominated convergence gives $\|\iota\tilde{v}_k - f\|_{L^1(\gamma_n)} = \int |\tau_k - F|\,d\gamma_n \to 0$, i.e. $\iota\tilde{v}_k \to f$ in $L^1(\gamma_n)$, which is conclusion (2).

Part D: a rapidly converging subsequence. Since $\|\iota\tilde{v}_k - f\|_{L^1(\gamma_n)} \to 0$, for each $j \in \mathbb{N}$ we may choose an index $\sigma(j)$ with $\|\iota\tilde{v}_{\sigma(j)} - f\|_{L^1(\gamma_n)} \le \frac{1}{2}8^{-j}$. Put $v_j = \tilde{v}_{\sigma(j)}$. Conclusion (1) is inherited, and $\|\iota v_j - f\|_{L^1(\gamma_n)} \le \frac{1}{2}8^{-j} \to 0$ gives conclusion (2) for (v_j). By the triangle inequality and $8^{-(j+1)} \le 8^{-j}$,

$$\|\iota v_j - \iota v_{j+1}\|_{L^1(\gamma_n)} \le \|\iota v_j - f\|_{L^1(\gamma_n)} + \|\iota v_{j+1} - f\|_{L^1(\gamma_n)} \le \tfrac{1}{2}8^{-j} + \tfrac{1}{2}8^{-(j+1)} \le 8^{-j}. \tag{D}$$

Part E: the almost-everywhere limit. Put $E_j = \{x : 2^{-j} \le |R_\gamma v_j(x) - R_\gamma v_{j+1}(x)|\}$. By (B) with $\varepsilon = 2^{-j}$ and then (D),

$$\gamma_n(E_j) \le \frac{4\,\|\iota v_j - \iota v_{j+1}\|_{L^1(\gamma_n)}}{2^{-j}} \le 4 \cdot 2^j \cdot 8^{-j} = 4 \cdot 4^{-j}.$$

The series $\sum_j 4 \cdot 4^{-j}$ converges, so $\sum_j \gamma_n(E_j) < \infty$. By the Borel–Cantelli lemma, for γ_n-a.e. x the index set $\{j : x \in E_j\}$ is finite. Fix such an x and let B be an upper bound for that finite set. For every $j \ge B+1$ we then have $x \notin E_j$, i.e. $|R_\gamma v_j(x) - R_\gamma v_{j+1}(x)| < 2^{-j}$. Hence the series $\sum_j |R_\gamma v_j(x) - R_\gamma v_{j+1}(x)|$ has all but finitely many terms dominated by the convergent geometric series $\sum_j 2^{-j}$, so it converges; a sequence in the complete space $\mathbb{R}^n$ whose consecutive distances form a convergent series is Cauchy, hence converges. Define $g(x)$ to be this limit at every x for which $(R_\gamma v_j(x))_j$ converges, and $g(x) = 0$ elsewhere. By construction $R_\gamma v_j \to g$ pointwise γ_n-a.e., which is conclusion (3); and g, being a γ_n-a.e. pointwise limit of the measurable functions $R_\gamma v_j$, is measurable, so it determines an element of $\mathcal{M}(\gamma_n; \mathbb{R}^n)$.

Part F: every approximating sequence has the same limit in measure. Since γ_n is a finite measure (`GaussIsProb`), the a.e. convergence of conclusion (3) implies $R_\gamma v_k \to g$ in measure; that is, for every $\delta > 0$,

$$\gamma_n\big(\{x : \delta \le |R_\gamma v_k(x) - g(x)|\}\big) \longrightarrow 0. \tag{F}$$

Now let (w_k) be any sequence in $L^2(\gamma_n)$ with $\iota w_k \to f$ in $L^1(\gamma_n)$, and fix $\varepsilon > 0$. By the triangle inequality in $\mathbb{R}^n$, if $\varepsilon \le |R_\gamma w_k(x) - g(x)|$ then at least one of $|R_\gamma w_k(x) - R_\gamma v_k(x)|$ and $|R_\gamma v_k(x) - g(x)|$ is at least $\varepsilon/2$; hence

$$\{x : \varepsilon \le |R_\gamma w_k(x) - g(x)|\} \subseteq \{x : \tfrac{\varepsilon}{2} \le |R_\gamma w_k(x) - R_\gamma v_k(x)|\} \cup \{x : \tfrac{\varepsilon}{2} \le |R_\gamma v_k(x) - g(x)|\},$$

and by monotonicity and subadditivity of γ_n,

$$\gamma_n\big(\{x : \varepsilon \le |R_\gamma w_k(x) - g(x)|\}\big) \le \gamma_n\big(\{x : \tfrac{\varepsilon}{2} \le |R_\gamma w_k(x) - R_\gamma v_k(x)|\}\big) + \gamma_n\big(\{x : \tfrac{\varepsilon}{2} \le |R_\gamma v_k(x) - g(x)|\}\big).$$

The second summand tends to 0 by (F) with $\delta = \varepsilon/2$. For the first, $\|\iota w_k - \iota v_k\|_{L^1(\gamma_n)} \to 0$, because both ιw_k and ιv_k converge to f in $L^1(\gamma_n)$; so (B) with $\varepsilon/2$ bounds the first summand by $8\,\|\iota w_k - \iota v_k\|_{L^1(\gamma_n)}/\varepsilon \to 0$. Both summands are nonnegative, so their sum tends to 0 and the left-hand side, squeezed between 0 and that sum, tends to 0 as well. This is conclusion (4). □

2.99. `L1Extension`.

Lemma 2.99 (Existence of the L^1 extension satisfying the weak bound)**.** *There is a linear map*

$$T : L^1(\gamma_n; \mathbb{R}) \longrightarrow \mathcal{M}(\gamma_n; \mathbb{R}^n),$$

where $\mathcal{M}(\gamma_n; \mathbb{R}^n)$ denotes the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions, such that

1. *T agrees with R_γ on $L^2(\gamma_n)$: for every $v \in L^2(\gamma_n)$ one has $T(\iota v) = R_\gamma v$ almost everywhere, with ι the inclusion of `ToL1` and R_γ as in `RieszTransform`; and*
2. *T satisfies the weak bound*

$$t\,\gamma_n\big(\{x : |Tf(x)| > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)} \qquad \text{for every } f \in L^1(\gamma_n) \text{ and } t > 0.$$

Proof. Step 0: a Cauchy-in-measure estimate. Let $v, w \in L^2(\gamma_n)$ and $\varepsilon > 0$. By `RieszTransform` the map R_γ is linear on $L^2(\gamma_n)$, so $R_\gamma v - R_\gamma w = R_\gamma(v - w)$, and `MainBound` applied to the real L^2 function $v - w$ at the level ε gives

$$\gamma_n\big(\{|R_\gamma v - R_\gamma w| > \varepsilon\}\big) \le \frac{2}{\varepsilon}\,\|\iota(v - w)\|_{L^1(\gamma_n)}. \tag{1}$$

Step 1: truncations approximate in L^1. Let $f \in L^1(\gamma_n)$ and put $f_k = \max\{-k, \min\{f, k\}\}$ for $k \in \mathbb{N}$. Each f_k is measurable with $|f_k| \le k$, hence bounded, hence in $L^2(\gamma_n)$ because γ_n is a probability measure (`GaussIsProb`). Moreover $|f_k| \le |f|$ and $f_k \to f$ pointwise, so $|f - f_k| \le 2|f| \in L^1(\gamma_n)$ and $|f - f_k| \to 0$ pointwise; dominated convergence gives $\|f - f_k\|_{L^1(\gamma_n)} \to 0$. In particular $(\iota f_k)_k$ is Cauchy in $L^1(\gamma_n)$.

Step 2: an almost-everywhere convergent subsequence. Using Step 1, choose indices $k_1 < k_2 < \cdots$ with $\|f_{k_{j+1}} - f_{k_j}\|_{L^1(\gamma_n)} \le 2^{-3j}$ for every $j \ge 1$ (possible since the sequence is Cauchy in L^1). Applying (1) with $v = f_{k_{j+1}}$, $w = f_{k_j}$ and $\varepsilon = 2^{-j}$,

$$\gamma_n\big(\{|R_\gamma f_{k_{j+1}} - R_\gamma f_{k_j}| > 2^{-j}\}\big) \le 2 \cdot 2^{j} \cdot 2^{-3j} = 2^{1-2j}.$$

The right-hand side is summable in j, so by the Borel–Cantelli lemma the set of x lying in infinitely many of the sets $E_j = \{|R_\gamma f_{k_{j+1}} - R_\gamma f_{k_j}| > 2^{-j}\}$ is γ_n-null. Hence for γ_n-a.e. x there is $J(x)$ with $|R_\gamma f_{k_{j+1}}(x) - R_\gamma f_{k_j}(x)| \le 2^{-j}$ for all $j \ge J(x)$; the tail sums $\sum_{j \ge J} 2^{-j}$ are finite, so $(R_\gamma f_{k_j}(x))_j$ is a Cauchy sequence in $\mathbb{R}^n$ and converges to a finite limit. Define Tf to be the γ_n-a.e. pointwise limit of $R_\gamma f_{k_j}$ (set to 0 on the null set where convergence fails); it is a measurable $\mathbb{R}^n$-valued function, hence an element of $\mathcal{M}(\gamma_n; \mathbb{R}^n)$.

Step 3: convergence in measure of the full sequence, and well-definedness. By Step 2 and Egorov-free elementary reasoning, $R_\gamma f_{k_j} \to Tf$ γ_n-a.e., hence in measure (on the finite measure space γ_n, a.e. convergence implies convergence in measure). Combined with (1) and Step 1,

which make $(R_\gamma f_k)_k$ Cauchy in measure, the full sequence $R_\gamma f_k$ converges to Tf in measure: given $\varepsilon > 0$,

$$\gamma_n(\{|R_\gamma f_k - Tf| > 2\varepsilon\}) \le \gamma_n(\{|R_\gamma f_k - R_\gamma f_{k_j}| > \varepsilon\}) + \gamma_n(\{|R_\gamma f_{k_j} - Tf| > \varepsilon\}),$$

and both terms can be made small, the first by (1) and Step 1, the second by a.e. convergence.

If $(\tilde f_k) \subseteq L^2(\gamma_n)$ is any other sequence with $\iota \tilde f_k \to f$ in $L^1(\gamma_n)$, then (1) applied to the pair $f_k, \tilde f_k$ gives $\gamma_n(\{|R_\gamma f_k - R_\gamma \tilde f_k| > \varepsilon\}) \le \frac{2}{\varepsilon}\|\iota f_k - \iota \tilde f_k\|_{L^1} \to 0$, so $R_\gamma \tilde f_k \to Tf$ in measure as well, and a limit in measure is unique up to a null set. Hence Tf does not depend on the approximating sequence.

Step 4: linearity. Let $f, h \in L^1(\gamma_n)$ and $\lambda \in \mathbb{R}$, and let $(f_k), (h_k)$ be the truncation sequences of Step 1. Then $(f_k + h_k)$ and (λf_k) are sequences in $L^2(\gamma_n)$ whose inclusions converge in $L^1(\gamma_n)$ to $f + h$ and λf respectively. By linearity of R_γ (`RieszTransform`), $R_\gamma(f_k + h_k) = R_\gamma f_k + R_\gamma h_k$ and $R_\gamma(\lambda f_k) = \lambda R_\gamma f_k$. Convergence in measure is compatible with addition and scalar multiplication, so the left sides converge in measure to $T(f + h)$ and $T(\lambda f)$ by Step 3, while the right sides converge in measure to $Tf + Th$ and λTf. Uniqueness of limits in measure gives $T(f + h) = Tf + Th$ and $T(\lambda f) = \lambda Tf$.

Step 5: agreement with R_γ on L^2. Let $v \in L^2(\gamma_n)$. The constant sequence $\tilde f_k = v$ lies in $L^2(\gamma_n)$ and satisfies $\iota \tilde f_k \to \iota v$ in $L^1(\gamma_n)$ trivially. By the independence established in Step 3, $T(\iota v) = \lim_k R_\gamma \tilde f_k = R_\gamma v$ γ_n-a.e.

Step 6: the weak bound. Fix $f \in L^1(\gamma_n)$ and $t > 0$, and let (f_{k_j}) be the subsequence of Step 2, so $R_\gamma f_{k_j} \to Tf$ γ_n-a.e. At a point x with $|Tf(x)| > t$ there is a strict margin $|Tf(x)| - t > 0$, and $|R_\gamma f_{k_j}(x)| \to |Tf(x)|$, so $|R_\gamma f_{k_j}(x)| > t$ for all large j. Hence γ_n-a.e.

$$\mathbf{1}_{\{|Tf|>t\}} \le \liminf_{j\to\infty} \mathbf{1}_{\{|R_\gamma f_{k_j}|>t\}}.$$

Fatou's lemma, applied to these nonnegative measurable functions, gives

$$\gamma_n(\{|Tf| > t\}) \le \liminf_j \gamma_n\big(\{|R_\gamma f_{k_j}| > t\}\big).$$

By `MainBound` applied to f_{k_j} at level t, each term is at most $\frac{2}{t}\|\iota f_{k_j}\|_{L^1(\gamma_n)}$. By Step 1, $\|\iota f_{k_j}\|_{L^1(\gamma_n)} \to \|f\|_{L^1(\gamma_n)}$, so

$$\gamma_n(\{|Tf| > t\}) \le \frac{2}{t}\,\|f\|_{L^1(\gamma_n)}.$$

Multiplying by $t > 0$ gives the asserted bound. □

2.100. `GaussLebesgueBallComparison`.

Lemma 2.100 (Comparison of γ_n with Lebesgue measure on a ball, and the null-set equivalence)**.** *Fix $n \in \mathbb{N}_0$, let λ denote Lebesgue measure on $\mathbb{R}^n$ and let γ_n be the Gaussian measure of `GaussMeasure`. For $R \in \mathbb{R}$ write $a_R = \pi^{-n/2} e^{-R^2}$. Then:*

(1) *for* every *set $S \subseteq \mathbb{R}^n$, measurable or not,*

$$\gamma_n(S) \le \pi^{-n/2}\,\lambda(S);$$

(2) *for every $R \in \mathbb{R}$ and every measurable $E \subseteq \overline{B}(0, R)$,*

$$a_R\,\lambda(E) \le \gamma_n(E);$$

(3) *for every $R, \rho \in \mathbb{R}$ and every $x \in \mathbb{R}^n$ with $|x| \le R$ and $0 < \rho \le 2$,*

$$a_{R+2}\,\lambda\big(\overline{B}(x,\rho)\big) \le \gamma_n\big(\overline{B}(x,\rho)\big);$$

(4) *for every set $A \subseteq \mathbb{R}^n$, measurable or not,*

$$\gamma_n(A) = 0 \iff \lambda(A) = 0.$$

Here $\overline{B}(x,\rho)$ is the closed ball, and both γ_n and λ are applied to arbitrary sets as outer measures, i.e. through $\mu(S) = \inf\{\mu(F) : S \subseteq F,\ F \text{ measurable}\}$. All products and comparisons are read in $[0, +\infty]$.

What this adds to `GaussMeasure`, which only says what γ_n is. *Clause (1) is a global one-sided comparison, valid for all sets, and it is the only direction that is globally true. Clause (2) is the*

reverse comparison, which is false *globally — the ratio $\gamma_n(E)/\lambda(E)$ tends to 0 as E escapes to infinity — and becomes true, with a quantitative constant, exactly because E is confined to a ball; it is this localization that removes any need for covering theory adapted to the non-doubling measure γ_n, by reducing a local Gaussian averaging argument to a Lebesgue one. Clause (3) is the form in which a covering argument uses clause (2): the ball $\overline{B}(x,\rho)$ is not centred at the origin, and the enlargement from R to $R+2$ is exactly what a cap of 2 on the radius of the balls costs. Clause (4) is a statement about* all *sets and is not a comparison of sizes at all: it says the two measures have the same null sets, so that "λ-almost everywhere" and "γ_n-almost everywhere" are interchangeable, and so that an almost-everywhere identity between continuous functions for one measure is one for the other. None of the four is the density formula, and none follows from it without the argument below.*

The constants are explicit and dimension-dependent. *Both $\pi^{-n/2}$ and $a_R = \pi^{-n/2}e^{-R^2}$ depend on n, and a_R degrades as R grows; no uniformity in n or in R is claimed.*

Proof. By `GaussMeasure`, γ_n is Lebesgue measure weighted by the density

$$\varrho(y) = \pi^{-n/2}e^{-|y|^2},$$

that is, $\gamma_n(E) = \int_E \varrho\, d\lambda$ for every measurable $E \subseteq \mathbb{R}^n$, the integral being the lower Lebesgue integral of a nonnegative function. Note $0 < \varrho(y) \le \pi^{-n/2}$ for every y, since $0 < e^{-|y|^2} \le 1$, and $\varrho(y) \ge \pi^{-n/2}e^{-R^2} = a_R$ whenever $|y| \le R$, since $t \mapsto e^{-t}$ is decreasing and $|y|^2 \le R^2$ in that case.

Step 1: clause (1) for measurable sets. Let E be measurable. By monotonicity of the integral in the integrand,

$$\gamma_n(E) = \int_E \varrho\, d\lambda \le \int_E \pi^{-n/2}\, d\lambda = \pi^{-n/2}\lambda(E).$$

Step 2: clause (1) for arbitrary sets. Let $S \subseteq \mathbb{R}^n$ be arbitrary and let F be any measurable set with $S \subseteq F$. By monotonicity of γ_n and Step 1,

$$\gamma_n(S) \le \gamma_n(F) \le \pi^{-n/2}\lambda(F).$$

Taking the infimum over all such F — the family is nonempty, since $F = \mathbb{R}^n$ qualifies — and using that multiplication by the finite constant $\pi^{-n/2}$ commutes with infima of nonempty families in $[0,+\infty]$, we get

$$\gamma_n(S) \le \pi^{-n/2}\inf\{\lambda(F) : S \subseteq F,\ F \text{ measurable}\} = \pi^{-n/2}\lambda(S),$$

the last equality being the displayed description of λ on arbitrary sets. This is clause (1).

Step 3: clause (2). Let $R \in \mathbb{R}$ and let E be measurable with $E \subseteq \overline{B}(0,R)$. If $R < 0$ then $\overline{B}(0,R) = \emptyset$, so $E = \emptyset$ and both sides vanish. Assume $R \ge 0$. Every $y \in E$ satisfies $|y| \le R$, hence $\varrho(y) \ge a_R$ there, and therefore

$$\gamma_n(E) = \int_E \varrho\, d\lambda \ge \int_E a_R\, d\lambda = a_R\, \lambda(E),$$

again by monotonicity of the integral in the integrand. This is clause (2).

Step 4: clause (3). Let x with $|x| \le R$ and $0 < \rho \le 2$. For every $y \in \overline{B}(x,\rho)$ the triangle inequality gives

$$|y| \le |x| + |y - x| \le R + \rho \le R + 2,$$

so $\overline{B}(x,\rho) \subseteq \overline{B}(0,R+2)$. The closed ball is a closed, hence measurable, set. Clause (2) applied with $R+2$ in place of R and $E = \overline{B}(x,\rho)$ gives $a_{R+2}\lambda(\overline{B}(x,\rho)) \le \gamma_n(\overline{B}(x,\rho))$, which is clause (3).

Step 5: clause (4), the direction $\lambda(A) = 0 \Rightarrow \gamma_n(A) = 0$. Immediate from clause (1): $\gamma_n(A) \le \pi^{-n/2} \cdot 0 = 0$.

Step 6: clause (4), the direction $\gamma_n(A) = 0 \Rightarrow \lambda(A) = 0$. Suppose $\gamma_n(A) = 0$. Since $\gamma_n(A)$ is the infimum of $\gamma_n(F)$ over measurable $F \supseteq A$, there is a measurable F with $A \subseteq F$ and $\gamma_n(F) = 0$. (Choose measurable $F_k \supseteq A$ with $\gamma_n(F_k) \le 1/k$ and set $F = \bigcap_k F_k$, which is measurable, contains A, and has $\gamma_n(F) \le \gamma_n(F_k) \le 1/k$ for every k, hence $\gamma_n(F) = 0$.)

Fix $m \in \mathbb{N}$ and put $F'_m = F \cap \overline{B}(0,m)$, a measurable subset of $\overline{B}(0,m)$. By clause (2) with $R = m$, and by monotonicity of γ_n,

$$a_m\, \lambda(F'_m) \le \gamma_n(F'_m) \le \gamma_n(F) = 0.$$

Since $a_m = \pi^{-n/2} e^{-m^2} > 0$, this forces $\lambda(F'_m) = 0$. Now $\mathbb{R}^n = \bigcup_{m\in\mathbb{N}} \overline{B}(0,m)$, so $F = \bigcup_{m\in\mathbb{N}} F'_m$, and countable subadditivity of λ (shared context) gives $\lambda(F) \le \sum_m \lambda(F'_m) = 0$. Finally $\lambda(A) \le \lambda(F) = 0$ by monotonicity. This completes clause (4), and the proof. □

2.101. `CoordinateEndpointLevelSetSmall`.

Lemma 2.101 (The weak-type level set of a coordinate of Th occupies at most a quarter of a small ball)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$ and let γ_n be the Gaussian measure of* `GaussMeasure`*. Let*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of* `ToL1` *and R_γ the transform of* `RieszTransform`*; and*

(T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

Let $R \ge 1$. Then there is a constant $K \in (0,\infty)$, depending only on n, j and R, such that the following holds.

Let $h \in L^1(\gamma_n)$, written also for its canonical representative, let $x \in \mathbb{R}^n$ and let ε and Φ be real, and suppose

$$|x| \le R, \qquad 0 < \varepsilon \le 1, \qquad 0 \le \Phi, \qquad \int_{\mathbb{R}^n} |h|\, d\gamma_n \;\le\; \Phi \cdot \lambda\big(B(x,\varepsilon)\big).$$

Then

$$\gamma_n\Big(\big\{z \in B(x,\tfrac12\varepsilon) : \big|(Th)_j(z)\big| > K\Phi\big\}\Big) \;\le\; \tfrac14\,\gamma_n\big(B(x,\tfrac12\varepsilon)\big),$$

where $(Th)_j$ denotes the jth coordinate of the canonical representative of Th, and $\lambda(B(x,\varepsilon))$ is read as the real number it is, the ball being bounded and hence of finite measure.

The threshold is $K\Phi$ and not a named constant, deliberately. *A consumer arrives with a specific threshold built from its own bookkeeping — in the intended application, $8\cdot 2^n \pi^{-n/2} e^{(R+2)^2}\Phi$ — and the only thing that matters about it is that it is a large enough multiple of Φ. Exporting the multiple existentially rather than writing the consumer's expression into this statement keeps the node free of arithmetic that is none of its business, and the existential constant absorbs it at no cost. One may take $K = 8\cdot 2^n/a$ with $a = \pi^{-n/2}e^{-(R+2)^2}$.*

The degenerate case $\Phi = 0$ is included and is not vacuous. *When $\Phi = 0$ the hypothesis forces $\int |h|\, d\gamma_n = 0$, hence $h = 0$ in $L^1(\gamma_n)$, and the conclusion asserts that $\{z \in B(x,\frac12\varepsilon) : |(Th)_j(z)| > 0\}$ is γ_n-null. That is true but it needs the* linearity *of T — it is $T0 = 0$ that makes it true — and not (T2), which is unavailable at threshold 0. So the statement is not merely the weak-type bound rearranged: it covers a case the weak-type bound cannot reach.*

Both the hypothesis and the conclusion are insensitive to the representative. *Changing the representative of h on a γ_n-null set changes neither $\int |h|\, d\gamma_n$ nor the class Th, and changing the representative of Th on a null set changes neither the level set's γ_n-measure nor the conclusion; so fixing the canonical representatives is notation, not an extra hypothesis.*

The constant depends on R, necessarily, and no uniformity is claimed. *K degrades in R through a, the lower bound for the Gaussian density on $\overline{B}(0,R+2)$, and in n through both a and the factor 2^n coming from the ratio $\lambda(B(x,\frac12\varepsilon))/\lambda(B(x,\varepsilon))$.*

Proof. Put $a = \pi^{-n/2}e^{-(R+2)^2} > 0$ and $K = 8\cdot 2^n/a \in (0,\infty)$, which depends only on n, j and R. Fix h, x, ε, Φ and the hypotheses, and write $V = \lambda(B(x,\varepsilon)) \in [0,\infty)$, finite because the ball is bounded, and $B' = B(x,\frac12\varepsilon)$.

Step 1: three facts about the balls. Lebesgue measure on $\mathbb{R}^n$ is an additive Haar measure and $\mathbb{R}^n$ has real dimension n, so $\lambda(B(x,r)) = r^n\lambda(B(0,1))$ for every $r \ge 0$ (shared context; the

hypothesis $n \geq 1$ is what makes $\mathbb{R}^n$ nontrivial, as that statement requires). Taking $r = \frac{1}{2}\varepsilon$ and $r = \varepsilon$ and dividing,

$$\lambda(B') = 2^{-n}\, V. \tag{1}$$

Next, B' is measurable and $B' \subseteq \overline{B}(0, R+2)$: for $y \in B'$ we have $|y| - |x| \leq |y - x| < \frac{1}{2}\varepsilon \leq \frac{1}{2}$, so $|y| \leq R + \frac{1}{2} \leq R + 2$. Hence clause (2) of `GaussLebesgueBallComparison`, applied with $R + 2$ in place of its R and with $E = B'$, gives

$$a\, \lambda(B') \leq \gamma_n(B'). \tag{2}$$

Combining (1) and (2) and dividing by 4,

$$a\, 2^{-n}\, V/4 \;\leq\; \tfrac{1}{4}\gamma_n(B'). \tag{3}$$

Finally, the $L^1(\gamma_n)$ norm of h is the integral of the absolute value of its canonical representative, so the hypothesis reads

$$\|h\|_{L^1(\gamma_n)} \leq \Phi\, V. \tag{4}$$

Step 2: the case $\Phi = 0$. Then (4) gives $\|h\|_{L^1(\gamma_n)} \leq 0$, and a norm is nonnegative, so $\|h\|_{L^1(\gamma_n)} = 0$ and therefore $h = 0$ in $L^1(\gamma_n)$. Since T is linear, $Th = T0 = 0$, so the canonical representative of Th is 0 γ_n-almost everywhere. The threshold is $K\Phi = 0$, so the level set in question is $\{z \in B' : |(Th)_j(z)| > 0\}$, and every z in it has $(Th)_j(z) \neq 0$, hence $(Th)(z) \neq 0$. So the level set is contained in the γ_n-null set $\{z : (Th)(z) \neq 0\}$ and is therefore itself γ_n-null, which is at most $\frac{1}{4}\gamma_n(B')$.

Step 3: the case $\Phi > 0$. Then $t := K\Phi > 0$. Write

$$E_1 = \{z \in B' : |(Th)_j(z)| > t\}, \qquad F = \{z : |(Th)_j(z)| > t\},$$

so $E_1 \subseteq F$ and hence $\gamma_n(E_1) \leq \gamma_n(F)$ by monotonicity of the measure. Applying `CoordinateEndpointBound` to this T with its two properties, which are (T1) and (T2) and hold by assumption, to the element h, to the coordinate j and to this $t > 0$,

$$t\, \gamma_n(F) \leq 2\, \|h\|_{L^1(\gamma_n)}.$$

Since $t > 0$ is finite and nonzero, multiplying through by t^{-1} gives $\gamma_n(F) \leq 2\|h\|_{L^1(\gamma_n)}/t$. By (4) and then the definition of K,

$$\frac{2\|h\|_{L^1(\gamma_n)}}{t} \leq \frac{2\,\Phi V}{K\Phi} = \frac{2V}{K} = \frac{2V\, a}{8 \cdot 2^n} = \frac{a\, 2^{-n} V}{4},$$

the cancellation of Φ being legitimate because $\Phi > 0$. Chaining this with $\gamma_n(E_1) \leq \gamma_n(F)$ and with (3),

$$\gamma_n(E_1) \leq \frac{a\, 2^{-n} V}{4} \leq \tfrac{1}{4}\gamma_n(B'),$$

which is the assertion. □

2.102. `LebesgueDilation`.

Lemma 2.102 (Scaling of Lebesgue measure under a dilation, and null sets under its inverse)**.** *Fix $n \in \mathbb{N}_0$ and write λ for Lebesgue measure on $\mathbb{R}^n$, applied to arbitrary subsets as an outer measure. For a real number c and a set $A \subseteq \mathbb{R}^n$ write $cA = \{c\, a : a \in A\}$ for the image of A under the dilation $z \mapsto cz$. Then:*

(1) *for every $c \in \mathbb{R}$ and every $A \subseteq \mathbb{R}^n$,*

$$\lambda(cA) = |c|^{\,n}\, \lambda(A),$$

an identity in $[0, +\infty]$; and

(2) *for every $c \in \mathbb{R}$ with $c \neq 0$ and every $N \subseteq \mathbb{R}^n$ with $\lambda(N) = 0$, the preimage of N under the dilation is again null:*

$$\lambda\big(\{z \in \mathbb{R}^n : cz \in N\}\big) = 0.$$

No measurability hypothesis, and no restriction on c or on n. *Clause (1) is asserted for every set, measurable or not, with λ read as an outer measure, which is the convention this tablet uses throughout; it holds in particular at $c = 0$, where for $A \neq \emptyset$ both sides read $0^n\lambda(A)$ against $\lambda(\{0\})$, and at $n = 0$, where both sides are $\lambda(\mathbb{R}^0)$ because $|c|^0 = 1$ and the dilation of the one-point space is itself. Clause (2) needs $c \neq 0$, since at $c = 0$ the preimage of a null set containing the origin is all of $\mathbb{R}^n$.*

Why clause (2) is stated separately rather than left to the reader. *Clause (1) is about the forward image and clause (2) about the preimage, and the two consumers in this tablet use different ones: a covering argument compares the measures of two concentric balls and needs the image form, while a change-of-variables argument needs to know that an exceptional set pulled back along $z \mapsto cz$ is still exceptional. Deriving (2) from (1) would mean rewriting the preimage as $c^{-1}N$ and invoking (1) at c^{-1}, a step whose only content is set-theoretic bookkeeping; stating it here means neither consumer carries that bookkeeping.*

This node exists to supply a classical fact that is not in the shared background. *It is purely Euclidean: no Gaussian weight appears, and it cites no other tablet node. The paper uses the dilation rule implicitly whenever it changes variables in the Mehler averaging formula, where the substitution $y = rx + \tau z$ has linear part τ times the identity.*

Proof. Write $E = \mathbb{R}^n$ with its Euclidean structure, so E is a finite-dimensional real normed space of dimension n and λ is an additive Haar measure on it.

Step 1 (clause (1)). For an additive Haar measure μ on a finite-dimensional real normed space E, the behaviour of μ under the dilation $z \mapsto cz$ is the classical scaling law

$$\mu(cA) = |c|^{\dim E}\,\mu(A) \qquad \text{for every } c \in \mathbb{R} \text{ and every } A \subseteq E,$$

read in $[0, +\infty]$ and valid for arbitrary A, μ being applied as an outer measure. This is the standard scaling property of Haar measure under the scalar action: the pushforward of μ along the linear automorphism $z \mapsto cz$ (for $c \neq 0$) is again a translation-invariant measure, hence a constant multiple of μ, and the constant is the absolute value of the determinant of that automorphism, namely $|c|^{\dim E}$; for $c = 0$ the image cA is $\{0\}$ when $A \neq \emptyset$ and $\emptyset$ when $A = \emptyset$, and the identity holds in both cases because $0^{\dim E}\mu(A)$ and $\mu(\{0\})$ are both 0 when $\dim E \geq 1$, while for $\dim E = 0$ the space is the single point 0 and both sides equal $\mu(E)$. Applying this with $\mu = \lambda$ and $\dim E = n$ gives clause (1).

Step 2 (clause (2)). Let $c \neq 0$ and let N satisfy $\lambda(N) = 0$. The dilation $z \mapsto cz$ is then a linear automorphism of $\mathbb{R}^n$, and the same scaling law applied to the preimage gives

$$\lambda\big(\{z : cz \in N\}\big) = |c|^{-n}\,\lambda(N),$$

the exponent being negated because the preimage along $z \mapsto cz$ is the image along $z \mapsto c^{-1}z$: indeed $\{z : cz \in N\} = c^{-1}N$, since $cz \in N$ holds exactly when $z = c^{-1}(cz)$ with $cz \in N$. Clause (1) at the scalar c^{-1} therefore evaluates the left-hand side as $|c^{-1}|^n\lambda(N) = |c|^{-n}\lambda(N)$. Since $\lambda(N) = 0$ and $|c|^{-n}$ is a finite nonnegative real, the product is 0, which is clause (2). □

2.103. `LocalMaximalWeakType`.

Lemma 2.103 (Weak-type $(1,1)$ bound for the local maximal function)**.** *Fix $n \in \mathbb{N}_0$ and let λ denote Lebesgue measure on $\mathbb{R}^n$. Let $w : \mathbb{R}^n \to \mathbb{R}$ be almost everywhere equal, for λ, to a Borel measurable function, and let $t \in (0, +\infty]$. Then*

$$\lambda\big(\{\, x \in \mathbb{R}^n : M_2 w(x) > t \,\}\big) \;\leq\; \frac{4^n}{t} \int_{\mathbb{R}^n} |w(y)|\, dy,$$

where M_2 is the local maximal function of `LocalMaximalFunction`, the integral is the lower Lebesgue integral of the nonnegative function $|w|$, taken in $[0, +\infty]$, and both sides and the quotient are read in $[0, +\infty]$, with the conventions $a/(+\infty) = 0$ and, for $a > 0$, $a/0 = +\infty$.

Two points about the shape of the statement. First, no measurability is asserted for the level set $\{M_2 w > t\}$, and none is needed: λ is applied to it as an outer measure, which is defined on every subset of $\mathbb{R}^n$ and is monotone and countably subadditive there. This is what lets consumers apply the bound to a level set of a maximal function without first proving that set measurable.

Second, no integrability hypothesis is placed on w: if $\int |w| = +\infty$ the right-hand side is $+\infty$ and the inequality is vacuous, while the content of the lemma is the case $\int |w| < \infty$. The constant 4^n is explicit rather than existentially quantified; it is the τ^n produced by a Vitali covering argument at $\tau = 4$, and it is not claimed to be optimal.

No Gaussian measure occurs here. *Every measure and every integral in the statement is Lebesgue, with no Gaussian weight anywhere, exactly as in `LocalMaximalFunction`; the statement is a classical fact about Euclidean space recorded here because it is not available in the ambient library.*

Proof. Write $E = \{x \in \mathbb{R}^n : M_2 w(x) > t\}$ and $I = \int_{\mathbb{R}^n} |w(y)|\, dy \in [0, +\infty]$, and recall from `LocalMaximalFunction` that

$$M_2 w(x) = \sup_{\rho \in (0,2]} \frac{1}{\lambda(\overline{B}(x,\rho))} \int_{\overline{B}(x,\rho)} |w|\, d\lambda. \tag{1}$$

Step 0: reduction to a measurable w. By hypothesis there is a Borel measurable $\widetilde{w} : \mathbb{R}^n \to \mathbb{R}$ with $w = \widetilde{w}$ λ-almost everywhere. Two nonnegative functions that agree λ-almost everywhere have the same lower Lebesgue integral against λ and against every restriction of λ to a measurable set, since a λ-null set is null for each such restriction. Hence $\int_{\overline{B}(x,\rho)} |w| = \int_{\overline{B}(x,\rho)} |\widetilde{w}|$ for every x and ρ, so by (1) $M_2 w = M_2 \widetilde{w}$ pointwise on $\mathbb{R}^n$, and also $I = \int |\widetilde{w}|$. Both sides of the asserted inequality are therefore unchanged if w is replaced by $\widetilde{w}$, and we may and do assume from now on that w is measurable.

Step 1: closed balls have positive finite volume, and the dilation rule. We claim that

$$0 < \lambda\big(\overline{B}(x,\rho)\big) < \infty \qquad \text{for all } x \in \mathbb{R}^n,\ \rho > 0. \tag{2}$$

Finiteness holds because $\overline{B}(x,\rho)$ is compact and a compact subset of $\mathbb{R}^n$ has finite Lebesgue measure (shared context). Positivity holds because $\overline{B}(x,\rho)$ contains the open ball $B(x,\rho)$, which is nonempty (it contains x) and open, and Lebesgue measure assigns strictly positive mass to every nonempty open subset of $\mathbb{R}^n$. Both arguments are valid for every $n \in \mathbb{N}_0$: when $n = 0$ the space $\mathbb{R}^0$ is a single point and every ball of nonnegative radius is the whole space, which is nonempty, open and compact.

We claim next the dilation rule

$$\lambda\big(\overline{B}(x,c\rho)\big) = c^n\, \lambda\big(\overline{B}(x,\rho)\big), \qquad c > 0,\ \rho \geq 0,\ x \in \mathbb{R}^n. \tag{3}$$

The ball $\overline{B}(x,c\rho)$ is the image of $\overline{B}(x,\rho)$ under the affine map $y \mapsto x + c(y-x)$, which is the composition of the translation by $-x$, the dilation $y \mapsto cy$ and the translation by x; concretely, $\overline{B}(x,c\rho) = x + c\,\big(\overline{B}(x,\rho) - x\big)$, since $|y-x| \leq c\rho$ holds exactly when $y = x + cz$ with $|z| \leq \rho$, c being positive. Lebesgue measure is translation invariant (shared context), so the two translations leave the measure unchanged: $\lambda(\overline{B}(x,c\rho)) = \lambda\big(c\,(\overline{B}(x,\rho) - x)\big)$ and $\lambda(\overline{B}(x,\rho) - x) = \lambda(\overline{B}(x,\rho))$. For the dilation itself, clause (1) of `LebesgueDilation` gives $\lambda(cA) = |c|^n \lambda(A)$ for every $c \in \mathbb{R}$ and every $A \subseteq \mathbb{R}^n$, measurable or not; at $c > 0$ one has $|c| = c$, so applying it with $A = \overline{B}(x,\rho) - x$ yields $\lambda(\overline{B}(x,c\rho)) = c^n \lambda(\overline{B}(x,\rho))$, which is (3). Because that clause is stated for arbitrary sets and for every $c \in \mathbb{R}$ and every $n \in \mathbb{N}_0$, no measurability side condition and no separate $n = 0$ case is needed; at $n = 0$ it reads $\lambda(\mathbb{R}^0) = c^0 \lambda(\mathbb{R}^0)$, which is consistent with both balls being the whole space.

Step 2: the case $t = +\infty$, and selection of radii. If $t = +\infty$ then $M_2 w(x) > t$ is impossible, because $M_2 w(x) \in [0, +\infty]$; so $E = \emptyset$, $\lambda(E) = 0$, and the inequality holds. Assume from now on $0 < t < \infty$.

Let $x \in E$. By (1) and the definition of a supremum there is $\rho_x \in (0, 2]$ with

$$t < \frac{1}{\lambda(\overline{B}(x,\rho_x))} \int_{\overline{B}(x,\rho_x)} |w|\, d\lambda.$$

By (2) the number $\lambda(\overline{B}(x,\rho_x))$ lies in $(0,\infty)$, so multiplying through by it is legitimate in $[0,+\infty]$ and gives

$$t\,\lambda\big(\overline{B}(x,\rho_x)\big) < \int_{\overline{B}(x,\rho_x)} |w|\, d\lambda; \qquad \text{hence} \qquad \lambda\big(\overline{B}(x,\rho_x)\big) \leq \frac{1}{t} \int_{\overline{B}(x,\rho_x)} |w|\, d\lambda, \tag{4}$$

the second form following by dividing by $t \in (0,\infty)$.

Step 3: the Vitali covering step. Consider the family of closed balls $\overline{B}(x,\rho_x)$ indexed by $x \in E$. All the radii satisfy $\rho_x \le 2$, so the family has a uniform upper bound on its radii. By the Vitali covering theorem for closed balls (classical; the version used is: given a family of closed balls $\overline{B}(x_a,r_a)$, $a \in T$, with $r_a \le R$ for all a, and given $\tau > 3$, there is $U \subseteq T$ such that the balls $\overline{B}(x_a,r_a)$, $a \in U$, are pairwise disjoint and every $\overline{B}(x_a,r_a)$, $a \in T$, is contained in $\overline{B}(x_b,\tau r_b)$ for some $b \in U$), applied with $R = 2$ and $\tau = 4$, there is $U \subseteq E$ such that

(a) the balls $\overline{B}(b,\rho_b)$, $b \in U$, are pairwise disjoint; and
(b) for every $x \in E$ there is $b \in U$ with $\overline{B}(x,\rho_x) \subseteq \overline{B}(b,4\rho_b)$.

Since $x \in \overline{B}(x,\rho_x)$ for each $x \in E$, item (b) gives

$$E \subseteq \bigcup_{b\in U} \overline{B}(b,4\rho_b). \tag{5}$$

Step 4: U is countable. The sets $\overline{B}(b,\rho_b)$, $b \in U$, are pairwise disjoint Borel sets (disjointness is item (a) of Step 3; they are Borel because closed balls are closed), each of strictly positive measure by (2), and λ is σ-finite (shared context). A pairwise disjoint family of measurable sets of positive measure in a σ-finite measure space is countable. In detail: write $\mathbb{R}^n = \bigcup_{k\in\mathbb{N}} B(0,k)$, each $B(0,k)$ having finite measure because it is contained in the compact set $\overline{B}(0,k)$ (shared context), and for $k,m \in \mathbb{N}$ let $U_{k,m} = \{b \in U : \lambda(\overline{B}(b,\rho_b) \cap B(0,k)) > 1/m\}$. For fixed k,m the set $U_{k,m}$ is finite, since any N of its members contribute disjoint subsets of $B(0,k)$ of measure more than $1/m$ each, forcing $N/m \le \lambda(B(0,k)) < \infty$ and hence $N \le m\,\lambda(B(0,k))$. Every $b \in U$ lies in some $U_{k,m}$: since $\lambda(\overline{B}(b,\rho_b)) > 0$ and $\overline{B}(b,\rho_b) \cap B(0,k) \uparrow \overline{B}(b,\rho_b)$ as $k \to \infty$, some k has $\lambda(\overline{B}(b,\rho_b) \cap B(0,k)) > 0$, and then some m has $1/m$ below that value. So $U = \bigcup_{k,m} U_{k,m}$ is a countable union of finite sets, hence countable.

Step 5: summation. By (5), monotonicity of λ as an outer measure, countable subadditivity over the countable family U (Step 4), and (3) with $c = 4$,

$$\lambda(E) \le \lambda\Big(\bigcup_{b\in U} \overline{B}(b,4\rho_b)\Big) \le \sum_{b\in U} \lambda\big(\overline{B}(b,4\rho_b)\big) = 4^n \sum_{b\in U} \lambda\big(\overline{B}(b,\rho_b)\big),$$

all sums being sums of countable families of elements of $[0,+\infty]$. By (4), applied to each $b \in U \subseteq E$, and then by countable additivity of the integral of the nonnegative measurable function $|w|$ over the pairwise disjoint Borel sets $\overline{B}(b,\rho_b)$, $b \in U$, together with monotonicity of that integral in the domain,

$$4^n \sum_{b\in U} \lambda\big(\overline{B}(b,\rho_b)\big) \le \frac{4^n}{t} \sum_{b\in U} \int_{\overline{B}(b,\rho_b)} |w|\,d\lambda = \frac{4^n}{t} \int_{\bigcup_{b\in U} \overline{B}(b,\rho_b)} |w|\,d\lambda \le \frac{4^n}{t}\, I.$$

Combining the two displays gives $\lambda(E) \le 4^n I/t$, which is the assertion. □

2.104. `LocalMaximalWeakL2`.

Lemma 2.104 (Weak-type $(2,2)$ bound for the local maximal function)**.** *Fix $n \in \mathbb{N}_0$ and let λ denote Lebesgue measure on $\mathbb{R}^n$. For $w : \mathbb{R}^n \to \mathbb{R}$ write*

$$\mathcal{A}(w) = \sup_{0<\sigma\le+\infty} \sigma^2\, \lambda\big(\{\, y \in \mathbb{R}^n : |w(y)| > \sigma \,\}\big) \;\in\; [0,+\infty],$$

the square of the weak-L^2 quasinorm of w, the supremum being over σ in the extended half-line and each comparison $|w(y)| > \sigma$ read in $[0,+\infty]$. Let w be almost everywhere equal, for λ, to a Borel measurable function, and let $s \in (0,+\infty]$. Then

$$\lambda\big(\{\, x \in \mathbb{R}^n : M_2 w(x) > s \,\}\big) \;\le\; \frac{16 \cdot 4^n\, \mathcal{A}(w)}{s^2},$$

with M_2 the local maximal function of `LocalMaximalFunction`, and with both sides and the quotient read in $[0,+\infty]$ under the conventions $a/(+\infty) = 0$ and, for $a > 0$, $a/0 = +\infty$.

Three remarks on the shape. First, only the exponent $p = 2$ is stated, which is the only exponent used downstream, and it is stated in weak-type *form: the hypothesis on w is a bound on its distribution function, not an L^2 bound, and the conclusion is a bound on a distribution*

function, not an L^2 bound on $M_2 w$. This matters, because it is the resulting s^{-2} – not an s^{-1} or an L^2 norm – that preserves the weak-$(1,1)$ exponent when this lemma is combined with an endpoint estimate downstream. Second, $\sigma = +\infty$ is included in the supremum only for notational uniformity: w is real valued, so $\{|w| > +\infty\}$ is empty and that term of the supremum is 0. Third, as in `LocalMaximalWeakType`*, λ is applied to the level set $\{M_2 w > s\}$ as an outer measure, so no measurability of that set is asserted or needed; and no finiteness hypothesis is placed on $\mathcal{A}(w)$, the case $\mathcal{A}(w) = +\infty$ being vacuous.*

No Gaussian measure occurs here. *Every measure in the statement is Lebesgue measure on $\mathbb{R}^n$, with no Gaussian weight in any integrand and no Gaussian measure in any level set, exactly as in* `LocalMaximalFunction` *and* `LocalMaximalWeakType`*. The constant $16 \cdot 4^n$ is explicit and is not claimed to be optimal.*

Proof. Write $A = \mathcal{A}(w)$ and $\mu(\sigma) = \lambda(\{y : |w(y)| > \sigma\})$ for $\sigma \in (0, +\infty]$, so that by the definition of the supremum

$$\sigma^2 \, \mu(\sigma) \le A \qquad \text{for every } \sigma \in (0, +\infty]. \tag{1}$$

Step 0: two reductions, and the ball volumes. By hypothesis there is a Borel measurable $\widetilde{w}$ with $w = \widetilde{w}$ λ-almost everywhere. Two nonnegative functions agreeing λ-almost everywhere have the same lower Lebesgue integral against λ and against the restriction of λ to any measurable set, a λ-null set being null for each such restriction; so by the defining formula of `LocalMaximalFunction` we get $M_2 w = M_2 \widetilde{w}$ pointwise on $\mathbb{R}^n$. Also, for each σ the sets $\{|w| > \sigma\}$ and $\{|\widetilde{w}| > \sigma\}$ differ only inside the λ-null set $\{w \neq \widetilde{w}\}$, and the outer measure λ is unchanged by modifying a set inside a null set (by monotonicity and subadditivity, shared context), so $\mu(\sigma)$ and hence A are unchanged as well. Neither side of the asserted inequality is therefore affected, and we may assume w Borel measurable.

We record once, for use in Step 1, that

$$0 < \lambda\big(\overline{B}(x, \rho)\big) < \infty \qquad \text{for every } x \in \mathbb{R}^n \text{ and } \rho > 0. \tag{$*$}$$

The closed ball is compact, hence of finite Lebesgue measure (shared context), and it contains the nonempty open ball $B(x, \rho)$, to which Lebesgue measure assigns strictly positive mass. Both arguments are valid for every $n \in \mathbb{N}_0$: when $n = 0$ the space $\mathbb{R}^0$ is a single point and every ball of nonnegative radius is the whole space, which is nonempty, open and compact.

If $s = +\infty$ then $M_2 w(x) > s$ is impossible, since $M_2 w$ takes values in $[0, +\infty]$, so the left-hand side is 0 and the inequality holds. Assume from now on $0 < s < \infty$, and set

$$\lambda_0 = \frac{s}{2} \in (0, \infty), \qquad w_1 = w \, \mathbf{1}_{\{|w| > \lambda_0\}}.$$

The set $\{|w| > \lambda_0\}$ is Borel, so w_1 is Borel measurable.

Step 1: $M_2 w \le M_2 w_1 + \lambda_0$ pointwise. For every y we have $|w(y)| \le |w_1(y)| + \lambda_0$: if $|w(y)| > \lambda_0$ then $w_1(y) = w(y)$ and the inequality is clear, and otherwise $|w(y)| \le \lambda_0$ while $|w_1(y)| \ge 0$. Fix x and $\rho \in (0, 2]$ and write $B = \overline{B}(x, \rho)$. Integrating the pointwise inequality over B and using additivity and monotonicity of the lower Lebesgue integral of nonnegative functions,

$$\int_B |w| \, d\lambda \le \int_B |w_1| \, d\lambda + \lambda_0 \, \lambda(B).$$

By $(*)$ we have $0 < \lambda(B) < \infty$, so we may multiply by $\lambda(B)^{-1} \in (0, \infty)$ and use $\lambda(B)^{-1} \lambda(B) = 1$ to get

$$\frac{1}{\lambda(B)} \int_B |w| \, d\lambda \le \frac{1}{\lambda(B)} \int_B |w_1| \, d\lambda + \lambda_0 \le M_2 w_1(x) + \lambda_0,$$

the last step by the definition of $M_2 w_1(x)$ as a supremum over $\rho \in (0, 2]$ (`LocalMaximalFunction`). Taking the supremum over $\rho \in (0, 2]$ on the left gives $M_2 w(x) \le M_2 w_1(x) + \lambda_0$.

Consequently, if $M_2 w(x) > s = \lambda_0 + \lambda_0$, then $M_2 w_1(x) > \lambda_0$: otherwise $M_2 w_1(x) \le \lambda_0$ would give $M_2 w(x) \le \lambda_0 + \lambda_0 = s$. Therefore

$$\{\, x : M_2 w(x) > s \,\} \subseteq \{\, x : M_2 w_1(x) > \lambda_0 \,\}. \tag{2}$$

Step 2: the L^1 norm of w_1 by dyadic decomposition. We claim

$$\int_{\mathbb{R}^n} |w_1|\,d\lambda \le \frac{4A}{\lambda_0}. \tag{3}$$

For $k \in \mathbb{N}_0$ put $\sigma_k = 2^k\lambda_0 \in (0,\infty)$ and

$$E_k = \{\, y : \sigma_k < |w(y)| \le \sigma_{k+1} \,\},$$

a Borel set. The E_k are pairwise disjoint, and their union is $\{|w| > \lambda_0\}$. Indeed, each E_k is contained in that set because $\sigma_k \ge \sigma_0 = \lambda_0$; conversely, if $|w(y)| > \lambda_0$ then, w being real valued, $|w(y)| < \infty$, while $\sigma_k \to \infty$, so there is a least $k \in \mathbb{N}_0$ with $|w(y)| \le \sigma_{k+1}$, and for that k one has $|w(y)| > \sigma_k$ — for $k = 0$ by hypothesis, and for $k \ge 1$ by minimality of k. Disjointness is clear since the intervals $(\sigma_k, \sigma_{k+1}]$ are pairwise disjoint.

Since w_1 vanishes off $\{|w| > \lambda_0\}$ and equals w on it, countable additivity of the integral of the nonnegative Borel function $|w|$ over the pairwise disjoint Borel sets E_k gives

$$\int_{\mathbb{R}^n} |w_1|\,d\lambda = \int_{\{|w|>\lambda_0\}} |w|\,d\lambda = \sum_{k\ge 0} \int_{E_k} |w|\,d\lambda \le \sum_{k\ge 0} \sigma_{k+1}\,\lambda(E_k),$$

using $|w| \le \sigma_{k+1}$ on E_k. Now $E_k \subseteq \{|w| > \sigma_k\}$, so $\lambda(E_k) \le \mu(\sigma_k)$, and by (1) together with $\sigma_k \in (0,\infty)$ we may divide and obtain $\mu(\sigma_k) \le A/\sigma_k^2$. Hence, using $\sigma_{k+1} = 2\sigma_k$,

$$\sigma_{k+1}\,\lambda(E_k) \le \frac{2\sigma_k A}{\sigma_k^2} = \frac{2A}{\sigma_k} = \frac{2A}{\lambda_0}\,2^{-k},$$

and summing the geometric series $\sum_{k\ge 0} 2^{-k} = 2$ in $[0,+\infty]$,

$$\int_{\mathbb{R}^n} |w_1|\,d\lambda \le \frac{2A}{\lambda_0} \sum_{k\ge 0} 2^{-k} = \frac{4A}{\lambda_0},$$

which is (3). (If $A = +\infty$ the bound is trivially true, and the computation above is still valid in $[0,+\infty]$.)

Step 3: conclusion. Apply `LocalMaximalWeakType` to the Borel measurable function w_1 – which is in particular almost everywhere equal to a Borel measurable function – and to the level $t = \lambda_0 \in (0,\infty)$, which is nonzero as required:

$$\lambda\big(\{x : M_2 w_1(x) > \lambda_0\}\big) \le \frac{4^n}{\lambda_0} \int_{\mathbb{R}^n} |w_1|\,d\lambda.$$

Combining with (2), monotonicity of the outer measure λ, and (3),

$$\lambda\big(\{x : M_2 w(x) > s\}\big) \le \lambda\big(\{x : M_2 w_1(x) > \lambda_0\}\big) \le \frac{4^n}{\lambda_0} \cdot \frac{4A}{\lambda_0} = \frac{4 \cdot 4^n\,A}{\lambda_0^2}.$$

Finally $\lambda_0^2 = s^2/4$, so $4/\lambda_0^2 = 16/s^2$ and the right-hand side is $16 \cdot 4^n A/s^2$, as asserted. □

2.105. `CoordinateEndpointMaximalSquareBound`.

Lemma 2.105 (Weak bound for the square of the local maximal function of a coordinate endpoint function)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

Let $R \in \mathbb{R}$, put $\overline{B} = \overline{B}(0, R+2)$, let M_2 be the local maximal function of `LocalMaximalFunction`*, and for $f \in L^1(\gamma_n)$ — written also for its canonical representative — put*

$$u_f = \big|(Tf)_j\big|^{1/2}\, \mathbf{1}_{\overline{B}} : \mathbb{R}^n \to \mathbb{R}.$$

Then for every $f \in L^1(\gamma_n)$ and every $s \in (0, +\infty]$,

$$\gamma_n\Big(\big\{\, x \in \mathbb{R}^n :\ M_2(u_f)(x)^2 > s \,\big\}\Big) \ \le\ \frac{32 \cdot 4^n\, e^{(R+2)^2}\, \|f\|_{L^1(\gamma_n)}}{s},$$

both sides read in $[0, +\infty]$, with $M_2(u_f)(x)^2$ the square taken there. Here $\|f\|_{L^1(\gamma_n)} = \int_{\mathbb{R}^n} |f|\, d\gamma_n$, the norm of the class of f written as the integral of the absolute value of its canonical representative; the two are the same number, and it is that integral the displayed bound uses.

Why the square appears in the statement. *The intended consumer reaches this term as $\{x : \Psi(x)^2 > s\}$ with $\Psi = M_2(u_f)$, because its own input bounds a quantity by $\Psi^2 + \Phi + N$ and it splits that sum at the level s; so the squared form is the shape in which the set arises, and passing to the unsquared level $\{M_2(u_f) > \sqrt{s}\}$ is part of the work done here rather than something the consumer must do. The square is taken in $[0, +\infty]$: $M_2(u_f)(x)$ is a supremum of averages which is not known in advance to be finite, exactly as in* `LocalMaximalFunction`*.*

The threshold is extended-real, and the endpoint is a triviality. *At $s = +\infty$ the level set is empty, nothing in $[0, +\infty]$ exceeding $+\infty$, while the right-hand side is a finite quantity divided by $+\infty$, hence 0; the assertion there is $0 \le 0$ and the content is at finite s. The threshold is excluded only at $s = 0$, where the right-hand side would be $+\infty$ unless $\|f\|_{L^1(\gamma_n)} = 0$ and the assertion would fail for f of norm zero with $M_2(u_f)$ not identically 0 — it is $0/0$ that is being avoided, not a loss of content.*

No measurability of the level set is asserted or needed. *γ_n is applied to the level set as an outer measure, which is defined on every subset and is monotone there; and the two facts chained below — the weak bound of* `LocalMaximalWeakL2` *and the comparison $\gamma_n \le \pi^{-n/2}\lambda$ of* `GaussLebesgueBallComparison` *— are themselves stated for arbitrary sets. Bounding the outer measure is the stronger assertion, so a consumer that knows the set to be measurable loses nothing.*

The constant, and what it does not claim. *$32 \cdot 4^n e^{(R+2)^2}$ depends only on n and R: the 4^n is the Vitali constant carried by* `LocalMaximalWeakL2`*, and $e^{(R+2)^2}$ enters through $a^{-1} = \pi^{n/2} e^{(R+2)^2}$, the reciprocal of the lower bound $a = \pi^{-n/2} e^{-(R+2)^2}$ for the Gaussian density on $\overline{B}$. No uniformity in n or R is claimed. The quantity bounded is coordinatewise in j; nothing here asserts anything about the full vector Tf.*

The hypothetical style. *(T1) and (T2) are written out in full and not imported by reference; the lemma is a statement about every T satisfying them, as* `CoordinateEndpointBound` *is, and it is not vacuous because* `L1Extension` *exhibits such a T.*

Proof. Write $p = \pi^{-n/2}$, $q = \pi^{n/2}$ and $a = \pi^{-n/2} e^{-(R+2)^2}$, all in $(0, \infty)$, so that

$$p\, q = 1 \qquad \text{and} \qquad a \cdot \big(q\, e^{(R+2)^2}\big) = 1, \tag{1}$$

the first because $\pi^{-n/2} = (\pi^{n/2})^{-1}$ and $\pi^{n/2} \neq 0$, and the second by multiplying the first by $e^{-(R+2)^2} e^{(R+2)^2} = e^{-(R+2)^2 + (R+2)^2} = e^0 = 1$. Fix $f \in L^1(\gamma_n)$, abbreviate $u = u_f$ and $\overline{B} = \overline{B}(0, R+2)$, and put $N = \|f\|_{L^1(\gamma_n)} = \int |f|\, d\gamma_n \in [0, \infty)$.

Step 1: u is measurable, nonnegative, and supported in $\overline{B}$. The canonical representative of a γ_n-almost-everywhere equivalence class of measurable functions is measurable, so $(Tf)_j$ is measurable. The map $r \mapsto |r|^{1/2}$ is continuous on $\mathbb{R}$, hence Borel, so $|(Tf)_j|^{1/2}$ is measurable; and $\overline{B}$ is closed, hence measurable, so $u = |(Tf)_j|^{1/2} \mathbf{1}_{\overline{B}}$ is measurable. Since $|(Tf)_j|^{1/2} \ge 0$ we have $u \ge 0$ everywhere, and $u(y) = 0$ for $y \notin \overline{B}$.

Step 2: the uniform bound on the weak-L^2 quantity. Put

$$\mathcal{A} = \sup_{0 < \sigma \le +\infty} \sigma^2\, \lambda\big(\{y : |u(y)| > \sigma\}\big) \ \in\ [0, +\infty].$$

We claim

$$\mathcal{A} \;\le\; 2\,q\,e^{(R+2)^2}\,N. \tag{2}$$

It suffices to bound each term of the supremum by the right-hand side, since that right-hand side does not depend on σ. (The uniformity is the point: a bound still depending on σ would not bound the supremum.)

If $\sigma = +\infty$ then $\{y : |u(y)| > \sigma\}$ is empty, u being real valued so that $|u(y)| \le +\infty$ always, and the term is $(+\infty)\cdot 0 = 0$, which is $\le$ the right-hand side.

So let σ be a real number with $\sigma > 0$ and put $E_\sigma = \{y : |u(y)| > \sigma\}$. By Step 1, $|u| = u$, so E_σ is measurable, being a strict superlevel set of a measurable function. Also $E_\sigma \subseteq \overline{B}$: if $y \notin \overline{B}$ then $u(y) = 0$ by Step 1, so $|u(y)| = 0 \not> \sigma$. Next,

$$E_\sigma \;\subseteq\; \big\{y : |(Tf)_j(y)| > \sigma^2\big\}. \tag{3}$$

Indeed let $y \in E_\sigma$. Then $y \in \overline{B}$ by the previous sentence, so $u(y) = |(Tf)_j(y)|^{1/2}$, and $|u(y)| = u(y) > \sigma$ gives $|(Tf)_j(y)|^{1/2} > \sigma > 0$; squaring the two sides of this inequality between nonnegative reals and using $\big(|(Tf)_j(y)|^{1/2}\big)^2 = |(Tf)_j(y)|$, which holds because $|(Tf)_j(y)| \ge 0$, gives $|(Tf)_j(y)| > \sigma^2$.

Now E_σ is measurable and contained in $\overline{B} = \overline{B}(0, R+2)$, so clause (2) of `GaussLebesgueBallComparison`, *instantiated at $R+2$ in place of its R and at $E := E_\sigma$*, gives

$$a\,\lambda(E_\sigma) \;\le\; \gamma_n(E_\sigma). \tag{4}$$

And `CoordinateEndpointBound` — applied to this T with its two properties, which are (T1) and (T2) and hold by assumption, to this f, to the coordinate j and to the real level $t := \sigma^2 > 0$ — gives

$$\sigma^2\,\gamma_n\big(\{y : |(Tf)_j(y)| > \sigma^2\}\big) \;\le\; 2\,\|f\|_{L^1(\gamma_n)} \;=\; 2N. \tag{5}$$

Multiplying (4) by $\sigma^2 \in (0,\infty)$, then using monotonicity of γ_n as an outer measure together with (3), then (5):

$$a\cdot\sigma^2\lambda(E_\sigma) \;\le\; \sigma^2\gamma_n(E_\sigma) \;\le\; \sigma^2\gamma_n\big(\{y : |(Tf)_j(y)| > \sigma^2\}\big) \;\le\; 2N \;=\; a\cdot\big(2\,q\,e^{(R+2)^2}N\big),$$

the last equality being (1) multiplied by $2N$. Cancelling the factor $a \in (0,\infty)$, which is legitimate in $[0,+\infty]$ because a is neither 0 nor $+\infty$, gives $\sigma^2\lambda(E_\sigma) \le 2\,q\,e^{(R+2)^2}N$. As this holds for every σ, (2) follows.

Step 3: the level s, its square root, and the conclusion. Let $s \in (0,+\infty]$ and put $S = \{x : M_2(u)(x)^2 > s\}$.

If $s = +\infty$ then S is empty, since $M_2(u)(x)^2 \le +\infty$ for every x, so $\gamma_n(S) = 0$ and the assertion holds, the right-hand side being ≥ 0.

So let $s \in (0,\infty)$ and let $\sqrt{s} \in (0,\infty)$ be its positive square root, so $(\sqrt{s})^2 = s$ and $\sqrt{s} \ne 0$. Then

$$S \;\subseteq\; \big\{x : M_2(u)(x) > \sqrt{s}\big\}, \tag{6}$$

by contraposition: if $M_2(u)(x) \le \sqrt{s}$ then, squaring is monotone on $[0,+\infty]$, $M_2(u)(x)^2 \le (\sqrt{s})^2 = s$, so $x \notin S$.

Apply `LocalMaximalWeakL2` to u, which is measurable hence measurable up to a null set by Step 1, at the level $\sqrt{s} \ne 0$. Together with $(\sqrt{s})^2 = s$ it gives

$$\lambda\big(\{x : M_2(u)(x) > \sqrt{s}\}\big) \;\le\; \frac{16\cdot 4^n\,\mathcal{A}}{s}.$$

By monotonicity of λ as an outer measure and (6), the same bound holds for $\lambda(S)$. By clause (1) of `GaussLebesgueBallComparison`, which holds for *every* set, $\gamma_n(S) \le p\,\lambda(S)$. Combining these with (2) and using monotonicity of $r \mapsto p\cdot 16\cdot 4^n r/s$ on $[0,+\infty]$,

$$\gamma_n(S) \;\le\; p\,\lambda(S) \;\le\; \frac{p\cdot 16\cdot 4^n\,\mathcal{A}}{s} \;\le\; \frac{p\cdot 16\cdot 4^n\cdot 2\,q\,e^{(R+2)^2}N}{s} \;=\; \frac{32\cdot 4^n\,e^{(R+2)^2}N}{s},$$

the last step because $p \cdot 16 \cdot 4^n \cdot 2\,q\,e^{(R+2)^2} N = 32 \cdot 4^n\, e^{(R+2)^2} N \cdot (p\,q)$ and $p\,q = 1$ by (1). This is the assertion for s, and with the case $s = +\infty$ it holds for every $s \in (0, +\infty]$. □

2.106. `EnergyFiniteOfSqrtDomain`.

Lemma 2.106 (The operator domain sits inside the form domain: $\mathrm{Dom}(L^{1/2}) \subseteq \mathrm{Dom}(L^{1/4})$)**.** *Let $v \in \mathrm{Dom}(L^{1/2})$ in the sense of `FracPowDomain`. Then the fractional energy of `Energy` is finite,*

$$\mathcal{E}(v) = \sum_{\alpha \neq 0} |\alpha|^{1/2}\, |v_\alpha|^2 < \infty,$$

where v_α are the Hermite coefficients of `HermiteBasis` and $|\alpha|$ is the weight of `MultiIndexSum`. Equivalently, v belongs to the form domain $\mathcal{V} = \{w \in L^2(\gamma_n) : \mathcal{E}(w) < \infty\} = \mathrm{Dom}(L^{1/4})$ of `Energy`: the operator domain of $B = L^{1/2}$ is contained in the form domain of the energy. Only this inclusion is asserted; the reverse inclusion is false and is not claimed.

Proof. Write v_α for the Hermite coefficients of v relative to `HermiteBasis`.

Step 1: what the hypothesis says. By the criterion of `FracPowDomain` at $s = 1/2$, whose multiplier is $|\alpha|^{2s} = |\alpha|$, the hypothesis $v \in \mathrm{Dom}(L^{1/2})$ says exactly

$$\sum_{\alpha \neq 0} |\alpha|\, |v_\alpha|^2 < \infty. \tag{1}$$

All summands are nonnegative, since $|\alpha|$ is a nonnegative integer by `MultiIndexSum`, so by the extended-real bridge clause of the summable-families item of the shared context, (1) is equivalent to summability of the family $\big(|\alpha|\,|v_\alpha|^2\big)_{\alpha \neq 0}$ as a family of real numbers.

Step 2: $|\alpha| \geq 1$ for $\alpha \neq 0$. By `MultiIndexSum`, $|\alpha| = \sum_{j=1}^{n} \alpha_j$ with every $\alpha_j \in \mathbb{N}_0$. If $\alpha \neq 0$ then $\alpha_{j_0} \geq 1$ for some j_0, and all the remaining summands are ≥ 0, so $|\alpha| \geq \alpha_{j_0} \geq 1$.

Step 3: the termwise comparison. Let $\alpha \neq 0$ and put $t = |\alpha| \geq 1$ by Step 2. Then $t^{1/2} \geq 1$, because $t^{1/2} < 1$ would give $t = t^{1/2}\, t^{1/2} < 1$; and multiplying the inequality $1 \leq t^{1/2}$ by the nonnegative number $t^{1/2}$ gives $t^{1/2} \leq t$. Hence, multiplying by $|v_\alpha|^2 \geq 0$,

$$0 \leq |\alpha|^{1/2}\, |v_\alpha|^2 \leq |\alpha|\, |v_\alpha|^2 \qquad (\alpha \neq 0). \tag{2}$$

Step 4: conclusion. By Step 1 the majorant family in (2) is summable, so by the comparison clause of the summable-families item of the shared context the family $\big(|\alpha|^{1/2}|v_\alpha|^2\big)_{\alpha \neq 0}$ is summable as well, with

$$\sum_{\alpha \neq 0} |\alpha|^{1/2}\, |v_\alpha|^2 \leq \sum_{\alpha \neq 0} |\alpha|\, |v_\alpha|^2 < \infty.$$

Its summands being nonnegative, the extended-real bridge clause applies once more and says that the same sum, read in $[0, +\infty]$, is finite and equal to the real number just bounded. By `Energy` that extended-real sum is precisely $\mathcal{E}(v)$, so $\mathcal{E}(v) < \infty$. By the description of the form domain in `Energy` this is the assertion $v \in \mathcal{V} = \mathrm{Dom}(L^{1/4})$. □

2.107. `ExpDecayPowerBoundAndRpowAbsorption`.

Lemma 2.107 (Exponential decay by a negative power, and absorption of a negative rpow factor)**.** *Fix $n \geq 0$. Then both of the following hold.*

(1) *For every real $t > 0$,*

$$e^{-t} \leq (n+2)^{n+2}\, (t^{-1})^{n+2}.$$

(2) *For all reals r, c with $\frac{1}{2} \leq r < 1$ and $c > 0$,*

$$(1-r)^{-(n/2+3/2)} \Big((n+2)^{n+2} \big(c(1-r)\big)^{n+2} \Big) \leq (n+2)^{n+2} c^{n+2}.$$

Proof. Clause (1). Let $t > 0$ and put $u = t/(n+2)$, which is positive since $n + 2 > 0$. From $1 + u \leq e^u$ we get $u \leq e^u$, and raising both nonnegative sides to the natural power $n + 2$ gives $u^{n+2} \leq (e^u)^{n+2}$. Now $(e^u)^{n+2} = e^{(n+2)u} = e^t$, so

$$\frac{t^{\,n+2}}{(n+2)^{n+2}} \leq e^t.$$

Multiplying by $(n+2)^{n+2} > 0$ gives $t^{\,n+2} \le (n+2)^{n+2} e^t$, and dividing by the positive quantities e^t and $t^{\,n+2}$ rearranges this to $e^{-t} \le (n+2)^{n+2} t^{-(n+2)}$, which is clause (1) since $t^{-(n+2)} = (t^{-1})^{n+2}$.

Clause (2). Let $\frac{1}{2} \le r < 1$ and $c > 0$, and put $\tau = 1 - r$, so $0 < \tau \le 1$. Expanding $(c\tau)^{n+2} = c^{\,n+2} \tau^{\,n+2}$ and regrouping,

$$\tau^{-(n/2+3/2)} \Big((n+2)^{n+2} (c\tau)^{n+2} \Big) = \big((n+2)^{n+2} c^{\,n+2} \big) \, \big(\tau^{-(n/2+3/2)} \tau^{\,n+2} \big).$$

Reading the natural power τ^{n+2} as the real power $\tau^{(n+2)}$ and adding exponents, the second factor is $\tau^{-(n/2+3/2)+(n+2)}$, whose exponent $n/2 + 1/2$ is nonnegative. Since $0 < \tau \le 1$, a nonnegative real power of τ is at most 1, so that factor is at most 1. Multiplying by the nonnegative constant $(n+2)^{n+2} c^{\,n+2}$ gives clause (2). □

2.108. `GaussEuclideanMomentBounds.`

Lemma 2.108 (Elementary Gaussian moment bounds on $\mathbb{R}^n$)**.** *Fix $n \ge 1$ and work on $\mathbb{R}^n$ with Lebesgue measure. Then all of the following hold.*

(1) *For every $b > 0$ the function $w \mapsto e^{-b|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$.*

(2) *For every $b > 0$,*

$$\int_{\mathbb{R}^n} e^{-b|w|^2} \, dw = \Big(\frac{\pi}{b} \Big)^{n/2}.$$

(3) *For every $w \in \mathbb{R}^n$ and every coordinate j, $|w_j| \le |w|$.*

(4) *For every $b > 0$ and every $v \in \mathbb{R}^n$,*

$$|v| \, e^{-b|v|^2} \le \Big(1 + \frac{2}{b} \Big) \, e^{-(b/2)|v|^2}.$$

(5) *For every $b > 0$ the function $w \mapsto |w| \, e^{-b|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$, and*

$$\int_{\mathbb{R}^n} |w| \, e^{-b|w|^2} \, dw \le \Big(1 + \frac{2}{b} \Big) \Big(\frac{\pi}{b/2} \Big)^{n/2}.$$

(6) *For all $\sigma > 0$ and $b' > 0$,*

$$\pi^{-n/2} \, \sigma^{-n/2} \Big(\frac{\pi}{b'} \Big)^{n/2} = \Big(\frac{1}{\sigma b'} \Big)^{n/2}.$$

Proof. Each item is proved separately; items (4) and (5) use the earlier ones, and no item uses any later one.

(1). Fix $b > 0$. The Gaussian $w \mapsto e^{-b|w|^2}$ on the n-dimensional inner product space $\mathbb{R}^n$ is integrable: this is the real part of the standard complex Gaussian integrability statement for $w \mapsto e^{-b|w+c|^2}$ with complex parameter b of positive real part, specialised to $c = 0$ and to real $b > 0$. Since $e^{-b|w|^2}$ is real valued, taking real parts changes nothing, and integrability of the complex function gives integrability of this one.

(2). Fix $b > 0$. The standard Gaussian integral over a real inner product space of finite dimension d is $\int e^{-b|w|^2} \, dw = (\pi/b)^{d/2}$. Here the space is $\mathbb{R}^n$, whose dimension is $d = n$, which gives $(\pi/b)^{n/2}$.

(3). Fix w and j. Writing the Euclidean norm as $|w| = \big(\sum_i |w_i|^2 \big)^{1/2}$, the single term $|w_j|^2$ is at most the full sum $\sum_i |w_i|^2$, because every term is nonnegative. Taking square roots, which is monotone, gives

$$|w_j| = \big(|w_j|^2 \big)^{1/2} \le \Big(\sum_i |w_i|^2 \Big)^{1/2} = |w|.$$

(4). Fix $b > 0$ and v, and set $t = |v| \ge 0$ and $E = e^{-(b/2)t^2}$. Then $E > 0$, and $E \le 1$ because $(b/2)t^2 \ge 0$. Also $e^{-bt^2} = E^2$, since $-bt^2 = -(b/2)t^2 + (-(b/2)t^2)$. Next, from the elementary inequality $1 + u \le e^u$ applied at $u = (b/2)t^2$ we get $(b/2)t^2 \le e^{(b/2)t^2}$, and since $e^{(b/2)t^2} E = 1$ this yields

$$\big((b/2)t^2 \big) E \le 1.$$

Multiplying by $2/b > 0$ and simplifying gives $t^2E \le 2/b$. Now estimate, using $t \le 1 + t^2$ (equivalently $(t-1)^2 \ge 0$ together with $t \ge 0$) and $E^2 > 0$:

$$t\,e^{-bt^2} = t\,E^2 \le (1+t^2)E^2 = E^2 + (t^2E)E \le 1 \cdot E + \frac{2}{b}\,E = \Big(1 + \frac{2}{b}\Big)E,$$

where the last inequality used $E \le 1$ on the first summand and $t^2E \le 2/b$ on the second. This is the claim, since $E = e^{-(b/2)|v|^2}$.

(5). Fix $b > 0$ and put $b/2 > 0$. By (1) the function $w \mapsto e^{-(b/2)|w|^2}$ is integrable, hence so is its constant multiple $w \mapsto (1 + 2/b)e^{-(b/2)|w|^2}$; call that majorant M. The function $w \mapsto |w|e^{-b|w|^2}$ is continuous, hence strongly measurable, and by (4) it is pointwise dominated in absolute value by M (the function is nonnegative, so its absolute value is itself). Domination by an integrable majorant gives integrability, which is the first half of the claim. For the second half, monotonicity of the integral applied to the pointwise bound of (4), followed by (2) at rate $b/2$, gives

$$\int |w|e^{-b|w|^2}\,dw \le \int \Big(1 + \frac{2}{b}\Big)e^{-(b/2)|w|^2}\,dw = \Big(1 + \frac{2}{b}\Big)\int e^{-(b/2)|w|^2}\,dw = \Big(1 + \frac{2}{b}\Big)\Big(\frac{\pi}{b/2}\Big)^{n/2}.$$

(6). Fix $\sigma, b' > 0$. All bases below are positive, so the real power laws apply. Rewriting $-n/2 = -(n/2)$ and using $c^{-n/2} = (c^{n/2})^{-1}$ for $c > 0$, together with $(\pi/b')^{n/2} = \pi^{n/2}/(b')^{n/2}$ and $(1/(\sigma b'))^{n/2} = \big((\sigma b')^{n/2}\big)^{-1} = \big(\sigma^{n/2}(b')^{n/2}\big)^{-1}$, both sides become quotients of products of the positive quantities $\pi^{n/2}$, $\sigma^{n/2}$ and $(b')^{n/2}$. The left side is $\pi^{n/2}/\big(\pi^{n/2}\sigma^{n/2}(b')^{n/2}\big)$ and the right side is $1/\big(\sigma^{n/2}(b')^{n/2}\big)$; these agree because $\pi^{n/2} \ne 0$. □

2.109. `GaussSecondMomentIntegralBound`.

Lemma 2.109 (Second-moment Gaussian majorant and integral bound)**.** *Fix $n \ge 1$. Then for every $b > 0$ all of the following hold, the integrals being taken with respect to Lebesgue measure on $\mathbb{R}^n$.*

(1) *For every $v \in \mathbb{R}^n$,*
$$|v|^2 e^{-b|v|^2} \le \frac{2}{b}\,e^{-(b/2)|v|^2}.$$
(2) *The function $w \mapsto |w|^2 e^{-b|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$.*
(3) *Its integral satisfies*
$$\int_{\mathbb{R}^n} |w|^2 e^{-b|w|^2}\,dw \le \frac{2}{b}\Big(\frac{\pi}{b/2}\Big)^{n/2}.$$

Proof. Fix $b > 0$. Two facts come from `GaussEuclideanMomentBounds`: for every $\beta > 0$ the function $w \mapsto e^{-\beta|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$, and

$$\int_{\mathbb{R}^n} e^{-\beta|w|^2}\,dw = (\pi/\beta)^{n/2}. \tag{1}$$

Clause (1): the pointwise majorant. Fix v and put $u = (b/2)|v|^2 \ge 0$. From $1 + u \le e^u$ we get $u \le e^u$, and multiplying by $e^{-u} > 0$ gives

$$u\,e^{-u} \le 1. \tag{2}$$

Multiplying (2) by $2/b > 0$ and substituting $u = (b/2)|v|^2$ turns it into

$$|v|^2 e^{-(b/2)|v|^2} \le \frac{2}{b}. \tag{3}$$

Since $-b|v|^2 = -(b/2)|v|^2 - (b/2)|v|^2$, the exponential splits as $e^{-b|v|^2} = e^{-(b/2)|v|^2}e^{-(b/2)|v|^2}$, so multiplying (3) by $e^{-(b/2)|v|^2} > 0$ gives

$$|v|^2 e^{-b|v|^2} = \big(|v|^2 e^{-(b/2)|v|^2}\big)e^{-(b/2)|v|^2} \le \frac{2}{b}\,e^{-(b/2)|v|^2},$$

which is clause (1).

Clause (2): integrability. The function $w \mapsto |w|^2 e^{-b|w|^2}$ is continuous, hence measurable, and the function $w \mapsto (2/b)e^{-(b/2)|w|^2}$ is integrable, being a constant multiple of an integrable

function at the rate $b/2 > 0$. By clause (1) the former is bounded in absolute value by the latter at every point, both sides being nonnegative, so domination by an integrable function gives integrability.

Clause (3): the integral bound. Both functions of clause (2) are integrable, so monotonicity of the integral applies to the pointwise inequality of clause (1), and then (1) at the rate $b/2$ evaluates the majorant:

$$\int_{\mathbb{R}^n} |w|^2 e^{-b|w|^2}\,dw \le \int_{\mathbb{R}^n} \frac{2}{b}\, e^{-(b/2)|w|^2}\,dw = \frac{2}{b}\Big(\frac{\pi}{b/2}\Big)^{n/2},$$

the last step by pulling the constant $2/b$ out of the integral. That is clause (3). □

2.110. `GaussFirstMomentTranslationL1Bound.`

Lemma 2.110 (L^1 translation-difference bound for the coordinate-weighted Gaussian)**.** *Fix $n \ge 1$ and a coordinate j. For $b > 0$, $c \ge 0$ and $w \in \mathbb{R}^n$ write $W(w) = c\, w_j\, e^{-b|w|^2}$. Then for every $b > 0$, every $c \ge 0$ and every $d \in \mathbb{R}^n$,*

$$\int_{\mathbb{R}^n} \big|W(z+d) - W(z)\big|\,dz \le c\,|d|\,(\pi/b)^{n/2}\big(5 + 24\cdot 8^{n/2}\big),$$

the integral being taken with respect to Lebesgue measure.

Proof. Fix $b > 0$, $c \ge 0$ and d, and write $E(w) = e^{-b|w|^2}$, so that $W(w) = c\, w_j\, E(w)$.

Step 0: the Gaussian toolkit. Four of the facts we need are already available. By `GaussEuclideanMomentBound` for every $\beta > 0$ the function $w \mapsto e^{-\beta|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$ with

$$\int_{\mathbb{R}^n} e^{-\beta|w|^2}\,dw = (\pi/\beta)^{n/2}, \tag{1}$$

and $|w_i| \le |w|$ for every w and every coordinate i. By `GaussSecondMomentIntegralBound`, for every $\beta > 0$ and every w

$$|w|^2 e^{-\beta|w|^2} \le \frac{2}{\beta}\, e^{-(\beta/2)|w|^2}, \tag{2}$$

and $w \mapsto |w|^2 e^{-\beta|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$ with

$$\int_{\mathbb{R}^n} |w|^2 e^{-\beta|w|^2}\,dw \le \frac{2}{\beta}\,(2\pi/\beta)^{n/2}, \tag{3}$$

the displayed constant because $\pi/(\beta/2) = 2\pi/\beta$. We prove the one remaining fact here.

Finally, for all reals a, b', m with $m \le a$ and $m \le b'$,

$$\big|e^{-a} - e^{-b'}\big| \le |a - b'|\, e^{-m}. \tag{4}$$

By symmetry assume $a \le b'$. Then $e^{-b'} \le e^{-a}$, so the left side is $e^{-a} - e^{-b'} = e^{-a}\big(1 - e^{-(b'-a)}\big)$; and $1 - e^{-t} \le t$ for $t \ge 0$ by the exponential inequality with $t = b' - a \ge 0$; so the left side is at most $e^{-a}(b' - a) \le e^{-m}|a - b'|$, using $e^{-a} \le e^{-m}$.

Step 1: the pointwise majorant. We claim that for every z,

$$\big|W(z+d) - W(z)\big| \le c\,|d|\Big(3E(z+d) + 2E(z) + 3b\,|z|^2 e^{-(b/4)|z|^2}\Big) =: M(z). \tag{5}$$

Since $(z+d)_j = z_j + d_j$,

$$W(z+d) - W(z) = c\,d_j\,E(z+d) + c\,z_j\big(E(z+d) - E(z)\big),$$

so by the triangle inequality, $|d_j| \le |d|$ and $E > 0$,

$$\big|W(z+d) - W(z)\big| \le c\,|d|\,E(z+d) + c\,|z_j|\,\big|E(z+d) - E(z)\big|. \tag{6}$$

If $|z| \le 2|d|$, then $|z_j| \le |z| \le 2|d|$ and $|E(z+d) - E(z)| \le E(z+d) + E(z)$, so the second term of (6) is at most $2c|d|\big(E(z+d)+E(z)\big)$ and the right side of (6) is at most $c|d|\big(3E(z+d)+2E(z)\big)$, which is at most $M(z)$ because the remaining summand of M is nonnegative.

If $|z| > 2|d|$, put $a = b|z+d|^2$, $b' = b|z|^2$ and $m = (b/4)|z|^2$. Then $m \le b'$; and $|d| \le |z|/2$ gives $|z+d| \ge |z| - |d| \ge |z|/2$, so $|z+d|^2 \ge |z|^2/4$ and $m \le a$. Moreover $|z+d|^2 - |z|^2 = 2\langle z, d\rangle + |d|^2$, so by Cauchy–Schwarz and $|d|^2 \le (|z|/2)|d|$,

$$|a - b'| = b\big|2\langle z, d\rangle + |d|^2\big| \le b\big(2|z||d| + \tfrac{1}{2}|z||d|\big) \le 3b|z||d|.$$

By (4), $|E(z+d) - E(z)| \le 3b|z||d|\, e^{-(b/4)|z|^2}$, so with $|z_j| \le |z|$ the second term of (6) is at most $3bc|d|\,|z|^2 e^{-(b/4)|z|^2}$, and the right side of (6) is at most $c|d|\big(E(z+d) + 3b|z|^2 e^{-(b/4)|z|^2}\big) \le M(z)$. This proves (5) in both cases.

Step 2: integrating the majorant. By (1), the translation invariance of Lebesgue measure, and (3) at $b/4$, each of the three summands of M is integrable and

$$\begin{aligned}\int_{\mathbb{R}^n} M &= c|d|\Big(3(\pi/b)^{n/2} + 2(\pi/b)^{n/2} + 3b\int_{\mathbb{R}^n} |z|^2 e^{-(b/4)|z|^2}\,dz\Big)\\ &\le c|d|\Big(5(\pi/b)^{n/2} + 3b\cdot\frac{8}{b}\,(8\pi/b)^{n/2}\Big) = c|d|\,(\pi/b)^{n/2}\big(5 + 24\cdot 8^{n/2}\big),\end{aligned}$$

using $(8\pi/b)^{n/2} = 8^{n/2}(\pi/b)^{n/2}$.

Finally $z \mapsto W(z+d) - W(z)$ is integrable. Indeed $|W(w)| \le c\,|w|\,E(w)$ by the coordinate bound, and $|w|E(w) \le \frac{1}{2}\big(1 + |w|^2\big)E(w)$, which is integrable by (1) and (3); so W is integrable, being continuous and dominated by an integrable function, and $z \mapsto W(z+d)$ is integrable by translation invariance. Hence, by (5) and monotonicity of the integral,

$$\int_{\mathbb{R}^n} \big|W(z+d) - W(z)\big|\,dz \le \int_{\mathbb{R}^n} M \le c|d|\,(\pi/b)^{n/2}\big(5 + 24\cdot 8^{n/2}\big),$$

which is the stated bound. □

2.111. `GaussLebesgueL1DensityTransfer`.

Lemma 2.111 (Gaussian-to-Lebesgue transfer of an L^1 bound through the density)**.** *Fix $n \ge 1$, let λ denote Lebesgue measure on $\mathbb{R}^n$, and let $v \in L^1(\gamma_n)$. Then the function $y \mapsto e^{-|y|^2}\,|v(y)|$ is Lebesgue integrable on $\mathbb{R}^n$, and*

$$\int_{\mathbb{R}^n} e^{-|y|^2}\,|v(y)|\,d\lambda(y) \le \pi^{n/2}\int_{\mathbb{R}^n} |v|\,d\gamma_n.$$

Proof. Write $\varrho(y) = \pi^{-n/2}e^{-|y|^2}$, so that by `GaussMeasure` the measure γ_n is λ weighted by ϱ. The inequality is in fact proved as an equality and then weakened, so it suffices to produce the equality.

Step 1: measurability against λ. Being an element of $L^1(\gamma_n)$, v has a γ_n-almost-everywhere strongly measurable representative. Moreover λ is absolutely continuous with respect to γ_n: by clause (4) of `GaussLebesgueBallComparison`, a γ_n-null set is λ-null. Absolute continuity carries a γ_n-almost-everywhere equality to a λ-almost-everywhere one, so the same representative witnesses that v is λ-almost-everywhere strongly measurable. The density $y \mapsto \varrho(y)$ is continuous, hence λ-measurable.

Step 2: the identity between the two upper integrals. Since $\gamma_n = \varrho\,\lambda$, the change-of-density formula for upper integrals of nonnegative functions gives, for the λ-measurable density ϱ and the λ-measurable $\|v\|$,

$$\int_{\mathbb{R}^n}^{*} \|v(y)\|\,d\gamma_n(y) = \int_{\mathbb{R}^n}^{*} \varrho(y)\,\|v(y)\|\,d\lambda(y).$$

Now $\varrho(y) = \big(\pi^{n/2}\big)^{-1}e^{-|y|^2}$, because $\pi^{-n/2} = (\pi^{n/2})^{-1}$ for the positive base π, so for every y

$$\pi^{n/2}\cdot\varrho(y)\,\|v(y)\| = e^{-|y|^2}\,|v(y)| = \big\|e^{-|y|^2}v(y)\big\|,$$

the last equality because $e^{-|y|^2} > 0$, so the absolute value of the product is the product of $e^{-|y|^2}$ with $|v(y)|$. Pulling the constant $\pi^{n/2}$ out of the upper integral, which is legitimate as it is finite

and positive, we obtain

$$\int_{\mathbb{R}^n}^{*} \left\| e^{-|y|^2} v(y) \right\| d\lambda(y) = \pi^{n/2} \int_{\mathbb{R}^n}^{*} \|v(y)\| \, d\gamma_n(y). \tag{1}$$

Step 3: both sides are finite. The right-hand upper integral of (1) is the $L^1(\gamma_n)$ norm of v, which is finite because $v \in L^1(\gamma_n)$. Multiplying by the finite constant $\pi^{n/2}$, the left-hand side of (1) is finite too.

Step 4: integrability. The function $y \mapsto e^{-|y|^2} v(y)$ is λ-almost-everywhere strongly measurable, being the product of the continuous function $y \mapsto e^{-|y|^2}$ — a composition of the exponential with $y \mapsto -|y|^2$ — and the λ-almost-everywhere strongly measurable v of Step 1. Together with the finiteness of its upper integral from Step 3, that is exactly Lebesgue integrability. Since $e^{-|y|^2} > 0$ for every y we have $e^{-|y|^2}|v(y)| = \left|e^{-|y|^2} v(y)\right|$ pointwise, and the absolute value of an integrable function is integrable, so $y \mapsto e^{-|y|^2}|v(y)|$ is Lebesgue integrable. That is the first assertion.

Step 5: the bound. For a real-valued integrable function the Bochner integral of its norm and the upper integral of its norm agree after the passage from extended reals, so applying that to $y \mapsto e^{-|y|^2} v(y)$ against λ on the left of (1), and to v against γ_n on the right — both integrable, the first by Step 4 and the second because $v \in L^1(\gamma_n)$ — turns (1) into an identity between nonnegative real numbers:

$$\int_{\mathbb{R}^n} \left|e^{-|y|^2} v(y)\right| d\lambda(y) = \pi^{n/2} \int_{\mathbb{R}^n} |v| \, d\gamma_n.$$

Both sides are nonnegative, so the comparison of their images under $t \mapsto \max(t, 0)$ is an equality of reals rather than merely of extended reals. Rewriting the left-hand integrand as $e^{-|y|^2}|v(y)|$, again by positivity of $e^{-|y|^2}$, gives

$$\int_{\mathbb{R}^n} e^{-|y|^2} |v(y)| \, d\lambda(y) = \pi^{n/2} \int_{\mathbb{R}^n} |v| \, d\gamma_n,$$

and in particular the stated inequality. □

2.112. `MehlerNumeratorCoordinateBallBound`.

Lemma 2.112 (Coordinate bound on the Mehler numerator over a ball)**.** *Fix a dimension n, a coordinate index $j \in \{0, \dots, n-1\}$ and a real number R. Then for all $x, y \in \mathbb{R}^n$ with $\|x\| \le R$, and for every real r with $0 \le r \le 1$,*

$$|y_j - r\, x_j| \le \|y\| + R,$$

where $\|\cdot\|$ is the Euclidean norm and v_j denotes the j-th coordinate of v. No sign or size condition on R is assumed: for $R < 0$ the hypothesis $\|x\| \le R$ cannot hold and the assertion is vacuous, and for $R \ge 0$ the proof below uses only $\|x\| \le R$ itself.

Proof. Fix x, y with $\|x\| \le R$, and fix r with $0 \le r \le 1$. The bound follows from four steps, each applied to the quantity $|y_j - r\, x_j|$.

Step 1: the triangle inequality. Writing $y_j - r\, x_j$ as $y_j + (-(r\, x_j))$ and applying $|a+b| \le |a| + |b|$ with $a = y_j$ and $b = -(r\, x_j)$, and using $|-(r\, x_j)| = |r\, x_j|$,

$$|y_j - r\, x_j| \le |y_j| + |r\, x_j|.$$

Since $|r\, x_j| = |r|\, |x_j|$ and $r \ge 0$ gives $|r| = r$, the right-hand side is $|y_j| + r\, |x_j|$.

Step 2: discarding r. From $r \le 1$ and $|x_j| \ge 0$ we get $r\, |x_j| \le 1 \cdot |x_j| = |x_j|$, multiplying the inequality $r \le 1$ by the nonnegative number $|x_j|$.

Step 3: a coordinate is dominated by the norm. For any $v \in \mathbb{R}^n$ and any index i we have $|v_i| \le \|v\|$, since $\|v\|^2 = \sum_k |v_k|^2 \ge |v_i|^2$ and both sides of $|v_i| \le \|v\|$ are nonnegative. Applying this to $v = y$, $i = j$ gives $|y_j| \le \|y\|$, and to $v = x$, $i = j$ gives $|x_j| \le \|x\|$.

Step 4: the hypothesis. Chaining steps 2 and 3 with $\|x\| \le R$,

$$r\, |x_j| \le |x_j| \le \|x\| \le R.$$

Substituting this and $|y_j| \le \|y\|$ into the bound from step 1 gives $|y_j - r\,x_j| \le \|y\| + R$, as claimed. □

2.113. `GaussRieszKernelSingularRangeSliceBound`.

Lemma 2.113 (A constant bound on the singular range $r \in [\frac{1}{2}, 1)$, at normalised radii)**.** *Fix $n \ge 1$ and $j \in \{1, \dots, n\}$. For $x, y \in \mathbb{R}^n$ and $r \in (0,1)$ put*

$$\Psi_{x,y}(r) = \frac{y_j - r x_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|x - ry|^2}{1-r^2}\Big),$$

the reflected integrand of the raw Gaussian Riesz kernel. Let R', ρ' and τ_0 be reals with

$$1 \le R', \qquad 0 < \rho', \qquad 0 < \tau_0 \le \tfrac{1}{2}, \qquad \tau_0 \le \frac{\rho'}{8R'}.$$

Then there is a constant $C_2 > 0$ such that

$$\big|\Psi_{x,y}(r)\big| \le C_2 \qquad \text{whenever } |x| \le R', \quad \rho' \le |x-y|, \quad \tfrac{1}{2} \le r < 1.$$

Proof. Put $m = n+2 \in \mathbb{N}$, and throughout write $\tau = 1 - r \in (0, \frac{1}{2}]$ for $r \in [\frac{1}{2}, 1)$ and $d = |x - ry|$. Note $0 < R'$, since $1 \le R'$. We verify the assertion with

$$C_2 = \underbrace{5R'\,m^m \Big(\frac{8}{\rho'^2}\Big)^m}_{C_1'} + \underbrace{5R'\,\tau_0^{-(n+3)/2}}_{C_2'} + \underbrace{\tfrac{5}{4}\,m^m\,32^m}_{C_3'} \;>\; 0,$$

each of the three summands being positive.

Step 1: an elementary decay inequality. For every $t > 0$,

$$e^{-t} \le \frac{m^m}{t^m}. \tag{1}$$

This is clause (1) of `ExpDecayPowerBoundAndRpowAbsorption`, with $m = n + 2$.

Step 2: the prefactor. Let $\frac{1}{2} \le r < 1$. Then $1 - r^2 = \tau(1 + r) \ge \tau > 0$, and since $\log s \le s - 1$ for $s > 0$ applied at $s = r$ gives $\log r \le r - 1$, also $-\log r \ge 1 - r = \tau > 0$. As $t \mapsto t^{n/2+1}$ and $\sqrt{\cdot}$ are nondecreasing on $[0, \infty)$,

$$(1-r^2)^{n/2+1}\sqrt{-\log r} \;\ge\; \tau^{n/2+1}\sqrt{\tau} \;=\; \tau^{(n+3)/2} > 0,$$

the last equality because $\tau^a \tau^{1/2} = \tau^{a+1/2}$ for $\tau > 0$. Taking reciprocals,

$$\frac{1}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \;\le\; \tau^{-(n+3)/2}. \tag{2}$$

Step 3: the absorption of the surviving power. For $\frac{1}{2} \le r < 1$ and every $c > 0$,

$$\tau^{-(n+3)/2}\,\big(m^m\,(c\tau)^m\big) \;\le\; m^m c^m. \tag{3}$$

This is clause (2) of `ExpDecayPowerBoundAndRpowAbsorption`, again with $m = n + 2$.

Step 4: the exponent, and the two cases. Since $1 - r^2 = \tau(1 + r) \le 2\tau$ and $1 - r^2 > 0$ and $d^2 \ge 0$, we have $d^2/(1-r^2) \ge d^2/(2\tau)$, hence

$$\exp\Big(-\frac{d^2}{1-r^2}\Big) \le \exp\Big(-\frac{d^2}{2\tau}\Big). \tag{4}$$

Also, for $0 \le r \le 1$,

$$|y_j - r x_j| \le |y_j| + r|x_j| \le |y| + |x| \le |y| + R', \tag{5}$$

using $|v_i| \le |v|$ for every coordinate, $r \le 1$ and $|x| \le R'$.

Now fix x, y with $|x| \le R'$ and $\rho' \le |x - y|$, and fix $r \in [\frac{1}{2}, 1)$. Because the prefactor and the exponential are positive, $|\Psi_{x,y}(r)| = |y_j - r x_j| \cdot \big((1-r^2)^{n/2+1}\sqrt{-\log r}\big)^{-1} \cdot e^{-d^2/(1-r^2)}$. We distinguish two cases.

Case A: $|y| \le 4R'$. Then (5) gives $|y_j - r x_j| \le 5R'$. Writing $x - ry = (x - y) + \tau y$ we get, from $|x - y| \le |x - ry| + \tau|y|$ and $|y| \le 4R'$,

$$d \;\ge\; |x - y| - \tau|y| \;\ge\; \rho' - 4R'\tau. \tag{6}$$

Sub-case $\tau \le \tau_0$. Then $\tau \le \rho'/(8R')$, so $4R'\tau \le \rho'/2$ and (6) gives $d \ge \rho'/2 > 0$. Hence $d^2 \ge \rho'^2/4$ and $d^2/(2\tau) \ge \rho'^2/(8\tau) > 0$, so by (4) and then (1) at $t = \rho'^2/(8\tau)$,

$$\exp\Big(-\frac{d^2}{1-r^2}\Big) \le e^{-\rho'^2/(8\tau)} \le m^m\Big(\frac{8\tau}{\rho'^2}\Big)^m = m^m\Big(\frac{8}{\rho'^2}\cdot\tau\Big)^m.$$

Combining with $|y_j - rx_j| \le 5R'$, with (2), and then with (3) at $c = 8/\rho'^2 > 0$,

$$|\Psi_{x,y}(r)| \le 5R'\cdot\tau^{-(n+3)/2}\cdot m^m\Big(\frac{8}{\rho'^2}\tau\Big)^m \le 5R'\,m^m\Big(\frac{8}{\rho'^2}\Big)^m = C_1' \le C_2,$$

the last step because $C_2', C_3' \ge 0$.

Sub-case $\tau > \tau_0$. Here we discard the Gaussian: $e^{-d^2/(1-r^2)} \le 1$, because $d^2/(1-r^2) \ge 0$. And $\tau \ge \tau_0 > 0$ with the exponent $-(n+3)/2 \le 0$ gives $\tau^{-(n+3)/2} \le \tau_0^{-(n+3)/2}$ — for $0 < \tau_0 \le \tau$ one has $\tau_0^{(n+3)/2} \le \tau^{(n+3)/2}$, and inverting reverses the inequality. So by (2) and (5),

$$|\Psi_{x,y}(r)| \le 5R'\,\tau_0^{-(n+3)/2} = C_2' \le C_2.$$

Case B: $|y| > 4R'$. Then $|y| > 4R' \ge 4 > 1$, and (5) gives $|y_j - rx_j| \le |y| + R' \le \frac{5}{4}|y|$. From $|ry| - |x| \le |x - ry|$ and $r \ge \frac{1}{2}$, $|x| \le R' < |y|/4$,

$$d \;\ge\; r|y| - |x| \;\ge\; \tfrac{1}{2}|y| - \tfrac{1}{4}|y| \;=\; \tfrac{1}{4}|y| > 0,$$

so $d^2 \ge |y|^2/16$ and $d^2/(2\tau) \ge |y|^2/(32\tau) > 0$. By (4) and (1) at $t = |y|^2/(32\tau)$,

$$\exp\Big(-\frac{d^2}{1-r^2}\Big) \le m^m\Big(\frac{32}{|y|^2}\cdot\tau\Big)^m,$$

and so by (2) and (3) at $c = 32/|y|^2 > 0$,

$$|\Psi_{x,y}(r)| \le \tfrac{5}{4}|y|\cdot m^m\Big(\frac{32}{|y|^2}\Big)^m = \tfrac{5}{4}\,m^m\,32^m\cdot\frac{|y|}{|y|^{2m}}.$$

Finally $|y| > 1$ and $2m \ge 1$ give $|y| = |y|^1 \le |y|^{2m}$, so the last fraction is at most 1 and $|\Psi_{x,y}(r)| \le \frac{5}{4}m^m 32^m = C_3' \le C_2$.

In all cases $|\Psi_{x,y}(r)| \le C_2$, with C_2 depending only on n, j, R', ρ' and τ_0. □

2.114. `GaussRieszKernelEndpointSliceBound`.

Lemma 2.114 (A constant bound for the reflected kernel integrand near $r = 1$)**.** *Fix $n \ge 1$ and $j \in \{1, \dots, n\}$. For $x, y \in \mathbb{R}^n$ and $r \in (0,1)$ put*

$$\Psi_{x,y}(r) = \frac{y_j - rx_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\exp\Big(-\frac{|x-ry|^2}{1-r^2}\Big),$$

the reflected *integrand of the raw Gaussian Riesz kernel: it is obtained from the kernel's own r-integrand by interchanging the roles of x and y inside the Gaussian exponent, and it is the quantity the splitting identity $|rx-y|^2/(1-r^2) = (|y|^2-|x|^2)+|x-ry|^2/(1-r^2)$ produces. Then for all $R > 0$ and $\rho > 0$ there is a constant $C > 0$, depending only on n, j, R and ρ, such that*

$$\big|\Psi_{x,y}(r)\big| \le C \qquad \text{whenever } |x| \le R, \quad |x-y| \ge \rho, \quad \tfrac{1}{2} \le r < 1.$$

The range $\frac{1}{2} \le r < 1$ is the one that matters: as $r \uparrow 1$ both $(1-r^2)^{-n/2-1}$ and $(-\log r)^{-1/2}$ blow up, and the assertion is that the Gaussian factor more than compensates, uniformly in x and y in the stated range. That is where the hypothesis $|x-y| \ge \rho$ is used, and it is the only place it is needed: at $r = 1$ the exponent is $-|x-y|^2/0^+$, so the Gaussian decay is available exactly because x and y are separated. The hypothesis $|x| \le R$ enters through the numerator and through the competition between $|x|$ and $|y|$.

The bound is a pointwise *constant bound, not a bound on $\int_{1/2}^1 |\Psi_{x,y}|$, and it is deliberately so: the route below produces the pointwise statement with no extra work, and the consumer needs a pointwise majorant in order to get integrability on $(0,1)$ at the same time as the size bound. Integrating the constant over a set of measure $\frac{1}{2}$ recovers the integral form at once. No uniformity in n, R or ρ is claimed, and nothing here feeds an endpoint constant.*

Proof. Fix $n \geq 1$ and j. Let $R, \rho > 0$ be given and put

$$R' = \max(R, 1), \qquad \rho' = \min(\rho, 1), \qquad \tau_0 = \min\Big(\tfrac{1}{2},\ \frac{\rho'}{8R'}\Big).$$

Then $1 \leq R'$ and $R \leq R'$; and $0 < \rho' \leq 1$ with $\rho' \leq \rho$. In particular $0 < R'$, so $\rho'/(8R') > 0$, whence $\tau_0 > 0$ as the minimum of two positive numbers, while $\tau_0 \leq \frac{1}{2}$ and $\tau_0 \leq \rho'/(8R')$ hold by the definition of the minimum. So R', ρ' and τ_0 satisfy the hypotheses of `GaussRieszKernelSingularRangeSlic` which therefore supplies a constant $C_2 > 0$ such that

$$\big|\Psi_{x,y}(r)\big| \leq C_2 \qquad \text{whenever } |x| \leq R', \quad \rho' \leq |x - y|, \quad \tfrac{1}{2} \leq r < 1.$$

It remains to see that this constant works for (R, ρ). Any pair (x, y) with $|x| \leq R$ satisfies $|x| \leq R'$ because $R \leq R'$, and any pair with $|x - y| \geq \rho$ satisfies $|x - y| \geq \rho'$ because $\rho' \leq \rho$. Hence $C = C_2$ has the required property, and it depends only on n, j, R' and ρ', hence only on n, j, R and ρ. □

2.115. `GaussianUnitIntegralLinearFactorBound.`

Lemma 2.115 (A linear factor against a Gaussian integral over the unit interval)**.** *For all real $t \geq 0$ and $R' \geq 0$,*

$$(t + R') \int_0^1 e^{-(t^2/4)r^2}\, dr \leq 5R' + \tfrac{5}{2}\sqrt{\pi},$$

the integral being taken with respect to Lebesgue measure on $(0, 1)$.

Proof. Fix $t \geq 0$ and $R' \geq 0$, so $t + R' \geq 0$, and recall $\sqrt{\pi} \geq 0$. The integrand is positive, so the integral is nonnegative. Split on the size of t relative to $4R'$.

Case $t \leq 4R'$. On $(0, 1)$ we have $(t^2/4)r^2 \geq 0$, so $e^{-(t^2/4)r^2} \leq 1$; since the constant 1 is integrable on $(0, 1)$, which has Lebesgue measure 1, monotonicity of the integral gives $\int_0^1 e^{-(t^2/4)r^2}\, dr \leq 1$. Multiplying by $t + R' \geq 0$,

$$(t + R') \int_0^1 e^{-(t^2/4)r^2}\, dr \leq t + R' \leq 5R' \leq 5R' + \tfrac{5}{2}\sqrt{\pi},$$

the middle inequality because $t \leq 4R'$ and the last because $\sqrt{\pi} \geq 0$.

Case $t > 4R'$. Then $t > 4R' \geq 0$, so $t > 0$ and $b := t^2/4 > 0$. Extending the integral from $(0, 1)$ to $\mathbb{R}$ is legitimate because the integrand is nonnegative and $r \mapsto e^{-br^2}$ is integrable on $\mathbb{R}$, and the Gaussian integral evaluates to

$$\int_{\mathbb{R}} e^{-br^2}\, dr = \sqrt{\pi/b} = \sqrt{(2/t)^2\pi} = (2/t)\sqrt{\pi},$$

using $\pi/(t^2/4) = (2/t)^2\pi$ and $2/t > 0$. Hence $\int_0^1 e^{-(t^2/4)r^2}\, dr \leq (2/t)\sqrt{\pi}$. Next,

$$(t + R')\,(2/t) = 2 + 2R'/t \leq \tfrac{5}{2},$$

because $2R'/t \leq \frac{1}{2}$ is equivalent, after multiplying by $t > 0$, to $4R' \leq t$, which is the case hypothesis. Therefore, multiplying the integral bound by $t + R' \geq 0$ and then using $\sqrt{\pi} \geq 0$,

$$(t + R') \int_0^1 e^{-(t^2/4)r^2}\, dr \leq \big((t + R')(2/t)\big)\sqrt{\pi} \leq \tfrac{5}{2}\sqrt{\pi} \leq 5R' + \tfrac{5}{2}\sqrt{\pi},$$

the last step because $5R' \geq 0$. This is the claim in both cases. □

2.116. `GaussRieszKernelFarSliceMajorant.`

Lemma 2.116 (The Gaussian far majorant for $K_{\gamma,j}$ off the diagonal)**.** *Fix $n \geq 1$ and $j \in \{1, \ldots, n\}$, let λ be Lebesgue measure, and let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of* `GaussRieszKernel`*, so that $K_{\gamma,j}(x, y) = c_n \int_0^1 \Phi_{x,y}(r)\, dr$ with $c_n = \sqrt{2}\,\pi^{-(n+1)/2}$ and*

$$\Phi_{x,y}(r) = \frac{y_j - r x_j}{(1 - r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|rx - y|^2}{1 - r^2}\Big), \qquad r \in (0, 1).$$

Then for all $R > 0$ and $\rho > 0$ there is a constant $C > 0$, depending only on n, j, R and ρ, such that for all $x, y \in \mathbb{R}^n$ with $|x| \leq R$ and $|x - y| \geq \rho$:

(1) $\Phi_{x,y}$ *is Lebesgue integrable on* $(0,1)$*; and*
(2) $\big|K_{\gamma,j}(x,y)\big| \le C\, e^{-|y|^2}$.

Why the integrability assertion is part of the content. $K_{\gamma,j}(x,y)$ *is defined in* `GaussRieszKernel` *as a Bochner integral over* $(0,1)$*, with the convention that the value is* 0 *when the integrand is not integrable; and* 0 *satisfies every upper bound. A bare size bound* $|K_{\gamma,j}| \le Ce^{-|y|^2}$ *would therefore be provable without establishing anything about the kernel, and would be useless to a consumer that needs the defining integral to converge. Clause (1) is what rules that out, so it is asserted here rather than assumed.*

The majorant carries no polynomial factor in $|y|$. *The right-hand side of (2) is* $Ce^{-|y|^2}$ *with* C *independent of* y*. That exact pairing with the Gaussian density* $\pi^{-n/2}e^{-|y|^2}$ *is what later lets a truncated kernel integral be dominated by* $\|f\|_{L^1(\gamma_n)}$*; a residual factor* $|y|^k$ *would break it. The constant does depend on* R *and on* ρ*, and no uniformity in* n*, in* R *or in* ρ *is claimed.*

The reference paper derives $K_{\gamma,j}$ *and fixes the prefactor* c_n *— it is explicit that an unnormalized kernel is a different family of operators, so* c_n *is not cosmetic — but it proves no off-diagonal majorant for it. Clauses (1) and (2) are therefore tablet-side rigorization with no paper proof to transcribe: what the paper supplies is the kernel, and what is needed downstream, for the* L^1 *identification, is that the kernel is an honest function off the diagonal with Gaussian decay in* y*.*

Proof. Fix $n \ge 1$ and j, and let $R, \rho > 0$. Put $R' = \max(R,1) \ge 1$ and $\rho' = \min(\rho,1) \in (0,1]$; a pair (x,y) with $|x| \le R$, $|x-y| \ge \rho$ also satisfies $|x| \le R'$, $|x-y| \ge \rho'$, so it suffices to produce a constant for (R',ρ'). Write

$$\Psi_{x,y}(r) = \frac{y_j - r x_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|x-ry|^2}{1-r^2}\Big), \qquad r \in (0,1),$$

for the reflected integrand. Throughout, $r \in (0,1)$ gives $1-r^2 > 0$ (because $r^2 < 1$) and $-\log r > 0$, so every denominator below is nonzero and the prefactor $\big((1-r^2)^{n/2+1}\sqrt{-\log r}\big)^{-1}$ is positive.

Step 0: the splitting identity. For all x, y and all r,

$$|rx - y|^2 - (1-r^2)\big(|y|^2 - |x|^2\big) = |x - ry|^2, \tag{1}$$

since expanding both sides in the inner product gives $|x|^2 - 2r\langle x,y\rangle + r^2|y|^2$ for each. Consequently, for $r \in (0,1)$, dividing (1) by $1-r^2 > 0$ and rearranging,

$$-\frac{|rx-y|^2}{1-r^2} = \big(|x|^2 - |y|^2\big) + \Big(-\frac{|x-ry|^2}{1-r^2}\Big),$$

and therefore, since $e^{a+b} = e^a e^b$,

$$\Phi_{x,y}(r) = e^{|x|^2} e^{-|y|^2}\, \Psi_{x,y}(r) \qquad (r \in (0,1)). \tag{2}$$

Step 1: the prefactor on $(0, \tfrac12]$. For $0 < r \le \tfrac12$ we have $1 - r^2 \ge \tfrac34$ and $r \le \tfrac12$, so $\log r \le \log \tfrac12 = -\log 2$, i.e. $-\log r \ge \log 2 > 0$. Hence, by monotonicity of $t \mapsto t^{n/2+1}$ on $[0,\infty)$ and of $\sqrt{\cdot}$,

$$(1-r^2)^{n/2+1}\sqrt{-\log r} \;\ge\; (\tfrac34)^{n/2+1}\sqrt{\log 2} \;>\; 0,$$

so with $K = \big((\tfrac34)^{n/2+1}\sqrt{\log 2}\big)^{-1} > 0$, a constant depending only on n,

$$\frac{1}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \le K \qquad (0 < r \le \tfrac12). \tag{3}$$

Step 2: the bound on $(0, \tfrac12]$. First, an elementary inequality: for every $t \ge 0$,

$$\big((t - R')^+\big)^2 \;\ge\; \frac{t^2}{4} - R'^2, \tag{4}$$

where $s^+ = \max(s,0)$. If $t \le 2R'$ the right-hand side is $\le R'^2 - R'^2 = 0$ and the left-hand side is ≥ 0; if $t > 2R'$ then $t - R' > t/2 > 0$, so the left-hand side is $> t^2/4 \ge t^2/4 - R'^2$.

Now fix x, y with $|x| \le R'$ and let $0 < r \le \frac{1}{2}$. From $|ry| - |x| \le |x - ry|$ and $|x - ry| \ge 0$ we get $|x - ry| \ge \big(r|y| - R'\big)^+$, so squaring (both sides nonnegative) and using (4) at $t = r|y| \ge 0$,

$$|x - ry|^2 \;\ge\; \frac{r^2|y|^2}{4} - R'^2.$$

Also $0 < 1 - r^2 \le 1$ gives $-|x - ry|^2/(1 - r^2) \le -|x - ry|^2$. Combining,

$$\exp\Big(-\frac{|x - ry|^2}{1 - r^2}\Big) \le e^{-|x-ry|^2} \le e^{R'^2} \exp\Big(-\frac{|y|^2}{4} r^2\Big).$$

With (3) and the numerator bound $|y_j - r x_j| \le |y| + R'$ (which holds because $|v_i| \le |v|$ for every coordinate, $r \le 1$ and $|x| \le R'$), and since all three factors are nonnegative,

$$\big|\Psi_{x,y}(r)\big| \le K\,(|y| + R')\,e^{R'^2} \exp\Big(-\frac{|y|^2}{4} r^2\Big) \qquad (0 < r \le \tfrac{1}{2}). \tag{5}$$

Step 3: the bound on $[\frac{1}{2}, 1)$. By `GaussRieszKernelEndpointSliceBound` applied to R' and ρ' there is a constant $C_2 > 0$, depending only on n, j, R' and ρ', with

$$\big|\Psi_{x,y}(r)\big| \le C_2 \qquad \text{whenever } |x| \le R',\ |x - y| \ge \rho',\ \tfrac{1}{2} \le r < 1. \tag{6}$$

Step 4: one majorant on all of $(0, 1)$. Fix x, y with $|x| \le R'$ and $|x - y| \ge \rho'$, and set

$$B_1 = K\,e^{R'^2}\,5R' + C_2 \ge 0, \qquad B_2 = K\,(|y| + R')\,e^{R'^2} \ge 0, \qquad b = \frac{|y|^2}{4} \ge 0,$$

$$M(r) = B_1 + B_2\,e^{-br^2}.$$

Then

$$\big|\Psi_{x,y}(r)\big| \le M(r) \qquad (r \in (0, 1)), \tag{7}$$

because for $r \le \frac{1}{2}$ the bound (5) is exactly $B_2 e^{-br^2}$ and $B_1 \ge 0$, while for $r > \frac{1}{2}$ the bound (6) gives $C_2 \le B_1$ and $B_2 e^{-br^2} \ge 0$. Since $e^{-br^2} \le 1$, (7) also gives the cruder constant bound $|\Psi_{x,y}| \le B_1 + B_2$ on $(0, 1)$.

$\Psi_{x,y}$ is Borel measurable on $\mathbb{R}$ (it is built from $r \mapsto r$, log, $\sqrt{\cdot}$, the real power $t \mapsto t^{n/2+1}$, the norm, division and exp, all Borel), and $\lambda\big((0, 1)\big) = 1 < \infty$, so the crude constant bound makes $\Psi_{x,y}$ Lebesgue integrable on $(0, 1)$; by (2), $\Phi_{x,y}$ is a constant multiple of $\Psi_{x,y}$ and is therefore integrable on $(0, 1)$ as well. This is clause (1).

Step 5: the integral of the majorant. M is continuous and bounded by $B_1 + B_2$ on $(0, 1)$, hence integrable there, and by additivity and $\lambda((0, 1)) = 1$,

$$\int_0^1 M\,d\lambda = B_1 + B_2 \int_0^1 e^{-br^2}\,dr.$$

We bound the last term by $K\,e^{R'^2}\big(5R' + \frac{5}{2}\sqrt{\pi}\big)$. By `GaussianUnitIntegralLinearFactorBound`, applied with $t = |y| \ge 0$ and the present $R' \ge 0$, and with $b = |y|^2/4$,

$$(|y| + R') \int_0^1 e^{-br^2}\,dr \le 5R' + \tfrac{5}{2}\sqrt{\pi},$$

the cited node proving this by the case split on whether $|y| \le 4R'$. Since $B_2 = Ke^{R'^2}(|y| + R')$, multiplying by the nonnegative constant $Ke^{R'^2}$ gives $B_2 \int_0^1 e^{-br^2} dr \le Ke^{R'^2}\big(5R' + \frac{5}{2}\sqrt{\pi}\big)$. Adding the nonnegative missing summand,

$$\int_0^1 M\,d\lambda \;\le\; C_0 := K\,e^{R'^2} 5R' + C_2 + K\,e^{R'^2}\big(5R' + \tfrac{5}{2}\sqrt{\pi}\big). \tag{8}$$

Step 6: clause (2). By (7), (8) and $|\int g| \le \int |g|$,

$$\Big|\int_0^1 \Psi_{x,y}\,d\lambda\Big| \le \int_0^1 |\Psi_{x,y}|\,d\lambda \le \int_0^1 M\,d\lambda \le C_0.$$

By (2) and linearity of the integral for the constant $e^{|x|^2}e^{-|y|^2}$, $\int_0^1 \Phi_{x,y} = e^{|x|^2}e^{-|y|^2}\int_0^1 \Psi_{x,y}$, so by the definition of $K_{\gamma,j}$ in `GaussRieszKernel`, and since $c_n > 0$, $e^{|x|^2} > 0$ and $e^{-|y|^2} > 0$,

$$\big|K_{\gamma,j}(x,y)\big| = c_n\, e^{|x|^2}e^{-|y|^2}\left|\int_0^1 \Psi_{x,y}\, d\lambda\right| \le c_n\, e^{R'^2}\, C_0\, e^{-|y|^2},$$

using $|x| \le R'$ and the monotonicity of exp to replace $e^{|x|^2}$ by $e^{R'^2}$. Setting $C = c_n\, e^{R'^2} C_0 > 0$, which depends only on n, j, R' and ρ' and hence only on n, j, R and ρ, gives clause (2). □

2.117. `GaussRieszKernelFarMajorant`.

Lemma 2.117 (A Gaussian far majorant for $K_{\gamma,j}$, and well-definedness of its hard truncations)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ denote Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of `GaussRieszKernel`, so that $K_{\gamma,j}(x,y) = \sqrt{2}\,\pi^{-(n+1)/2}\int_0^1 \Phi_{x,y}(r)\,dr$ with*

$$\Phi_{x,y}(r) = \frac{y_j - r x_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big), \qquad r \in (0,1).$$

For $\varepsilon > 0$ and $f : \mathbb{R}^n \to \mathbb{R}$ write

$$T_{j,\varepsilon} f(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\, f(y)\, dy,$$

the hard truncation at radius ε of the jth coordinate, the integral being against Lebesgue measure in y. Below, "$f \in L^1(\gamma_n)$" means that the function *$f : \mathbb{R}^n \to \mathbb{R}$ is γ_n-integrable; f is a genuine function and not an equivalence class, because $T_{j,\varepsilon}f(x)$ is formed from its pointwise values. Then:*

(1) *for all $R > 0$ and $\rho > 0$ there is a constant $C > 0$, depending only on n, j, R and ρ, such that*
   (a) *for all $x, y \in \mathbb{R}^n$ with $|x| \le R$ and $|x-y| \ge \rho$, the function $\Phi_{x,y}$ is Lebesgue integrable on $(0,1)$ and*
   $$\big|K_{\gamma,j}(x,y)\big| \le C\, e^{-|y|^2};$$
   (b) *for every $f \in L^1(\gamma_n)$ and every x with $|x| \le R$, the function $y \mapsto K_{\gamma,j}(x,y)f(y)$ is Lebesgue integrable on $\{y : |x-y| > \rho\}$ and*
   $$\big|T_{j,\rho}f(x)\big| \le C\,\|f\|_{L^1(\gamma_n)}, \qquad \|f\|_{L^1(\gamma_n)} = \int_{\mathbb{R}^n} |f|\, d\gamma_n;$$
(2) *for every $f \in L^1(\gamma_n)$ and every $\varepsilon > 0$ the function $x \mapsto T_{j,\varepsilon}f(x)$ is Borel measurable on $\mathbb{R}^n$; and*
(3) *for every $f \in L^1(\gamma_n)$ and every $x \in \mathbb{R}^n$ the function $\varepsilon \mapsto T_{j,\varepsilon}f(x)$ is continuous on $(0,\infty)$.*

Why clause (1)(a) carries the integrability assertion, and (1)(b) its own. *$K_{\gamma,j}(x,y)$ is defined in `GaussRieszKernel` as a Bochner integral over $(0,1)$, with the convention that the value is 0 when the integrand is not integrable; and 0 satisfies every upper bound. A bare size bound $|K_{\gamma,j}| \le Ce^{-|y|^2}$ would therefore be provable without establishing anything about the kernel, and would be useless to a consumer that needs the integral to converge. The same applies one level up: $T_{j,\rho}f(x)$ is itself a Bochner integral in y, so the bound on it is stated only alongside the assertion that its integrand is integrable. Both integrability assertions are part of the content, not hypotheses. The clauses (2) and (3) need no such companion: measurability and continuity are not quantities with a conventional value.*

The majorant carries no polynomial factor in $|y|$. *The right-hand side of (1)(a) is $Ce^{-|y|^2}$ with C independent of y. This is what makes (1)(b) true with $\|f\|_{L^1(\gamma_n)}$ on the right, since $\|f\|_{L^1(\gamma_n)} = \pi^{-n/2}\int |f(y)|e^{-|y|^2}\,dy$ pairs exactly with $e^{-|y|^2}$; a residual factor $|y|^k$ would break that pairing. The constant C does depend on R and on ρ, and no uniformity in n, in R or in ρ is claimed.*

Notation. *$T_{j,\varepsilon}$ is the jth coordinate of the hard truncation T_ε displayed in the reference paper. The paper's maximal truncation is $\sup_{\varepsilon>0}|T_\varepsilon f(x)|$ with the* Euclidean vector *norm, which is a different object from the coordinatewise $\sup_{\varepsilon>0}|T_{j,\varepsilon}f(x)|$; only the coordinatewise truncations appear here, and the two are reconciled later by summing over the finitely many coordinates. Nothing below asserts anything about the vector object.*

Proof. Throughout $n \geq 1$ (the index j ranges over a nonempty set), $c_n = \sqrt{2}\,\pi^{-(n+1)/2}$, and λ is Lebesgue measure.

Step 0: the splitting identity. For all $x, y \in \mathbb{R}^n$ and $r \in (0,1)$,

$$\frac{|rx-y|^2}{1-r^2} = \big(|y|^2 - |x|^2\big) + \frac{|x-ry|^2}{1-r^2}. \tag{1}$$

Indeed, multiplying by $1-r^2 > 0$, the claim is $|rx-y|^2 - (1-r^2)(|y|^2-|x|^2) = |x-ry|^2$, and expanding both sides in the inner product gives $|x|^2 - 2r\langle x,y\rangle + r^2|y|^2$ for each. Consequently, writing

$$\Psi_{x,y}(r) = \frac{y_j - rx_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\exp\Big(-\frac{|x-ry|^2}{1-r^2}\Big),$$

we have the pointwise identity $\Phi_{x,y}(r) = e^{|x|^2}e^{-|y|^2}\,\Psi_{x,y}(r)$ on $(0,1)$. Since the factor $e^{|x|^2}e^{-|y|^2}$ is a positive constant in r, $\Phi_{x,y}$ is Lebesgue integrable on $(0,1)$ if and only if $\Psi_{x,y}$ is, and in that case

$$K_{\gamma,j}(x,y) = c_n\, e^{|x|^2}e^{-|y|^2}\int_0^1 \Psi_{x,y}(r)\,dr. \tag{2}$$

Step 1: a reduction. Let $R, \rho > 0$ be given and put $R' = \max(R,1) \geq 1$ and $\rho' = \min(\rho,1) \in (0,1]$. Any pair (x,y) with $|x| \leq R$ and $|x-y| \geq \rho$ also satisfies $|x| \leq R'$ and $|x-y| \geq \rho'$, and $\|f\|_{L^1(\gamma_n)}$ does not involve R or ρ; so a constant that works for (R',ρ') works for (R,ρ). We may therefore assume

$$R \geq 1, \qquad 0 < \rho \leq 1.$$

Fix such R, ρ and fix x, y with $|x| \leq R$ and $|x-y| \geq \rho$. We bound $\int_0^1 |\Psi_{x,y}|\,d\lambda$ by a constant $C_0 = C_0(n,R,\rho)$ that does not depend on x or y. We will use throughout that $|y_j - rx_j| \leq |y| + |x| \leq |y| + R$ for $r \in (0,1)$.

Step 2: the range $r \in (0, \frac{1}{2}]$. Here $1 - r^2 \in [\frac{3}{4}, 1)$, so $(1-r^2)^{-n/2-1} \leq (4/3)^{n/2+1}$; and $-\log r \geq \log 2$, so $(-\log r)^{-1/2} \leq (\log 2)^{-1/2}$. Also $1 - r^2 \leq 1$ gives $\exp(-|x-ry|^2/(1-r^2)) \leq \exp(-|x-ry|^2)$, and $|x-ry| \geq r|y| - |x| \geq r|y| - R$, so with $s^+ = \max(s,0)$,

$$|\Psi_{x,y}(r)| \leq (4/3)^{n/2+1}(\log 2)^{-1/2}\,(|y|+R)\,\exp\big(-((r|y|-R)^+)^2\big).$$

If $|y| \leq 4R$ then $|y| + R \leq 5R$ and, bounding the exponential by 1,

$$\int_0^{1/2}(|y|+R)\exp\big(-((r|y|-R)^+)^2\big)\,dr \leq \tfrac{5}{2}R.$$

If $|y| > 4R$ then $|y| > 0$ and the change of variables $t = r|y|$ (a dilation of $\mathbb{R}$, under which Lebesgue measure scales by $|y|$) gives

$$\int_0^{1/2}(|y|+R)\exp\big(-((r|y|-R)^+)^2\big)\,dr = \frac{|y|+R}{|y|}\int_0^{|y|/2}\exp\big(-((t-R)^+)^2\big)\,dt \leq \tfrac{5}{4}\Big(R + \tfrac{\sqrt{\pi}}{2}\Big),$$

using $(|y|+R)/|y| = 1 + R/|y| \leq \frac{5}{4}$ and $\int_0^\infty \exp(-((t-R)^+)^2)\,dt = \int_0^R 1\,dt + \int_R^\infty e^{-(t-R)^2}dt = R + \frac{\sqrt{\pi}}{2}$, the last integral being the half-line Gaussian integral (shared context: the scalar Gaussian integral together with the invariance of Lebesgue measure under the reflection $t \mapsto -t$, which makes the two half-lines contribute equally). In either case

$$\int_0^{1/2}|\Psi_{x,y}|\,d\lambda \leq (4/3)^{n/2+1}(\log 2)^{-1/2}\Big(\tfrac{5}{2}R + \tfrac{5}{4}R + \tfrac{5\sqrt{\pi}}{8}\Big) =: C_1(n,R). \tag{3}$$

Step 3: the range $r \in [\frac{1}{2}, 1)$. Write $\tau = 1 - r \in (0, \frac{1}{2}]$. Then $1 - r^2 = \tau(1 + r) \in [\tau, 2\tau]$, and $\log r = \log(1 - \tau) \le -\tau$ gives $-\log r \ge \tau$. Hence

$$\frac{1}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \le \tau^{-n/2-3/2}, \qquad \exp\Big(-\frac{|x-ry|^2}{1-r^2}\Big) \le \exp\Big(-\frac{|x-ry|^2}{2\tau}\Big).$$

We use repeatedly the elementary identity, valid for $a > 1$ and $b > 0$,

$$\int_0^\infty \tau^{-a} e^{-b/\tau}\, d\tau = b^{1-a}\,\Gamma(a-1) < \infty, \tag{4}$$

obtained from the substitution $\tau = b/u$, under which $d\tau = (b/u^2)\, du$ and the integral becomes $b^{1-a}\int_0^\infty u^{a-2}e^{-u}\, du$; the latter converges because $a - 2 > -1$ and because e^{-u} decays at infinity. We apply (4) with $a = \frac{n+3}{2} > 1$, so $a - 1 = \frac{n+1}{2}$.

Case $|y| \le 4R$. Then $|y_j - rx_j| \le 5R$. Since $x - ry = (x - y) + \tau y$ we get $|x - ry| \ge |x - y| - \tau|y| \ge \rho - 4R\tau$. Put $\tau_0 = \min(\frac{1}{2},\ \rho/(8R)) > 0$. For $0 < \tau \le \tau_0$ we have $4R\tau \le \rho/2$, hence $|x - ry| \ge \rho/2$ and $\exp(-|x - ry|^2/(2\tau)) \le \exp(-\rho^2/(8\tau))$, so by (4) with $b = \rho^2/8$,

$$\int_{\{0<\tau\le\tau_0\}} 5R\,\tau^{-n/2-3/2} e^{-\rho^2/(8\tau)}\, d\tau \le 5R\,(\rho^2/8)^{-(n+1)/2}\,\Gamma\big(\tfrac{n+1}{2}\big).$$

For $\tau_0 < \tau \le \frac{1}{2}$ we bound the exponential by 1 and $\tau^{-n/2-3/2} \le \tau_0^{-n/2-3/2}$, so that part of the integral is at most $\frac{5}{2}R\,\tau_0^{-n/2-3/2}$.

Case $|y| > 4R$. Then $|y_j - rx_j| \le |y| + R \le \frac{5}{4}|y|$, and for $r \ge \frac{1}{2}$, $|x - ry| \ge r|y| - |x| \ge \frac{1}{2}|y| - R > \frac{1}{2}|y| - \frac{1}{4}|y| = \frac{1}{4}|y|$, so $\exp(-|x - ry|^2/(2\tau)) \le \exp(-|y|^2/(32\tau))$. By (4) with $b = |y|^2/32$,

$$\int_{\{0<\tau\le 1/2\}} \tfrac{5}{4}|y|\,\tau^{-n/2-3/2} e^{-|y|^2/(32\tau)}\, d\tau \le \tfrac{5}{4}|y|\,(|y|^2/32)^{-(n+1)/2}\Gamma\big(\tfrac{n+1}{2}\big) = \tfrac{5}{4}\,32^{(n+1)/2}\Gamma\big(\tfrac{n+1}{2}\big)\,|y|^{-n},$$

and $|y| > 4R \ge 4 > 1$ with $n \ge 1$ gives $|y|^{-n} \le 1$.

Adding the two cases (only one occurs, but the sum dominates either) and substituting $\tau = 1 - r$, which is an isometry of $\mathbb{R}$ composed with a translation and therefore preserves Lebesgue measure (shared context),

$$\int_{1/2}^1 |\Psi_{x,y}|\, d\lambda \le C_2(n, R, \rho), \tag{5}$$

where $C_2 = 5R(\rho^2/8)^{-(n+1)/2}\Gamma(\frac{n+1}{2}) + \frac{5}{2}R\tau_0^{-n/2-3/2} + \frac{5}{4}\,32^{(n+1)/2}\Gamma(\frac{n+1}{2})$ depends only on n, R and ρ.

Step 4: clause (1)(a). By (3) and (5), $\int_0^1 |\Psi_{x,y}|\, d\lambda \le C_0 := C_1 + C_2$, which is finite and independent of x and y in the range considered. The function $\Psi_{x,y}$ is continuous on $(0, 1)$, hence Borel measurable there, so it is Lebesgue integrable on $(0, 1)$; by Step 0 so is $\Phi_{x,y}$. By (2) and $|x| \le R$,

$$|K_{\gamma,j}(x, y)| \le c_n\, e^{R^2} e^{-|y|^2} \int_0^1 |\Psi_{x,y}|\, d\lambda \le c_n\, e^{R^2} C_0\, e^{-|y|^2}.$$

Set $C_{\mathrm{maj}} = c_n e^{R^2} C_0 > 0$.

Step 5: joint measurability of the kernel. The map $(x, y, r) \mapsto \Phi_{x,y}(r)$ is continuous on $\mathbb{R}^n \times \mathbb{R}^n \times (0, 1)$: on that domain $1 - r^2 \in (0, 1]$ and $-\log r \in (0, \infty)$, so every factor is a composition of continuous functions with nonvanishing denominators. Hence it is Borel measurable for the product σ-algebra. A partial Bochner integral of a jointly measurable function is measurable in the remaining parameters, with the convention value 0 on the parameters whose slice is not integrable (this is the measurability half of the Fubini package, and uses only σ-finiteness of Lebesgue measure on $(0, 1)$, shared context). Therefore

$$(x, y) \longmapsto K_{\gamma,j}(x, y) \quad \text{is Borel measurable on } \mathbb{R}^n \times \mathbb{R}^n. \tag{6}$$

Step 6: clause (1)(b). Let $f \in L^1(\gamma_n)$ and let x satisfy $|x| \le R$. Being γ_n-integrable, f agrees γ_n-almost everywhere with a Borel measurable function $\widetilde{f}$; by clause (4) of GaussLebesgueBallComparison

the measures γ_n and λ have the same null sets, so $f = \widetilde{f}$ also λ-almost everywhere, and in particular f is λ-measurable up to a null set. By (6) the function $y \mapsto K_{\gamma,j}(x,y)$ is Borel, so $y \mapsto K_{\gamma,j}(x,y)f(y)$ is λ-measurable up to a null set, and the same is true of its restriction to $\{y : |x-y| > \rho\}$.

On that set $|x-y| \geq \rho$, so clause (1)(a) applies and gives $|K_{\gamma,j}(x,y)f(y)| \leq C_{\mathrm{maj}} e^{-|y|^2}|f(y)|$. By `GaussMeasure`, γ_n is λ weighted by $\pi^{-n/2}e^{-|y|^2}$, so

$$\int_{\mathbb{R}^n} e^{-|y|^2}|f(y)|\,dy = \pi^{n/2}\int_{\mathbb{R}^n}|f|\,d\gamma_n = \pi^{n/2}\|f\|_{L^1(\gamma_n)} < \infty, \tag{7}$$

the finiteness because $f \in L^1(\gamma_n)$. Hence $y \mapsto K_{\gamma,j}(x,y)f(y)$ is dominated on $\{|x-y| > \rho\}$ by an integrable function, so it is Lebesgue integrable there, and

$$|T_{j,\rho}f(x)| \leq \int_{\{|x-y|>\rho\}} |K_{\gamma,j}(x,y)|\,|f(y)|\,dy \leq C_{\mathrm{maj}}\,\pi^{n/2}\,\|f\|_{L^1(\gamma_n)}.$$

Taking $C = \max\bigl(C_{\mathrm{maj}},\ C_{\mathrm{maj}}\pi^{n/2}\bigr) > 0$ makes both (1)(a) and (1)(b) hold with the single constant C, which depends only on n, j, R and ρ. This proves clause (1).

Step 7: clause (2). Fix $f \in L^1(\gamma_n)$ and $\varepsilon > 0$, and let $\widetilde{f}$ be a Borel function with $f = \widetilde{f}$ λ-almost everywhere (Step 6). For every x the integrands $y \mapsto \mathbf{1}_{\{|x-y|>\varepsilon\}}K_{\gamma,j}(x,y)f(y)$ and the same with $\widetilde{f}$ agree λ-almost everywhere, so they have the same integral; it therefore suffices to prove the measurability for $\widetilde{f}$. The set $\{(x,y) : |x-y| > \varepsilon\}$ is open in $\mathbb{R}^n \times \mathbb{R}^n$, hence Borel, so by (6) the function

$$(x,y) \longmapsto \mathbf{1}_{\{|x-y|>\varepsilon\}}\,K_{\gamma,j}(x,y)\,\widetilde{f}(y)$$

is Borel on the product. Applying again the measurability half of the Fubini package (Step 5), this time integrating out y against the σ-finite measure λ, the function $x \mapsto \int \mathbf{1}_{\{|x-y|>\varepsilon\}}K_{\gamma,j}(x,y)\widetilde{f}(y)\,dy = T_{j,\varepsilon}\widetilde{f}(x)$ is Borel measurable.

Step 8: clause (3). Fix $f \in L^1(\gamma_n)$ and $x \in \mathbb{R}^n$, and fix $\varepsilon_* \in (0,\infty)$; we show continuity at ε_* relative to $(0,\infty)$. Put $R_x = \max(|x|,1)$ and $\rho_* = \varepsilon_*/2 > 0$, and let C be the constant of clause (1) for the pair (R_x,ρ_*). By clause (1)(a) and (7) the function

$$g(y) = \mathbf{1}_{\{|x-y|>\rho_*\}}\,C\,e^{-|y|^2}\,|f(y)|$$

is Lebesgue integrable, and for every $\varepsilon \geq \rho_*$,

$$\bigl|\mathbf{1}_{\{|x-y|>\varepsilon\}}K_{\gamma,j}(x,y)f(y)\bigr| \leq g(y) \qquad \text{for every } y,$$

because $\{|x-y| > \varepsilon\} \subseteq \{|x-y| > \rho_*\}$ and clause (1)(a) applies on the latter set.

Let (ε_k) be any sequence in $(0,\infty)$ with $\varepsilon_k \to \varepsilon_*$; discarding finitely many terms we may assume $\varepsilon_k \geq \rho_*$ for all k. For every y with $|x-y| \neq \varepsilon_*$ we have $\mathbf{1}_{\{|x-y|>\varepsilon_k\}} \to \mathbf{1}_{\{|x-y|>\varepsilon_*\}}$: if $|x-y| > \varepsilon_*$ then $|x-y| > \varepsilon_k$ for large k, and if $|x-y| < \varepsilon_*$ then $|x-y| < \varepsilon_k$ for large k. The exceptional set $\{y : |x-y| = \varepsilon_*\}$ is a sphere in $\mathbb{R}^n$ with $n \geq 1$, hence Lebesgue-null. So the integrands converge λ-almost everywhere and are dominated by the integrable g; dominated convergence (shared context) gives $T_{j,\varepsilon_k}f(x) \to T_{j,\varepsilon_*}f(x)$. Since the sequence was arbitrary and $(0,\infty)$ is a metric space, $\varepsilon \mapsto T_{j,\varepsilon}f(x)$ is continuous at ε_* relative to $(0,\infty)$; as ε_* was arbitrary, it is continuous on $(0,\infty)$. □

2.118. `MehlerLebesgueKernel`.

Lemma 2.118 (The Lebesgue kernel of the Ornstein–Uhlenbeck semigroup)**.** *Fix $s > 0$ and write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} = 1 - r^2 \in (0,1),$$

and define the Lebesgue kernel

$$p_s(x,y) = \frac{1}{\pi^{n/2}\,\sigma^{n/2}}\exp\Bigl(-\frac{|rx-y|^2}{\sigma}\Bigr), \qquad x,y \in \mathbb{R}^n.$$

Then for every $v \in L^2(\gamma_n)$ and γ_n-almost every $x \in \mathbb{R}^n$:

(1) *the function $y \mapsto p_s(x,y)\,v(y)$ is integrable on $\mathbb{R}^n$ for* Lebesgue *measure; and*

(2) $(P_s v)(x) = \displaystyle\int_{\mathbb{R}^n} p_s(x,y)\, v(y)\, dy,$

with P_s the Ornstein–Uhlenbeck semigroup of `OUSemigroup` *and γ_n as in* `GaussMeasure`*.*

This is the reference paper's displayed Lebesgue-kernel formula. The integration in (2) is against Lebesgue measure dy, not against $d\gamma_n(y)$: that is the whole point of the statement, and it is what distinguishes p_s from the averaging form of `MehlerFormula`*, which integrates a contracted and noise-shifted argument against γ_n. The two clauses are asserted for γ_n-almost every x and not for every x, matching the strength of* `MehlerFormula`*: for a general $v \in L^2(\gamma_n)$ the left-hand side is an almost-everywhere equivalence class and an every-x assertion would be meaningless. Pointwise statements are recovered later only for inputs for which both sides have continuous representatives.*

Proof. Fix $s > 0$ and put $r = e^{-s}$, $\sigma = 1 - r^2$ and $\tau = \sqrt{\sigma}$. Since $s > 0$ we have $0 < r < 1$, hence $0 < \sigma < 1$ and $0 < \tau < 1$; in particular $\tau > 0$, which is what the change of variables below needs. Note also $\sigma = 1 - e^{-2s}$, because $r^2 = e^{-2s}$.

Step 1: the averaging form. By `MehlerFormula`, applied to this s and this v, there is a γ_n-null set off which both of its first two clauses hold; fix x outside it. Clause (1) of that node says that

$$z \longmapsto v\big(rx + \tau z\big)$$

is γ_n-integrable, and clause (2) says

$$(P_s v)(x) = \int_{\mathbb{R}^n} v\big(rx + \tau z\big)\, d\gamma_n(z). \tag{1}$$

(The node writes the contraction factor as e^{-s} and the spreading factor as $\sqrt{1 - e^{-2s}}$, which are exactly r and τ.)

Step 2: unfolding γ_n. By `GaussMeasure`, γ_n is Lebesgue measure weighted by the density $\pi^{-n/2} e^{-|z|^2}$. Hence the integral in (1) is the Lebesgue integral

$$(P_s v)(x) = \int_{\mathbb{R}^n} v\big(rx + \tau z\big)\, \pi^{-n/2} e^{-|z|^2}\, dz, \tag{2}$$

and the integrability statement of Step 1 is Lebesgue integrability of the integrand of (2).

Step 3: the affine change of variables. Let $\Psi(z) = rx + \tau z$, a bijection of $\mathbb{R}^n$ with inverse $\Psi^{-1}(y) = \tau^{-1}(y - rx)$; both are continuous, hence Borel measurable. We claim that the push-forward of Lebesgue measure λ along Ψ is $\tau^{-n}\lambda$. Indeed, for a Borel set A,

$$(\Psi_*\lambda)(A) = \lambda\big(\Psi^{-1}(A)\big) = \lambda\big(\tau^{-1}(A - rx)\big) = \tau^{-n}\lambda(A - rx) = \tau^{-n}\lambda(A),$$

using in the third equality that Lebesgue measure of a dilated set scales by the nth power of the dilation factor, $\lambda(cB) = c^n \lambda(B)$ for $c > 0$ (here with $c = \tau^{-1}$), and in the fourth the translation invariance of Lebesgue measure (shared context).

Consequently, for any Borel $g : \mathbb{R}^n \to \mathbb{R}$, the function $g \circ \Psi$ is λ-integrable if and only if g is $(\Psi_*\lambda)$-integrable, i.e. if and only if g is λ-integrable (the two measures differ by the positive finite factor τ^{-n}), and in that case the change-of-variables formula for a pushforward gives

$$\int g(\Psi(z))\, dz = \int g\, d(\Psi_*\lambda) = \tau^{-n} \int g(y)\, dy. \tag{3}$$

Step 4: identifying the kernel. Apply (3) to

$$g(y) = v(y)\, \pi^{-n/2} \exp\big(-\tau^{-2}|y - rx|^2\big).$$

Then $g(\Psi(z)) = v(rx + \tau z)\, \pi^{-n/2} \exp(-\tau^{-2}|\tau z|^2) = v(rx + \tau z)\, \pi^{-n/2} e^{-|z|^2}$, which is exactly the integrand of (2). By Step 1 that function is λ-integrable, so by the equivalence in Step 3 g is λ-integrable, and (3) together with (2) gives

$$(P_s v)(x) = \tau^{-n} \int_{\mathbb{R}^n} v(y)\, \pi^{-n/2} \exp\Big(-\frac{|y - rx|^2}{\tau^2}\Big) dy.$$

Finally $\tau^{-n} = \sigma^{-n/2}$ and $\tau^2 = \sigma$, and $|y - rx| = |rx - y|$, so the integrand equals

$$\pi^{-n/2}\sigma^{-n/2}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big)v(y) = p_s(x,y)\,v(y).$$

Thus $y \mapsto p_s(x,y)v(y)$ is Lebesgue integrable, which is clause (1), and $(P_s v)(x) = \int p_s(x,y)v(y)\,dy$, which is clause (2). Since x was an arbitrary point outside a fixed γ_n-null set, both clauses hold for γ_n-almost every x. □

2.119. `MehlerKernelXDerivative`.

Lemma 2.119 (The x-derivative of the Lebesgue kernel)**.** *Fix $s > 0$ and write $r = e^{-s} \in (0,1)$ and $\sigma = 1 - e^{-2s} = 1 - r^2 \in (0,1)$, and let*

$$p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big)$$

be the Lebesgue kernel of `MehlerLebesgueKernel`. Then for every $x, y \in \mathbb{R}^n$ and every $j \in \{1,\dots,n\}$ the function $w \mapsto p_s(w,y)$ is differentiable at x, and

$$D_j^x\, p_s(x,y) = \sqrt{2}\,r\,\frac{y_j - rx_j}{\sigma}\;p_s(x,y),$$

where $D_j = 2^{-1/2}\partial_j$ is the energy-normalized partial derivative, so that the left-hand side means $2^{-1/2}\,\partial_{x_j} p_s(x,y)$.

The normalization matters and is the easiest thing here to misread. *In the Euclidean normalization the same computation reads $\partial_{x_j} p_s = 2r(y_j - rx_j)\sigma^{-1}p_s$, which is the form the reference paper uses when it derives the normalized kernel; the $\sqrt{2}$ above is that 2 multiplied by the $2^{-1/2}$ of $D = 2^{-1/2}\nabla$, the convention of `GaussD` that the whole tablet uses. The two displays are the same identity, not two different constants.*

The statement is also exactly the appendix's derivative identity in different letters. With the appendix's reparametrization $\sigma = 1 - r^2$, $d = (1-r)x$, $z = y - x$ and $q_\sigma(z) = (\pi\sigma)^{-n/2}e^{-|z|^2/\sigma}$ one has $z + d = y - rx$, hence $p_s(x,y) = q_\sigma(z+d)$ and $\sqrt{2}\,r\,(y_j - rx_j)\sigma^{-1}p_s(x,y) = \sqrt{2}\,r\,V_{j,\sigma}(z+d)$ with $V_{j,\sigma}(z) = (z_j/\sigma)q_\sigma(z)$. So the appendix's displayed derivative identity is this node, and the appendix's (σ, d, z, q_σ) notation is a change of letters to be introduced locally where it is used, not a second fact.

Unlike `MehlerLebesgueKernel`, this is an assertion at every *(x,y): no almost-everywhere qualifier appears, because p_s is an explicit smooth function of its first argument and no equivalence class is involved.*

Proof. Fix $s > 0$, $y \in \mathbb{R}^n$ and j. Since $s > 0$ we have $0 < r < 1$, hence $0 < \sigma = 1 - r^2 < 1$; in particular $\sigma \neq 0$, so every division by σ below is legitimate, and the constant $c = \pi^{-n/2}\sigma^{-n/2}$ is a well-defined positive real.

Step 1: the exponent is a polynomial. Write $A(w) = -\sigma^{-1}|rw - y|^2$, so that $p_s(w,y) = c\,e^{A(w)}$. In coordinates,

$$A(w) = -\sigma^{-1}\sum_{i=1}^{n}(r\,w_i - y_i)^2,$$

a polynomial in the coordinates $w_1,\dots,w_n$; polynomials are differentiable everywhere, and the jth partial derivative is computed termwise. Only the $i = j$ term depends on w_j, and by the chain rule for $t \mapsto (rt - y_j)^2$,

$$\partial_{w_j} A(w) = -\sigma^{-1}\cdot 2\,(rw_j - y_j)\cdot r = \frac{2r\,(y_j - rw_j)}{\sigma},$$

the last step using $-(rw_j - y_j) = y_j - rw_j$.

Step 2: differentiating the exponential. The real exponential is differentiable with derivative itself, so by the chain rule $w \mapsto c\,e^{A(w)}$ is differentiable at every w, in particular at x, which is the first assertion. Its jth partial derivative is

$$\partial_{w_j}\big(c\,e^{A(w)}\big) = c\,e^{A(w)}\,\partial_{w_j}A(w) = p_s(w,y)\cdot\frac{2r\,(y_j - rw_j)}{\sigma}.$$

Evaluating at $w = x$,

$$\partial_{x_j} p_s(x,y) = \frac{2r\,(y_j - r x_j)}{\sigma}\, p_s(x,y). \tag{1}$$

This is the Euclidean-normalized form quoted in the statement, and it is the identity the reference paper displays when it passes from the semigroup to the normalized kernel.

Step 3: passing to the D normalization. By `GaussD` the energy-normalized derivative is $D = 2^{-1/2}\nabla$, so $D_j^x p_s(x,y) = 2^{-1/2}\partial_{x_j} p_s(x,y)$. Multiplying (1) by $2^{-1/2}$ and using $2^{-1/2}\cdot 2 = 2^{1/2} = \sqrt{2}$,

$$D_j^x p_s(x,y) = 2^{-1/2}\cdot \frac{2r\,(y_j - r x_j)}{\sigma}\, p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y),$$

which is the assertion. □

2.120. `MehlerKernelSliceVanishing`.

Lemma 2.120 (Vanishing of the fixed-time Mehler gradient slice)**.** *Fix $n \geq 1$, $j \in \{1,\dots,n\}$ and $x \in \mathbb{R}^n$. For $s > 0$ write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} \in (0,1), \qquad p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}} \exp\Big(-\frac{|rx - y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, as evaluated in `MehlerKernelXDerivative` (the displayed right-hand side is the closed-form expression that node proves $2^{-1/2}\partial_{x_j} p_s(x,y)$ equals, and it is that explicit expression, not a derivative, that occurs here). Then for every $s > 0$ the function $y \mapsto D_j^x p_s(x,y)$ is Lebesgue integrable on $\mathbb{R}^n$ and

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)\, dy = 0.$$

This is the fixed-time half of the reference paper's appendix cancellation computation: the appendix isolates the odd part of the heat-kernel derivative and records that the function $V_{j,\sigma}(z) = (z_j/\sigma) q_\sigma(z)$ appearing in it is odd in z_j, so that its integral over all of $\mathbb{R}^n$ vanishes. The statement above is exactly that vanishing, together with the absolute convergence that makes it meaningful, at one fixed s. Nothing is asserted here about any truncation $\{|x-y| > \varepsilon\}$, nor about the normalized Gaussian Riesz kernel obtained by integrating in s, nor about any limit as $\varepsilon \downarrow 0$; and no claim of uniformity in s is made. Every constant occurring in the proof is allowed to depend on n, on j, on x and on s, as the appendix header permits.

Proof. Throughout, λ denotes Lebesgue measure on $\mathbb{R}^n$, and $n \geq 1$, j, x and $s > 0$ are fixed. Since $s > 0$ we have $e^{-2s} < 1$, so $\sigma = 1 - e^{-2s} > 0$; also $r = e^{-s} > 0$.

Step 0: translating the integrand to the origin. Define

$$G(w) = \sqrt{2}\, r\, \frac{w_j}{\sigma}\cdot \frac{1}{\pi^{n/2}\sigma^{n/2}} e^{-|w|^2/\sigma}, \qquad w \in \mathbb{R}^n.$$

For every $w \in \mathbb{R}^n$ we claim $D_j^x p_s(x,\, rx + w) = G(w)$. Indeed, putting $y = rx + w$ we have $y_j - r x_j = (rx)_j + w_j - r x_j = w_j$, because the jth coordinate of rx is $r x_j$; and $rx - y = rx - (rx + w) = -w$, so $|rx - y| = |-w| = |w|$. Substituting these two identities into the displayed formula for $D_j^x p_s$ gives exactly $G(w)$.

Step 1: a Gaussian majorant for G. We first record the elementary inequality

$$u\, e^{-u} \leq 1 \qquad (u \geq 0). \tag{1}$$

Indeed $1 + u \leq e^u$ for every real u (shared context, the exponential inequality), so $u \leq e^u$, and multiplying by $e^{-u} > 0$ gives (1).

Fix w and put $t = |w| \geq 0$ and $E = e^{-t^2/(2\sigma)}$. Then $0 < E \leq 1$, since $t^2/(2\sigma) \geq 0$, and $e^{-t^2/\sigma} = E \cdot E$ because $-t^2/\sigma = -t^2/(2\sigma) - t^2/(2\sigma)$. Applying (1) with $u = t^2/(2\sigma)$ gives $\big(t^2/(2\sigma)\big)E \leq 1$, i.e.

$$t^2 E \leq 2\sigma. \tag{2}$$

Also $t \leq 1 + t^2$, since $(t-1)^2 \geq 0$ gives $2t \leq 1 + t^2$ and hence $t \leq (1+t^2)/2 \leq 1 + t^2$. Therefore, using $E > 0$, then $E \leq 1$ and (2),

$$t\, e^{-t^2/\sigma} = t\, E\, E \leq (1+t^2) E\, E = E\, E + (t^2 E) E \leq E + 2\sigma E = (1 + 2\sigma) E,$$

that is,

$$|w|\, e^{-|w|^2/\sigma} \leq (1 + 2\sigma)\, e^{-|w|^2/(2\sigma)} \qquad (w \in \mathbb{R}^n). \tag{3}$$

Next, $|w_j| \leq |w|$ for every w and every j: writing $|w| = \big(\sum_i |w_i|^2\big)^{1/2}$, the sum of nonnegative terms is at least its jth term, so $|w_j|^2 \leq |w|^2$, and the square root is monotone.

Finally, $e^{-|w|^2/(2\sigma)}$ is Lebesgue integrable on $\mathbb{R}^n$: with $b = 1/(2\sigma) > 0$ we have $e^{-b|w|^2} = \prod_{i=1}^n e^{-b w_i^2}$, Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of one-dimensional Lebesgue measure, the integrand is nonnegative and continuous hence product-measurable, so by Tonelli (shared context) and the scalar Gaussian integral (shared context, $\int_{\mathbb{R}} e^{-bv^2} dv = \sqrt{\pi/b}$),

$$\int_{\mathbb{R}^n} e^{-b|w|^2}\, dw = (\pi/b)^{n/2} < \infty. \tag{4}$$

Combining, G is continuous, hence Borel measurable, and for every w

$$|G(w)| = \frac{\sqrt{2}\, r}{\sigma} \cdot \frac{1}{\pi^{n/2}\sigma^{n/2}}\, |w_j|\, e^{-|w|^2/\sigma} \leq \frac{\sqrt{2}\, r}{\sigma} \cdot \frac{1}{\pi^{n/2}\sigma^{n/2}}\, (1 + 2\sigma)\, e^{-|w|^2/(2\sigma)},$$

all the displayed factors $\sqrt{2}$, r, σ^{-1}, $\pi^{-n/2}$, $\sigma^{-n/2}$ and $e^{-|w|^2/\sigma}$ being strictly positive. The right-hand side is a constant multiple of the integrable function of (4), so by the domination criterion for integrability (shared context) G is Lebesgue integrable on $\mathbb{R}^n$.

Step 2: integrability of the slice. The map $w \mapsto rx + w$ is a translation, and Lebesgue measure is translation invariant (shared context), so a function is λ-integrable if and only if its composition with that translation is. By Step 0 the composition of $y \mapsto D_j^x p_s(x, y)$ with $w \mapsto rx + w$ is G, which is λ-integrable by Step 1. Hence $y \mapsto D_j^x p_s(x, y)$ is λ-integrable on $\mathbb{R}^n$, which is the first assertion.

Step 3: oddness, and the vanishing. For every w we have $(-w)_j = -w_j$ and $|-w| = |w|$, so

$$G(-w) = \sqrt{2}\, r\, \frac{-w_j}{\sigma} \cdot \frac{1}{\pi^{n/2}\sigma^{n/2}} e^{-|w|^2/\sigma} = -G(w),$$

i.e. G is odd. This is the appendix's observation that $V_{j,\sigma}$ is odd, in the form in which it is used: the linear factor changes sign and the Gaussian factor, depending on w only through $|w|$, does not. Lebesgue measure on $\mathbb{R}^n$ is invariant under the reflection $w \mapsto -w$ (shared context), so for the λ-integrable function G,

$$\int_{\mathbb{R}^n} G(w)\, dw = \int_{\mathbb{R}^n} G(-w)\, dw = \int_{\mathbb{R}^n} -G(w)\, dw = -\int_{\mathbb{R}^n} G(w)\, dw,$$

the middle equality by oddness and the last by linearity of the integral, which applies because G is integrable. A real number equal to its own negative is 0, so

$$\int_{\mathbb{R}^n} G(w)\, dw = 0. \tag{5}$$

By translation invariance of λ once more and Step 0,

$$\int_{\mathbb{R}^n} D_j^x p_s(x, y)\, dy = \int_{\mathbb{R}^n} D_j^x p_s(x,\, rx + w)\, dw = \int_{\mathbb{R}^n} G(w)\, dw = 0,$$

both integrals converging absolutely by Step 2 and Step 1. This is the second assertion. □

2.121. `PVTruncatedSliceMajorant`.

Lemma 2.121 (The ε-independent majorant for the truncated Mehler gradient slice)**.** *Fix $n \geq 1$, $j \in \{1, \dots, n\}$ and $x \in \mathbb{R}^n$. For $s > 0$ write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} \in (0,1), \qquad p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}} \exp\Big(-\frac{|rx-y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, as evaluated in `MehlerKernelXDerivative` *(the displayed right-hand side is the closed-form expression that node proves $2^{-1/2}\partial_{x_j} p_s(x,y)$ equals, and it is that explicit expression, not a derivative, that occurs here). For $\varepsilon > 0$ put $\Omega_\varepsilon = \{\, y \in \mathbb{R}^n : |x-y| > \varepsilon \,\}$ and*

$$I_{\varepsilon,s} = \int_{\Omega_\varepsilon} D_j^x p_s(x,y)\, dy.$$

Then:

(1) *there is a constant $C \geq 0$ such that*

$$\big|I_{\varepsilon,s}\big| \leq C\,|x|\, e^{-s} \qquad \text{for every } \varepsilon > 0 \text{ and every } s > 0;$$

(2) *for every $s > 0$, $\lim_{\varepsilon \downarrow 0} I_{\varepsilon,s} = 0$, the limit being taken along $\varepsilon \to 0$ with $\varepsilon > 0$.*

The force of clause (1) is the quantifier order: C is chosen after n, j and x and so may depend on all three, but before *ε and s, so it is independent of both. That independence is the whole point — the truncated integrals blow up like $\sigma^{-1/2}$ if the cancellation between the two translates is not used, and the bound above is uniform in the truncation parameter and decays in s. This is the constant the reference paper's appendix header licenses: it may depend on n, it serves only to identify two realizations of one operator, and no claim of uniformity in n is made or used here. Clause (2) is the fixed-time limit, with no claim of uniformity in s. Nothing is asserted about the normalized Gaussian Riesz kernel, about any time integral, or about any exchange of the two integrations.*

Proof. Throughout, λ denotes Lebesgue measure on $\mathbb{R}^n$, and $n \geq 1$, j and x are fixed. We take

$$C = \sqrt{2}\,\big(5 + 24 \cdot 8^{n/2}\big) \geq 0,$$

which depends on n only.

Step 0: a Gaussian toolkit. First, for every real $u \geq 0$,

$$u\, e^{-u} \leq 1. \tag{0}$$

Indeed $1 + u \leq e^u$ (shared context, the exponential inequality), so $u \leq e^u$, and multiplying by $e^{-u} > 0$ gives (0).

Second, for every $\beta > 0$ the function $w \mapsto e^{-\beta|w|^2}$ is Lebesgue integrable on $\mathbb{R}^n$ with

$$\int_{\mathbb{R}^n} e^{-\beta|w|^2}\, dw = (\pi/\beta)^{n/2}. \tag{1}$$

Indeed $e^{-\beta|w|^2} = \prod_{i=1}^n e^{-\beta w_i^2}$, Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of one-dimensional Lebesgue measure, and the integrand is nonnegative and continuous, hence product-measurable; so Tonelli (shared context) and the scalar Gaussian integral (shared context, $\int_{\mathbb{R}} e^{-\beta v^2} dv = \sqrt{\pi/\beta}$) give (1), the value being finite.

Third, for every $\beta > 0$ and every w,

$$|w|^2 e^{-\beta|w|^2} \leq \frac{2}{\beta}\, e^{-(\beta/2)|w|^2}, \tag{2}$$

whence $w \mapsto |w|^2 e^{-\beta|w|^2}$ is integrable, being continuous and dominated by an integrable function (shared context), and

$$\int_{\mathbb{R}^n} |w|^2 e^{-\beta|w|^2}\, dw \le \frac{2}{\beta}\,(2\pi/\beta)^{n/2}. \tag{3}$$

For (2): put $u = (\beta/2)|w|^2 \ge 0$; then $e^{-\beta|w|^2} = e^{-u}e^{-(\beta/2)|w|^2}$, and (0) gives $u\,e^{-u} \le 1$, i.e. $|w|^2 e^{-u} \le 2/\beta$ after multiplying by $2/\beta > 0$; multiplying by $e^{-(\beta/2)|w|^2} > 0$ gives (2). Then (3) follows by integrating (2) and using (1) at $\beta/2$.

Fourth, for all reals a, b, m with $m \le a$ and $m \le b$,

$$\left|e^{-a} - e^{-b}\right| \le |a-b|\, e^{-m}. \tag{4}$$

By symmetry assume $a \le b$. Then $e^{-b} \le e^{-a}$, so the left side is $e^{-a} - e^{-b} = e^{-a}\big(1 - e^{-(b-a)}\big)$; and $1 - e^{-t} \le t$ for $t \ge 0$ by the exponential inequality again, with $t = b - a \ge 0$; so the left side is at most $e^{-a}(b-a) \le e^{-m}|a-b|$, using $e^{-a} \le e^{-m}$.

Fifth, $|w_j| \le |w|$ for every w: writing $|w| = (\sum_i |w_i|^2)^{1/2}$, a sum of nonnegative terms is at least its jth term, so $|w_j|^2 \le |w|^2$, and the square root is monotone.

Step 1: the integrand after translation. Fix $s > 0$ and put $\beta = 1/\sigma > 0$,

$$c = \frac{\sqrt{2}\, r\, \pi^{-n/2}\sigma^{-n/2}}{\sigma} \ge 0, \qquad d = (1-r)x, \qquad W(w) = c\, w_j\, e^{-\beta|w|^2}.$$

For every w we have $(rx + w)_j - rx_j = w_j$ and $rx - (rx + w) = -w$, so $|rx - (rx + w)| = |w|$ and hence

$$D_j^x p_s(x,\, rx + w) = W(w). \tag{5}$$

Note $|d| = (1-r)|x|$ since $1 - r > 0$, and $\sigma = 1 - r^2 = (1-r)(1+r)$.

Step 2: the pointwise majorant for the translation difference. We claim that for every z,

$$\big|W(z+d) - W(z)\big| \le c\,|d|\Big(3e^{-\beta|z+d|^2} + 2e^{-\beta|z|^2} + 3\beta\,|z|^2 e^{-(\beta/4)|z|^2}\Big) =: M(z). \tag{6}$$

Write $E(w) = e^{-\beta|w|^2}$. Since $(z+d)_j = z_j + d_j$,

$$W(z+d) - W(z) = c\,d_j\,E(z+d) + c\,z_j\big(E(z+d) - E(z)\big),$$

so by the triangle inequality, $|d_j| \le |d|$ and $E > 0$,

$$\big|W(z+d) - W(z)\big| \le c\,|d|\,E(z+d) + c\,|z_j|\,\big|E(z+d) - E(z)\big|. \tag{7}$$

If $|z| \le 2|d|$, then $|z_j| \le |z| \le 2|d|$ and $|E(z+d) - E(z)| \le E(z+d) + E(z)$, so the second term of (7) is at most $2c|d|(E(z+d) + E(z))$ and the right side of (7) is at most $c|d|(3E(z+d) + 2E(z))$, which is at most $M(z)$ because the remaining term of M is nonnegative.

If $|z| > 2|d|$, put $a = \beta|z+d|^2$, $b = \beta|z|^2$ and $m = (\beta/4)|z|^2$. Then $m \le b$; and $|d| \le |z|/2$ gives $|z+d| \ge |z| - |d| \ge |z|/2$, so $|z+d|^2 \ge |z|^2/4$ and $m \le a$. Moreover $|z+d|^2 - |z|^2 = 2\langle z, d\rangle + |d|^2$, so by Cauchy–Schwarz (shared context) and $|d|^2 \le (|z|/2)|d|$,

$$|a - b| = \beta\big|2\langle z, d\rangle + |d|^2\big| \le \beta\big(2|z||d| + \tfrac{1}{2}|z||d|\big) \le 3\beta|z||d|.$$

By (4), $|E(z+d) - E(z)| \le 3\beta|z||d|\, e^{-(\beta/4)|z|^2}$, so with $|z_j| \le |z|$ the second term of (7) is at most $3\beta c|d|\,|z|^2 e^{-(\beta/4)|z|^2}$, and the right side of (7) is at most $c|d|\big(E(z+d) + 3\beta|z|^2 e^{-(\beta/4)|z|^2}\big) \le M(z)$. This proves (6) in both cases.

Step 3: integrating the majorant. By (1), translation invariance of λ (shared context) and (3) at $\beta/4$, each of the three summands of M is integrable and

$$\begin{aligned}\int_{\mathbb{R}^n} M &= c|d|\Big(3(\pi/\beta)^{n/2} + 2(\pi/\beta)^{n/2} + 3\beta\int |z|^2 e^{-(\beta/4)|z|^2} dz\Big) \\ &\le c|d|\Big(5(\pi/\beta)^{n/2} + 3\beta\cdot\frac{8}{\beta}\,(8\pi/\beta)^{n/2}\Big) = c|d|\,(\pi/\beta)^{n/2}\big(5 + 24\cdot 8^{n/2}\big),\end{aligned}$$

using $(8\pi/\beta)^{n/2} = 8^{n/2}(\pi/\beta)^{n/2}$. The function $z \mapsto W(z+d) - W(z)$ is integrable: W is integrable because, by (5) and translation invariance, it is the composition of $y \mapsto D_j^x p_s(x, y)$ with a translation, and that function is integrable by clause (1) of `MehlerKernelSliceVanishing`; and

$z \mapsto W(z+d)$ is then integrable by translation invariance again. Hence, by (6) and monotonicity of the integral,

$$\int_{\mathbb{R}^n} \big|W(z+d) - W(z)\big|\, dz \le c|d|\,(\pi/\beta)^{n/2}\big(5 + 24\cdot 8^{n/2}\big). \tag{8}$$

Steps 2 and 3 together are `GaussFirstMomentTranslationL1Bound`, which states exactly the bound (8) for every $b > 0$, every $c \ge 0$ and every d, and is what the formalization invokes here; the Gaussian toolkit of Step 0 that it needs is in turn supplied to it by `GaussEuclideanMomentBounds`.

Step 4: two translations and the oddness. Fix $\varepsilon > 0$. The set Ω_ε is open, hence measurable, $y \mapsto |x - y|$ being continuous. Translating by rx: the preimage of Ω_ε under $w \mapsto rx + w$ is $\{w : |d - w| > \varepsilon\}$, because $x - (rx + w) = d - w$; so by translation invariance of λ and (5),

$$I_{\varepsilon,s} = \int_{\{|d-w|>\varepsilon\}} W(w)\, dw.$$

Translating by d: the preimage of $\{|d - w| > \varepsilon\}$ under $z \mapsto z + d$ is $\{z : |z| > \varepsilon\}$, because $d - (z + d) = -z$ and $|-z| = |z|$; so

$$I_{\varepsilon,s} = \int_{\{|z|>\varepsilon\}} W(z+d)\, dz.$$

Next, W is odd: $(-z)_j = -z_j$ and $|-z| = |z|$ give $W(-z) = -W(z)$. The set $\{|z| > \varepsilon\}$ is invariant under $z \mapsto -z$, again because $|-z| = |z|$, and Lebesgue measure is invariant under that reflection (shared context); so, W being integrable,

$$\int_{\{|z|>\varepsilon\}} W = \int_{\{|z|>\varepsilon\}} W(-z)\, dz = -\int_{\{|z|>\varepsilon\}} W,$$

whence $\int_{\{|z|>\varepsilon\}} W = 0$. Subtracting,

$$I_{\varepsilon,s} = \int_{\{|z|>\varepsilon\}} \big(W(z+d) - W(z)\big)\, dz,$$

both terms being integrable, and therefore, by (8) and the fact that the integral of a nonnegative integrable function over a measurable set is at most its integral over $\mathbb{R}^n$,

$$\big|I_{\varepsilon,s}\big| \le \int_{\mathbb{R}^n} \big|W(z+d) - W(z)\big|\, dz \le c|d|\,(\pi/\beta)^{n/2}\big(5 + 24\cdot 8^{n/2}\big). \tag{9}$$

Step 5: the constant, and clause (1). Since $\beta = 1/\sigma$ we have $\pi/\beta = \pi\sigma$ and therefore $(\pi/\beta)^{n/2} = \pi^{n/2}\sigma^{n/2}$, so

$$c\,(\pi/\beta)^{n/2} = \frac{\sqrt{2}\,r}{\sigma}\,\big(\pi^{-n/2}\pi^{n/2}\big)\big(\sigma^{-n/2}\sigma^{n/2}\big) = \frac{\sqrt{2}\,r}{\sigma}.$$

Using $|d| = (1 - r)|x|$ and $\sigma = (1 - r)(1 + r)$ with $1 - r > 0$,

$$\frac{\sqrt{2}\,r}{\sigma}\,|d| = \sqrt{2}\,|x|\,r \cdot \frac{1}{1+r} \le \sqrt{2}\,|x|\,r,$$

because $0 \le 1/(1 + r) \le 1$ and $\sqrt{2}|x|r \ge 0$. Substituting into (9),

$$\big|I_{\varepsilon,s}\big| \le \sqrt{2}\,\big(5 + 24\cdot 8^{n/2}\big)\,|x|\,e^{-s} = C\,|x|\,e^{-s},$$

for every $\varepsilon > 0$ and every $s > 0$, which is clause (1).

Step 6: clause (2). Fix $s > 0$. By clause (1) of `MehlerKernelSliceVanishing` the function $y \mapsto D^x_j p_s(x, y)$ is integrable on $\mathbb{R}^n$ with integral 0, so splitting the integral over Ω_ε and its complement,

$$I_{\varepsilon,s} = -\int_{\Omega_\varepsilon^c} D^x_j p_s(x,y)\, dy = -\int_{\mathbb{R}^n} \mathbf{1}_{\Omega_\varepsilon^c}(y)\, D^x_j p_s(x,y)\, dy,$$

where $\Omega_\varepsilon^c = \{y : |x - y| \le \varepsilon\}$. For each $\varepsilon > 0$ the integrand is measurable, being the product of a measurable indicator with an integrable function, and it is dominated in absolute value by $|D^x_j p_s(x, \cdot)|$, which is integrable and independent of ε. Moreover, for every $y \ne x$ we have $|x - y| > 0$, and $\{\varepsilon : \varepsilon < |x - y|\}$ is a neighbourhood of 0, so for all ε in a neighbourhood

of 0 within $(0,\infty)$ we have $\varepsilon < |x-y|$, i.e. $y \notin \Omega_\varepsilon^c$ and the integrand vanishes at y; hence the integrand tends to 0 as $\varepsilon \downarrow 0$ for every such y. Since $n \geq 1$, the singleton $\{x\}$ is λ-null, so this holds λ-almost everywhere. Dominated convergence along the filter of neighbourhoods of 0 within $(0,\infty)$ (shared context; that filter is countably generated, $\mathbb{R}$ being first countable) therefore gives

$$\int_{\mathbb{R}^n} \mathbf{1}_{\Omega_\varepsilon^c}(y)\, D_j^x p_s(x,y)\, dy \longrightarrow \int_{\mathbb{R}^n} 0\, dy = 0 \qquad (\varepsilon \downarrow 0),$$

and hence $I_{\varepsilon,s} \to 0$, which is clause (2). □

2.122. `PVKernelTimeIntegral`.

Lemma 2.122 (The normalized Gaussian Riesz kernel as a time integral)**.** *Fix $n \geq 1$, $j \in \{1,\dots,n\}$ and $x, y \in \mathbb{R}^n$. For $s > 0$ write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} \in (0,1), \qquad p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}} \exp\Big(-\frac{|rx-y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, as evaluated in `MehlerKernelXDerivative` (the displayed right-hand side is the closed-form expression that node proves $2^{-1/2}\partial_{x_j} p_s(x,y)$ equals, and it is that explicit expression, not a derivative, that occurs here). Let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of `GaussRieszKernel`, so that by definition

$$K_{\gamma,j}(x,y) = \sqrt{2}\,\pi^{-(n+1)/2} \int_{(0,1)} h(\varrho)\, d\varrho, \qquad h(\varrho) = \frac{y_j - \varrho x_j}{(1-\varrho^2)^{n/2+1}\sqrt{-\log \varrho}} \exp\Big(-\frac{|\varrho x - y|^2}{1-\varrho^2}\Big),$$

the integral being a Lebesgue integral over $(0,1)$. Then:

1. *the function $s \mapsto s^{-1/2} D_j^x p_s(x,y)$ is Lebesgue integrable on $(0,\infty)$ if and only if h is Lebesgue integrable on $(0,1)$; and*
2. $$K_{\gamma,j}(x,y) = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{D_j^x p_s(x,y)}{\sqrt{s}}\, ds.$$

Both clauses are read under the convention — the one `GaussRieszKernel` already uses for its own defining integral — that the Lebesgue integral of a non-integrable integrand is 0. Clause (2) is therefore asserted for every x and y with no integrability hypothesis, and clause (1) is exactly what keeps that convention from hiding anything: it says the two integrals are simultaneously genuine or simultaneously conventional, so a consumer that needs to know which case it is in can read it off from either side. The identity is the change of variables $\varrho = e^{-s}$ and nothing more; no estimate, no uniformity in x or y, and no truncation $\{|x-y| > \varepsilon\}$ occurs here. The reference paper writes the time-integral form of the kernel directly, so clause (2) is the rigorization of a step the paper performs silently, and clause (1) is the bookkeeping that rigorization needs.

Proof. Throughout, $n \geq 1$, j, x and y are fixed, λ denotes Lebesgue measure, and we abbreviate

$$A = \sqrt{2}\,\pi^{-(n+1)/2} > 0, \qquad H(s) = \frac{D_j^x p_s(x,y)}{\sqrt{s}} \quad (s > 0),$$

so that $K_{\gamma,j}(x,y) = A \int_{(0,1)} h\, d\lambda$ by the definition recalled in the statement, and the two assertions concern H and h.

Step 1: the substitution is a change of variables. Let $\psi(s) = e^{-s}$. We check three things about ψ on $(0,\infty)$.

First, $\psi\big((0,\infty)\big) = (0,1)$. If $s > 0$ then $e^{-s} > 0$ and $e^{-s} < 1$ because $-s < 0$ and the exponential is < 1 exactly on the negatives. Conversely if $0 < \varrho < 1$ then $\log \varrho < 0$, so $s = -\log \varrho > 0$ and $\psi(s) = e^{\log \varrho} = \varrho$.

Second, ψ is differentiable at every s with $\psi'(s) = -e^{-s}$, by the chain rule applied to the exponential and to $s \mapsto -s$; in particular ψ has that derivative within $(0,\infty)$ at each of its points, and $|\psi'(s)| = e^{-s}$ since $e^{-s} > 0$.

Third, ψ is injective on $(0,\infty)$: if $e^{-a}=e^{-b}$ then $-a=-b$ by injectivity of the exponential, so $a=b$.

Since $(0,\infty)$ is measurable, the one-dimensional change-of-variables theorem for an injective differentiable map (shared context) applies to ψ on $(0,\infty)$ and gives, for every Borel $G:\mathbb{R}\to\mathbb{R}$,

$$\int_{(0,1)} G\,d\lambda = \int_{(0,\infty)} |\psi'(s)|\,G(\psi(s))\,ds = \int_{(0,\infty)} e^{-s}\,G(e^{-s})\,ds, \tag{1}$$

and, in its integrability form, G is Lebesgue integrable on $(0,1)$ if and only if $s\mapsto e^{-s}G(e^{-s})$ is Lebesgue integrable on $(0,\infty)$. Both forms hold with no integrability hypothesis, the two sides of (1) being simultaneously genuine or simultaneously 0 by the convention.

Step 2: the pointwise identity under the substitution. We claim that for every $s>0$,

$$e^{-s}\,h(e^{-s}) = \frac{1}{\sqrt{\pi}\,A}\,H(s). \tag{2}$$

Fix $s>0$ and put $r=e^{-s}>0$ and $\sigma=1-e^{-2s}$. Then $\sigma>0$, since $-2s<0$ gives $e^{-2s}<1$; and

$$1-r^2 = 1-e^{-s}e^{-s} = 1-e^{-2s} = \sigma, \qquad -\log r = -\log e^{-s} = s,$$

the first because $e^{-s}e^{-s}=e^{-s-s}=e^{-2s}$ and the second because $\log e^{t}=t$. Substituting these into the formula for h,

$$h(e^{-s}) = \frac{y_j - r x_j}{\sigma^{n/2+1}\sqrt{s}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big).$$

Next, two identities for the powers. Since $\sigma>0$,

$$\sigma^{-n/2} = \big(\sigma^{n/2}\big)^{-1}, \qquad \sigma^{n/2+1} = \sigma^{n/2}\,\sigma,$$

the first because $t\mapsto\sigma^{t}$ sends negation to inversion and the second because it sends addition to multiplication, with $\sigma^1=\sigma$. And since $\pi>0$,

$$\sqrt{\pi}\,\pi^{-(n+1)/2} = \pi^{1/2}\pi^{-(n+1)/2} = \pi^{1/2-(n+1)/2} = \pi^{-n/2},$$

so that

$$\sqrt{\pi}\,A = \sqrt{2}\,\pi^{-n/2}. \tag{3}$$

Now expand H:

$$H(s) = \frac{1}{\sqrt{s}}\cdot\sqrt{2}\,r\,\frac{y_j-rx_j}{\sigma}\cdot\pi^{-n/2}\sigma^{-n/2}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big).$$

Using $\sigma^{-n/2}=(\sigma^{n/2})^{-1}$ and $\sigma^{n/2+1}=\sigma^{n/2}\sigma$, the factor $\sigma^{-1}\sigma^{-n/2}$ equals $(\sigma^{n/2+1})^{-1}$, so

$$H(s) = \sqrt{2}\,\pi^{-n/2}\cdot r\,\frac{y_j-rx_j}{\sigma^{n/2+1}\sqrt{s}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big) = \sqrt{2}\,\pi^{-n/2}\,e^{-s}h(e^{-s}),$$

and dividing by $\sqrt{2}\,\pi^{-n/2}\neq 0$ and using (3) gives (2). All the divisors occurring here — $\sqrt{s}$, $\sqrt{2}$, $\pi^{-n/2}$, $\sigma^{n/2}$ and σ — are nonzero, being positive.

Step 3: clause (1). By the integrability form of (1) applied to $G=h$, the function h is integrable on $(0,1)$ if and only if $s\mapsto e^{-s}h(e^{-s})$ is integrable on $(0,\infty)$. By (2) the latter function equals $(\sqrt{\pi}A)^{-1}H$ on $(0,\infty)$, and $(\sqrt{\pi}A)^{-1}$ is a nonzero constant, so it is integrable on $(0,\infty)$ if and only if H is: a nonzero constant multiple of an integrable function is integrable, in either direction, and two functions agreeing at every point of $(0,\infty)$ are integrable there together. This is clause (1).

Step 4: clause (2). By (1) applied to $G=h$, then (2) at every point of $(0,\infty)$, then linearity of the integral for the constant $(\sqrt{\pi}A)^{-1}$,

$$\int_{(0,1)} h\,d\lambda = \int_{(0,\infty)} e^{-s}h(e^{-s})\,ds = \int_{(0,\infty)} \frac{1}{\sqrt{\pi}\,A}H(s)\,ds = \frac{1}{\sqrt{\pi}\,A}\int_{(0,\infty)} H(s)\,ds.$$

Multiplying by A and using $K_{\gamma,j}(x,y) = A\int_{(0,1)} h\,d\lambda$ from the definition,

$$K_{\gamma,j}(x,y) = A\cdot\frac{1}{\sqrt{\pi}\,A}\int_{(0,\infty)} H(s)\,ds = \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{D_j^x p_s(x,y)}{\sqrt{s}}\,ds,$$

since $\sqrt{\pi}\,A \neq 0$. This is clause (2). Note that no integrability was assumed at any point: the change of variables (1), the pointwise identity (2) and the extraction of the constant all hold under the convention, both sides being 0 together when neither integrand is integrable. □

2.123. `SubordinationWeightIntegral.`

Lemma 2.123 (The total mass of the subordination weight)**.** *The function*

$$s \longmapsto \frac{e^{-s}}{\sqrt{s}}$$

is Lebesgue integrable on $(0,\infty)$*, and*

$$\int_0^\infty \frac{e^{-s}}{\sqrt{s}}\,ds = \sqrt{\pi}.$$

This is the scalar normalization constant of the subordination formula $L^{-1/2}\Pi_0 = \pi^{-1/2}\int_0^\infty s^{-1/2}P_s\Pi_0\,ds$ *used throughout the tablet's kernel layer: it is the total mass of the weight* $s^{-1/2}e^{-s}$ *against which every time integral in that layer is dominated, and it is the reason the prefactor in the subordination formula is* $\pi^{-1/2}$ *and not some other constant. Equivalently it is the classical evaluation* $\Gamma(1/2) = \sqrt{\pi}$*. No Gaussian measure on* $\mathbb{R}^n$ *occurs here: the statement is a one-dimensional Lebesgue integral of an explicit scalar function, and it is recorded as its own node because several nodes of the kernel layer need it as a majorant and as a normalization.*

Proof. Write $g(s) = s^{-1/2}e^{-s}$ for $s > 0$. The function g is continuous on $(0,\infty)$, hence Borel measurable there, and it is nonnegative.

Step 1: the substitution $s = v^2$. Let $\psi : (0,\infty) \to (0,\infty)$, $\psi(v) = v^2$. It is a bijection of $(0,\infty)$ onto $(0,\infty)$ with inverse $s \mapsto \sqrt{s}$, it is continuously differentiable, and $\psi'(v) = 2v > 0$ on $(0,\infty)$. By the one-dimensional change-of-variables formula for a continuously differentiable bijection with nonvanishing derivative, applied to the nonnegative Borel function g,

$$\int_{(0,\infty)} g(s)\,ds = \int_{(0,\infty)} g(\psi(v))\,\psi'(v)\,dv, \tag{1}$$

both sides being elements of $[0,+\infty]$.

For $v > 0$ we have $\sqrt{\psi(v)} = \sqrt{v^2} = v$ because $v > 0$, so

$$g(\psi(v))\,\psi'(v) = \frac{e^{-v^2}}{v}\cdot 2v = 2\,e^{-v^2}.$$

Hence (1) reads

$$\int_{(0,\infty)} \frac{e^{-s}}{\sqrt{s}}\,ds = 2\int_{(0,\infty)} e^{-v^2}\,dv. \tag{2}$$

Step 2: evaluating the half-line Gaussian integral. The set $\{0\}$ is Lebesgue-null, so

$$\int_{\mathbb{R}} e^{-v^2}\,dv = \int_{(-\infty,0)} e^{-v^2}\,dv + \int_{(0,\infty)} e^{-v^2}\,dv,$$

the integrand being nonnegative and measurable, so that the additivity of the integral over a partition into measurable pieces applies in $[0,+\infty]$. By the invariance of Lebesgue measure under the reflection $v \mapsto -v$ (shared context), and since $e^{-(-v)^2} = e^{-v^2}$, the two half-line integrals are equal. By the scalar Gaussian integral (shared context), $\int_{\mathbb{R}} e^{-v^2}\,dv = \sqrt{\pi}$. Therefore

$$\int_{(0,\infty)} e^{-v^2}\,dv = \frac{\sqrt{\pi}}{2}.$$

Step 3: conclusion. Substituting into (2),

$$\int_{(0,\infty)} \frac{e^{-s}}{\sqrt{s}}\, ds = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}.$$

In particular this integral of the nonnegative measurable function g is finite, so g is Lebesgue integrable on $(0,\infty)$, and its integral has the stated value. □

2.124. `PVTruncatedSliceEnormBound`.

Lemma 2.124 (The per-time-slice bound for the truncated Mehler gradient kernel)**.** *Fix $n \geq 1$. Let j be a coordinate, $x \in \mathbb{R}^n$, let $\varepsilon > 0$ and $s > 0$, and write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} \in (0,1], \qquad \Omega_\varepsilon = \{y \in \mathbb{R}^n : \varepsilon < |x - y|\}.$$

Let $D : \mathbb{R}^n \to \mathbb{R}$ be the slice

$$D(y) = \frac{\sqrt{2}\, r\, (y_j - r x_j)}{\sigma}\, \pi^{-n/2} \sigma^{-n/2} \exp\Big(-\frac{|rx - y|^2}{\sigma}\Big),$$

written out as the hypothesis of the lemma rather than abbreviated, so that the statement applies to any family presented in that form. Then both of the following hold.

(1) *If $|x - rx| \leq \varepsilon/2$, then*

$$\int_{\Omega_\varepsilon}^{*} \big|D(y)\big|\, dy \leq \frac{40\sqrt{2} \cdot 4^{n/2}}{\varepsilon^2}.$$

(2) *Without any hypothesis on $|x - rx|$,*

$$\int_{\Omega_\varepsilon}^{*} \big|D(y)\big|\, dy \leq \frac{3\sqrt{2} \cdot 2^{n/2}\, e^{-s}}{\sigma}.$$

Proof. Throughout $b, b' > 0$ denote arbitrary positive rates and $|v|$ is the Euclidean norm on $\mathbb{R}^n$. Note $r > 0$ and $\sigma > 0$, because $-s < 0$ and $-2s < 0$.

Step 0: the inputs. Four facts come from `GaussEuclideanMomentBounds`: for every $\beta > 0$ the function $w \mapsto e^{-\beta|w|^2}$ is integrable with

$$\int_{\mathbb{R}^n} e^{-\beta|w|^2}\, dw = (\pi/\beta)^{n/2}; \tag{1}$$

for every $\beta > 0$ and every w,

$$|w|\, e^{-\beta|w|^2} \leq \Big(1 + \frac{2}{\beta}\Big) e^{-(\beta/2)|w|^2}; \tag{2}$$

the function $w \mapsto |w| e^{-\beta|w|^2}$ is integrable with

$$\int_{\mathbb{R}^n} |w|\, e^{-\beta|w|^2}\, dw \leq \Big(1 + \frac{2}{\beta}\Big)\Big(\frac{\pi}{\beta/2}\Big)^{n/2}; \tag{3}$$

and, for all $\sigma', \beta > 0$,

$$\pi^{-n/2}(\sigma')^{-n/2}(\pi/\beta)^{n/2} = \big(1/(\sigma'\beta)\big)^{n/2}; \tag{4}$$

together with the coordinate bound $|w_i| \leq |w|$ for every w and every coordinate i. One further fact we prove here: for every $t > 0$,

$$e^{-t} \leq \frac{1}{t}. \tag{5}$$

Indeed $1 + t \leq e^t$ gives $t \leq e^t$, so $t\, e^{-t} \leq e^t e^{-t} = 1$, and dividing by $t > 0$ gives (5).

Step 1: the translation and the pointwise majorant. Put

$$d = x - rx, \qquad b = \frac{1}{\sigma} > 0, \qquad c = \frac{\sqrt{2}\, r}{\sigma}\, \pi^{-n/2} \sigma^{-n/2} > 0.$$

The translation $T(w) = rx + w$ is measure preserving on $\mathbb{R}^n$ and is a measurable embedding, and

$$T^{-1}(\Omega_\varepsilon) = \{w : \varepsilon < |x - (rx + w)|\} = \{w : \varepsilon < |d - w|\} =: R,$$

because $x-(rx+w)=d-w$. Hence for every nonnegative extended-real G,

$$\int_R^* G(rx+w)\,dw=\int_{\Omega_\varepsilon}^* G(y)\,dy. \tag{6}$$

Moreover, at $y=rx+w$ we have $y_j-rx_j=w_j$ and $|rx-y|=|-w|=|w|$, so $D(rx+w)=c\,w_j\,e^{-b|w|^2}$, and since $|w_j|\le|w|$,

$$\big|D(rx+w)\big|\le c\,|w|\,e^{-b|w|^2}\qquad(w\in\mathbb{R}^n). \tag{7}$$

Step 2: the small-time bound, clause (1). Suppose $|d|\le\varepsilon/2$. If $w\in R$ then $\varepsilon<|d-w|\le|d|+|w|$, so $|w|>\varepsilon/2$. For such w, splitting $e^{-b|w|^2}=e^{-(b/2)|w|^2}e^{-(b/2)|w|^2}$, bounding the first factor together with $|w|$ by (2) at the rate $b/2$, and the remaining factor by $e^{-(b/2)(\varepsilon/2)^2}=e^{-b\varepsilon^2/8}$, we get

$$c\,|w|\,e^{-b|w|^2}\le C\,e^{-(b/4)|w|^2},\qquad C=c\,e^{-b\varepsilon^2/8}\Big(1+\frac{4}{b}\Big)\ge 0.$$

Hence by (7), monotonicity of the upper integral in the integrand and in the region, (6), and then (1) at the rate $b/4$,

$$\int_{\Omega_\varepsilon}^*\big|D(y)\big|\,dy=\int_R^*\big|D(rx+w)\big|\,dw\le\int_{\mathbb{R}^n}C\,e^{-(b/4)|w|^2}\,dw=C\Big(\frac{\pi}{b/4}\Big)^{n/2}.$$

By (4) with $\beta=b/4=1/(4\sigma)$ the product $\pi^{-n/2}\sigma^{-n/2}(\pi/(b/4))^{n/2}$ is $4^{n/2}$, so

$$C\Big(\frac{\pi}{b/4}\Big)^{n/2}=\frac{\sqrt{2}\,r}{\sigma}\,4^{n/2}\,e^{-\varepsilon^2/(8\sigma)}(1+4\sigma),$$

using $4/b=4\sigma$ and $b\varepsilon^2/8=\varepsilon^2/(8\sigma)$. Finally (5) at $t=\varepsilon^2/(8\sigma)>0$ gives $e^{-\varepsilon^2/(8\sigma)}\le 8\sigma/\varepsilon^2$, which cancels the one power of σ^{-1}, and $r<1$, $\sigma\le 1$ give $1+4\sigma\le 5$; therefore

$$\int_{\Omega_\varepsilon}^*\big|D(y)\big|\,dy\le\frac{40\sqrt{2}\cdot 4^{n/2}}{\varepsilon^2},$$

which is clause (1). This is the paper's "exponential decay in ε^2/σ dominates the powers of σ^{-1}", with the single power σ^{-1} named and cancelled.

Step 3: the bound valid at every time, clause (2). Without any hypothesis on $|d|$, discard the region: by (7), (6), (3) at the rate b, and (4) with $\beta=b/2=1/(2\sigma)$,

$$\int_{\Omega_\varepsilon}^*\big|D(y)\big|\,dy\le c\int_{\mathbb{R}^n}|w|e^{-b|w|^2}\,dw\le c\Big(1+\frac{2}{b}\Big)\Big(\frac{\pi}{b/2}\Big)^{n/2}=\frac{\sqrt{2}\,r}{\sigma}\,2^{n/2}(1+2\sigma),$$

and $\sigma\le 1$ gives $1+2\sigma\le 3$, so

$$\int_{\Omega_\varepsilon}^*\big|D(y)\big|\,dy\le\frac{3\sqrt{2}\cdot 2^{n/2}\,e^{-s}}{\sigma},$$

which is clause (2). □

2.125. `PVTruncatedTimeSliceFinite`.

Lemma 2.125 (The Tonelli input for the truncated principal-value kernel integral)**.** *Fix $n\ge 1$, $j\in\{1,\dots,n\}$, $x\in\mathbb{R}^n$ and $\varepsilon>0$. For $s>0$ write*

$$r=e^{-s}\in(0,1),\qquad\sigma=1-e^{-2s}\in(0,1],\qquad p_s(x,y)=\frac{1}{\pi^{n/2}\sigma^{n/2}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y)=\sqrt{2}\,r\,\frac{y_j-rx_j}{\sigma}\,p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, written in the closed form that the surrounding nodes of this layer all use: the displayed right-hand side is that explicit expression,

and it is that expression, not a derivative, that occurs in the statement below. Let $\Omega_\varepsilon = \{y \in \mathbb{R}^n : |x-y| > \varepsilon\}$ *and let* λ *be Lebesgue measure. Then*

$$\int_{(0,\infty)}^{*} \frac{1}{\sqrt{s}} \Big(\int_{\Omega_\varepsilon}^{*} \big| D_j^x p_s(x,y) \big| \, dy \Big) ds \; < \; \infty,$$

where $\int^*$ *denotes the upper (lower Lebesgue) integral of a nonnegative extended-real integrand, taken in* $[0,\infty]$.

The quantification matters. Both integrals are upper integrals of nonnegative quantities, so each is defined for every s *with no integrability hypothesis and neither can be* 0 *by default: in particular the inner quantity is* $+\infty$*, not* 0*, at any* s *for which* $|D_j^x p_s(x,\cdot)|$ *fails to be integrable on* Ω_ε*, and the finiteness asserted here therefore genuinely controls every* $s > 0$ *at once. This is exactly the hypothesis that Tonelli's theorem consumes to license Fubini on* $\Omega_\varepsilon \times (0,\infty)$ *at fixed* x *and* ε*, which is why the statement is phrased with* $\int^*$ *rather than with Bochner integrals. No uniformity in* ε *is claimed, and none is needed: the constants below blow up as* $\varepsilon \downarrow 0$.

The reference paper asserts this step in two sentences — at small s *one has* $|d| < \varepsilon/2$*, so the substitution* $w = z + d$ *leaves a Gaussian tail with* $|w| > \varepsilon/2$ *whose "exponential decay in* ε^2/σ *dominates the powers of* σ^{-1}*", while for* s *bounded away from zero one uses* $\int |D_j^x p_s(x,y)|\, dy \le C_n e^{-s}/\sqrt{1-e^{-2s}}$ *— and concludes that "these bounds prove absolute integrability of the double integral for fixed* x, ε*". This node is that assertion made quantitative: the threshold separating the two regimes is named, the "dominates" is replaced by the inequality* $e^{-t} \le 1/t$ *applied at* $t = \varepsilon^2/(8\sigma)$*, and the constants are written out. In the large-*s *regime we prove the weaker bound* $\int |D_j^x p_s(x,y)|\, dy \le C_n e^{-s}/\sigma$*, with* σ *in place of* $\sqrt{\sigma}$*; since the paper uses its own large-*s *bound only where* σ *is bounded below, the weaker form suffices there and nothing downstream sees the difference. All constants are explicitly dimension-dependent and local to this node.*

Proof. Throughout, n, j, x and $\varepsilon > 0$ are fixed, and $b, b' > 0$ denote arbitrary positive rates. We write $|v|$ for the Euclidean norm on $\mathbb{R}^n$.

Step 1: the Gaussian integral at an arbitrary rate. For every $b > 0$ the function $v \mapsto e^{-b|v|^2}$ is Lebesgue integrable on $\mathbb{R}^n$ and

$$\int_{\mathbb{R}^n} e^{-b|v|^2}\, dv = \Big(\frac{\pi}{b}\Big)^{n/2}. \tag{1}$$

Both are the shared-context Gaussian integral on a finite-dimensional inner product space, read at the rate b and with $\dim \mathbb{R}^n = n$.

Step 2: the half-rate majorants. The inequality $1 + u \le e^u$, at $u = t$ and at $u = -t$, gives two facts we use repeatedly. First, for $t > 0$,

$$e^{-t} \le \frac{1}{t}, \tag{2}$$

because $t \le e^t$ gives $te^{-t} \le e^t e^{-t} = 1$. Second, for $b > 0$ and $v \in \mathbb{R}^n$, writing $E = e^{-(b/2)|v|^2} \in (0,1]$ we have $e^{-b|v|^2} = E^2$ and, by (2) applied at $t = (b/2)|v|^2$ when $v \ne 0$ and trivially when $v = 0$, $|v|^2 E \le 2/b$; hence from $|v| \le 1 + |v|^2$,

$$|v|\, e^{-b|v|^2} = |v|\, E \cdot E \le (1 + |v|^2) E \cdot E = E^2 + (|v|^2 E) E \le \Big(1 + \frac{2}{b}\Big) e^{-(b/2)|v|^2}. \tag{3}$$

In particular $v \mapsto |v| e^{-b|v|^2}$ is dominated by an integrable function and is continuous, so it is integrable, and integrating (3) and using (1) at the rate $b/2$,

$$\int_{\mathbb{R}^n} |v|\, e^{-b|v|^2}\, dv \le \Big(1 + \frac{2}{b}\Big)\Big(\frac{2\pi}{b}\Big)^{n/2}. \tag{4}$$

Step 3: the normalization identity. For $\sigma > 0$ and $b' > 0$,

$$\pi^{-n/2} \sigma^{-n/2} \Big(\frac{\pi}{b'}\Big)^{n/2} = \Big(\frac{1}{\sigma b'}\Big)^{n/2}, \tag{5}$$

since $\pi^{-n/2}\pi^{n/2} = 1$ and $\sigma^{-n/2}(b')^{-n/2} = (\sigma b')^{-n/2}$. Taking $b' = 1/(4\sigma)$ and $b' = 1/(2\sigma)$ turns the right-hand side into $4^{n/2}$ and $2^{n/2}$ respectively; these are the only two instances used below.

Step 4: the translation and the pointwise majorant. Fix $s > 0$ and keep $r = e^{-s} \in (0,1)$ and $\sigma = 1 - e^{-2s} \in (0,1]$, both positive because $-s < 0$ and $-2s < 0$. Put

$$d = x - rx, \qquad b = \frac{1}{\sigma} > 0, \qquad c = \frac{\sqrt{2}\,r}{\sigma}\,\pi^{-n/2}\sigma^{-n/2} > 0.$$

The translation $T(w) = rx + w$ is measure preserving on $\mathbb{R}^n$ and is a measurable embedding, and

$$T^{-1}(\Omega_\varepsilon) = \{w : \varepsilon < |x - (rx + w)|\} = \{w : \varepsilon < |d - w|\} =: R,$$

because $x - (rx + w) = d - w$. Hence for every nonnegative extended-real G,

$$\int_R^* G(rx + w)\,dw = \int_{\Omega_\varepsilon}^* G(y)\,dy. \tag{6}$$

Moreover, at $y = rx + w$ we have $y_j - rx_j = w_j$ and $|rx - y| = |-w| = |w|$, so

$$D_j^x p_s(x, rx + w) = c\,w_j\,e^{-b|w|^2},$$

and since $|w_j| \le |w|$,

$$\left|D_j^x p_s(x, rx + w)\right| \le c\,|w|\,e^{-b|w|^2} \qquad (w \in \mathbb{R}^n). \tag{7}$$

Step 5: the small-time bound. Suppose in addition that $|d| \le \varepsilon/2$. If $w \in R$ then $\varepsilon < |d - w| \le |d| + |w|$, so $|w| > \varepsilon/2$. For such w, splitting $e^{-b|w|^2} = e^{-(b/2)|w|^2} e^{-(b/2)|w|^2}$, bounding the first factor by (3) at the rate $b/2$ and the second by $e^{-(b/2)(\varepsilon/2)^2} = e^{-b\varepsilon^2/8}$, we get

$$c\,|w|\,e^{-b|w|^2} \le C\,e^{-(b/4)|w|^2}, \qquad C = c\,e^{-b\varepsilon^2/8}\Big(1 + \frac{4}{b}\Big) \ge 0.$$

Hence by (7), monotonicity of the upper integral in the integrand and in the region, (6), and then (1) at the rate $b/4$,

$$\int_{\Omega_\varepsilon}^* \left|D_j^x p_s(x,y)\right| dy = \int_R^* \left|D_j^x p_s(x, rx + w)\right| dw \le \int_{\mathbb{R}^n} C\,e^{-(b/4)|w|^2}\,dw = C\Big(\frac{\pi}{b/4}\Big)^{n/2}.$$

By (5) with $b' = b/4 = 1/(4\sigma)$ the product $\pi^{-n/2}\sigma^{-n/2}(\pi/(b/4))^{n/2}$ is $4^{n/2}$, so

$$C\Big(\frac{\pi}{b/4}\Big)^{n/2} = \frac{\sqrt{2}\,r}{\sigma}\,4^{n/2}\,e^{-\varepsilon^2/(8\sigma)}(1 + 4\sigma),$$

using $4/b = 4\sigma$ and $b\varepsilon^2/8 = \varepsilon^2/(8\sigma)$. Finally (2) at $t = \varepsilon^2/(8\sigma) > 0$ gives $e^{-\varepsilon^2/(8\sigma)} \le 8\sigma/\varepsilon^2$, which cancels the one power of σ^{-1}, and $r < 1$, $\sigma \le 1$ give $1 + 4\sigma \le 5$; therefore

$$|d| \le \frac{\varepsilon}{2} \quad\Longrightarrow\quad \int_{\Omega_\varepsilon}^* \left|D_j^x p_s(x,y)\right| dy \le \frac{40\sqrt{2}\cdot 4^{n/2}}{\varepsilon^2} =: K_1. \tag{8}$$

This is the paper's "exponential decay in ε^2/σ dominates the powers of σ^{-1}", with the single power σ^{-1} named and cancelled.

Step 5′: the bound valid at every time. Without any hypothesis on $|d|$, discard the region: by (7), (6), (4) at the rate b, and (5) with $b' = b/2 = 1/(2\sigma)$,

$$\int_{\Omega_\varepsilon}^* \left|D_j^x p_s(x,y)\right| dy \le c\int_{\mathbb{R}^n} |w| e^{-b|w|^2}\,dw \le c\Big(1 + \frac{2}{b}\Big)\Big(\frac{\pi}{b/2}\Big)^{n/2} = \frac{\sqrt{2}\,r}{\sigma}\,2^{n/2}(1 + 2\sigma),$$

and $1 + 2\sigma \le 3$, so

$$\int_{\Omega_\varepsilon}^* \left|D_j^x p_s(x,y)\right| dy \le \frac{3\sqrt{2}\cdot 2^{n/2} e^{-s}}{\sigma}. \tag{9}$$

Steps 4, 5 and 5′ together are `PVTruncatedSliceEnormBound`, which states exactly the pair (8) and (9) for every j, x, $\varepsilon > 0$, $s > 0$ and every slice D presented in the displayed form, and is what the formalization invokes here; the Gaussian inputs of Steps 1 to 3 reach it from `GaussEuclideanMomentBounds`.

Step 6: the threshold. Put

$$s_0 = \min\Big(1,\ \frac{\varepsilon}{2(1+|x|)}\Big) \in (0,1], \qquad \sigma_0 = 1 - e^{-2s_0} > 0.$$

If $0 < s \le s_0$ then, since $d = (1-e^{-s})x$ with $0 \le 1-e^{-s} \le s$ (the right inequality is $1-s \le e^{-s}$, an instance of $1+u \le e^u$ at $u=-s$),

$$|d| = (1-e^{-s})|x| \le s|x| \le \frac{\varepsilon|x|}{2(1+|x|)} \le \frac{\varepsilon}{2},$$

so (8) applies. If $s_0 \le s$ then $e^{-2s} \le e^{-2s_0}$ gives $\sigma \ge \sigma_0$, so (9) gives

$$s_0 \le s \quad\Longrightarrow\quad \int_{\Omega_\varepsilon}^{*} \big|D_j^x p_s(x,y)\big|\,dy \le K_2\, e^{-s}, \qquad K_2 = \frac{3\sqrt{2}\cdot 2^{n/2}}{\sigma_0}. \tag{10}$$

Step 7: the time integral. Two weights are integrable. First, $s \mapsto s^{-1/2}$ is integrable on $(0,1]$, being the restriction of $s \mapsto s^a$ with $a = -1/2 > -1$, hence integrable on $(0, s_0]$ since $s_0 \le 1$; let $W_1 < \infty$ be its upper integral there. Second, $s \mapsto s^{-1/2}e^{-s}$ is integrable on $(0,\infty)$ by `SubordinationWeightIntegral`, hence on (s_0,∞); let $W_2 < \infty$ be its upper integral there. Now $(0,\infty) = (0,s_0] \cup (s_0,\infty)$ is a disjoint union of measurable sets, so the upper integral over $(0,\infty)$ is the sum of the two, and by (8) on the first piece and (10) on the second,

$$\int_{(0,s_0]}^{*} \frac{1}{\sqrt{s}}\Big(\int_{\Omega_\varepsilon}^{*} |D_j^x p_s|\,dy\Big)ds \le K_1 W_1 < \infty,$$

$$\int_{(s_0,\infty)}^{*} \frac{1}{\sqrt{s}}\Big(\int_{\Omega_\varepsilon}^{*} |D_j^x p_s|\,dy\Big)ds \le K_2 \int_{(s_0,\infty)}^{*} \frac{e^{-s}}{\sqrt{s}}\,ds = K_2 W_2 < \infty,$$

the constants K_1, K_2, W_1, W_2 being finite and nonnegative. Adding the two bounds gives the assertion. Note that the two regimes are genuinely both needed: (8) fails for large s, because there $|d|$ is comparable to $|x|$ and the region $\{|d-w| > \varepsilon\}$ no longer forces $|w|$ to be large, while (9) is useless as $s \downarrow 0$, where σ^{-1} blows up like $(2s)^{-1}$ and $s^{-1/2}\sigma^{-1}$ is not integrable at 0. □

2.126. `PVKernelConstantCancellation`.

Lemma 2.126 (Principal-value cancellation of the Gaussian Riesz kernel on constants)**.** *Fix $n \ge 1$, $j \in \{1,\dots,n\}$ and $x \in \mathbb{R}^n$. For $s > 0$ write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} = 1 - r^2 \in (0,1), \qquad p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}} \exp\Big(-\frac{|rx-y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, as evaluated in `MehlerKernelXDerivative` (the displayed right-hand side is the closed-form expression that node proves $2^{-1/2}\partial_{x_j} p_s(x,y)$ equals, and it is that explicit expression, not a derivative, that occurs below). Let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of `GaussRieszKernel`. Then:

(1) *for every $s > 0$ the function $y \mapsto D_j^x p_s(x,y)$ is Lebesgue integrable on $\mathbb{R}^n$ and*

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)\,dy = 0;$$

(2) *for every $\varepsilon > 0$ the function $y \mapsto K_{\gamma,j}(x,y)$ is Lebesgue integrable on the exterior region $\{y : |x-y| > \varepsilon\}$; and*

(3)

$$\lim_{\varepsilon \downarrow 0} \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,dy = 0,$$

the limit being taken along $\varepsilon \to 0$ with $\varepsilon > 0$.

Clause (3) is the reference paper's appendix display asserting that the truncated kernel integrals of the constant function 1 *tend to zero, i.e. that no diagonal multiplication term is left undetermined by the principal-value realization; clauses (1) and (2) are the two well-definedness facts the appendix establishes on the way, and both are used again downstream, clause (1) in the subtracted-input step and clause (2) to know that the truncated integrals of the appendix's final splitting are genuine Lebesgue integrals. Every constant occurring in the proof is allowed to depend on* n*, on* j *and on* x*; as the appendix header instructs, these constants serve only to identify two realizations of one operator and no claim of uniformity in* n *is made or used.*

Proof. Throughout, λ denotes Lebesgue measure on $\mathbb{R}^n$ and $n \geq 1$, j and x are fixed. Fix also, for the whole proof, the abbreviations: for $s > 0$,

$$r = e^{-s} \in (0,1), \quad \sigma = 1 - r^2 \in (0,1), \quad d = (1-r)x,$$

and, for $z \in \mathbb{R}^n$,

$$q_\sigma(z) = (\pi\sigma)^{-n/2} e^{-|z|^2/\sigma}, \qquad V_{j,\sigma}(z) = \frac{z_j}{\sigma}\, q_\sigma(z).$$

These are the appendix's letters; they are local notation for this proof and not new objects. Note $\sigma = 1 - e^{-2s}$ because $r^2 = e^{-2s}$, and that $0 < r < 1$ and $0 < \sigma < 1$ because $s > 0$.

Step 0: the kernel derivative in the appendix's letters. For $y \in \mathbb{R}^n$ put $z = y - x$. Then

$$z + d = (y - x) + (1 - r)x = y - rx,$$

so $|z + d| = |y - rx| = |rx - y|$ and therefore $q_\sigma(z + d) = (\pi\sigma)^{-n/2} e^{-|rx-y|^2/\sigma} = p_s(x, y)$, while $(z + d)_j = y_j - rx_j$. Consequently

$$D_j^x p_s(x,y) = \sqrt{2}\, r\, \frac{(z+d)_j}{\sigma}\, q_\sigma(z+d) = \sqrt{2}\, r\, V_{j,\sigma}(z+d), \qquad z = y - x. \tag{0}$$

In particular $y \mapsto D_j^x p_s(x, y)$ is continuous, hence Borel measurable.

Step 1: Gaussian scaling identities. For $c > 0$ the dilation $w \mapsto cw$ of $\mathbb{R}^n$ multiplies Lebesgue measure by c^n, $\lambda(cB) = c^n \lambda(B)$ for Borel B; hence for every Borel $h : \mathbb{R}^n \to [0, +\infty]$,

$$\int_{\mathbb{R}^n} h(z)\, dz = c^n \int_{\mathbb{R}^n} h(cw)\, dw. \tag{1}$$

By `GaussMeasure` the measure γ_n is Lebesgue measure with density $\pi^{-n/2} e^{-|w|^2}$, and by `GaussIsProb` it is a probability measure; hence

$$\int_{\mathbb{R}^n} \pi^{-n/2} e^{-|w|^2}\, dw = 1. \tag{2}$$

Put

$$a_n = \int_{\mathbb{R}^n} |w_j|\, d\gamma_n(w), \qquad b_n = \int_{\mathbb{R}^n} |w_j|\, |w|\, d\gamma_n(w).$$

Both are finite: $|w_j| \leq |w| \leq 1 + |w|^2$ and $|w_j||w| \leq |w|^2$, and $1 + |w|^2$ and $|w|^2$ are polynomials in the coordinates of w, so finiteness follows from the Gaussian exponential moments item of the shared context (with $t = 0$) together with `GaussMeasure`.

Apply (1) with $c = \sqrt{\sigma}$ and the three choices $h = q_\sigma$, $h = |z_j| q_\sigma$ and $h = |z_j|\, |z|\, q_\sigma$, using $q_\sigma(\sqrt{\sigma} w) = (\pi\sigma)^{-n/2} e^{-|w|^2}$, $|(\sqrt{\sigma} w)_j| = \sqrt{\sigma}|w_j|$ and $|\sqrt{\sigma} w| = \sqrt{\sigma}|w|$, and then (2):

$$\int_{\mathbb{R}^n} q_\sigma(z)\, dz = \sigma^{n/2} (\pi\sigma)^{-n/2} \int_{\mathbb{R}^n} e^{-|w|^2} dw = 1, \tag{3a}$$

$$\int_{\mathbb{R}^n} |z_j|\, q_\sigma(z)\, dz = \sqrt{\sigma}\, a_n, \tag{3b}$$

$$\int_{\mathbb{R}^n} |z_j|\, |z|\, q_\sigma(z)\, dz = \sigma\, b_n. \tag{3c}$$

From (3*b*),

$$\int_{\mathbb{R}^n} |V_{j,\sigma}(z)|\, dz = \frac{1}{\sigma} \int_{\mathbb{R}^n} |z_j| q_\sigma(z)\, dz = \frac{a_n}{\sqrt{\sigma}} < \infty, \tag{4}$$

so $V_{j,\sigma}$ is Lebesgue integrable. By (0) and the translation invariance of Lebesgue measure (shared context), applied first to $y = x + z$ and then to $z \mapsto z + d$,

$$\int_{\mathbb{R}^n} \left|D_j^x p_s(x,y)\right| dy = \sqrt{2}\, r \int_{\mathbb{R}^n} |V_{j,\sigma}(z+d)|\, dz = \sqrt{2}\, r\, \frac{a_n}{\sqrt{\sigma}} < \infty. \tag{5}$$

In particular $y \mapsto D_j^x p_s(x,y)$ is Lebesgue integrable on $\mathbb{R}^n$, which is the integrability assertion of clause (1).

Step 2: the gradient and translation bounds for $V_{j,\sigma}$. The function $z \mapsto -|z|^2/\sigma$ is a polynomial in the coordinates with gradient $-2z/\sigma$, so by the chain rule q_σ is smooth with $\nabla q_\sigma(z) = -\frac{2z}{\sigma} q_\sigma(z)$, and therefore $V_{j,\sigma}$ is smooth with

$$\nabla V_{j,\sigma}(z) = \frac{q_\sigma(z)}{\sigma}\, e_j - \frac{2z_j}{\sigma^2}\, q_\sigma(z)\, z,$$

e_j being the jth standard basis vector. By the triangle inequality and $|e_j| = 1$,

$$|\nabla V_{j,\sigma}(z)| \le \frac{q_\sigma(z)}{\sigma} + \frac{2|z_j|\,|z|}{\sigma^2} q_\sigma(z),$$

so integrating and using $(3a)$ and $(3c)$,

$$\int_{\mathbb{R}^n} |\nabla V_{j,\sigma}(z)|\, dz \le \frac{1}{\sigma} + \frac{2}{\sigma^2}\, \sigma\, b_n = \frac{1 + 2b_n}{\sigma} =: \frac{C_n^{(1)}}{\sigma}. \tag{6}$$

Next, for any $e \in \mathbb{R}^n$ and any z, the function $t \mapsto V_{j,\sigma}(z + te)$ is smooth on $[0,1]$, so by the fundamental theorem of calculus

$$V_{j,\sigma}(z+e) - V_{j,\sigma}(z) = \int_0^1 \nabla V_{j,\sigma}(z+te) \cdot e\, dt,$$

whence $|V_{j,\sigma}(z+e) - V_{j,\sigma}(z)| \le |e| \int_0^1 |\nabla V_{j,\sigma}(z+te)|\, dt$ by Cauchy–Schwarz (shared context) applied pointwise in t. The function $(z,t) \mapsto |\nabla V_{j,\sigma}(z+te)|$ is continuous, hence product-measurable and nonnegative, so Tonelli (shared context) and the translation invariance of λ for each fixed t give

$$\int_{\mathbb{R}^n} \left|V_{j,\sigma}(z+e) - V_{j,\sigma}(z)\right| dz \le |e| \int_0^1 \int_{\mathbb{R}^n} |\nabla V_{j,\sigma}(z+te)|\, dz\, dt \le \frac{C_n^{(1)}\, |e|}{\sigma}. \tag{7}$$

Step 3: oddness, and clause (1). Let $R_j : \mathbb{R}^n \to \mathbb{R}^n$ be the reflection in the jth coordinate, $R_j z = z - 2z_j e_j$; it is linear, satisfies $R_j \circ R_j = \mathrm{id}$ and preserves the Euclidean norm. Lebesgue measure on $\mathbb{R}^n$ is the n-fold product of one-dimensional Lebesgue measure, and R_j acts as the reflection $v \mapsto -v$ in the jth factor and as the identity in the others; one-dimensional Lebesgue measure is invariant under that reflection (shared context), and a product of measure-preserving maps of the factors preserves the product measure, so λ is invariant under R_j. Since $|R_j z| = |z|$ we have $q_\sigma \circ R_j = q_\sigma$, while $(R_j z)_j = -z_j$; hence

$$V_{j,\sigma} \circ R_j = -V_{j,\sigma}.$$

Let $A \subseteq \mathbb{R}^n$ be Borel with $R_j(A) = A$; both $A = \mathbb{R}^n$ and $A = \{|z| > \varepsilon\}$ for $\varepsilon > 0$ have this property, the latter because $|R_j z| = |z|$. Since $V_{j,\sigma}$ is integrable by (4), invariance of λ under R_j gives

$$\int_A V_{j,\sigma}(z)\, dz = \int_A V_{j,\sigma}(R_j z)\, dz = -\int_A V_{j,\sigma}(z)\, dz,$$

so

$$\int_{\mathbb{R}^n} V_{j,\sigma}\, dz = 0 \qquad \text{and} \qquad \int_{|z|>\varepsilon} V_{j,\sigma}\, dz = 0 \quad (\varepsilon > 0). \tag{8}$$

Now by (0), the translation invariance of λ (first $y = x + z$, then $z \mapsto z + d$) and the first identity of (8),

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)\, dy = \sqrt{2}\, r \int_{\mathbb{R}^n} V_{j,\sigma}(z+d)\, dz = \sqrt{2}\, r \int_{\mathbb{R}^n} V_{j,\sigma}(w)\, dw = 0,$$

all the integrals being absolutely convergent by (4) and (5). Together with (5) this is clause (1).

Step 4: the ε-independent majorant. For $\varepsilon > 0$ and $s > 0$ set $\Omega_\varepsilon = \{y \in \mathbb{R}^n : |x-y| > \varepsilon\}$ and

$$I_{\varepsilon,s} = \int_{\Omega_\varepsilon} D_j^x p_s(x,y)\, dy,$$

which is a well-defined real number by (5). Translating $y = x + z$, so that $|x-y| = |z|$, and using (0) and then the second identity of (8),

$$I_{\varepsilon,s} = \sqrt{2}\, r \int_{|z|>\varepsilon} V_{j,\sigma}(z+d)\, dz = \sqrt{2}\, r \int_{|z|>\varepsilon} \big[V_{j,\sigma}(z+d) - V_{j,\sigma}(z)\big]\, dz,$$

both integrands being integrable by (4) and (7). Hence by (7) with $e = d$,

$$|I_{\varepsilon,s}| \le \sqrt{2}\, r\, \frac{C_n^{(1)}|d|}{\sigma}.$$

Since $|d| = (1-r)|x|$ and $\sigma = 1 - r^2 = (1-r)(1+r)$ with $1 - r > 0$,

$$\frac{r|d|}{\sigma} = \frac{r(1-r)|x|}{(1-r)(1+r)} = |x|\, \frac{e^{-s}}{1+e^{-s}} \le |x|\, e^{-s},$$

which is the appendix's displayed majorant. Therefore, with $C_n^{(2)} = \sqrt{2}\, C_n^{(1)}$,

$$|I_{\varepsilon,s}| \le C_n^{(2)}\, |x|\, e^{-s} \qquad \text{for every } \varepsilon > 0 \text{ and every } s > 0. \tag{9}$$

Moreover, for each fixed $s > 0$,

$$\lim_{\varepsilon \downarrow 0} I_{\varepsilon,s} = 0. \tag{10}$$

Indeed, by clause (1) already proved, $|I_{\varepsilon,s}| = \big|\int_{\mathbb{R}^n} D_j^x p_s(x,y)\, dy - \int_{|x-y|\le\varepsilon} D_j^x p_s(x,y)\, dy\big| = \big|\int_{|x-y|\le\varepsilon} D_j^x p_s(x,y)\, dy\big| \le \int_{\mathbb{R}^n} \mathbf{1}_{\{|x-y|\le\varepsilon\}} |D_j^x p_s(x,y)|\, dy$. Given any sequence $\varepsilon_k \downarrow 0$ of positive reals, the integrands $\mathbf{1}_{\{|x-y|\le\varepsilon_k\}}|D_j^x p_s(x,y)|$ converge pointwise to $\mathbf{1}_{\{y=x\}}|D_j^x p_s(x,y)|$, which vanishes λ-almost everywhere since $\{x\}$ is Lebesgue-null, and they are dominated by the integrable function $|D_j^x p_s(x,\cdot)|$ of (5); dominated convergence (shared context) gives $I_{\varepsilon_k,s} \to 0$. As the sequence was arbitrary and $\mathbb{R}$ is a metric space, so that convergence along $\varepsilon \downarrow 0$ is equivalent to convergence along every such sequence, (10) follows.

Step 5: $K_{\gamma,j}$ as a time integral. We claim that for every $y \in \mathbb{R}^n$,

$$K_{\gamma,j}(x,y) = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{D_j^x p_s(x,y)}{\sqrt{s}}\, ds, \tag{11}$$

with the convention — the one `GaussRieszKernel` uses for its own defining integral — that an integral of a non-integrable integrand is 0; moreover the integrand on the right is Lebesgue integrable on $(0,\infty)$ if and only if the integrand in the definition of $K_{\gamma,j}(x,y)$ is Lebesgue integrable on $(0,1)$.

To see this, expand the right-hand integrand. With $r = e^{-s}$ and $\sigma = 1 - r^2$,

$$\frac{1}{\sqrt{\pi}}\, \frac{D_j^x p_s(x,y)}{\sqrt{s}} = \frac{\sqrt{2}}{\pi^{(n+1)/2}}\, \frac{r\,(y_j - r x_j)}{\sqrt{s}\,(1-r^2)^{n/2+1}}\, \exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big),$$

where we used $p_s(x,y) = \pi^{-n/2}(1-r^2)^{-n/2} e^{-|rx-y|^2/(1-r^2)}$ and $\pi^{-1/2}\pi^{-n/2} = \pi^{-(n+1)/2}$. Let

$$h(\varrho) = \frac{y_j - \varrho x_j}{(1-\varrho^2)^{n/2+1}\sqrt{-\log\varrho}}\, \exp\Big(-\frac{|\varrho x - y|^2}{1-\varrho^2}\Big), \qquad \varrho \in (0,1),$$

so that by `GaussRieszKernel`, $K_{\gamma,j}(x,y) = \sqrt{2}\,\pi^{-(n+1)/2} \int_{(0,1)} h(\varrho)\, d\varrho$. The map $\psi(s) = e^{-s}$ is a continuously differentiable bijection of $(0,\infty)$ onto $(0,1)$ with $\psi'(s) = -e^{-s} \ne 0$ and inverse $\varrho \mapsto -\log\varrho$; in particular $-\log\psi(s) = s$, so $\sqrt{-\log\psi(s)} = \sqrt{s}$, and $|\psi'(s)| = e^{-s}$. By the one-dimensional change-of-variables formula for such a bijection, which inserts the factor $|\psi'|$, applied first to $|h|$ (a nonnegative Borel function, both sides then being elements of $[0,+\infty]$) and then, when those are finite, to h itself,

$$\int_{(0,1)} h(\varrho)\, d\varrho = \int_{(0,\infty)} h(e^{-s})\, e^{-s}\, ds,$$

with the left integrand Lebesgue integrable exactly when the right one is. A direct substitution gives

$$h(e^{-s})\,e^{-s} = \frac{r\,(y_j - rx_j)}{(1-r^2)^{n/2+1}\sqrt{s}}\exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big),$$

which is exactly $\pi^{(n+1)/2}2^{-1/2}$ times the displayed right-hand integrand above. Multiplying through by $\sqrt{2}\,\pi^{-(n+1)/2}$ yields (11) together with the stated equivalence of integrability; and when neither integrand is integrable both sides of (11) are 0 by the convention.

Step 6: Fubini on $\Omega_\varepsilon \times (0,\infty)$, and clause (2). Fix $\varepsilon > 0$ and put $F(y,s) = s^{-1/2}D_j^x p_s(x,y)$ for $(y,s) \in \Omega_\varepsilon \times (0,\infty)$. It is continuous in (y,s), hence product-measurable. We bound $\int_{\Omega_\varepsilon} |F(y,s)|\,dy$ in two regimes. Put

$$s_0 = \min\Big(1,\ \frac{\varepsilon}{2(1+|x|)}\Big) > 0.$$

Small s. Let $0 < s \le s_0$. Since $1 - e^{-s} \le s$ we get $|d| = (1-r)|x| \le s|x| \le \varepsilon|x|/(2(1+|x|)) \le \varepsilon/2$. Translating $y = x + z$ and then $w = z + d$, the region $\{|z| > \varepsilon\}$ becomes $\{|w-d| > \varepsilon\}$, and $|w-d| > \varepsilon$ with $|d| \le \varepsilon/2$ forces $|w| \ge |w-d| - |d| > \varepsilon/2$. Hence, using (0) and $|w_j| \le |w|$,

$$\int_{\Omega_\varepsilon} |D_j^x p_s(x,y)|\,dy \le \sqrt{2}\,r \int_{|w|>\varepsilon/2} \frac{|w|}{\sigma} q_\sigma(w)\,dw.$$

By (1) with $c = \sqrt{\sigma}$ this equals $\sqrt{2}\,\frac{r}{\sqrt{\sigma}}\int_{|u|>T} |u|\,\pi^{-n/2}e^{-|u|^2}\,du$ with $T = \varepsilon/(2\sqrt{\sigma})$. For $|u| > T$ we have $e^{-|u|^2} = e^{-|u|^2/2}e^{-|u|^2/2} \le e^{-T^2/2}e^{-|u|^2/2}$, so

$$\int_{|u|>T} |u|\,\pi^{-n/2}e^{-|u|^2}\,du \le \kappa_n\,e^{-T^2/2}, \qquad \kappa_n := \pi^{-n/2}\int_{\mathbb{R}^n} |u|\,e^{-|u|^2/2}\,du,$$

and $\kappa_n < \infty$: by (1) with $c = \sqrt{2}$, $\int |u|e^{-|u|^2/2}du = 2^{n/2}\sqrt{2}\int |v|e^{-|v|^2}dv$, finite by the Gaussian exponential moments item of the shared context (with $t = 0$ and the polynomial $1 + |v|^2 \ge |v|$). Finally $e^{-t} \le 1/t$ for $t > 0$ with $t = \varepsilon^2/(8\sigma)$ gives $e^{-T^2/2} = e^{-\varepsilon^2/(8\sigma)} \le 8\sigma/\varepsilon^2$, so, using $r \le 1$ and $\sigma \le 1$,

$$\int_{\Omega_\varepsilon} |D_j^x p_s(x,y)|\,dy \le \frac{8\sqrt{2}\,\kappa_n\,r\sqrt{\sigma}}{\varepsilon^2} \le \frac{8\sqrt{2}\,\kappa_n}{\varepsilon^2} \qquad (0 < s \le s_0). \tag{12}$$

Large s. Let $s \ge s_0$. Then $\sigma = 1 - e^{-2s} \ge 1 - e^{-2s_0} > 0$, so with $c_0 = (1-e^{-2s_0})^{-1/2} < \infty$, (5) gives

$$\int_{\Omega_\varepsilon} |D_j^x p_s(x,y)|\,dy \le \int_{\mathbb{R}^n} |D_j^x p_s(x,y)|\,dy \le \sqrt{2}\,a_n\,c_0\,e^{-s} \qquad (s \ge s_0). \tag{13}$$

Combining, and using $\int_0^{s_0} s^{-1/2}\,ds = 2\sqrt{s_0} < \infty$ for the first regime and `SubordinationWeightIntegral` for the second,

$$\int_{(0,\infty)}\int_{\Omega_\varepsilon} |F(y,s)|\,dy\,ds \le \frac{8\sqrt{2}\,\kappa_n}{\varepsilon^2}\cdot 2\sqrt{s_0} + \sqrt{2}\,a_n\,c_0\int_0^\infty \frac{e^{-s}}{\sqrt{s}}\,ds < \infty,$$

the iterated integral of the nonnegative product-measurable $|F|$ being the product integral by Tonelli (shared context). Hence F is integrable on $\Omega_\varepsilon \times (0,\infty)$, and Fubini (shared context) applies: for λ-almost every $y \in \Omega_\varepsilon$ the slice $s \mapsto F(y,s)$ is integrable on $(0,\infty)$, the function $y \mapsto \int_0^\infty F(y,s)\,ds$ — read with the same convention, so that it is defined at *every* y, with value 0 at the exceptional ones — is Lebesgue integrable on Ω_ε, and

$$\int_{\Omega_\varepsilon}\Big(\int_0^\infty F(y,s)\,ds\Big)dy = \int_0^\infty\Big(\int_{\Omega_\varepsilon} F(y,s)\,dy\Big)ds = \int_0^\infty \frac{I_{\varepsilon,s}}{\sqrt{s}}\,ds. \tag{14}$$

By (11), $\int_0^\infty F(y,s)\,ds = \sqrt{\pi}\,K_{\gamma,j}(x,y)$ for every y — for those y at which the s-integrand is integrable this is (11) itself, and at the remaining y both sides are 0 by the convention. So

$y \mapsto \sqrt{\pi} K_{\gamma,j}(x,y)$ is Lebesgue integrable on Ω_ε, which is clause (2), and dividing (14) by $\sqrt{\pi}$,

$$\int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,dy = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{I_{\varepsilon,s}}{\sqrt{s}}\,ds. \tag{15}$$

Step 7: dominated convergence in ε, and clause (3). By Fubini in Step 6 the function $s \mapsto I_{\varepsilon,s}$ is measurable on $(0,\infty)$ for each $\varepsilon > 0$, and by (9)

$$\left|\frac{I_{\varepsilon,s}}{\sqrt{s}}\right| \le C_n^{(2)}|x|\,\frac{e^{-s}}{\sqrt{s}} \qquad \text{for all } \varepsilon > 0,\ s > 0,$$

the majorant being Lebesgue integrable on $(0,\infty)$ by `SubordinationWeightIntegral`. Let $\varepsilon_k \downarrow 0$ be any sequence of positive reals. By (10), $I_{\varepsilon_k,s} \to 0$ for every fixed $s > 0$, so dominated convergence (shared context, which is stated for sequences) gives $\int_0^\infty s^{-1/2} I_{\varepsilon_k,s}\,ds \to 0$. Since the sequence was arbitrary and $\mathbb{R}$ is a metric space, the limit along $\varepsilon \downarrow 0$ exists and equals 0. By (15) this is clause (3). □

2.127. `SemigroupGradientCommute`.

Lemma 2.127 (Differentiating Mehler's formula: $\partial_j P_s f = e^{-s} P_s(\partial_j f)$ on the smooth core)**.** *Fix $s > 0$ and write $r = e^{-s} \in (0,1)$ and $\tau = \sqrt{1-e^{-2s}} \in (0,1)$. For a function $g : \mathbb{R}^n \to \mathbb{R}$ put*

$$(A_s g)(x) = \int_{\mathbb{R}^n} g\big(rx + \tau z\big)\,d\gamma_n(z),$$

the Mehler average of `MehlerFormula`, with γ_n as in `GaussMeasure`. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support and let $j \in \{1,\dots,n\}$. Then:

1. *for every x, the function $z \mapsto f(rx+\tau z)$ is γ_n-integrable;*
2. *for every x, the function $z \mapsto (\partial_j f)(rx+\tau z)$ is γ_n-integrable;*
3. *the function $A_s(\partial_j f)$ is continuous on $\mathbb{R}^n$; and*
4. *for every x the function $A_s f$ is differentiable at x, and*

$$\partial_j (A_s f)(x) = e^{-s}\,\big(A_s(\partial_j f)\big)(x).$$

Clause (4) is the reference paper's one-clause remark that, for $f \in C_c^\infty$, differentiation of Mehler's formula gives $DP_s f = e^{-s} P_s Df$. The statement is in the Euclidean normalization ∂_j rather than in the energy normalization $D_j = 2^{-1/2}\partial_j$, and the two are the same identity: multiplying both sides of (4) by the constant $2^{-1/2}$ turns ∂_j into D_j on the left and $\partial_j f$ into $D_j f$ on the right, with no change of constant. The connection to the semigroup itself is `MehlerFormula`: for every $v \in L^2(\gamma_n)$ the class $P_s v$ agrees γ_n-almost everywhere with A_s applied to a representative of v, so $A_s f$ is a representative of $P_s f$ for the class of f and $A_s(\partial_j f)$ one of $P_s(\partial_j f)$.

Clauses (1)–(3) are stated rather than left implicit because every later use needs them: (1) and (2) make the two averages genuine integrals rather than the zero value the Bochner convention assigns to a non-integrable integrand, and (3), combined with (4) applied at each coordinate, is what makes $A_s f$ continuously differentiable and therefore available for ordinary integration by parts.

Proof. Fix $s > 0$, so $0 < r = e^{-s} < 1$, hence $0 < 1 - e^{-2s} < 1$ and $0 < \tau < 1$; in particular $r > 0$ and $\tau > 0$.

Step 0: f and its first derivatives are bounded. The function f is continuous with compact support, so it is bounded: writing $S = \operatorname{supp} f$, which is compact, $|f|$ attains a maximum on S by the extreme value theorem (shared context) and vanishes off S; put $M_0 = \sup_{\mathbb{R}^n} |f| < \infty$. Since f is smooth, its total derivative $Df : \mathbb{R}^n \to (\mathbb{R}^n)^*$ is continuous, and it vanishes off S: if $x \notin S$ then f vanishes identically on an open neighbourhood of x, so all its derivatives vanish at x. Hence $\operatorname{supp}(Df) \subseteq S$ is compact and, by the same extreme value argument applied to the continuous function $y \mapsto \|Df(y)\|$, $M = \sup_{\mathbb{R}^n} \|Df\| < \infty$. In particular $M_1 := \sup_{\mathbb{R}^n} |\partial_j f| \le M < \infty$, and Df, being continuous with compact support, is uniformly continuous on $\mathbb{R}^n$ (shared context: a continuous function with compact support is uniformly continuous, by Heine–Cantor on a compact neighbourhood of its support together with vanishing outside the support).

Step 1: clauses (1) and (2). Fix x. The map $z \mapsto rx + \tau z$ is continuous, so $z \mapsto f(rx + \tau z)$ is continuous, hence measurable, and it is bounded by M_0. By `GaussIsProb` the measure γ_n is a probability measure, hence finite, and a bounded measurable function on a finite measure space is integrable. This is clause (1). The same argument with $\partial_j f$, which is continuous and bounded by M_1 by Step 0, gives clause (2).

Step 2: clause (3). Let $x_k \to x$ in $\mathbb{R}^n$. For every fixed z we have $rx_k + \tau z \to rx + \tau z$, so $(\partial_j f)(rx_k + \tau z) \to (\partial_j f)(rx + \tau z)$ by continuity of $\partial_j f$. All these functions of z are bounded in absolute value by the constant M_1, which is γ_n-integrable because γ_n is finite (`GaussIsProb`). Dominated convergence (shared context) gives $(A_s(\partial_j f))(x_k) \to (A_s(\partial_j f))(x)$. As $\mathbb{R}^n$ is a metric space, sequential continuity is continuity, which is clause (3).

Step 3: differentiation under the integral sign. Fix x and define the linear map

$$\Lambda_x : \mathbb{R}^n \to \mathbb{R}, \qquad \Lambda_x(h) = r \int_{\mathbb{R}^n} Df(rx + \tau z)[h]\, d\gamma_n(z).$$

This is well defined and bounded: for each h the integrand is continuous in z and bounded by $M\,|h|$, hence γ_n-integrable as in Step 1, and linearity in h follows from linearity of $Df(y)$ at each y together with linearity of the integral; moreover $|\Lambda_x(h)| \le rM|h|$, so Λ_x is a bounded linear functional.

We claim $A_s f$ is differentiable at x with derivative Λ_x. Let $\varepsilon > 0$. By the uniform continuity of Df (Step 0) choose $\delta > 0$ with $\|Df(a) - Df(b)\| \le \varepsilon$ whenever $|a - b| \le \delta$. Let $h \in \mathbb{R}^n$ with $0 < |h| \le \delta/r$ (recall $r > 0$). For each fixed z, apply the fundamental theorem of calculus to the smooth function $t \mapsto f(rx + \tau z + t\,rh)$ on $[0, 1]$:

$$f\big(r(x + h) + \tau z\big) - f(rx + \tau z) = \int_0^1 Df(rx + \tau z + t\,rh)[rh]\, dt,$$

using $r(x + h) + \tau z = (rx + \tau z) + rh$. Subtracting $Df(rx + \tau z)[rh] = \int_0^1 Df(rx + \tau z)[rh]\, dt$,

$$\Big|f\big(r(x+h)+\tau z\big) - f(rx+\tau z) - Df(rx+\tau z)[rh]\Big| \le \int_0^1 \big\|Df(rx+\tau z+t\,rh) - Df(rx+\tau z)\big\|\,|rh|\,dt \le \varepsilon\, r|h|,$$

because $|t\,rh| \le r|h| \le \delta$ for $t \in [0, 1]$. This bound is uniform in z. Integrating in z against the probability measure γ_n — each of the three terms being γ_n-integrable, the first two by Step 1 and the third by the first paragraph of this step — and using that $|\int \cdot| \le \int |\cdot|$ together with $\gamma_n(\mathbb{R}^n) = 1$,

$$\big|(A_s f)(x + h) - (A_s f)(x) - \Lambda_x(h)\big| \le \varepsilon\, r\,|h|.$$

Since $\varepsilon > 0$ was arbitrary, the difference quotient error is $o(|h|)$, i.e. $A_s f$ is differentiable at x with derivative Λ_x.

Step 4: clause (4). Evaluating the derivative at the jth standard basis vector e_j and using $Df(y)[e_j] = (\partial_j f)(y)$,

$$\partial_j (A_s f)(x) = \Lambda_x(e_j) = r \int_{\mathbb{R}^n} (\partial_j f)(rx + \tau z)\, d\gamma_n(z) = e^{-s}\,\big(A_s(\partial_j f)\big)(x),$$

which is clause (4); the factor $r = e^{-s}$ is exactly the inner derivative of the affine map $x \mapsto rx + \tau z$, which is where the paper's e^{-s} comes from. □

2.128. `MehlerGradientSubordination`.

Lemma 2.128 (Absolute convergence and continuity of the subordinated Mehler gradient average)**.** *For $s > 0$ write $r_s = e^{-s}$ and $\tau_s = \sqrt{1 - e^{-2s}}$, and for $g : \mathbb{R}^n \to \mathbb{R}$ write $(A_s g)(x) = \int g(r_s x + \tau_s z)\, d\gamma_n(z)$ for the Mehler average of `MehlerFormula`. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $j \in \{1, \dots, n\}$, and put $D_j = 2^{-1/2}\partial_j$, the jth component of the energy-normalized gradient $D = 2^{-1/2}\nabla$, so that $(A_s(D_j f))(x) = 2^{-1/2}\big(A_s(\partial_j f)\big)(x)$. Then:*

(1) *for every $x \in \mathbb{R}^n$ the function*

$$s \;\longmapsto\; s^{-1/2}\, e^{-s}\,\big(A_s(D_j f)\big)(x)$$

is Lebesgue integrable on $(0, \infty)$; and

(2) *the function*

$$x \longmapsto \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2}\, e^{-s}\, \big(A_s(D_j f)\big)(x)\, ds$$

is continuous on $\mathbb{R}^n$.

These two assertions are the reference paper's remark that the subordination integral for $R_\gamma f$ *in the gradient form converges absolutely at every* x*, because* $|P_s Df(x)| \le \|Df\|_\infty$*, and provides a continuous representative. Both are consequences of that one sup-norm bound, which is uniform in* x *and in* s*; no identification of the resulting function with* $R_\gamma f$ *is asserted here. No square-integrability hypothesis on* f *is needed or imposed, since neither clause mentions* $L^2(\gamma_n)$*: the Mehler average is taken of the honest function* $\partial_j f$*, not of an* $L^2(\gamma_n)$ *class. The factor* $(\sqrt{\pi})^{-1}$ *is carried in clause (2) for agreement with the normalization of the subordination formula; it is a nonzero constant and does not affect continuity.*

Proof. Write $S = \operatorname{supp} f$, a compact set.

Step 0: the sup-norm bound on the gradient. For each index i the function $\partial_i f$ is continuous, f being smooth, and it vanishes off S: a point $x \notin S$ has an open neighbourhood on which f vanishes identically, hence on which all derivatives of f vanish. If $S = \emptyset$ then $\partial_i f \equiv 0$; otherwise $|\partial_i f|$, continuous on the nonempty compact set S, attains a maximum there by the extreme value theorem (shared context, "Compact sets: extreme values and finite Lebesgue measure", clause (1)). In either case

$$M_1 = \sup_{y \in \mathbb{R}^n} \big|\partial_j f(y)\big| < \infty.$$

By `GaussIsProb` the measure γ_n is a probability measure. Fix $s > 0$ and x. By clause (2) of `SemigroupGradientCommute`, applied at this s, to this f — smooth with compact support — and at the index j, the function $z \mapsto \partial_j f(r_s x + \tau_s z)$ is γ_n-integrable, so $(A_s(\partial_j f))(x)$ is a well-defined real number; and since the integrand is bounded in absolute value by the constant M_1 while $\gamma_n(\mathbb{R}^n) = 1$, monotonicity of the integral gives

$$\big|\big(A_s(\partial_j f)\big)(x)\big| \le M_1, \qquad \text{hence} \qquad \big|\big(A_s(D_j f)\big)(x)\big| \le 2^{-1/2} M_1, \tag{0}$$

for every $s > 0$ and every x. Both bounds are uniform in s and in x; this single estimate is all that is used below.

Step 1: joint continuity of the average. The map $(s, x) \mapsto \big(A_s(\partial_j f)\big)(x)$ is continuous on $(0, \infty) \times \mathbb{R}^n$. Indeed, let $(s_k, x_k) \to (s, x)$ in $(0, \infty) \times \mathbb{R}^n$. The scalar functions $t \mapsto r_t = e^{-t}$ and $t \mapsto \tau_t = \sqrt{1 - e^{-2t}}$ are continuous on $(0, \infty)$ — the exponential is continuous, and the square root is continuous on $[0, \infty)$ while $1 - e^{-2t} > 0$ for $t > 0$ — so $r_{s_k} \to r_s$ and $\tau_{s_k} \to \tau_s$. Hence for each fixed z we have $r_{s_k} x_k + \tau_{s_k} z \to r_s x + \tau_s z$, and therefore $\partial_j f(r_{s_k} x_k + \tau_{s_k} z) \to \partial_j f(r_s x + \tau_s z)$ by continuity of $\partial_j f$. Every one of these integrands is bounded in absolute value by the constant M_1, which is γ_n-integrable because γ_n is a probability measure (`GaussIsProb`), and each is γ_n-integrable by clause (2) of `SemigroupGradientCommute` as in Step 0. Dominated convergence (shared context) therefore gives $\big(A_{s_k}(\partial_j f)\big)(x_k) \to \big(A_s(\partial_j f)\big)(x)$. Sequential continuity is continuity on the metric space $(0, \infty) \times \mathbb{R}^n$, so the claim follows.

In particular, for each fixed x the map $s \mapsto (A_s(D_j f))(x)$ is continuous on $(0, \infty)$, and for each fixed $s > 0$ the map $x \mapsto (A_s(D_j f))(x)$ is continuous on $\mathbb{R}^n$ — the latter being also exactly clause (3) of `SemigroupGradientCommute` up to the constant factor $2^{-1/2}$.

Step 2: the scalar majorant. By clause (1) of `SubordinationWeightIntegral` the function $s \mapsto s^{-1/2} e^{-s}$ is Lebesgue integrable on $(0, \infty)$; hence so is its constant multiple

$$b(s) = 2^{-1/2} M_1\, s^{-1/2} e^{-s}. \tag{1}$$

By (0), for every $x \in \mathbb{R}^n$ and every $s > 0$,

$$\big|s^{-1/2} e^{-s} \big(A_s(D_j f)\big)(x)\big| = s^{-1/2} e^{-s}\, \big|\big(A_s(D_j f)\big)(x)\big| \le b(s), \tag{2}$$

using that $s^{-1/2} > 0$ and $e^{-s} > 0$ for $s > 0$. The majorant b does not depend on x.

Step 3: clause (1). Fix x. By Step 1 the function $s \mapsto (A_s(D_j f))(x)$ is continuous, hence Lebesgue measurable, on $(0, \infty)$; the factors $s \mapsto s^{-1/2}$ and $s \mapsto e^{-s}$ are continuous there as

well, so the function $s \mapsto s^{-1/2}e^{-s}(A_s(D_jf))(x)$ is measurable on $(0,\infty)$. By (2) its absolute value is dominated by the integrable function b of (1). A measurable function dominated in absolute value by an integrable function is integrable (shared context), so the function is Lebesgue integrable on $(0,\infty)$. This is clause (1), and it makes the integral appearing in clause (2) a well-defined real number for every x.

Step 4: clause (2). Let $x_k \to x$ in $\mathbb{R}^n$. For each fixed $s > 0$, Step 1 gives $\big(A_s(D_jf)\big)(x_k) \to \big(A_s(D_jf)\big)(x)$, so the integrands

$$s \longmapsto s^{-1/2}e^{-s}\big(A_s(D_jf)\big)(x_k)$$

converge pointwise on $(0,\infty)$ to $s \mapsto s^{-1/2}e^{-s}(A_s(D_jf))(x)$. Each of them is measurable by Step 3 and, by (2), is dominated in absolute value by the single x-independent integrable function b. Dominated convergence (shared context) therefore gives

$$\int_0^\infty s^{-1/2}e^{-s}\big(A_s(D_jf)\big)(x_k)\,ds \;\longrightarrow\; \int_0^\infty s^{-1/2}e^{-s}\big(A_s(D_jf)\big)(x)\,ds.$$

Multiplying by the constant $(\sqrt{\pi})^{-1}$ preserves the convergence, so the function of clause (2) is sequentially continuous at x. Since x was arbitrary and $\mathbb{R}^n$ is a metric space, sequential continuity is continuity, which is clause (2). □

2.129. `MehlerFixedTimeTesting`.

Lemma 2.129 (The fixed-time testing identity for the centered Ornstein–Uhlenbeck evolution)**.** *For $s > 0$ write $r_s = e^{-s}$ and $\tau_s = \sqrt{1-e^{-2s}}$, and for $g : \mathbb{R}^n \to \mathbb{R}$ write $(A_sg)(x) = \int g(r_sx + \tau_sz)\,d\gamma_n(z)$ for the Mehler average of `MehlerFormula`. Let λ denote Lebesgue measure on $\mathbb{R}^n$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support and square γ_n-integrable, and write $u \in L^2(\gamma_n)$ for the class it determines. Let $\Pi_0 = L^0$ be the mean-zero projection of `FracPow` and let P_s be the Ornstein–Uhlenbeck semigroup of `OUSemigroup`. Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $i \in \{1,\dots,n\}$ and let $s > 0$. Then:*

(1) *the function $x \mapsto (P_s\Pi_0u)(x)\,\partial_i\varphi(x)$ is λ-integrable;*
(2) *the function $x \mapsto \big(A_s(\partial_if)\big)(x)\,\varphi(x)$ is λ-integrable; and*
(3)

$$\int_{\mathbb{R}^n} (P_s\Pi_0u)(x)\,\partial_i\varphi(x)\,d\lambda(x) = -e^{-s}\int_{\mathbb{R}^n} \big(A_s(\partial_if)\big)(x)\,\varphi(x)\,d\lambda(x).$$

Here $(P_s\Pi_0u)(x)$ means the value at x of the function to which the $L^2(\gamma_n)$ class $P_s\Pi_0u$ is coerced. Such a coercion is determined only up to a γ_n-null set, but γ_n and λ have the same null sets on $\mathbb{R}^n$ — this is recorded as a separate fact below and used in the proof — so neither the integrability in clause (1) nor the value of the integral in clause (3) depends on which coercion is taken.

This is the reference paper's fixed-time half of the derivation of the semigroup formula for R_γ: the paper differentiates Mehler's formula to get $DP_sf = e^{-s}P_sDf$ at each fixed s, and only afterwards passes to the time integral. The identity above is that fixed-time statement in tested form, with the derivative moved onto the test function φ by integration by parts and with the mean-zero projection already accounted for; the factor e^{-s} is exactly the one the paper's fixed-time identity carries. No time integral, no subordination weight and no Riesz transform occurs here, and nothing is asserted about the s-dependence of either side.

Proof. Throughout, λ is Lebesgue measure and all integrals written $\int_{\mathbb{R}^n}\cdots d\lambda$ are against λ; the Mehler averages are against γ_n in the inner variable. Since $s > 0$ we have $e^{-2s} < 1$, so $1 - e^{-2s} > 0$ and therefore

$$\tau_s = \sqrt{1-e^{-2s}} > 0, \qquad \text{in particular } \tau_s \neq 0. \tag{0}$$

Step 1: γ_n and λ have the same null sets. By clause (4) of `GaussLebesgueBallComparison`, for every $A \subseteq \mathbb{R}^n$ one has $\gamma_n(A) = 0$ if and only if $\lambda(A) = 0$. Consequently a property holding γ_n-almost everywhere holds λ-almost everywhere and conversely: the exceptional set is the same

set in both cases, and it is null for one measure exactly when it is null for the other. We use this in both directions below without further comment.

Step 2: regularity of the two Mehler averages. Apply `SemigroupGradientCommute` at this s, to this f — smooth with compact support — and at the index i. Its clause (3) says that $A_s(\partial_i f)$ is continuous on $\mathbb{R}^n$. Its clause (4) says that $A_s f$ is differentiable at every point x and that

$$\partial_i(A_s f)(x) = e^{-s}\big(A_s(\partial_i f)\big)(x) \qquad \text{for every } x. \tag{1}$$

Being differentiable at every point, $A_s f$ is in particular continuous. By (1) the function $x \mapsto \partial_i(A_s f)(x)$ is the constant e^{-s} times the continuous function $A_s(\partial_i f)$, hence is itself continuous.

Also, φ being smooth, $\partial_i\varphi$ is continuous; and $\partial_i\varphi$ vanishes off the compact set $\operatorname{supp}\varphi$, because a point outside that set has an open neighbourhood on which φ vanishes identically and hence on which all derivatives of φ vanish. So $\partial_i\varphi$ is continuous with compact support.

Step 3: the three integrability statements. A continuous function with compact support on $\mathbb{R}^n$ is λ-integrable (shared context). The products

$$(A_s f)\cdot\partial_i\varphi, \qquad c\cdot\partial_i\varphi \quad (c\in\mathbb{R} \text{ constant}), \qquad \big(A_s(\partial_i f)\big)\cdot\varphi$$

are each continuous, by Step 2, and each has compact support — the first two because $\partial_i\varphi$ does, the third because φ does — so all three are λ-integrable. The third is clause (2). Being integrable, the difference $\big((A_s f)-c\big)\cdot\partial_i\varphi$ is integrable as well, for any constant c.

Step 4: the Mehler average of the class is the Mehler average of f. Let $N = \{\, y\in\mathbb{R}^n : u(y)\neq f(y)\,\}$, where u is read through its coercion. By construction of the class of f we have $u = f$ γ_n-almost everywhere, i.e. $\gamma_n(N)=0$, so $\lambda(N)=0$ by Step 1. Fix $x\in\mathbb{R}^n$ and put $\Psi(z) = r_s x + \tau_s z$. Then

$$\Psi^{-1}(N) = \{\, z : r_s x+\tau_s z\in N\,\} = \big(z\mapsto\tau_s z\big)^{-1}\big(\{\, y : r_s x + y\in N\,\}\big),$$

the two descriptions being the same set by definition of preimage. Lebesgue measure is invariant under translation (shared context), so $\lambda(\{y : r_s x+y\in N\}) = \lambda(N) = 0$. By (0) we have $\tau_s\neq 0$, so clause (2) of `LebesgueDilation`, applied with the scalar $c=\tau_s$ and the λ-null set $\{y : r_s x+y\in N\}$, gives $\lambda(\Psi^{-1}(N))=0$, hence $\gamma_n(\Psi^{-1}(N))=0$ by Step 1. Outside $\Psi^{-1}(N)$ the two integrands $z\mapsto u(\Psi(z))$ and $z\mapsto f(\Psi(z))$ agree, so they agree γ_n-almost everywhere and their γ_n-integrals coincide:

$$\int u(r_s x+\tau_s z)\,d\gamma_n(z) = (A_s f)(x) \qquad \text{for every } x. \tag{2}$$

By clause (2) of `MehlerFormula`, applied at this s to $v=u$, we have $(P_s u)(x) = \int u(r_s x+\tau_s z)\,d\gamma_n(z)$ for γ_n-almost every x. Combining with (2),

$$(P_s u)(x) = (A_s f)(x) \qquad \text{for } \gamma_n\text{-a.e. } x, \text{ hence for } \lambda\text{-a.e. } x. \tag{3}$$

Step 5: the projection costs only a constant. Let $c = u_0$ be the Hermite coefficient of u at the multi-index $\alpha = 0$, relative to the basis of `HermiteBasis`; recall from that node that the coefficient map $v\mapsto (v_\alpha)_\alpha$ is an isometric isomorphism onto $\ell^2(\mathbb{N}_0^n)$, in particular injective. By clause (1) of `FracPowLinear` at exponent 0, every element of $L^2(\gamma_n)$ lies in $\mathrm{Dom}(L^0)$, so $\Pi_0 u$ is defined; and by `FracPow` at exponent 0 its coefficients are

$$(\Pi_0 u)_\alpha = \begin{cases} 0, & \alpha = 0,\\ u_\alpha, & \alpha\neq 0,\end{cases}$$

the nonzero case using $|\alpha|^0 = 1$. Write h_0 for the basis vector at $\alpha=0$; by clause (2) of `HermiteBasisVectorFacts` its coefficients are $(h_0)_\alpha = 1$ at $\alpha=0$ and 0 at $\alpha\neq 0$, after identifying h_0 with the corresponding Hermite element by clause (1) of that node. Now apply `OUSemigroupSpectral`, valid for every $s\geq 0$, to $\Pi_0 u$ and to u: the coefficients of $P_s\Pi_0 u$ are $e^{-s|\alpha|}(\Pi_0 u)_\alpha$, and those of $P_s u - c\,h_0$ are $e^{-s|\alpha|}u_\alpha - c\,(h_0)_\alpha$. At $\alpha=0$ we have $|0|=0$, so the first is $e^0\cdot 0 = 0$ and the second is $e^0 u_0 - c = 0$; at $\alpha\neq 0$ the first is $e^{-s|\alpha|}u_\alpha$ and the second is $e^{-s|\alpha|}u_\alpha - 0$. The two coefficient families agree, so by injectivity of the coefficient map

$$P_s\Pi_0 u = P_s u - c\,h_0. \tag{4}$$

By HermiteL2ZeroAEOne the coercion of h_0 agrees γ_n-almost everywhere with the constant function 1 — again using clause (1) of HermiteBasisVectorFacts to identify h_0 with that Hermite element — so taking coercions in (4) and using (3),

$$(P_s\Pi_0 u)(x) = (A_s f)(x) - c \qquad \text{for } \lambda\text{-a.e. } x. \tag{5}$$

By Step 3 the right-hand side of (5), multiplied by $\partial_i\varphi$, is λ-integrable; since two functions agreeing λ-almost everywhere are integrable together, clause (1) follows from (5).

Step 6: the constant term integrates to zero. We claim $\int_{\mathbb{R}^n} \partial_i\varphi \, d\lambda = 0$. Apply CompIntegrationByParts at the index $j := i$ with the following data. Take $q = \mathrm{id}_{\mathbb{R}}$ and $q' \equiv 1$: then q is differentiable at every t with derivative $q'(t) = 1$, and q', being constant, is continuous. Take $w \equiv 1$, the constant function on $\mathbb{R}^n$: a constant function is differentiable at every point with vanishing derivative, so $\partial_i w(x) = 0$ for every x and $x \mapsto \partial_i w(x)$ is the zero function, which is continuous. Take the test function to be φ, which is smooth with compact support. Both hypotheses on w hold, and the conclusion of that node reads

$$\int_{\mathbb{R}^n} q(w(x))\, \partial_i\varphi(x)\, d\lambda(x) = -\int_{\mathbb{R}^n} \big(q'(w(x))\, \partial_i w(x)\big)\varphi(x)\, d\lambda(x).$$

Here $q(w(x)) = 1$ and $q'(w(x))\, \partial_i w(x) = 1 \cdot 0 = 0$ for every x, so the left side is $\int \partial_i\varphi \, d\lambda$ and the right side is $-\int 0 \, d\lambda = 0$. Hence

$$\int_{\mathbb{R}^n} c\, \partial_i\varphi \, d\lambda = c \int_{\mathbb{R}^n} \partial_i\varphi \, d\lambda = 0. \tag{6}$$

Step 7: integration by parts at fixed time. Apply CompIntegrationByParts again at the index $j := i$, now with $w = A_s f$, with $q = \mathrm{id}_{\mathbb{R}}$ and $q' \equiv 1$ as in Step 6, and with the test function φ. The two hypotheses that node places on w are differentiability at every point and continuity of the single partial derivative $\partial_i w$; both were verified in Step 2. Since $q(w(x)) = (A_s f)(x)$ and $q'(w(x))\, \partial_i w(x) = \partial_i (A_s f)(x)$ for every x, its conclusion reads

$$\int_{\mathbb{R}^n} (A_s f)\, \partial_i\varphi \, d\lambda = -\int_{\mathbb{R}^n} \partial_i (A_s f)\, \varphi \, d\lambda = -e^{-s} \int_{\mathbb{R}^n} \big(A_s(\partial_i f)\big)\varphi \, d\lambda,$$

the last equality by (1) together with linearity of the integral, legitimate because $\big(A_s(\partial_i f)\big)\varphi$ is integrable by Step 3. There is no boundary term because φ is compactly supported.

Step 8: conclusion. By (5) the integrands $(P_s\Pi_0 u)\, \partial_i\varphi$ and $\big((A_s f) - c\big)\partial_i\varphi$ agree λ-almost everywhere, so their integrals coincide. Both $(A_s f)\, \partial_i\varphi$ and $c\, \partial_i\varphi$ are integrable by Step 3, so the integral of their difference splits, and

$$\int_{\mathbb{R}^n} (P_s\Pi_0 u)\, \partial_i\varphi \, d\lambda = \int_{\mathbb{R}^n} (A_s f)\, \partial_i\varphi \, d\lambda - \int_{\mathbb{R}^n} c\, \partial_i\varphi \, d\lambda = -e^{-s} \int_{\mathbb{R}^n} \big(A_s(\partial_i f)\big)\varphi \, d\lambda,$$

using Step 7 for the first term and (6) for the second. This is clause (3). □

2.130. SubordinationPseudoInverse.

Lemma 2.130 (The subordination representation of the pseudoinverse $L^{-1/2}\Pi_0$)**.** *Let $f \in L^2(\gamma_n)$, let $\Pi_0 = L^0$ be the mean-zero projection and $L^{-1/2}\Pi_0$ the pseudoinverse, both as in FracPow, and let P_s be the Ornstein–Uhlenbeck semigroup of OUSemigroup. Then:*

(1) *the $L^2(\gamma_n)$-valued function $s \mapsto s^{-1/2}\, P_s\Pi_0 f$ is Bochner integrable on $(0, \infty)$; and*
(2)

$$L^{-1/2}\Pi_0 f = \frac{1}{\sqrt{\pi}} \int_0^\infty P_s\Pi_0 f \, \frac{ds}{\sqrt{s}},$$

the right-hand side being the Bochner integral of clause (1), taken in $L^2(\gamma_n)$.

This is the reference paper's displayed convergent Bochner integral for the pseudoinverse. The paper writes the integrand as $P_s f - \mathbb{E}(f)\mathbf{1}$ and remarks that this equals $P_s\Pi_0 f$; the form used here is the latter, which is the one the spectral description of Π_0 in FracPow makes available directly, and the identification of the two forms is neither asserted nor used below. The projection is what makes clause (1) true at all: without removing the constant mode the integrand has L^2

norm bounded below by $|\mathbb{E}(f)|$ for every s, and $\int_1^\infty s^{-1/2}ds = \infty$, so the integral would diverge at infinity whenever $\mathbb{E}(f) \neq 0$ — exactly the point the paper makes.

The statement is for every $f \in L^2(\gamma_n)$ with no domain hypothesis, which is legitimate because $L^{-1/2}\Pi_0$ is everywhere defined: the exponent $-1/2$ is ≤ 0, so every element of $L^2(\gamma_n)$ lies in $\mathrm{Dom}(L^{-1/2})$.

Proof. Write $c_\alpha = \langle f, h_\alpha\rangle$ for the Hermite coefficients of f relative to the basis of `HermiteBasis`, and recall from that node that the coefficient map $v \mapsto (v_\alpha)_\alpha$ is an isometric isomorphism of $L^2(\gamma_n)$ onto $\ell^2(\mathbb{N}_0^n)$; in particular it is injective, and Parseval's identity $\|v\|_2^2 = \sum_\alpha |v_\alpha|^2$ holds.

Step 0: the coefficients of $\Pi_0 f$ and of $P_s\Pi_0 f$. By `FracPow` at $s = 0$, the element $\Pi_0 f$ is the element of $L^2(\gamma_n)$ whose Hermite coefficients are 0 at $\alpha = 0$ and $|\alpha|^0 c_\alpha = c_\alpha$ at $\alpha \neq 0$. By `OUSemigroupSpectral`, valid for every $s \geq 0$, the coefficients of $P_s\Pi_0 f$ are therefore

$$(P_s\Pi_0 f)_\alpha = \begin{cases} 0, & \alpha = 0, \\ e^{-s|\alpha|}c_\alpha, & \alpha \neq 0. \end{cases} \tag{1}$$

Step 1: the exponential norm bound. Let $s \geq 0$. By `OneLeMultiIndexSum` we have $|\alpha| \geq 1$ for every $\alpha \neq 0$, so $-s|\alpha| \leq -s$ and hence $e^{-s|\alpha|} \leq e^{-s}$, the exponential being increasing. By (1) and Parseval,

$$\|P_s\Pi_0 f\|_2^2 = \sum_{\alpha\neq 0} e^{-2s|\alpha|}c_\alpha^2 \leq e^{-2s}\sum_{\alpha\neq 0} c_\alpha^2 \leq e^{-2s}\sum_\alpha c_\alpha^2 = e^{-2s}\|f\|_2^2,$$

all terms being nonnegative. Taking square roots,

$$\|P_s\Pi_0 f\|_2 \leq e^{-s}\,\|f\|_2 \qquad (s \geq 0). \tag{2}$$

Step 2: the scalar integral. For every $\lambda > 0$,

$$\int_0^\infty s^{-1/2}e^{-\lambda s}\,ds = \sqrt{\pi/\lambda}. \tag{3}$$

Indeed, substitute $s = u^2$, a C^1 increasing bijection of $(0,\infty)$ onto itself with $ds = 2u\,du$ and $s^{-1/2} = u^{-1}$ for $u > 0$; the integrand is nonnegative and measurable, so the change of variables is valid in $[0,+\infty]$ and gives

$$\int_0^\infty s^{-1/2}e^{-\lambda s}\,ds = \int_0^\infty u^{-1}e^{-\lambda u^2}\,2u\,du = 2\int_0^\infty e^{-\lambda u^2}\,du = \int_{\mathbb{R}} e^{-\lambda u^2}\,du = \sqrt{\pi/\lambda},$$

the third equality because $u \mapsto e^{-\lambda u^2}$ is even, and the fourth by the scalar Gaussian integral of the shared context. In particular (3) with $\lambda = 1$ gives $\int_0^\infty s^{-1/2}e^{-s}ds = \sqrt{\pi} < \infty$.

Step 3: continuity in s, hence strong measurability. We check that $s \mapsto P_s\Pi_0 f$ is continuous from $[0,\infty)$ to $L^2(\gamma_n)$. Let $s \geq 0$ and $\varepsilon > 0$. Since $\sum_\alpha c_\alpha^2 = \|f\|_2^2 < \infty$, choose a finite set $S \subseteq \mathbb{N}_0^n$ with $\sum_{\alpha\notin S} c_\alpha^2 < \varepsilon^2/8$. By (1) and Parseval, for $t \geq 0$,

$$\|P_s\Pi_0 f - P_t\Pi_0 f\|_2^2 = \sum_{\alpha\neq 0}\bigl(e^{-s|\alpha|} - e^{-t|\alpha|}\bigr)^2 c_\alpha^2.$$

Split the sum at S. For $\alpha \notin S$ use $|e^{-s|\alpha|} - e^{-t|\alpha|}| \leq 2$, since both exponentials lie in $[0,1]$; that part is at most $4\sum_{\alpha\notin S} c_\alpha^2 < \varepsilon^2/2$. The remaining sum is finite, and each of its finitely many terms is a continuous function of t vanishing at $t = s$, so there is $\delta > 0$ with that part $< \varepsilon^2/2$ whenever $|t - s| < \delta$. Hence $\|P_s\Pi_0 f - P_t\Pi_0 f\|_2 < \varepsilon$ for $|t - s| < \delta$, which is the claimed continuity. Consequently $s \mapsto s^{-1/2}P_s\Pi_0 f$ is continuous on $(0,\infty)$, $s \mapsto s^{-1/2}$ being continuous and nonvanishing there; a continuous function into the separable Banach space $L^2(\gamma_n)$ — separable because `HermiteBasis` exhibits a countable orthonormal basis — is strongly measurable.

Step 4: clause (1). By Step 3 the integrand is strongly measurable on $(0,\infty)$, and by (2) its norm satisfies $\|s^{-1/2}P_s\Pi_0 f\|_2 \leq s^{-1/2}e^{-s}\|f\|_2$, whose integral over $(0,\infty)$ is $\sqrt{\pi}\,\|f\|_2 < \infty$ by Step 2. Bochner integrability of a function into a Banach space *means* strong measurability together with an integrable norm — this is the definition of the notion, not a criterion imported

from elsewhere — so the function is Bochner integrable on $(0,\infty)$. This is clause (1), and it makes the integral in clause (2) a well-defined element of $L^2(\gamma_n)$.

Step 5: the coefficients of the integral. Fix $\alpha \in \mathbb{N}_0^n$ and let $\Lambda_\alpha(v) = \langle v, h_\alpha\rangle$. By Cauchy–Schwarz (shared context) Λ_α is a continuous linear functional on $L^2(\gamma_n)$: it is linear in v, and $|\Lambda_\alpha(v)| \le \|v\|_2\|h_\alpha\|_2 = \|v\|_2$ by orthonormality of the basis (`HermiteBasis`). Writing $I = \int_0^\infty s^{-1/2} P_s\Pi_0 f\, ds$, apply mathlib's `ContinuousLinearMap.integral_comp_commSL`, which commutes a continuous linear map with a Bochner integral, with $X = (0,\infty)$ carrying Lebesgue measure, with $E = L^2(\gamma_n)$ — a real Banach space, being a Hilbert space — with the integrand $F(s) = s^{-1/2}P_s\Pi_0 f$, Bochner integrable on $(0,\infty)$ by Step 4, and with the functional Λ_α. Hence

$$\Lambda_\alpha(I) = \int_0^\infty s^{-1/2}\,(P_s\Pi_0 f)_\alpha\, ds.$$

If $\alpha = 0$ the integrand vanishes identically by (1), so $\Lambda_0(I) = 0$. If $\alpha \neq 0$ then $|\alpha| \ge 1 > 0$ by `OneLeMultiIndexSum`, and (1) together with (3) at $\lambda = |\alpha|$ gives

$$\Lambda_\alpha(I) = c_\alpha \int_0^\infty s^{-1/2} e^{-s|\alpha|}\, ds = c_\alpha\sqrt{\pi/|\alpha|} = \sqrt{\pi}\,|\alpha|^{-1/2}c_\alpha,$$

the scalar integral being absolutely convergent by Step 2, which is what lets it be pulled out of the pairing. Multiplying by $(\sqrt{\pi})^{-1}$, the element $(\sqrt{\pi})^{-1}I$ has Hermite coefficients 0 at $\alpha = 0$ and $|\alpha|^{-1/2}c_\alpha$ at $\alpha \neq 0$.

Step 6: conclusion. Since $-1/2 \le 0$, clause (1) of `FracPowLinear` at $s = -1/2$ gives $f \in \mathrm{Dom}(L^{-1/2})$, so by `FracPow` the element $L^{-1/2}\Pi_0 f$ is the element of $L^2(\gamma_n)$ whose Hermite coefficients are 0 at $\alpha = 0$ and $|\alpha|^{-1/2}c_\alpha$ at $\alpha \neq 0$. By Step 5 the element $(\sqrt{\pi})^{-1}I$ has the same coefficients, and the coefficient map is injective by `HermiteBasis`. Therefore $L^{-1/2}\Pi_0 f = (\sqrt{\pi})^{-1}I$, which is clause (2). □

2.131. `RieszPseudoInverseTestedIdentity`.

Lemma 2.131 (The tested identity for the pseudoinverse of the smooth core)**.** *Fix $n \ge 1$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support and with $f \in L^2(\gamma_n)$, and let $u \in L^2(\gamma_n)$ be the class of f. Let λ denote Lebesgue measure on $\mathbb{R}^n$, write P_s for the Ornstein–Uhlenbeck semigroup of `OUSemigroup`, Π_0 for the projection off the zeroth Hermite mode, and $w = L^{-1/2}u$ for the pseudoinverse of `FracPow` at exponent $-1/2$. Suppose given*

(a) *a family $A : (0,\infty)\times\{1,\dots,n\}\times\mathbb{R}^n \to \mathbb{R}$ with*

$$A_s(\partial_i f)(x) := A(s,i,x) = \int_{\mathbb{R}^n} \partial_i f\big(e^{-s}x + \sqrt{1-e^{-2s}}\,z\big)\, d\gamma_n(z),$$

the Mehler average of $\partial_i f$ at time s;

(b) *a family $\Phi : \{1,\dots,n\}\times\mathbb{R}^n \to \mathbb{R}$ with*

$$\Phi_i(x) = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2}\Big(e^{-s}\big(2^{-1/2}A_s(\partial_i f)(x)\big)\Big)\, ds,$$

its subordination integral;

(c) *a constant N with $|A_s(\partial_i f)(x)| \le N$ for all s, i and x;*

(d) *joint continuity of $(s,x) \mapsto A_s(\partial_i f)(x)$ on $(0,\infty)\times\mathbb{R}^n$, for each i;*

(e) *integrability of $s \mapsto s^{-1/2}e^{-s}$ on $(0,\infty)$; and*

(f) *that every γ_n-almost-everywhere statement holds λ-almost everywhere.*

Then for every index i and every smooth compactly supported $\psi : \mathbb{R}^n \to \mathbb{R}$,

$$\int_{\mathbb{R}^n} w(x)\,\partial_i\psi(x)\, d\lambda(x) = -\int_{\mathbb{R}^n}\big(\sqrt{2}\,\Phi_i(x)\big)\,\psi(x)\, d\lambda(x).$$

Proof. Fix an index i and a smooth compactly supported ψ, and set

$$\Lambda(v) = \int_{\mathbb{R}^n} v(x)\,\partial_i\psi(x)\, d\lambda(x), \qquad v \in L^2(\gamma_n).$$

Step 1: Λ is a bounded linear functional on $L^2(\gamma_n)$. Let $K = \operatorname{supp}\psi$, a compact set. The function $\partial_i\psi$ is continuous, ψ being smooth, and vanishes off K, because a point outside K has

an open neighbourhood on which ψ vanishes identically and hence on which all its derivatives vanish. So $|\partial_i\psi|$ attains a maximum on K by the extreme value theorem when $K \neq \emptyset$, and $\partial_i\psi \equiv 0$ when $K = \emptyset$; in either case $\|\partial_i\psi\|_\infty < \infty$. By clause (3) of `GaussL2LocallyIntegrable` applied to K there is $C \geq 0$, depending on K but not on the function, such that every $g \in L^2(\gamma_n)$ is Lebesgue integrable on K with $\int_K |g|\,d\lambda \leq C\,\|g\|_{L^2(\gamma_n)}$. Hence $v\,\partial_i\psi$ is λ-integrable and

$$|\Lambda(v)| \leq \|\partial_i\psi\|_\infty \int_K |v|\,d\lambda \leq \|\partial_i\psi\|_\infty\, C\,\|v\|_{L^2(\gamma_n)}.$$

Λ is linear by linearity of the integral, and well defined on $L^2(\gamma_n)$ classes: two representatives agree γ_n-a.e., hence λ-a.e. by (f), so the Lebesgue integral is unchanged.

Step 2: pushing Λ inside the subordination integral. By clauses (1) and (2) of `SubordinationPseudoInverse` the $L^2(\gamma_n)$-valued map $s \mapsto s^{-1/2}P_s\Pi_0 u$ is Bochner integrable on $(0,\infty)$ and $w = (\sqrt{\pi})^{-1}\int_0^\infty s^{-1/2}P_s\Pi_0 u\,ds$. Apply mathlib's `ContinuousLinearMap.integral_comp_commSL`, which commutes a continuous linear map with a Bochner integral, with $X = (0,\infty)$ carrying Lebesgue measure, with $E = L^2(\gamma_n)$ — a real Banach space, being a Hilbert space — with $F(s) = s^{-1/2}P_s\Pi_0 u$, Bochner integrable by the clause just quoted, and with the continuous linear functional Λ of Step 1. It gives $\int_0^\infty \Lambda(s^{-1/2}P_s\Pi_0 u)\,ds = \Lambda\big(\int_0^\infty s^{-1/2}P_s\Pi_0 u\,ds\big)$. Multiplying by $(\sqrt{\pi})^{-1}$ and using linearity of Λ both to pull that constant out and to pull the scalar $s^{-1/2}$ out of $\Lambda(s^{-1/2}P_s\Pi_0 u)$,

$$\Lambda(w) = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2}\,\Lambda(P_s\Pi_0 u)\,ds. \tag{1}$$

Step 3: the projection may be dropped inside Λ. Let $c = \langle u, h_0\rangle$ be the zeroth Hermite coefficient of u with respect to `HermiteBasis`. By `FracPow` at exponent 0, the element $\Pi_0 u$ has coefficients 0 at $\alpha = 0$ and u_α at $\alpha \neq 0$; the element $u - c\,h_0$ has, by linearity of the coefficient map and orthonormality of the basis, coefficients $u_0 - c = 0$ at $\alpha = 0$ and u_α at $\alpha \neq 0$. The coefficient map is injective, so $\Pi_0 u = u - c\,h_0$. Applying `OUSemigroupSpectral` to both sides, the coefficients of $P_s\Pi_0 u$ are 0 at $\alpha = 0$ and $e^{-s|\alpha|}u_\alpha$ at $\alpha \neq 0$, and those of $P_s u - c\,h_0$ are $u_0 - c = 0$ at $\alpha = 0$ and $e^{-s|\alpha|}u_\alpha$ at $\alpha \neq 0$; by injectivity again $P_s\Pi_0 u = P_s u - c\,h_0$.

By `HermiteL2ZeroAEOne` the class h_0 agrees γ_n-a.e. with the constant function 1, hence λ-a.e. by (f), so $\Lambda(c\,h_0) = c\int_{\mathbb{R}^n}\partial_i\psi\,d\lambda$. That integral vanishes, by `CompIntegrationByParts` at the index $j := i$ with the following data. Take $q = \mathrm{id}_{\mathbb{R}}$ and $q' \equiv 1$: then q is differentiable at every t with derivative 1, and q', being constant, is continuous. Take the inner function to be $\equiv 1$, the constant function on $\mathbb{R}^n$: it is differentiable everywhere with vanishing derivative, so its ith partial is the zero function, which is continuous. Take the test function to be ψ. The conclusion of that node reads $\int q(1)\,\partial_i\psi\,d\lambda = -\int\big(q'(1)\cdot 0\big)\psi\,d\lambda$, i.e. $\int\partial_i\psi\,d\lambda = 0$. Hence $\Lambda(c\,h_0) = 0$ and

$$\Lambda(P_s\Pi_0 u) = \Lambda(P_s u) \qquad \text{for every } s > 0. \tag{2}$$

Step 4: the identity at a fixed time. Fix $s > 0$. Since u is the class of f and, by `MehlerFixedTimeTesting`, the coercion of $P_s u$ agrees γ_n-a.e. with $A_s f$ — hence λ-a.e. by (f) — we get $\Lambda(P_s u) = \int (A_s f)\,\partial_i\psi\,d\lambda$. Apply `SemigroupGradientCommute` at this s, to this f, and at the index i: its clause (4) says $A_s f$ is differentiable at every x with $\partial_i(A_s f)(x) = e^{-s}A_s(\partial_i f)(x)$, and its clause (3) says $A_s(\partial_i f)$ is continuous. Hence $x \mapsto \partial_i(A_s f)(x)$, the constant e^{-s} times a continuous function, is continuous.

These are exactly the two hypotheses `CompIntegrationByParts` places on its inner function at the index $j := i$: differentiability everywhere, and continuity of the single partial ∂_i. Apply it with inner function $A_s f$, with $q = \mathrm{id}_{\mathbb{R}}$, $q' \equiv 1$, and test function ψ. Its conclusion reads

$$\int (A_s f)\,\partial_i\psi\,d\lambda = -\int \partial_i(A_s f)\,\psi\,d\lambda = -e^{-s}\int A_s(\partial_i f)\,\psi\,d\lambda,$$

the last equality by clause (4). Convergence of both integrals is part of that node's conclusion, and there is no boundary term because ψ is compactly supported. Combining with (2), $\Lambda(P_s\Pi_0 u) = -e^{-s}\int A_s(\partial_i f)\,\psi\,d\lambda$.

Step 5: Fubini in (s,x). Substituting the last display into (1),

$$\Lambda(w) = -\frac{1}{\sqrt{\pi}}\int_0^\infty\!\!\int_{\mathbb{R}^n} s^{-1/2}e^{-s}\,A_s(\partial_i f)(x)\,\psi(x)\,d\lambda(x)\,ds.$$

The integrand is jointly measurable: $(s,x) \mapsto A_s(\partial_i f)(x)$ is jointly continuous by (d), and $(s,x) \mapsto s^{-1/2}e^{-s}\psi(x)$ is continuous. It is absolutely integrable, since by (c), the vanishing of ψ off the compact set K, the finiteness of $\lambda(K)$ and (e),

$$\int_0^\infty \int \left| s^{-1/2}e^{-s} A_s(\partial_i f)(x)\, \psi(x) \right| d\lambda(x)\, ds \le N\, \|\psi\|_\infty\, \lambda(K) \int_0^\infty s^{-1/2}e^{-s}\, ds < \infty.$$

Both factor measures are σ-finite, so Fubini permits exchanging the order:

$$\Lambda(w) = -\int_{\mathbb{R}^n} \psi(x) \Big[\frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2}e^{-s} A_s(\partial_i f)(x)\, ds \Big] d\lambda(x) = -\int_{\mathbb{R}^n} \sqrt{2}\, \Phi_i(x)\, \psi(x)\, d\lambda(x),$$

the last equality because, by (b), $\Phi_i(x)$ is $(\sqrt{\pi})^{-1} \int_0^\infty s^{-1/2}e^{-s} 2^{-1/2} A_s(\partial_i f)(x)\, ds$, so multiplying by $\sqrt{2}$ removes the factor $2^{-1/2}$. Since $\Lambda(w) = \int w\, \partial_i \psi\, d\lambda$, this is the stated identity. □

2.132. `RieszSemigroupPointwise`.

Lemma 2.132 (The semigroup formula for R_γ on the smooth core, with a continuous representative)**.** *For $s > 0$ write $r_s = e^{-s}$, $\tau_s = \sqrt{1 - e^{-2s}}$, and for $g : \mathbb{R}^n \to \mathbb{R}$ write $(A_s g)(x) = \int g(r_s x + \tau_s z)\, d\gamma_n(z)$ for the Mehler average of `MehlerFormula`. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support and square γ_n-integrable — the last hypothesis is carried explicitly but is automatic here, such an f being bounded and γ_n finite — and write $u \in L^2(\gamma_n)$ for the class it determines. Let $j \in \{1, \dots, n\}$. Put*

$$\Phi_j(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-s} \big(A_s(D_j f)\big)(x)\, \frac{ds}{\sqrt{s}}, \qquad D_j = 2^{-1/2}\partial_j,$$

so that the inner average is $2^{-1/2}(A_s(\partial_j f))(x)$, with $D = 2^{-1/2}\nabla$ the energy-normalized gradient of `GaussD`. Then:

1. *for every $x \in \mathbb{R}^n$ the integral defining $\Phi_j(x)$ converges absolutely, i.e. $s \mapsto s^{-1/2}e^{-s}(A_s(D_j f))(x)$ is integrable on $(0, \infty)$;*
2. *Φ_j is continuous on $\mathbb{R}^n$; and*
3. *Φ_j is a representative of the jth coordinate of $R_\gamma u$:*

$$\Phi_j = \big(R_\gamma u\big)_j \qquad \gamma_n\text{-almost everywhere},$$

with R_γ as in `RieszTransform`.

This is the reference paper's displayed semigroup formula for R_γ together with the sentence that follows it: the second integral there converges absolutely at every x, since $|P_s D f(x)| \le \|Df\|_\infty$, and provides a continuous representative of $R_\gamma f$. Clauses (1) and (2) are that sentence and clause (3) is the identification. All three are asserted at every *x except (3), which is an identity between an everywhere-defined continuous function and an almost-everywhere class and cannot be more than an almost-everywhere identity; that is exactly the strength needed downstream, where the point is to have a canonical continuous function to evaluate pointwise and to know that it represents the transform.*

Route. *The reference paper derives the formula by truncating the time integral and invoking closedness of D together with L^2 convergence. The proof below reaches the same conclusion differently, by testing the subordination formula against a fixed smooth compactly supported weight: for each such weight the pairing is a bounded linear functional on $L^2(\gamma_n)$, so it passes inside the Bochner integral, and the per-time identity is then ordinary integration by parts for a smooth function. No closedness of D and no continuity of $s \mapsto P_s v$ in L^2 is used.*

Proof. With $u \in L^2(\gamma_n)$ the class of f, write $\Pi_0 = L^0$ and $L^{-1/2}\Pi_0$ for the operators of `FracPow`, and $w = L^{-1/2}\Pi_0 u$. Throughout, λ denotes Lebesgue measure on $\mathbb{R}^n$; the integrals in the definition of a weak Euclidean gradient in `HasGaussWeakGrad` are taken against λ, not against γ_n.

Step 0: bounds. Write $S = \operatorname{supp} f$, a compact set. The function f is continuous, being smooth, and vanishes off S, so if $S = \emptyset$ then $f \equiv 0$; otherwise $|f|$, being continuous on the nonempty compact set S, attains a maximum there by the extreme value theorem (shared context, "Compact sets: extreme values and finite Lebesgue measure", clause (1)). In either

case $M_0 = \sup_{\mathbb{R}^n} |f| < \infty$. The same argument applies to each partial derivative: for every index i the function $\partial_i f$ is continuous, f being smooth, and it vanishes off S, because a point $x \notin S$ has an open neighbourhood on which f vanishes identically and hence on which all its derivatives vanish; so by the extreme value theorem applied to $|\partial_i f|$ on S (or trivially when $S = \emptyset$), $M^{(i)} = \sup_{\mathbb{R}^n} |\partial_i f| < \infty$. Write $M_1 = M^{(j)}$. By `GaussIsProb` the measure γ_n is a probability measure, so for every $s > 0$ and every x, using clause (2) of `SemigroupGradientCommute` for integrability,

$$\big|(A_s(\partial_j f))(x)\big| \le M_1, \qquad \big|(A_s f)(x)\big| \le M_0. \tag{0}$$

Step 1: the scalar time integral. $\int_0^\infty s^{-1/2} e^{-s}\, ds = \sqrt{\pi}$. Substituting $s = v^2$, a C^1 increasing bijection of $(0,\infty)$ onto itself with $ds = 2v\, dv$ and $s^{-1/2} = v^{-1}$, and noting the integrand is nonnegative and measurable so the change of variables is valid in $[0, +\infty]$,

$$\int_0^\infty s^{-1/2} e^{-s} ds = \int_0^\infty v^{-1} e^{-v^2} 2v\, dv = 2\int_0^\infty e^{-v^2} dv = \int_{\mathbb{R}} e^{-v^2} dv = \sqrt{\pi},$$

by evenness and the scalar Gaussian integral of the shared context.

Step 2: clause (1). Fix x. The map $s \mapsto (A_s(\partial_j f))(x)$ is continuous on $(0,\infty)$: if $s_k \to s$ in $(0,\infty)$ then $r_{s_k} \to r_s$ and $\tau_{s_k} \to \tau_s$ by continuity of the exponential and of the square root at $1 - e^{-2s} > 0$, so for each z the integrand converges by continuity of $\partial_j f$, and all integrands are bounded by the constant M_1, which is γ_n-integrable since γ_n is finite; dominated convergence (shared context) applies. Hence the integrand of $\Phi_j(x)$ is continuous, so measurable, on $(0,\infty)$, and by (0) it is bounded in absolute value by $2^{-1/2} M_1\, s^{-1/2} e^{-s}$, which is integrable there by Step 1. So the integral converges absolutely, which is clause (1); and

$$|\Phi_j(x)| \le \frac{1}{\sqrt{\pi}} \cdot 2^{-1/2} M_1 \cdot \sqrt{\pi} = 2^{-1/2} M_1 \qquad \text{for every } x. \tag{1}$$

Step 3: clause (2). Let $x_k \to x$. For each fixed $s > 0$, clause (3) of `SemigroupGradientCommute` says $A_s(\partial_j f)$ is continuous, so the integrands converge pointwise in s; they are dominated by $2^{-1/2} M_1 s^{-1/2} e^{-s}$, integrable by Step 1. Dominated convergence gives $\Phi_j(x_k) \to \Phi_j(x)$, and sequential continuity is continuity on the metric space $\mathbb{R}^n$.

Step 4: γ_n and λ have the same null sets. By `GaussMeasure`, $\gamma_n(A) = \int_A \rho\, d\lambda$ with $\rho(x) = \pi^{-n/2} e^{-|x|^2}$, and $\rho(x) > 0$ for every x. If $\lambda(A) = 0$ then the integral of any function over A vanishes, so $\gamma_n(A) = 0$. Conversely, suppose $\gamma_n(A) = 0$ and write $A = \bigcup_{m \ge 1} A_m$ with $A_m = A \cap \{x : \rho(x) > 1/m\}$, a countable increasing union since $\rho > 0$ everywhere. Then $0 = \gamma_n(A) \ge \gamma_n(A_m) \ge \lambda(A_m)/m$, so $\lambda(A_m) = 0$ for each m, and $\lambda(A) = 0$ by countable subadditivity (shared context). Hence the two measures have exactly the same null sets, and the phrases "γ_n-almost everywhere" and "λ-almost everywhere" are interchangeable from here on.

Step 5: the Mehler average of the class equals the Mehler average of f. Fix $s > 0$ and x. The coercion of u to a function agrees with f off a γ_n-null set N, by construction of the class of f; by Step 4, $\lambda(N) = 0$. Put $\Psi(z) = r_s x + \tau_s z$. Then

$$\Psi^{-1}(N) = \{z : r_s x + \tau_s z \in N\} = \{z : \tau_s z \in N - r_s x\},$$

where $N - r_s x$ is the translate of N by $-r_s x$. Lebesgue measure is translation invariant (shared context), so $\lambda(N - r_s x) = \lambda(N) = 0$. Since $\tau_s = \sqrt{1 - e^{-2s}} > 0$ for $s > 0$, in particular $\tau_s \ne 0$, clause (2) of `LebesgueDilation` applied with the scalar $c = \tau_s$ and the λ-null set $N - r_s x$ gives $\lambda(\Psi^{-1}(N)) = 0$. By Step 4 again $\gamma_n(\Psi^{-1}(N)) = 0$, so the two integrands $z \mapsto u(r_s x + \tau_s z)$ and $z \mapsto f(r_s x + \tau_s z)$ agree γ_n-almost everywhere and their integrals coincide. Combining with the averaging formula of `MehlerFormula` applied to u at this s — that node states, for every $s > 0$ and every $v \in L^2(\gamma_n)$, that $P_s v(x) = \int v(r_s x + \tau_s z)\, d\gamma_n(z)$ for γ_n-a.e. x, the inner integral being absolutely convergent for γ_n-a.e. x — we get

$$(P_s u)(x) = (A_s f)(x) \qquad \text{for } \gamma_n\text{-a.e. } x, \text{ hence for } \lambda\text{-a.e. } x. \tag{2}$$

Steps 6–10: the tested identity, by citation. Fix an index $i \in \{1, \dots, n\}$ and a smooth compactly supported $\varphi : \mathbb{R}^n \to \mathbb{R}$. Beyond the data already assembled, one further fact is

needed: the map $(s,x)\mapsto (A_s(\partial_i f))(x)$ is jointly continuous on $(0,\infty)\times\mathbb{R}^n$. This is dominated continuity. For each fixed z the integrand $\partial_i f\big(e^{-s}x+\sqrt{1-e^{-2s}}\,z\big)$ is continuous in (s,x), since $\partial_i f$ is continuous and $(s,x)\mapsto e^{-s}x+\sqrt{1-e^{-2s}}\,z$ is; it is bounded in absolute value by the constant M_1 of (0), uniformly in (s,x,z); and that constant is γ_n-integrable because γ_n is a finite measure. Dominated continuity (shared context) therefore makes the integral jointly continuous in (s,x).

With that, `RieszPseudoInverseTestedIdentity` applies. Its hypotheses are exactly the data in hand: f smooth, compactly supported and in $L^2(\gamma_n)$, with u its class; the defining formulas for A and for Φ; the uniform bound M_1 on $A_s(\partial_i f)$ from (0); the joint continuity just established; the integrability of $s^{-1/2}e^{-s}$ on $(0,\infty)$ from Step 1; and the identification of the γ_n-null and λ-null sets from Step 4. Its conclusion is the identity displayed below, for every such i and φ.

Step 11: the weak gradient of w, and clause (3). Let $\vec{\Phi}:\mathbb{R}^n\to\mathbb{R}^n$ have coordinates $\vec{\Phi}(x)_i=\Phi_i(x)$. Each Φ_i is continuous by Step 3 and bounded by $2^{-1/2}M^{(i)}$ by (1), so $\vec{\Phi}$ is continuous and bounded; since γ_n is finite (`GaussIsProb`), $\vec{\Phi}$ determines an element $\Psi\in L^2(\gamma_n;\mathbb{R}^n)$ whose coercion agrees with $\vec{\Phi}$ γ_n-a.e., hence λ-a.e. by Step 4.

The identity supplied by that node holds for an arbitrary index i and an arbitrary smooth compactly supported φ, and reads

$$\int w(x)\,\partial_i\varphi(x)\,d\lambda(x)=-\int\big(\sqrt{2}\,\vec{\Phi}(x)\big)_i\,\varphi(x)\,d\lambda(x),$$

which is the defining identity of `HasGaussWeakGrad` for the pair $(w,\sqrt{2}\,\vec{\Phi})$. Replacing $\vec{\Phi}$ by the coercion of Ψ changes the right-hand side on a λ-null set only, so the same identity holds with $\sqrt{2}\,\Psi$, i.e. $\sqrt{2}\,\Psi$ is a weak Euclidean gradient of w.

On the other hand, `PseudoInverseSqrt` gives $w=L^{-1/2}\Pi_0 u\in\mathrm{Dom}(L^{1/2})$, so `SqrtDomainSobolev` puts w in $W^{1,2}(\gamma_n)$, i.e. in the set of `SobolevW12`; by `GaussD` the element Dw satisfies the same defining identity with $\sqrt{2}\,Dw$ in place of $\sqrt{2}\,\Psi$. Both $\sqrt{2}\,\Psi$ and $\sqrt{2}\,Dw$ are elements of $L^2(\gamma_n;\mathbb{R}^n)$ and both are weak Euclidean gradients of w, so `GaussWeakGradUnique` gives $\sqrt{2}\,\Psi=\sqrt{2}\,Dw$, hence $\Psi=Dw$.

Finally, by `RieszTransform`, $R_\gamma u=D\big(L^{-1/2}\Pi_0 u\big)=Dw=\Psi$. Taking jth coordinates and recalling that the coercion of Ψ agrees γ_n-a.e. with $\vec{\Phi}$, we get $(R_\gamma u)_j=\Phi_j$ γ_n-almost everywhere, which is clause (3). □

2.133. `PVSubtractedSliceBound`.

Lemma 2.133 (The subtracted slice bound for the Mehler gradient kernel)**.** *Fix $n\ge 1$, $j\in\{1,\dots,n\}$ and $x\in\mathbb{R}^n$. For $s>0$ write*

$$r=e^{-s}\in(0,1),\qquad \sigma=1-e^{-2s}\in(0,1],\qquad p_s(x,y)=\frac{1}{\pi^{n/2}\sigma^{n/2}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big),$$

and let

$$D_j^x p_s(x,y)=\sqrt{2}\,r\,\frac{y_j-rx_j}{\sigma}\,p_s(x,y)$$

be the energy-normalized x-derivative of the Lebesgue kernel, written in the closed form that the surrounding nodes of this layer all use: the displayed right-hand side is that explicit expression, and it is that expression, not a derivative, that occurs below. Let λ be Lebesgue measure.

Then there is a constant $C\ge 0$ such that for every function $f:\mathbb{R}^n\to\mathbb{R}$, every $L\ge 0$ satisfying

$$(*)\qquad |f(y)-f(x)|\le L\,|y-x|\qquad \text{for all } y\in\mathbb{R}^n,$$

and every $s>0$,

$$\int_{\mathbb{R}^n}^{*}\big|D_j^x p_s(x,y)\big|\,|f(y)-f(x)|\,dy\;\le\;C\,L\,e^{-s}\,(1+|x|),$$

where $\int^*$ denotes the upper (lower Lebesgue) integral of a nonnegative extended-real integrand, taken in $[0,+\infty]$. The constant is quantified before f, L and s, so it is independent of all three; the witness exhibited in the proof is $C=3\sqrt{2}\cdot 2^{n/2}$, which depends only on n.

Three points about the shape of the statement. First, the hypothesis $(*)$ is the only property of f that is used, and it is exactly the mean-value bound that a Lipschitz function — in particular a smooth compactly supported one, with $L = \sup|\nabla f|$ — satisfies; stating the bound against $(*)$ rather than against smoothness keeps the estimate and the regularity of f in separate nodes, and is what the consumer supplies. Second, no measurability or integrability hypothesis on f is needed or asserted, because the left-hand side is an upper integral of a nonnegative quantity: it is defined for every f and every $s > 0$, and it is $+\infty$, not 0, whenever the integrand fails to be integrable. So the bound genuinely forces integrability rather than being satisfied by a conventional value — which is the whole point of the statement, since its consumer is a Tonelli argument on $\mathbb{R}^n \times (0,\infty)$ that needs the finiteness at every s at once. Third, the bound $C\,L\,e^{-s}(1+|x|)$ is ε-free and integrable against $ds/\sqrt{s}$ on $(0,\infty)$, which is the property the consumer uses; no uniformity in n is claimed and nothing here feeds an endpoint constant.

This is the reference paper's appendix bound: $|f(y) - f(x)| \le L_f|y - x|$ together with the substitution $w = z + d$ gives $\int |D_j^x p_s(x,y)||f(y) - f(x)|\,dy \le C_n L_f r(1 + |d|/\sqrt{\sigma}) \le C_n L_f e^{-s}(1 + |x|)$, with $|d| = (1-r)|x|$ and $(1-r)/\sqrt{1-r^2} \le 1$. The proof below is that argument with the two Gaussian moments taken directly at the rate $1/\sigma$ instead of being rescaled to the unit Gaussian, so that the paper's means a_n and b_n are never named. Taking the moments at the rate $1/\sigma$ costs a factor $\sigma^{-1/2}$ in the first-moment bound, and the taming factor correspondingly becomes $|d|/\sigma = |x|/(1+r) \le |x|$ in place of $|d|/\sqrt{\sigma} \le |x|$; the end served is the same displayed bound, so the constant is the only thing that changes.

Proof. Throughout $n \ge 1$, j and x are fixed, and $b, b' > 0$ denote arbitrary positive rates. We will verify the assertion with

$$C = 3\sqrt{2} \cdot 2^{n/2} \ge 0.$$

So let f, $L \ge 0$ satisfying $(*)$, and $s > 0$ be given, and keep $r = e^{-s}$, $\sigma = 1 - e^{-2s}$, $b = 1/\sigma$ and

$$d = x - rx, \qquad c = \frac{\sqrt{2}\,r}{\sigma}\,\pi^{-n/2}\sigma^{-n/2}.$$

Step 0: the elementary quantities. Since $-s < 0$ we have $0 < r < 1$, and since $e^{-s}e^{-s} = e^{-2s}$ we have $r^2 = e^{-2s}$, so

$$\sigma = 1 - r^2 = (1-r)(1+r), \tag{1}$$

whence $0 < \sigma$ (a product of two positive numbers) and $\sigma \le 1$ (because $(1-r)(1+r) = 1 - r^2 \le 1$). Thus $b = 1/\sigma > 0$, $\sigma b = 1$, and $c > 0$, the factors $\pi^{-n/2}$ and $\sigma^{-n/2}$ being positive powers of positive numbers. Also $d = (1-r)x$, so

$$|d| = (1-r)\,|x|. \tag{2}$$

Step 1: the Gaussian integral at an arbitrary rate. For every $b > 0$ the function $v \mapsto e^{-b|v|^2}$ is Lebesgue integrable on $\mathbb{R}^n$ and

$$\int_{\mathbb{R}^n} e^{-b|v|^2}\,dv = \left(\frac{\pi}{b}\right)^{n/2}. \tag{3}$$

Both are the shared-context Gaussian integral on a finite-dimensional inner product space, read at the rate b and with $\dim \mathbb{R}^n = n$.

Step 2: the two half-rate moment majorants. From $1 + u \le e^u$ at $u = (b/2)|v|^2$ we get $(b/2)|v|^2 \le e^{(b/2)|v|^2}$, hence, multiplying by $E := e^{-(b/2)|v|^2} \in (0,1]$ and using $e^{(b/2)|v|^2}E = 1$,

$$|v|^2\,E \le \frac{2}{b}. \tag{4}$$

By `GaussSecondMomentIntegralBound`, at every rate $b > 0$ the second-moment majorant

$$|v|^2\,e^{-b|v|^2} \le \frac{2}{b}\,e^{-(b/2)|v|^2} \tag{5}$$

holds for every v, and $v \mapsto |v|^2 e^{-b|v|^2}$ is Lebesgue integrable with integral at most $\frac{2}{b}(\pi/(b/2))^{n/2}$. For the first moment, (4) together with $|v| \le 1 + |v|^2$ gives

$$|v|\, e^{-b|v|^2} \le (1+|v|^2)E \cdot E = E^2 + (|v|^2 E)E \le \Big(1 + \frac{2}{b}\Big) e^{-(b/2)|v|^2}. \tag{6}$$

The function $v \mapsto |v| e^{-b|v|^2}$ is continuous and, by (6), dominated in absolute value by a constant multiple of the integrable function $e^{-(b/2)|v|^2}$ of Step 1; hence it is Lebesgue integrable, and integrating (6) and using (3) at the rate $b/2$, together with the integral bound just cited,

$$\int_{\mathbb{R}^n} |v|^2 e^{-b|v|^2} dv \le \frac{2}{b}\Big(\frac{\pi}{b/2}\Big)^{n/2}, \qquad \int_{\mathbb{R}^n} |v|\, e^{-b|v|^2} dv \le \Big(1 + \frac{2}{b}\Big)\Big(\frac{\pi}{b/2}\Big)^{n/2}. \tag{7}$$

Step 3: the normalization identity. For $\sigma > 0$ and $b' > 0$,

$$\pi^{-n/2}\sigma^{-n/2}\Big(\frac{\pi}{b'}\Big)^{n/2} = \Big(\frac{1}{\sigma b'}\Big)^{n/2}, \tag{8}$$

since $\pi^{-n/2}\pi^{n/2} = 1$ and $\sigma^{-n/2}(b')^{-n/2} = (\sigma b')^{-n/2}$. With $b' = b/2 = 1/(2\sigma)$ we have $\sigma b' = 1/2$, so the right-hand side is $2^{n/2}$, and therefore

$$c\Big(\frac{\pi}{b/2}\Big)^{n/2} = \frac{\sqrt{2}\, r}{\sigma} \cdot 2^{n/2}. \tag{9}$$

Step 4: the translation and the pointwise majorant. The translation $T(w) = rx + w$ preserves λ (shared context) and is a measurable embedding, so for every nonnegative extended-real G on $\mathbb{R}^n$,

$$\int_{\mathbb{R}^n}^{*} G(rx+w)\, dw = \int_{\mathbb{R}^n}^{*} G(y)\, dy. \tag{10}$$

At $y = rx + w$ we have $y_j - rx_j = w_j$ and $|rx - y| = |-w| = |w|$, so by the definition of p_s and of c,

$$D_j^x p_s(x, rx+w) = c\, w_j\, e^{-b|w|^2},$$

the exponent being $-|w|^2/\sigma = -b|w|^2$. Also $y - x = w - d$, so $|y-x| \le |w| + |d|$, and therefore by $(*)$

$$|f(rx+w) - f(x)| \le L\big(|w| + |d|\big).$$

Multiplying the two displays, and using $|w_j| \le |w|$ together with $c > 0$, $e^{-b|w|^2} > 0$, $L \ge 0$ and $|d| \ge 0$,

$$\begin{aligned} \big|D_j^x p_s(x, rx+w)\big|\, |f(rx+w) - f(x)| &= c\, |w_j|\, e^{-b|w|^2}\, |f(rx+w) - f(x)| \\ &\le c\, |w|\, e^{-b|w|^2} \cdot L\big(|w| + |d|\big) \\ &= cL\, |w|^2 e^{-b|w|^2} + cL|d|\, |w|\, e^{-b|w|^2} =: H(w). \end{aligned} \tag{11}$$

Step 5: integrating the majorant. By Step 2 the function H is a nonnegative linear combination, with the nonnegative coefficients cL and $cL|d|$, of the two integrable functions of (7); hence H is Lebesgue integrable and, by additivity and homogeneity of the integral and then (7),

$$\int_{\mathbb{R}^n} H = cL \int |w|^2 e^{-b|w|^2} + cL|d| \int |w| e^{-b|w|^2} \le cL\, \frac{2}{b}\Big(\frac{\pi}{b/2}\Big)^{n/2} + cL|d|\Big(1 + \frac{2}{b}\Big)\Big(\frac{\pi}{b/2}\Big)^{n/2}.$$

Since $2/b = 2\sigma$, factoring out $c(\pi/(b/2))^{n/2}$ and using (9),

$$\int_{\mathbb{R}^n} H \le \frac{\sqrt{2}\, r}{\sigma}\, 2^{n/2}\, L\,\big(2\sigma + |d|(1+2\sigma)\big). \tag{12}$$

Because $H \ge 0$ is integrable, its upper integral equals its Lebesgue integral; so combining (10), (11) and monotonicity of the upper integral,

$$\int_{\mathbb{R}^n}^{*} \big|D_j^x p_s(x,y)\big|\, |f(y) - f(x)|\, dy = \int_{\mathbb{R}^n}^{*} \big|D_j^x p_s(x, rx+w)\big|\, |f(rx+w) - f(x)|\, dw \le \int_{\mathbb{R}^n} H. \tag{13}$$

Step 6: the taming factor. We claim

$$2\sigma + |d|(1+2\sigma) \le 3\sigma(1+|x|). \tag{14}$$

Indeed, substituting (2) and then (1) for the single factor σ multiplying $|x|$ on the right,

$$3\sigma(1+|x|) - \big(2\sigma + (1-r)|x|(1+2\sigma)\big) = \sigma + (1-r)\,|x|\,\big(3(1+r) - (1+2\sigma)\big),$$

an identity between polynomials in r and $|x|$ once σ is replaced by $(1-r)(1+r)$ throughout: both sides reduce to $\sigma+(1-r)|x|(2+3r)-2\sigma(1-r)|x|$, using $3(1-r)(1+r)-(1-r) = (1-r)(2+3r)$. Now $1-r \ge 0$, $|x| \ge 0$ and $3(1+r)-(1+2\sigma) = 2+3r-2\sigma \ge 3r > 0$ because $\sigma \le 1$; so the right-hand side is at least $\sigma > 0$, which gives (14).

Finally, combining (13), (12) and (14), and using that $\sqrt{2}\,r\sigma^{-1}2^{n/2}L \ge 0$,

$$\int_{\mathbb{R}^n}^{*} \big|D_j^x p_s(x,y)\big|\,|f(y)-f(x)|\,dy \le \frac{\sqrt{2}\,r}{\sigma}\,2^{n/2}\,L\cdot 3\sigma(1+|x|) = 3\sqrt{2}\cdot 2^{n/2}\,L\,e^{-s}\,(1+|x|),$$

the last step cancelling $\sigma \ne 0$ and recalling $r = e^{-s}$. This is the assertion with $C = 3\sqrt{2}\cdot 2^{n/2}$, a constant depending only on n and in particular independent of f, L and s. □

2.134. `PVSemigroupKernelDerivative`.

Lemma 2.134 (The kernel form of the Mehler gradient: $\int D_j^x p_s(x,y) f(y)\,dy = e^{-s}(A_s(D_j f))(x)$). *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $x \in \mathbb{R}^n$, and let $s > 0$. Write*

$$r = e^{-s} \in (0,1), \qquad \sigma = 1 - e^{-2s} \in (0,1], \qquad \tau = \sqrt{\sigma},$$

$$p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}}\exp\Big(-\frac{|rx-y|^2}{\sigma}\Big), \qquad D_j^x p_s(x,y) = \sqrt{2}\,r\,\frac{y_j - r x_j}{\sigma}\,p_s(x,y),$$

the latter being the energy-normalized x-derivative of the Lebesgue kernel in the closed form the surrounding nodes of this layer all use: the displayed right-hand side is that explicit expression, and it is that expression, not a derivative, that occurs below. Let λ be Lebesgue measure, let γ_n be the Gaussian measure of `GaussMeasure`*, let ∂_j denote the Fréchet derivative evaluated at the jth standard basis vector, and let*

$$(A_s g)(x) = \int_{\mathbb{R}^n} g(rx + \tau z)\,d\gamma_n(z)$$

be the Mehler average. Then:

(1) *the function $y \mapsto D_j^x p_s(x,y)\,f(y)$ is Lebesgue integrable on $\mathbb{R}^n$; and*
(2)

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)\,f(y)\,dy = e^{-s}\,\frac{1}{\sqrt{2}}\big(A_s(\partial_j f)\big)(x).$$

This is the pointwise, kernel-level form of the reference paper's identity $DP_s f = e^{-s}P_s Df$ for $f \in C_c^\infty$, which the paper obtains by "differentiation of Mehler's formula" and then uses, together with the substitution $r = e^{-s}$, to read off the off-diagonal Lebesgue kernel of the Gaussian Riesz transform. Clause (1) is the absolute convergence that makes clause (2) an identity of real numbers rather than of conventional values: it is asserted here, and not left to the Bochner convention, precisely because clause (2)'s consumer splits the integral $\int D_j^x p_s(x,y)(f(y)-f(x))\,dy$ into two pieces and needs each to converge.

Both sides are honest pointwise quantities at the given x and s; nothing here is almost-everywhere, and no L^2 class or representative of f occurs. The constant $2^{-1/2}$ is the energy normalization $D_j = 2^{-1/2}\partial_j$ carried by both sides, and σ, τ are positive, so every divisor below is nonzero.

Proof. Fix n, j, f, x and $s > 0$, and keep r, σ, τ as in the statement. Since $-s < 0$ and $-2s < 0$ we have $0 < r < 1$ and $0 < e^{-2s} < 1$, so $0 < \sigma \le 1$, and hence $\tau > 0$ with $\tau^2 = \sigma$ and

$$\tau^n = \sigma^{n/2}, \tag{1}$$

the latter because $\tau = \sigma^{1/2}$ and $(\sigma^{1/2})^n = \sigma^{n/2}$ for $\sigma > 0$. Write

$$A = \pi^{-n/2}\sigma^{-n/2} > 0, \qquad w(y) = -\sigma^{-1}|rx - y|^2, \qquad g(y) = \partial_j f(y),$$

so that $p_s(x,y) = A\, e^{w(y)}$ and

$$\sigma^{n/2} A = \pi^{-n/2}. \tag{2}$$

Step 1: clause (1). By clause (1) of `PVKernelConstantCancellation`, applied at this s, the function $y \mapsto D_j^x p_s(x,y)$ is Lebesgue integrable on $\mathbb{R}^n$. Also f is continuous with compact support, so it is bounded: there is M with $|f(y)| \le M$ for every y (the shared-context bound for a continuous function with compact support). The product $y \mapsto D_j^x p_s(x,y) f(y)$ is therefore measurable — a product of a measurable and a continuous function — and dominated in absolute value by the integrable function $M\,|D_j^x p_s(x,\cdot)|$, since $|D_j^x p_s(x,y) f(y)| = |D_j^x p_s(x,y)|\,|f(y)| \le |D_j^x p_s(x,y)|\,M$. Hence it is Lebesgue integrable. This is clause (1).

Step 2: the jth partial of the kernel exponent. The map $y \mapsto rx - y$ is differentiable at every y with derivative $-\mathrm{id}$, so $y \mapsto |rx-y|^2$ is differentiable at every y with derivative $v \mapsto 2\langle rx - y, -v\rangle$, by the chain rule applied to the squared norm (whose derivative at a point u is $v \mapsto 2\langle u, v\rangle$). Multiplying by the constant $-\sigma^{-1}$, the function w is differentiable at every y and

$$\partial_j w(y) = -\sigma^{-1} \cdot 2\langle rx - y, -e_j\rangle = \frac{2\langle rx - y, e_j\rangle}{\sigma} = \frac{2\,(rx_j - y_j)}{\sigma}. \tag{3}$$

In particular $y \mapsto \partial_j w(y)$ is continuous, being an affine function of the single coordinate y_j divided by the nonzero constant σ.

Step 3: integration by parts in y. Apply `CompIntegrationByParts` with $q = q' = \exp$ (the exponential is differentiable at every real number with derivative itself, and that derivative is continuous), with the coordinate index j, with the function w of Step 2 — which is differentiable everywhere and has continuous jth partial, exactly the hypothesis that node requires — and with the test function $\varphi = f$, which is smooth with compact support. That node's identity reads

$$\int_{\mathbb{R}^n} e^{w(y)}\, \partial_j f(y)\, dy = -\int_{\mathbb{R}^n} \big(e^{w(y)}\, \partial_j w(y)\big) f(y)\, dy,$$

both integrals converging. Substituting (3) into the right-hand side and then using $y_j - rx_j = -(rx_j - y_j)$, so that the integrands differ by a sign,

$$\int_{\mathbb{R}^n} \Big(e^{w(y)}\, \frac{2\,(y_j - rx_j)}{\sigma}\Big) f(y)\, dy = -\int_{\mathbb{R}^n} \Big(e^{w(y)}\, \frac{2\,(rx_j - y_j)}{\sigma}\Big) f(y)\, dy = \int_{\mathbb{R}^n} e^{w(y)}\, g(y)\, dy. \tag{4}$$

Step 4: the Mehler average is the Lebesgue-kernel average. We claim

$$\big(A_s g\big)(x) = \int_{\mathbb{R}^n} g(rx + \tau z)\, d\gamma_n(z) = A \int_{\mathbb{R}^n} e^{w(y)}\, g(y)\, dy. \tag{5}$$

Put $\Psi(y) = A\, e^{w(y)} g(y)$, so that the right-hand side of (5) is $\int \Psi$.

By `GaussMeasure`, γ_n is λ weighted by the density $\rho(z) = \pi^{-n/2} e^{-|z|^2}$, which is finite and nonnegative at every point; so for every h, $\int h\, d\gamma_n = \int \rho(z) h(z)\, d\lambda(z)$, the weighted-integral identity of the shared context. Hence

$$\int_{\mathbb{R}^n} g(rx + \tau z)\, d\gamma_n(z) = \int_{\mathbb{R}^n} \pi^{-n/2} e^{-|z|^2}\, g(rx + \tau z)\, dz.$$

Next, for every z,

$$|rx - (rx + \tau z)|^2 = |-\tau z|^2 = \tau^2 |z|^2 = \sigma |z|^2, \qquad \text{so} \qquad w(rx + \tau z) = -\sigma^{-1}\sigma |z|^2 = -|z|^2,$$

whence $\Psi(rx + \tau z) = A\, e^{-|z|^2} g(rx + \tau z)$ and therefore, by (2),

$$\pi^{-n/2} e^{-|z|^2} g(rx + \tau z) = \sigma^{n/2} A\, e^{-|z|^2} g(rx + \tau z) = \sigma^{n/2}\, \Psi(rx + \tau z) \qquad (z \in \mathbb{R}^n). \tag{6}$$

Finally, the dilation $z \mapsto \tau z$ scales Lebesgue measure by τ^n and translation by rx preserves it, so, in the integral form of those two shared-context facts — both of which are identities valid without

any integrability hypothesis, the two sides being simultaneously genuine or simultaneously 0 by the convention —

$$\int_{\mathbb{R}^n} \Psi(rx+\tau z)\,dz = |\tau^n|^{-1}\int_{\mathbb{R}^n} \Psi(rx+v)\,dv = \big(\sigma^{n/2}\big)^{-1}\int_{\mathbb{R}^n}\Psi(y)\,dy, \tag{7}$$

using (1) and $\tau^n > 0$ in the last step. Combining the three displays, (6) and (7) give

$$\int g(rx+\tau z)\,d\gamma_n(z) = \sigma^{n/2}\int \Psi(rx+\tau z)\,dz = \sigma^{n/2}\big(\sigma^{n/2}\big)^{-1}\int\Psi = \int\Psi,$$

which is (5) after cancelling $\sigma^{n/2}\neq 0$ and pulling the constant A out of $\int\Psi$.

Step 5: assembly. For every y, since $p_s(x,y) = A\,e^{w(y)}$ and $\sqrt{2}/2 = 2^{-1/2}$,

$$D_j^x p_s(x,y)f(y) = \sqrt{2}\,r\,\frac{y_j - rx_j}{\sigma}\,A\,e^{w(y)}f(y) = \frac{\sqrt{2}\,r}{2}\,A\cdot\Big(e^{w(y)}\,\frac{2\,(y_j-rx_j)}{\sigma}\Big)f(y),$$

the two sides differing only by the regrouping of the nonzero constants $\sqrt{2}$, r, 2, A and σ. Integrating in y and pulling the constant $\frac{\sqrt{2}\,r}{2}A$ out of the integral, then using (4) and then (5),

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)f(y)\,dy = \frac{\sqrt{2}\,r}{2}A\int e^{w(y)}\Big(\frac{2(y_j-rx_j)}{\sigma}\Big)f(y)\,dy = \frac{\sqrt{2}\,r}{2}\,A\int e^{w(y)}g(y)\,dy = \frac{\sqrt{2}\,r}{2}\big(A_s g\big)(x).$$

Since $\sqrt{2}\cdot\sqrt{2} = 2$ gives $\sqrt{2}/2 = 1/\sqrt{2}$, and $r = e^{-s}$, the right-hand side is $e^{-s}\,2^{-1/2}\big(A_s(\partial_j f)\big)(x)$. This is clause (2). □

2.135. `PVCoreSmooth`.

Lemma 2.135 (Agreement of the kernel and spectral realizations on a smooth core)**.** *Fix $n \geq 1$ and $j \in \{1,\dots,n\}$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, and let $x \in \mathbb{R}^n$. For $s > 0$ write*

$$r = e^{-s}, \qquad \tau = \sqrt{1-e^{-2s}}, \qquad (A_s g)(x) = \int_{\mathbb{R}^n} g(rx+\tau z)\,d\gamma_n(z)$$

for the Mehler average of `MehlerFormula`, let $D_j = 2^{-1/2}\partial_j$ be the energy-normalized partial derivative of `GaussD`, and put

$$\Phi_j(x) = \frac{1}{\sqrt{\pi}}\int_0^\infty e^{-s}\,\big(A_s(D_j f)\big)(x)\,\frac{ds}{\sqrt{s}} = \frac{1}{\sqrt{\pi}}\int_0^\infty e^{-s}\,\frac{1}{\sqrt{2}}\big(A_s(\partial_j f)\big)(x)\,\frac{ds}{\sqrt{s}},$$

the function of `RieszSemigroupPointwise`. Let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of `GaussRieszKernel`. Then:

(1) *for every $\varepsilon > 0$ the function $y \mapsto K_{\gamma,j}(x,y)f(y)$ is Lebesgue integrable on the exterior region $\{y : |x-y| > \varepsilon\}$; and*

(2)

$$\lim_{\varepsilon\downarrow 0}\int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,f(y)\,dy \;=\; \Phi_j(x),$$

the limit being taken along $\varepsilon \to 0$ with $\varepsilon > 0$.

This is the reference paper's appendix lemma: the truncated kernel integrals of a smooth compactly supported input converge, at every *x and not merely almost everywhere, to the Gaussian Riesz transform of that input, the right-hand side being read as the continuous semigroup representative. That reading is exactly what Φ_j is: by `RieszSemigroupPointwise` the integral defining $\Phi_j(x)$ converges absolutely at every x (clause (1)), Φ_j is continuous on $\mathbb{R}^n$ (clause (2)), and Φ_j represents the jth coordinate of $R_\gamma u$, where $u \in L^2(\gamma_n)$ is the class of f (clause (3)). The every-x strength of the present statement is what later nodes consume, so it is not weakened to an almost-everywhere assertion; it is available because both sides are honest pointwise quantities, the left-hand side by clause (1) above and the right-hand side by the continuity of Φ_j.*

Every constant appearing in the proof may depend on n, on j, on x and on f. As the appendix header instructs, these constants serve only to identify two realizations of one operator, and nothing below is uniform in n or feeds any endpoint constant.

Proof. Throughout, λ is Lebesgue measure on $\mathbb{R}^n$, γ_n is the Gaussian measure of `GaussMeasure`, and $n \geq 1$, j, f and x are fixed. For $s > 0$ we keep the abbreviations

$$r = e^{-s} \in (0,1), \quad \sigma = 1 - r^2 = 1 - e^{-2s} \in (0,1), \quad \tau = \sqrt{\sigma}, \quad d = (1-r)x,$$

and we write

$$p_s(x,y) = \frac{1}{\pi^{n/2}\sigma^{n/2}} \exp\Big(-\frac{|rx-y|^2}{\sigma}\Big), \qquad q_\sigma(z) = (\pi\sigma)^{-n/2} e^{-|z|^2/\sigma}, \qquad V_{j,\sigma}(z) = \frac{z_j}{\sigma} q_\sigma(z),$$

so that, exactly as in `PVKernelConstantCancellation`, with $z = y - x$ we have $z + d = y - rx$, $p_s(x,y) = q_\sigma(z+d)$ and

$$D_j^x p_s(x,y) := \frac{1}{\sqrt{2}}\, \partial_{x_j} p_s(x,y) = \sqrt{2}\, r\, \frac{y_j - r x_j}{\sigma}\, p_s(x,y) = \sqrt{2}\, r\, V_{j,\sigma}(z+d), \tag{0}$$

the middle equality being `MehlerKernelXDerivative`. Finally put

$$\Phi^{(s)}(w) = \int_{\mathbb{R}^n} p_s(w,y) f(y)\, dy, \qquad w \in \mathbb{R}^n.$$

Step 0: elementary bounds on f. Let $S = \operatorname{supp} f$, a compact set, and let $M_0 = \sup_{\mathbb{R}^n} |f|$ and $L_f = \sup_{\mathbb{R}^n} |\nabla f|$. Both are finite: f and $|\nabla f|$ are continuous and vanish off S (at a point outside S the function f vanishes on a neighbourhood, hence so do all its derivatives), so the extreme value theorem on S (shared context) bounds them. Also $\lambda(S) < \infty$, a compact set having finite Lebesgue measure (shared context). By the mean value inequality applied along the segment from x to y — $t \mapsto f(x + t(y-x))$ is smooth with derivative $\nabla f(x + t(y-x)) \cdot (y-x)$, so $|f(y) - f(x)| \leq \int_0^1 |\nabla f(x + t(y-x))|\, |y-x|\, dt$ — we get

$$|f(y) - f(x)| \leq L_f\, |y - x| \qquad \text{for all } y \in \mathbb{R}^n. \tag{1}$$

Finally f is bounded and continuous and γ_n is a probability measure (`GaussIsProb`), so f is square γ_n-integrable and determines a class $u \in L^2(\gamma_n)$; this is the u of the statement, and it is the hypothesis under which `RieszSemigroupPointwise` is applied below.

Step 1: γ_n and λ have the same null sets, and γ_n has full support. By `GaussMeasure`, γ_n is λ weighted by the density $\rho(y) = \pi^{-n/2} e^{-|y|^2}$, i.e. $\gamma_n(A) = \int_A \rho\, d\lambda$ for measurable A. The density is finite and strictly positive at every point.

If $\lambda(A) = 0$ then $\int_A \rho\, d\lambda = 0$, so $\gamma_n(A) = 0$. Conversely suppose $\gamma_n(A) = 0$ and put $A_m = A \cap \{\rho > 1/m\}$ for $m \in \mathbb{N}$. Then $\lambda(A_m)/m \leq \int_{A_m} \rho\, d\lambda \leq \gamma_n(A) = 0$, so $\lambda(A_m) = 0$; since $\rho > 0$ everywhere we have $A = \bigcup_m A_m$, and countable subadditivity (shared context) gives $\lambda(A) = 0$. So the two measures have the same null sets.

If $U \subseteq \mathbb{R}^n$ is open and nonempty it contains a ball, so $\lambda(U) > 0$, hence $\gamma_n(U) > 0$ by the previous paragraph. Consequently:

(2) if $g_1, g_2 : \mathbb{R}^n \to \mathbb{R}$ are continuous and $g_1 = g_2$ γ_n-almost everywhere, then $g_1 = g_2$ everywhere.

Indeed, if $g_1(w_0) \neq g_2(w_0)$ then $|g_1 - g_2| > 0$ on some open ball U around w_0 by continuity of $g_1 - g_2$, and $\gamma_n(U) > 0$ contradicts $\gamma_n(\{g_1 \neq g_2\}) = 0$.

Step 2: the kernel form of the semigroup derivative. Fix $s > 0$.

(a) $\Phi^{(s)}$ is well defined and continuous on $\mathbb{R}^n$. For each w the function $y \mapsto p_s(w,y) f(y)$ is continuous, vanishes off S and is bounded in absolute value by $\pi^{-n/2}\sigma^{-n/2} M_0$, because the exponential factor of p_s is at most 1; since $\lambda(S) < \infty$ it is Lebesgue integrable, so $\Phi^{(s)}(w)$ is a well-defined real number. If $w_k \to w$ then $p_s(w_k, y) f(y) \to p_s(w,y) f(y)$ for every y, by continuity of p_s in its first argument, and all these functions are dominated by the integrable function $\pi^{-n/2}\sigma^{-n/2} M_0 \mathbf{1}_S$; dominated convergence (shared context) gives $\Phi^{(s)}(w_k) \to \Phi^{(s)}(w)$. As $\mathbb{R}^n$ is a metric space, $\Phi^{(s)}$ is continuous.

(b) $\Phi^{(s)} = A_s f$ everywhere on $\mathbb{R}^n$. Let $\tilde{u}$ denote the function underlying the class u of Step 0, the same representative that `MehlerFormula` and `MehlerLebesgueKernel` use. By clause (2)

of `MehlerFormula` applied to s and u, and clause (2) of `MehlerLebesgueKernel` applied to the same s and u, there is a γ_n-null set off which both

$$(P_s u)(w) = \int_{\mathbb{R}^n} \tilde{u}(rw + \tau z)\, d\gamma_n(z) \qquad \text{and} \qquad (P_s u)(w) = \int_{\mathbb{R}^n} p_s(w,y)\, \tilde{u}(y)\, dy$$

hold; hence for γ_n-almost every w,

$$\int_{\mathbb{R}^n} p_s(w,y)\, \tilde{u}(y)\, dy = \int_{\mathbb{R}^n} \tilde{u}(rw + \tau z)\, d\gamma_n(z). \tag{3}$$

Now replace $\tilde{u}$ by f on both sides of (3), for *every* w. Let $N = \{\tilde{u} \neq f\}$; then $\gamma_n(N) = 0$, since $\tilde{u}$ is a representative of the class of f, and therefore $\lambda(N) = 0$ by Step 1. On the left, the integrands $p_s(w,\cdot)\tilde{u}$ and $p_s(w,\cdot)f$ differ only on N, a λ-null set, so the two Lebesgue integrals agree for every w (and in particular the second is absolutely convergent by (a)). On the right, fix w and note that $\{z : rw + \tau z \in N\} = \tau^{-1}(N - rw)$, which is λ-null because λ is translation invariant (shared context) and the dilation $v \mapsto \tau^{-1}v$ multiplies λ by τ^{-n}; hence it is γ_n-null by Step 1, and the two γ_n-integrals agree for every w (the one with f being absolutely convergent by clause (1) of `SemigroupGradientCommute`). Therefore

$$\Phi^{(s)}(w) = (A_s f)(w) \qquad \text{for } \gamma_n\text{-almost every } w.$$

Both sides are continuous: the left by (a), and the right because clause (4) of `SemigroupGradientCommute` asserts that $A_s f$ is differentiable at every point, hence continuous there. By (2) of Step 1 they agree at every w.

(c) the kernel form. Set $G(w,y) = 2r(y_j - rw_j)\sigma^{-1}p_s(w,y)$; by `MehlerKernelXDerivative` the map $w \mapsto p_s(w,y)$ is differentiable at every w and $2^{-1/2}G(w,y) = D_j^x p_s(w,y)$, i.e. $G(w,y) = \partial_{w_j} p_s(w,y)$ — multiply that node's displayed identity by $\sqrt{2}$ and use $\sqrt{2}\cdot\sqrt{2} = 2$. The function G is continuous on $\mathbb{R}^n \times \mathbb{R}^n$, being a product and composition of continuous functions.

We claim

$$\lim_{t\to 0} \frac{\Phi^{(s)}(x + te_j) - \Phi^{(s)}(x)}{t} = \int_{\mathbb{R}^n} G(x,y) f(y)\, dy, \tag{4}$$

e_j being the jth standard basis vector. Let $\eta > 0$. The set $\overline{B}(x,1) \times S$ is compact, so G is uniformly continuous on it (Heine–Cantor); choose $\delta \in (0,1]$ with $|G(w,y) - G(x,y)| \le \eta$ whenever $w \in \overline{B}(x,1)$, $|w - x| \le \delta$ and $y \in S$. Fix $y \in S$ and t with $0 < |t| \le \delta$. The function $t' \mapsto p_s(x + t'e_j, y)$ is differentiable on $\mathbb{R}$ with derivative $t' \mapsto G(x + t'e_j, y)$, which is continuous; so by the fundamental theorem of calculus

$$p_s(x + te_j, y) - p_s(x,y) = \int_0^t G(x + t'e_j, y)\, dt',$$

and subtracting $t\, G(x,y) = \int_0^t G(x,y)\, dt'$,

$$\left| p_s(x + te_j, y) - p_s(x,y) - t\, G(x,y) \right| \le |t|\, \eta,$$

because every $x + t'e_j$ with t' between 0 and t lies in $\overline{B}(x,1)$ at distance at most δ from x. Multiplying by $f(y)$, which is bounded by M_0 and vanishes off S, and integrating over y — the three functions $y \mapsto p_s(x + te_j, y)f(y)$, $y \mapsto p_s(x,y)f(y)$ and $y \mapsto G(x,y)f(y)$ are continuous, vanish off S and are bounded, hence integrable, the last because $G(x,\cdot)$ is continuous — we obtain

$$\left| \Phi^{(s)}(x + te_j) - \Phi^{(s)}(x) - t \int G(x,y) f(y)\, dy \right| \le |t|\, \eta\, M_0\, \lambda(S).$$

Dividing by $|t|$ shows that the difference quotient in (4) differs from $\int G(x,y)f(y)\, dy$ by at most $\eta M_0 \lambda(S)$ for every t with $0 < |t| \le \delta$. As $\eta > 0$ was arbitrary, (4) follows.

On the other hand, by (b) the functions $\Phi^{(s)}$ and $A_s f$ coincide, so the limit in (4) is also the derivative at $t = 0$ of $t \mapsto (A_s f)(x + te_j)$. By clause (4) of `SemigroupGradientCommute`, $A_s f$

is differentiable at x and the value of its derivative at e_j — which is precisely that derivative along the line — equals $e^{-s}\int(\partial_j f)(rx+\tau z)\,d\gamma_n(z) = e^{-s}\big(A_s(\partial_j f)\big)(x)$. Hence

$$\int_{\mathbb{R}^n} \partial_{x_j} p_s(x,y)\, f(y)\, dy = e^{-s}\big(A_s(\partial_j f)\big)(x).$$

Multiplying by $2^{-1/2}$ and using $D_j^x p_s = 2^{-1/2}\partial_{x_j} p_s$ and $D_j = 2^{-1/2}\partial_j$ (GaussD),

$$\int_{\mathbb{R}^n} D_j^x p_s(x,y)\, f(y)\, dy = e^{-s}\,\frac{1}{\sqrt{2}}\big(A_s(\partial_j f)\big)(x) = e^{-s}\big(A_s(D_j f)\big)(x) =: \Theta(s). \tag{5}$$

The integral on the left is absolutely convergent, since $|D_j^x p_s(x,\cdot)|$ is Lebesgue integrable by clause (1) of PVKernelConstantCancellation and $|f| \le M_0$.

Step 3: the subtracted bound. We claim that there is a constant C_n, depending only on n, with

$$\int_{\mathbb{R}^n} \big|D_j^x p_s(x,y)\big|\,|f(y)-f(x)|\,dy \le C_n\, L_f\, e^{-s}\,(1+|x|) \qquad \text{for every } s>0. \tag{6}$$

First record two Gaussian moment identities. The dilation $v \mapsto cv$ multiplies λ by c^n, so for every Borel $h:\mathbb{R}^n \to [0,+\infty]$ and every $c>0$, $\int h(z)\,dz = c^n \int h(cv)\,dv$; applying this with $c=\sqrt{\sigma}$ and $h = |z_j|q_\sigma$, respectively $h = |z_j|\,|z|\,q_\sigma$, and using $q_\sigma(\sqrt{\sigma}v) = (\pi\sigma)^{-n/2}e^{-|v|^2}$ together with GaussMeasure,

$$\int_{\mathbb{R}^n} |z_j|\,q_\sigma(z)\,dz = \sqrt{\sigma}\,a_n, \qquad \int_{\mathbb{R}^n} |z_j|\,|z|\,q_\sigma(z)\,dz = \sigma\, b_n, \tag{7}$$

where $a_n = \int |v_j|\,d\gamma_n(v)$ and $b_n = \int |v_j|\,|v|\,d\gamma_n(v)$ are finite, since $|v_j| \le |v| \le 1+|v|^2$ and $|v_j||v| \le |v|^2$ are dominated by polynomials and the Gaussian exponential moments item of the shared context (with $t=0$) applies.

Now use (1) and (0), substitute $y = x+z$ and then $w = z+d$ (two translations, which preserve λ; shared context), and use $|z| = |w-d| \le |w|+|d|$:

$$\begin{aligned}
\int_{\mathbb{R}^n} |D_j^x p_s(x,y)|\,|f(y)-f(x)|\,dy &\le L_f\sqrt{2}\,r\int_{\mathbb{R}^n} |V_{j,\sigma}(z+d)|\,|z|\,dz \\
&\le \frac{\sqrt{2}\,r\,L_f}{\sigma}\int_{\mathbb{R}^n} |w_j|\,\big(|w|+|d|\big)\,q_\sigma(w)\,dw \\
&= \frac{\sqrt{2}\,r\,L_f}{\sigma}\big(\sigma b_n + |d|\sqrt{\sigma}a_n\big) = \sqrt{2}\,L_f\,r\Big(b_n + a_n\frac{|d|}{\sqrt{\sigma}}\Big),
\end{aligned}$$

by (7). Since $|d| = (1-r)|x|$ and $\sigma = (1-r)(1+r)$ with $0<r<1$,

$$\frac{|d|}{\sqrt{\sigma}} = |x|\sqrt{\frac{1-r}{1+r}} \le |x|,$$

so the right-hand side is at most $\sqrt{2}\,L_f\,e^{-s}(b_n + a_n|x|) \le C_n L_f e^{-s}(1+|x|)$ with $C_n = \sqrt{2}\max(a_n,b_n)$. This is (6).

Step 4: Fubini in (y,s), and the subtracted identity. We first record the time-integral representation of the kernel: for every $y \in \mathbb{R}^n$,

$$K_{\gamma,j}(x,y) = \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{D_j^x p_s(x,y)}{\sqrt{s}}\,ds, \tag{8}$$

with the convention — the one GaussRieszKernel uses for its own defining integral — that an integral of a non-integrable integrand is 0, and with the s-integrand Lebesgue integrable on $(0,\infty)$ exactly when the integrand of that defining integral is Lebesgue integrable on $(0,1)$. Indeed, by (0) and $p_s(x,y) = \pi^{-n/2}(1-r^2)^{-n/2}e^{-|rx-y|^2/(1-r^2)}$,

$$\frac{1}{\sqrt{\pi}}\,\frac{D_j^x p_s(x,y)}{\sqrt{s}} = \frac{\sqrt{2}}{\pi^{(n+1)/2}}\,\frac{r\,(y_j - rx_j)}{\sqrt{s}\,(1-r^2)^{n/2+1}}\exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big),$$

while `GaussRieszKernel` gives $K_{\gamma,j}(x,y) = \sqrt{2}\,\pi^{-(n+1)/2}\int_{(0,1)} h(\varrho)\,d\varrho$ with

$$h(\varrho) = \frac{y_j - \varrho x_j}{(1-\varrho^2)^{n/2+1}\sqrt{-\log\varrho}}\exp\Big(-\frac{|\varrho x - y|^2}{1-\varrho^2}\Big).$$

The map $\psi(s) = e^{-s}$ is a continuously differentiable bijection of $(0,\infty)$ onto $(0,1)$ with $|\psi'(s)| = e^{-s} \neq 0$ and inverse $\varrho \mapsto -\log\varrho$, so that $\sqrt{-\log\psi(s)} = \sqrt{s}$. The one-dimensional change-of-variables formula for such a bijection, which inserts the factor $|\psi'|$, applied first to the nonnegative Borel function $|h|$ (an identity in $[0,+\infty]$, so that integrability of one side is equivalent to integrability of the other) and then to h itself, gives $\int_{(0,1)} h(\varrho)\,d\varrho = \int_{(0,\infty)} h(e^{-s})e^{-s}\,ds$; and

$$h(e^{-s})\,e^{-s} = \frac{r\,(y_j - r x_j)}{(1-r^2)^{n/2+1}\sqrt{s}}\exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big),$$

which is $\pi^{(n+1)/2}2^{-1/2}$ times the displayed integrand above. Multiplying by $\sqrt{2}\pi^{-(n+1)/2}$ gives (8); and when neither integrand is integrable both sides of (8) are 0 by the convention.

Now put $F(y,s) = s^{-1/2}D_j^x p_s(x,y)\big(f(y)-f(x)\big)$ on $\mathbb{R}^n\times(0,\infty)$. It is continuous in (y,s), hence product-measurable. By Tonelli (shared context), (6) and `SubordinationWeightIntegral`,

$$\iint |F| \le \int_0^\infty \frac{C_n L_f(1+|x|)e^{-s}}{\sqrt{s}}\,ds = C_n L_f(1+|x|)\sqrt{\pi} < \infty,$$

so F is integrable on the product and Fubini (shared context) applies:

$$\int_{\mathbb{R}^n}\Big(\int_0^\infty F(y,s)\,ds\Big)dy = \int_0^\infty\Big(\int_{\mathbb{R}^n} F(y,s)\,dy\Big)ds, \tag{9}$$

the inner integral on the left being Lebesgue integrable in y and the inner integral on the right Lebesgue integrable in s.

The inner integral on the left is, by (8), $(f(y)-f(x))\sqrt{\pi}\,K_{\gamma,j}(x,y)$ for every y — at the y for which the s-integrand is integrable this is (8), and at the remaining y both sides are 0 by the convention. So $y\mapsto K_{\gamma,j}(x,y)(f(y)-f(x))$ is Lebesgue integrable on $\mathbb{R}^n$. The inner integral on the right is, for each $s>0$,

$$\int_{\mathbb{R}^n} F(y,s)\,dy = \frac{1}{\sqrt{s}}\Big(\int D_j^x p_s(x,y)f(y)\,dy - f(x)\int D_j^x p_s(x,y)\,dy\Big) = \frac{\Theta(s)}{\sqrt{s}},$$

splitting the integral — legitimate because both pieces are absolutely convergent, by clause (1) of `PVKernelConstantCancellation` and $|f|\le M_0$ — and then using (5) for the first piece and clause (1) of `PVKernelConstantCancellation` again, which says $\int D_j^x p_s(x,y)\,dy = 0$, for the second. Therefore (9) reads

$$\sqrt{\pi}\int_{\mathbb{R}^n} K_{\gamma,j}(x,y)\big(f(y)-f(x)\big)\,dy = \int_0^\infty \frac{\Theta(s)}{\sqrt{s}}\,ds = \sqrt{\pi}\,\Phi_j(x),$$

the last equality being the definition of Φ_j together with (5); the s-integral converges absolutely, which is also clause (1) of `RieszSemigroupPointwise`. Dividing by $\sqrt{\pi}$,

$$\int_{\mathbb{R}^n} K_{\gamma,j}(x,y)\big(f(y)-f(x)\big)\,dy = \Phi_j(x), \tag{10}$$

the left-hand side being an absolutely convergent Lebesgue integral. This is the appendix's subtracted identity.

Step 5: splitting the truncated integrals, and the limit. Fix $\varepsilon>0$ and write $\Omega_\varepsilon = \{y : |x-y| > \varepsilon\}$. By clause (2) of `PVKernelConstantCancellation`, $K_{\gamma,j}(x,\cdot)$ is Lebesgue integrable on Ω_ε; since f is continuous with $|f|\le M_0$, the product $K_{\gamma,j}(x,\cdot)f$ is measurable and dominated by $M_0|K_{\gamma,j}(x,\cdot)|$ there, hence integrable on Ω_ε. This is clause (1) of the lemma. By (10) the function $K_{\gamma,j}(x,\cdot)(f-f(x))$ is integrable on $\mathbb{R}^n$, hence on Ω_ε, and therefore

$$\int_{\Omega_\varepsilon} K_{\gamma,j}(x,y)f(y)\,dy = \int_{\Omega_\varepsilon} K_{\gamma,j}(x,y)\big(f(y)-f(x)\big)\,dy + f(x)\int_{\Omega_\varepsilon} K_{\gamma,j}(x,y)\,dy, \tag{11}$$

all three integrals being absolutely convergent.

Let $\varepsilon_k \downarrow 0$ be any sequence of positive reals. The second term of (11) tends to 0 by clause (3) of `PVKernelConstantCancellation`. For the first, the functions $\mathbf{1}_{\Omega_{\varepsilon_k}} K_{\gamma,j}(x,\cdot)(f - f(x))$ converge pointwise on $\mathbb{R}^n \setminus \{x\}$ — hence λ-almost everywhere, a single point being λ-null — to $K_{\gamma,j}(x,\cdot)(f - f(x))$, and they are dominated in absolute value by that same integrable function; dominated convergence (shared context) gives

$$\int_{\Omega_{\varepsilon_k}} K_{\gamma,j}(x,y)(f(y) - f(x))\,dy \longrightarrow \int_{\mathbb{R}^n} K_{\gamma,j}(x,y)(f(y) - f(x))\,dy = \Phi_j(x)$$

by (10). Hence $\int_{\Omega_{\varepsilon_k}} K_{\gamma,j}(x,y) f(y)\,dy \to \Phi_j(x)$. Since the sequence (ε_k) was arbitrary and $\mathbb{R}$ is a metric space, so that convergence along $\varepsilon \downarrow 0$ is equivalent to convergence along every such sequence, clause (2) follows. □

2.136. `SmoothCompactDenseL1Gauss`.

Lemma 2.136 (Smooth compactly supported functions are dense in $L^1(\gamma_n)$)**.** *Fix $n \in \mathbb{N}_0$, let λ be Lebesgue measure on $\mathbb{R}^n$ and let γ_n be the Gaussian measure of `GaussMeasure`. Let $g \in L^1(\gamma_n)$, and write g also for its canonical representative, a function $\mathbb{R}^n \to \mathbb{R}$. Let $\delta > 0$. Then there is a function $\varphi : \mathbb{R}^n \to \mathbb{R}$ which is smooth, has compact support and is γ_n-integrable, such that*

$$\|g - \varphi\|_{L^1(\gamma_n)} = \int_{\mathbb{R}^n} |g(y) - \varphi(y)|\,d\gamma_n(y) < \delta,$$

the integral being read in $[0,+\infty]$ as the $L^1(\gamma_n)$ seminorm of the difference, so that the inequality asserts in particular that it is finite.

This is the density statement the reference paper uses when it writes "choose $f_k \in C_c^\infty$ with $f_k \to f$ in $L^1(\gamma)$": applying the lemma with $\delta = 1/(k+1)$ produces exactly such a sequence. It is stated plainly, with no side condition on the support of φ beyond compactness, because that is the form the paper uses and the form that is reusable; callers that need φ to vanish on a prescribed ball obtain that for themselves by multiplying by a cut-off, which costs nothing because the function being approximated already vanishes there.

Why the tablet's existing approximation results do not give this. *Two of them are nearby and neither suffices, which is why this statement is separate from both; they are named here for contrast only, are not used in the proof below, and are correspondingly neither cited as dependencies nor imported. `L1ApproxByL2Truncations` produces elements of $L^2(\gamma_n)$ whose $L^1(\gamma_n)$ images converge to g; those are equivalence classes carrying no smoothness and no support control whatever. `MollifierL2Local` concludes convergence of mollifications in L^2 of* Lebesgue *measure restricted to a compact set — a local statement, about a different measure and a different exponent. Neither yields a C_c^∞ approximant in $L^1(\gamma_n)$.*

Proof. Write $\varrho(y) = \pi^{-n/2} e^{-|y|^2}$, which is smooth, strictly positive and bounded above by $\pi^{-n/2}$ at every point of $\mathbb{R}^n$. By `GaussMeasure`, γ_n is λ weighted by ϱ, so that for every measurable $w : \mathbb{R}^n \to \mathbb{R}$

$$\int_{\mathbb{R}^n} |w|\,d\gamma_n = \int_{\mathbb{R}^n} |w(y)|\,\varrho(y)\,d\lambda(y), \tag{1}$$

both sides read in $[0,+\infty]$.

Step 0: measurability bookkeeping. The canonical representative g is γ_n-almost everywhere equal to a Borel function $\widetilde{g}$. By clause (4) of `GaussLebesgueBallComparison` the measures γ_n and λ have the same null sets, so $g = \widetilde{g}$ also λ-almost everywhere. Replacing g by $\widetilde{g}$ changes neither side of the asserted inequality — the integrand $|g - \varphi|$ changes only on a γ_n-null set — so we may and do assume g Borel measurable. Being the representative of an element of $L^1(\gamma_n)$, it satisfies $\int |g|\,d\gamma_n < \infty$.

Step 1: transport to $L^1(\lambda)$. Put $h(y) = g(y)\varrho(y)$, a Borel function. By (1), $\int |h|\,d\lambda = \int |g|\,d\gamma_n < \infty$, so $h \in L^1(\lambda)$.

Step 2: a continuous compactly supported approximant to h. By the density of compactly supported continuous functions in L^p of a regular Borel measure (shared context, applied to

$\mu = \lambda$, $p = 1$ and the function h) there is $u : \mathbb{R}^n \to \mathbb{R}$ which is continuous, has compact support, lies in $L^1(\lambda)$ and satisfies

$$\int_{\mathbb{R}^n} |h - u| \, d\lambda \le \frac{\delta}{3}. \tag{2}$$

The same shared-context item records that such a u is bounded and uniformly continuous on $\mathbb{R}^n$.

Step 3: a smooth compactly supported approximant to u. For $\varepsilon > 0$ fix $\rho_\varepsilon : \mathbb{R}^n \to \mathbb{R}$ which is smooth, nonnegative, has $\operatorname{supp} \rho_\varepsilon \subseteq \overline{B}(0, \varepsilon)$, and satisfies $\int \rho_\varepsilon \, d\lambda = 1$. Such a family exists. For $n \ge 1$: take the standard radial bump b at centre 0 with inner radius $\varepsilon/2$ and outer radius ε, so that b is smooth, takes values in $[0, 1]$, equals 1 on $\overline{B}(0, \varepsilon/2)$ and is supported in $B(0, \varepsilon)$; then $\int b \, d\lambda > 0$, because b is nonnegative, continuous and equal to 1 on a ball of positive Lebesgue measure, and $\rho_\varepsilon = b / \int b \, d\lambda$ has all four properties. For $n = 0$ the space $\mathbb{R}^0$ is a single point of unit λ-mass and $\rho_\varepsilon \equiv 1$ has all four properties. Set

$$u_\varepsilon(x) = (\rho_\varepsilon * u)(x) = \int_{\mathbb{R}^n} \rho_\varepsilon(t) \, u(x - t) \, d\lambda(t).$$

By the shared-context item on smoothness of a mollification, u_ε is defined at every x and is smooth.

Support. Let $S = \operatorname{supp} u$, compact, and let $K = \{x : \operatorname{dist}(x, S) \le 1\}$, which is closed and bounded, hence compact, and has $\lambda(K) < \infty$ (shared context). If $0 < \varepsilon \le 1$ and $x \notin K$ then for every t with $|t| \le \varepsilon$ we have $x - t \notin S$, so $u(x - t) = 0$; since ρ_ε vanishes off $\overline{B}(0, \varepsilon)$, the integrand above vanishes identically and $u_\varepsilon(x) = 0$. Thus $\{u_\varepsilon \ne 0\} \subseteq K$, and as K is closed, $\operatorname{supp} u_\varepsilon \subseteq K$ is compact.

Uniform convergence. Since $\int \rho_\varepsilon \, d\lambda = 1$ we have $u(x) = \int \rho_\varepsilon(t) u(x) \, d\lambda(t)$, so for every x

$$|u_\varepsilon(x) - u(x)| = \Big| \int \rho_\varepsilon(t) \big(u(x - t) - u(x)\big) d\lambda(t) \Big| \le \sup_{|t| \le \varepsilon} \sup_{w \in \mathbb{R}^n} |u(w - t) - u(w)|,$$

using $\rho_\varepsilon \ge 0$, $\int \rho_\varepsilon = 1$ and that ρ_ε vanishes for $|t| > \varepsilon$. By the uniform continuity of u (Step 2), given $\eta > 0$ there is $\varepsilon \in (0, 1]$ with $|u(w - t) - u(w)| \le \eta$ for all w and all $|t| \le \varepsilon$, and then $\sup_x |u_\varepsilon(x) - u(x)| \le \eta$. Since $u_\varepsilon - u$ vanishes off K,

$$\int_{\mathbb{R}^n} |u_\varepsilon - u| \, d\lambda \le \eta \, \lambda(K).$$

Choose $\eta > 0$ with $\eta \, \lambda(K) \le \delta/3$ — possible because $\lambda(K) < \infty$ — and let $v = u_\varepsilon$ for the corresponding ε. Then v is smooth with compact support and

$$\int_{\mathbb{R}^n} |u - v| \, d\lambda \le \frac{\delta}{3}. \tag{3}$$

Step 4: transport back, and conclusion. Define

$$\varphi(y) = \pi^{n/2} e^{|y|^2} \, v(y), \qquad \text{so that} \qquad \varphi(y) \, \varrho(y) = v(y) \text{ for every } y.$$

The function $y \mapsto \pi^{n/2} e^{|y|^2}$ is smooth and nowhere zero, so φ is smooth and $\{\varphi \ne 0\} = \{v \ne 0\}$, whence $\operatorname{supp} \varphi = \operatorname{supp} v$ is compact. Being continuous with compact support, φ is bounded; and γ_n is a probability measure, hence finite (`GaussIsProb`), so a bounded measurable function is γ_n-integrable: $\varphi \in L^1(\gamma_n)$.

Finally, by (1) applied to $w = g - \varphi$, then $(g(y) - \varphi(y))\varrho(y) = h(y) - v(y)$, then the triangle inequality in $L^1(\lambda)$ together with (2) and (3),

$$\int_{\mathbb{R}^n} |g - \varphi| \, d\gamma_n = \int_{\mathbb{R}^n} |h - v| \, d\lambda \le \int_{\mathbb{R}^n} |h - u| \, d\lambda + \int_{\mathbb{R}^n} |u - v| \, d\lambda \le \frac{\delta}{3} + \frac{\delta}{3} < \delta.$$

This is the assertion. □

2.137. `GaussRieszKernelTruncationRedundant`.

Lemma 2.137 (The hard truncation is redundant when the input vanishes near the ball)**.** *Fix $n \geq 1$ and $j \in \{1, \dots, n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of `GaussRieszKernel`. Fix $x_0 \in \mathbb{R}^n$ and $\rho > 0$, and for $z \in \mathbb{R}^n$ write*

$$\Omega_z = \{y \in \mathbb{R}^n : |z - y| > \rho/4\}.$$

Let $h : \mathbb{R}^n \to \mathbb{R}$ be γ_n-integrable and suppose

$$h = 0 \quad \lambda\text{-almost everywhere on } B(x_0, \tfrac{3\rho}{4}).$$

Then for every $z \in B(x_0, \rho/2)$ *the function $y \mapsto K_{\gamma,j}(z, y)h(y)$ is Lebesgue integrable on $\mathbb{R}^n$, and*

$$\int_{\mathbb{R}^n} K_{\gamma,j}(z, y)\, h(y)\, dy = \int_{\Omega_z} K_{\gamma,j}(z, y)\, h(y)\, dy.$$

Both conclusions are content, and the first is not an afterthought. *The two displayed integrals are Bochner integrals, each equal to* 0 *by convention on a non-integrable integrand; an equality between them could therefore hold with both sides conventional and assert nothing about the kernel. The integrability assertion is what rules that out, and it is stated for* every z *in the ball rather than almost every one, so that a consumer may evaluate either side pointwise.*

The two radii are not interchangeable. *The hypothesis is vanishing on the ball of radius $3\rho/4$ and the conclusion holds for centres in the ball of radius $\rho/2$, with the truncation taken at $\rho/4$; the three are tied by $\rho/4 + \rho/2 = 3\rho/4$, which is exactly what makes the discarded region lie inside the region where h vanishes. Weakening the hypothesis to vanishing on $B(x_0, \rho/2)$, or widening the conclusion to centres in $B(x_0, 3\rho/4)$, would break the proof and the statement.*

The statement is deliberately about an arbitrary such h rather than about one particular function: its consumer applies it both to a given $L^1(\gamma_n)$ input and to each member of a sequence of smooth approximants to that input, and those are different functions sharing only the vanishing hypothesis.

Proof. Put $R = \|x_0\| + \rho + 1 > 0$, and let $C > 0$ be the constant supplied by clause (1) of `GaussRieszKernelFarMajorant` for the pair $(R, \rho/4)$, both of which are positive. Fix $z \in B(x_0, \rho/2)$, so that $|z - x_0| < \rho/2$.

Step 1: z lies in the ball the far majorant needs. By the triangle inequality, $|z| = |(z - x_0) + x_0| \leq |z - x_0| + |x_0| < \rho/2 + \|x_0\|$, and $\rho/2 + \|x_0\| \leq \|x_0\| + \rho + 1 = R$ because $\rho/2 \leq \rho + 1$. So $|z| \leq R$.

Step 2: integrability on Ω_z. The set $\Omega_z = \{y : \rho/4 < |z - y|\}$ is measurable, being the preimage of an open half-line under the continuous map $y \mapsto |z - y|$. Clause (1)(b) of `GaussRieszKernelFarMajorant`, applied to the γ_n-integrable function h and to this z — legitimate by Step 1 — states that $y \mapsto K_{\gamma,j}(z, y)h(y)$ is Lebesgue integrable on $\{y : \rho/4 < |z - y|\} = \Omega_z$. (That clause also supplies a bound on the integral, which is not needed here.)

Step 3: the integrand vanishes λ-almost everywhere off Ω_z. Let $y \notin \Omega_z$, that is $|z - y| \leq \rho/4$. Then, using $|y - z| = |z - y|$,

$$|y - x_0| = |(y - z) + (z - x_0)| \leq |y - z| + |z - x_0| < \tfrac{\rho}{4} + \tfrac{\rho}{2} = \tfrac{3\rho}{4},$$

so $y \in B(x_0, 3\rho/4)$. By hypothesis h vanishes λ-almost everywhere on that ball; hence for λ-almost every y, if $y \notin \Omega_z$ then $h(y) = 0$ and therefore $K_{\gamma,j}(z, y)h(y) = 0$.

Step 4: conclusion. Write $F(y) = K_{\gamma,j}(z, y)h(y)$ and let $\mathbf{1}_{\Omega_z} F$ be the function equal to F on Ω_z and to 0 elsewhere. Since Ω_z is measurable, $\mathbf{1}_{\Omega_z} F$ is Lebesgue integrable on $\mathbb{R}^n$ if and only if F is Lebesgue integrable on Ω_z (shared context: the indicator criterion for integrability on a measurable set); by Step 2 it therefore is. By Step 3, $F = \mathbf{1}_{\Omega_z} F$ λ-almost everywhere: the two agree at every point of Ω_z by definition, and at almost every point off Ω_z both vanish. Two functions agreeing almost everywhere are integrable together, so F is Lebesgue integrable on $\mathbb{R}^n$. This is the first conclusion.

For the second, Step 3 says precisely that F vanishes almost everywhere on the complement of Ω_z, and for such an F the integral over Ω_z equals the integral over the whole space (shared

context; this identity needs no integrability hypothesis, both sides being read with the convention, and it is applied here to an F already known integrable). That is the displayed equality, read from right to left. □

2.138. `GaussRieszKernelSmoothOffDiagonal`.

Lemma 2.138 (The untruncated kernel integral of a smooth input off its support)**.** *Fix $n \geq 1$ and $j \in \{1, \dots, n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of* `GaussMeasure`*, and let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of* `GaussRieszKernel`*. Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $z \in \mathbb{R}^n$, let $\sigma > 0$, and suppose that*

$$\varphi(y) = 0 \qquad \text{whenever } |z - y| \leq \sigma.$$

Write, for $s > 0$,

$$(A_s w)(z) = \int_{\mathbb{R}^n} w\big(e^{-s} z + \sqrt{1 - e^{-2s}}\, \zeta\big)\, d\gamma_n(\zeta)$$

for the Mehler average, and

$$\Phi_j^{\varphi}(z) = \frac{1}{\sqrt{\pi}} \int_0^{\infty} \frac{1}{\sqrt{s}}\, e^{-s}\, \frac{1}{\sqrt{2}} \big(A_s(\partial_j \varphi)\big)(z)\, ds,$$

which is the quantity appearing as the limit in clause (2) of `PVCoreSmooth` *at the point z. Then the function $y \mapsto K_{\gamma,j}(z, y)\varphi(y)$ is Lebesgue integrable on $\mathbb{R}^n$, and*

$$\Phi_j^{\varphi}(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z, y)\, \varphi(y)\, dy.$$

This is an off-diagonal identification and not a principal value. *No limit in a truncation radius is taken in the conclusion: the point of the hypothesis on φ is that on the range $0 < \varepsilon \leq \sigma$ the truncated integrals do not depend on ε at all, so the limit that* `PVCoreSmooth` *exhibits is attained, and is attained by the integral over all of $\mathbb{R}^n$. The reference paper licenses exactly this much when it says that for compactly supported smooth inputs and a point off their support the integrals may be interchanged directly, and immediately adds that this is an off-diagonal kernel identification rather than a principal-value identification; the present statement stays on the first side of that line.*

The integrability assertion is content. *The right-hand side is a Bochner integral, equal to 0 by convention on a non-integrable integrand; without the integrability assertion the displayed identity could hold with the right-hand side conventional. It is asserted for the given z with no exceptional set, because φ is an honest function and the hypothesis is pointwise.*

Proof. Write $F(y) = K_{\gamma,j}(z, y)\varphi(y)$ and, for $\varepsilon > 0$, $\Omega_\varepsilon = \{y : |z - y| > \varepsilon\}$, a measurable set (the preimage of an open half-line under the continuous map $y \mapsto |z - y|$).

Step 1: for every $\varepsilon \in (0, \sigma]$, F is integrable on $\mathbb{R}^n$ and its integral is the truncated one. Fix such an ε. Clause (1) of `PVCoreSmooth`, applied to φ (smooth with compact support), to j, to the point z and to this $\varepsilon > 0$, states that F is Lebesgue integrable on Ω_ε.

Next, F vanishes off Ω_ε: if $y \notin \Omega_\varepsilon$ then $|z - y| \leq \varepsilon \leq \sigma$, so $\varphi(y) = 0$ by hypothesis and hence $F(y) = 0$. This holds at every such y, not merely almost every one.

Let $\mathbf{1}_{\Omega_\varepsilon} F$ be the function equal to F on Ω_ε and to 0 elsewhere. Since Ω_ε is measurable, $\mathbf{1}_{\Omega_\varepsilon} F$ is Lebesgue integrable on $\mathbb{R}^n$ if and only if F is Lebesgue integrable on Ω_ε (shared context: the indicator criterion for integrability on a measurable set), so it is. And $F = \mathbf{1}_{\Omega_\varepsilon} F$ everywhere: the two agree on Ω_ε by definition and both vanish off it by the previous paragraph. Two functions that agree are integrable together, so F is Lebesgue integrable on $\mathbb{R}^n$. Finally, a function vanishing (almost) everywhere off a set has the same integral over that set as over the whole space (shared context), so

$$\int_{\Omega_\varepsilon} F\, d\lambda = \int_{\mathbb{R}^n} F\, d\lambda \qquad (0 < \varepsilon \leq \sigma). \tag{1}$$

Taking $\varepsilon = \sigma$, which is admissible since $\sigma > 0$, gives the first conclusion.

Step 2: the truncated integral is eventually constant, so the limit is that constant. Consider the function $G(\varepsilon) = \int_{\Omega_\varepsilon} F\, d\lambda$ on $(0, \infty)$, and the filter of neighbourhoods of 0 within $(0, \infty)$.

Every ε in the intersection of $(0,\infty)$ with the neighbourhood $(-\infty,\sigma)$ of 0 satisfies $0<\varepsilon<\sigma$, and that intersection is a member of the filter; so by (1),

$$G(\varepsilon)=\int_{\mathbb{R}^n} F\,d\lambda \qquad \text{for all } \varepsilon \text{ in a member of the filter,}$$

i.e. G agrees with the constant function $\int_{\mathbb{R}^n} F\,d\lambda$ along that filter.

Clause (2) of `PVCoreSmooth`, applied to the same data, states that $G(\varepsilon)$ tends to $\Phi_j^{\varphi}(z)$ as $\varepsilon\downarrow 0$, i.e. along that same filter. Two functions agreeing along a filter have the same limits, so the constant function $\int_{\mathbb{R}^n} F\,d\lambda$ also tends to $\Phi_j^{\varphi}(z)$ along it. But a constant function tends to its own value along every filter, and the filter of neighbourhoods of 0 within $(0,\infty)$ is not the trivial filter (every such neighbourhood contains points of $(0,\infty)$, since 0 is a limit point of $(0,\infty)$), so limits along it are unique. Therefore

$$\Phi_j^{\varphi}(z)=\int_{\mathbb{R}^n} F\,d\lambda,$$

which is the second conclusion. □

2.139. `GaussRieszOffDiagonalL1.`

Lemma 2.139 (Off-diagonal kernel representation of the L^1 extension)**.** *Fix $n\ge 1$ and $j\in\{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel`. Let*

$$T : L^1(\gamma_n)\longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

- (T1) *$T(\iota v)=R_\gamma v$ for every $v\in L^2(\gamma_n)$, where $\iota : L^2(\gamma_n)\to L^1(\gamma_n)$ is the inclusion of `ToL1` and R_γ is the Riesz transform of `RieszTransform`; and*
- (T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n}>t\}\big)\le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f\in L^1(\gamma_n;\mathbb{R})$ and every real $t>0$, the distribution function being that of the Euclidean norm of the vector $Tf(x)$.*

Let $g\in L^1(\gamma_n)$, write g also for its canonical representative, let $x_0\in\mathbb{R}^n$ and $\rho>0$, and assume that

$$g=0 \quad \gamma_n\text{-almost everywhere on } B(x_0,\rho).$$

Then:

1. *for* every *$z\in B(x_0,\rho/2)$ the function $y\mapsto K_{\gamma,j}(z,y)\,g(y)$ is Lebesgue integrable on $\mathbb{R}^n$; and*
2. *for γ_n-almost every z, if $z\in B(x_0,\rho/2)$ then*

$$(Tg)_j(z)=\int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\,g(y)\,dy.$$

The hypothetical style, and why it is not a citation of hypotheses. *(T1) and (T2) are written out in full above; they are not imported by reference from any other node, and the lemma is a statement about every T satisfying them. This is the same style as `CoordinateEndpointBound`, with which the lemma composes directly: an actual such T exists by `L1Extension`, and every consumer that has one in hand can feed it to both nodes at once.*

What this adds, and what it is not. *The reference paper's own interchange of the time and space integrals is licensed only "for compactly supported smooth inputs and x off their support". Clause (2) upgrades that to an arbitrary $L^1(\gamma_n)$ input, at the price of holding z in a ball on which the input vanishes and of weakening "every z" to "almost every z" — the latter being unavoidable, since Tg is only an almost-everywhere equivalence class. It is emphatically* not *a principal-value identification: no limit in the truncation radius occurs, because on the range considered the truncated integrals are eventually independent of the radius.*

Clause (1) is stated and is not an afterthought. *The integral in clause (2) is a Bochner integral, whose value is 0 by convention on a non-integrable integrand; without clause (1) the identity could be read as holding at points where the right-hand side is that conventional value.*

Clause (1) also holds at every *point of the ball, not merely almost every one, which is what lets a consumer evaluate the right-hand side pointwise.*

Proof. Throughout, $B(a,s)$ and $\overline{B}(a,s)$ are the open and closed Euclidean balls, λ is Lebesgue measure, and we write $\|h\|_1 = \int |h|\, d\gamma_n$ for a γ_n-integrable function h and also for the norm of the corresponding element of $L^1(\gamma_n)$; the two agree, because the canonical representative of an element of $L^1(\gamma_n)$ realises its norm as that integral. Set

$$R = \|x_0\| + \rho + 1 > 0,$$

so that every $z \in B(x_0, \rho/2)$ has $|z| \le \|x_0\| + \rho/2 \le R$, and let $C > 0$ be the constant supplied by clause (1) of `GaussRieszKernelFarMajorant` for the pair $(R, \rho/4)$, both of which are positive. For $z \in \mathbb{R}^n$ write $\Omega_z = \{y : |z - y| > \rho/4\}$.

Step 0: the vanishing holds λ-almost everywhere as well. The hypothesis says that the set $N = \{y \in B(x_0,\rho) : g(y) \neq 0\}$ is contained in a γ_n-null set. By clause (4) of `GaussLebesgueBallComparison`, γ_n and λ have the same null sets, so N is contained in a λ-null set too; that is, $g = 0$ λ-almost everywhere on $B(x_0,\rho)$.

Step 1: two separation estimates. Let $z \in B(x_0, \rho/2)$.

(a) If $y \notin B(x_0,\rho)$ then $|z - y| \ge |x_0 - y| - |x_0 - z| \ge \rho - \rho/2 = \rho/2 > \rho/4$, so $y \in \Omega_z$. Equivalently, $y \notin \Omega_z$ implies $y \in B(x_0,\rho)$.

(b) If $|z-y| \le \rho/4$ then $|y - x_0| \le |y-z| + |z-x_0| < \rho/4 + \rho/2 = 3\rho/4$, so $y \in B(x_0, 3\rho/4) \subseteq B(x_0, 7\rho/8)$.

Step 2: clause (1), and the truncated form of the integral. We prove the following statement, which we will use both for g and for the approximants of Step 4. *Let $h : \mathbb{R}^n \to \mathbb{R}$ be γ_n-integrable and suppose $h = 0$ λ-almost everywhere on $B(x_0, 3\rho/4)$. Then for every $z \in B(x_0,\rho/2)$ the function $y \mapsto K_{\gamma,j}(z,y)h(y)$ is Lebesgue integrable on $\mathbb{R}^n$ and*

$$\int_{\mathbb{R}^n} K_{\gamma,j}(z,y)h(y)\, dy = \int_{\Omega_z} K_{\gamma,j}(z,y)h(y)\, dy. \tag{1}$$

Indeed, fix such an h and such a z. Being γ_n-integrable, h satisfies the hypothesis of clause (1)(b) of `GaussRieszKernelFarMajorant`, and $|z| \le R$, so that clause gives that $y \mapsto K_{\gamma,j}(z,y)h(y)$ is Lebesgue integrable on Ω_z. By Step 1(b), every $y \notin \Omega_z$ — that is, every y with $|z-y| \le \rho/4$ — lies in $B(x_0, 3\rho/4)$, where h vanishes λ-almost everywhere. Hence $K_{\gamma,j}(z,\cdot)h$ vanishes λ-almost everywhere on the complement of Ω_z. A function integrable on a measurable set and vanishing almost everywhere off it is integrable on the whole space, with the same integral; this gives both the integrability on $\mathbb{R}^n$ and (1).

Applying this with $h = g$, which is γ_n-integrable and vanishes λ-almost everywhere on $B(x_0,\rho) \supseteq B(x_0, 3\rho/4)$ by Step 0, proves clause (1) of the lemma.

Step 3: the smooth off-diagonal identification, derived inline. Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support, let $z \in \mathbb{R}^n$, and suppose $\varphi(y) = 0$ whenever $|z - y| \le \sigma$, for some $\sigma > 0$. Write

$$\Phi_j^{\varphi}(z) = \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-s}\, \frac{1}{\sqrt{2}} \big(A_s(\partial_j \varphi)\big)(z)\, \frac{ds}{\sqrt{s}}, \qquad (A_s w)(z) = \int w\big(e^{-s} z + \sqrt{1 - e^{-2s}}\, \zeta\big) d\gamma_n(\zeta),$$

for the subordination integral appearing on the right-hand side of `PVCoreSmooth` and of `RieszSemigroupPointwi`. We claim

$$y \mapsto K_{\gamma,j}(z,y)\varphi(y) \text{ is Lebesgue integrable on } \mathbb{R}^n, \quad \text{and} \quad \Phi_j^{\varphi}(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\varphi(y)\, dy. \tag{2}$$

Indeed, let $0 < \varepsilon \le \sigma$. Clause (1) of `PVCoreSmooth`, applied to φ, j, z and this ε, says that $y \mapsto K_{\gamma,j}(z,y)\varphi(y)$ is Lebesgue integrable on $\{y : |z-y| > \varepsilon\}$. Its complement is contained in $\{y : |z-y| \le \sigma\}$, where φ vanishes identically, so — by the same elementary fact used in Step 2 — the function is integrable on $\mathbb{R}^n$ and

$$\int_{\{|z-y|>\varepsilon\}} K_{\gamma,j}(z,y)\varphi(y)\, dy = \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\varphi(y)\, dy \qquad (0 < \varepsilon \le \sigma).$$

Thus the function $\varepsilon \mapsto \int_{\{|z-y|>\varepsilon\}} K_{\gamma,j}(z,y)\varphi(y)\,dy$ is constant on $(0,\sigma]$, so its limit as $\varepsilon \downarrow 0$ exists and equals that constant. Clause (2) of `PVCoreSmooth` identifies that limit as $\Phi_j^{\varphi}(z)$. This is (2).

Step 4: the approximating sequence. First fix a cut-off. Since $0 < 7\rho/8 < 15\rho/16$, there is a smooth bump function $b : \mathbb{R}^n \to [0,1]$ with $b = 1$ on $\overline{B}(x_0, 7\rho/8)$ and $\operatorname{supp} b \subseteq B(x_0, 15\rho/16)$: this is the standard radial bump at centre x_0 with inner radius $7\rho/8$ and outer radius $15\rho/16$, which exists for any pair of radii $0 < r_{\text{in}} < r_{\text{out}}$. Set $\chi = 1 - b$, so that

$$\chi \text{ is smooth}, \quad 0 \le \chi \le 1, \quad \chi = 0 \text{ on } \overline{B}(x_0, \tfrac{7\rho}{8}), \quad \chi = 1 \text{ off } B(x_0, \tfrac{15\rho}{16}),$$

the last because b vanishes off its support. Let $k \in \mathbb{N}$. By `SmoothCompactDenseL1Gauss` applied to g and $\delta = 1/(k+1) > 0$ there is a smooth, compactly supported, γ_n-integrable φ_k with $\|g - \varphi_k\|_1 < 1/(k+1)$; that node states the bound for the $L^1(\gamma_n)$ seminorm read in $[0,+\infty]$, and the bound in particular makes it finite, so $g - \varphi_k$ is γ_n-integrable and the seminorm is the ordinary integral $\int |g - \varphi_k|\, d\gamma_n$. Put $\psi_k = \chi\,\varphi_k$. Then:

- ψ_k is smooth, being a product of smooth functions, and $\{\psi_k \neq 0\} \subseteq \{\varphi_k \neq 0\}$, so $\operatorname{supp}\psi_k \subseteq \operatorname{supp}\varphi_k$ is compact;
- $\psi_k = 0$ on $\overline{B}(x_0, 7\rho/8)$, hence in particular λ-almost everywhere on $B(x_0, 3\rho/4)$;
- ψ_k is bounded (continuous with compact support) and γ_n is a probability measure, hence finite (`GaussIsProb`), so ψ_k is γ_n-integrable and also square γ_n-integrable.

Moreover

$$\|g - \psi_k\|_1 \le \int |g - \chi g|\, d\gamma_n + \int |\chi|\,|g - \varphi_k|\, d\gamma_n \le \int |(1-\chi)g|\, d\gamma_n + \|g - \varphi_k\|_1,$$

using $0 \le \chi \le 1$. The first term vanishes: $1 - \chi$ is 0 off $B(x_0, 15\rho/16)$, so $(1-\chi)g$ vanishes off $B(x_0, 15\rho/16) \subseteq B(x_0, \rho)$, and on $B(x_0, \rho)$ we have $g = 0$ γ_n-almost everywhere by hypothesis; so $(1-\chi)g = 0$ γ_n-almost everywhere. Therefore

$$\|g - \psi_k\|_1 < \frac{1}{k+1}. \tag{3}$$

Write $u_k \in L^2(\gamma_n)$ for the class of ψ_k and $\iota\psi_k = \iota u_k \in L^1(\gamma_n)$ for its image under the inclusion of `ToL1`; by the definition of that inclusion the canonical representative of $\iota\psi_k$ equals ψ_k γ_n-almost everywhere, so $\|g - \iota\psi_k\|_{L^1(\gamma_n)} = \|g - \psi_k\|_1 < 1/(k+1)$.

Step 5: the right-hand side converges, uniformly on the ball. Let $z \in B(x_0, \rho/2)$. By Step 4 the function ψ_k satisfies the hypotheses of Step 2, so (1) holds for it; and (1) holds for g by Step 2. Both $K_{\gamma,j}(z,\cdot)\psi_k$ and $K_{\gamma,j}(z,\cdot)g$ are integrable on Ω_z, so by linearity of the integral there and by clause (1)(b) of `GaussRieszKernelFarMajorant` applied to the γ_n-integrable function $\psi_k - g$ at the point z (recall $|z| \le R$),

$$\Big|\int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\psi_k(y)\,dy - \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)g(y)\,dy\Big| = \Big|\int_{\Omega_z} K_{\gamma,j}(z,y)\big(\psi_k - g\big)(y)\,dy\Big| \le C\,\|\psi_k - g\|_1 < \frac{C}{k+1}, \tag{4}$$

the bound being independent of $z \in B(x_0, \rho/2)$.

Step 6: the left-hand side converges in measure. Let $t > 0$ and $k \in \mathbb{N}$. Applying `CoordinateEndpointBound` — to this T, with the two properties it requires, which are (T1) and (T2) and hold by assumption — to the element $g - \iota\psi_k$ of $L^1(\gamma_n)$, to the coordinate j and to t,

$$t\,\gamma_n\big(\{z : |\,(T(g - \iota\psi_k))_j(z)\,| > t\}\big) \le 2\,\|g - \iota\psi_k\|_{L^1(\gamma_n)} < \frac{2}{k+1}.$$

Since T is linear, $T(g - \iota\psi_k) = Tg - T(\iota\psi_k)$ as almost-everywhere classes, and the canonical representative of a difference of such classes is γ_n-almost everywhere the difference of the representatives. The two sets $\{z : |(T(g - \iota\psi_k))_j(z)| > t\}$ and $\{z : |(Tg)_j(z) - (T(\iota\psi_k))_j(z)| > t\}$ therefore differ inside a γ_n-null set and have the same γ_n-measure, so

$$t\,\gamma_n\big(\{z : |\,(Tg)_j(z) - (T(\iota\psi_k))_j(z)\,| > t\}\big) < \frac{2}{k+1}.$$

Hence for each fixed $t > 0$ the measures $\gamma_n(\{|(Tg)_j - (T(\iota\psi_k))_j| > t\})$ tend to 0 as $k \to \infty$; that is, $(T(\iota\psi_k))_j \to (Tg)_j$ in γ_n-measure. Since γ_n is a finite measure (`GaussIsProb`), a sequence converging in measure has a subsequence converging almost everywhere; fix $k_1 < k_2 < \cdots$ with

$$(T(\iota\psi_{k_m}))_j(z) \longrightarrow (Tg)_j(z) \qquad \text{for } \gamma_n\text{-almost every } z. \tag{5}$$

Step 7: identification of each approximant, and the conclusion. Fix m. The function ψ_{k_m} is smooth with compact support, and for every $z \in B(x_0, \rho/2)$ it vanishes on $\{y : |z - y| \le \rho/4\}$, because by Step 1(b) such a y lies in $B(x_0, 7\rho/8)$ where ψ_{k_m} is zero. So Step 3 applies with $\sigma = \rho/4$ and gives

$$\Phi_j^{\psi_{k_m}}(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\psi_{k_m}(y)\,dy \qquad \text{for every } z \in B(x_0, \rho/2). \tag{6}$$

On the other hand, clause (3) of `RieszSemigroupPointwise`, applied to the smooth compactly supported ψ_{k_m} (which is square γ_n-integrable by Step 4) and to j, states that $\Phi_j^{\psi_{k_m}}$ agrees γ_n-almost everywhere with the jth coordinate of $R_\gamma u_{k_m}$, u_{k_m} being the $L^2(\gamma_n)$ class of ψ_{k_m}. By (T1), $T(\iota u_{k_m}) = R_\gamma u_{k_m}$ as almost-everywhere classes, so their representatives agree γ_n-almost everywhere. Combining,

$$(T(\iota\psi_{k_m}))_j(z) = \Phi_j^{\psi_{k_m}}(z) \qquad \text{for } \gamma_n\text{-almost every } z. \tag{7}$$

Now intersect countably many sets of full γ_n-measure: the one from (5), and for each m the one from (7). A countable intersection of conull sets is conull. Let z belong to it and suppose $z \in B(x_0, \rho/2)$. By (7) and (6), $(T(\iota\psi_{k_m}))_j(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\psi_{k_m}(y)\,dy$ for every m. Letting $m \to \infty$, the left-hand side tends to $(Tg)_j(z)$ by (5), and the right-hand side tends to $\int_{\mathbb{R}^n} K_{\gamma,j}(z,y)g(y)\,dy$ by (4), since $C/(k_m + 1) \to 0$. A convergent sequence of reals has a unique limit, so

$$(Tg)_j(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)\,g(y)\,dy.$$

As this holds for every z in a set of full γ_n-measure, clause (2) follows. □

2.140. `TauPowerIntegralBounds`.

Lemma 2.140 (Two integral bounds for negative powers of τ)**.** *Let λ be Lebesgue measure on $\mathbb{R}$. Then:*

1. *for every real $a > 1$ there is a constant $K = K(a) > 0$ such that for every real $b > 0$ the function $\tau \mapsto \tau^{-a}e^{-b/\tau}$ is Lebesgue integrable on $(0, \infty)$ and*
$$\int_0^\infty \tau^{-a}e^{-b/\tau}\,d\tau \le K\,b^{1-a};$$
2. *for all real $q > 1$ and c with $0 < c \le 1$, the function $\tau \mapsto \tau^{-q}$ is Lebesgue integrable on $(c, 1)$ and*
$$\int_c^1 \tau^{-q}\,d\tau \le \frac{c^{1-q}}{q-1}.$$

Here τ^{-a}, τ^{-q} and b^{1-a}, c^{1-q} are real powers of positive reals.

Why integrability is part of both statements. *Each displayed integral is a Bochner integral, whose value is 0 by convention when the integrand is not integrable, and 0 satisfies every upper bound; so either inequality on its own would be provable with no information about the integrand. The integrability assertions remove that, and they are also what a consumer needs in order to split the integral over a subinterval or add two such bounds.*

Why clause (1) is a bound with an unnamed constant and not the exact value. *The constant may be taken to be $K(a) = \int_0^\infty u^{-a}e^{-1/u}\,du$, which the proof below shows is finite and positive and which the substitution $u = 1/s$ identifies with $\Gamma(a-1)$; the proof in fact gives equality, $\int_0^\infty \tau^{-a}e^{-b/\tau}d\tau = K(a)\,b^{1-a}$. The clause is stated as an inequality with an existentially quantified K because that is all any consumer uses: what matters is the exact power b^{1-a} of b, which is what converts a Gaussian decay rate into a negative power of a distance, while the value of K*

is absorbed into a constant depending only on the dimension. An inequality with an existential constant is weaker than the identity, so nothing is claimed here that the identity does not give.

The dependence on b is exact, and that is the point. *Replacing b^{1-a} by any larger power of b — for instance by the bound $\int_0^1 \tau^{-a}e^{-b/\tau}d\tau \le C_m b^{-m}$ with $m > a-1$, which follows from the elementary decay bound alone without any change of variables — would be useless to a consumer, because the exponent of b is what fixes the exponent of the distance in the final estimate. The change of variables is therefore not an optimization but the content of the clause.*

Proof. An elementary decay bound. For every integer $m \ge 1$ and every real $t > 0$,

$$e^{-t} \le m^m\, t^{-m}. \tag{†}$$

Indeed $1 + s \le e^s$ for all real s; taking $s = t/m$ gives $t/m \le 1 + t/m \le e^{t/m}$. Both sides are nonnegative, so raising to the mth power preserves the inequality: $(t/m)^m \le (e^{t/m})^m = e^t$, that is $t^m/m^m \le e^t$. Since $t^m > 0$ and $e^t > 0$, rearranging gives $e^{-t} = 1/e^t \le m^m/t^m$, which is (†).

Clause (1): integrability. Fix $a > 1$ and choose an integer $m \ge 1$ with $a \le m$ (for instance $m = \lceil a \rceil$, which is at least 1 because $a > 1$). Fix $b > 0$. The function $\tau \mapsto \tau^{-a}e^{-b/\tau}$ is continuous on $(0,\infty)$, since $\tau \mapsto \tau^{-a}$ is continuous where $\tau \ne 0$ and $\tau \mapsto -b/\tau$ is continuous there, so it is Borel measurable on every subset of $(0,\infty)$. We bound it on two pieces whose union is $(0,\infty)$.

On $(0,1)$: applying (†) with $t = b/\tau > 0$ gives

$$e^{-b/\tau} \le m^m \Big(\frac{b}{\tau}\Big)^{-m} = m^m \frac{\tau^m}{b^m} = \frac{m^m}{b^m}\,\tau^m,$$

so, multiplying by $\tau^{-a} > 0$ and using $\tau^{-a}\tau^m = \tau^{m-a}$,

$$\tau^{-a}e^{-b/\tau} \le \frac{m^m}{b^m}\,\tau^{m-a} \qquad (0 < \tau < 1).$$

Since $a \le m$ we have $m - a \ge 0 > -1$, and $\tau \mapsto \tau^s$ is Lebesgue integrable on $(0,1)$ exactly when $s > -1$; hence the right-hand side is integrable on $(0,1)$. A measurable function dominated in absolute value by an integrable function is integrable, and the integrand is nonnegative, so $\tau \mapsto \tau^{-a}e^{-b/\tau}$ is integrable on $(0,1)$.

On $[1,\infty)$: for $\tau \ge 1$ we have $b/\tau \ge 0$, hence $-b/\tau \le 0$ and $e^{-b/\tau} \le 1$, so $\tau^{-a}e^{-b/\tau} \le \tau^{-a}$. Since $-a < -1$, the function $\tau \mapsto \tau^{-a}$ is Lebesgue integrable on $(1,\infty)$; as the single point $\{1\}$ is Lebesgue-null, integrability on $(1,\infty)$ and on $[1,\infty)$ are equivalent. Domination again gives integrability of $\tau \mapsto \tau^{-a}e^{-b/\tau}$ on $[1,\infty)$.

As $(0,\infty) = (0,1) \cup [1,\infty)$, the function is integrable on $(0,\infty)$.

Clause (1): the constant and the scaling. Set

$$K = \int_0^\infty u^{-a}e^{-1/u}\,du,$$

which is the case $b = 1$ just treated, hence a finite real number. It is positive: the integrand $u \mapsto u^{-a}e^{-1/u}$ is strictly positive at every $u > 0$, so its support contains $(0,1)$, a set of Lebesgue measure $1 > 0$; a nonnegative integrable function whose support has positive measure has positive integral.

Now fix $b > 0$ and let $g(\tau) = \tau^{-a}e^{-b/\tau}$. The map $v \mapsto bv$ is a scaling of $\mathbb{R}$ by the positive factor b which carries $(0,\infty)$ onto $(0,\infty)$ and multiplies Lebesgue measure by b, so

$$\int_0^\infty g(bv)\,dv = b^{-1}\int_0^\infty g(\tau)\,d\tau.$$

For $v > 0$,

$$g(bv) = (bv)^{-a}\exp\Big(-\frac{b}{bv}\Big) = b^{-a}v^{-a}e^{-1/v},$$

using $(bv)^{-a} = b^{-a}v^{-a}$ for $b, v > 0$ and $b/(bv) = 1/v$. Therefore

$$b^{-1}\int_0^\infty \tau^{-a}e^{-b/\tau}\,d\tau = \int_0^\infty b^{-a}v^{-a}e^{-1/v}\,dv = b^{-a}K,$$

and multiplying by $b > 0$,

$$\int_0^\infty \tau^{-a} e^{-b/\tau}\, d\tau = b \cdot b^{-a} K = b^{1-a} K,$$

using $b^1 b^{-a} = b^{1-a}$. In particular the asserted inequality holds, with equality, and K depends only on a.

Clause (2). Let $q > 1$ and $0 < c \le 1$. On the closed interval $[c, 1]$ the function $\tau \mapsto \tau^{-q}$ is continuous, because $c > 0$ so $\tau \neq 0$ throughout; a continuous function on a compact interval is Lebesgue integrable there, and $(c, 1) \subseteq [c, 1]$, so it is integrable on $(c, 1)$.

For the value: $0 \notin [c, 1]$ because $c > 0$, and $-q \neq -1$ because $q > 1$, so the power function $\tau \mapsto \tau^{-q}$ has the antiderivative $\tau \mapsto \tau^{-q+1}/(-q+1)$ on $[c, 1]$ and the fundamental theorem of calculus gives

$$\int_c^1 \tau^{-q}\, d\tau = \frac{1^{-q+1} - c^{-q+1}}{-q+1} = \frac{1 - c^{1-q}}{1-q}.$$

Since $1 - q = -(q-1)$,

$$\frac{1 - c^{1-q}}{1-q} = \frac{c^{1-q} - 1}{q-1},$$

and because $q - 1 > 0$ and $c^{1-q} - 1 \le c^{1-q}$, dividing the latter inequality by $q - 1$ gives

$$\int_c^1 \tau^{-q}\, d\tau = \frac{c^{1-q} - 1}{q-1} \le \frac{c^{1-q}}{q-1},$$

which is clause (2). □

2.141. `GaussRieszKernelNearSliceBounds`.

Lemma 2.141 (Pointwise slice bounds and the slice gradient for $K_{\gamma,j}$)**.** *Fix $n \ge 1$ and $j \in \{1, \dots, n\}$, and let*

$$\Phi_{w,y}(r) = \frac{y_j - r w_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|rw - y|^2}{1-r^2}\Big), \qquad w, y \in \mathbb{R}^n,\ r \in (0,1),$$

be the r-integrand of the normalized Gaussian Riesz kernel of `GaussRieszKernel`*. Let $M \ge 3$ and let $y \in \mathbb{R}^n$ satisfy $|y| \le M$. Write $\tau = 1 - r$, so that $\tau \in (0,1)$ for $r \in (0,1)$. Then:*

(1) *for all $w \in \mathbb{R}^n$ and all $r \in (0,1)$,*

$$\frac{|rw - y|^2}{1-r^2} = \big(|y|^2 - |w|^2\big) + \frac{|w - ry|^2}{1-r^2};$$

(2) *for all $w \in \mathbb{R}^n$ with $|w| \le M$ and all $r \in (0,1)$,*

$$\big|\Phi_{w,y}(r)\big| \le e^{M^2} \big(|w - y| + \tau M\big)\, \tau^{-n/2-3/2} \exp\Big(-\frac{|w - ry|^2}{2\tau}\Big);$$

(3) *there is a map G assigning to each $w \in \mathbb{R}^n$ and each $r \in (0,1)$ a vector $G(w,r) \in \mathbb{R}^n$ such that*
   (a) *for every $w \in \mathbb{R}^n$ the map $r \mapsto G(w,r)$ is continuous on $(0,1)$; and*
   (b) *for all $w \in \mathbb{R}^n$ with $|w| \le M$ and all $r \in (0,1)$, the map $v \mapsto \Phi_{v,y}(r)$ is (Fréchet) differentiable at w with derivative the continuous linear functional $h \mapsto \langle G(w,r), h\rangle$, and*

$$|G(w,r)| \le e^{M^2}\, \tau^{-n/2-3/2} \Big(1 + \frac{2\big(|w - y| + \tau M\big)^2}{\tau}\Big) \exp\Big(-\frac{|w - ry|^2}{2\tau}\Big).$$

Why clause (3) produces the gradient rather than bounding a chosen derivative. *A bound on a derivative operator selected by convention where a function is not differentiable is vacuous wherever differentiability fails. Clause (3)(b) therefore asserts genuine differentiability at w and bounds the vector that represents the actual derivative there. Clause (3)(a) records that the representing vector may be chosen to depend continuously on r; this is what makes $r \mapsto G(w,r)$ measurable on $(0,1)$, which is one of the hypotheses of the theorem on differentiation under the integral sign that a consumer applies to $\int_0^1 \Phi_{w,y}(r)\, dr$.*

Clause (1) carries no hypothesis on w, y or M. *It is an algebraic identity valid for every $w, y \in \mathbb{R}^n$ and every $r \in (0,1)$; the hypotheses $M \geq 3$ and $|y| \leq M$ are used only in clauses (2) and (3). Clause (1) is stated because it is the step that makes the factor $e^{|w|^2-|y|^2}$ visible, and it is what both of the other clauses are proved from.*

The bounds are stated for a general w, not at a single base point. *A consumer of clause (3) needs a bound on $|G(w,r)|$ that is valid simultaneously at every w of a ball, with the dependence on w exhibited through $|w-y|$, in order to produce a majorant of $G(\cdot, r)$ that is independent of w on that ball. A version pinned at one point would not serve that use.*

Proof. Throughout fix $r \in (0,1)$ and write $\tau = 1 - r \in (0,1)$,

$$c = (1-r^2)^{n/2+1}\sqrt{-\log r}, \qquad A(w,r) = \exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big),$$

so that $\Phi_{w,y}(r) = (y_j - rw_j)\,A(w,r)/c$. Note $1 - r^2 = \tau(1+r)$, so $0 < \tau \leq 1 - r^2 \leq 2\tau$; and $\log r \leq r - 1 = -\tau$, so $-\log r \geq \tau > 0$. In particular $\sqrt{-\log r} > 0$ and $c > 0$, and $A(w,r) > 0$.

Clause (1). Expanding squares,

$$|rw - y|^2 = r^2|w|^2 - 2r\langle w, y\rangle + |y|^2, \qquad |w - ry|^2 = |w|^2 - 2r\langle w, y\rangle + r^2|y|^2.$$

Hence

$$\begin{aligned}(1-r^2)\big(|y|^2 - |w|^2\big) + |w - ry|^2 &= |y|^2 - r^2|y|^2 - |w|^2 + r^2|w|^2 + |w|^2 - 2r\langle w, y\rangle + r^2|y|^2 \\ &= |y|^2 + r^2|w|^2 - 2r\langle w, y\rangle = |rw - y|^2.\end{aligned}$$

Dividing by $1 - r^2 > 0$ gives clause (1).

The common estimate $(*)$. We claim that for $|w| \leq M$ and $r \in (0,1)$,

$$\frac{A(w,r)}{c} \leq e^{M^2}\,\tau^{-n/2-3/2}\exp\Big(-\frac{|w-ry|^2}{2\tau}\Big). \tag{$*$}$$

For the numerator, clause (1) gives $-|rw-y|^2/(1-r^2) = |w|^2 - |y|^2 - |w-ry|^2/(1-r^2)$, so

$$A(w,r) = e^{|w|^2-|y|^2}\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big).$$

Since $|w| \leq M$ and $|y|^2 \geq 0$ we have $|w|^2 - |y|^2 \leq M^2$, so $e^{|w|^2-|y|^2} \leq e^{M^2}$ by monotonicity of exp. Since $0 < 1 - r^2 \leq 2\tau$ and $|w - ry|^2 \geq 0$ we have $|w-ry|^2/(2\tau) \leq |w-ry|^2/(1-r^2)$, hence $\exp(-|w-ry|^2/(1-r^2)) \leq \exp(-|w-ry|^2/(2\tau))$. Multiplying the two (both factors being positive) bounds $A(w,r)$ by $e^{M^2}\exp(-|w-ry|^2/(2\tau))$.

For the denominator, $0 < \tau \leq 1 - r^2$ and $n/2 + 1 \geq 0$ give $\tau^{n/2+1} \leq (1-r^2)^{n/2+1}$ by monotonicity of $t \mapsto t^{n/2+1}$ on $[0,\infty)$; and $0 \leq \tau \leq -\log r$ gives $\tau^{1/2} = \sqrt{\tau} \leq \sqrt{-\log r}$. Both factors on the left being nonnegative, multiplying the two inequalities gives $\tau^{n/2+3/2} = \tau^{n/2+1}\tau^{1/2} \leq c$, and $\tau^{n/2+3/2} > 0$, so $c^{-1} \leq \tau^{-n/2-3/2}$. Multiplying the numerator and denominator bounds (all quantities being nonnegative) gives $(*)$.

The two numerator bounds. Let $|w| \leq M$. Since $y_j - rw_j = (y_j - w_j) + \tau w_j$ and $|y_j - w_j| = |(y-w)_j| \leq |y - w| = |w - y|$ and $|\tau w_j| = \tau|w_j| \leq \tau|w| \leq \tau M$,

$$|y_j - rw_j| \leq |w - y| + \tau M. \tag{6a}$$

Since $rw - y = (w - y) - \tau w$ and $|\tau w| = \tau|w| \leq \tau M$,

$$|rw - y| \leq |w - y| + \tau M. \tag{6b}$$

Clause (2). Because $c > 0$ and $A(w,r) > 0$,

$$\big|\Phi_{w,y}(r)\big| = \big|y_j - rw_j\big| \cdot \frac{A(w,r)}{c}.$$

Both factors are nonnegative, and the right-hand sides of (6a) and of $(*)$ are nonnegative, so multiplying (6a) by $(*)$ gives clause (2).

Clause (3): the gradient. Fix $r \in (0,1)$. The map $v \mapsto y_j - rv_j$ is affine, hence differentiable at every w with derivative $h \mapsto -rh_j = \langle -re_j, h\rangle$, where e_j is the jth standard basis vector (so $h_j = \langle e_j, h\rangle$ and $|e_j| = 1$). The map $v \mapsto rv - y$ is affine with derivative $h \mapsto rh$, so $v \mapsto |rv - y|^2$

is differentiable at w with derivative $h \mapsto 2\langle rw-y, rh\rangle = 2r\langle rw-y, h\rangle$; dividing by the constant $1-r^2$, negating, and composing with exp, the map $v \mapsto A(v,r)$ is differentiable at w with derivative

$$h \mapsto A(w,r)\cdot\Big(-\frac{2r}{1-r^2}\Big)\langle rw-y, h\rangle.$$

By the product rule and division by the constant c, the map $v \mapsto \Phi_{v,y}(r)$ is differentiable at every $w \in \mathbb{R}^n$, with derivative

$$h \mapsto \frac{A(w,r)}{c}\Big[-r\langle e_j, h\rangle - (y_j - rw_j)\frac{2r}{1-r^2}\langle rw-y, h\rangle\Big] = \langle G(w,r), h\rangle,$$

where we define

$$G(w,r) := \frac{A(w,r)}{c}\Big(-r\,e_j - (y_j - rw_j)\frac{2r}{1-r^2}\,(rw-y)\Big).$$

This is a continuous linear functional of h, being an inner product against a fixed vector. So clause (3)(b)'s differentiability assertion holds, with this G, for every w — in particular for every w with $|w| \le M$.

Clause (3)(a): continuity in r. Fix $w \in \mathbb{R}^n$ and $r \in (0,1)$. On $(0,1)$ the functions $r \mapsto 1-r^2$ and $r \mapsto -\log r$ are continuous and strictly positive (as noted at the start), so $r \mapsto (1-r^2)^{n/2+1}$ and $r \mapsto \sqrt{-\log r}$ are continuous and strictly positive, and hence so is $r \mapsto c$. Also $r \mapsto |rw-y|^2$, $r \mapsto y_j - rw_j$ and $r \mapsto rw-y$ are continuous, so $r \mapsto A(w,r)$ is continuous, and $r \mapsto 2r/(1-r^2)$ is continuous because its denominator does not vanish. A quotient of continuous functions with nonvanishing denominator is continuous, and sums, products and scalar multiples of continuous maps are continuous; therefore $r \mapsto G(w,r)$ is continuous on $(0,1)$.

Clause (3)(b): the bound. Let $|w| \le M$. By the triangle inequality and homogeneity of the norm,

$$|G(w,r)| \le \frac{A(w,r)}{c}\Big(r\,|e_j| + |y_j - rw_j|\,\frac{2r}{1-r^2}\,|rw-y|\Big),$$

using that $A(w,r)/c > 0$ and $2r/(1-r^2) > 0$. Now $|e_j| = 1$ and $r \le 1$, so the first summand is at most 1. For the second, $2r/(1-r^2) \le 2/\tau$: indeed both denominators are positive and $2r\tau \le 2\tau(1+r) = 2(1-r^2)$ because $r \le 1+r$. Combining this with $(6a)$ and $(6b)$, and using that all the factors involved are nonnegative,

$$|y_j - rw_j|\,\frac{2r}{1-r^2}\,|rw-y| \le \big(|w-y| + \tau M\big)\frac{2}{\tau}\big(|w-y| + \tau M\big) = \frac{2\big(|w-y| + \tau M\big)^2}{\tau}.$$

Hence

$$|G(w,r)| \le \frac{A(w,r)}{c}\Big(1 + \frac{2(|w-y| + \tau M)^2}{\tau}\Big),$$

and since the bracket is nonnegative, $(*)$ bounds the right-hand side by

$$e^{M^2}\,\tau^{-n/2-3/2}\Big(1 + \frac{2(|w-y| + \tau M)^2}{\tau}\Big)\exp\Big(-\frac{|w-ry|^2}{2\tau}\Big),$$

which is clause (3)(b)'s bound. □

2.142. `RpowPowerReductionsToNegPower`.

Lemma 2.142 (Reduction of the near-diagonal power factors to a single negative power of d)**.** *Fix $n \ge 1$ with $1 \le n$ as a real number, and let $M \ge 3$ and $0 < d \le 2$. Write $a = n/2 + 3/2$. Then all five of the following hold, every power being a real power.*

(1) $\big(\frac{9}{128}d^2\big)^{1-a} = \big(\frac{9}{128}\big)^{1-a} d^{-(n+1)}$.

(2) $8d^2\big(\frac{9}{128}d^2\big)^{1-(n/2+5/2)} = 8\big(\frac{9}{128}\big)^{-a} d^{-(n+1)}$.

(3) $\big(d/(8M/3)\big)^{-a} \le (8M/3)^a\, 2^{-a+(n+1)}\, d^{-(n+1)}$.

(4) $8d^2\big(d/(8M/3)\big)^{-(n/2+5/2)} \le 8\,(8M/3)^{n/2+5/2}\, 2^{1-a+(n+1)}\, d^{-(n+1)}$.

(5) $4M^2\big(d/(8M/3)\big)^{-(n/2+1/2)} \le 4M^2(8M/3)^{n/2+1/2}\, 2^{1-a+(n+1)}\, d^{-(n+1)}$.

Proof. Four elementary rules are used throughout; in each, $d > 0$ so every real power of d is defined and positive.

(i) Splitting a scaled square. For $\kappa > 0$ and real s, $(\kappa d^2)^{-s} = \kappa^{-s} d^{-2s}$: the product rule for real powers of positive bases gives $(\kappa d^2)^{-s} = \kappa^{-s}(d^2)^{-s}$, and reading d^2 as the real power d^2 and composing exponents gives $(d^2)^{-s} = d^{-2s}$.

(ii) Splitting a quotient. For $u > 0$ and real s, $(d/u)^{-s} = d^{-s}u^s$, by the quotient rule for real powers of positive bases together with $u^{-s} = (u^s)^{-1}$.

(iii) Absorbing a square. For real β, $d^2 d^\beta = d^{\beta+2}$, by adding exponents.

(iv) The scale comparison. If $\beta + \gamma \geq 0$ then $d^\beta \leq 2^{\beta+\gamma} d^{-\gamma}$ for every $0 < d \leq 2$. Indeed $d^{\beta+\gamma} \leq 2^{\beta+\gamma}$ because $d \leq 2$ and the exponent is nonnegative, and multiplying by $d^{-\gamma} > 0$ gives the claim after adding exponents on the left.

Clauses (1) and (2). Since $1 - a = -(n/2 + 1/2)$, rule (i) with $\kappa = \frac{9}{128}$ and $s = n/2 + 1/2$ gives $(\frac{9}{128} d^2)^{1-a} = (\frac{9}{128})^{1-a} d^{-(n+1)}$, because $2s = n + 1$. This is clause (1). For clause (2), $1 - (n/2 + 5/2) = -a$, so rule (i) with $s = a$ gives $(\frac{9}{128} d^2)^{1-(n/2+5/2)} = (\frac{9}{128})^{-a} d^{-(n+3)}$, as $2a = n + 3$; multiplying by $8d^2$ and applying rule (iii) with $\beta = -(n + 3)$, whose effect is $d^2 d^{-(n+3)} = d^{-(n+1)}$, gives clause (2).

Clauses (3), (4) and (5). Each starts from rule (ii) with $u = 8M/3 > 0$, which is positive because $M \geq 3$. For clause (3), rule (ii) with $s = a$ turns the left side into $d^{-a}(8M/3)^a$, and rule (iv) with $\beta = -a$ and $\gamma = n + 1$ applies because $-a + (n + 1) = n/2 - 1/2 \geq 0$ by $n \geq 1$; multiplying the resulting bound $d^{-a} \leq 2^{-a+(n+1)} d^{-(n+1)}$ by the nonnegative constant $(8M/3)^a$ gives clause (3).

For clause (4), rule (ii) with $s = n/2 + 5/2$ gives $d^{-(n/2+5/2)}(8M/3)^{n/2+5/2}$; rule (iii) with $\beta = -(n/2 + 5/2)$ turns $d^2 d^{-(n/2+5/2)}$ into d^{1-a}, and rule (iv) with $\beta = 1 - a$ and $\gamma = n + 1$ applies because $1 - a + (n + 1) = n/2 + 1/2 \geq 0$. Multiplying by the nonnegative constant $8(8M/3)^{n/2+5/2}$ gives clause (4).

For clause (5), rule (ii) with $s = n/2+1/2$ gives $d^{-(n/2+1/2)}(8M/3)^{n/2+1/2}$, and $-(n/2+1/2) = 1 - a$, so rule (iv) applies exactly as in clause (4); multiplying by the nonnegative constant $4M^2(8M/3)^{n/2+1/2}$ gives clause (5). □

2.143. `GaussRieszKernelNearBallUniformTauMajorant.`

Lemma 2.143 (A w-uniform τ-majorant on a ball for the near-diagonal gradient)**.** *Fix $n \geq 1$ and $M \geq 3$. Let $x, y \in \mathbb{R}^n$ with $|y| \leq M$ and $d := |x - y| > 0$, and put $U = B(x, d/4)$ and $\tau'' = 3d/(8M)$. Let $G : \mathbb{R}^n \times (0, 1) \to \mathbb{R}^n$ satisfy, for every w with $|w| \leq M$ and every $r \in (0, 1)$,*

$$|G(w, r)| \leq e^{M^2}(1 - r)^{-n/2-3/2}\Big(1 + \frac{2\big(|w - y| + (1 - r)M\big)^2}{1 - r}\Big) \exp\Big(-\frac{|w - ry|^2}{2(1 - r)}\Big),$$

and suppose every $w \in U$ satisfies $|w| \leq M$ and $\frac{3}{4}d \leq |w - y| \leq \frac{5}{4}d$. Then for every $w \in U$ and every $\tau \in (0, 1)$,

$$|G(w, 1-\tau)| \leq e^{M^2}\Big(\tau^{-n/2-3/2}+8d^2\,\tau^{-n/2-5/2}\Big)e^{-9d^2/(128\tau)}+e^{M^2}\Big((\tau'')^{-n/2-3/2}+8d^2(\tau'')^{-n/2-5/2}+4M^2(\tau'')^{-n/2-}$$

Proof. Fix $w \in U$ and $\tau \in (0, 1)$, and write $r = 1 - \tau \in (0, 1)$. By hypothesis $|w| \leq M$ and $\frac{3}{4}d \leq |w-y| \leq \frac{5}{4}d$. Both groups on the right are nonnegative, so it suffices to bound $|G(w, 1-\tau)|$ by one of them in each of two complementary cases.

Step 0: the separation estimate. Since $w - (1 - \tau)y = (w - y) + \tau y$ and $|y| \leq M$, the triangle inequality in the form $|a + b| \geq |a| - |b|$ gives

$$|w - (1 - \tau)y| \geq |w - y| - \tau M \geq \tfrac{3}{4}d - \tau M. \tag{1}$$

Applying the hypothesis on G at this w and $r = 1 - \tau$, and using $|w - y| \leq \frac{5}{4}d$ in the increasing factor,

$$|G(w, 1 - \tau)| \leq e^{M^2}\tau^{-n/2-3/2}\Big(1 + \frac{2\big(\frac{5}{4}d + \tau M\big)^2}{\tau}\Big) \exp\Big(-\frac{|w - (1 - \tau)y|^2}{2\tau}\Big). \tag{2}$$

Step 1: small τ. Suppose $\tau \le \tau'' = 3d/(8M)$. Multiplying by $8M/3 > 0$ gives $\tau M \le \frac{3}{8}d$, so by (1) we get $|w-(1-\tau)y| \ge \frac{3}{4}d - \frac{3}{8}d = \frac{3}{8}d \ge 0$; squaring the nonnegative sides and dividing by $2\tau > 0$,

$$\exp\Big(-\frac{|w-(1-\tau)y|^2}{2\tau}\Big) \le \exp\Big(-\frac{9d^2}{128\tau}\Big).$$

Also $\frac{5}{4}d + \tau M \le \frac{5}{4}d + \frac{3}{8}d = \frac{13}{8}d \le 2d$, so $2(\frac{5}{4}d + \tau M)^2 \le 8d^2$ and the bracket in (2) is at most $1 + 8d^2/\tau$. Hence (2) gives

$$|G(w,1-\tau)| \le e^{M^2}\big(\tau^{-n/2-3/2} + 8d^2\tau^{-n/2-5/2}\big)e^{-9d^2/(128\tau)},$$

using $\tau^{-n/2-3/2}/\tau = \tau^{-n/2-5/2}$. This is the first group, and the second is nonnegative, so the claim holds in this case.

Step 2: large τ. Suppose $\tau'' < \tau$. Bound the exponential in (2) by 1, legitimate because its argument is nonpositive, and use $(\frac{5}{4}d + \tau M)^2 \le 2(\frac{5}{4}d)^2 + 2(\tau M)^2 \le 4d^2 + 2\tau^2M^2$, which follows from $(p+q)^2 \le 2p^2 + 2q^2$ and $2(\frac{5}{4})^2 = \frac{25}{8} \le 4$. Then the bracket in (2) is at most $1 + (8d^2 + 4\tau^2M^2)/\tau$, so

$$|G(w,1-\tau)| \le e^{M^2}\big(\tau^{-n/2-3/2} + 8d^2\tau^{-n/2-5/2} + 4M^2\tau^{-n/2-1/2}\big),$$

using $\tau^{-n/2-3/2}\tau = \tau^{-n/2-1/2}$. Each exponent here is negative and $\tau'' \le \tau$ with $\tau'' > 0$, so $t \mapsto t^{-s}$ being nonincreasing for $s \ge 0$ gives $\tau^{-s} \le (\tau'')^{-s}$ for each of $s = \frac{n}{2}+\frac{3}{2},\ \frac{n}{2}+\frac{5}{2},\ \frac{n}{2}+\frac{1}{2}$. Hence the displayed bound is at most the second group, and the first group is nonnegative, so the claim holds in this case too. □

2.144. `OneSubReflectionIntegralTransfer.`

Lemma 2.144 (Reflection $r \leftrightarrow 1-r$ transfers set integrals and integrability between $(0,c)$ and $(1-c,1)$)**.** *Both of the following hold, for every function $g : \mathbb{R} \to \mathbb{R}$ and every real number c.*

(1) *(Integral identity)*

$$\int_{(1-c,\,1)} g(1-r)\,dr = \int_{(0,\,c)} g(\tau)\,d\tau,$$

both integrals being taken with respect to Lebesgue measure on $\mathbb{R}$ over the indicated open interval.

(2) *(Integrability transfer) If g is integrable on $(0,c)$ for Lebesgue measure, then $r \mapsto g(1-r)$ is integrable on $(1-c,1)$ for Lebesgue measure.*

No measurability or integrability hypothesis on g is assumed in clause (1), and no sign condition on c is assumed in either clause.

Proof. Write $T : \mathbb{R} \to \mathbb{R}$, $T(t) = 1-t$, and let λ denote Lebesgue measure on $\mathbb{R}$. The proof rests on three properties of T, each established before either clause is addressed; both clauses then follow in one step each.

T preserves λ. Lebesgue measure on $\mathbb{R}$ is invariant under translation and under the reflection $x \mapsto -x$, hence under $t \mapsto 1-t$, which is the composition of the two. So T is a measure-preserving map of $(\mathbb{R},\lambda)$ to itself: $\lambda(T^{-1}(A)) = \lambda(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}$.

T is a measurable embedding. T is an involution, $T(T(t)) = t$, and it is continuous with continuous inverse T itself, so it is a homeomorphism of $\mathbb{R}$. In particular T is injective, Borel measurable, and carries measurable sets to measurable sets; that is, T is a measurable embedding.

The preimage computation. Fix $c \in \mathbb{R}$. Then

$$T^{-1}\big((0,c)\big) = (1-c,\,1).$$

Indeed, for $t \in \mathbb{R}$ the condition $T(t) \in (0,c)$ reads $0 < 1-t$ together with $1-t < c$. The first inequality is equivalent to $t < 1$ and the second to $1-c < t$, so $T(t) \in (0,c)$ holds if and only if $1-c < t < 1$, i.e. $t \in (1-c,1)$. Each of the four implications is a rearrangement of a linear inequality, so both inclusions hold and the sets are equal. No sign condition on c enters: if $c \le 0$ then $(0,c)$ is empty and so is $(1-c,1)$, because then $1-c \ge 1$, and the displayed identity reads $\emptyset = \emptyset$.

Clause (1). For a measure-preserving measurable embedding T of $(\mathbb{R},\lambda)$, every $h:\mathbb{R}\to\mathbb{R}$ and every measurable A,

$$\int_{T^{-1}(A)} h(T(t))\,d\lambda(t)=\int_A h(\tau)\,d\lambda(\tau),$$

because the image under T of λ restricted to $T^{-1}(A)$ is λ restricted to A, and a set integral depends only on that restricted measure. Apply this with $h=g$ and $A=(0,c)$ and substitute the preimage computation: since $T(t)=1-t$, the left-hand side is $\int_{(1-c,1)} g(1-r)\,dr$ and the right-hand side is $\int_{(0,c)} g(\tau)\,d\tau$, which is clause (1). The identity needs no hypothesis on g: with the convention that the integral of a non-integrable function is 0, the two sides are simultaneously given by a genuine integral or simultaneously 0, because $g\circ T$ fails to be integrable on $T^{-1}(A)$ exactly when g fails to be integrable on A — which is clause (2) read in both directions.

Clause (2). The same two properties of T give, for every $h:\mathbb{R}\to\mathbb{R}$ and every measurable A, that $h\circ T$ is integrable on $T^{-1}(A)$ if and only if h is integrable on A: the transported measures agree, so the integrals of $|h\circ T|$ over $T^{-1}(A)$ and of $|h|$ over A coincide in $[0,+\infty]$ and are finite together. Taking $h=g$ and $A=(0,c)$, and using the forward direction only, integrability of g on $(0,c)$ gives integrability of $t\mapsto g(1-t)$ on $T^{-1}((0,c))$, which by the preimage computation is $(1-c,1)$. That is clause (2). □

2.145. `GaussRieszKernelNearFDerivBound`.

Lemma 2.145 (Near-diagonal derivative bound for $K_{\gamma,j}$ by differentiation under the integral sign)**.** *Fix $n\ge 1$, a coordinate j, a radius $R>0$, and set $M=R+3$. Write $c_n=\sqrt{2}\,\pi^{-(n+1)/2}$ and, for $w,y\in\mathbb{R}^n$ and $r\in(0,1)$,*

$$\Phi_{w,y}(r)=\frac{y_j-rw_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big),$$

so that $K_{\gamma,j}(w,y)=c_n\int_0^1\Phi_{w,y}(r)\,dr$ as in `GaussRieszKernel`. Then there is a constant $C>0$, depending only on n and M, with the following property. Let $x,y\in\mathbb{R}^n$ satisfy $|x|\le R+1$, $|y|\le M$ and $0<|x-y|\le 2$, and suppose $r\mapsto\Phi_{x,y}(r)$ is integrable on $(0,1)$. Then there is a continuous linear functional g on $\mathbb{R}^n$ such that $w\mapsto K_{\gamma,j}(w,y)$ is Fréchet differentiable at x with derivative g, and

$$\|g\|\le C\,|x-y|^{-(n+1)}.$$

Proof. Throughout $\tau=1-r$, so $r\in(0,1)$ corresponds to $\tau\in(0,1)$, and the substitution $r\mapsto 1-r$ preserves Lebesgue measure on $\mathbb{R}$, being a reflection followed by a translation. Note $M=R+3\ge 3$ and $M>0$.

Step 1: the splitting identity and the elementary τ-bounds. For all $w,y\in\mathbb{R}^n$ and $r\in(0,1)$,

$$\frac{|rw-y|^2}{1-r^2}=\big(|y|^2-|w|^2\big)+\frac{|w-ry|^2}{1-r^2},\tag{1}$$

since after multiplying by $1-r^2>0$ both $|rw-y|^2-(1-r^2)(|y|^2-|w|^2)$ and $|w-ry|^2$ expand to $|w|^2-2r\langle w,y\rangle+r^2|y|^2$. Hence, whenever $|w|\le M$,

$$\exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big)=e^{|w|^2}e^{-|y|^2}\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big)\le e^{M^2}\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big),\tag{2}$$

using $e^{-|y|^2}\le 1$. Also $1-r^2=\tau(1+r)\in[\tau,2\tau]$, and $\log r=\log(1-\tau)\le-\tau$ gives $-\log r\ge\tau$, so

$$\frac{1}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\le\tau^{-n/2-3/2},\qquad\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big)\le\exp\Big(-\frac{|w-ry|^2}{2\tau}\Big).\tag{3}$$

We use two integral facts, both supplied by `TauPowerIntegralBounds`: for $a>1$ there is $K_a>0$ with

$$\int_0^\infty\tau^{-a}e^{-b/\tau}\,d\tau\le K_a\,b^{1-a}<\infty\qquad(b>0),\tag{4}$$

and for $0 < a \le 1$ and real $p \le q$ with $q > 1$,

$$\int_a^1 \tau^{-p}\,d\tau \le \int_a^1 \tau^{-q}\,d\tau = \frac{a^{1-q}-1}{q-1} \le \frac{a^{1-q}}{q-1}, \tag{5}$$

the first inequality because $\tau^{-p} \le \tau^{-q}$ on $(0,1]$.

Step 2: the pointwise gradient bound. Fix y with $|y| \le M$, let w satisfy $|w| \le M$, and put $\rho = |w-y|$. Since $y_j - rw_j = (y_j - w_j) + \tau w_j$ and $y - rw = (y-w) + \tau w$,

$$|y_j - rw_j| \le \rho + \tau M, \qquad |rw - y| \le \rho + \tau M. \tag{6}$$

For each fixed $r \in (0,1)$ the map $w \mapsto \Phi_{w,y}(r)$ is smooth on $\mathbb{R}^n$: it is a product of a polynomial in w with the exponential of a polynomial in w, divided by the positive constants $(1-r^2)^{n/2+1}$ and $\sqrt{-\log r}$; this per-slice smoothness, together with the continuity in r used below, is the content of `GaussRieszKernelNearSliceBounds`. Differentiating, with e_j the jth standard basis vector,

$$\nabla_w \Phi_{w,y}(r) = \frac{\exp\bigl(-|rw-y|^2/(1-r^2)\bigr)}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\Bigl[-r\,e_j - (y_j - rw_j)\,\frac{2r(rw-y)}{1-r^2}\Bigr],$$

because $\nabla_w(y_j - rw_j) = -re_j$ and $\nabla_w\bigl(-|rw-y|^2/(1-r^2)\bigr) = -2r(rw-y)/(1-r^2)$. Using $r \le 1$, $1-r^2 \ge \tau$, and then (2), (3) and (6),

$$|\nabla_w \Phi_{w,y}(r)| \le e^{M^2}\,\tau^{-n/2-3/2}\Bigl(1 + \frac{2(\rho+\tau M)^2}{\tau}\Bigr)\exp\Bigl(-\frac{|w-ry|^2}{2\tau}\Bigr). \tag{7}$$

Step 3: a majorant uniform on a ball around x. Let x, y be as in the statement and put $d = |x-y| \in (0,2]$. Then $|x| \le R+1 \le M$, and $U = B(x, d/4)$ is an open convex set. For $w \in U$ we have $|w| \le |x| + \frac12 \le R + \frac32 \le M$, and $|w-y| \ge d - d/4 = \frac34 d > 0$ while $|w-y| \le \frac54 d$. Also $w - ry = (w-y) + \tau y$ gives $|w-ry| \ge |w-y| - \tau|y| \ge \frac34 d - \tau M$. Applying (7) at w, whose $\rho = |w-y|$ is at most $\frac54 d$, and noting that the right-hand side of (7) is increasing in ρ and decreasing in $|w-ry|$, `GaussRieszKernelNearBallUniformTauMajorant` removes the w-dependence: for all $w \in U$ and all $\tau \in (0,1)$

$$|\nabla_w \Phi_{w,y}(r)| \le H(\tau) := e^{M^2}\tau^{-n/2-3/2}\Bigl(1 + \frac{2(\frac54 d + \tau M)^2}{\tau}\Bigr)\exp\Bigl(-\frac{\bigl((\frac34 d - \tau M)^+\bigr)^2}{2\tau}\Bigr),$$

where $s^+ = \max(s,0)$; the right-hand side no longer depends on w. Put $\tau'' = 3d/(8M)$, which satisfies $\tau'' \le \frac14 < 1$ because $d \le 2$ and $M \ge 3$.

By that node, for $0 < \tau \le \tau''$ one has $\tau M \le \frac38 d$, hence $\frac34 d - \tau M \ge \frac38 d$ and $\frac54 d + \tau M \le 2d$, so

$$H(\tau) \le e^{M^2}\Bigl(\tau^{-n/2-3/2} + 8d^2\tau^{-n/2-5/2}\Bigr)e^{-9d^2/(128\tau)},$$

and (4) with $b = 9d^2/128$, applied at $a = \frac{n+3}{2}$ and at $a = \frac{n+5}{2}$ (both > 1), bounds $\int_{\{0<\tau\le\tau''\}} H$ by a constant depending only on n and M times d^{-n-1}: the two values of b^{1-a} are supplied by clauses (1) and (2) of `RpowPowerReductionsToNegPower`.

For $\tau'' < \tau < 1$ the same node discards the exponential and uses $(\frac54 d + \tau M)^2 \le 4d^2 + 2\tau^2 M^2$, so $H(\tau) \le e^{M^2}\bigl(\tau^{-n/2-3/2} + 8d^2\tau^{-n/2-5/2} + 4M^2\tau^{-n/2-1/2}\bigr)$. By (5), with $q = \frac{n+3}{2}$, $q = \frac{n+5}{2}$ and $q = \frac{n+2}{2}$ respectively (each > 1, and in the third case $p = \frac{n+1}{2} \le q$),

$$\int_{\tau''}^1 H \le e^{M^2}\Bigl[\tfrac{2}{n+1}(\tau'')^{-(n+1)/2} + \tfrac{16d^2}{n+3}(\tau'')^{-(n+3)/2} + \tfrac{8M^2}{n}(\tau'')^{-n/2}\Bigr].$$

Substituting $\tau'' = 3d/(8M)$ and using $0 < d \le 2$, each of the three terms reduces to a constant multiple, depending only on n and M, of d^{-n-1}: these are clauses (3), (4) and (5) of `RpowPowerReductionsToNegPower`, one per power of τ'', each absorbing it into $d^{-(n+1)}$ at the cost of an explicit constant. Altogether there is $\kappa(n,M) > 0$ with

$$\int_0^1 H(\tau)\,d\tau \le \kappa(n,M)\,d^{-n-1} < \infty. \tag{8}$$

Step 4: differentiation under the integral sign. Consider $F(w,r)=\Phi_{w,y}(r)$ on $U\times(0,1)$. Then: (i) for each $r\in(0,1)$, $F(\cdot,r)$ is differentiable on U, by Step 2; (ii) for each $w\in U$, both $F(w,\cdot)$ and $\nabla_w F(w,\cdot)$ are measurable on $(0,1)$, being continuous there, and $F(x,\cdot)$ is integrable on $(0,1)$ by the hypothesis of the lemma; (iii) $|\nabla_w F(w,r)|\le H(\tau)$ for all $w\in U$ and $r\in(0,1)$, with H integrable on $(0,1)$ by (8). By the classical theorem on differentiation under the integral sign — for a convex open parameter set U, hypotheses (i)–(iii) imply that $w\mapsto\int_0^1 F(w,r)\,dr$ is differentiable on U with derivative $h\mapsto\int_0^1\nabla_w F(w,r)\cdot h\,dr$ — the map $w\mapsto\int_0^1\Phi_{w,y}(r)\,dr$ is differentiable at x, hence so is $w\mapsto K_{\gamma,j}(w,y)=c_n\int_0^1\Phi_{w,y}(r)\,dr$, and its derivative g at x is the continuous linear functional $h\mapsto c_n\int_0^1\nabla_x\Phi_{x,y}(r)\cdot h\,dr$. Its operator norm is $c_n\big|\int_0^1\nabla_x\Phi_{x,y}(r)\,dr\big|$, which by (8) is at most

$$c_n\int_0^1|\nabla_x\Phi_{x,y}(r)|\,dr\le c_n\,\kappa(n,M)\,d^{-n-1}.$$

So $C:=c_n\,\kappa(n,M)$, which depends only on n and M, has the required property. □

2.146. `GaussRieszKernelNearReflectedSliceMajorants.`

Lemma 2.146 (The two reflected pointwise majorants for the Mehler slice)**.** *Fix $n\ge1$, a coordinate j, and $M\ge3$. For $w,y\in\mathbb{R}^n$ and $r\in(0,1)$ write*

$$\Phi_{w,y}(r)=\frac{y_j-rw_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big),$$

and for $x,y\in\mathbb{R}^n$ put $d=|x-y|$. Let $|x|\le M$, $|y|\le M$ and $d\le2$. Then both of the following hold, with $\tau=1-r$.

(1) *For every τ with $0<\tau\le d/(2M)$,*

$$\big|\Phi_{x,y}(1-\tau)\big|\le e^{M^2}\,\tfrac{3}{2}d\,\tau^{-n/2-3/2}e^{-d^2/(8\tau)}.$$

(2) *For every $\tau\in(0,1)$,*

$$\big|\Phi_{x,y}(1-\tau)\big|\le e^{M^2}d\,\tau^{-n/2-3/2}+e^{M^2}M\,\tau^{-(n+2)/2}.$$

Proof. Throughout $\tau=1-r\in(0,1)$, so $r\in(0,1)$, and $d=|x-y|$.

Step 0: the two inputs. By `GaussRieszKernelNearSliceBounds`, applied at the base point $w=x$ (legitimate since $|x|\le M$ and $|y|\le M$), for every $\tau\in(0,1)$

$$\big|\Phi_{x,y}(1-\tau)\big|\le e^{M^2}\,\big(d+\tau M\big)\,\tau^{-n/2-3/2}\exp\Big(-\frac{|x-(1-\tau)y|^2}{2\tau}\Big).\tag{1}$$

Second, since $x-(1-\tau)y=(x-y)+\tau y$ and $|y|\le M$, the triangle inequality in the form $|a+b|\ge|a|-|b|$ gives

$$|x-(1-\tau)y|\ge|x-y|-\tau|y|\ge d-\tau M.\tag{2}$$

Step 1: clause (1), the near majorant. Let $0<\tau\le d/(2M)$. Multiplying that inequality by $2M>0$ gives $2M\tau\le d$, hence $\tau M\le d/2$ and therefore $d+\tau M\le\frac{3}{2}d$. By (2) also $|x-(1-\tau)y|\ge d-\tau M\ge d-d/2=d/2\ge0$, so squaring the nonnegative sides gives $|x-(1-\tau)y|^2\ge d^2/4$ and hence, dividing by $2\tau>0$ and using monotonicity of $t\mapsto e^{-t}$,

$$\exp\Big(-\frac{|x-(1-\tau)y|^2}{2\tau}\Big)\le\exp\Big(-\frac{d^2}{8\tau}\Big).$$

Substituting both estimates into (1), the prefactor $e^{M^2}(d+\tau M)\tau^{-n/2-3/2}$ being nonnegative and the exponential positive, yields clause (1).

Step 2: clause (2), the majorant valid at every time. Let $\tau\in(0,1)$. Bound the exponential in (1) by 1, which is legitimate because $-|x-(1-\tau)y|^2/(2\tau)\le0$, and expand the prefactor:

$$(d+\tau M)\,\tau^{-n/2-3/2}=d\,\tau^{-n/2-3/2}+M\,\tau^{1-(n/2+3/2)}=d\,\tau^{-n/2-3/2}+M\,\tau^{-(n+1)/2}.$$

Finally $0<\tau<1$ and $(n+1)/2\le(n+2)/2$ give $\tau^{-(n+1)/2}\le\tau^{-(n+2)/2}$, since $s\mapsto\tau^{-s}$ is nondecreasing for $\tau\le1$. Multiplying by $e^{M^2}M\ge0$ and adding the first term gives clause (2). □

2.147. `GaussRieszKernelNearSliceL1Bound`.

Lemma 2.147 (Near-diagonal L^1 bound in r for the Mehler slice)**.** *Fix $n \geq 1$, a coordinate j, and $M \geq 3$. For $w, y \in \mathbb{R}^n$ and $r \in (0,1)$ write*

$$\Phi_{w,y}(r) = \frac{y_j - r w_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big).$$

Then there is a constant $A > 0$, depending only on n and M, such that for all $x, y \in \mathbb{R}^n$ with $|x| \leq M$, $|y| \leq M$ and $0 < |x-y| \leq 2$, the function $r \mapsto \Phi_{x,y}(r)$ is Lebesgue integrable on $(0,1)$ and

$$\int_0^1 \big|\Phi_{x,y}(r)\big|\, dr \leq A\,|x-y|^{-n}.$$

Proof. Throughout $\tau = 1 - r$, so $r \in (0,1)$ corresponds to $\tau \in (0,1)$, and the substitution $r \mapsto 1 - r$ preserves Lebesgue measure on $\mathbb{R}$, being a reflection followed by a translation. Fix x, y as in the statement and put $d = |x - y| \in (0, 2]$.

The pointwise majorant. By `GaussRieszKernelNearSliceBounds`, applied at the base point $w = x$ (legitimate since $|x| \leq M$ and $|y| \leq M$), for every $r \in (0,1)$

$$\big|\Phi_{x,y}(r)\big| \leq e^{M^2}\,\big(d + \tau M\big)\,\tau^{-n/2-3/2} \exp\Big(-\frac{|x-ry|^2}{2\tau}\Big). \tag{1}$$

That same node also supplies the continuity of $r \mapsto \Phi_{x,y}(r)$ on $(0,1)$, so $\Phi_{x,y}$ is Borel measurable there and it suffices to bound $\int_0^1 |\Phi_{x,y}|$ to obtain integrability as well as the stated estimate. Since $x - ry = (x-y) + \tau y$ and $|y| \leq M$,

$$|x - ry| \geq |x-y| - \tau|y| \geq d - \tau M. \tag{2}$$

The two τ-integral facts. Both are supplied by `TauPowerIntegralBounds`. For $a > 1$ there is $K_a > 0$, depending only on a, with

$$\int_0^\infty \tau^{-a} e^{-b/\tau}\, d\tau \leq K_a\, b^{1-a} < \infty \qquad (b > 0), \tag{3}$$

and for $0 < c \leq 1$ and real $p \leq q$ with $q > 1$,

$$\int_c^1 \tau^{-p}\, d\tau \leq \int_c^1 \tau^{-q}\, d\tau = \frac{c^{1-q} - 1}{q-1} \leq \frac{c^{1-q}}{q-1}, \tag{4}$$

the first inequality because $\tau^{-p} \leq \tau^{-q}$ on $(0,1]$.

The split. Put $\tau' = d/(2M)$. Since $d \leq 2$ and $M \geq 3$ we have $\tau' \leq \frac{1}{3} < 1$, so τ' splits $(0,1)$ into $(0,\tau']$ and $(\tau', 1)$.

By `GaussRieszKernelNearReflectedSliceMajorants`, which derives both reflected pointwise majorants from (1) and (2), for $0 < \tau \leq \tau'$

$$\big|\Phi_{x,y}\big| \leq e^{M^2}\,\tfrac{3}{2} d\,\tau^{-n/2-3/2} e^{-d^2/(8\tau)},$$

and (3) with $a = \frac{n+3}{2} > 1$ and $b = d^2/8$ bounds the contribution of this part by

$$e^{M^2}\,\tfrac{3}{2} d\, K_{(n+3)/2}\,(d^2/8)^{-(n+1)/2} = e^{M^2}\,\tfrac{3}{2}\, K_{(n+3)/2}\, 8^{(n+1)/2}\, d^{-n},$$

using $d\,(d^2)^{-(n+1)/2} = d \cdot d^{-(n+1)} = d^{-n}$.

For $\tau' < \tau < 1$ the same node's second majorant gives $|\Phi_{x,y}| \leq e^{M^2} d\,\tau^{-n/2-3/2} + e^{M^2} M\,\tau^{-(n+2)/2}$. By (4) with $p = q = \frac{n+3}{2} > 1$ for the first term, and with $p = q = \frac{n+2}{2} > 1$ for the second,

$$\int_{\tau'}^1 d\,\tau^{-n/2-3/2}\, d\tau \leq \frac{2d}{n+1}(\tau')^{-(n+1)/2} = \frac{2(2M)^{(n+1)/2}}{n+1}\, d^{(1-n)/2},$$

$$\int_{\tau'}^1 M\,\tau^{-n/2-1/2}\, d\tau \leq \frac{2M}{n}(\tau')^{-n/2} = \frac{2M(2M)^{n/2}}{n}\, d^{-n/2},$$

where in each case $(\tau')^{-s} = (2M)^s d^{-s}$. Since $0 < d \leq 2$ we have $d^{(1-n)/2} = d^{-n} d^{(n+1)/2} \leq 2^{(n+1)/2} d^{-n}$ and $d^{-n/2} = d^{-n} d^{n/2} \leq 2^{n/2} d^{-n}$, both displayed powers of d being positive.

Adding the three contributions, each of which is a constant depending only on n and M times d^{-n}, there is $A = A(n, M) > 0$ with

$$\int_0^1 \big|\Phi_{x,y}(r)\big|\, dr \le A\, d^{-n} < \infty.$$

Finiteness of this integral, together with the measurability recorded above, is exactly Lebesgue integrability of $\Phi_{x,y}$ on $(0,1)$, and the displayed inequality is the stated bound. □

2.148. `GaussRieszKernelNearBounds`.

Lemma 2.148 (Near-diagonal size, gradient and Hörmander bounds for $K_{\gamma,j}$)**.** *Fix $n \ge 1$ and $j \in \{1, \dots, n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, and let $K_{\gamma,j}$ be the normalized Gaussian Riesz kernel of* `GaussRieszKernel`*, so that $K_{\gamma,j}(x,y) = \sqrt{2}\,\pi^{-(n+1)/2} \int_0^1 \Phi_{x,y}(r)\, dr$ with*

$$\Phi_{x,y}(r) = \frac{y_j - r x_j}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \exp\Big(-\frac{|rx-y|^2}{1-r^2}\Big), \qquad r \in (0,1).$$

Let $R > 0$. Then there is a constant $C > 0$, depending only on n, j and R, such that:

(1) *for all $x, y \in \mathbb{R}^n$ with $|x| \le R+1$ and $0 < |x-y| \le 2$,*
   (a) *$\Phi_{x,y}$ is Lebesgue integrable on $(0,1)$;*
   (b) *$|K_{\gamma,j}(x,y)| \le \dfrac{C}{|x-y|^n}$; and*
   (c) *the function $w \mapsto K_{\gamma,j}(w,y)$ is (Fréchet) differentiable at x, and its derivative g there — a continuous linear functional on $\mathbb{R}^n$ — satisfies $\|g\| \le \dfrac{C}{|x-y|^{n+1}}$;*

(2) *(Hörmander estimate) for all $x, y, z \in \mathbb{R}^n$ with $|x| \le R$, $0 < |x-y| \le 1$ and $|x-z| \le \frac{1}{2}|x-y|$,*

$$\big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big| \le \frac{C\,|x-z|}{|x-y|^{n+1}}.$$

Since $|x-y| > 0$ throughout, the quotients $C/|x-y|^n$ and $C/|x-y|^{n+1}$ are the usual negative powers $C|x-y|^{-n}$ and $C|x-y|^{-n-1}$.

Why (1)(a) and the differentiability in (1)(c) are part of the statement, and not merely of the proof. *$K_{\gamma,j}(x,y)$ is a Bochner integral over $(0,1)$ whose value is 0 by convention when the integrand is not integrable, and 0 satisfies every upper bound; so a bare size bound $|K_{\gamma,j}| \le C|x-y|^{-n}$ would be provable with no information about the kernel. Clause (1)(a) removes that, and the reference paper's own display of k_j^{raw} carries the qualifier $x \ne y$, so the integrability assertion records what the paper intends rather than strengthening it. Symmetrically, a gradient bound phrased as a bound on a* derivative operator *chosen by convention where the function is not differentiable would be vacuous wherever differentiability fails, and would not support a mean value argument. Clause (1)(c) therefore asserts genuine differentiability and bounds the actual derivative.*

Why the Hörmander clause carries its own hypotheses. *It is not a formal consequence of (1)(c): the mean value inequality needs a* uniform *bound on the derivative along the whole segment from x to z, whereas (1)(c) gives a bound varying with the base point. The hypothesis $|x-z| \le \frac{1}{2}|x-y|$ is exactly what keeps that segment at distance at least $\frac{1}{2}|x-y|$ from y, so that the varying bound is uniformly comparable to $|x-y|^{-n-1}$ there; the hypotheses $|x| \le R$ and $|x-y| \le 1$ keep the whole segment inside the range of clause (1).*

The constant depends on R, necessarily. *After the splitting of the exponent used below, the kernel carries a factor $e^{|x|^2-|y|^2}$, which is bounded only on bounded sets of x; on the range considered it is at most $e^{(R+3)^2}$. Neither bound is stated with a dimension-only constant, and no uniformity in R is claimed.*

Proof. Write $c_n = \sqrt{2}\,\pi^{-(n+1)/2}$ and $M = R + 3 \ge 3$. Throughout $\tau = 1 - r$, so that $r \in (0,1)$ corresponds to $\tau \in (0,1)$, and the substitution $r \mapsto 1 - r$ preserves Lebesgue measure on $\mathbb{R}$ (it is a reflection followed by a translation; shared context).

Step 0: the splitting identity and the elementary τ-bounds. For all $w, y \in \mathbb{R}^n$ and $r \in (0,1)$,

$$\frac{|rw-y|^2}{1-r^2} = \big(|y|^2 - |w|^2\big) + \frac{|w-ry|^2}{1-r^2}, \tag{1}$$

since after multiplying by $1-r^2>0$ both $|rw-y|^2-(1-r^2)(|y|^2-|w|^2)$ and $|w-ry|^2$ expand to $|w|^2-2r\langle w,y\rangle + r^2|y|^2$. Hence, whenever $|w|\le M$,

$$\exp\Big(-\frac{|rw-y|^2}{1-r^2}\Big) = e^{|w|^2}e^{-|y|^2}\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big) \le e^{M^2}\exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big), \tag{2}$$

using $e^{-|y|^2}\le 1$. Also $1-r^2=\tau(1+r)\in[\tau,2\tau]$ and $\log r = \log(1-\tau)\le -\tau$, so $-\log r\ge\tau$ and

$$\frac{1}{(1-r^2)^{n/2+1}\sqrt{-\log r}} \le \tau^{-n/2-3/2}, \qquad \exp\Big(-\frac{|w-ry|^2}{1-r^2}\Big)\le \exp\Big(-\frac{|w-ry|^2}{2\tau}\Big). \tag{3}$$

We record two integral facts. For $a>1$ and $b>0$, the substitution $\tau = b/u$ (so $d\tau = (b/u^2)du$) gives

$$\int_0^\infty \tau^{-a}e^{-b/\tau}\,d\tau = b^{1-a}\int_0^\infty u^{a-2}e^{-u}\,du = b^{1-a}\,\Gamma(a-1) < \infty, \tag{4}$$

the last integral converging because $a-2>-1$ and e^{-u} decays. And for $0<a\le 1$ and real $p\le q$ with $q>1$, using $\tau^{-p}\le\tau^{-q}$ on $(0,1]$,

$$\int_a^1 \tau^{-p}\,d\tau \le \int_a^1 \tau^{-q}\,d\tau = \frac{a^{1-q}-1}{q-1} \le \frac{a^{1-q}}{q-1}. \tag{5}$$

Step 1: pointwise bounds on Φ and on its w-gradient. Fix y and let w satisfy $|w|\le M$ and $|y|\le M$; write $d=|w-y|$. Since $y_j - rw_j = (y_j-w_j)+\tau w_j$ and $y-rw=(y-w)+\tau w$,

$$|y_j - rw_j| \le d+\tau M, \qquad |rw-y|\le d+\tau M. \tag{6}$$

Combining (2), (3) and (6),

$$|\Phi_{w,y}(r)| \le e^{M^2}\,(d+\tau M)\,\tau^{-n/2-3/2}\exp\Big(-\frac{|w-ry|^2}{2\tau}\Big). \tag{7}$$

For each fixed $r\in(0,1)$ the map $w\mapsto\Phi_{w,y}(r)$ is smooth on $\mathbb{R}^n$: it is a product of a polynomial in w with the exponential of a polynomial in w, divided by the positive constants $(1-r^2)^{n/2+1}$ and $\sqrt{-\log r}$. Differentiating, with e_j the jth standard basis vector,

$$\nabla_w\Phi_{w,y}(r) = \frac{\exp\big(-|rw-y|^2/(1-r^2)\big)}{(1-r^2)^{n/2+1}\sqrt{-\log r}}\Big[-r\,e_j - (y_j-rw_j)\,\frac{2r(rw-y)}{1-r^2}\Big],$$

because $\nabla_w(y_j-rw_j) = -re_j$ and $\nabla_w\big(-|rw-y|^2/(1-r^2)\big) = -2r(rw-y)/(1-r^2)$. Using $r\le 1$, $1-r^2\ge\tau$, (2), (3) and (6),

$$|\nabla_w\Phi_{w,y}(r)| \le e^{M^2}\,\tau^{-n/2-3/2}\Big(1+\frac{2(d+\tau M)^2}{\tau}\Big)\exp\Big(-\frac{|w-ry|^2}{2\tau}\Big). \tag{8}$$

Step 2: the size bound and clause (1)(a)–(1)(b). Let $|x|\le R+1$ and $0<d:=|x-y|\le 2$. Then $|x|\le M$ and $|y|\le |x|+d\le R+3=M$, so (7) applies with $w=x$. Moreover $x-ry=(x-y)+\tau y$ gives $|x-ry|\ge d-\tau M$. Put $\tau' = d/(2M)$; since $d\le 2$ and $M\ge 3$ we have $\tau'\le\frac13<1$.

For $0<\tau\le\tau'$ we have $\tau M\le d/2$, hence $d+\tau M\le\frac32 d$ and $|x-ry|\ge d/2$, so (7) gives $|\Phi_{x,y}|\le e^{M^2}\frac32 d\,\tau^{-n/2-3/2}e^{-d^2/(8\tau)}$, and by (4) with $a=\frac{n+3}{2}$ and $b=d^2/8$,

$$\int_{\{0<\tau\le\tau'\}}|\Phi_{x,y}|\,d\tau \le e^{M^2}\tfrac32 d\,(d^2/8)^{-(n+1)/2}\Gamma\big(\tfrac{n+1}{2}\big) = e^{M^2}\tfrac32\,8^{(n+1)/2}\Gamma\big(\tfrac{n+1}{2}\big)\,d^{-n}.$$

For $\tau'<\tau<1$ we discard the exponential in (7), bounding it by 1, and expand the prefactor: $(d+\tau M)\tau^{-n/2-3/2} = d\,\tau^{-n/2-3/2} + M\,\tau^{-n/2-1/2}$, so that $|\Phi_{x,y}|\le e^{M^2}\big(d\,\tau^{-n/2-3/2}+M\,\tau^{-n/2-1/2}\big)$. By (5) with $p=q=\frac{n+3}{2}>1$ and with $p=\frac{n+1}{2}\le q=\frac{n+2}{2}>1$ respectively,

$$\int_{\tau'}^1 d\,\tau^{-n/2-3/2}\,d\tau \le \frac{2d}{n+1}(\tau')^{-(n+1)/2} = \frac{2(2M)^{(n+1)/2}}{n+1}\,d^{(1-n)/2}, \qquad \int_{\tau'}^1 M\,\tau^{-n/2-1/2}\,d\tau \le \frac{2M(2M)^{n/2}}{n}\,d^{-n/2}.$$

Since $0 < d \le 2$ we have $d^{(1-n)/2} = d^{-n} d^{(n+1)/2} \le 2^{(n+1)/2} d^{-n}$ and $d^{-n/2} = d^{-n} d^{n/2} \le 2^{n/2} d^{-n}$. Adding the three contributions,

$$\int_0^1 |\Phi_{x,y}(r)|\, dr \le \kappa_1(n, M)\, d^{-n} < \infty, \tag{9}$$

with κ_1 depending only on n and M. The function $\Phi_{x,y}$ is continuous on $(0,1)$, hence Borel there, so (9) says it is Lebesgue integrable on $(0,1)$: that is clause (1)(a). And then

$$|K_{\gamma,j}(x,y)| = c_n \Big| \int_0^1 \Phi_{x,y} \Big| \le c_n \int_0^1 |\Phi_{x,y}| \le c_n \kappa_1\, d^{-n},$$

which is clause (1)(b) with the constant $C_1 := c_n \kappa_1$. The estimate (9) on the slice integral, which is what carries clause (1)(a) and the size bound behind (1)(b), is `GaussRieszKernelNearSliceL1Bound`; it states exactly this bound for every x, y with $|x|, |y| \le M$ and $0 < |x - y| \le 2$, and is what the formalization invokes at this point.

Step 3: a uniform majorant near x, and clause (1)(c). Keep x, y, d as in Step 2 and set $U = B(x, d/4)$, an open convex set. For $w \in U$: $|w| \le |x| + \frac{1}{2} \le R + \frac{3}{2} \le M$; and $|w - y| \ge d - d/4 = \frac{3}{4} d > 0$ while $|w - y| \le \frac{5}{4} d \le \frac{5}{2}$. Also $w - ry = (w - y) + \tau y$ gives $|w - ry| \ge |w - y| - \tau |y| \ge \frac{3}{4} d - \tau M$. Now apply (8) at the point w, where its "d" is $|w - y| \le \frac{5}{4} d$: the right-hand side of (8) is increasing in that quantity and decreasing in $|w - ry|$, so for all $w \in U$ and all $\tau \in (0,1)$

$$|\nabla_w \Phi_{w,y}(r)| \le H(\tau) := e^{M^2} \tau^{-n/2-3/2} \Big(1 + \frac{2(\frac{5}{4} d + \tau M)^2}{\tau}\Big) \exp\Big(-\frac{\big((\frac{3}{4} d - \tau M)^+\big)^2}{2\tau}\Big),$$

where $s^+ = \max(s, 0)$; note the right-hand side no longer depends on w. Put $\tau'' = 3d/(8M)$, which satisfies $\tau'' \le \frac{1}{4} < 1$ because $d \le 2$ and $M \ge 3$.

For $0 < \tau \le \tau''$ we have $\tau M \le \frac{3}{8} d$, hence $\frac{3}{4} d - \tau M \ge \frac{3}{8} d$ and $\frac{5}{4} d + \tau M \le 2d$, so

$$H(\tau) \le e^{M^2} \Big(\tau^{-n/2-3/2} + 8 d^2 \tau^{-n/2-5/2}\Big) e^{-9d^2/(128\tau)},$$

and (4) with $b = 9d^2/128$, applied with $a = \frac{n+3}{2}$ and with $a = \frac{n+5}{2}$, bounds $\int_{\{0 < \tau \le \tau''\}} H$ by

$$e^{M^2} \Big[(9/128)^{-(n+1)/2} \Gamma\big(\tfrac{n+1}{2}\big) + 8\, (9/128)^{-(n+3)/2} \Gamma\big(\tfrac{n+3}{2}\big) \Big]\, d^{-n-1},$$

because $b^{1-a} = (9/128)^{-(n+1)/2} (d^2)^{-(n+1)/2} = (9/128)^{-(n+1)/2} d^{-n-1}$ when $a = \frac{n+3}{2}$, and $8 d^2\, b^{1-a} = 8\, (9/128)^{-(n+3)/2} d^2\, d^{-(n+3)} = 8\, (9/128)^{-(n+3)/2} d^{-n-1}$ when $a = \frac{n+5}{2}$.

For $\tau'' < \tau < 1$ we discard the exponential and use $(\frac{5}{4} d + \tau M)^2 \le 4d^2 + 2\tau^2 M^2$, so $H(\tau) \le e^{M^2} \big(\tau^{-n/2-3/2} + 8 d^2 \tau^{-n/2-5/2} + 4 M^2 \tau^{-n/2-1/2}\big)$. By (5), with $q = \frac{n+3}{2}$, $q = \frac{n+5}{2}$ and $q = \frac{n+2}{2}$ respectively (each > 1, and in the third case $p = \frac{n+1}{2} \le q$),

$$\int_{\tau''}^1 H \le e^{M^2} \Big[\tfrac{2}{n+1} (\tau'')^{-(n+1)/2} + \tfrac{16 d^2}{n+3} (\tau'')^{-(n+3)/2} + \tfrac{8 M^2}{n} (\tau'')^{-n/2} \Big].$$

Substituting $\tau'' = 3d/(8M)$ and using $0 < d \le 2$ once more, the three terms are at most constant multiples (depending only on n and M) of $d^{-(n+1)/2}$, $d^{2-(n+3)/2} = d^{(1-n)/2}$ and $d^{-n/2}$, and each of these is at most $2^{(3n+2)/2} d^{-n-1}$, because $d^{-(n+1)/2} = d^{-n-1} d^{(n+1)/2}$, $d^{(1-n)/2} = d^{-n-1} d^{(n+3)/2}$ and $d^{-n/2} = d^{-n-1} d^{(3n+2)/2}$, with all the displayed positive powers of d bounded by $2^{(3n+2)/2}$. Altogether

$$\int_0^1 H(\tau)\, d\tau \le \kappa_2(n, M)\, d^{-n-1} < \infty. \tag{10}$$

We now differentiate under the integral sign. Consider $F(w, r) = \Phi_{w,y}(r)$ on $U \times (0,1)$. Then: (i) for each $r \in (0,1)$, $F(\cdot, r)$ is differentiable on U (Step 1); (ii) for each $w \in U$, both $F(w, \cdot)$ and $\nabla_w F(w, \cdot)$ are measurable on $(0,1)$, being continuous there, and $F(x, \cdot)$ is integrable by (9); (iii) $|\nabla_w F(w, r)| \le H(\tau)$ for all $w \in U$ and $r \in (0,1)$, with H integrable on $(0,1)$ by (10). By the classical theorem on differentiation under the integral sign — for a convex open parameter set U, hypotheses (i)–(iii) imply that $w \mapsto \int_0^1 F(w, r)\, dr$ is differentiable on U with

derivative $h \mapsto \int_0^1 \nabla_w F(w,r)\cdot h\,dr$ — the function $w \mapsto \int_0^1 \Phi_{w,y}(r)\,dr$ is differentiable at x, hence so is $w \mapsto K_{\gamma,j}(w,y) = c_n \int_0^1 \Phi_{w,y}(r)\,dr$, and its derivative g at x is the linear functional $h \mapsto c_n \int_0^1 \nabla_x \Phi_{x,y}(r)\cdot h\,dr$. Its operator norm is $c_n\big|\int_0^1 \nabla_x \Phi_{x,y}(r)\,dr\big|$, which by (10) is at most

$$c_n \int_0^1 |\nabla_x \Phi_{x,y}(r)|\,dr \le c_n\,\kappa_2\,d^{-n-1}.$$

This is clause (1)(c) with the constant $C_2 := c_n\kappa_2$. The content of this step — the uniform majorant H together with the differentiation under the integral sign that it licenses, and the resulting bound on $\|g\|$ — is exactly `GaussRieszKernelNearFDerivBound`, which states this conclusion for every x, y in the stated range and is what the formalization invokes at this point.

Step 4: clause (2). Let $|x| \le R$, $0 < d := |x-y| \le 1$ and $|x-z| \le d/2$, and let $S = \{x + t(z-x) : t \in [0,1]\}$ be the segment from x to z, a convex set. For $w \in S$ we have $|w-x| \le |z-x| \le d/2 \le \frac{1}{2}$, hence

$$|w| \le |x| + \tfrac{1}{2} \le R+1, \qquad |w-y| \ge d - \tfrac{d}{2} = \tfrac{d}{2} > 0, \qquad |w-y| \le d + \tfrac{1}{2} \le \tfrac{3}{2} \le 2.$$

So every $w \in S$ satisfies the hypotheses of clause (1), and clause (1)(c) gives, for each such w, that $v \mapsto K_{\gamma,j}(v,y)$ is differentiable at w with derivative g_w satisfying

$$\|g_w\| \le \frac{C_2}{|w-y|^{n+1}} \le \frac{C_2}{(d/2)^{n+1}} = \frac{2^{n+1}C_2}{d^{n+1}}.$$

Differentiability at w implies differentiability within S at w with the same derivative. By the mean value inequality for a function differentiable on a convex set with derivative bounded in operator norm by a constant there — applied to $v \mapsto K_{\gamma,j}(v,y)$ on S, with the uniform bound $2^{n+1}C_2 d^{-n-1}$, and to the two points $x, z \in S$ —

$$\big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big| \le \frac{2^{n+1}C_2}{d^{n+1}}\,|x-z|,$$

which is clause (2) with the constant $2^{n+1}C_2$.

Conclusion. Take $C = \max\big(C_1,\ C_2,\ 2^{n+1}C_2\big) > 0$, which depends only on n, j and R. All the bounds proved above hold a fortiori with C in place of their individual constants, so clauses (1) and (2) hold with this single C. □

2.149. `GaussRieszKernelHormanderDyadicBound`.

Lemma 2.149 (Dyadic summation of the Hörmander estimate against a local maximal bound)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$ and let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel`. Let $R \ge 1$. Then there is a constant $C \in (0,\infty)$, read in $[0,+\infty]$, depending only on n, j and R, such that the following holds.*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be any function, let $\Phi \in [0,+\infty]$, let $x, z \in \mathbb{R}^n$ and let ε be real, and suppose

$$|x| \le R, \qquad 0 < \varepsilon \le 1, \qquad |x-z| < \tfrac{1}{2}\varepsilon,$$

and that

$$\text{(H)} \qquad \int^*_{\overline{B}(x,\rho)} |f|\,d\lambda \ \le\ \Phi \cdot \lambda\big(\overline{B}(x,\rho)\big) \qquad \text{for every real } \rho \in (0,2].$$

Then

$$\int^*_{\{y\,:\,\varepsilon<|x-y|\le 1\}} \big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big|\,|f(y)|\,d\lambda(y) \ \le\ C\cdot\Phi.$$

Here $\int^*$ denotes the lower Lebesgue integral of a $[0,+\infty]$-valued function, and all three displayed quantities are read in $[0,+\infty]$.

No measurability or integrability hypothesis is imposed, and none is needed. *Both the hypothesis (H) and the conclusion are lower Lebesgue integrals of nonnegative extended-real integrands, which are defined for every function; the monotonicity, constant-pulling, domain-enlargement and countable subadditivity properties used in the proof hold for lower Lebesgue integrals without measurability. This is the same discipline that makes `LocalMaximalFunction` extended-real valued and defined for every f, and it is what makes (H) exactly the shape a consumer has*

available: (H) *with* Φ *the value of that maximal function at* x *is precisely what the definition of the maximal function as a supremum over* $\rho \in (0,2]$ *supplies, after multiplying by the finite positive number* $\lambda(\overline{B}(x,\rho))$.

The radius cap 2 in (H) and the radius cap 1 in the region are matched, with no margin. *The dyadic annuli used below have outer radii* $2^{k+1}\varepsilon$, *and the largest one meeting the region* $\{\varepsilon < |x-y| \le 1\}$ *has* $2^k\varepsilon < 1$, *hence outer radius* $2^{k+1}\varepsilon < 2$. *So* (H) *is invoked only at radii in* $(0,2]$, *and it would not suffice if the region were enlarged beyond* $|x-y| \le 1$. *The cap* $|x-y| \le 1$ *is in turn exactly the range on which the Hörmander estimate of* `GaussRieszKernelNearBounds` *is available.*

The constant depends on R, necessarily. C *may be taken to be* $C_h\, 2^n\, \omega_n$ *with* C_h *the constant of* `GaussRieszKernelNearBounds` *at* R *and* $\omega_n = \lambda(B(0,1))$; C_h *degrades in* R *and no uniformity in* n, j *or* R *is claimed.*

Proof. Let $C_h > 0$ be the constant supplied by `GaussRieszKernelNearBounds` at R; its hypothesis $R > 0$ holds because $R \ge 1$. Put $\omega_n = \lambda(B(0,1))$, which lies in $(0,\infty)$: the open unit ball is nonempty and open, so of positive measure, and is bounded, so of finite measure (shared context). Set $C = C_h 2^n \omega_n$, which lies in $(0,\infty)$ and depends only on n, j and R. We show the asserted inequality with this C.

Fix f, Φ, x, z, ε and the hypotheses, and abbreviate

$$A = \{y : \varepsilon < |x-y| \le 1\}, \qquad D_k = \{y : 2^k\varepsilon < |x-y| \le 2^{k+1}\varepsilon\} \quad (k \in \mathbb{N}_0), \qquad F(y) = \big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big|\,|f$$

Step 1: the dyadic annuli cover A. We claim $A = \bigcup_{k\ge 0}(D_k \cap A)$. The inclusion $\supseteq$ is immediate. For $\subseteq$, let $y \in A$. The set of $k \in \mathbb{N}_0$ with $|x-y| \le 2^{k+1}\varepsilon$ is nonempty: choose a natural number $m > 1/\varepsilon$; then $m \le 2^m$, so $1/\varepsilon < 2^m$, so $1 < 2^m\varepsilon \le 2^{m+1}\varepsilon$, and $|x-y| \le 1$ gives $|x-y| \le 2^{m+1}\varepsilon$. Let k be the least such natural number; we check $2^k\varepsilon < |x-y|$, which with $|x-y| \le 2^{k+1}\varepsilon$ puts $y \in D_k$ and hence $y \in D_k \cap A$. If $k = 0$ then $2^0\varepsilon = \varepsilon < |x-y|$ because $y \in A$. If $k = m+1$ for some $m \in \mathbb{N}_0$ then minimality of k gives $\neg\,(|x-y| \le 2^{m+1}\varepsilon)$, that is $2^k\varepsilon = 2^{m+1}\varepsilon < |x-y|$.

Step 2: countable subadditivity. By Step 1 and countable subadditivity of the lower Lebesgue integral over a countable union,

$$\int_A^* F\,d\lambda \;\le\; \sum_{k\ge 0} \int_{D_k\cap A}^* F\,d\lambda. \tag{1}$$

Step 3: the empty annuli. Suppose $2^{k+1}\varepsilon > 2$. Since $2^{k+1} = 2\cdot 2^k$ this gives $2^k\varepsilon > 1$, so every $y \in D_k$ has $|x-y| > 2^k\varepsilon > 1$ and therefore $y \notin A$. Hence $D_k \cap A = \emptyset$ and $\int_{D_k\cap A}^* F\,d\lambda = 0$, which is at most $(C\Phi)\cdot 2^{-(k+1)}$.

Step 4: the live annuli. Suppose now $2^{k+1}\varepsilon \le 2$, and let $y \in D_k \cap A$. Then $|x| \le R$; $0 < |x-y|$, because $|x-y| > \varepsilon > 0$; $|x-y| \le 1$, because $y \in A$; and $|x-z| < \varepsilon/2 < |x-y|/2$, because $\varepsilon < |x-y|$. These are exactly the hypotheses of clause (2) of `GaussRieszKernelNearBounds` at R, which therefore gives

$$\big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big| \le \frac{C_h|x-z|}{|x-y|^{n+1}} \le \frac{C_h\,\varepsilon/2}{(2^k\varepsilon)^{n+1}},$$

the second inequality because $|x-z| < \varepsilon/2$ in the numerator and, in the denominator, $0 < 2^k\varepsilon < |x-y|$ together with monotonicity of $t \mapsto t^{n+1}$ on $[0,\infty)$. Multiplying by $|f(y)| \ge 0$,

$$F(y) \le \frac{C_h\,\varepsilon/2}{(2^k\varepsilon)^{n+1}}\,|f(y)| \qquad (y \in D_k \cap A).$$

Integrating this pointwise bound over $D_k\cap A$, pulling the finite constant out of the lower Lebesgue integral, and then enlarging the domain of integration using $D_k \cap A \subseteq \overline{B}(x, 2^{k+1}\varepsilon)$ — which holds because $y \in D_k$ gives $|x-y| \le 2^{k+1}\varepsilon$ and $|x-y| = |y-x|$ — we get

$$\int_{D_k\cap A}^* F\,d\lambda \le \frac{C_h\,\varepsilon/2}{(2^k\varepsilon)^{n+1}} \int_{\overline{B}(x,2^{k+1}\varepsilon)}^* |f|\,d\lambda.$$

Since $0 < 2^{k+1}\varepsilon \le 2$, hypothesis (H) applies at $\rho = 2^{k+1}\varepsilon$ and bounds the last integral by $\Phi\,\lambda(\overline{B}(x, 2^{k+1}\varepsilon))$. Lebesgue measure on $\mathbb{R}^n$ is an additive Haar measure and $\mathbb{R}^n$ has real dimension n, so $\lambda(\overline{B}(x,r)) = r^n\,\omega_n$ for every $r \ge 0$ (shared context). Therefore

$$\int_{D_k \cap A}^{*} F\,d\lambda \le \frac{C_h\,\varepsilon/2}{(2^k\varepsilon)^{\,n+1}}\,(2^{k+1}\varepsilon)^n\,\omega_n\,\Phi.$$

The numerical factor simplifies: writing $2^{k+1} = 2\cdot 2^k$,

$$\frac{\varepsilon/2}{(2^k\varepsilon)^{\,n+1}}(2^{k+1}\varepsilon)^n = \frac{\varepsilon}{2}\,2^{-k(n+1)}\varepsilon^{-(n+1)}\,2^{(k+1)n}\varepsilon^{\,n} = \frac{1}{2}\,2^{\,n-k} = 2^{\,n}\,2^{-(k+1)},$$

since the powers of ε cancel $(1-(n+1)+n=0)$ and $-k(n+1)+(k+1)n = n-k$. Hence

$$\int_{D_k \cap A}^{*} F\,d\lambda \le C_h 2^{\,n}\omega_n\,\Phi\cdot 2^{-(k+1)} = (C\Phi)\cdot 2^{-(k+1)}. \tag{2}$$

Step 5: summation. By Step 3 and Step 4, (2) holds for every $k \in \mathbb{N}_0$. Substituting into (1) and summing the geometric series $\sum_{k\ge 0} 2^{-(k+1)} = \frac{1}{2}\big(1-\frac{1}{2}\big)^{-1} = 1$ in $[0,+\infty]$,

$$\int_{A}^{*} F\,d\lambda \le \sum_{k\ge 0}(C\Phi)\cdot 2^{-(k+1)} = (C\Phi)\sum_{k\ge 0} 2^{-(k+1)} = C\Phi,$$

which is the assertion. □

2.150. `GaussRieszKernelTruncationDifference`.

Lemma 2.150 (The truncated kernel integral varies little over a small ball)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of* `GaussMeasure` *and let $K_{\gamma,j}$ be the kernel of* `GaussRieszKernel`*. Let $R \ge 1$. Then there is a constant $C \in (0,\infty)$, depending only on n, j and R, such that the following holds.*

Let $g : \mathbb{R}^n \to \mathbb{R}$, let Φ and N be real, let $x, z \in \mathbb{R}^n$ and let ε be real, and suppose

$$|x| \le R, \qquad 0 < \varepsilon \le 1, \qquad |x - z| < \tfrac{1}{2}\varepsilon, \qquad 0 \le \Phi,$$

and that

(a) *g is γ_n-integrable;*
(b) *$g = 0$ λ-almost everywhere on $\{y : |x-y| \le \varepsilon\}$;*
(c) *$\displaystyle\int_{\overline{B}(x,\rho)}^{*} |g|\,d\lambda \le \Phi\cdot\lambda\big(\overline{B}(x,\rho)\big)$ for every real $\rho \in (0,2]$, the left side being the lower Lebesgue integral in $[0,+\infty]$ and the right side read there too; and*
(d) *$y \mapsto e^{-|y|^2}|g(y)|$ is Lebesgue integrable on $\mathbb{R}^n$, with $\displaystyle\int_{\mathbb{R}^n} e^{-|y|^2}|g(y)|\,d\lambda(y) \le N$.*

Then

$$\Big|\int_{\mathbb{R}^n} K_{\gamma,j}(x,y)g(y)\,dy - \int_{\mathbb{R}^n} K_{\gamma,j}(z,y)g(y)\,dy\Big| \;\le\; C\,(\Phi + N),$$

both integrals being absolutely convergent Bochner integrals.

No hypothetical operator is needed here, and that is not an oversight. *A reader coming from a Cotlar-type argument will expect the two whole-space integrals to be identified with coordinates of a transform, and hence expect an abstract T with its two properties to be carried along. They are not needed: the only thing required of the two integrals is that they converge absolutely, and that comes from clause (1)(b) of* `GaussRieszKernelFarMajorant` *at the pairs (R,ε) and $(R+1,\varepsilon/2)$, whose constants are irrelevant here because only the integrability half of that clause is used. A consumer that also wants to identify these integrals with $(Tg)_j$ must invoke* `GaussRieszOffDiagonalL1` *separately; that identification is not part of this lemma.*

Why (c) is a lower Lebesgue integral, and why (d) is a hypothesis. *Hypothesis (c) is stated with $\int^*$ because that is the shape a consumer has available: it is what the definition of the local maximal function of* `LocalMaximalFunction` *as a supremum over $\rho \in (0,2]$ supplies, after multiplying by the finite positive number $\lambda(\overline{B}(x,\rho))$. Hypothesis (d) is assumed rather than derived from (a) because the Gaussian-to-Lebesgue density conversion that supplies it —*

$\int e^{-|y|^2}|g|\,d\lambda = \pi^{n/2}\int |g|\,d\gamma_n$ — is used by the intended consumer in a second, independent place as well, so it belongs in the consumer once rather than being re-derived here.

The constant depends on R, necessarily. *C may be taken to be $\max(C_d,\, C_a + C_b)$ with C_d the constant of `GaussRieszKernelHormanderDyadicBound` at R and C_a, C_b the constants of clause (1) of `GaussRieszKernelFarMajorant` at the pairs $(R,1)$ and $(R+1,\frac{1}{2})$; all three degrade in R, and no uniformity in n, j or R is claimed. Note that C does not depend on ε: that is the point of the lemma, and it is why the two far-field constants are taken at the fixed radii 1 and $\frac{1}{2}$ rather than at ε and $\varepsilon/2$.*

Proof. Let $C_d > 0$ be the constant supplied by `GaussRieszKernelHormanderDyadicBound` at R, and let $C_a, C_b > 0$ be the constants supplied by clause (1) of `GaussRieszKernelFarMajorant` at the pairs $(R,1)$ and $(R+1,\frac{1}{2})$ respectively; the hypotheses $R > 0$, $1 > 0$, $R+1 > 0$ and $\frac{1}{2} > 0$ all hold because $R \geq 1$. Set $C = \max(C_d,\, C_a + C_b) \in (0,\infty)$, which depends only on n, j and R. Fix g, Φ, N, x, z, ε and the hypotheses.

Step 0: two preliminaries. Since $e^{-|y|^2}|g(y)| \geq 0$ for every y, its integral is nonnegative, so (d) gives $N \geq 0$. And $|z| - |x| \leq |z - x| = |x - z| < \varepsilon/2 \leq \frac{1}{2}$, so

$$|z| \leq R + 1. \tag{1}$$

Step 1: three regions. Put

$$S = \{y : |x-y| > \varepsilon\}, \qquad A_1 = \{y : \varepsilon < |x-y| \leq 1\}, \qquad A_2 = \{y : |x-y| > 1\}.$$

All three are Borel, being preimages of an interval under the continuous map $y \mapsto |x-y|$. They satisfy $S = A_1 \cup A_2$ with $A_1 \cap A_2 = \emptyset$: the union is contained in S because $\varepsilon \leq 1$, the reverse inclusion holds by splitting on whether $|x-y| \leq 1$, and the intersection is empty because $y \in A_1$ gives $|x-y| \leq 1$ while $y \in A_2$ gives $|x-y| > 1$.

Step 2: both kernel products are integrable on S. By (a), g satisfies the hypothesis of clause (1)(b) of `GaussRieszKernelFarMajorant`. Applying that clause at the pair (R,ε) — legitimate since $\varepsilon > 0$ — to g and to the point x, which has $|x| \leq R$, gives that $y \mapsto K_{\gamma,j}(x,y)g(y)$ is Lebesgue integrable on $\{y : |x-y| > \varepsilon\} = S$. Applying the same clause at the pair $(R+1,\varepsilon/2)$ to g and to the point z, which has $|z| \leq R+1$ by (1), gives that $y \mapsto K_{\gamma,j}(z,y)g(y)$ is Lebesgue integrable on $\{y : |z-y| > \varepsilon/2\}$. Now

$$S \subseteq \{y : |z-y| > \varepsilon/2\}, \tag{2}$$

because for $y \in S$, writing $x - y = (x - z) + (z - y)$ and using the triangle inequality, $|z-y| \geq |x-y| - |x-z| > \varepsilon - \varepsilon/2 = \varepsilon/2$. A function integrable on a set is integrable on every measurable subset of it, so $y \mapsto K_{\gamma,j}(z,y)g(y)$ is Lebesgue integrable on S as well, and both products are integrable on each of $A_1 \subseteq S$ and $A_2 \subseteq S$.

Step 3: the whole-space integrals collapse onto S. Let w be either x or z. The complement of S is exactly $\{y : |x-y| \leq \varepsilon\}$, on which g vanishes λ-almost everywhere by (b); hence $K_{\gamma,j}(w,y)g(y) = 0$ for λ-almost every $y \notin S$. A function integrable on a measurable set and vanishing almost everywhere off it is integrable on the whole space with the same integral, so by Step 2

$$\int_{\mathbb{R}^n} K_{\gamma,j}(w,y)g(y)\,dy = \int_S K_{\gamma,j}(w,y)g(y)\,dy. \tag{3}$$

In particular both whole-space integrals in the conclusion are absolutely convergent.

Step 4: the splitting. By Step 1, Step 2 and additivity of the integral over a disjoint union of measurable sets, $\int_S = \int_{A_1} + \int_{A_2}$ for each of the two products. Subtracting and using linearity of the integral on A_1 — legitimate because both products are integrable there — we get from (3)

$$\int K_{\gamma,j}(x,y)g\,dy - \int K_{\gamma,j}(z,y)g\,dy = \int_{A_1} \big(K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big)g(y)\,dy + \Big(\int_{A_2} K_{\gamma,j}(x,y)g\,dy - \int_{A_2} K_{\gamma,j}(z,y)g\,d \tag{4}$$

Step 5: the near annulus. The hypotheses of `GaussRieszKernelHormanderDyadicBound` at R are met with this g, with Φ read in $[0,+\infty]$, and with these x, z, ε: that node requires $|x| \leq R$,

$0 < \varepsilon \le 1$, $|x - z| < \varepsilon/2$ and exactly hypothesis (c). It therefore gives

$$\int_{A_1}^{*} \big|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big|\,|g(y)|\,d\lambda(y) \le C_d\,\Phi$$

in $[0,+\infty]$. For any function h and any measurable set E one has $\big|\int_E h\,d\lambda\big| \le \int_E^{*} |h|\,d\lambda$, the right side read in $[0,+\infty]$ and the inequality understood after passing the right side to $[0,\infty)$ when it is finite; here it is finite, being at most $C_d\Phi$ with C_d, Φ real and $\Phi \ge 0$. Applying this with $E = A_1$ and $h(y) = (K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y))g(y)$, whose absolute value is $|K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)|\,|g(y)|$,

$$\Big|\int_{A_1} \big(K_{\gamma,j}(x,y) - K_{\gamma,j}(z,y)\big)g(y)\,dy\Big| \le C_d\,\Phi. \tag{5}$$

Step 6: the far annulus. Let $y \in A_2$, so $|x-y| > 1$ and in particular $1 \le |x-y|$. Since $|x| \le R$, clause (1)(a) of `GaussRieszKernelFarMajorant` at the pair $(R,1)$ gives $|K_{\gamma,j}(x,y)| \le C_a e^{-|y|^2}$, hence $|K_{\gamma,j}(x,y)g(y)| \le C_a\,e^{-|y|^2}|g(y)|$. Both of these are integrable on A_2 — the left by Step 2 and the right by (d) — so, integrating the pointwise bound and then using that $e^{-|y|^2}|g(y)| \ge 0$ is integrable on $\mathbb{R}^n$, so that its integral over the subset A_2 is at most its integral over $\mathbb{R}^n$,

$$\Big|\int_{A_2} K_{\gamma,j}(x,y)g(y)\,dy\Big| \le \int_{A_2} \big|K_{\gamma,j}(x,y)g(y)\big|\,d\lambda \le C_a \int_{\mathbb{R}^n} e^{-|y|^2}|g|\,d\lambda \le C_a\,N, \tag{6}$$

the last step by (d). Symmetrically, for $y \in A_2$ we have $|z| \le R+1$ by (1) and

$$|z-y| \ge |x-y| - |x-z| > 1 - \tfrac{1}{2}\varepsilon \ge 1 - \tfrac{1}{2} = \tfrac{1}{2},$$

using $\varepsilon \le 1$; so clause (1)(a) at the pair $(R+1, \frac{1}{2})$ gives $|K_{\gamma,j}(z,y)| \le C_b e^{-|y|^2}$, and the same three steps give

$$\Big|\int_{A_2} K_{\gamma,j}(z,y)g(y)\,dy\Big| \le C_b\,N. \tag{7}$$

Step 7: conclusion. Taking absolute values in (4) and using the triangle inequality twice, then (5), (6) and (7),

$$\Big|\int K_{\gamma,j}(x,y)g\,dy - \int K_{\gamma,j}(z,y)g\,dy\Big| \le C_d\Phi + \big(C_aN + C_bN\big).$$

Finally $C_d \le C$ and $C_a + C_b \le C$ by definition of C, and $\Phi \ge 0$ and $N \ge 0$, so $C_d\Phi \le C\Phi$ and $(C_a + C_b)N \le CN$, whence the right-hand side is at most $C(\Phi + N)$. □

2.151. `LocalMaximalLevelSetSmall`.

Lemma 2.151 (A Chebyshev level set of a locally averaged function occupies at most a quarter of a small ball)**.** *Fix $n \ge 1$, let λ be Lebesgue measure on $\mathbb{R}^n$ and let γ_n be the Gaussian measure of `GaussMeasure`. Let $R \ge 1$. Then there is a constant $K \in (0,\infty)$, depending only on n and R, such that the following holds.*

Let $u : \mathbb{R}^n \to \mathbb{R}$, let $x \in \mathbb{R}^n$ and let ε and Ψ be real, and suppose

$$|x| \le R, \qquad 0 < \varepsilon \le 1, \qquad 0 \le \Psi,$$

and that

(a) *$u \ge 0$ everywhere;*
(b) *u is Lebesgue-measurable up to a null set; and*
(c) *$\int_{\overline{B}(x,\varepsilon/2)}^{*} u\,d\lambda \;\le\; \Psi \cdot \lambda\big(\overline{B}(x,\varepsilon/2)\big)$, the left side being the lower Lebesgue integral in $[0,+\infty]$ and the right side read there too.*

Then

$$\gamma_n\Big(\big\{z \in B(x,\tfrac{1}{2}\varepsilon) : u(z) > K\Psi\big\}\Big) \;\le\; \tfrac{1}{4}\,\gamma_n\big(B(x,\tfrac{1}{2}\varepsilon)\big).$$

Why the statement is about u and not about u^2. *The intended consumer applies this with $u = |v|^{1/2}$ for a function v whose level sets at height $(K\Psi)^2$ it wants to control, since $|v(z)| > (K\Psi)^2$ holds exactly when $|v(z)|^{1/2} > K\Psi$, the square root being increasing on $[0,\infty)$. That equivalence*

is one line in the consumer, and the square root is an artefact of how the consumer reaches its threshold rather than anything this lemma needs; so the statement is given for u directly.

Why (c) is a lower Lebesgue integral and (b) is nonetheless required. *Hypothesis (c) is stated with $\int^*$ because that is the shape a consumer has available: it is what the definition of the local maximal function of* `LocalMaximalFunction` *as a supremum over radii in $(0,2]$ supplies, after multiplying by the finite positive number $\lambda(\overline{B}(x,\varepsilon/2))$. Unlike (c), the Chebyshev step in the proof does need (b): it compares the integral of u over the level set with the constant $K\Psi$ there, and that comparison is made on the measure restricted to the level set, which requires the level set to be measurable up to a null set. Hypothesis (b) is what supplies that, and a consumer has it, the canonical representative of an almost-everywhere equivalence class being measurable up to a null set.*

The degenerate case $\Psi = 0$ is included. *When $\Psi = 0$ the hypothesis forces the lower integral of u over $\overline{B}(x,\varepsilon/2)$ to vanish, hence $u = 0$ λ-almost everywhere there by (a), so the level set $\{z \in B(x,\frac{1}{2}\varepsilon) : u(z) > 0\}$ is λ-null and therefore γ_n-null, the two measures having the same null sets by clause (4) of* `GaussLebesgueBallComparison`*. That case is not reachable by Chebyshev, which needs a positive threshold.*

The constant depends on R, necessarily. *One may take $K = 4e^{(R+2)^2}$, which is $4p/a$ for $p = \pi^{-n/2}$ the upper bound for the Gaussian density and $a = \pi^{-n/2}e^{-(R+2)^2}$ its lower bound on $\overline{B}(0,R+2)$; the dimensional factors cancel, so K happens not to depend on n, but no uniformity in R is claimed.*

Proof. Put $p = \pi^{-n/2} > 0$ and $a = \pi^{-n/2}e^{-(R+2)^2} > 0$, so that $p/a = e^{(R+2)^2}$, and set $K = 4e^{(R+2)^2} = 4p/a \in (0,\infty)$. Fix u, x, ε, Ψ and the hypotheses, and write $B' = B(x,\frac{1}{2}\varepsilon)$ and $S = \overline{B}(x,\frac{1}{2}\varepsilon)$, so $B' \subseteq S$.

Step 1: the two balls have the same volume, and $\gamma_n(B')$ is bounded below. Lebesgue measure on $\mathbb{R}^n$ is an additive Haar measure and $\mathbb{R}^n$ has real dimension n, so both $\lambda(\overline{B}(x,r))$ and $\lambda(B(x,r))$ equal $r^n\lambda(B(0,1))$ for every $r \geq 0$ (shared context; $n \geq 1$ is what makes $\mathbb{R}^n$ nontrivial, as those statements require). Hence

$$\lambda(S) = \lambda(B'). \tag{1}$$

Also B' is measurable and $B' \subseteq \overline{B}(0,R+2)$: for $y \in B'$, $|y| - |x| \leq |y-x| < \frac{1}{2}\varepsilon \leq \frac{1}{2}$, so $|y| \leq R + \frac{1}{2} \leq R+2$. So clause (2) of `GaussLebesgueBallComparison`, applied with $R+2$ in place of its R and with $E = B'$, gives

$$a\,\lambda(B') \leq \gamma_n(B'). \tag{2}$$

Finally, since $u \geq 0$ by (a), the extended-real norm of $u(y)$ is $u(y)$ itself for every y; we use this silently below when passing between $\int^* u$ and $\int^* \|u\|$.

Step 2: the case $\Psi = 0$. Then the right-hand side of (c) is 0, so $\int_S^* u\,d\lambda = 0$. A nonnegative function with vanishing lower Lebesgue integral vanishes almost everywhere, so $u = 0$ λ-almost everywhere on S; that is, for λ-almost every y, $y \in S$ implies $u(y) = 0$. The threshold is $K\Psi = 0$, so the level set is $E_2 = \{z \in B' : u(z) > 0\}$. Every $z \in E_2$ lies in $B' \subseteq S$ and has $u(z) \neq 0$, so E_2 is contained in the λ-null set of points where that implication fails; hence $\lambda(E_2) = 0$, and by clause (4) of `GaussLebesgueBallComparison` — which says γ_n and λ have the same null sets — also $\gamma_n(E_2) = 0$, which is at most $\frac{1}{4}\gamma_n(B')$.

Step 3: the case $\Psi > 0$. Then $t := K\Psi > 0$. Put $E_2 = \{z \in B' : u(z) > t\}$. By (b), E_2 is the intersection of the measurable set B' with a preimage under u of an open half-line, so it is measurable up to a λ-null set.

Chebyshev's inequality for λ on E_2: the constant function t is at most u at every point of E_2, so integrating that pointwise bound over E_2 and evaluating the left integral,

$$t\,\lambda(E_2) = \int_{E_2}^* t\,d\lambda \leq \int_{E_2}^* u\,d\lambda \leq \int_S^* u\,d\lambda,$$

the last step by $E_2 \subseteq B' \subseteq S$ and monotonicity of the lower Lebesgue integral in the domain. By (c) and then (1),

$$t\,\lambda(E_2) \leq \Psi\,\lambda(S) = \Psi\,\lambda(B').$$

Since $t > 0$ is finite and nonzero, multiplying through by t^{-1},

$$\lambda(E_2) \le \frac{\Psi}{t}\,\lambda(B'). \tag{3}$$

Now pass to γ_n. Clause (1) of `GaussLebesgueBallComparison`, which holds for every set, gives $\gamma_n(E_2) \le p\,\lambda(E_2)$. With (3) and the definition of K,

$$\gamma_n(E_2) \le p\,\frac{\Psi}{K\Psi}\,\lambda(B') = \frac{p}{K}\,\lambda(B') = \frac{p\,a}{4p}\,\lambda(B') = \frac{a}{4}\,\lambda(B'),$$

the cancellation of Ψ being legitimate because $\Psi > 0$. Finally (2) gives $a\,\lambda(B') \le \gamma_n(B')$, so dividing by 4,

$$\gamma_n(E_2) \le \tfrac{1}{4}\,a\,\lambda(B') \le \tfrac{1}{4}\gamma_n(B'),$$

which is the assertion. □

2.152. `GaussRieszCotlarLocal`.

Lemma 2.152 (Cotlar's inequality for $K_{\gamma,j}$, localized to a ball)**.** *Fix $n \ge 1$ and $j \in \{1,\dots,n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel` and let M_2 be the local maximal function of `LocalMaximalFunction`. Let*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

For $f \in L^1(\gamma_n)$, written also for its canonical representative, put

$$T_{j,\varepsilon}f(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)f(y)\,dy, \qquad T_{j,*}f(x) = \sup_{\varepsilon>0}\big|T_{j,\varepsilon}f(x)\big| \;\in\; [0,+\infty],$$

the y-integral against Lebesgue measure and the supremum taken in the extended nonnegative reals. Let $R \ge 1$. Then there is a constant $C \in (0,\infty)$, depending only on n, j and R, such that for every $f \in L^1(\gamma_n)$ and every *x with $|x| \le R$,*

$$T_{j,*}f(x) \;\le\; C\Big[\,M_2\big(|(Tf)_j|^{1/2}\,\mathbf{1}_{\overline{B}(0,R+2)}\big)(x)^2 + M_2\big(f\,\mathbf{1}_{\overline{B}(0,R+2)}\big)(x) + \|f\|_{L^1(\gamma_n)}\Big],$$

the right-hand side read in $[0,+\infty]$, where $(Tf)_j$ denotes the jth coordinate of the canonical representative of Tf.

Every x, not almost every x. *The conclusion holds at every point of the closed ball, with no exceptional set. This matters downstream: the consumer integrates the right-hand side in x against weak-type bounds for M_2, and an almost-everywhere conclusion in x would still suffice there, but the stronger form costs nothing and is what the argument produces. The exceptional sets that do arise — three of them — are all in the auxiliary variable z over which a point is selected inside the proof, and they are discarded there. Note also that both maximal functions and the level sets below are unchanged if the representative of Tf or of f is modified on a null set, so the choice of the canonical representative is a matter of fixing notation and not an extra hypothesis.*

Why both sides are read in $[0,+\infty]$. *M_2 is extended-real valued by `LocalMaximalFunction`, and $T_{j,*}f(x)$ is a supremum over $\varepsilon > 0$ of real numbers which is not known in advance to be finite: the bound $|T_{j,\varepsilon}f(x)| \le C(\varepsilon)\|f\|_{L^1(\gamma_n)}$ available from the far majorant degrades as $\varepsilon \downarrow 0$. Taking the supremum in $[0,+\infty]$ rather than as a real supremum is what keeps the inequality from being satisfiable by a conventional value where the family is unbounded; it is the same discipline that makes `LocalMaximalFunction` extended-real valued. Each individual $T_{j,\varepsilon}f(x)$ is a genuine absolutely convergent integral, by clause (1)(b) of `GaussRieszKernelFarMajorant`.*

The constant depends on R, necessarily, and the statement is coordinatewise. *C degrades in R through the far-majorant and near-diagonal constants and through $a_{R+2} = \pi^{-n/2}e^{-(R+2)^2}$, the*

lower bound for the Gaussian density on $\overline{B}(0,R+2)$; no uniformity in n, j or R is claimed. The object bounded is the coordinatewise maximal truncation $\sup_\varepsilon |T_{j,\varepsilon}f|$, not the reference paper's $\sup_\varepsilon |T_\varepsilon f|_{\mathbb{R}^n}$, which is a different quantity; nothing here asserts anything about the latter.

The hypothetical style. *(T1) and (T2) are written out in full and are not imported by reference; the lemma is a statement about every T satisfying them, exactly as* `CoordinateEndpointBound` *and* `GaussRieszOffDiagonalL1` *are, and it is not vacuous because* `L1Extension` *exhibits such a T.*

Proof. Fix $R \geq 1$ and j. Write $p = \pi^{-n/2}$, $a = \pi^{-n/2}e^{-(R+2)^2} > 0$, and $\omega_n = \lambda(\overline{B}(0,1))$, which lies in $(0,\infty)$ because the closed unit ball is compact (so of finite measure, shared context) and contains a nonempty open set (so of positive measure).

Fix once and for all the following three constants.

- $C_a > 0$: the constant supplied by clause (1) of `GaussRieszKernelFarMajorant` for the pair $(R,1)$.
- $C_b > 0$: the constant supplied by the same clause for the pair $(R+1, \frac{1}{2})$.
- $C_h > 0$: the constant supplied by `GaussRieszKernelNearBounds` at R (its hypothesis $R > 0$ holds since $R \geq 1$).

Set

$$c = \max\Big(1,\ (C_a + C_b)\pi^{n/2},\ C_h\omega_n 2^n + 8\cdot 2^n p\, a^{-1},\ 16\, p^2 a^{-2}\Big) \in [1,\infty),$$

which depends only on n, j and R, and let C be c read in $(0,\infty)$. We show the asserted inequality with this C.

Fix $f \in L^1(\gamma_n)$ and x with $|x| \leq R$, and abbreviate

$$\Phi = M_2\big(f\mathbf{1}_{\overline{B}(0,R+2)}\big)(x), \quad \Psi = M_2\big(|(Tf)_j|^{1/2}\mathbf{1}_{\overline{B}(0,R+2)}\big)(x), \quad N = \|f\|_{L^1(\gamma_n)} = \int |f|\, d\gamma_n < \infty.$$

Write $w = f\mathbf{1}_{\overline{B}(0,R+2)}$ and $u = |(Tf)_j|^{1/2}\mathbf{1}_{\overline{B}(0,R+2)}$ for the two functions whose maximal functions these are.

Step 0: the infinite cases, and the reduction to a real bound. If $\Phi = +\infty$ or $\Psi = +\infty$ then the bracket on the right is $+\infty$ and, since $C > 0$, the right-hand side is $+\infty$ and there is nothing to prove. So assume $\Phi, \Psi < \infty$ and write $\varphi, \psi \geq 0$ for the corresponding reals. The embedding of $[0,\infty)$ into $[0,+\infty]$ is an order isomorphism onto its image and respects sums, products and squares, so $\Psi^2 + \Phi + N$ is the image of the real $\psi^2 + \varphi + N$ and $C\,[\Psi^2 + \Phi + N]$ is the image of $c(\psi^2 + \varphi + N)$; and the supremum defining $T_{j,*}f(x)$ is the supremum in $[0,+\infty]$ of the images of the reals $|T_{j,\varepsilon}f(x)|$. It therefore suffices to prove

$$\big|T_{j,\varepsilon}f(x)\big| \leq c\,\big(\psi^2 + \varphi + N\big) \qquad \text{for every } \varepsilon > 0, \tag{$\star$}$$

and then take the supremum over ε on the left.

Step 1: two consequences of finiteness of Φ and Ψ. By `LocalMaximalFunction`, for every $\rho \in (0,2]$

$$\frac{1}{\lambda(\overline{B}(x,\rho))}\int_{\overline{B}(x,\rho)} |w|\, d\lambda \leq \Phi, \qquad \frac{1}{\lambda(\overline{B}(x,\rho))}\int_{\overline{B}(x,\rho)} |u|\, d\lambda \leq \Psi,$$

the integrals being lower Lebesgue integrals in $[0,+\infty]$ and $\lambda(\overline{B}(x,\rho)) \in (0,\infty)$ for the same two reasons as ω_n above. Multiplying by that finite positive number and using $\Phi, \Psi < \infty$,

$$\int_{\overline{B}(x,\rho)} |w|\, d\lambda \leq \varphi\,\lambda(\overline{B}(x,\rho)) < \infty, \qquad \int_{\overline{B}(x,\rho)} |u|\, d\lambda \leq \psi\,\lambda(\overline{B}(x,\rho)) < \infty \tag{1}$$

for every $\rho \in (0,2]$; in particular w and u are Lebesgue integrable on every such ball. We also record the dilation identity $\lambda(\overline{B}(x,r)) = \omega_n r^n$ for $r \geq 0$, which holds for every $n \in \mathbb{N}_0$: the closed ball $\overline{B}(x,r)$ is the image of $\overline{B}(0,1)$ under $v \mapsto x + rv$, Lebesgue measure is translation invariant (shared context) and satisfies $\lambda(rA) = r^n\lambda(A)$.

Step 2: the Gaussian/Lebesgue conversions used below. By `GaussMeasure`, γ_n is λ weighted by $\varrho(y) = \pi^{-n/2}e^{-|y|^2}$, so for every measurable $v \geq 0$

$$\int v\, d\gamma_n = \int v(y)\varrho(y)\, d\lambda(y). \tag{2}$$

Two consequences. First, taking $v = |f|$ and using $\varrho(y) = p\,e^{-|y|^2}$,

$$\int_{\mathbb{R}^n} e^{-|y|^2}|f(y)|\,d\lambda(y) = \pi^{n/2}N < \infty. \tag{3}$$

The integrability of $y \mapsto e^{-|y|^2}|f(y)|$ against λ together with the value in (3) is GaussLebesgueL1DensityTransf stated there for an arbitrary element of $L^1(\gamma_n)$ and applied here to f; that node is what the formalization invokes at this point. Second, $\varrho \geq a$ on $\overline{B}(0, R+2)$ and $\varrho \leq p$ everywhere, so for every measurable $v \geq 0$ vanishing off $\overline{B}(0, R+2)$,

$$a \int v\,d\lambda \leq \int v\,d\gamma_n \leq p \int v\,d\lambda. \tag{4}$$

In particular, with $v = |w|$, $\int_{\overline{B}(0,R+2)} |f|\,d\lambda \leq a^{-1}N < \infty$, so f is Lebesgue integrable on $\overline{B}(0, R+2)$.

Step 3: the case $\varepsilon \geq 1$. Let $\varepsilon \geq 1$. By clause (1)(b) of GaussRieszKernelFarMajorant at the pair $(R, 1)$ — applicable since f is γ_n-integrable and $|x| \leq R$ — the function $y \mapsto K_{\gamma,j}(x, y)f(y)$ is Lebesgue integrable on $\{y : |x-y| > 1\}$, hence on the subset $\{y : |x-y| > \varepsilon\}$. By clause (1)(a) of the same node at the same pair, $|K_{\gamma,j}(x, y)| \leq C_a e^{-|y|^2}$ whenever $|x| \leq R$ and $|x - y| \geq 1$, which holds throughout that subset. Hence, by (3),

$$|T_{j,\varepsilon}f(x)| \leq \int_{\{|x-y|>\varepsilon\}} |K_{\gamma,j}(x, y)|\,|f(y)|\,d\lambda(y) \leq C_a \int e^{-|y|^2}|f|\,d\lambda = C_a\pi^{n/2}N \leq c\,N,$$

which is $(\star)$ for such ε. For the rest of the proof fix ε with $0 < \varepsilon \leq 1$.

Step 4: the splitting of f, and the two inner balls. Put

$$B = \overline{B}(x, \varepsilon), \qquad B' = B(x, \varepsilon/2),$$

and let $h, g \in L^1(\gamma_n)$ be the classes of the functions $f\mathbf{1}_B$ and $f\mathbf{1}_{\mathbb{R}^n\setminus B}$ respectively; both are γ_n-integrable because they are dominated by $|f|$. Their canonical representatives agree γ_n-almost everywhere, hence λ-almost everywhere by clause (4) of GaussLebesgueBallComparison, with those two functions, so we may and do use $f\mathbf{1}_B$ and $f\mathbf{1}_{\mathbb{R}^n\setminus B}$ as the representatives. Since $f = f\mathbf{1}_B + f\mathbf{1}_{\mathbb{R}^n\setminus B}$ pointwise, $f = h + g$ in $L^1(\gamma_n)$.

Note $B' \subseteq B \subseteq \overline{B}(0, R+2)$, because $|x| \leq R$ and $\varepsilon \leq 1$; and g vanishes identically on B, hence γ_n-almost everywhere on the open ball $B(x, \varepsilon)$.

Step 5: the two integral representations at x and on B'. Apply GaussRieszOffDiagonalL1 to T (with its two properties, which are (T1) and (T2) and hold by assumption), to g, to j, and with $x_0 = x$ and $\rho = \varepsilon > 0$; its vanishing hypothesis was verified at the end of Step 4. Its clause (1) gives that for *every* $z \in B(x, \varepsilon/2) = B'$ the function $y \mapsto K_{\gamma,j}(z, y)g(y)$ is Lebesgue integrable on $\mathbb{R}^n$, and its clause (2) gives a γ_n-null set N_1 off which

$$z \in B' \implies (Tg)_j(z) = \int_{\mathbb{R}^n} K_{\gamma,j}(z, y)g(y)\,dy. \tag{5}$$

Since $\varepsilon > 0$ we have $x \in B'$, so clause (1) applies at $z = x$ as well. Moreover $\mathbb{R}^n \setminus B = \{y : |x - y| > \varepsilon\}$ exactly, so $K_{\gamma,j}(x, \cdot)g$ is the restriction of $K_{\gamma,j}(x, \cdot)f$ to that set, extended by zero; since it is Lebesgue integrable on $\mathbb{R}^n$ by the previous paragraph,

$$T_{j,\varepsilon}f(x) = \int_{\mathbb{R}^n} K_{\gamma,j}(x, y)g(y)\,dy, \tag{6}$$

both sides being absolutely convergent.

Step 6: the difference estimate, uniformly over $z \in B'$. We claim that for every $z \in B'$,

$$\left|\int K_{\gamma,j}(x, y)g(y)\,dy - \int K_{\gamma,j}(z, y)g(y)\,dy\right| \leq C_h\omega_n 2^n\varphi + (C_a + C_b)\pi^{n/2}N. \tag{7}$$

Fix $z \in B'$, so $|x - z| < \varepsilon/2 \leq 1/2$ and $|z| \leq |x| + 1/2 \leq R + 1$. Both integrals in (7) are absolutely convergent by Step 5, and g vanishes off $\{y : |x - y| > \varepsilon\} = A_1 \cup A_2$, where

$$A_1 = \{y : \varepsilon < |x - y| \leq 1\}, \qquad A_2 = \{y : |x - y| > 1\},$$

disjoint. So the left-hand side of (7) is at most

$$\int_{A_1} \big|K_{\gamma,j}(x,y)-K_{\gamma,j}(z,y)\big|\,|f(y)|\,d\lambda(y) \;+\; \int_{A_2} |K_{\gamma,j}(x,y)||f|\,d\lambda \;+\; \int_{A_2} |K_{\gamma,j}(z,y)||f|\,d\lambda.$$

Here we used that the difference of the two absolutely convergent integrals is $\int (K_{\gamma,j}(x,y)-K_{\gamma,j}(z,y))g(y)\,dy$, that g vanishes off $A_1\cup A_2$ and equals f on it, and that on A_2 the integrand is bounded in absolute value by the sum of the two separate absolute values. All three integrals are finite: on A_1 the integrand is $|K_{\gamma,j}(x,y)g(y)-K_{\gamma,j}(z,y)g(y)|$, dominated by $|K_{\gamma,j}(x,\cdot)g|+|K_{\gamma,j}(z,\cdot)g|$, and on A_2 the two integrands are $|K_{\gamma,j}(x,\cdot)g|$ and $|K_{\gamma,j}(z,\cdot)g|$; both of those functions are Lebesgue integrable on $\mathbb{R}^n$ by Step 5.

The far part A_2. For $y\in A_2$: $|x|\le R$ and $|x-y|\ge 1$, so clause (1)(a) of `GaussRieszKernelFarMajorant` at $(R,1)$ gives $|K_{\gamma,j}(x,y)|\le C_a e^{-|y|^2}$; and $|z-y|\ge |x-y|-|x-z|>1-\frac12=\frac12$ with $|z|\le R+1$, so clause (1)(a) at $(R+1,\frac12)$ gives $|K_{\gamma,j}(z,y)|\le C_b e^{-|y|^2}$. With (3), the two A_2 terms are together at most $(C_a+C_b)\pi^{n/2}N$.

The near part A_1, by the Hörmander estimate. Let $y\in A_1$. Then $|x|\le R$, $0<\varepsilon<|x-y|\le 1$ and $|x-z|<\varepsilon/2<|x-y|/2$, which are exactly the hypotheses of clause (2) of `GaussRieszKernelNearBounds` at R; it gives

$$\big|K_{\gamma,j}(x,y)-K_{\gamma,j}(z,y)\big|\le \frac{C_h\,|x-z|}{|x-y|^{n+1}}\le \frac{C_h\,\varepsilon/2}{|x-y|^{n+1}}.$$

Decompose A_1 dyadically. For $k\in\mathbb{N}_0$ let $D_k=\{y: 2^k\varepsilon<|x-y|\le 2^{k+1}\varepsilon\}$; these are pairwise disjoint, and every $y\in A_1$ lies in exactly one of them, namely the one with k the largest integer ≥ 0 with $2^k\varepsilon<|x-y|$ (such a k exists because $|x-y|>\varepsilon=2^0\varepsilon$ and $2^k\varepsilon\to\infty$). Let $\mathcal{K}$ be the set of $k\in\mathbb{N}_0$ with $D_k\cap A_1\neq\emptyset$; for $k\in\mathcal{K}$ we have $2^k\varepsilon<1$, hence

$$r_k:=2^{k+1}\varepsilon<2,$$

so that $r_k\in(0,2]$ and (1) applies at radius r_k. This is the only place the radius cap of `LocalMaximalFunction` is used, and it is used with no margin: the cap is 2 and the Hörmander estimate is available only for $|x-y|\le 1$, which is exactly what forces $r_k<2$.

For $k\in\mathcal{K}$, on D_k we have $|x-y|^{-n-1}\le (2^k\varepsilon)^{-n-1}$ and $D_k\subseteq\overline{B}(x,r_k)$. Also $\overline{B}(x,r_k)\subseteq\overline{B}(0,R+2)$, since $|x|\le R$ and $r_k\le 2$; hence $f=w$ on $\overline{B}(x,r_k)$ and, by (1) and the dilation identity of Step 1,

$$\int_{D_k}|f|\,d\lambda\le\int_{\overline{B}(x,r_k)}|w|\,d\lambda\le\varphi\,\omega_n r_k^n.$$

Therefore

$$\int_{D_k\cap A_1}\big|K_{\gamma,j}(x,y)-K_{\gamma,j}(z,y)\big|\,|f|\,d\lambda\le C_h\frac{\varepsilon}{2}\,(2^k\varepsilon)^{-n-1}\varphi\,\omega_n(2^{k+1}\varepsilon)^n=C_h\,\omega_n\,2^{n-1}\,2^{-k}\varphi,$$

because $(2^k\varepsilon)^{-n-1}(2^{k+1}\varepsilon)^n=2^{n-k}\varepsilon^{-1}$ and $(\varepsilon/2)\cdot 2^{n-k}\varepsilon^{-1}=2^{n-k-1}$. The sets $D_k\cap A_1$, $k\in\mathcal{K}$, are pairwise disjoint, measurable and cover A_1, so by countable additivity the integral over A_1 is the sum of the integrals over them. Summing the displayed bound over $k\in\mathcal{K}$ and enlarging to all $k\ge 0$ (the terms are nonnegative), $\sum_{k\ge0}2^{-k}=2$ gives

$$\int_{A_1}\big|K_{\gamma,j}(x,y)-K_{\gamma,j}(z,y)\big|\,|f|\,d\lambda\le C_h\omega_n 2^n\varphi.$$

Adding the two parts gives (7).

Step 7: selection of a good z. Put

$$\lambda_1=8\cdot 2^n p\,a^{-1}\varphi\ge 0,\qquad \lambda_2=\big(4p\,a^{-1}\psi\big)^2\ge 0,$$

and

$$E_1=\{z\in B': |(Th)_j(z)|>\lambda_1\},\qquad E_2=\{z\in B': |(Tf)_j(z)|>\lambda_2\}.$$

We claim $\gamma_n(E_i)\le\gamma_n(B')/4$ for $i=1,2$.

First record two measure facts. By the dilation identity, $\lambda(B)=\omega_n\varepsilon^n$; and $B'=B(x,\varepsilon/2)$ differs from $\overline{B}(x,\varepsilon/2)$ by the sphere of radius $\varepsilon/2$, which is Lebesgue-null in $\mathbb{R}^n$ for $n\ge1$, so

$\lambda(B') = \omega_n(\varepsilon/2)^n = 2^{-n}\lambda(B)$. Also B' is measurable and $B' \subseteq \overline{B}(0, R+2)$, so clause (2) of `GaussLebesgueBallComparison`, applied with $R+2$ in place of its R, gives

$$\gamma_n(B') \geq a\,\lambda(B') = a\,2^{-n}\lambda(B) > 0. \tag{8}$$

The set E_1. Suppose first $\varphi > 0$, so $\lambda_1 > 0$. By `CoordinateEndpointBound`, applied to T with its two properties, to the element h, to the coordinate j and to $t = \lambda_1$,

$$\gamma_n(E_1) \leq \gamma_n\big(\{z : |(Th)_j(z)| > \lambda_1\}\big) \leq \frac{2\|h\|_{L^1(\gamma_n)}}{\lambda_1}.$$

Now $\|h\|_{L^1(\gamma_n)} = \int_B |f|\,d\gamma_n$, and $B \subseteq \overline{B}(0, R+2)$ so $f = w$ on B; by the upper half of (4) and then (1) at radius $\varepsilon \in (0, 2]$,

$$\|h\|_{L^1(\gamma_n)} \leq p \int_B |w|\,d\lambda \leq p\,\varphi\,\lambda(B).$$

Hence

$$\gamma_n(E_1) \leq \frac{2p\varphi\lambda(B)}{8 \cdot 2^n p\,a^{-1}\varphi} = \frac{a\,2^{-n}\lambda(B)}{4} \leq \frac{\gamma_n(B')}{4}$$

by (8). Suppose instead $\varphi = 0$. Then by (1) at radius ε, $\int_B |w|\,d\lambda = 0$, so $w = 0$ and hence $f = 0$ λ-almost everywhere on B; therefore $f\mathbf{1}_B = 0$ λ-almost everywhere, hence γ_n-almost everywhere by clause (4) of `GaussLebesgueBallComparison`, so $h = 0$ in $L^1(\gamma_n)$. By linearity $Th = T0 = 0$, so $(Th)_j = 0$ γ_n-almost everywhere and $E_1 = \{z \in B' : |(Th)_j(z)| > 0\}$ is γ_n-null. In both cases $\gamma_n(E_1) \leq \gamma_n(B')/4$.

The set E_2. Since $B' \subseteq \overline{B}(0, R+2)$, on B' we have $u = |(Tf)_j|^{1/2}$. Note that u is measurable, the canonical representative of an almost-everywhere class being measurable, so E_1 and E_2 are measurable sets. Suppose first $\psi > 0$, so $\lambda_2 > 0$ and $\lambda_2^{1/2} = 4p\,a^{-1}\psi$. For $z \in B'$, $|(Tf)_j(z)| > \lambda_2$ holds exactly when $u(z) > \lambda_2^{1/2}$, the square root being increasing on $[0, \infty)$. By Chebyshev's inequality for λ applied to the nonnegative function u on B' — that is, $\lambda_2^{1/2}\lambda(\{z \in B' : u(z) > \lambda_2^{1/2}\}) \leq \int_{B'} u\,d\lambda$, which is immediate from $\lambda_2^{1/2}\mathbf{1}_{\{u > \lambda_2^{1/2}\}} \leq u$ on B' — together with $B' \subseteq \overline{B}(x, \varepsilon/2)$ and (1) at radius $\varepsilon/2 \in (0, 2]$,

$$\lambda(E_2) \leq \frac{1}{\lambda_2^{1/2}} \int_{\overline{B}(x,\varepsilon/2)} u\,d\lambda \leq \frac{\psi\,\lambda(\overline{B}(x, \varepsilon/2))}{\lambda_2^{1/2}} = \frac{\psi\,2^{-n}\lambda(B)}{4p\,a^{-1}\psi} = \frac{a\,2^{-n}\lambda(B)}{4p}.$$

By clause (1) of `GaussLebesgueBallComparison`, which holds for every set, $\gamma_n(E_2) \leq p\,\lambda(E_2) \leq a2^{-n}\lambda(B)/4 \leq \gamma_n(B')/4$ by (8). Suppose instead $\psi = 0$. Then by (1) at radius $\varepsilon/2$, $\int_{\overline{B}(x,\varepsilon/2)} u\,d\lambda = 0$, so $u = 0$ λ-almost everywhere on B', i.e. $(Tf)_j = 0$ λ-almost everywhere on B', hence γ_n-almost everywhere there by clause (4) of `GaussLebesgueBallComparison`; so $E_2 = \{z \in B' : |(Tf)_j(z)| > 0\}$ is γ_n-null. In both cases $\gamma_n(E_2) \leq \gamma_n(B')/4$.

Two further null sets, both free. N_1 is the γ_n-null set of (5). And since T is linear, $Tg = Tf - Th$ as almost-everywhere classes, so the canonical representative of Tg agrees γ_n-almost everywhere with the difference of those of Tf and Th; let N_2 be the γ_n-null set off which

$$(Tg)_j(z) = (Tf)_j(z) - (Th)_j(z). \tag{9}$$

The good set is nonempty. Let $G = B' \setminus (E_1 \cup E_2 \cup N_1 \cup N_2)$. By subadditivity (shared context),

$$\gamma_n(G) \geq \gamma_n(B') - \gamma_n(E_1) - \gamma_n(E_2) - \gamma_n(N_1) - \gamma_n(N_2) \geq \gamma_n(B') - \tfrac{1}{4}\gamma_n(B') - \tfrac{1}{4}\gamma_n(B') = \tfrac{1}{2}\gamma_n(B') > 0$$

by (8), the subtraction being legitimate because $\gamma_n(B') \leq p\,\lambda(B') < \infty$ by clause (1) of `GaussLebesgueBallComparison`. Hence $G \neq \emptyset$; fix $z \in G$.

Step 8: conclusion. By (6), the triangle inequality, (7) and then (5) (legitimate since $z \in B'$ and $z \notin N_1$),

$$|T_{j,\varepsilon}f(x)| \leq \left|\int K_{\gamma,j}(x,y)g\,dy - \int K_{\gamma,j}(z,y)g\,dy\right| + \left|\int K_{\gamma,j}(z,y)g\,dy\right| \leq C_h\omega_n 2^n\varphi + (C_a + C_b)\pi^{n/2}N + \big|(Tg)_j(z)\big|.$$

By (9) (as $z \notin N_2$) and then $z \notin E_1 \cup E_2$,

$$|(Tg)_j(z)| \le |(Tf)_j(z)| + |(Th)_j(z)| \le \lambda_2 + \lambda_1 = 16p^2a^{-2}\psi^2 + 8\cdot 2^n p\, a^{-1}\varphi.$$

Combining, and using the definition of c,

$$|T_{j,\varepsilon}f(x)| \le \big(C_h\omega_n 2^n + 8\cdot 2^n p\, a^{-1}\big)\varphi + 16p^2a^{-2}\psi^2 + (C_a + C_b)\pi^{n/2}N \le c\,(\varphi + \psi^2 + N).$$

This is $(\star)$ for $0 < \varepsilon \le 1$; with Step 3 it holds for every $\varepsilon > 0$. Taking the supremum over $\varepsilon > 0$ in $[0, +\infty]$ gives $T_{j,*}f(x) \le c(\psi^2 + \varphi + N) = C\,[\Psi^2 + \Phi + N]$, which is the assertion. □

2.153. `GaussMaximalTruncationLocal`.

Lemma 2.153 (Localized weak-type $(1,1)$ bound for the maximal truncation)**.** *Fix $n \ge 1$ and $j \in \{1, \dots, n\}$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of* `GaussMeasure`*, let $K_{\gamma,j}$ be the kernel of* `GaussRieszKernel`*, and let*

$$T : L^1(\gamma_n; \mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n; \mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of* `ToL1` *and R_γ the transform of* `RieszTransform`*; and*

(T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n; \mathbb{R})$ and every real $t > 0$.*

For $f \in L^1(\gamma_n)$, written also for its canonical representative, put

$$T_{j,\varepsilon}f(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)f(y)\,dy, \qquad T_{j,*}f(x) = \sup_{\varepsilon>0}\big|T_{j,\varepsilon}f(x)\big| \ \in\ [0,+\infty],$$

the y-integral against Lebesgue measure and the supremum taken in the extended nonnegative reals. Let $R \ge 1$. Then there is a constant $C \in (0,\infty)$, depending only on n, j and R, such that for every $f \in L^1(\gamma_n)$ and every $t \in (0, +\infty]$,

$$\gamma_n\Big(\big\{\, x \in \mathbb{R}^n :\ |x| \le R \ \ \text{and}\ \ T_{j,*}f(x) > t \,\big\}\Big) \ \le\ \frac{C}{t}\,\|f\|_{L^1(\gamma_n)},$$

the right-hand side read in $[0, +\infty]$.

This is the reference paper's displayed maximal estimate, localized. *The paper states it with no restriction on x and with a constant $C_{n,j}$ depending only on the dimension and the coordinate; the present node restricts x to the closed ball of radius R and allows the constant to depend on R as well. That is the whole of the departure, and it is recorded, with its return-to-faithful argument, in the deviation file* `reference/maximal-truncation-localization.tex`*, which this node claims. Nothing else about the statement differs: the operator, the truncations, the exponent and the shape $C\|f\|_1/t$ are the paper's.*

The supremum and the threshold are extended-real, and this is load-bearing. *$T_{j,*}f(x)$ is a supremum over $\varepsilon > 0$ of real numbers which is not known in advance to be finite — the far-majorant bound $|T_{j,\varepsilon}f(x)| \le C(\varepsilon)\|f\|_{L^1(\gamma_n)}$ degrades as $\varepsilon \downarrow 0$ — so it is taken in $[0, +\infty]$, exactly as in* `GaussRieszCotlarLocal`*, whose left-hand side this is verbatim. A real supremum would take a conventional value where the family is unbounded, which is precisely where the estimate carries content. Correspondingly the threshold t ranges over $(0, +\infty]$: at $t = +\infty$ the level set is empty and the assertion is the triviality $0 \le 0$, and the content is at finite t.*

No measurability of the level set is asserted or needed. *γ_n is applied to the level set as an outer measure, which is defined on every subset and is monotone and countably subadditive there. That is enough for the proof below, because every inequality it chains together — the weak-type bounds for M_2 and the comparison $\gamma_n \le \pi^{-n/2}\lambda$ — is itself stated for arbitrary sets. Bounding the outer measure is the stronger assertion, so a consumer that does know the set to be measurable loses nothing.*

The constant depends on R, and the object is coordinatewise. *C degrades in R through the Cotlar constant and through $a_{R+2} = \pi^{-n/2}e^{-(R+2)^2}$, the lower bound for the Gaussian density on $\overline{B}(0, R+2)$; no uniformity in n, j or R is claimed, and the paper explicitly licenses the*

dimension dependence at this step. The quantity bounded is the coordinatewise maximal truncation $\sup_\varepsilon |T_{j,\varepsilon} f|$*, not the paper's vector maximal truncation* $\sup_\varepsilon |T_\varepsilon f|_{\mathbb{R}^n}$*; nothing here asserts anything about the latter.*

The hypothetical style. *(T1) and (T2) are written out in full and not imported by reference; the lemma is a statement about every* T *satisfying them, as* `CoordinateEndpointBound` *and* `GaussRieszCotlarLocal` *are, and it is not vacuous because* `L1Extension` *exhibits such a* T*.*

Proof. Write $p = \pi^{-n/2}$, $a = \pi^{-n/2} e^{-(R+2)^2} \in (0,\infty)$ — so that $a^{-1} = \pi^{n/2} e^{(R+2)^2}$ — and $E = e^{(R+2)^2}$. Throughout, $\overline{B} = \overline{B}(0, R+2)$.

Step 0: the Cotlar constant. Apply `GaussRieszCotlarLocal` to this T (with the two properties it requires, which are (T1) and (T2) and hold by assumption), to j, and to this R, whose hypothesis $R \geq 1$ holds. It supplies a constant $C_0 \in (0,\infty)$ — in particular $C_0 \neq 0$ and $C_0 \neq +\infty$ — such that for every $f \in L^1(\gamma_n)$ and every x with $|x| \leq R$,

$$T_{j,*} f(x) \;\leq\; C_0 \big[\, \Psi_f(x)^2 + \Phi_f(x) + N_f \,\big], \tag{1}$$

where, with M_2 the local maximal function of `LocalMaximalFunction`,

$$\Phi_f = M_2\big(w_f\big), \quad w_f = f\mathbf{1}_{\overline{B}}; \qquad \Psi_f = M_2\big(u_f\big), \quad u_f = |(Tf)_j|^{1/2}\mathbf{1}_{\overline{B}}; \qquad N_f = \|f\|_{L^1(\gamma_n)}.$$

Crucially C_0 does not depend on f: it is produced before f is quantified. Define

$$C = 3\,C_0\big(1 + 33 \cdot 4^n E\big).$$

Then $C \in (0,\infty)$ and C depends only on n, j and R. We show the lemma with this C.

Fix $f \in L^1(\gamma_n)$ and $t \in (0,+\infty]$, and abbreviate $\Phi = \Phi_f$, $\Psi = \Psi_f$, $N = N_f$, $w = w_f$, $u = u_f$. Note $N < \infty$. Put

$$s = \frac{t}{3C_0},$$

which is nonzero because $t \neq 0$ and $3C_0 < \infty$.

Step 1: measurability of the two inputs. The canonical representative of an almost-everywhere class is measurable, so $(Tf)_j$ is measurable; hence $u = |(Tf)_j|^{1/2}\mathbf{1}_{\overline{B}}$ is measurable, the square root being continuous on $[0,\infty)$ and $\overline{B}$ closed. For w: the canonical representative f is γ_n-almost everywhere equal to a measurable function, and by clause (4) of `GaussLebesgueBallComparison` the measures γ_n and λ have the same null sets, so f is λ-almost everywhere equal to a measurable function; multiplying by the indicator of the measurable set $\overline{B}$ preserves this. Thus w and u are Lebesgue-almost-everywhere measurable, which is the hypothesis both weak-type nodes require.

Step 2: the level set is covered by three sets. Let

$$\mathcal{E} = \big\{x : |x| \leq R \text{ and } T_{j,*} f(x) > t\big\}, \qquad S_1 = \{x : \Phi(x) > s\}, \quad S_2 = \{x : \Psi(x)^2 > s\}, \quad S_3 = \{x : N > s\}.$$

We claim $\mathcal{E} \subseteq S_1 \cup S_2 \cup S_3$. Indeed, let $x \in \mathcal{E}$ and suppose x lay in none of the three. Then $\Psi(x)^2 \leq s$, $\Phi(x) \leq s$ and $N \leq s$, so $\Psi(x)^2 + \Phi(x) + N \leq 3s = 3\,t/(3C_0) = t/C_0$, the cancellation of the factor 3 being legitimate in $[0,+\infty]$ because $3 \in (0,\infty)$. Multiplying by $C_0 \in (0,\infty)$ and using (1), $T_{j,*} f(x) \leq C_0\,(t/C_0) = t$, contradicting $T_{j,*} f(x) > t$. Hence by monotonicity and subadditivity of the outer measure γ_n (shared context),

$$\gamma_n(\mathcal{E}) \leq \gamma_n(S_1) + \gamma_n(S_2) + \gamma_n(S_3). \tag{2}$$

Note that S_3 does not depend on x: it is $\mathbb{R}^n$ if $N > s$ and $\emptyset$ otherwise.

Step 3: the term S_1*.* By `LocalMaximalWeakType`, applied to w (measurable up to a null set by Step 1) at the level $s \neq 0$,

$$\lambda(S_1) \leq \frac{4^n}{s} \int_{\mathbb{R}^n} |w|\, d\lambda.$$

By `GaussMeasure`, γ_n is λ weighted by the density $\varrho(y) = \pi^{-n/2} e^{-|y|^2}$, and $\varrho \geq a$ on $\overline{B}$ because $|y| \leq R+2$ there; since w vanishes off $\overline{B}$,

$$a \int_{\mathbb{R}^n} |w|\, d\lambda = a \int_{\overline{B}} |f|\, d\lambda \leq \int_{\overline{B}} |f|\, \varrho\, d\lambda = \int_{\overline{B}} |f|\, d\gamma_n \leq N,$$

so $\int |w|\,d\lambda \le a^{-1}N = \pi^{n/2}E\,N$. Finally, by clause (1) of GaussLebesgueBallComparison, which holds for *every* set, $\gamma_n(S_1) \le p\,\lambda(S_1)$. Combining, and using $p\,\pi^{n/2} = 1$,

$$\gamma_n(S_1) \le \frac{p\,4^n\pi^{n/2}E\,N}{s} = \frac{4^n E\,N}{s} = \frac{3C_0\,4^n E\,N}{t}. \tag{3}$$

Step 4: the term S_2. Let $\sqrt{s} \in (0,+\infty]$ be the square root of s, i.e. the unique element of $[0,+\infty]$ whose square is s; it is nonzero because s is. Squaring is strictly monotone on $[0,+\infty]$, so $\Psi(x)^2 > s = (\sqrt{s})^2$ forces $\Psi(x) > \sqrt{s}$, whence $S_2 \subseteq \{x : \Psi(x) > \sqrt{s}\}$. By LocalMaximalWeakL2, applied to u (measurable by Step 1) at the level $\sqrt{s} \neq 0$, and using $(\sqrt{s})^2 = s$,

$$\lambda(S_2) \le \frac{16\cdot 4^n}{s}\,\mathcal{A}, \qquad \mathcal{A} = \sup_{0<\sigma\le+\infty} \sigma^2\,\lambda\big(\{y : |u(y)| > \sigma\}\big). \tag{4}$$

We claim the *uniform* bound

$$\mathcal{A} \le 2\,a^{-1}N = 2\,\pi^{n/2}E\,N. \tag{5}$$

It suffices to bound each term of the supremum by the right-hand side. If $\sigma = +\infty$ the set $\{|u| > \sigma\}$ is empty, u being real valued, and the term is $(+\infty)\cdot 0 = 0$. So let σ be a real number with $\sigma > 0$ and put

$$E_\sigma = \{y : |u(y)| > \sigma\} = \{y \in \overline{B} : |(Tf)_j(y)| > \sigma^2\},$$

the second description because $u = |(Tf)_j|^{1/2}$ on $\overline{B}$ and 0 off it, and because $r \mapsto r^{1/2}$ is strictly increasing on $[0,\infty)$. The set E_σ is measurable (Step 1) and contained in $\overline{B} = \overline{B}(0,R+2)$, so clause (2) of GaussLebesgueBallComparison, *instantiated at $R+2$ in place of its R*, gives $a\,\lambda(E_\sigma) \le \gamma_n(E_\sigma)$, hence $\lambda(E_\sigma) \le a^{-1}\gamma_n(E_\sigma)$ since $a \in (0,\infty)$. Next, $E_\sigma \subseteq \{y : |(Tf)_j(y)| > \sigma^2\}$, and CoordinateEndpointBound — applied to this T with its two properties, to f, to the coordinate j and to the real level $\sigma^2 > 0$ — gives

$$\sigma^2\,\gamma_n\big(\{y : |(Tf)_j(y)| > \sigma^2\}\big) \le 2N.$$

Therefore

$$\sigma^2\lambda(E_\sigma) \le a^{-1}\,\sigma^2\gamma_n(E_\sigma) \le a^{-1}\,\sigma^2\gamma_n\big(\{|(Tf)_j| > \sigma^2\}\big) \le 2\,a^{-1}N,$$

a bound independent of σ, which is (5). (The uniformity is the point: a bound still depending on σ would not bound the supremum.)

Feeding (5) into (4) and applying clause (1) of GaussLebesgueBallComparison again,

$$\gamma_n(S_2) \le p\,\lambda(S_2) \le \frac{p\cdot 16\cdot 4^n\cdot 2\pi^{n/2}E\,N}{s} = \frac{32\cdot 4^n E\,N}{s} = \frac{96\,C_0\,4^n E\,N}{t}. \tag{6}$$

Step 5: the term S_3, and the conclusion. If $N \le s$ then $S_3 = \emptyset$ and $\gamma_n(S_3) = 0$. Otherwise $N > s$; then $s \neq +\infty$ (nothing exceeds $+\infty$), so $s \in (0,\infty)$ and $N/s > 1$, while $\gamma_n(S_3) = \gamma_n(\mathbb{R}^n) = 1$ because γ_n is a probability measure (GaussIsProb). In both cases

$$\gamma_n(S_3) \le \frac{N}{s} = \frac{3C_0\,N}{t}. \tag{7}$$

Adding (3), (6) and (7) into (2),

$$\gamma_n(\mathcal{E}) \le \big(3C_0\,4^n E + 96\,C_0\,4^n E + 3C_0\big)\frac{N}{t} = 3C_0\big(1 + 33\cdot 4^n E\big)\frac{N}{t} = \frac{C\,N}{t},$$

which is the assertion. □

2.154. GaussRieszTruncationSmoothSplit.

Lemma 2.154 (Smooth approximation of an $L^1(\gamma)$ datum, and the three-term splitting of every truncated Riesz integral)**.** *Fix $n \ge 1$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of GaussMeasure, and let $K_{\gamma,j}$ be the kernel of GaussRieszKernel. For a γ_n-integrable $h : \mathbb{R}^n \to \mathbb{R}$, a coordinate $j \in \{1,\dots,n\}$, a point $x \in \mathbb{R}^n$ and a real $\varepsilon > 0$ write*

$$T_{j,\varepsilon}h(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,h(y)\,dy,$$

the y-integral against λ; for an element of $L^1(\gamma_n)$ this is read through its canonical representative.

Let $f \in L^1(\gamma_n)$ and let $k \in \mathbb{N}$. Then there exist a function $\varphi : \mathbb{R}^n \to \mathbb{R}$ and an element $g \in L^1(\gamma_n)$ such that

(i) *φ is smooth, has compact support, and is γ_n-integrable;*

(ii) $\displaystyle\int_{\mathbb{R}^n} |g|\, d\gamma_n \;<\; \frac{1}{k+1}$, *the integral of the canonical representative of g; and*

(iii) *$f = \varphi + g$ holds γ_n-almost everywhere, as an identity between the canonical representative of f, the function φ, and the canonical representative of g; equivalently $f = [\varphi] + g$ in $L^1(\gamma_n)$, where $[\varphi]$ is the class of φ, which exists by (i); and*

(iv) *for every coordinate j, every $x \in \mathbb{R}^n$ and every real $\varepsilon > 0$, each of the three functions*

$$y \mapsto K_{\gamma,j}(x,y)f(y), \qquad y \mapsto K_{\gamma,j}(x,y)\varphi(y), \qquad y \mapsto K_{\gamma,j}(x,y)g(y)$$

is Lebesgue integrable on $\{y : |x-y| > \varepsilon\}$, and

$$T_{j,\varepsilon}f(x) \;=\; T_{j,\varepsilon}\varphi(x) + T_{j,\varepsilon}g(x).$$

Why g is an element of $L^1(\gamma_n)$ and not the function $f - \varphi$. *The intended consumer feeds g to a weak-type estimate whose input is an $L^1(\gamma_n)$ element, so what it needs is a class, not a representative; and it needs the splitting (iv) stated for* that class's own *canonical representative, since that is the function whose truncations and maximal truncation the estimate controls. Stating (iv) with g rather than with $f - \varphi$ is therefore not a convenience: it discharges here, once, the reconciliation between the class $f - [\varphi]$ and the pointwise difference $f - \varphi$, which otherwise has to be carried through every later step. The two agree γ_n-almost everywhere and hence, by clause (4) of `GaussLebesgueBallComparison`, λ-almost everywhere, which is what makes the substitution legitimate inside a Lebesgue integral.*

Why (iii) is stated between representatives rather than between classes. *The two formulations are equivalent — two elements of $L^1(\gamma_n)$ whose representatives agree γ_n-almost everywhere are equal — and the representative form is the one a consumer can use without first naming the class $[\varphi]$, which needs the membership (i) as a datum. A consumer that wants the class form reads it off: (iii) says that g and $f - \varphi$ agree γ_n-almost everywhere, so $g = f - [\varphi]$ in $L^1(\gamma_n)$, and hence $\|g\|_{L^1(\gamma_n)} = \|f - [\varphi]\|_{L^1(\gamma_n)}$ and, for any linear T on $L^1(\gamma_n)$, $Tg = Tf - T[\varphi]$.*

The finiteness in (iv) is at every point, with no exceptional set. *Each of the three integrands is asserted integrable on $\{|x-y| > \varepsilon\}$ for* every *x and every $\varepsilon > 0$, so each truncation is a finite real number everywhere, and the identity in (iv) is an identity between finite reals rather than an almost-everywhere statement. This is available because the far majorant of `GaussRieszKernelFarMajorant` is a genuine pointwise kernel bound off the diagonal; no maximal-function or almost-everywhere finiteness input is used or needed.*

What the bound in (ii) does and does not say. *The constant $1/(k+1)$ is the only role k plays: the statement is an approximation statement at the single accuracy $1/(k+1)$, and φ and g both depend on k. Nothing is claimed about how φ or g vary with k, and in particular no uniformity in k and no convergence as $k \to \infty$ is asserted; a consumer that wants a sequence applies the lemma once for each k. Note also that φ and g do not depend on the coordinate j: j is quantified inside (iv), after both have been produced.*

Proof. Step 1: the approximant and the class g. Apply `SmoothCompactDenseL1Gauss` to f with $\delta = 1/(k+1) > 0$. It produces a function $\varphi : \mathbb{R}^n \to \mathbb{R}$ which is smooth, has compact support, is γ_n-integrable, and satisfies

$$\| f - \varphi \|_{L^1(\gamma_n)} < \frac{1}{k+1}, \tag{1}$$

where $f - \varphi$ means the pointwise difference of the canonical representative of f and φ, and the norm is the $L^1(\gamma_n)$ norm of that function. This gives (i).

Let $[\varphi] \in L^1(\gamma_n)$ be the class of φ, which exists because φ is γ_n-integrable, and put

$$g = f - [\varphi] \;\in\; L^1(\gamma_n).$$

The canonical representative of g agrees γ_n-almost everywhere with the pointwise difference $f - \varphi$: the representative of a difference of classes agrees almost everywhere with the difference

of the representatives, and the representative of $[\varphi]$ agrees almost everywhere with φ. Hence

$$g = f - \varphi \quad \gamma_n\text{-almost everywhere.} \tag{2}$$

Rearranging (2) pointwise at each of the γ_n-almost every y at which it holds gives $f(y) = \varphi(y) + g(y)$ there, which is (iii).

Since the $L^1(\gamma_n)$ norm of a function is unchanged by a γ_n-almost-everywhere modification, (2) and (1) give $\|g\|_{L^1(\gamma_n)} < 1/(k+1)$; and $\|g\|_{L^1(\gamma_n)} = \int |g|\, d\gamma_n$, the integral of the canonical representative. That is (ii).

Step 2: every truncation is finite, and the splitting. Fix j, $x \in \mathbb{R}^n$ and a real $\varepsilon > 0$, and put $R_x = \max(|x|, 1)$, so that $R_x > 0$ and $|x| \le R_x$. Apply clause (1) of `GaussRieszKernelFarMajorant` with $R := R_x > 0$ and $\rho := \varepsilon > 0$. Its second part states that for every $h \in L^1(\gamma_n)$ — more precisely for every h that is γ_n-integrable — and every x' with $|x'| \le R_x$, the function $y \mapsto K_{\gamma,j}(x', y)h(y)$ is Lebesgue integrable on $\{y : |x' - y| > \varepsilon\}$. Taking $x' = x$, which is legitimate because $|x| \le R_x$, and taking h in turn to be the canonical representative of f, then φ, then the canonical representative of g — all three being γ_n-integrable, the first and third as representatives of elements of $L^1(\gamma_n)$ and the second by (i) — gives the three integrability assertions of (iv). In particular each of $T_{j,\varepsilon}f(x)$, $T_{j,\varepsilon}\varphi(x)$ and $T_{j,\varepsilon}g(x)$ is a finite real number.

For the identity, note first that by clause (4) of `GaussLebesgueBallComparison` a subset of $\mathbb{R}^n$ is γ_n-null if and only if it is λ-null, so the γ_n-almost-everywhere identity (2) is also a λ-almost-everywhere identity:

$$g(y) = f(y) - \varphi(y) \quad \text{for } \lambda\text{-almost every } y. \tag{3}$$

Consequently, for λ-almost every y,

$$K_{\gamma,j}(x,y)f(y) = K_{\gamma,j}(x,y)\varphi(y) + K_{\gamma,j}(x,y)\big(f(y) - \varphi(y)\big) = K_{\gamma,j}(x,y)\varphi(y) + K_{\gamma,j}(x,y)g(y),$$

the first equality being the distributive law at the point y and the second substituting (3). The set $\{y : |x - y| > \varepsilon\}$ is Lebesgue measurable, being the strict superlevel set of the continuous function $y \mapsto |x - y|$, so two functions agreeing λ-almost everywhere have the same integral over it; hence

$$T_{j,\varepsilon}f(x) = \int_{|x-y|>\varepsilon} \big(K_{\gamma,j}(x,y)\varphi(y) + K_{\gamma,j}(x,y)g(y)\big)\,dy.$$

Both summands are Lebesgue integrable on $\{y : |x - y| > \varepsilon\}$ by the previous paragraph, so the integral of the sum is the sum of the integrals:

$$T_{j,\varepsilon}f(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\varphi(y)\,dy + \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)g(y)\,dy = T_{j,\varepsilon}\varphi(x) + T_{j,\varepsilon}g(x).$$

As j, x and $\varepsilon > 0$ were arbitrary, this is (iv). □

2.155. `GaussRieszPrincipalValueExists`.

Lemma 2.155 (The principal value of a truncated Gaussian Riesz integral exists almost everywhere)**.** *Fix $n \ge 1$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel`. Let*

$$T : L^1(\gamma_n; \mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n; \mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(T2) *$t\,\gamma_n\big(\{x : |Th(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|h\|_{L^1(\gamma_n)}$ for every $h \in L^1(\gamma_n; \mathbb{R})$ and every real $t > 0$.*

For a γ_n-integrable h, a coordinate j, a point x and a real $\varepsilon > 0$ write $T_{j,\varepsilon}h(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)h(y)\,dy$, the y-integral against λ; for an element of $L^1(\gamma_n)$ this is read through its canonical representative.

Let $f \in L^1(\gamma_n)$ and let $j \in \{1, \dots, n\}$. Then for γ_n-almost every $x \in \mathbb{R}^n$ there is a real number L with

$$\lim_{\varepsilon \downarrow 0} T_{j,\varepsilon}f(x) \;=\; L,$$

the limit taken through $\varepsilon \in (0,\infty)$ *as* $\varepsilon \to 0$.

Existence only, and deliberately so. *The statement asserts that the limit exists; it says nothing about its value, and in particular does not identify it with any coordinate of* Tf. *That identification is a separate assertion requiring separate inputs, and keeping it out of this statement is what makes the two independently usable: a consumer that wants the value obtains* L *here and identifies it elsewhere. Correspondingly no function* $x \mapsto L(x)$ *is produced: the conclusion is an existential at each good point, so no choice is made and no total function enters the interface.*

Why the conclusion is almost-everywhere. *The exceptional set is genuine and not an artefact. It is the set where the oscillation of* $\varepsilon \mapsto T_{j,\varepsilon}f(x)$ *at* 0 *fails to vanish, and it is controlled only in measure, through the weak-type bound of* `GaussMaximalTruncationLocal` *applied to the errors of a sequence of smooth approximants. By contrast each individual* $T_{j,\varepsilon}f(x)$ *is a finite real number at* every x, *by* `GaussRieszTruncationSmoothSplit`; *it is only the limit that needs an exceptional set.*

Both (T1) and (T2) are assumed, because the input carries both. *The weak-type bound of* `GaussMaximalTruncationLocal` *is a statement about every* T *satisfying the two, so this lemma is too. It is not vacuous because* `L1Extension` *exhibits such a* T.

Proof. Step 1: the approximants, their errors, and the splitting. For each $k \in \mathbb{N}$ apply `GaussRieszTruncationSm` to f and k. It produces a function φ_k and an element $g_k \in L^1(\gamma_n)$ such that φ_k is smooth with compact support and γ_n-integrable, such that

$$\int_{\mathbb{R}^n} |g_k|\, d\gamma_n < \frac{1}{k+1}, \tag{1}$$

and such that for every j, every x and every real $\varepsilon > 0$ the three integrands are Lebesgue integrable on $\{y : |x-y| > \varepsilon\}$ and

$$T_{j,\varepsilon}f(x) = T_{j,\varepsilon}\varphi_k(x) + T_{j,\varepsilon}g_k(x). \tag{2}$$

All quantities appearing in (2) are finite real numbers, at every x and every $\varepsilon > 0$.

Step 2: off a γ_n*-null set, the errors' maximal truncations can be made arbitrarily small.* We claim that for γ_n-almost every $x \in \mathbb{R}^n$,

$$\text{for every } m \in \mathbb{N} \text{ there is } k \in \mathbb{N} \text{ with } \big|T_{j,\varepsilon}g_k(x)\big| \le \frac{1}{m+1} \text{ for every real } \varepsilon > 0. \tag{3}$$

Let B be the set of x at which (3) fails. If $x \in B$ there is $m \in \mathbb{N}$ such that for every k the bound fails, i.e. for every k there is a real $\varepsilon > 0$ with $|T_{j,\varepsilon}g_k(x)| > 1/(m+1)$; and since ofReal is strictly increasing on $[0,\infty)$ and a single term is at most the supremum,

$$\operatorname{ofReal}\Big(\frac{1}{m+1}\Big) \;<\; \sup_{\varepsilon>0} \operatorname{ofReal}\big|T_{j,\varepsilon}g_k(x)\big| \qquad \text{for every } k,$$

the supremum taken in $[0,+\infty]$. Also $|x| \le r+1$ for some $r \in \mathbb{N}$, for instance $r = \lceil |x| \rceil$. Hence

$$B \;\subseteq\; \bigcup_{m\in\mathbb{N}} \bigcup_{r\in\mathbb{N}} A_{r,m}, \qquad A_{r,m} = \Big\{x : |x| \le r+1 \text{ and } \operatorname{ofReal}\big(\tfrac{1}{m+1}\big) < \sup_{\varepsilon>0} \operatorname{ofReal}\big|T_{j,\varepsilon}g_i(x)\big| \text{ for every } i\Big\}.$$

By countable subadditivity of the outer measure γ_n it therefore suffices to show $\gamma_n(A_{r,m}) = 0$ for each fixed r and m.

Fix r and m. Apply `GaussMaximalTruncationLocal` to this T with its two properties, which are (T1) and (T2) and hold by assumption, to the coordinate j, and to the radius $R := r+1$, whose hypothesis $R \ge 1$ holds because $r \ge 0$. It supplies a constant $C_r \in (0,\infty)$ — in particular $C_r \ne 0$ and $C_r \ne +\infty$, and produced before its input is quantified, hence the same for every input — such that for every $h \in L^1(\gamma_n)$ and every $t \in (0,+\infty]$,

$$\gamma_n\Big(\Big\{x : |x| \le r+1 \text{ and } t < \sup_{\varepsilon>0} \operatorname{ofReal}\big|T_{j,\varepsilon}h(x)\big|\Big\}\Big) \;\le\; \frac{C_r \cdot \operatorname{ofReal}\big(\int |h|\, d\gamma_n\big)}{t}. \tag{4}$$

Put $t_m = \operatorname{ofReal}(1/(m+1))$, which is nonzero because $1/(m+1) > 0$, and $D = C_r/t_m$, which is not $+\infty$ because C_r is not and t_m is not 0. For each k we have $A_{r,m} \subseteq \{x : |x| \le r+1 \text{ and } t_m < \sup_{\varepsilon>0} \operatorname{ofReal}|T_{j,\varepsilon}g_k(x)|\}$, by taking $i = k$ in the defining condition. So by monotonicity, then

(4) with $h = g_k$ and $t = t_m$, then (1) and the monotonicity of ofReal and of multiplication in $[0, +\infty]$,

$$\gamma_n(A_{r,m}) \;\le\; \frac{C_r \cdot \mathrm{ofReal}\big(\int |g_k|\, d\gamma_n\big)}{t_m} \;=\; D \cdot \mathrm{ofReal}\Big(\int |g_k|\, d\gamma_n\Big) \;\le\; D \cdot \mathrm{ofReal}\Big(\frac{1}{k+1}\Big),$$

the middle equality being $ab/c = (a/c)b$ in $[0, +\infty]$. This holds for every k. Now $1/(k+1) \to 0$ as $k \to \infty$, so $\mathrm{ofReal}(1/(k+1)) \to 0$ in $[0, +\infty]$, and since $D \neq +\infty$ also $D \cdot \mathrm{ofReal}(1/(k+1)) \to D \cdot 0 = 0$. A quantity in $[0, +\infty]$ bounded above by every term of a sequence converging to 0 is at most 0, so $\gamma_n(A_{r,m}) = 0$, proving (3).

Step 3: at a point where (3) *holds the truncations are Cauchy.* Let x be such a point. We claim that for every real $\eta > 0$ there is a real $\rho > 0$ with

$$\big|T_{j,\varepsilon} f(x) - T_{j,\delta} f(x)\big| \le \eta \qquad \text{for all } \varepsilon, \delta \in (0, \rho). \tag{5}$$

Fix $\eta > 0$. Choose $m \in \mathbb{N}$ with $1/(m+1) < \eta/8$, which is possible since $\eta/8 > 0$, and let k be as in (3) for this m, so

$$\big|T_{j,\varepsilon} g_k(x)\big| \le \frac{1}{m+1} \qquad \text{for every real } \varepsilon > 0. \tag{6}$$

Since φ_k is smooth with compact support, `PVCoreSmooth` applies to it at the coordinate j and the point x, and its clause (2) gives a real number Φ with $T_{j,\varepsilon}\varphi_k(x) \to \Phi$ as $\varepsilon \downarrow 0$. By the definition of that limit there is a real $\rho > 0$ such that

$$\big|T_{j,\varepsilon}\varphi_k(x) - \Phi\big| < \frac{\eta}{8} \qquad \text{for every } \varepsilon \in (0, \rho). \tag{7}$$

Let $\varepsilon, \delta \in (0, \rho)$. Writing $a = T_{j,\varepsilon}\varphi_k(x)$, $b = T_{j,\varepsilon} g_k(x)$, $c = T_{j,\delta}\varphi_k(x)$, $d = T_{j,\delta} g_k(x)$, all finite reals by Step 1, the identity $a + b - (c + d) = \big((a - \Phi) + (\Phi - c)\big) + \big(b + (-d)\big)$ and the triangle inequality for the absolute value give

$$|a + b - (c + d)| \;\le\; |a - \Phi| + |c - \Phi| + |b| + |d|,$$

using $|\Phi - c| = |c - \Phi|$ and $|-d| = |d|$. By (2) at ε and at δ the left-hand side is $|T_{j,\varepsilon} f(x) - T_{j,\delta} f(x)|$, and by (7) and (6) the right-hand side is at most

$$\frac{\eta}{8} + \frac{\eta}{8} + \frac{1}{m+1} + \frac{1}{m+1} \;<\; \frac{\eta}{8} + \frac{\eta}{8} + \frac{\eta}{8} + \frac{\eta}{8} \;=\; \frac{\eta}{2} \;\le\; \eta.$$

This is (5).

Step 4: a Cauchy family in a complete space converges. Still with x as in Step 3, let $\mathcal{F}$ be the filter of neighbourhoods of 0 relative to $(0, \infty)$, i.e. the filter generated by the sets $(0, \rho)$ for $\rho > 0$; it is not the trivial filter, because every such set is nonempty. Let $\mathcal{G}$ be the image of $\mathcal{F}$ under $\varepsilon \mapsto T_{j,\varepsilon} f(x)$, a filter on $\mathbb{R}$, which is again nontrivial. We check that $\mathcal{G}$ is Cauchy, i.e. that for every real $\eta > 0$ there is a set $S \in \mathcal{G}$ all of whose pairs of points are at distance less than η. Given $\eta > 0$, apply (5) with $\eta/2$ to get $\rho > 0$, and take $S = \{T_{j,\varepsilon} f(x) : \varepsilon \in (0, \rho)\}$. Then $S \in \mathcal{G}$, because $(0, \rho)$ belongs to $\mathcal{F}$ and is contained in the preimage of S; and any two points of S are $T_{j,\varepsilon} f(x)$ and $T_{j,\delta} f(x)$ for some $\varepsilon, \delta \in (0, \rho)$, hence at distance at most $\eta/2 < \eta$ by (5). Since $\mathbb{R}$ is complete (Preamble), the Cauchy filter $\mathcal{G}$ converges: there is a real L with $\mathcal{G}$ finer than the neighbourhood filter of L, which is exactly the assertion that $T_{j,\varepsilon} f(x) \to L$ as $\varepsilon \downarrow 0$.

As the set of x for which (3) holds is of full γ_n-measure by Step 2, this completes the proof. □

2.156. `SmoothCoreTruncationBridge`.

Lemma 2.156 (On a smooth compactly supported datum the truncated Riesz integrals converge to the operator)**.** *Fix $n \ge 1$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel`. Let*

$$T : L^1(\gamma_n; \mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n; \mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`.*

Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth with compact support and γ_n-integrable, let $[\varphi] \in L^1(\gamma_n)$ be its class, and let $j \in \{1, \dots, n\}$. Then for γ_n-almost every $x \in \mathbb{R}^n$,

$$\lim_{\varepsilon \downarrow 0} \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,\varphi(y)\,dy \;=\; \big(T[\varphi]\big)_j(x),$$

the limit taken through $\varepsilon \in (0,\infty)$ as $\varepsilon \to 0$, the y-integral against λ, and the right-hand side the jth coordinate of the canonical representative of $T[\varphi]$.

Only (T1) is assumed, deliberately. *The companion hypothesis $t\,\gamma_n(\{|Tf| > t\}) \le 2\|f\|_{L^1(\gamma_n)}$ that accompanies (T1) elsewhere in the tablet is not used here and is therefore not assumed: on the class of a smooth compactly supported function, (T1) alone already determines T, because such a function lies in $L^2(\gamma_n)$ and so its L^1 class is in the range of ι. The statement is correspondingly a statement about every T satisfying (T1), and it is not vacuous because `L1Extension` exhibits such a T.*

Why the conclusion is almost-everywhere and the limit itself is not. *The limit on the left exists at* every *x, with no exceptional set, and equals the explicit Mehler–subordination integral $\Phi_j(\varphi)(x)$ of `PVCoreSmooth`. The almost-everywhere qualifier attaches only to the identification of that integral with $(T[\varphi])_j$: the right-hand side is the coordinate of an almost-everywhere equivalence class, so no pointwise assertion about it is available or meaningful. A consumer therefore gets convergence everywhere and the value almost everywhere.*

What is combined here, and why none of the pieces says it. `PVCoreSmooth` *identifies the limit with $\Phi_j(\varphi)$ and mentions no operator;* `RieszSemigroupPointwise` *identifies $\Phi_j(\varphi)$ with $(R_\gamma u)_j$ for u the $L^2(\gamma_n)$ class of φ and mentions no truncation; and (T1) carries $R_\gamma u$ to $T[\varphi]$. The conclusion relates the truncated integrals to T, which none of the three does on its own.*

Proof. (a) φ lies in $L^2(\gamma_n)$. Being smooth, φ is continuous; having compact support, it is bounded (Preamble: a continuous function with compact support attains a maximum of its absolute value on the compact support and vanishes off it), so there is a real C with $|\varphi(y)| \le C$ for every y. Since γ_n is a probability measure, hence finite, a bounded measurable function lies in $L^p(\gamma_n)$ for every p; in particular

$$\int_{\mathbb{R}^n} |\varphi|^2\,d\gamma_n \le C^2 < \infty,$$

so $\varphi \in L^2(\gamma_n)$. Let $u \in L^2(\gamma_n)$ be its class.

(b) The spectral identification of the core. Apply `RieszSemigroupPointwise` to φ — smooth, compactly supported, and square γ_n-integrable by (a) — at the coordinate j, with its $L^2(\gamma_n)$ membership hypothesis discharged by (a) and its class taken to be u. Its clause (3) states

$$\Phi_j(\varphi) = \big(R_\gamma u\big)_j \qquad \gamma_n\text{-almost everywhere}, \tag{1}$$

where $\Phi_j(\varphi)$ denotes the function

$$x \;\longmapsto\; \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{1}{\sqrt{s}} \cdot e^{-s} \cdot \frac{1}{\sqrt{2}} \int_{\mathbb{R}^n} \partial_j\varphi\big(e^{-s}x + \sqrt{1-e^{-2s}}\,z\big)\,d\gamma_n(z)\,ds$$

appearing in that node's statement, and $(R_\gamma u)_j$ is the jth coordinate of the canonical representative of $R_\gamma u$.

(c) (T1) carries $R_\gamma u$ to $T[\varphi]$. First, $\iota u = [\varphi]$ as elements of $L^1(\gamma_n)$. Indeed ιu is by the definition in `ToL1` the $L^1(\gamma_n)$ class of the canonical representative of u; that representative agrees γ_n-almost everywhere with φ, because u is the class of φ; and two elements of $L^1(\gamma_n)$ whose representatives agree γ_n-almost everywhere are equal. Hence, by (T1) applied to $v := u$,

$$T[\varphi] \;=\; T(\iota u) \;=\; R_\gamma u \tag{2}$$

as elements of $\mathcal{M}(\gamma_n; \mathbb{R}^n)$. Being equal elements, they have the same canonical representative, so $(T[\varphi])_j(x) = (R_\gamma u)_j(x)$ for *every* x.

(d) Conclusion. Let N be the γ_n-null set off which (1) holds, and let $x \notin N$. Since φ is smooth with compact support, `PVCoreSmooth` applies to it at the coordinate j and the point x,

and its clause (2) gives

$$\lim_{\varepsilon\downarrow 0}\int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\,\varphi(y)\,dy \;=\; \Phi_j(\varphi)(x),$$

a limit of real numbers. By (1), which holds at x because $x\notin N$, the right-hand side equals $(R_\gamma u)_j(x)$, and by (2) that equals $(T[\varphi])_j(x)$. So the displayed limit equals $(T[\varphi])_j(x)$ for every $x\notin N$, which is the assertion. □

2.157. `GaussRieszPrincipalValueIsOperator`.

Lemma 2.157 (Any limit of the truncated Gaussian Riesz integrals is the corresponding coordinate of the operator)**.** *Fix $n\ge 1$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the kernel of `GaussRieszKernel`. Let*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v\in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(T2) *$t\,\gamma_n\big(\{x : |Th(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|h\|_{L^1(\gamma_n)}$ for every $h\in L^1(\gamma_n;\mathbb{R})$ and every real $t>0$.*

For a γ_n-integrable h, a coordinate j, a point x and a real $\varepsilon>0$ write $T_{j,\varepsilon}h(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)h(y)\,dy$, the y-integral against λ; for an element of $L^1(\gamma_n)$ this is read through its canonical representative.

Let $f\in L^1(\gamma_n)$ and let $j\in\{1,\dots,n\}$. Then for γ_n-almost every $x\in\mathbb{R}^n$ the following holds: for every real number L,

$$\lim_{\varepsilon\downarrow 0} T_{j,\varepsilon}f(x) = L \qquad\Longrightarrow\qquad L = (Tf)_j(x),$$

the limit taken through $\varepsilon\in(0,\infty)$ as $\varepsilon\to 0$ and $(Tf)_j(x)$ the jth coordinate of the canonical representative of Tf.

Identification only, and the implication form is the point. *The statement asserts nothing about whether the limit exists: it is vacuously true at any x where no L is a limit. That is deliberate. Existence of the limit is a separate assertion with separate inputs, and a statement combining the two would be the conclusion that a covering node draws from this one together with that one, so stating it here would leave that node nothing to do. Keeping identification and existence apart is what makes each of them independently citable.*

Why quantifying over all L loses nothing. *Limits along $\varepsilon\downarrow 0$ in $\mathbb{R}$ are unique, so at a given x there is at most one such L and the universally quantified form says exactly "the limit, if it exists, is $(Tf)_j(x)$". The quantified form is used because it needs no existence hypothesis and no partial function.*

Why the conclusion is almost-everywhere. *Two exceptional sets are unavoidable, and both are controlled only in measure. The first is where the maximal truncation of the approximation error is large, controlled by `GaussMaximalTruncationLocal`; the second is where T of the error is large, controlled by `CoordinateEndpointBound`. Neither is empty for a general $f\in L^1(\gamma_n)$, and the right-hand side $(Tf)_j$ is in any case a coordinate of an almost-everywhere equivalence class, so no pointwise statement about it is available.*

Both (T1) and (T2) are assumed, because the inputs carry both. *It is a statement about every T satisfying the two, and it is not vacuous because `L1Extension` exhibits such a T.*

Proof. Step 1: the approximants, and the error as a difference of classes. For each $k\in\mathbb{N}$ apply `GaussRieszTruncationSmoothSplit` to f and k. It produces φ_k and $g_k\in L^1(\gamma_n)$ with φ_k smooth, compactly supported and γ_n-integrable; with

$$\int_{\mathbb{R}^n} |g_k|\,d\gamma_n < \frac{1}{k+1}; \tag{1}$$

with $f = \varphi_k + g_k$ γ_n-almost everywhere; and with, for every x and every real $\varepsilon > 0$, each of the three integrands Lebesgue integrable on $\{y : |x - y| > \varepsilon\}$ and

$$T_{j,\varepsilon} f(x) = T_{j,\varepsilon}\varphi_k(x) + T_{j,\varepsilon} g_k(x). \tag{2}$$

Write $\psi_k \in L^1(\gamma_n)$ for the class of φ_k, which exists because φ_k is γ_n-integrable. The third clause says that the canonical representative of g_k agrees γ_n-almost everywhere with $f - \varphi_k$, hence with the canonical representative of $f - \psi_k$; and two elements of $L^1(\gamma_n)$ whose representatives agree almost everywhere are equal, so

$$g_k = f - \psi_k \quad \text{in } L^1(\gamma_n). \tag{3}$$

Also $\|g_k\|_{L^1(\gamma_n)} = \int |g_k| \, d\gamma_n < 1/(k+1)$ by (1).

Step 2: two almost-everywhere facts, for all k at once. For each k, `SmoothCoreTruncationBridge` applied to this T with its property (T1), to φ_k — smooth, compactly supported, γ_n-integrable — and to the coordinate j, gives a γ_n-null set off which

$$\lim_{\varepsilon \downarrow 0} T_{j,\varepsilon}\varphi_k(x) = (T\psi_k)_j(x). \tag{4}$$

And T is linear, so by (3), $Tg_k = Tf - T\psi_k$ in $\mathcal{M}(\gamma_n; \mathbb{R}^n)$; the canonical representative of a difference of classes agrees γ_n-almost everywhere with the difference of the representatives, so there is a γ_n-null set off which

$$(Tg_k)_j(x) = (Tf)_j(x) - (T\psi_k)_j(x). \tag{5}$$

A countable union of γ_n-null sets is γ_n-null, so there is a single γ_n-null set N off which (4) and (5) hold *for every* $k \in \mathbb{N}$. It therefore suffices to show that the set

$$B = \big\{x : \text{some real } L \text{ is a limit of } T_{j,\varepsilon} f(x) \text{ and } L \neq (Tf)_j(x)\big\}$$

satisfies $\gamma_n(B \setminus N) = 0$, since then $\gamma_n(B) \le \gamma_n(B \setminus N) + \gamma_n(N) = 0$ by subadditivity.

Step 3: a two-way split in measure. Note first that the split is two-way and not three-way: (4) identifies the limit of the smooth part *as* $(T\psi_k)_j(x)$ directly, so no separate set on which that identification fails has to be introduced.

Let $x \in B \setminus N$, with witness L. Then $|L - (Tf)_j(x)| > 0$, so there is $m \in \mathbb{N}$ with

$$2 \cdot \frac{1}{m+1} < \big|L - (Tf)_j(x)\big|, \tag{6}$$

and there is $r \in \mathbb{N}$ with $|x| \le r + 1$, for instance $r = \lceil |x| \rceil$. Fix any k. By (5) and the triangle inequality,

$$\big|L - (Tf)_j(x)\big| \le \big|L - (T\psi_k)_j(x)\big| + \big|(Tg_k)_j(x)\big|,$$

so by (6) at least one of the two terms on the right exceeds $1/(m+1)$.

Suppose $|(Tg_k)_j(x)| > 1/(m+1)$. Then x lies in $F_{m,k} = \{x : |(Tg_k)_j(x)| > 1/(m+1)\}$.

Suppose instead $|L - (T\psi_k)_j(x)| > 1/(m+1)$. By (2) at each $\varepsilon > 0$, $T_{j,\varepsilon} g_k(x) = T_{j,\varepsilon} f(x) - T_{j,\varepsilon}\varphi_k(x)$, and both terms on the right converge as $\varepsilon \downarrow 0$ — the first to L by the choice of L, the second to $(T\psi_k)_j(x)$ by (4), available because $x \notin N$. Hence $T_{j,\varepsilon} g_k(x) \to L - (T\psi_k)_j(x)$ and so $|T_{j,\varepsilon} g_k(x)| \to |L - (T\psi_k)_j(x)| > 1/(m+1)$. A convergent family whose limit exceeds $1/(m+1)$ eventually exceeds it, and every neighbourhood of 0 meets $(0, \infty)$, so there is a real $\varepsilon > 0$ with $|T_{j,\varepsilon} g_k(x)| > 1/(m+1)$. Since a single term is at most the supremum and ofReal is strictly increasing on $[0, \infty)$,

$$\operatorname{ofReal}\Big(\frac{1}{m+1}\Big) < \sup_{\varepsilon > 0} \operatorname{ofReal}\big|T_{j,\varepsilon} g_k(x)\big|,$$

the supremum in $[0, +\infty]$; together with $|x| \le r + 1$ this puts x in

$$E_{r,m,k} = \Big\{x : |x| \le r + 1 \text{ and } \operatorname{ofReal}\big(\tfrac{1}{m+1}\big) < \sup_{\varepsilon > 0} \operatorname{ofReal}\big|T_{j,\varepsilon} g_k(x)\big|\Big\}.$$

As k was arbitrary, $x \in \bigcap_k \big(E_{r,m,k} \cup F_{m,k}\big)$, and therefore

$$B \setminus N \subseteq \bigcup_{m \in \mathbb{N}} \bigcup_{r \in \mathbb{N}} \bigcap_{k \in \mathbb{N}} \big(E_{r,m,k} \cup F_{m,k}\big).$$

By countable subadditivity it suffices to show $\gamma_n\big(\bigcap_k (E_{r,m,k} \cup F_{m,k})\big) = 0$ for each fixed r and m.

Fix r and m and put $t_m = \mathrm{ofReal}(1/(m+1)) \neq 0$. Apply `GaussMaximalTruncationLocal` to this T with its two properties, to the coordinate j and to the radius $R := r + 1 \geq 1$; it supplies $C_r \in (0,\infty)$, produced before its input is quantified, with $\gamma_n(E_{r,m,k}) \leq C_r \cdot \mathrm{ofReal}\big(\int |g_k| d\gamma_n\big)/t_m$ for every k. Apply `CoordinateEndpointBound` to this T with its two properties, to the input g_k, the coordinate j and the real level $1/(m+1) > 0$; it gives $t_m \cdot \gamma_n(F_{m,k}) \leq \mathrm{ofReal}(2\|g_k\|_{L^1(\gamma_n)})$, hence $\gamma_n(F_{m,k}) \leq \mathrm{ofReal}(2\|g_k\|_{L^1(\gamma_n)})/t_m$ because t_m is neither 0 nor $+\infty$. Let $D = C_r/t_m + \mathrm{ofReal}(2)/t_m$, which is not $+\infty$ because C_r is not and t_m is not 0. Using $ab/c = (a/c)b$ in $[0,+\infty]$, then (1) and Step 1's norm identity, both bounds become at most $D \cdot \mathrm{ofReal}(1/(k+1))$. So for every k, by monotonicity and subadditivity,

$$\gamma_n\Big(\bigcap_i (E_{r,m,i} \cup F_{m,i})\Big) \leq \gamma_n(E_{r,m,k}) + \gamma_n(F_{m,k}) \leq (D+D) \cdot \mathrm{ofReal}\Big(\frac{1}{k+1}\Big).$$

Now $1/(k+1) \to 0$, so $\mathrm{ofReal}(1/(k+1)) \to 0$ in $[0,+\infty]$, and since $D + D \neq +\infty$ also $(D+D) \cdot \mathrm{ofReal}(1/(k+1)) \to 0$. A quantity in $[0,+\infty]$ bounded above by every term of a sequence converging to 0 is at most 0, so the intersection is γ_n-null, completing the proof. □

3. Main results

3.1. `SqrtDomainSobolev` (paper, Theorem 3.1, $\mathrm{Dom}(B) \subseteq W^{1,2}(\gamma)$).

Lemma 3.1 (The operator domain of $L^{1/2}$ sits inside $W^{1,2}(\gamma_n)$, isometrically)**.** *Let $v \in \mathrm{Dom}(L^{1/2})$ in the sense of `FracPowDomain`. Then $v \in W^{1,2}(\gamma_n)$ in the sense of `SobolevW12`, and*

$$\|L^{1/2}v\|_{L^2(\gamma_n)} = \|Dv\|_{L^2(\gamma_n;\mathbb{R}^n)}, \qquad \textit{equivalently} \qquad \|L^{1/2}v\|_2^2 = \tfrac{1}{2}\int |\nabla v|^2 \, d\gamma_n,$$

with $L^{1/2}$ as in `FracPow` and $D = 2^{-1/2}\nabla$ as in `GaussD`. Only this inclusion and this norm identity are asserted; the reverse inclusion $W^{1,2}(\gamma_n) \subseteq \mathrm{Dom}(L^{1/2})$ is not claimed here and is not used anywhere below.

Proof. Write v_α for the Hermite coefficients of v relative to `HermiteBasis`, so $v_\alpha = \langle v, h_\alpha\rangle$. By `FracPowDomain` with $s = 1/2$,

$$\sum_{\alpha\neq 0} |\alpha|\,|v_\alpha|^2 < \infty. \tag{1}$$

Step 1: the gradient of a single Hermite function. Let $\alpha \in \mathbb{N}_0^n$. By `HermiteFunDeriv`, $D_j h_\alpha = \sqrt{\alpha_j}\, h_{\alpha-e_j}$ for each j, as an identity between polynomials; a classical derivative of a polynomial is in particular a weak Euclidean derivative in the sense of `HasGaussWeakGrad`, because the defining identity is ordinary integration by parts against a smooth compactly supported test function. All the functions h_β lie in $L^2(\gamma_n)$ by `HermiteFunMemLp`, so $Dh_\alpha \in L^2(\gamma_n;\mathbb{R}^n)$ and $h_\alpha \in W^{1,2}(\gamma_n)$ by `SobolevW12`.

Step 2: the Gram matrix of the gradients. Let $\alpha, \beta \in \mathbb{N}_0^n$. The inner product of $L^2(\gamma_n;\mathbb{R}^n)$ pairs matching coordinates, so by Step 1

$$\langle Dh_\alpha, Dh_\beta\rangle = \sum_{j=1}^n \big\langle D_j h_\alpha,\, D_j h_\beta\big\rangle = \sum_{j=1}^n \sqrt{\alpha_j}\,\sqrt{\beta_j}\,\big\langle h_{\alpha-e_j},\, h_{\beta-e_j}\big\rangle.$$

By `HermiteL2Orthonormal` the inner product in the jth summand equals 1 when $\alpha - e_j = \beta - e_j$ and 0 otherwise. We evaluate the jth summand by first disposing of the degenerate indices, because under the convention of `MultiIndexSum` — the jth entry is left at 0 when $\alpha_j = 0$ — the decremented multi-indices can coincide even for $\alpha \neq \beta$: for instance $n = 2$, $j = 1$, $\alpha = (0,5)$, $\beta = (1,5)$ give $\alpha - e_1 = \beta - e_1 = (0,5)$. So it is the prefactor, not the Gram entry, that must kill those summands.

(2a) If $\alpha_j = 0$ or $\beta_j = 0$, the jth summand is 0. Its prefactor is $\sqrt{\alpha_j}\sqrt{\beta_j}$ and one of the two factors is $\sqrt{0} = 0$, so the prefactor is 0; the Gram entry is a real number, hence the product vanishes whatever that entry is.

(2b) If $\alpha_j \geq 1$ and $\beta_j \geq 1$, then $\alpha - e_j = \beta - e_j$ forces $\alpha = \beta$. By `MultiIndexSum` the multi-index $\alpha - e_j$ agrees with α at every entry $i \neq j$ and has jth entry $\alpha_j - 1$, and likewise for β; the convention plays no role here, since both jth entries are at least 1 and the decrement is therefore genuine. So $\alpha - e_j = \beta - e_j$ gives $\alpha_i = \beta_i$ for every $i \neq j$ together with $\alpha_j - 1 = \beta_j - 1$, and the latter yields $\alpha_j = \beta_j$ because both α_j and β_j are at least 1. Hence $\alpha = \beta$.

Suppose now $\alpha \neq \beta$, and let j be arbitrary. If $\alpha_j = 0$ or $\beta_j = 0$, the jth summand is 0 by (2a). Otherwise $\alpha_j \geq 1$ and $\beta_j \geq 1$, and then $\alpha - e_j \neq \beta - e_j$ — equality would force $\alpha = \beta$ by (2b), contradicting $\alpha \neq \beta$ — so the Gram entry is 0 and the jth summand is 0 again. Every summand vanishes, so $\langle Dh_\alpha, Dh_\beta \rangle = 0$.

Suppose instead $\alpha = \beta$. The jth summand is $\sqrt{\alpha_j}\sqrt{\alpha_j}\,\langle h_{\alpha - e_j}, h_{\alpha - e_j}\rangle$, and this equals α_j: if $\alpha_j \geq 1$ the two decremented indices are literally the same multi-index, so the Gram entry is 1 and $\sqrt{\alpha_j}\sqrt{\alpha_j} = \alpha_j$; if $\alpha_j = 0$ the summand is 0 by (2a) and $\alpha_j = 0$ as well. Summing over j and using `MultiIndexSum` gives $\sum_{j=1}^{n} \alpha_j = |\alpha|$. Altogether

$$\langle Dh_\alpha, Dh_\beta \rangle = \begin{cases} |\alpha|, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases} \tag{2}$$

Step 3: the identity on finite Hermite sums. Let $S \subseteq \mathbb{N}_0^n$ be finite and $v_S = \sum_{\alpha \in S} v_\alpha h_\alpha$.

(3a) The gradient side. By Step 1 each h_α lies in $W^{1,2}(\gamma_n)$, so by induction on the finite set S, using the two-term additivity clause and the homogeneity clause of `GaussDLinear` at each step, $v_S \in W^{1,2}(\gamma_n)$ and $Dv_S = \sum_{\alpha \in S} v_\alpha\, Dh_\alpha$; the base case $S = \emptyset$ gives $v_\emptyset = 0$, and taking $c = 0$ in the third clause of `GaussDLinear` puts 0 in $W^{1,2}(\gamma_n)$ with $D0 = 0$. Expanding the squared norm of this finite sum bilinearly and inserting (2),

$$\|Dv_S\|^2 = \sum_{\alpha,\beta \in S} v_\alpha v_\beta \langle Dh_\alpha, Dh_\beta \rangle = \sum_{\alpha \in S} |\alpha|\, |v_\alpha|^2,$$

the off-diagonal terms vanishing by the second case of (2). The term $\alpha = 0$ carries the factor $|0| = 0$ by `MultiIndexSum` and may be dropped, so

$$\|Dv_S\|^2 = \sum_{\alpha \in S,\ \alpha \neq 0} |\alpha|\, |v_\alpha|^2. \tag{3}$$

We now compute $\|L^{1/2} v_S\|_2^2$ and obtain the same quantity. This is done through the Hermite coefficients of v_S, and not by invoking additivity or homogeneity of $L^{1/2}$: by `FracPow` the map $L^{1/2}$ is given by its series only on $\mathrm{Dom}(L^{1/2})$ and is 0 off that set, so membership must be established before the series may be used at all.

(3b) The coefficients of v_S. By `HermiteBasis` the βth Hermite coefficient of an element of $L^2(\gamma_n)$ is its inner product with h_β. Hence, by linearity of the inner product in its second argument over the finite sum defining v_S, and by `HermiteL2Orthonormal`,

$$(v_S)_\beta = \langle h_\beta, v_S \rangle = \sum_{\alpha \in S} v_\alpha \langle h_\beta, h_\alpha \rangle = \begin{cases} v_\beta, & \beta \in S, \\ 0, & \beta \notin S. \end{cases}$$

(3c) $v_S \in \mathrm{Dom}(L^{1/2})$. By (3b) the family $\beta \mapsto |\beta|\, |(v_S)_\beta|^2$ vanishes at every $\beta \notin S$, so

$$\textstyle\sum_{\beta \neq 0} |\beta|\, |(v_S)_\beta|^2 = \sum_{\beta \in S,\ \beta \neq 0} |\beta|\, |v_\beta|^2,$$

a sum of finitely many real numbers, hence finite. By the $s = 1/2$ criterion of `FracPowDomain` this says exactly $v_S \in \mathrm{Dom}(L^{1/2})$.

(3d) The coefficients of $L^{1/2} v_S$, and its norm. Since v_S lies in $\mathrm{Dom}(L^{1/2})$ by (3c), `FracPow` applies to v_S and states that $L^{1/2} v_S$ is the element of $L^2(\gamma_n)$ whose βth Hermite coefficient is 0 at $\beta = 0$ and $|\beta|^{1/2} (v_S)_\beta$ at $\beta \neq 0$; by (3b) that coefficient is $|\beta|^{1/2} v_\beta$ for $\beta \in S$ with $\beta \neq 0$, and 0 for every other β. Parseval's identity of `HermiteBasis` therefore gives

$$\|L^{1/2} v_S\|_2^2 = \sum_{\beta \in S,\ \beta \neq 0} \left(|\beta|^{1/2} v_\beta\right)^2 = \sum_{\beta \in S,\ \beta \neq 0} |\beta|\, |v_\beta|^2. \tag{4}$$

Comparing (3) and (4), $\|Dv_S\| = \|L^{1/2}v_S\|_2$ for every finite $S \subseteq \mathbb{N}_0^n$. That the number $|\alpha|$ carried by these two expressions really is the eigenvalue of the Ornstein–Uhlenbeck differential operator $-\frac{1}{2}\Delta + x\cdot\nabla$ on h_α, so that the spectral multiplier $|\alpha|^{1/2}$ of `FracPow` is the square root of that eigenvalue, is the content of `HermiteFunOU`.

Step 4: passing to the limit on the gradient side. Let $S_N = \{\alpha \in \mathbb{N}_0^n : |\alpha| \le N\}$ and $v_N = v_{S_N}$. Each S_N is finite, since $|\alpha| \le N$ forces $\alpha_j \le N$ for every j, so $S_N \subseteq \{0,1,\dots,N\}^n$. Moreover every finite subset of $\mathbb{N}_0^n$ is contained in some S_N, namely for N the maximum of $|\alpha|$ over that subset; consequently, for any family (c_α) of nonnegative reals with $\sum_\alpha c_\alpha < \infty$, the partial sums $\sum_{\alpha\in S_N} c_\alpha$ increase to $\sum_\alpha c_\alpha$, and therefore the tails $\sum_{\alpha\notin S_N} c_\alpha$ tend to 0 as $N\to\infty$. We use this twice.

First, $\|v - v_N\|_2^2 = \sum_{|\alpha|>N} |v_\alpha|^2$ by (3b) and Parseval (`HermiteBasis`), and $\sum_\alpha |v_\alpha|^2 = \|v\|_2^2 < \infty$ by Parseval again, so $v_N \to v$ in $L^2(\gamma_n)$.

Second, let $M < N$. Then $S_M \subseteq S_N$ and $v_N - v_M = \sum_{\alpha\in S_N\setminus S_M} v_\alpha h_\alpha$, which is $v_{S_N\setminus S_M}$ for the finite set $S_N \setminus S_M$; by `GaussDLinear` (additivity and homogeneity, applied as in (3a)) we have $Dv_N - Dv_M = Dv_{S_N\setminus S_M}$, so (3) gives

$$\|Dv_N - Dv_M\|^2 = \sum_{M<|\alpha|\le N} |\alpha|\,|v_\alpha|^2 \le \sum_{\alpha\notin S_M,\ \alpha\neq 0} |\alpha|\,|v_\alpha|^2,$$

the inequality because the left sum is over a subset of the index set of the right sum and all terms are nonnegative. By (1) the family $\alpha \mapsto |\alpha|\,|v_\alpha|^2$ restricted to $\alpha \neq 0$ is summable with nonnegative terms, so its tails tend to 0; hence $(Dv_N)_N$ is Cauchy in $L^2(\gamma_n;\mathbb{R}^n)$, which is complete, and we let Γ be its limit.

By Step 1 and (3a) each $\sqrt{2}\,Dv_N$ is a weak Euclidean gradient of v_N in the sense of `HasGaussWeakGrad`, and $\sqrt{2}\,Dv_N \to \sqrt{2}\,\Gamma$ in $L^2(\gamma_n;\mathbb{R}^n)$. Applying `GaussWeakGradClosed` to the sequences $v_N \to v$ and $\sqrt{2}\,Dv_N \to \sqrt{2}\,\Gamma$, the field $\sqrt{2}\,\Gamma$ is a weak Euclidean gradient of v. Since $\Gamma \in L^2(\gamma_n;\mathbb{R}^n)$, the criterion of `SobolevW12` is met, so $v \in W^{1,2}(\gamma_n)$, which is the first assertion; and $Dv = \Gamma$ by `GaussD` together with `GaussWeakGradUnique`, since both $\sqrt{2}\,Dv$ and $\sqrt{2}\,\Gamma$ are weak Euclidean gradients of v.

Step 5: passing to the limit on the spectral side, and the norm identity. By hypothesis $v \in \mathrm{Dom}(L^{1/2})$, so `FracPow` applies to v and the βth Hermite coefficient of $L^{1/2}v$ is 0 at $\beta = 0$ and $|\beta|^{1/2}v_\beta$ at $\beta \neq 0$. By (3d) with $S = S_N$, the βth coefficient of $L^{1/2}v_N$ is $|\beta|^{1/2}v_\beta$ for $\beta \in S_N$ with $\beta \neq 0$ and 0 otherwise. The coefficient map of `HermiteBasis` is linear, being an isometric isomorphism onto $\ell^2(\mathbb{N}_0^n)$, so the coefficient family of $L^{1/2}v - L^{1/2}v_N$ is $|\beta|^{1/2}v_\beta$ at those $\beta \neq 0$ with $\beta \notin S_N$ and 0 at all other β. Parseval's identity of `HermiteBasis` then gives

$$\|L^{1/2}v - L^{1/2}v_N\|_2^2 = \sum_{\alpha\notin S_N,\ \alpha\neq 0} |\alpha|\,|v_\alpha|^2,$$

which tends to 0 as $N\to\infty$ by (1) and the tail statement of Step 4. Hence $L^{1/2}v_N \to L^{1/2}v$ in $L^2(\gamma_n)$.

The norm is continuous, so from $Dv_N \to Dv$ (Step 4, since $Dv = \Gamma$ is the limit of Dv_N) and $L^{1/2}v_N \to L^{1/2}v$ we may pass to the limit in the equality $\|Dv_N\| = \|L^{1/2}v_N\|_2$ of Step 3 and conclude

$$\|Dv\| = \lim_{N\to\infty}\|Dv_N\| = \lim_{N\to\infty}\|L^{1/2}v_N\|_2 = \|L^{1/2}v\|_2,$$

which is the second assertion. Finally $D = 2^{-1/2}\nabla$ by `GaussD`, so $\|Dv\|^2 = \frac{1}{2}\int|\nabla v|^2\,d\gamma_n$, giving the second displayed form in the statement. □

3.2. `ObstacleDecomposition` (paper, Theorem 3.1, existence and minimization).

Theorem 3.2 (Gaussian obstacle decomposition)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., put $m = \int F\,d\gamma_n$, and let $a > m$. Then there exist $u \in \mathrm{Dom}(B) = \mathrm{Dom}(L^{1/2})$ and $g \in$*

$L^2(\gamma_n) \cap L^\infty(\gamma_n)$ such that, with $B = L^{1/2}$ as in `FracPow` *and D as in* `GaussD`*,*

$$u \geq 0 \text{ a.e.}, \qquad F = g + Bu \quad \text{in } L^2(\gamma_n),$$
$$\min\{F, a\} \leq g \leq a \quad \text{a.e.},$$
$$\int g \, d\gamma_n = m.$$

Moreover, with $\Omega = \{u > 0\}$,

$$g = a \quad \text{a.e. on } \Omega,$$
$$a\,\gamma_n(\Omega) \leq m,$$
$$Du = 0 \quad \text{a.e. on } \Omega^c.$$

Finally u lies in the cone $\mathcal{K}$ of `ObstacleCone` *and is the unique minimizer over $\mathcal{K}$ of the functional J of* `ObstacleFunctional`*.*

Proof. First note $m \geq 0$, since $F \geq 0$ a.e.; hence $a > m \geq 0$ gives $a > 0$.

Step 1 (the potential). By `ObstacleMinimizerExists` there is $u \in \mathcal{K}$ minimizing J over $\mathcal{K}$. By `ObstacleCone` this gives $u \geq 0$ γ_n-a.e. and $\mathcal{E}(u) < \infty$. By `ObstacleMinimizerUnique`, any $u' \in \mathcal{K}$ that also minimizes J over $\mathcal{K}$ equals u; this is the uniqueness assertion.

Step 2 (the multiplier as a function). By `MultiplierL2Representation`, applicable since $F \geq 0$ a.e., $a > 0$ and u minimizes J, there is $\eta_2 \in L^2(\gamma_n)$ with $\langle \eta, \varphi \rangle = \int \eta_2 \varphi \, d\gamma_n$ for every $\varphi \in \mathcal{V}$, where η is the multiplier functional of `ObstacleMultiplier`. Define

$$g = a\mathbf{1} - \eta_2 \in L^2(\gamma_n).$$

Step 3 (pointwise bounds on g). `MultiplierPointwiseBound` applies to the same data, all of it already in hand: $F \geq 0$ γ_n-a.e. by hypothesis and $a > 0$ as noted at the outset, $u \in \mathcal{K}$ minimizing J over $\mathcal{K}$ by Step 1, and η_2 representing η on $\mathcal{V}$ by Step 2. It gives $0 \leq \eta_2 \leq (a - F)^+$ γ_n-a.e. Hence a.e.

$$g = a - \eta_2 \leq a, \qquad g = a - \eta_2 \geq a - (a - F)^+ = \min\{F, a\},$$

the last identity because $a - (a - F)^+ = \min\{a, F\}$ for all real values. Together with $(a - F)^+ \leq a$ (valid since $F \geq 0$ a.e. and $a > 0$) this also gives $0 \leq g \leq a$ a.e., so $g \in L^\infty(\gamma_n)$.

Step 4 (the decomposition identity and the operator domain). By `DomainUpgrade`, applied with this u and η_2, we have $u \in \mathrm{Dom}(L^{1/2})$ and $L^{1/2}u = F - g$ in $L^2(\gamma_n)$, i.e. $F = g + Bu$.

Step 5 (mass). By `MassConservation`, $\int g \, d\gamma_n = \int (a - \eta_2) \, d\gamma_n = m$.

Step 6 (contact set). By `Complementarity`, $\eta_2 = 0$ γ_n-a.e. on $\Omega = \{u > 0\}$, hence $g = a$ a.e. on Ω.

Step 7 (measure of the contact set). The three hypotheses of `ExceptionalSetBound` hold: its item (1) is Step 5, its item (2) is Step 3 (the bound $0 \leq \eta_2 \leq (a - F)^+$), and its item (3) is Step 6. It therefore gives $a\,\gamma_n(\Omega) \leq m$.

Step 8 (locality of the gradient). By Step 4 and `SqrtDomainSobolev`, $u \in W^{1,2}(\gamma_n)$. Since $u \geq 0$ a.e. by Step 1, `GradientLocality` gives $Du = 0$ γ_n-a.e. on $\{u \leq 0\} = \Omega^c$.

Collecting Steps 1–8 gives all the assertions. □

3.3. `ComplexEndpointTheorem` (paper, Corollary 5.5).

Corollary 3.3 (Endpoint weak-type $(1,1)$ bound for the complex Gaussian Riesz transform)**.** *Let $\mathcal{M}(\gamma_n; \mathbb{C}^n)$ denote the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{C}^n$-valued functions on $\mathbb{R}^n$. There is exactly one $\mathbb{C}$-linear map*

$$T_{\mathbb{C}} : L^1(\gamma_n; \mathbb{C}) \longrightarrow \mathcal{M}(\gamma_n; \mathbb{C}^n)$$

with the following two properties:

(1) *$T_{\mathbb{C}}$ extends the complex transform of* `ComplexRieszTransform` *from L^2, i.e. $T_{\mathbb{C}}(\iota_{\mathbb{C}} v) = R^{\mathbb{C}}_\gamma v$ for every $v \in L^2(\gamma_n; \mathbb{C})$, with $\iota_{\mathbb{C}}$ the inclusion of* `ComplexToL1`*; and*

(2) $T_{\mathbb{C}}$ *satisfies the weak-type bound*

$$\sup_{t>0}\, t\,\gamma_n\big(\{x : |T_{\mathbb{C}}f(x)|_{\mathbb{C}^n} > t\}\big) \;\le\; \frac{8}{\pi}\,\|f\|_{L^1(\gamma_n;\mathbb{C})} \qquad \text{for every } f \in L^1(\gamma_n;\mathbb{C}).$$

The constant $8/\pi$ does not depend on n.

This is the reference paper's complex-valued corollary: its displayed estimate is clause (2), and the phrase "the complex-linear extension" naming the operator it is about is clauses (1) and the $\mathbb{C}$-linearity together, which the last sentence of its proof supplies as a unique continuous *complex-linear extension to L^1. Property (2) is precisely boundedness from $L^1(\gamma_n;\mathbb{C})$ into $L^{1,\infty}(\gamma_n;\mathbb{C}^n)$ with quasi-norm at most $8/\pi$, hence continuity of $T_{\mathbb{C}}$ for the weak L^1 quasi-norm. Both clauses are needed for the uniqueness assertion to have content: clause (2) alone is satisfied by the zero map, so it is the agreement with $R^{\mathbb{C}}_\gamma$ on L^2 in clause (1) that pins the operator down, and the bound in (2) that propagates that agreement from the L^2 functions to all of L^1. Uniqueness is asserted in the larger class of all $\mathbb{C}$-linear maps into $\mathcal{M}(\gamma_n;\mathbb{C}^n)$ satisfying (1) and (2), not merely among continuous ones.*

Proof. Existence. `ComplexL1Extension` supplies a $\mathbb{C}$-linear map $T_{\mathbb{C}} : L^1(\gamma_n;\mathbb{C}) \to \mathcal{M}(\gamma_n;\mathbb{C}^n)$ satisfying (1) and (2); the supremum form of (2) in the present statement is the same assertion as the per-level form there, since a family of real numbers is bounded by a constant exactly when its supremum is.

Uniqueness. Let T and T' both be $\mathbb{C}$-linear and satisfy (1) and (2), and fix $f \in L^1(\gamma_n;\mathbb{C})$.

Step 1: an approximating sequence. By clause (2) of `ComplexRieszL1Limit` applied to f there is a sequence (v_k) in $L^2(\gamma_n;\mathbb{C})$ with $\iota_{\mathbb{C}}v_k \to f$ in $L^1(\gamma_n;\mathbb{C})$, i.e. $\|f - \iota_{\mathbb{C}}v_k\|_{L^1} \to 0$.

Step 2: the two maps agree on the approximants. By clause (1) for T and for T', $T(\iota_{\mathbb{C}}v_k) = R^{\mathbb{C}}_\gamma v_k = T'(\iota_{\mathbb{C}}v_k)$ for every k. Writing $f = (f - \iota_{\mathbb{C}}v_k) + \iota_{\mathbb{C}}v_k$ and using $\mathbb{C}$-linearity of T and of T',

$$Tf - T'f = T\big(f - \iota_{\mathbb{C}}v_k\big) - T'\big(f - \iota_{\mathbb{C}}v_k\big) \qquad \text{for every } k.$$

Step 3: the level sets. Let $\varepsilon > 0$. If $|Tf(x) - T'f(x)|_{\mathbb{C}^n} > 2\varepsilon$ then, by Step 2 and the triangle inequality in $\mathbb{C}^n$, at least one of $|T(f - \iota_{\mathbb{C}}v_k)(x)| > \varepsilon$ and $|T'(f - \iota_{\mathbb{C}}v_k)(x)| > \varepsilon$ holds. Hence, by monotonicity and subadditivity of γ_n (shared context) and then clause (2) applied to each of T and T' with the input $f - \iota_{\mathbb{C}}v_k$ and the level ε,

$$\gamma_n\big(\{|Tf - T'f| > 2\varepsilon\}\big) \le \frac{8}{\pi\varepsilon}\|f - \iota_{\mathbb{C}}v_k\|_{L^1} + \frac{8}{\pi\varepsilon}\|f - \iota_{\mathbb{C}}v_k\|_{L^1} = \frac{16}{\pi\varepsilon}\,\|f - \iota_{\mathbb{C}}v_k\|_{L^1}.$$

Step 4: conclusion. The left-hand side of the last display does not depend on k, while by Step 1 the right-hand side tends to 0 as $k \to \infty$; a nonnegative constant bounded by a sequence tending to 0 is 0, so $\gamma_n(\{|Tf - T'f| > 2\varepsilon\}) = 0$ for every $\varepsilon > 0$. Taking $\varepsilon = 1/j$ for $j \ge 1$ and using

$$\{Tf \ne T'f\} = \bigcup_{j\ge1}\{|Tf - T'f| > 2/j\},$$

a countable union of γ_n-null sets is null by subadditivity, so $Tf = T'f$ γ_n-almost everywhere, i.e. $Tf = T'f$ in $\mathcal{M}(\gamma_n;\mathbb{C}^n)$. As $f \in L^1(\gamma_n;\mathbb{C})$ was arbitrary, $T = T'$. □

3.4. `EndpointTheorem` (paper, Theorem 5.1).

Theorem 3.4 (Endpoint weak-type $(1,1)$ bound for the Gaussian vector Riesz transform)**.** *Let $\mathcal{M}(\gamma_n;\mathbb{R}^n)$ denote the space of γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions on $\mathbb{R}^n$. There is exactly one linear map*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}(\gamma_n;\mathbb{R}^n)$$

with the following two properties:

(1) *T extends the transform of `RieszTransform` from L^2, i.e. $T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1`; and*

(2) *T satisfies the weak-type bound*

$$\sup_{t>0}\, t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \;\le\; 2\,\|f\|_{L^1(\gamma_n)} \qquad \text{for every } f \in L^1(\gamma_n;\mathbb{R}).$$

The constant 2 does not depend on n. Property (2) is precisely boundedness from $L^1(\gamma_n)$ into $L^{1,\infty}(\gamma_n;\mathbb{R}^n)$ with quasi-norm at most 2, hence continuity of T for the weak-L^1 quasi-norm; so the statement is the assertion that R_γ has a unique continuous linear extension from $L^1(\gamma_n)$ to $L^{1,\infty}(\gamma_n;\mathbb{R}^n)$ satisfying the bound with constant 2, and uniqueness is asserted in the larger class of all linear maps into $\mathcal{M}(\gamma_n;\mathbb{R}^n)$ satisfying (1) and (2).

Proof. Existence is `L1Extension`, which supplies a linear $T : L^1(\gamma_n) \to \mathcal{M}(\gamma_n;\mathbb{R}^n)$ satisfying (1) and (2).

For uniqueness, let T and T' both be linear and satisfy (1) and (2), and fix $f \in L^1(\gamma_n)$. Choose $f_k \in L^2(\gamma_n)$ with $f_k \to f$ in $L^1(\gamma_n)$: for instance the truncations $f_k = \max\{-k, \min\{f, k\}\}$, which are bounded, hence in $L^2(\gamma_n)$ since γ_n is a probability measure (`GaussIsProb`), and converge to f in L^1 by dominated convergence. (That $L^2(\gamma_n)$ is dense in $L^1(\gamma_n)$ also follows from `L2IntoL1` together with `HermiteL2Complete`; the explicit truncations suffice here.)

By (1), $T(\iota f_k) = R_\gamma f_k = T'(\iota f_k)$ for every k, so

$$Tf - T'f = T(f - \iota f_k) - T'(f - \iota f_k)$$

by linearity of T and T'. Let $\varepsilon > 0$. If $|Tf(x) - T'f(x)| > 2\varepsilon$ then at least one of $|T(f-\iota f_k)(x)| > \varepsilon$ and $|T'(f - \iota f_k)(x)| > \varepsilon$ holds, so by subadditivity and (2) applied to each of T and T' with the input $f - \iota f_k$ and the level ε,

$$\gamma_n\big(\{|Tf - T'f| > 2\varepsilon\}\big) \le \frac{2}{\varepsilon}\|f - \iota f_k\|_{L^1(\gamma_n)} + \frac{2}{\varepsilon}\|f - \iota f_k\|_{L^1(\gamma_n)} = \frac{4}{\varepsilon}\|f - \iota f_k\|_{L^1(\gamma_n)}.$$

Letting $k \to \infty$, the right-hand side tends to 0, so $\gamma_n(\{|Tf - T'f| > 2\varepsilon\}) = 0$ for every $\varepsilon > 0$. Taking $\varepsilon = 1/j$ and using that $\{Tf \neq T'f\} = \bigcup_{j\ge1}\{|Tf - T'f| > 2/j\}$ is a countable union of null sets, we get $Tf = T'f$ γ_n-almost everywhere, i.e. $Tf = T'f$ in $\mathcal{M}(\gamma_n;\mathbb{R}^n)$. As $f \in L^1(\gamma_n)$ was arbitrary, $T = T'$. □

3.5. `CoordinateEndpointBound` (paper, Corollary 5.4, coordinate bound).

Corollary 3.5 (Endpoint weak-type bound for a single coordinate $R_{\gamma,j}$)**.** *Let $\mathcal{M}(\gamma_n;\mathbb{R}^n)$ be the space of γ_n-a.e. equivalence classes of measurable $\mathbb{R}^n$-valued functions on $\mathbb{R}^n$, and let $T : L^1(\gamma_n;\mathbb{R}) \to \mathcal{M}(\gamma_n;\mathbb{R}^n)$ be any linear map with the two properties*

(1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*
(2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

Then for every $j \in \{1,\dots,n\}$, every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$, the jth coordinate function $x \mapsto (Tf(x))_j$ satisfies

$$t\,\gamma_n\big(\{x : |(Tf(x))_j| > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}.$$

This is the first assertion of the reference paper's corollary on coordinate and historical conventions, $\sup_{t>0} t\,\gamma_n(\{|R_{\gamma,j}f| > t\}) \le 2\|f\|_1$, stated per level t rather than as a supremum; the two forms are the same assertion, since a family of real numbers is bounded by a constant exactly when its supremum is. It differs from property (2) above, and hence from the endpoint theorem it is drawn from, in that the distribution function is taken for the scalar *function $(Tf)_j$ rather than for the Euclidean norm $|Tf|_{\mathbb{R}^n}$ of the whole vector; this is a strictly weaker conclusion for each fixed j and is the form in which the coordinate transforms $R_{\gamma,j}$ are usually quoted.*

Properties (1) and (2) are written out here rather than imported by citation. They are exactly the two properties for which `EndpointTheorem` asserts existence and uniqueness of a map T, so the universally quantified statement above is not vacuous: there is such a T, there is only one, and the statement about that unique extension of R_γ — the one the reference paper writes $R_{\gamma,j}$ for in its jth coordinate — is the case of the present statement at that T.

Proof. Fix j, f and $t > 0$, and choose a representative of Tf, so that $Tf(x)$ is a point of $\mathbb{R}^n$ for every x and $(Tf(x))_j$ is its jth coordinate.

Step 1: the pointwise bound. For every $x \in \mathbb{R}^n$ the jth coordinate of a vector is bounded in absolute value by its Euclidean norm:

$$|(Tf(x))_j| \;=\; \big((Tf(x))_j^2\big)^{1/2} \;\le\; \Big(\sum_{i=1}^{n}(Tf(x))_i^2\Big)^{1/2} \;=\; |Tf(x)|_{\mathbb{R}^n},$$

the inequality because the omitted summands $(Tf(x))_i^2$, $i \neq j$, are nonnegative and $s \mapsto s^{1/2}$ is nondecreasing on $[0,\infty)$. This is the pointwise inequality $|R_{\gamma,j}f| \le |R_\gamma f|_{\mathbb{R}^n}$ with which the reference paper begins its proof.

Step 2: inclusion of the level sets. If $t < |(Tf(x))_j|$ then $t < |Tf(x)|_{\mathbb{R}^n}$ by Step 1, so

$$\{x : |(Tf(x))_j| > t\} \subseteq \{x : |Tf(x)|_{\mathbb{R}^n} > t\}.$$

By monotonicity of a measure (shared context), $\gamma_n(\{|(Tf)_j| > t\}) \le \gamma_n(\{|Tf|_{\mathbb{R}^n} > t\})$.

Step 3: conclusion. Multiplying the inequality of Step 2 by $t > 0$ and then applying hypothesis (2) at this f and this t,

$$t\,\gamma_n\big(\{|(Tf)_j| > t\}\big) \;\le\; t\,\gamma_n\big(\{|Tf|_{\mathbb{R}^n} > t\}\big) \;\le\; 2\,\|f\|_{L^1(\gamma_n)},$$

which is the assertion. Both quantities are elements of $[0,+\infty]$ and the first inequality is multiplication of an inequality in $[0,+\infty]$ by the nonnegative constant t, which preserves the order. □

3.6. `ObstacleDecompositionUnique` (paper, Theorem 3.1, uniqueness).

Theorem 3.6 (Uniqueness of the pair in the Gaussian obstacle decomposition)**.** *Let $F \in L^2(\gamma_n)$ be nonnegative γ_n-a.e., put $m = \int F\,d\gamma_n$, and let $a > m$. Write $B = L^{1/2}$ for the operator of `FracPow`, whose domain $\mathrm{Dom}(B)$ is the case $s = 1/2$ of `FracPowDomain`. Suppose that for each $i \in \{1,2\}$ the pair (u_i, g_i) with $u_i \in \mathrm{Dom}(B)$ and $g_i \in L^2(\gamma_n)$ satisfies*

$$u_i \ge 0 \text{ a.e.}, \qquad F = g_i + Bu_i \;\text{ in } L^2(\gamma_n), \qquad g_i \le a \text{ a.e.}, \qquad g_i = a \text{ a.e. on } \{u_i > 0\}.$$

Then $u_1 = u_2$ and $g_1 = g_2$ in $L^2(\gamma_n)$.

The hypothesis list is exactly the reduced one from which the reference paper derives the uniqueness assertion of its obstacle-decomposition theorem: the mass identity $\int g_i\,d\gamma_n = m$, the lower bound $\min\{F,a\} \le g_i$, the measure bound $a\,\gamma_n(\{u_i > 0\}) \le m$ and the gradient-support identity $Du_i = 0$ a.e. off $\{u_i > 0\}$ are deliberately not *assumed here, so the conclusion applies to every pair meeting the four displayed conditions, in particular to any two pairs satisfying the full conclusion list of that theorem. The ambient hypotheses $F \ge 0$ a.e., $m = \int F$ and $a > m$ are those of the theorem; the first is carried along as part of the ambient setting and the last two are what make the strict height inequality $a - m > 0$ available.*

Finally, a remark on the way the reference paper phrases the potential's regularity. Its decomposition theorem produces u in "$\mathrm{Dom}(B) = W^{1,2}(\gamma_n)$". That equality is the background identity recalled in the paper's section on the Ornstein–Uhlenbeck operator, obtained there from the square-root part of the representation theorem; it is not an assertion that the decomposition theorem proves, and the decomposition argument consumes only the forward inclusion $\mathrm{Dom}(L^{1/2}) \subseteq W^{1,2}(\gamma_n)$ together with the norm identity $\|L^{1/2}v\|_2 = \|Dv\|_2$. Accordingly the present statement places the potentials in $\mathrm{Dom}(B) = \mathrm{Dom}(L^{1/2})$ only.

Proof. Throughout, $\mathcal{E}(\cdot)$ is the energy of `Energy`, $\mathcal{E}(\cdot,\cdot)$ the form of `EnergyBil`, $\mathcal{K}$ the cone of `ObstacleCone`, and J the functional of `ObstacleFunctional` built from the given F and a, so that $J(z) = \frac{1}{2}\mathcal{E}(z) + \int (a - F)\,z\,d\gamma_n$. By `GaussIsProb` the measure γ_n is a probability measure, so the constant function $a\mathbf{1}$ lies in $L^2(\gamma_n)$ and $a\mathbf{1} - F \in L^2(\gamma_n)$.

Fix $i \in \{1,2\}$ and abbreviate $u = u_i$, $g = g_i$. Put

$$\eta = a\mathbf{1} - g \in L^2(\gamma_n).$$

Steps 1–6 show that u minimizes J over $\mathcal{K}$; since i is arbitrary, this holds for both u_1 and u_2.

Step 1: $u \in \mathcal{K}$. By hypothesis $u \in \mathrm{Dom}(L^{1/2})$, so `EnergyFiniteOfSqrtDomain` gives $\mathcal{E}(u) < \infty$. Together with $u \ge 0$ γ_n-a.e., the defining description of the cone in `ObstacleCone` gives $u \in \mathcal{K}$.

Step 2: $\eta \geq 0$ a.e. and $\eta\, u = 0$ a.e. Since $g \leq a$ a.e., we get $\eta = a - g \geq 0$ a.e. For the product, let N_1, N_2, N_3 be γ_n-null sets off which, respectively, $u \geq 0$, $g \leq a$, and the implication $u(x) > 0 \Rightarrow g(x) = a$ holds. Their union N is null by subadditivity of a measure (shared context). Let $x \notin N$. If $u(x) = 0$ then $\eta(x)\, u(x) = 0$. Otherwise $u(x) \neq 0$, and $u(x) \geq 0$ gives $u(x) > 0$; then $g(x) = a$, so $\eta(x) = 0$ and again $\eta(x)\, u(x) = 0$. Hence $\eta\, u = 0$ γ_n-a.e.

Step 3: $Bu = F - g$ and the multiplier identity. The hypothesis $F = g + Bu$ is an identity in $L^2(\gamma_n)$, so $Bu = F - g$ there. Adding $a\mathbf{1} - F$ to both sides,

$$(\dagger) \qquad Bu + a\mathbf{1} - F = a\mathbf{1} - g = \eta \quad \text{in } L^2(\gamma_n).$$

Step 4: the quadratic expansion of J at u. Let $v \in \mathcal{K}$ be arbitrary and put $w = v - u \in L^2(\gamma_n)$. By Step 1 and `ObstacleCone` we have $\mathcal{E}(u) < \infty$ and $\mathcal{E}(v) < \infty$. Clause (4) of `EnergyFormBilinear`, applied first with $v_1 = u$ and $c = -1$, gives $\mathcal{E}((-1)\, u) < \infty$; applied again to the two finite-energy elements v and $(-1)u$ it gives $\mathcal{E}(w) = \mathcal{E}(v + (-1)u) < \infty$.

All three of $\mathcal{E}(u), \mathcal{E}(w), \mathcal{E}(v)$ being finite, with $v = u + w$, clause (3) of `EnergyFormBilinear` applies to every pair drawn from $\{u, w, v\}$. Additivity in the first argument, with second argument $v = u + w$, gives $\mathcal{E}(v, v) = \mathcal{E}(u, v) + \mathcal{E}(w, v)$. Symmetry and additivity again give $\mathcal{E}(u, v) = \mathcal{E}(v, u) = \mathcal{E}(u, u) + \mathcal{E}(w, u)$ and $\mathcal{E}(w, v) = \mathcal{E}(v, w) = \mathcal{E}(u, w) + \mathcal{E}(w, w)$. Using symmetry once more, $\mathcal{E}(w, u) = \mathcal{E}(u, w)$, so

$$\mathcal{E}(v, v) = \mathcal{E}(u, u) + 2\,\mathcal{E}(u, w) + \mathcal{E}(w, w).$$

By clause (2) of `EnergyFormBilinear`, $\mathcal{E}(z, z) = \mathcal{E}(z)$ as real numbers for each of $z \in \{u, w, v\}$, all three having finite energy. Hence

$$(3) \qquad \tfrac{1}{2}\big(\mathcal{E}(v) - \mathcal{E}(u)\big) = \mathcal{E}(u, w) + \tfrac{1}{2}\,\mathcal{E}(w).$$

The linear part of J is handled directly: $a\mathbf{1} - F$, u and v all lie in $L^2(\gamma_n)$, so the products $(a - F)u$ and $(a - F)v$ are γ_n-integrable by the integral form of Cauchy–Schwarz (shared context), and by linearity of the integral $\int (a - F)v - \int (a - F)u = \int (a - F)\, w$. Combining this with (3) and the formula for J in `ObstacleFunctional`,

$$(4) \qquad J(v) - J(u) = \tfrac{1}{2}\,\mathcal{E}(w) + \mathcal{E}(u, w) + \int (a - F)\, w \, d\gamma_n.$$

Step 5: the variational inequality. Since $u \in \mathrm{Dom}(L^{1/2})$ and $w \in L^2(\gamma_n)$ is arbitrary, `EnergyBilIsOperator` applies and gives $\mathcal{E}(u, w) = \langle Bu, w\rangle$, and $\langle Bu, w\rangle = \int (Bu)\, w \, d\gamma_n$ by `L2InnerGaussIntegral`. The products $(Bu)w$ and $(a - F)w$ are γ_n-integrable by Cauchy–Schwarz (shared context), all factors lying in $L^2(\gamma_n)$, so by linearity of the integral and (†),

$$\mathcal{E}(u, w) + \int (a - F)\, w \, d\gamma_n = \int \big(Bu + a - F\big)\, w \, d\gamma_n = \int \eta\, w \, d\gamma_n.$$

Both ηv and ηu are γ_n-integrable by Cauchy–Schwarz, so $\int \eta\, w = \int \eta\, v - \int \eta\, u$, and $\int \eta\, u = 0$ because $\eta u = 0$ a.e. by Step 2. Finally $\eta \geq 0$ a.e. by Step 2 and $v \geq 0$ a.e. because $v \in \mathcal{K}$ (`ObstacleCone`), so $\eta v \geq 0$ a.e. and therefore $\int \eta v \, d\gamma_n \geq 0$. Altogether

$$(5) \qquad \mathcal{E}(u, w) + \int (a - F)\, w \, d\gamma_n = \int \eta\, v \, d\gamma_n \geq 0.$$

Step 6: u minimizes J over $\mathcal{K}$. By `Energy` the energy is a sum of nonnegative terms, so $\mathcal{E}(w) \geq 0$, and it is a real number by Step 4. Substituting (5) into (4),

$$J(v) - J(u) = \tfrac{1}{2}\,\mathcal{E}(w) + \int \eta\, v \, d\gamma_n \geq 0$$

for every $v \in \mathcal{K}$. That is, $J(u) \leq J(v)$ for all $v \in \mathcal{K}$.

Step 7: conclusion. Steps 1 and 6 hold for $i = 1$ and $i = 2$, so $u_1, u_2 \in \mathcal{K}$ and each minimizes J over $\mathcal{K}$. The hypotheses of `ObstacleMinimizerUnique` are therefore met with this F, this $m = \int F \, d\gamma_n$ and this $a > m$, and that lemma gives $u_1 = u_2$. By Step 3 applied to each index, $g_i = F - Bu_i$, whence

$$g_1 = F - Bu_1 = F - Bu_2 = g_2.$$

This is the asserted uniqueness of the pair. □

3.7. `PVIdentification` (paper, Proposition 5.3).

Theorem 3.7 (Principal-value realization of the L^1 extension)**.** *Fix $n \geq 1$, let λ be Lebesgue measure on $\mathbb{R}^n$, let γ_n be the Gaussian measure of `GaussMeasure`, and let $K_{\gamma,j}$ be the* Lebesgue *Gaussian Riesz kernel of `GaussRieszKernel`. Let*

$$T : L^1(\gamma_n;\mathbb{R}) \longrightarrow \mathcal{M}\big(\gamma_n;\mathbb{R}^n\big)$$

be any *linear map into the γ_n-almost-everywhere equivalence classes of measurable $\mathbb{R}^n$-valued functions on $\mathbb{R}^n$ such that*

(T1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(T2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \leq 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

Let $f \in L^1(\gamma_n;\mathbb{R})$, written also for its canonical representative, and for $\varepsilon > 0$ and $j \in \{1,\dots,n\}$ write

$$T_{j,\varepsilon} f(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y)\, f(y)\, dy,$$

the y-integral taken against Lebesgue measure. Then:

(1) *for γ_n-almost every x, and for all n coordinates j simultaneously, the principal value $\lim_{\varepsilon\downarrow 0} T_{j,\varepsilon} f(x)$ exists in $\mathbb{R}$ and equals $\big(Tf(x)\big)_j$, the class Tf being evaluated through any representative, which is immaterial because the assertion is almost everywhere; and*

(2) *every realization of that principal value satisfies the full-vector endpoint bound with constant 2: if $F : \mathbb{R}^n \to \mathbb{R}^n$ is any function such that for γ_n-almost every x and every $j \in \{1,\dots,n\}$ one has $T_{j,\varepsilon} f(x) \to F(x)_j$ as $\varepsilon \downarrow 0$ — here F is a genuine function, not an equivalence class, and the conclusion is about that function — then*

$$t\,\gamma_n\big(\{x : |F(x)|_{\mathbb{R}^n} > t\}\big) \leq 2\,\|f\|_{L^1(\gamma_n)} \qquad \text{for every real } t > 0.$$

This is the reference paper's principal-value proposition, both of its assertions. *Clause (1) is the first: the truncated kernel integrals of an $L^1(\gamma_n)$ input converge almost everywhere, and the limit is the L^1 extension of the spectrally defined transform, with $K_{\gamma,j}$ the Lebesgue kernel as the paper specifies. Clause (2) is the paper's closing sentence, "Consequently, the same full-vector bound 2 holds for this principal-value realization on real inputs": it is carried here as a conjunct of the same statement rather than as a separate result, since it is a consequence of clause (1) and of (T2) and has no independent content. The paper's qualifier* on real inputs *is carried literally: f ranges over the real space $L^1(\gamma_n;\mathbb{R})$ and F is $\mathbb{R}^n$-valued, and nothing is asserted for complex inputs.*

The statement is unconditional. *It carries no hypothesis expressing the fixed-dimensional maximal estimate, no constant depending on a radius, and no restriction of x to a ball; those appear only inside the proof, through `GaussMaximalTruncationLocal`, and are consumed there. Accordingly this node claims no deviation.*

Why clause (2) quantifies over realizations F. *A principal value that exists only almost everywhere does not single out a function, so "this principal-value realization" is not one object but any function realizing the limit; the assertion is that every one of them obeys the bound, which is the strongest reading and the one a consumer can use. The quantifier is not vacuous: by clause (1) the canonical representative of Tf is such an F. Note also that no measurability of F, and none of the level set $\{|F| > t\}$, is assumed: γ_n is applied to that set as an outer measure, which is what (T2) also does, so clause (2) is exactly as strong as (T2) is for Tf itself.*

The hypothetical style, and where the constant 2 comes from. *(T1) and (T2) are written out in full and not imported by citation; the theorem is a statement about every T satisfying them, as `CoordinateEndpointBound` and `GaussMaximalTruncationLocal` are, and it is not vacuous because `L1Extension` exhibits such a T. The constant 2 in clause (2) is (T2)'s own constant, transported along the almost-everywhere identity of clause (1); it is not obtained by tracking any constant through the localized estimates used in the proof, and indeed those constants depend on n, on j and on a radius, while this one does not.*

Proof. Write $\gamma = \gamma_n$ and λ for Lebesgue measure on $\mathbb{R}^n$. Both are applied to arbitrary, not necessarily measurable, sets as outer measures; monotonicity and countable subadditivity, which are all that is used of them, hold in that generality (Preamble, monotonicity and subadditivity of a measure). For a function $g : \mathbb{R}^n \to \mathbb{R}$ and x at which the integrals converge put

$$T_{j,\varepsilon} g(x) = \int_{|x-y|>\varepsilon} K_{\gamma,j}(x,y) g(y)\, dy, \qquad T_{j,*} g(x) = \sup_{\varepsilon>0} \left| T_{j,\varepsilon} g(x) \right| \ \in\ [0,+\infty],$$

the supremum taken in the extended nonnegative reals, so that it is defined whether or not the family is bounded; for an element $g \in L^1(\gamma)$ these are read through its canonical representative. Four facts are used repeatedly and are recorded here in the exact form in which they are applied.

(A1) γ is a probability measure (`GaussIsProb`); in particular $\gamma(S) \le 1$ for every $S \subseteq \mathbb{R}^n$, and every bounded set has finite γ-measure.

(A2) For $h \in L^1(\gamma)$ one has $\|h\|_{L^1(\gamma)} = \int_{\mathbb{R}^n} |h|\, d\gamma < \infty$, the integral of the canonical representative.

(A3) By clause (4) of `GaussLebesgueBallComparison`, a subset of $\mathbb{R}^n$ is γ-null if and only if it is λ-null. Consequently two functions that agree γ-almost everywhere agree λ-almost everywhere, hence have the same Lebesgue integral over every region; in particular $T_{j,\varepsilon}$ is unchanged, *at every x and every $\varepsilon > 0$*, by a γ-almost-everywhere change of its integrand.

(A4) If $A \subseteq B \cup N$ with $\gamma(N) = 0$ then $\gamma(A) \le \gamma(B)$ (Preamble, monotonicity up to null sets).

We also use the elementary facts that $a \mapsto \mathrm{ofReal}(a)$ from $[0,\infty)$ to $[0,+\infty]$ is strictly increasing and additive, and that if $a, b \in [0,+\infty]$ satisfy $b \le \eta + a$ for *every* real $\eta > 0$, the sum read in $[0,+\infty]$, then $b \le a$. (For the latter: if $a = +\infty$ there is nothing to prove, and if $a < \infty$ and $a < b$, choose a real c with $a < c < b$ — take $c = a + 1$ when $b = +\infty$ and $c = (a+b)/2$ otherwise — and apply the hypothesis with $\eta = c - a > 0$ to get $b \le c < b$, a contradiction.)

Step 1: the approximating sequence. For each $k \in \mathbb{N}$ apply `SmoothCompactDenseL1Gauss` to f with $\delta = 1/(k+1) > 0$. It produces a function $\varphi_k : \mathbb{R}^n \to \mathbb{R}$ which is smooth, has compact support and is γ-integrable, with

$$\int_{\mathbb{R}^n} |f(y) - \varphi_k(y)|\, d\gamma(y) < \frac{1}{k+1}. \tag{1}$$

The sequence (φ_k) does not depend on the coordinate j. Let $\psi_k \in L^1(\gamma)$ be the class of φ_k. Then $f - \psi_k \in L^1(\gamma)$, its canonical representative agrees γ-almost everywhere with $f - \varphi_k$, and therefore by (A2) and (1)

$$\|f - \psi_k\|_{L^1(\gamma)} = \int_{\mathbb{R}^n} |f - \varphi_k|\, d\gamma < \frac{1}{k+1}. \tag{2}$$

By (A3) the canonical representative of $f - \psi_k$ agrees with $f - \varphi_k$ λ-almost everywhere, so

$$T_{j,\varepsilon}(f - \psi_k)(x) = T_{j,\varepsilon}(f - \varphi_k)(x) \qquad \text{for every } x \in \mathbb{R}^n,\ \varepsilon > 0,\ j, \tag{3}$$

and likewise $T_{j,*}(f - \psi_k)(x) = T_{j,*}(f - \varphi_k)(x)$ for every x.

Step 2: all truncated integrals converge, at every x and every ε. Fix $x \in \mathbb{R}^n$, $\varepsilon > 0$ and j, and put $R_x = \max(|x|, 1) > 0$. Apply clause (1) of `GaussRieszKernelFarMajorant` with $R := R_x > 0$ and $\rho := \varepsilon > 0$. It supplies a constant $C = C(n, j, R_x, \varepsilon) > 0$ such that for every $g \in L^1(\gamma)$ and every x' with $|x'| \le R_x$, the function $y \mapsto K_{\gamma,j}(x', y) g(y)$ is Lebesgue integrable on $\{y : |x' - y| > \varepsilon\}$ and

$$\left| \int_{|x'-y|>\varepsilon} K_{\gamma,j}(x', y) g(y)\, dy \right| \le C \int_{\mathbb{R}^n} |g|\, d\gamma. \tag{4}$$

Since $|x| \le R_x$, this applies at $x' = x$ to the three γ-integrable functions f, φ_k and $f - \varphi_k$ (the first is an $L^1(\gamma)$ representative, the second is γ-integrable by Step 1, the third is their difference). Hence $T_{j,\varepsilon} f(x)$, $T_{j,\varepsilon} \varphi_k(x)$ and $T_{j,\varepsilon}(f - \varphi_k)(x)$ are *finite real numbers* for every x, every $\varepsilon > 0$ and

every k, and since the integrands satisfy $K_{\gamma,j}(x,y)f(y) = K_{\gamma,j}(x,y)\varphi_k(y) + K_{\gamma,j}(x,y)(f-\varphi_k)(y)$ pointwise in y, additivity of the Lebesgue integral of integrable functions gives

$$T_{j,\varepsilon}f(x) = T_{j,\varepsilon}\varphi_k(x) + T_{j,\varepsilon}(f-\varphi_k)(x) \qquad \text{for every } x,\ \varepsilon>0,\ k. \tag{5}$$

This is the point at which the tablet is stronger than the paper, which obtains finiteness of its truncations from almost-everywhere finiteness of the maximal function; here every single truncation is finite at every point, with no exceptional set, so that citation is not needed and is not made.

Step 3: the smooth core. Fix k and j, and let $x \in \mathbb{R}^n$ be arbitrary. Put

$$\Phi_j^{(k)}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{e^{-s}}{\sqrt{s}} \cdot \frac{1}{\sqrt{2}} \int_{\mathbb{R}^n} \partial_j \varphi_k\big(e^{-s}x + \sqrt{1-e^{-2s}}\, z\big)\, d\gamma(z)\, ds,$$

which is the function Φ_j of `PVCoreSmooth` and of `RieszSemigroupPointwise` for the input φ_k, written out: the inner integral is the Mehler average of $\partial_j\varphi_k$ at time s and the factor $2^{-1/2}$ makes it the energy-normalized derivative D_j. Since φ_k is smooth with compact support, `PVCoreSmooth` applies to it at the coordinate j and the point x, and its clause (2) gives

$$\lim_{\varepsilon \downarrow 0} T_{j,\varepsilon}\varphi_k(x) = \Phi_j^{(k)}(x), \tag{6}$$

a limit of real numbers, valid at *every* x with no exceptional set.

Step 4: the bridge $\Phi_j^{(k)} = (T\psi_k)_j$ almost everywhere. This is a composition of three links, and none of them states it alone.

(a) φ_k is continuous with compact support, hence bounded: by the extreme value theorem (Preamble) it attains a maximum of $|\varphi_k|$ on the compact set $\operatorname{supp}\varphi_k$, and it vanishes off that set, so $|\varphi_k| \le M_k$ for some real M_k. Since γ is a probability measure by (A1), $\int |\varphi_k|^2\, d\gamma \le M_k^2 < \infty$, so $\varphi_k \in L^2(\gamma)$; write $u_k \in L^2(\gamma)$ for its class.

(b) Apply `RieszSemigroupPointwise` to φ_k — smooth, compactly supported, square γ-integrable by (a) — at the coordinate j. Its clause (3) says

$$\Phi_j^{(k)} = \big(R_\gamma u_k\big)_j \qquad \gamma\text{-almost everywhere.} \tag{7}$$

(c) $\iota u_k = \psi_k$ in $L^1(\gamma)$: the canonical representative of ιu_k agrees γ-almost everywhere with that of u_k, which agrees γ-almost everywhere with φ_k, and ψ_k is by definition the $L^1(\gamma)$-class of φ_k; two elements of $L^1(\gamma)$ whose representatives agree almost everywhere are equal. Hence hypothesis (T1) gives $T\psi_k = T(\iota u_k) = R_\gamma u_k$ as elements of $\mathcal{M}(\gamma;\mathbb{R}^n)$, so their canonical representatives agree γ-almost everywhere, and in particular so do their jth coordinates.

Combining (b) and (c),

$$\Phi_j^{(k)} = \big(T\psi_k\big)_j \qquad \gamma\text{-almost everywhere.} \tag{8}$$

Step 5: the oscillation is dominated by the maximal truncation of the error. Fix j. For $x \in \mathbb{R}^n$ and $\rho > 0$ put

$$S_\rho(x) = \sup_{0<\varepsilon,\delta<\rho} \big|T_{j,\varepsilon}f(x) - T_{j,\delta}f(x)\big|, \qquad S_\rho^{(k)}(x) = \sup_{0<\varepsilon,\delta<\rho} \big|T_{j,\varepsilon}\varphi_k(x) - T_{j,\delta}\varphi_k(x)\big|,$$

both suprema taken in $[0,+\infty]$, and

$$\omega_f(x) = \inf_{\rho>0} S_\rho(x) \ \in\ [0,+\infty].$$

$S_\rho(x)$ is nondecreasing in ρ, so $\omega_f(x)$ is the limit of $S_\rho(x)$ as $\rho \downarrow 0$; this is the paper's oscillation at zero.

Claim. For every $x \in \mathbb{R}^n$ and every $k \in \mathbb{N}$,

$$\omega_f(x) \ \le\ 2\, T_{j,*}(f-\psi_k)(x). \tag{9}$$

Indeed, fix $\rho > 0$ and $\varepsilon, \delta \in (0,\rho)$. Using (5) at ε and at δ, all six quantities involved being finite reals by Step 2, the triangle inequality gives

$$\big|T_{j,\varepsilon}f(x) - T_{j,\delta}f(x)\big| \le \big|T_{j,\varepsilon}\varphi_k(x) - T_{j,\delta}\varphi_k(x)\big| + \big|T_{j,\varepsilon}(f-\varphi_k)(x)\big| + \big|T_{j,\delta}(f-\varphi_k)(x)\big|.$$

By (3) the last two terms are $|T_{j,\varepsilon}(f-\psi_k)(x)|$ and $|T_{j,\delta}(f-\psi_k)(x)|$, each at most $T_{j,*}(f-\psi_k)(x)$ by definition of that supremum. Reading the inequality in $[0,+\infty]$ and bounding the first term by its supremum,

$$\big|T_{j,\varepsilon}f(x)-T_{j,\delta}f(x)\big| \le S_\rho^{(k)}(x)+2\,T_{j,*}(f-\psi_k)(x),$$

and the right-hand side does not involve ε or δ, so

$$S_\rho(x) \le S_\rho^{(k)}(x)+2\,T_{j,*}(f-\psi_k)(x) \qquad \text{for every } \rho>0. \tag{10}$$

Next, $\inf_{\rho>0} S_\rho^{(k)}(x)=0$. To see this, let $\eta>0$ and let L be the real limit (6); there is $\rho>0$ with $|T_{j,\varepsilon}\varphi_k(x)-L|<\eta/2$ for all $\varepsilon\in(0,\rho)$, whence $|T_{j,\varepsilon}\varphi_k(x)-T_{j,\delta}\varphi_k(x)|<\eta$ for all $\varepsilon,\delta\in(0,\rho)$ and therefore $S_\rho^{(k)}(x)\le\eta$. Feeding this ρ into (10) gives $\omega_f(x)\le S_\rho(x)\le\eta+2T_{j,*}(f-\psi_k)(x)$ for every real $\eta>0$, and the elementary fact recorded before Step 1 yields (9).

Step 6: $\omega_f=0$ almost everywhere. Still with j fixed, let R be an integer with $R\ge1$ and write $A_R=\{x:|x|\le R\}$. Apply `GaussMaximalTruncationLocal` to the same T, with the same properties (T1) and (T2) written out there, the same coordinate j, and this R. It supplies a constant $C_R\in(0,\infty)$, depending only on n, j and R — in particular produced before the input is quantified, hence the same for every input — such that

$$\gamma\big(\{x:|x|\le R \text{ and } T_{j,*}g(x)>t\}\big) \le \frac{C_R}{t}\,\|g\|_{L^1(\gamma)} \tag{11}$$

for every $g\in L^1(\gamma)$ and every $t\in(0,+\infty]$, the left-hand side being the γ-outer measure of a set about which no measurability is asserted and the right-hand side read in $[0,+\infty]$.

For $k,m\in\mathbb{N}$ put

$$N_{k,m}=\Big\{x:|x|\le R \text{ and } T_{j,*}(f-\psi_k)(x)>\tfrac{1}{m+1}\Big\}.$$

Applying (11) with $g=f-\psi_k$ and $t=1/(m+1)$, and using (2),

$$\gamma(N_{k,m}) \le C_R\,(m+1)\,\|f-\psi_k\|_{L^1(\gamma)} \le \frac{C_R\,(m+1)}{k+1}.$$

Let $N_m=\bigcap_{k\in\mathbb{N}}N_{k,m}$. By monotonicity $\gamma(N_m)\le\gamma(N_{k,m})\le C_R(m+1)/(k+1)$ for every k, and the right-hand side tends to 0 as $k\to\infty$ with m fixed, so $\gamma(N_m)=0$. Let $N_j^{(R)}=\bigcup_{m\in\mathbb{N}}N_m$; by countable subadditivity $\gamma\big(N_j^{(R)}\big)=0$.

Let $x\in A_R\setminus N_j^{(R)}$ and let $m\in\mathbb{N}$. Since $x\notin N_m$ there is k with $x\notin N_{k,m}$, and since $|x|\le R$ this forces $T_{j,*}(f-\psi_k)(x)\le 1/(m+1)$. By (9), $\omega_f(x)\le 2/(m+1)$. As m was arbitrary, $\omega_f(x)=0$.

Finally put $N_j^{\mathrm{osc}}=\bigcup_{R\in\mathbb{N},\,R\ge1}N_j^{(R)}$, a countable union of γ-null sets, so $\gamma(N_j^{\mathrm{osc}})=0$. Every $x\in\mathbb{R}^n$ lies in A_R for some integer $R\ge1$, so for every $x\notin N_j^{\mathrm{osc}}$ we have $\omega_f(x)=0$.

Step 7: the principal value exists off that null set. Let $x\notin N_j^{\mathrm{osc}}$, so $\inf_{\rho>0}S_\rho(x)=0$. For each $m\in\mathbb{N}$ choose $\rho_m>0$ with $S_{\rho_m}(x)<1/(m+1)$, and set $\sigma_m=\min\{\rho_0,\dots,\rho_m\}>0$; then (σ_m) is nonincreasing and, $S_\rho(x)$ being nondecreasing in ρ and $\sigma_m\le\rho_m$,

$$S_{\sigma_m}(x)\le S_{\rho_m}(x)<\frac{1}{m+1}. \tag{12}$$

Put $a_m=T_{j,\sigma_m/2}f(x)$, a finite real by Step 2. For $p\ge q$ both $\sigma_p/2$ and $\sigma_q/2$ lie in $(0,\sigma_q)$, because $\sigma_p\le\sigma_q$; hence $|a_p-a_q|\le S_{\sigma_q}(x)<1/(q+1)$ by (12). So (a_m) is a Cauchy sequence of reals and converges, $\mathbb{R}$ being complete; call its limit L. For $0<\varepsilon<\sigma_m$, both ε and $\sigma_m/2$ lie in $(0,\sigma_m)$, so $|T_{j,\varepsilon}f(x)-a_m|\le S_{\sigma_m}(x)<1/(m+1)$ and therefore

$$\big|T_{j,\varepsilon}f(x)-L\big|\le\frac{1}{m+1}+|a_m-L|.$$

Given $\eta>0$, choose m with $1/(m+1)<\eta/2$ and $|a_m-L|<\eta/2$; then $|T_{j,\varepsilon}f(x)-L|<\eta$ for every $\varepsilon\in(0,\sigma_m)$. Hence

$$T_jf(x):=\lim_{\varepsilon\downarrow0}T_{j,\varepsilon}f(x) \tag{13}$$

exists in $\mathbb{R}$ for every $x \notin N_j^{\mathrm{osc}}$.

Step 8: $T_j f = (Tf)_j$ almost everywhere. Fix j, an integer $R \geq 1$ and a real $t > 0$.

(a) Let $x \notin N_j^{\mathrm{osc}}$. By (5) and (3), $T_{j,\varepsilon}(f-\psi_k)(x) = T_{j,\varepsilon}f(x) - T_{j,\varepsilon}\varphi_k(x)$ for every $\varepsilon > 0$, and both terms on the right converge as $\varepsilon \downarrow 0$, by (13) and (6). Hence $T_{j,\varepsilon}(f-\psi_k)(x) \to T_j f(x) - \Phi_j^{(k)}(x)$, and so $|T_{j,\varepsilon}(f-\psi_k)(x)| \to |T_j f(x) - \Phi_j^{(k)}(x)|$. Now $|T_{j,\varepsilon}(f-\psi_k)(x)| \leq T_{j,*}(f-\psi_k)(x)$ for every ε. If $T_{j,*}(f-\psi_k)(x) = +\infty$ the bound

$$\left|T_j f(x) - \Phi_j^{(k)}(x)\right| \leq T_{j,*}(f-\psi_k)(x) \tag{14}$$

is trivial; otherwise $T_{j,*}(f-\psi_k)(x)$ is a real number M, every term of the convergent family is $\leq M$, and so is its limit. Thus (14) holds for every $x \notin N_j^{\mathrm{osc}}$ and every k. Consequently

$$E_1^{(k)} := \left\{x \in A_R \setminus N_j^{\mathrm{osc}} : \left|T_j f(x) - \Phi_j^{(k)}(x)\right| > \tfrac{t}{3}\right\} \subseteq \left\{x : |x| \leq R \text{ and } T_{j,*}(f-\psi_k)(x) > \tfrac{t}{3}\right\},$$

so (11) with $g = f - \psi_k$ and the level $t/3$, together with (2), gives $\gamma\big(E_1^{(k)}\big) \leq 3C_R/\big(t(k+1)\big)$.

(b) Let $E_2^{(k)} = \{x : \Phi_j^{(k)}(x) \neq (T\psi_k)_j(x)\}$. By (8), $\gamma\big(E_2^{(k)}\big) = 0$.

(c) T is linear, so $T(f-\psi_k) = Tf - T\psi_k$ in $\mathcal{M}(\gamma;\mathbb{R}^n)$; the canonical representative of a difference of classes agrees γ-almost everywhere with the difference of the canonical representatives, so there is a γ-null set N_k' off which $\big(T(f-\psi_k)\big)_j(x) = (Tf)_j(x) - (T\psi_k)_j(x)$. Apply `CoordinateEndpointBound` to the same T, with the same properties (T1) and (T2) written out there, to the input $f - \psi_k \in L^1(\gamma)$, the coordinate j and the level $t/3 > 0$:

$$\tfrac{t}{3}\,\gamma\Big(\big\{x : \big|\big(T(f-\psi_k)\big)_j(x)\big| > \tfrac{t}{3}\big\}\Big) \leq 2\,\|f-\psi_k\|_{L^1(\gamma)} \leq \frac{2}{k+1},$$

using (2). Since $t/3 \in (0,\infty)$ we may divide, and then (A4) with the null set N_k' transfers the bound to

$$E_3^{(k)} := \left\{x : \left|(Tf)_j(x) - (T\psi_k)_j(x)\right| > \tfrac{t}{3}\right\}, \qquad \gamma\big(E_3^{(k)}\big) \leq \frac{6}{t\,(k+1)}.$$

(d) Let $x \in A_R \setminus N_j^{\mathrm{osc}}$ with $|T_j f(x) - (Tf)_j(x)| > t$. If all three of $|T_j f(x) - \Phi_j^{(k)}(x)|$, $|\Phi_j^{(k)}(x) - (T\psi_k)_j(x)|$ and $|(T\psi_k)_j(x) - (Tf)_j(x)|$ were at most $t/3$, the triangle inequality would give $|T_j f(x) - (Tf)_j(x)| \leq t$; so at least one exceeds $t/3$, i.e. $x \in E_1^{(k)} \cup E_2^{(k)} \cup E_3^{(k)}$ (note that $|\Phi_j^{(k)}(x) - (T\psi_k)_j(x)| > t/3 > 0$ forces $x \in E_2^{(k)}$). By subadditivity and (a)–(c), for every k,

$$\gamma\Big(\big\{x \in A_R \setminus N_j^{\mathrm{osc}} : |T_j f(x) - (Tf)_j(x)| > t\big\}\Big) \leq \frac{3C_R}{t(k+1)} + 0 + \frac{6}{t(k+1)} = \frac{3C_R+6}{t\,(k+1)}.$$

The left-hand side does not depend on k, and the right-hand side tends to 0 as $k \to \infty$; a quantity in $[0,+\infty]$ bounded by every term of a sequence tending to 0 is 0. Hence that γ-measure is 0.

This is where the paper invokes uniqueness of a limit in measure. The three-way split above is the quantitative form of that argument and gives the conclusion directly from the two weak-type bounds, so no separate uniqueness statement is needed; the ball A_R has finite γ-measure by (A1), but that finiteness is not used, and the role of A_R here is solely that (11) is available only on it.

(e) Taking the union over the levels $t = 1/(m+1)$, $m \in \mathbb{N}$, over the integers $R \geq 1$, and with N_j^{osc} adjoined, countable subadditivity produces a γ-null set $N_j \supseteq N_j^{\mathrm{osc}}$ such that for every $x \notin N_j$ the limit $T_j f(x)$ of (13) exists and

$$T_j f(x) = (Tf)_j(x). \tag{15}$$

(Every $x \notin N_j^{\mathrm{osc}}$ with $T_j f(x) \neq (Tf)_j(x)$ lies in A_R for some integer $R \geq 1$ and satisfies $|T_j f(x) - (Tf)_j(x)| > 1/(m+1)$ for some m, so it lies in one of the countably many null sets just bounded.)

Step 9: the coordinate assembly, giving clause (1). Steps 5 to 8 were carried out for a fixed coordinate j, using the j-independent approximants of Step 1, and produced a γ-null set N_j. Doing this for each of the n coordinates and putting $N_* = \bigcup_{j=1}^n N_j$ — a finite union,

so $\gamma(N_*) = 0$ by subadditivity — we obtain: for every $x \notin N_*$ and every $j \in \{1, \dots, n\}$, the limit $\lim_{\varepsilon \downarrow 0} T_{j,\varepsilon} f(x)$ exists in $\mathbb{R}$ and equals $(Tf(x))_j$. This is clause (1), and the finiteness of the number of coordinates is exactly the paper's reason for the same step.

Step 10: clause (2). Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be as in clause (2), and let N_F be a γ-null set off which, for every j, $T_{j,\varepsilon} f(x) \to F(x)_j$ as $\varepsilon \downarrow 0$. Put $N' = N_* \cup N_F$, so $\gamma(N') = 0$. For $x \notin N'$ and each j the family $\varepsilon \mapsto T_{j,\varepsilon} f(x)$ converges, as $\varepsilon \downarrow 0$, both to $F(x)_j$ and to $(Tf(x))_j$ by Step 9. The filter of right-hand neighbourhoods of 0 is not the trivial filter, since every neighbourhood of 0 meets $(0, \infty)$, and $\mathbb{R}$ is Hausdorff, so a limit along it is unique; hence $F(x)_j = (Tf(x))_j$ for every j, and therefore $F(x) = Tf(x)$ as elements of $\mathbb{R}^n$, two vectors with the same n coordinates being equal.

Now fix a real $t > 0$. Then

$$\{x : |F(x)| > t\} \subseteq \{x : |Tf(x)| > t\} \cup N', \qquad \{x : |Tf(x)| > t\} \subseteq \{x : |F(x)| > t\} \cup N',$$

since the two functions agree off N'. By (A4) in both directions the two sets have the same γ-outer measure. Hypothesis (T2), applied to f and this t, gives $t\,\gamma(\{x : |Tf(x)| > t\}) \le 2\|f\|_{L^1(\gamma)}$, and therefore

$$t\,\gamma\big(\{x : |F(x)| > t\}\big) \le 2\,\|f\|_{L^1(\gamma)},$$

which is clause (2). The constant is (T2)'s, transported along the almost-everywhere identity $F = Tf$ of the previous paragraph; the constants C_R of (11), which depend on n, j and R, enter only through Steps 6 and 8 and do not appear in this bound. This is the paper's closing remark that the constant 2 is supplied by its endpoint theorem and not by the constants used for the identification. □

3.8. `SpectralNormalizationBound` (paper, Corollary 5.4, spectral normalization).

Corollary 3.8 (Endpoint bound for the historical spectral normalization $\mathcal{R} = -\sqrt{2}\,R_\gamma$)**.** *Let $\mathcal{M}(\gamma_n; \mathbb{R}^n)$ be the space of γ_n-a.e. equivalence classes of measurable $\mathbb{R}^n$-valued functions on $\mathbb{R}^n$, and let $T : L^1(\gamma_n; \mathbb{R}) \to \mathcal{M}(\gamma_n; \mathbb{R}^n)$ be any linear map with the two properties*

(1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n; \mathbb{R})$ and every real $t > 0$.*

Then the scalar multiple $(-\sqrt{2})\,T$, which is again a linear map $L^1(\gamma_n; \mathbb{R}) \to \mathcal{M}(\gamma_n; \mathbb{R}^n)$, satisfies

$$t\,\gamma_n\big(\{x : |((-\sqrt{2})\,T)f(x)|_{\mathbb{R}^n} > t\}\big) \;\le\; 2\sqrt{2}\,\|f\|_{L^1(\gamma_n)}$$

for every $f \in L^1(\gamma_n; \mathbb{R})$ and every real $t > 0$.

This is the second assertion of the reference paper's corollary on coordinate and historical conventions, its displayed bound $\sup_{t>0} t\,\gamma_n(\{|\mathcal{R}f| > t\}) \le 2\sqrt{2}\|f\|_1$ for the operator $\mathcal{R} = -\nabla L^{-1/2}\Pi_0$ of the historical Fabes–Gutiérrez–Scotto normalization, stated per level t rather than as a supremum. No new operator has to be introduced for it: the paper's own spectral computation identifies $\mathcal{R} = -\sqrt{2}\,R_\gamma$ coordinatewise, the factor $\sqrt{2}$ coming from $\partial_j = \sqrt{2}\,D_j$, so $\mathcal{R}$ is literally the scalar multiple $(-\sqrt{2})\,T$ of the extension T. The content is therefore a bound for a different operator *— one whose weak-type constant is $2\sqrt{2}$, not 2 — and not a restatement of property (2): the two inequalities have different left-hand sides and different constants, and neither constant can be replaced by the other.*

Properties (1) and (2) are written out here rather than imported by citation. They are exactly the two properties for which `EndpointTheorem` asserts existence and uniqueness of a map T, so the universally quantified statement above is not vacuous and its instance at that unique T is the paper's assertion about $\mathcal{R}$.

Proof. Fix $f \in L^1(\gamma_n; \mathbb{R})$ and a real $t > 0$, and write $\lambda = \sqrt{2} > 0$.

Step 1: the pointwise norm identity. Scalar multiplication in $\mathcal{M}(\gamma_n; \mathbb{R}^n)$ is defined representativewise, and T is linear in the sense that $((-\lambda)T)f = (-\lambda)\,(Tf)$ as elements of $\mathcal{M}(\gamma_n; \mathbb{R}^n)$; so for γ_n-almost every x,

$$|((-\lambda)T)f(x)|_{\mathbb{R}^n} = |(-\lambda)\,Tf(x)|_{\mathbb{R}^n} = |-\lambda|\;|Tf(x)|_{\mathbb{R}^n} = \lambda\,|Tf(x)|_{\mathbb{R}^n},$$

using absolute homogeneity of the Euclidean norm and $\lambda > 0$. (The identity holds almost everywhere rather than at every point because the displayed objects are a.e. equivalence classes; that is all that is used below, since the sets occurring are measured by γ_n.)

Step 2: the level set at t is a level set of Tf at t/λ. Let x be a point where the identity of Step 1 holds. Since $\lambda > 0$,

$$t < |((-\lambda)T)f(x)|_{\mathbb{R}^n} \iff t < \lambda\,|Tf(x)|_{\mathbb{R}^n} \iff \frac{t}{\lambda} < |Tf(x)|_{\mathbb{R}^n},$$

the last step being division of a strict inequality by the positive number λ. The two sets $\{x : |((-\lambda)T)f(x)|_{\mathbb{R}^n} > t\}$ and $\{x : |Tf(x)|_{\mathbb{R}^n} > t/\lambda\}$ therefore differ only inside a γ_n-null set, so they have the same γ_n-measure (monotonicity up to null sets, shared context, applied in both directions):

$$\gamma_n\big(\{|((-\lambda)T)f|_{\mathbb{R}^n} > t\}\big) = \gamma_n\big(\{|Tf|_{\mathbb{R}^n} > t/\lambda\}\big).$$

Step 3: homogeneity of the weak quasi-norm. Apply hypothesis (2) to this f at the level $t/\lambda > 0$:

$$\frac{t}{\lambda}\,\gamma_n\big(\{|Tf|_{\mathbb{R}^n} > t/\lambda\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}.$$

Multiplying by $\lambda > 0$ and using Step 2,

$$t\,\gamma_n\big(\{|((-\lambda)T)f|_{\mathbb{R}^n} > t\}\big) = \lambda \cdot \frac{t}{\lambda}\,\gamma_n\big(\{|Tf|_{\mathbb{R}^n} > t/\lambda\}\big) \le \lambda \cdot 2\,\|f\|_{L^1(\gamma_n)} = 2\sqrt{2}\,\|f\|_{L^1(\gamma_n)},$$

which is the assertion. This is exactly the step the reference paper calls the homogeneity of the weak-L^1 quasi-norm: rescaling the operator by λ rescales the level at which the endpoint bound is applied by $1/\lambda$ and the resulting constant by λ. □

3.9. `UnnormalizedMeasureInvariance` (paper, Corollary 5.4, unnormalized measure).

Corollary 3.9 (Both endpoint conventions are unchanged by rescaling the measure)**.** *Let $\mathcal{M}(\gamma_n;\mathbb{R}^n)$ be the space of γ_n-a.e. equivalence classes of measurable $\mathbb{R}^n$-valued functions on $\mathbb{R}^n$, and let $T : L^1(\gamma_n;\mathbb{R}) \to \mathcal{M}(\gamma_n;\mathbb{R}^n)$ be any linear map with the two properties*

(1) *$T(\iota v) = R_\gamma v$ for every $v \in L^2(\gamma_n)$, with ι the inclusion of `ToL1` and R_γ the transform of `RieszTransform`; and*

(2) *$t\,\gamma_n\big(\{x : |Tf(x)|_{\mathbb{R}^n} > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}$ for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$.*

Let $c > 0$ be real and let $\mu = c\,\gamma_n$ be the rescaled measure, so $\mu(A) = c\,\gamma_n(A)$ for every Borel A. Then for every $f \in L^1(\gamma_n;\mathbb{R})$ and every real $t > 0$, with the same *operator T:*

(a) *for every $j \in \{1,\dots,n\}$,*

$$t\,\mu\big(\{x : |(Tf(x))_j| > t\}\big) \le 2\int |f|\,d\mu;$$

(b)

$$t\,\mu\big(\{x : |((-\sqrt{2})\,T)f(x)|_{\mathbb{R}^n} > t\}\big) \le 2\sqrt{2}\int |f|\,d\mu.$$

This is the last sentence of the reference paper's corollary on coordinate and historical conventions: "Both statements remain unchanged if γ is replaced by the unnormalized measure $e^{-|x|^2}\,dx$, with the same operator." The two statements meant are exactly the two the paper displays — the coordinate bound with constant 2 and the historical bound for $\mathcal{R} = -\sqrt{2}\,R_\gamma$ with constant $2\sqrt{2}$ — and they reappear here as (a) and (b). Since γ_n has density $\pi^{-n/2}e^{-|x|^2}$ with respect to Lebesgue measure (`GaussMeasure`), the paper's unnormalized measure $e^{-|x|^2}\,dx$ is the case $c = \pi^{n/2}$; the statement is given for a general real $c > 0$ because the paper's own justification — that both the distribution measure and the input L^1 norm acquire the same factor — is the general scaling argument and uses nothing about the particular value $\pi^{n/2}$.

Two points of formulation. First, the right-hand sides are written as integrals $\int |f|\,d\mu$ rather than as $\|f\|_{L^1(\mu)}$: this keeps f an element of $L^1(\gamma_n)$ and T the same operator on the same space, which is what the paper's "with the same operator" asks for, and avoids re-erecting the Lebesgue space over μ. This is harmless, since $c > 0$ makes μ and γ_n mutually absolutely continuous, so they have the same null sets, the same a.e. equivalence classes, and the same integrable functions,

and $\int |f|\,d\mu = c\int |f|\,d\gamma_n = c\,\|f\|_{L^1(\gamma_n)}$ *is finite. Second, the content here is neither of the two bounds being rescaled: those hold over* γ_n*, while (a) and (b) are assertions about the measure* μ*, and what they record is precisely that the weak-type constants* 2 *and* $2\sqrt{2}$ *are invariant under the rescaling, i.e. that they are properties of the operator and not artifacts of the normalization of the underlying measure.*

Properties (1) and (2) are written out here rather than imported by citation. They are exactly the two properties for which `EndpointTheorem` *asserts existence and uniqueness of a map* T*, so the universally quantified statement above is not vacuous and its instance at that unique* T *is the paper's assertion.*

Proof. Fix $c > 0$, $f \in L^1(\gamma_n;\mathbb{R})$ and a real $t > 0$, and put $\mu = c\,\gamma_n$.

Step 1: both sides carry the same factor c*.* By definition of the rescaled measure, $\mu(A) = c\,\gamma_n(A)$ for every Borel set A, as an identity in $[0,+\infty]$. Likewise, integration against $c\,\gamma_n$ is c times integration against γ_n, so

$$\int |f|\,d\mu = c\int |f|\,d\gamma_n = c\,\|f\|_{L^1(\gamma_n)}, \tag{1}$$

the second equality being the definition of the $L^1(\gamma_n)$ norm of f as the integral of $|f|$ against γ_n; both quantities are finite because $f \in L^1(\gamma_n)$.

Step 2: part (a). Let $j \in \{1,\dots,n\}$. By `CoordinateEndpointBound`, applied to this T with its hypotheses (1) and (2) — which are the hypotheses of the present statement verbatim — and to this f, this j and this $t > 0$,

$$t\,\gamma_n\big(\{|(Tf)_j| > t\}\big) \le 2\,\|f\|_{L^1(\gamma_n)}.$$

Multiplying this inequality in $[0,+\infty]$ by the nonnegative constant c, which preserves the order, and using Step 1 on both sides,

$$t\,\mu\big(\{|(Tf)_j| > t\}\big) = c\,t\,\gamma_n\big(\{|(Tf)_j| > t\}\big) \le c\cdot 2\,\|f\|_{L^1(\gamma_n)} = 2\int |f|\,d\mu,$$

which is (a).

Step 3: part (b). By `SpectralNormalizationBound`, applied to the same T with the same two hypotheses and to this f and this $t > 0$,

$$t\,\gamma_n\big(\{|((-\sqrt{2})T)f|_{\mathbb{R}^n} > t\}\big) \le 2\sqrt{2}\,\|f\|_{L^1(\gamma_n)}.$$

Multiplying by $c \ge 0$ and using Step 1 on both sides exactly as in Step 2,

$$t\,\mu\big(\{|((-\sqrt{2})T)f|_{\mathbb{R}^n} > t\}\big) = c\,t\,\gamma_n\big(\{|((-\sqrt{2})T)f|_{\mathbb{R}^n} > t\}\big) \le c\cdot 2\sqrt{2}\,\|f\|_{L^1(\gamma_n)} = 2\sqrt{2}\int |f|\,d\mu,$$

which is (b). In both parts the factor c appears once on each side and cancels, which is the reference paper's "scaling the measure scales both sides by the same factor". □

Appendix A. Deviation from the paper

This appendix reproduces the deviation note that was reviewed during the formalization. Line numbers in it refer to the LaTeX source of the paper.

1. The affected node. This deviation is claimed by exactly one tablet node, `GaussMaximalTruncationLocal`, and by no other. No other node's statement or proof depends on it, and it relies on no other deviation: the return-to-faithful argument below uses only the reference paper and the tablet's own nodes.

2. The paper statement, and the departure. The reference paper displays, as (5.15) of the paper (paper lines 2123–2125), the fixed-dimensional maximal estimate it takes from Fabes, Gutiérrez and Scotto:

$$\gamma\Big(\Big\{\sup_{\varepsilon>0}|T_{j,\varepsilon}h| > t\Big\}\Big) \le \frac{C_{n,j}}{t}\,\|h\|_1, \qquad t > 0.$$

The level set here is taken over all of $\mathbb{R}^n$, and the constant depends only on n and j.

`GaussMaximalTruncationLocal` states instead: for every $R \geq 1$ there is $C = C(n, j, R)$ such that for every $f \in L^1(\gamma_n)$ and every $t \in (0, +\infty]$,

$$\gamma_n\Big(\Big\{x : |x| \leq R \text{ and } \sup_{\varepsilon>0} |T_{j,\varepsilon} f(x)| > t\Big\}\Big) \leq \frac{C}{t}\, \|f\|_{L^1(\gamma_n)}.$$

The departure is therefore exactly two things, and nothing else:

(1) the level set is intersected with the closed ball $\overline{B}(0, R)$; and
(2) the constant is allowed to depend on R in addition to n and j.

The operator, the truncations $T_{j,\varepsilon}$, the coordinate index j (the paper's display is already coordinatewise, so there is no change there), the class of inputs, the exponent, and the shape $C\|f\|_1/t$ are the paper's, unchanged. The departure is a *weakening* of an intermediate estimate: the localized statement is implied by the paper's for each R, so nothing is claimed that the paper does not already assert.

3. Why the departure is minor.
The paper licenses the dimension dependence outright. Immediately after the display, at line 2127, the paper writes:

> "No uniformity in n is needed in this step."

So the constant at this step is explicitly allowed to degrade in n, and the only genuine addition is the dependence on R.
The R-dependence is forced, and is confined to this step. The tablet's route to this estimate runs a Cotlar inequality, `GaussRieszCotlarLocal`, whose constant degrades in R through two places: the near-diagonal kernel estimates of `GaussRieszKernelNearBounds`, whose constant carries the factor $e^{|x|^2-|y|^2}$ bounded only on bounded sets of x; and the lower bound $a_{R+2} = \pi^{-n/2} e^{-(R+2)^2}$ for the Gaussian density on $\overline{B}(0, R+2)$, which is what lets a local Gaussian averaging argument be run against the *Lebesgue* Hardy–Littlewood maximal function and so avoids covering theory for the non-doubling measure γ_n. Both degrade as $|x|$ grows; recovering the global-in-x constant is precisely the theorem the paper cites to [FGS94] and is not reproved here.

4. Where the difference is absorbed. The estimate is used in the paper's proof of Proposition 5.3 of the paper exactly three times, at paper lines 2129–2156, and every one of the three is an almost-everywhere statement or a limit statement:

(1) $\gamma(\{\omega_f > t\}) \leq 2C_{n,j}\|f - f_k\|_1/t$, from which "letting $k \to \infty$ shows that $\omega_f = 0$ almost everywhere" (lines 2142–2144);
(2) almost-everywhere finiteness of the maximal truncation, "so the principal value exists and is finite" (lines 2145–2146);
(3) $\gamma(\{|T_j f - T_j f_k| > t\}) \leq C_{n,j}\|f - f_k\|_1/t$, i.e. convergence in measure (lines 2149–2153).

None of the three asserts anything at a single point, and none of the three retains the constant: in each, the constant is multiplied by a quantity tending to 0, or the conclusion is that a set is null.

The tablet's covering node `PVIdentification` therefore uses the localized estimate as follows. For each fixed $R \in \mathbb{N}$ it runs the three steps above *on* $\overline{B}(0, R)$, obtaining in each case an assertion that holds for γ_n-almost every $x \in \overline{B}(0, R)$. Since $\mathbb{R}^n = \bigcup_{R\in\mathbb{N}} \overline{B}(0, R)$ and a countable union of γ_n-null sets is γ_n-null, the three assertions then hold for γ_n-almost every $x \in \mathbb{R}^n$. Two further remarks make the absorption exact:

- Use (2) is not even needed in the tablet's route: clause (1)(b) of `GaussRieszKernelFarMajorant` already makes *every* $T_{j,\varepsilon} f(x)$ a finite number, so $\omega_f(x) = 0$ by itself gives the Cauchy property and a finite limit.
- Use (3) closes on $\overline{B}(0, R)$ because that set has finite γ_n-measure, so uniqueness of limits in measure applies there; it is the finiteness of the localizing set, not the size of the constant, that this step needs.

Consequently `PVIdentification`'s own statement is the paper's proposition verbatim: it is quantified over all of $\mathbb{R}^n$ and all of $L^1(\gamma_n)$, it contains no R, and it contains no constant coming from this estimate. No consumer inherits the localization.

The paper itself isolates the absorption point. At line 2158 the paper writes, of the identification argument:

> "The constant 2 is supplied by Theorem 5.1 of the paper, not by the constants $C_{n,j}$ used for this identification."

That sentence is the return-to-faithful argument in the paper's own words. The constants of the maximal estimate — here $C(n, j, R)$ — do only almost-everywhere and limit work; the one constant that survives into the proposition's final clause, the full-vector bound 2 for the principal-value realization, comes from Theorem 5.1 of the paper through `CoordinateEndpointBound`, which is a different source entirely and is untouched by this deviation.

5. What this deviation does *not* claim. The paper, at lines 2275–2289, is explicit about what is open:

> "Hard truncation does not preserve this identity, so the obstacle argument does not yield a weak-type $(1, 1)$ estimate for the Gaussian maximal truncation $R_{\gamma,*}$ ... A dimension-independent weak-type bound for either maximal truncation is a separate question, not a consequence of the present argument."

Nothing here contradicts or silently strengthens that passage. The passage denies two things: that a *dimension-independent* weak-type bound for the maximal truncation follows, and that the *obstacle argument* yields one.

`GaussMaximalTruncationLocal` claims neither. Its constant $C(n, j, R)$ degrades in n, in j and in R, so it is not dimension independent. And its proof runs no obstacle argument on truncations: it consumes that argument's finished output — the untruncated weak-type bound with constant 2, as `CoordinateEndpointBound` states it — as a black box inside a classical Cotlar argument, together with the Euclidean Hardy–Littlewood layer of `LocalMaximalWeakType` and `LocalMaximalWeakL2` and the kernel estimates of `GaussRieszKernelFarMajorant` and `GaussRieszKernelNearBounds`. The exact identity $R_\gamma L^{1/2} u = Du$, which the quoted passage says hard truncation destroys, is never applied to a truncated operator anywhere in that chain.